In [1]:
```
import os
import glob
import copy
import datetime
import matplotlib.pyplot as plt
from matplotlib import colors
from matplotlib import colormaps
import matplotlib.patches as patches
import numpy as np
import pandas as pd
import scipy.stats as stats

from sklearn import metrics
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

import gsf_ims_fitness as fitness

import pickle

import seaborn as sns
import cmocean
import palettable
sns.set()

from Bio.Seq import Seq

%load_ext autoreload
%autoreload 2

%matplotlib inline

%autosave 0

sns.set_style("white")
sns.set_style("ticks", {'xtick.direction':'in', 'xtick.top':True, 'ytick.directio

plt.rcParams['axes.labelsize'] = 16
plt.rcParams['xtick.labelsize'] = 14
plt.rcParams['ytick.labelsize'] = 14

plt.rcParams['legend.fontsize'] = 12
plt.rcParams['legend.edgecolor'] = 'k'
```

```
Autosave disabled
```

In [ ]:
```
```

In [2]:
```
notebook_directory = os.getcwd()
notebook_directory
```

Out[2]:
```
'C:\\Users\\djross\\Documents\\Jcloud\\GSF-IMS\\E-Coli\\pLMSF-lacI\\2022-11-08_
two-lig_two-sel_DNA-5-plates\\BarSeq\\HiSeq'
```

In [3]:
```
%%time
os.chdir(notebook_directory)
pickle_file = '2022-11-08_two-lig_two-sel_DNA-5-plates_single_amino_BarSeqFitness
print(pickle_file)

barcode_frame = pickle.load(open(pickle_file, 'rb'))

hiseq_count_frame = barcode_frame.barcode_frame
```

```
2022-11-08_two-lig_two-sel_DNA-5-plates_single_amino_BarSeqFitnessFrame.pkl
CPU times: total: 484 ms
Wall time: 672 ms
```

In [4]:
```
hiseq_count_frame.shape
```

Out[4]:
```
(978, 597)
```

In [5]:
```
len(np.unique(hiseq_count_frame.lacI_amino_seq))
```

Out[5]:
```
967
```

In [6]:
```
np.unique(hiseq_count_frame.lacI_amino_mutations)
```

Out[6]:
```
array([0, 1], dtype=int64)
```

In [7]:
```
df = hiseq_count_frame
df = df[df.lacI_amino_mutations==1]
len(df), len(np.unique(df.lacI_amino_seq))
```

Out[7]:
```
(966, 966)
```

In [8]:
```
df_wt = hiseq_count_frame
df_wt = df_wt[df_wt.lacI_amino_mutations==0]
df_wt = df_wt[df_wt.total_counts>20000]
print(len(df_wt))
y1_wt_arr = np.array([x for x in df_wt.GP_log_g_IPTG])[:,2,:]
y1var_wt_arr = np.array([x for x in df_wt.GP_log_g_IPTG_var])

y2_wt_arr = np.array([x for x in df_wt.GP_log_g_ONPF])[:,2,:]
y2var_wt_arr = np.array([x for x in df_wt.GP_log_g_ONPF_var])

y1_wt = np.average(y1_wt_arr, axis=0, weights=1/y1var_wt_arr)
y1err_wt = np.std(y1_wt_arr, axis=0)

y1err_wt = fitness.log_plot_errorbars(y1_wt, y1err_wt)
y1_wt = 10**y1_wt

y2_wt = np.average(y2_wt_arr, axis=0, weights=1/y2var_wt_arr)
y2err_wt = np.std(y2_wt_arr, axis=0)

y2err_wt = fitness.log_plot_errorbars(y2_wt, y2err_wt)
y2_wt = 10**y2_wt
```

```
11
```

In [ ]:

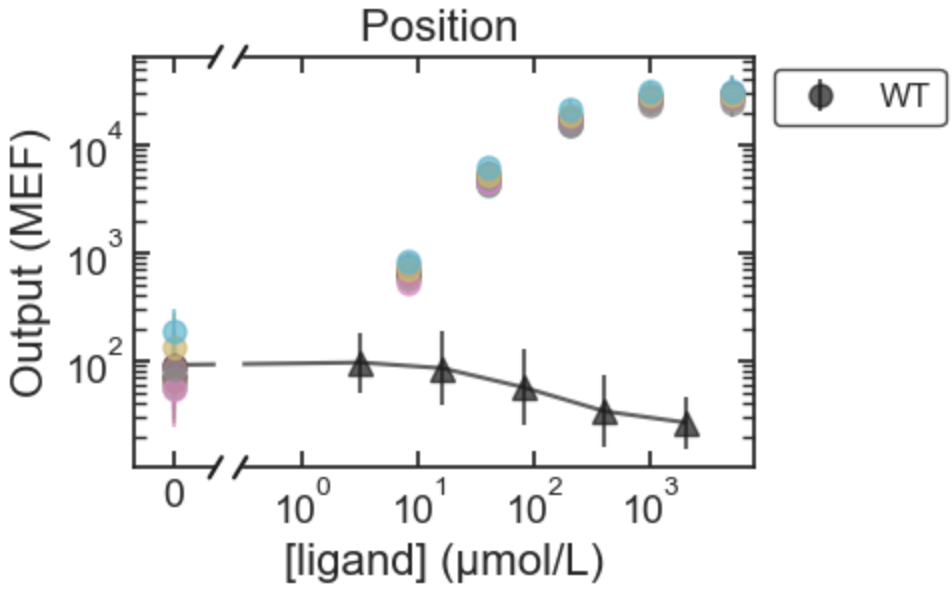
In [9]:
```
def add_break(ax, x_0=0.13, w=0.04, d_x=0.01, d_y=0.025, zorder=2000):
    rect = [(x_0, -0.025), w, 1.05]
    ax.add_patch(patches.Rectangle(*rect, transform=ax.transAxes, clip_on=False,
    zorder += 1
    for y_0 in [0, 1]:
        ax.plot([x_0-d_x, x_0+d_x], [y_0-d_y, y_0+d_y], transform=ax.transAxes, c
        ax.plot([x_0-d_x+w, x_0+d_x+w], [y_0-d_y, y_0+d_y], transform=ax.transAxe
```

In [ ]:

In [10]:
```
def plot_dose_response_by_position(pos, include_stops=False, plot_ligands=['IPTG'
    df = hiseq_count_frame#single_mutant_frame
    df = df[df.position==pos]
    if not include_stops:
        df = df[['*' not in x for x in df.mutation]]

    if len(df)>0:
        plt.rcParams["figure.figsize"] = [4, 8/3]
        fig, ax = plt.subplots()

        x1 = [0] + list(barcode_frame.inducer_conc_lists[0])
        x2 = [0] + list(barcode_frame.inducer_conc_lists[1])

        ax.errorbar(x1, y1_wt, y1err_wt, fmt=fmt1, color='k', label='WT', alpha=0
        ax.errorbar(x2, y2_wt, y2err_wt, fmt=fmt2, color='k', alpha=0.7, ms=9)

        for (ind, row), color in zip(df.iterrows(), sns.color_palette()):
            if 'IPTG' in plot_ligands:
                y1 = row.GP_log_g_IPTG[2]
                y1err = np.sqrt(row.GP_log_g_IPTG_var)
                y1err = fitness.log_plot_errorbars(y1, y1err)
                y1 = 10**y1
                ax.errorbar(x1, y1, y1err, fmt=fmt1, color=color, label=row.mutat

            if 'ONPF' in plot_ligands:
                y2 = row.GP_log_g_ONPF[2]
                y2err = np.sqrt(row.GP_log_g_ONPF_var)
                y2err = fitness.log_plot_errorbars(y2, y2err)
                y2 = 10**y2
                ax.errorbar(x2, y2, y2err, fmt=fmt2, color=color, alpha=0.7, ms=9
        ax.set_yscale('log')
        ax.set_xscale('symlog', linthresh=1)
        ax.legend(loc='upper left', bbox_to_anchor= (1.03, 0.97), ncol=1, bordera

        ax.set_xlim(-0.3, 8000)
        ax.set_title(f'Position {df.iloc[0].mutation[:-1]}', size=16)
        ax.set_xlabel('[ligand] (µmol/L)')
        ax.set_ylabel('Output (MEF)')

        add_break(ax)

        return fig, ax
```

In [ ]:
```
```

In [11]:
```
fig, ax = plot_dose_response_by_position(0, plot_ligands=['IPTG'], fmt1='o');
```



In [ ]:
```

```

In [12]:
```
plot_params = ['log_k_a_1_mut', 'log_k_i_1_mut', 'log_k_a_2_mut', 'log_k_i_2_mut'
               'log_k_ratio_1_mut', 'log_k_ratio_2_mut']
```

In [13]:
```
save_params = plot_params + ['log_k_a_1_wt', 'log_k_i_1_wt', 'log_k_a_2_wt', 'log
                             'delta_eps_AI_wt', 'delta_eps_RA_wt', 'log_y_max', '
save_params
```

Out[13]:
```
['log_k_a_1_mut',
 'log_k_i_1_mut',
 'log_k_a_2_mut',
 'log_k_i_2_mut',
 'delta_eps_AI_mut',
 'delta_eps_RA_mut',
 'log_k_ratio_1_mut',
 'log_k_ratio_2_mut',
 'log_k_a_1_wt',
 'log_k_i_1_wt',
 'log_k_a_2_wt',
 'log_k_i_2_wt',
 'delta_eps_AI_wt',
 'delta_eps_RA_wt',
 'log_y_max',
 'sigma']
```

In [ ]:
```

```

In [14]:
```
os.chdir(notebook_directory)
os.chdir('../../../fit_groq_seq_w_biophysical_model/2022-11-08/')
mut_list = glob.glob('*.csv')
os.chdir(notebook_directory)
mut_list = [x.replace('.csv', '') for x in mut_list]
```

In [15]:
```
[x for x in mut_list if int(x[1:-1])==1]
```

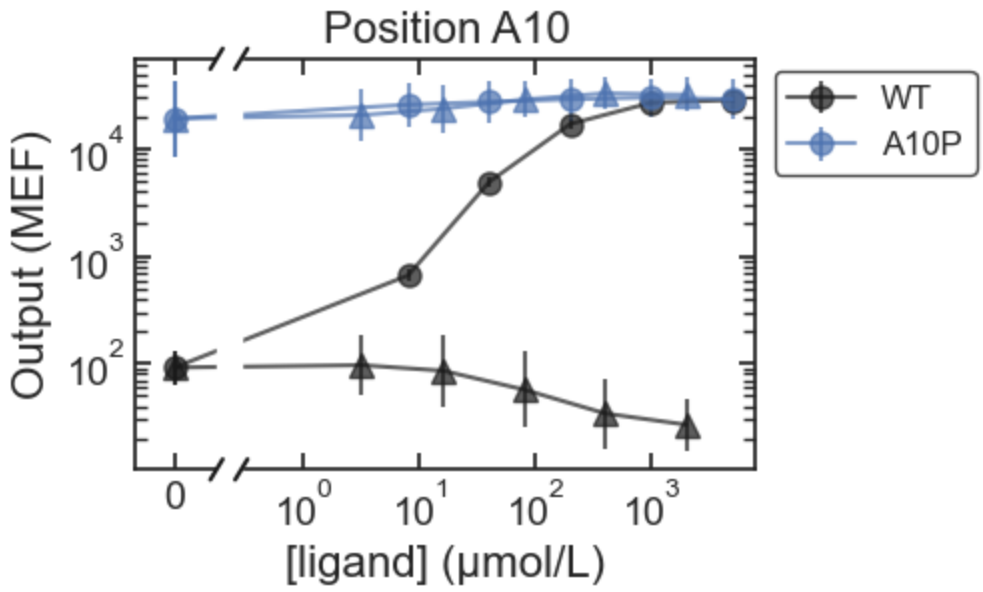
Out[15]:
```
['M1V', 'M1T', 'M1I']
```

In [ ]:

In [16]:
```
os.chdir(notebook_directory)
os.chdir('../../../fit_groq_seq_w_biophysical_model/2022-11-08/')
mut_df_dict = {}
for mut in mut_list:
    file = f'{mut}.csv'
    mut = mut.replace('stop','*')
    df = pd.read_csv(file)
    mut_df_dict[mut] = df
os.chdir(notebook_directory)
```

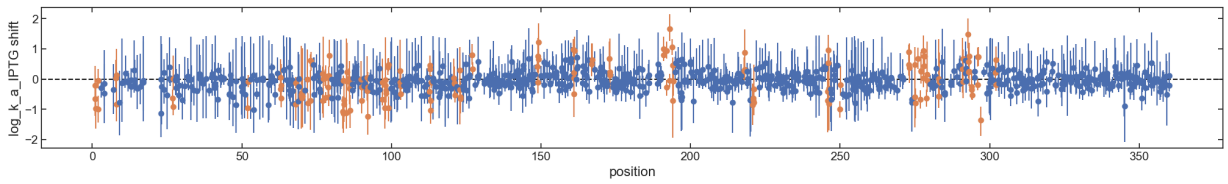
In [ ]:

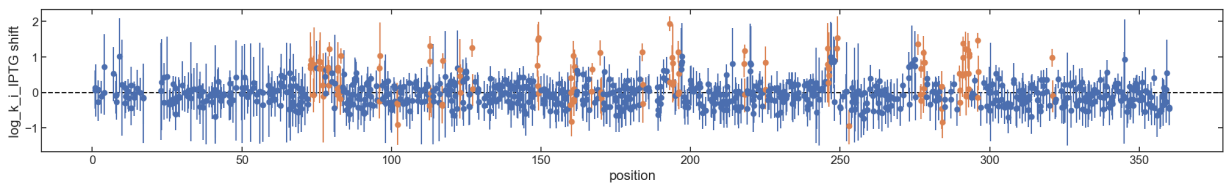
In [17]:
```
plot_dose_response_by_position(10);
```

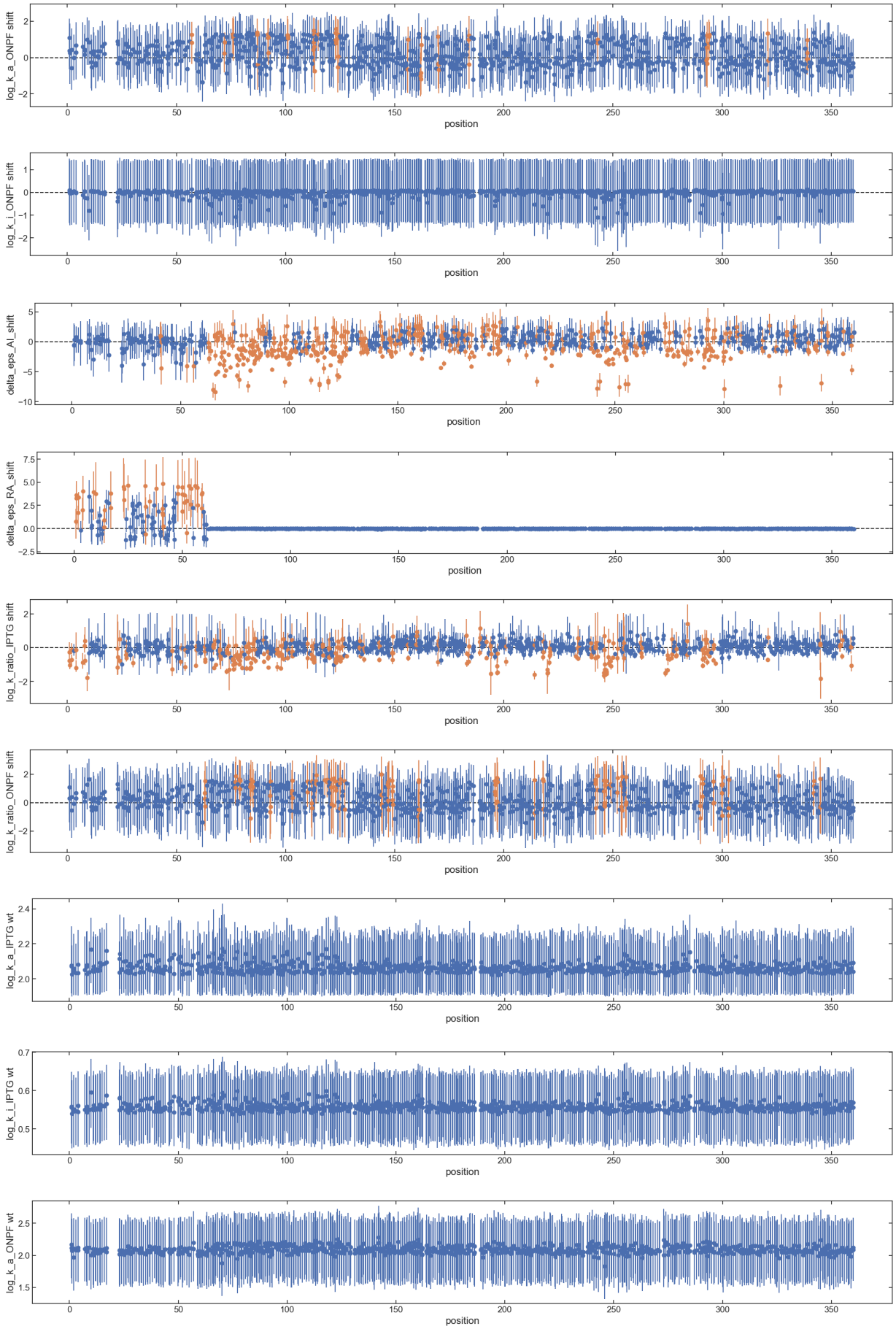


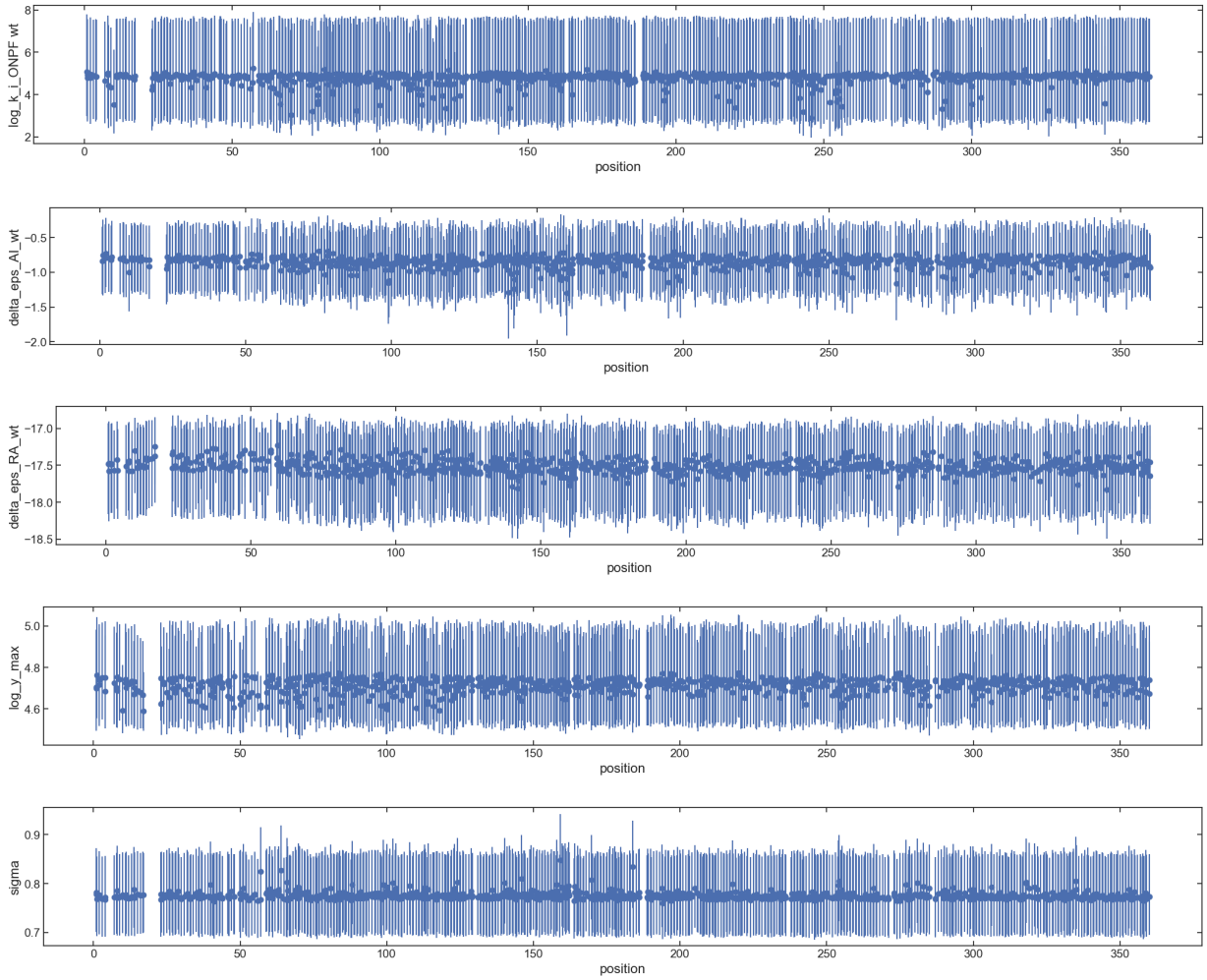
In [ ]:

In [18]:
```
plt.rcParams["figure.figsize"] = [25, 3]

spacing = 0.1
min_fold_shift = 2
sig_positions_dict = {}
for p in save_params:
    min_shift = np.log10(min_fold_shift) if 'log_k_' in p else np.log(min_fold_sh
    if p in save_params[:8]:
        sig_positions_dict[p] = []
    fig, ax = plt.subplots()
    y_label = p
    y_label = y_label.replace('_1_', '_IPTG ').replace('_2_', '_ONPF ').replace(
    for pos in [x for x in range(1, 361)]:
        x = []
        y = []
        yerr = []
        #pos_dict = {k:v for k, v in plot_fit_dict.items() if (int(k[1:-1])==pos)
        pos_mut_list = [x for x in mut_list if ('stop' not in x) and (int(x[1:-1
        span = spacing*len(pos_mut_list)
        for i, mut in enumerate(pos_mut_list):
            if '*' not in mut:
                df = mut_df_dict[mut]

                x.append(int(mut[1:-1]) - span/2 + spacing*i + spacing/2)
                y_samp = df[p]
                y.append(np.mean(y_samp))
                yerr.append(np.quantile(y_samp, [0.05, 0.95]))
        if len(x)>0:
            color = sns.color_palette()[0]
            if np.any(np.array(yerr).transpose()[0] > min_shift) or np.any(np.arr
                if 'shift' in y_label:
                    color = sns.color_palette()[1]
                if p in save_params[:8]:
                    sig_positions_dict[p].append(pos)
            yerr = np.abs(y - np.array(yerr).transpose())
            ax.errorbar(x, y, yerr, fmt='o', color=color)

    ax.set_ylabel(y_label)
    ax.set_xlabel('position')
    if 'shift' in y_label:
        xlim = ax.get_xlim()
        ax.set_xlim(xlim)
        ax.plot(xlim, [0]*2, '--k', zorder=-10)
    #break
```






log_k_a_ONPF shift
position
log_k_i_ONPF shift
position
delta_eps_AI_shift
position
delta_eps_RA_shift
position
log_k_ratio_IPTG shift
position
log_k_ratio_ONPF shift
position
log_k_a_IPTG wt
position
log_k_i_IPTG wt
position
log_k_a_ONPF wt
position


log_k_i_ONPF_wt
position
delta_eps_AI_wt
position
delta_eps_RA_wt
position
log_y_max
position
sigma
position

In [ ]:

In [19]:
```
pos_list = np.unique([int(x[1:-1]) for x in mut_list])
pos_list
```

Out[19]:
```
array([  1,   2,   3,   4,   7,   8,   9,  10,  11,  12,  13,  14,  15,
        16,  17,  23,  24,  25,  26,  27,  28,  29,  30,  31,  32,  33,
        34,  35,  36,  37,  38,  39,  40,  41,  42,  43,  44,  46,  47,
        48,  50,  51,  52,  53,  54,  55,  56,  57,  59,  60,  61,  62,
        63,  64,  65,  66,  67,  68,  69,  70,  71,  72,  73,  74,  75,
        76,  77,  78,  79,  80,  81,  82,  83,  84,  85,  86,  87,  88,
        89,  90,  91,  92,  93,  94,  95,  96,  97,  98,  99, 100, 101,
       102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114,
       115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127,
       128, 129, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141,
       142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154,
       155, 156, 157, 158, 159, 160, 161, 162, 164, 165, 166, 167, 168,
       169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181,
       182, 183, 184, 185, 186, 189, 190, 191, 192, 193, 194, 195, 196,
       197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209,
       210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222,
       223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235,
       236, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249,
       250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262,
       263, 264, 265, 266, 267, 268, 269, 270, 271, 273, 274, 275, 276,
       277, 278, 279, 280, 281, 282, 283, 284, 285, 287, 288, 289, 290,
       291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303,
       304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316,
       317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329,
       330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342,
       343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355,
       356, 357, 358, 359, 360])
```

In [20]:
```
pos_count_dict = np.unique([int(x[1:-1]) for x in mut_list], return_counts=True)
pos_count_dict = {p:c for p, c in zip(pos_count_dict[0], pos_count_dict[1])}
#pos_count_dict
```

In [ ]:
```

```

In [21]:
```
for k, v in sig_positions_dict.items():
    y_label = k
    y_label = y_label.replace('_1_', '_IPTG ').replace('_2_', '_ONPF ').replace(
    print(y_label)
    print(v)
    print()
```

```
log_k_a_IPTG shift
[1, 2, 8, 27, 52, 63, 68, 70, 72, 73, 77, 78, 79, 84, 85, 86, 90, 92, 97, 98, 9
9, 103, 113, 121, 123, 127, 149, 161, 167, 173, 191, 192, 193, 194, 218, 221, 2
46, 250, 273, 275, 276, 277, 278, 279, 283, 291, 293, 294, 296, 297, 302]

log_k_i_IPTG shift
[73, 74, 76, 77, 79, 82, 83, 96, 102, 113, 117, 123, 127, 149, 160, 161, 167, 1
70, 184, 193, 194, 196, 218, 225, 246, 249, 253, 276, 277, 278, 284, 290, 291,
292, 293, 296, 321]

log_k_a_ONPF shift
[57, 72, 76, 87, 92, 101, 113, 115, 123, 124, 156, 162, 170, 184, 243, 293, 29
4, 321, 339]

log_k_i_ONPF shift
[]

delta_eps_AI_shift
[41, 53, 56, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 76, 77, 78, 79, 8
0, 81, 82, 83, 84, 85, 86, 87, 88, 90, 92, 93, 94, 95, 96, 97, 98, 99, 100, 10
1, 103, 106, 108, 110, 111, 112, 113, 114, 115, 116, 118, 119, 120, 121, 122, 1
23, 124, 125, 126, 127, 133, 134, 135, 136, 142, 143, 146, 147, 149, 151, 155,
156, 158, 159, 160, 161, 168, 170, 171, 174, 176, 178, 181, 183, 184, 185, 189,
191, 192, 194, 195, 196, 197, 201, 205, 209, 214, 218, 220, 221, 226, 233, 234,
239, 240, 242, 243, 246, 247, 249, 250, 251, 252, 254, 255, 256, 259, 260, 263,
273, 274, 275, 276, 277, 279, 281, 282, 283, 284, 285, 288, 291, 292, 293, 294,
295, 296, 297, 300, 301, 303, 306, 308, 318, 321, 324, 326, 329, 333, 339, 342,
343, 345, 348, 355, 359]

delta_eps_RA_shift
[1, 2, 4, 9, 10, 14, 17, 23, 25, 33, 35, 38, 41, 48, 50, 51, 52, 53, 54, 56, 5
7, 59]

log_k_ratio_IPTG shift
[1, 2, 4, 7, 8, 9, 23, 24, 27, 30, 48, 52, 54, 59, 63, 67, 68, 70, 73, 74, 76,
77, 78, 79, 80, 82, 83, 84, 85, 86, 90, 92, 96, 97, 98, 99, 101, 103, 110, 112,
113, 117, 120, 123, 124, 125, 126, 128, 134, 136, 143, 149, 156, 160, 170, 183,
184, 189, 192, 194, 197, 201, 214, 218, 220, 221, 233, 234, 240, 242, 243, 246,
247, 248, 249, 250, 252, 255, 263, 274, 275, 276, 277, 279, 284, 291, 292, 293,
294, 296, 297, 321, 345, 354, 355, 359]

log_k_ratio_ONPF shift
[63, 77, 79, 84, 85, 93, 103, 110, 112, 114, 118, 119, 121, 122, 123, 125, 127,
144, 147, 149, 161, 196, 197, 214, 218, 242, 243, 246, 247, 249, 252, 254, 256,
290, 291, 296, 300, 303, 326, 342, 345]
```

In [ ]:

In [70]:
```
row_list = []
for pos in [x for x in range(1, 361)]:
    row_d = {'position':pos, 'mutation_count':pos_count_dict.get(pos, 0)}
    for k, v in sig_positions_dict.items():
        k = k.replace('_mut', '_shift').replace('_1', '_IPTG').replace('_2', '_ON
        row_d[k] = (pos in v)
    row = pd.Series(row_d)
    row_list.append(row)
    #if pos == 326:
    #    break
```

In [ ]:

In [71]:
```
out_frame = pd.DataFrame(row_list)
out_frame
```

Out[71]:

	position	mutation_count	log_k_a_IPTG_shift	log_k_i_IPTG_shift	log_k_a_ONPF_shift	log_k_i_
0	1	3	True	False	False	
1	2	2	True	False	False	
2	3	1	False	False	False	
3	4	2	False	False	False	
4	5	0	False	False	False	
...	...	...	...	...	...	
355	356	1	False	False	False	
356	357	3	False	False	False	
357	358	5	False	False	False	
358	359	3	False	False	False	
359	360	2	False	False	False	

360 rows × 10 columns

In [72]:
```
out_frame.columns
```

Out[72]:
```
Index(['position', 'mutation_count', 'log_k_a_IPTG_shift',
       'log_k_i_IPTG_shift', 'log_k_a_ONPF_shift', 'log_k_i_ONPF_shift',
       'delta_eps_AI_shift', 'delta_eps_RA_shift', 'log_k_ratio_IPTG_shift',
       'log_k_ratio_ONPF_shift'],
      dtype='object')
```

In [73]:
```
def apply_categories(df):
    df['IPTG'] = df.log_k_a_IPTG_shift|df.log_k_i_IPTG_shift|df.log_k_ratio_IPTG_
    df['ONPF'] = df.log_k_a_ONPF_shift|df.log_k_i_ONPF_shift|df.log_k_ratio_ONPF_
    df['MWC'] = df.delta_eps_AI_shift

    df['MWC_only'] = df['MWC']&~(df['IPTG']|df['ONPF'])

    df['MWC_IPTG'] = df['MWC']&(df['IPTG'])
    df['MWC_ONPF'] = df['MWC']&(df['ONPF'])
    df['MWC_IPTG_ONPF'] = df['MWC']&(df['IPTG'])&(df['ONPF'])

    df['IPTG_only'] = df['IPTG']&~(df['MWC']|df['ONPF'])
    df['ONPF_only'] = df['ONPF']&~(df['MWC']|df['IPTG'])

    df['Both_noMWC'] = (~df['MWC'])&(df['IPTG'])&(df['ONPF'])
    #return df
```

In [74]:
```
apply_categories(out_frame)
out_frame
```

Out[74]:

	position	mutation_count	log_k_a_IPTG_shift	log_k_i_IPTG_shift	log_k_a_ONPF_shift	log_k_i_
0	1	3	True	False	False	
1	2	2	True	False	False	
2	3	1	False	False	False	
3	4	2	False	False	False	
4	5	0	False	False	False	
...	...	...	...	...	...	
355	356	1	False	False	False	
356	357	3	False	False	False	
357	358	5	False	False	False	
358	359	3	False	False	False	
359	360	2	False	False	False	

360 rows × 20 columns

In [ ]:
```
```

In [75]:
```
os.chdir(notebook_directory)
os.chdir('../../../fit_groq_seq_w_biophysical_model/')
out_frame.to_csv('IPTG_ONPF_positions.csv', index=False)
```

In [ ]:
```
```

In [76]:
```
df = out_frame
df = df[df.ONPF_only]
df
```

Out[76]:

	position	mutation_count	log_k_a_IPTG_shift	log_k_i_IPTG_shift	log_k_a_ONPF_shift	log_k_i_
56	57	2	False	False	True	
143	144	5	False	False	False	
161	162	4	False	False	True	

In [77]:
```
mut_list[0]
```

Out[77]:
```
'V119I'
```

In [ ]:
```

```

In [83]:
```
# Output 0.05, 0.0, and 0.95 quantiles for every mutation
mut = mut_list[0]
df = mut_df_dict[mut]
quant_columns = [x for x in df.columns if '_mut' in x]
min_fold_shift = 2

row_list = []
for mut in mut_list:
    df = mut_df_dict[mut]
    row_d = {'mutation':mut, 'position':int(mut[1:-1])}
    for c in quant_columns:
        c2 = c.replace('_1', '_IPTG').replace('_2', '_ONPF')
        for q in [5, 50, 95]:
            row_d[f'{c2}_{q:02}'] = np.quantile(df[c], q/100)

    for c in quant_columns:
        c1 = c.replace('_1', '_IPTG').replace('_2', '_ONPF')
        c2 = c1.replace('_mut', '_shift')
        min_shift = np.log10(min_fold_shift) if 'log_k_' in c else np.log(min_fo
        row_d[c2] = (row_d[f'{c1}_05']>min_shift) or (row_d[f'{c1}_95']<-min_shi

    row_list.append(pd.Series(row_d))
    #break
out_mut_frame = pd.DataFrame(row_list)
apply_categories(out_mut_frame)
out_mut_frame = out_mut_frame.sort_values(by=['position', 'mutation'], ignore_in
out_mut_frame
```

Out[83]:

	mutation	position	log_k_a_IPTG_mut_05	log_k_a_IPTG_mut_50	log_k_a_IPTG_mut_95	log_k_i
0	M1I	1	-1.649228	-0.967614	-0.305219	
1	M1T	1	-1.220760	-0.698248	0.128159	
2	M1V	1	-0.636930	-0.256099	0.442714	
3	K2E	2	-1.434717	-0.988525	-0.552492	
4	K2R	2	-0.313470	-0.089149	0.341082	
...	...	...	...	...	...	
933	G359E	359	-0.284835	0.014223	0.829093	
934	G359R	359	-0.577390	-0.206972	0.201057	
935	G359W	359	-1.529151	-0.534439	0.590333	
936	Q360E	360	-0.257095	0.008067	0.881052	
937	Q360R	360	-0.412597	-0.202674	0.008972	

938 rows × 44 columns

In [ ]:

In [84]:
```
os.chdir(notebook_directory)
os.chdir('../../../fit_groq_seq_w_biophysical_model/')
out_mut_frame.to_csv('IPTG_ONPF_mutations.csv', index=False)
```

In [ ]:
```
```

In [85]:
```
df = out_mut_frame
df = df[df.ONPF_only]
df[['mutation', 'position'] + [x for x in df.columns if 'ONPF' in x]]
```

Out[85]:

	mutation	position	log_k_a_ONPF_mut_05	log_k_a_ONPF_mut_50	log_k_a_ONPF_mut_95	log_
100	A57S	57	0.464597	1.312875	1.979213	
268	L114I	114	0.039727	1.207665	2.237159	
360	P144S	144	0.093874	1.102575	2.150112	
419	S162Y	162	-1.994588	-1.187975	-0.332062	
749	K290I	290	-0.020619	0.956071	2.072317	

In [ ]:
```
```

In [ ]:
```
```

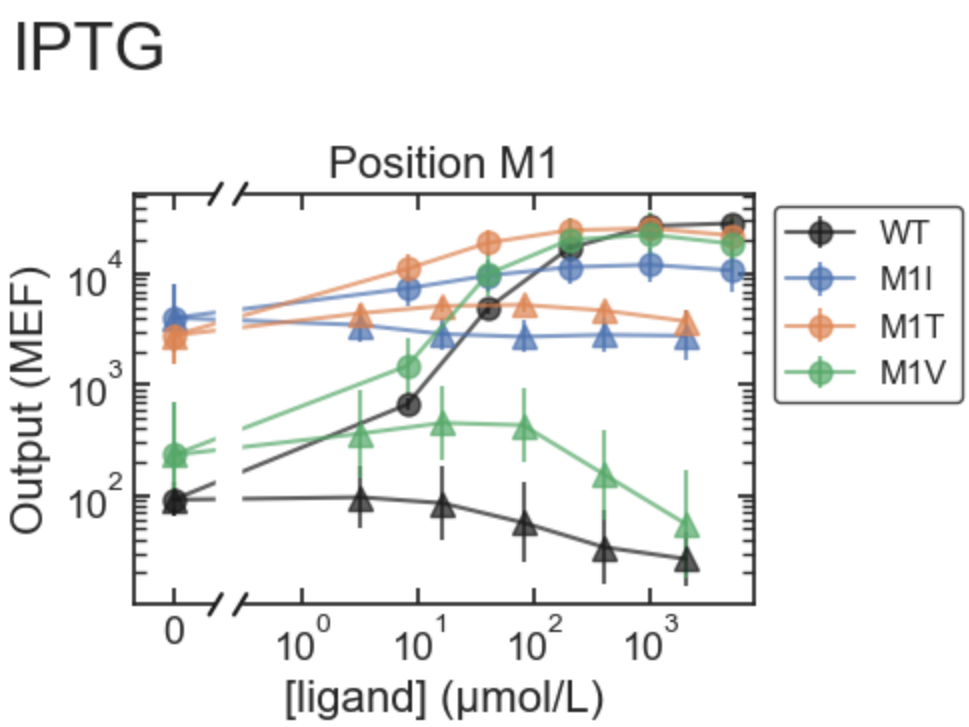
In [87]:
```
plot_categories = [x for x in out_frame.columns if '_shift' not in x]
plot_categories = [x for x in plot_categories if x not in ['position', 'mutation_
plot_categories
```

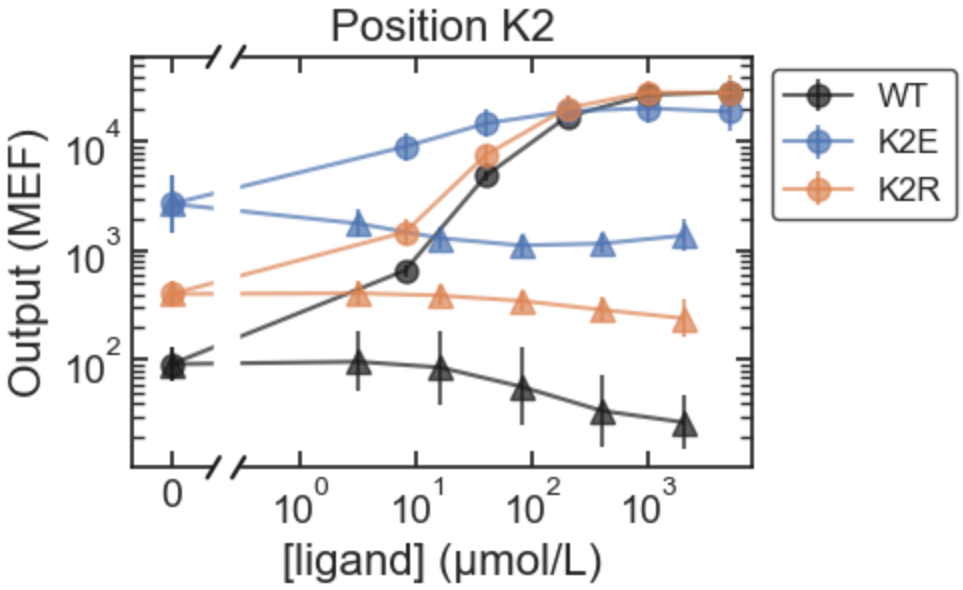
Out[87]:
```
['IPTG',
 'ONPF',
 'MWC',
 'MWC_only',
 'MWC_IPTG',
 'MWC_ONPF',
 'MWC_IPTG_ONPF',
 'IPTG_only',
 'ONPF_only',
 'Both_noMWC']
```

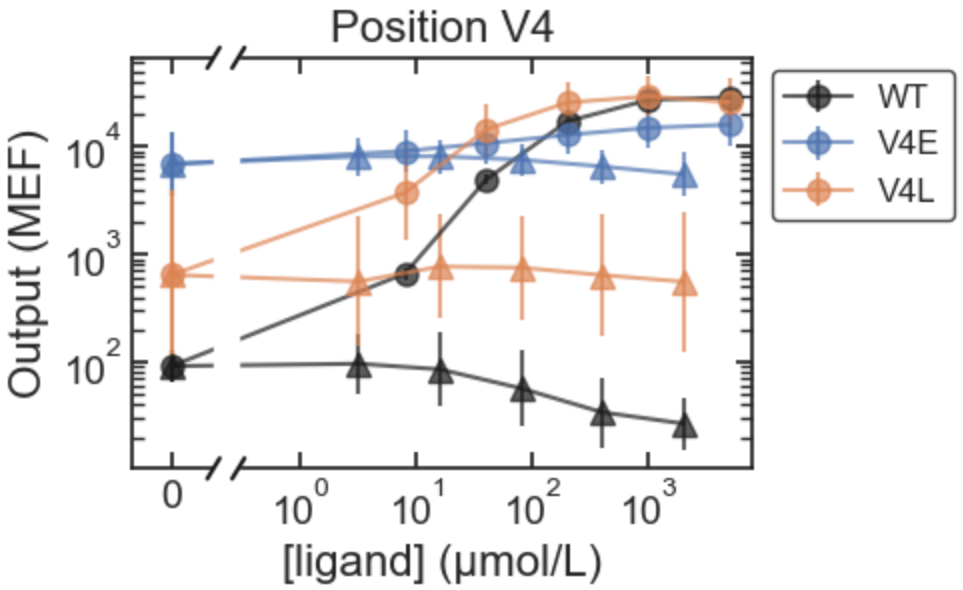
In [ ]:
```
```

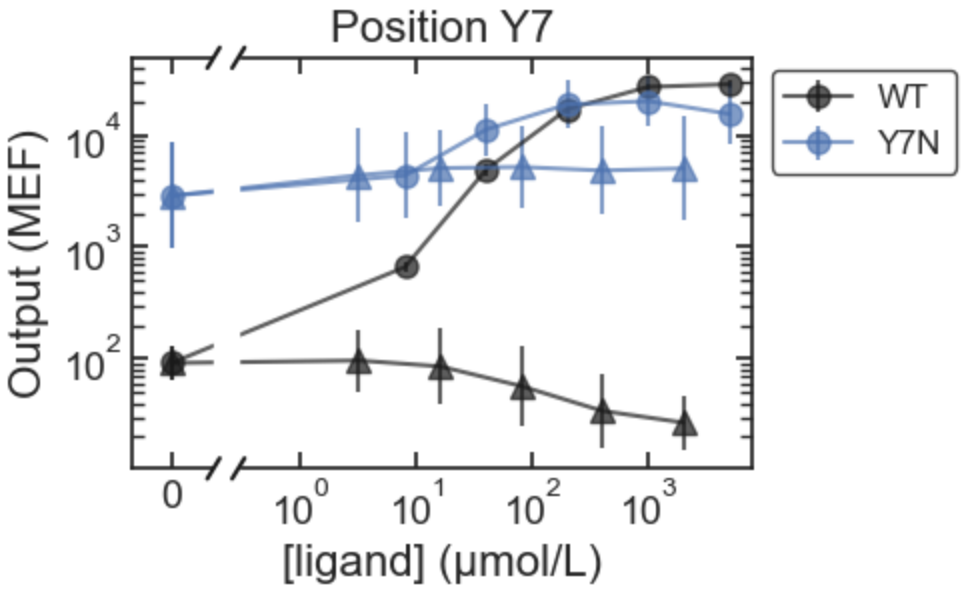
In [93]:
```
for c in plot_categories:
    df = out_frame
    df = df[df[c]]
    plt.rcParams["figure.figsize"] = [6, 0.2]
    fig, ax = plt.subplots()
    fig.suptitle(c, size=24)
    ax.set_visible(False)
    for pos in df.position:
        plot_dose_response_by_position(pos)
        #break
    #break
```

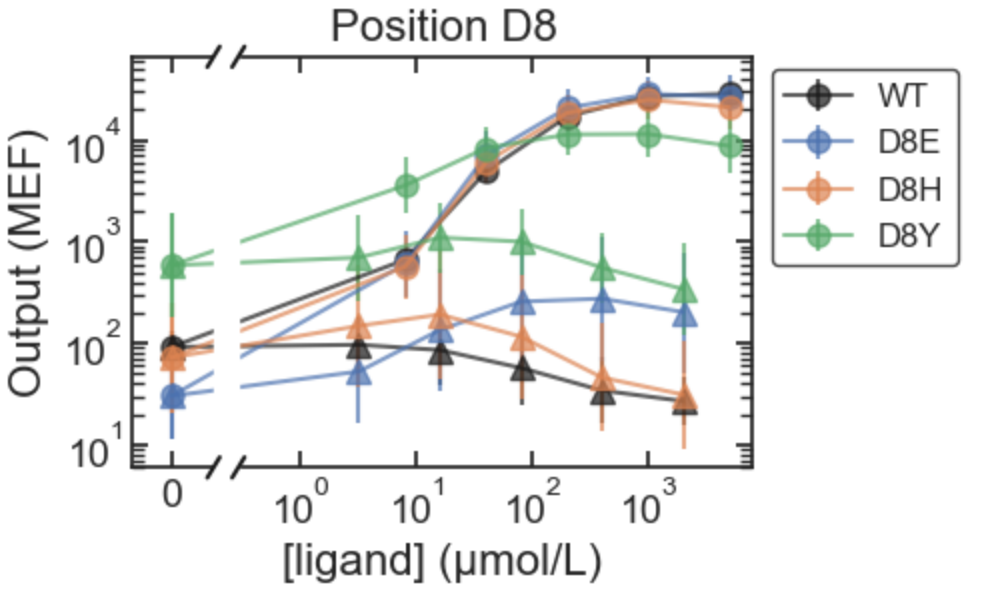
```
C:\Users\djross\AppData\Local\Temp\2\ipykernel_3888\2999772484.py:9: RuntimeWar
ning: More than 20 figures have been opened. Figures created through the pyplot
interface (`matplotlib.pyplot.figure`) are retained until explicitly closed and
may consume too much memory. (To control this warning, see the rcParam `figure.
max_open_warning`). Consider using `matplotlib.pyplot.close()`.
  fig, ax = plt.subplots()
```

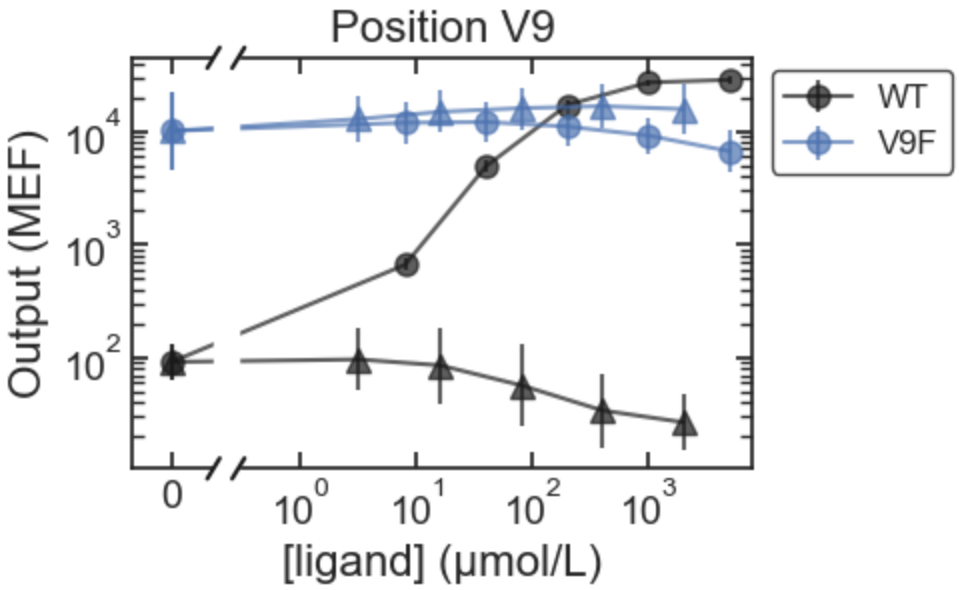


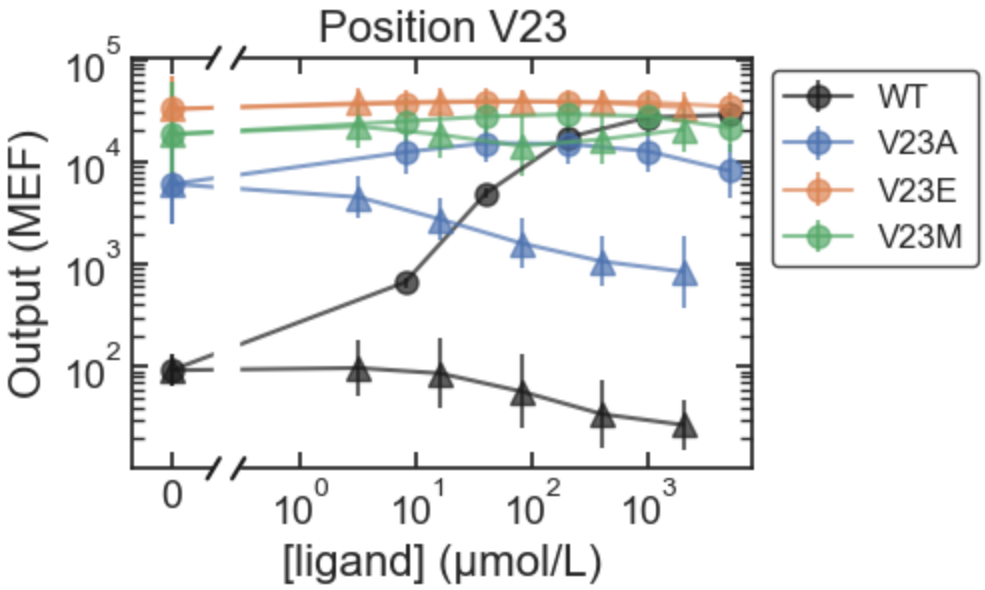

Position K2
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K2E
K2R

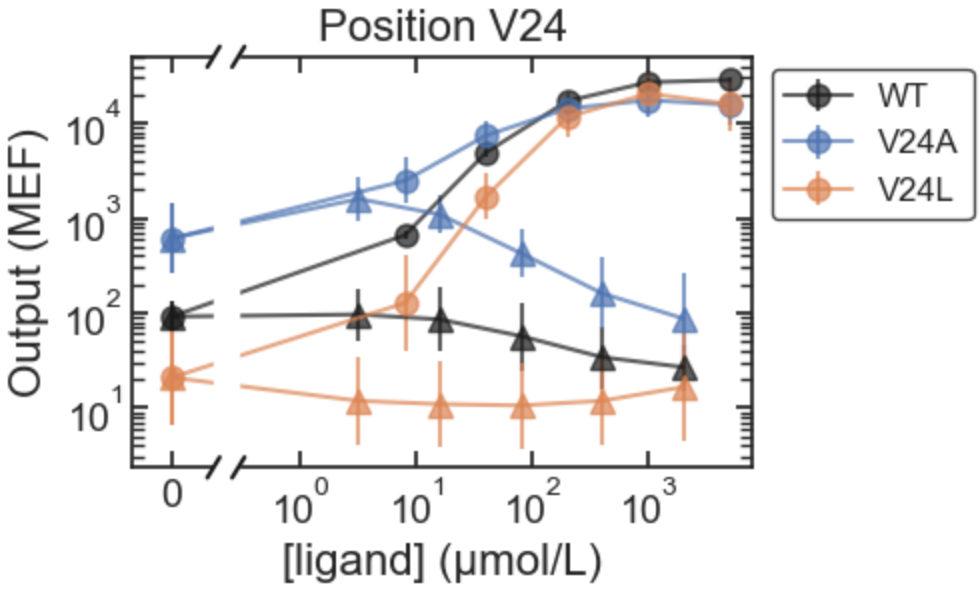

Position V4
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V4E
V4L

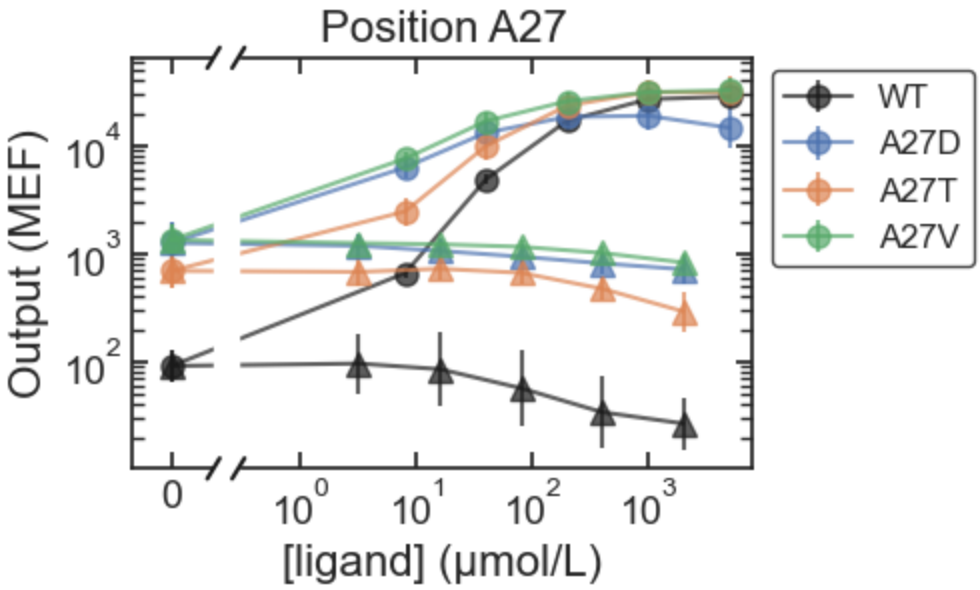

Position Y7
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Y7N

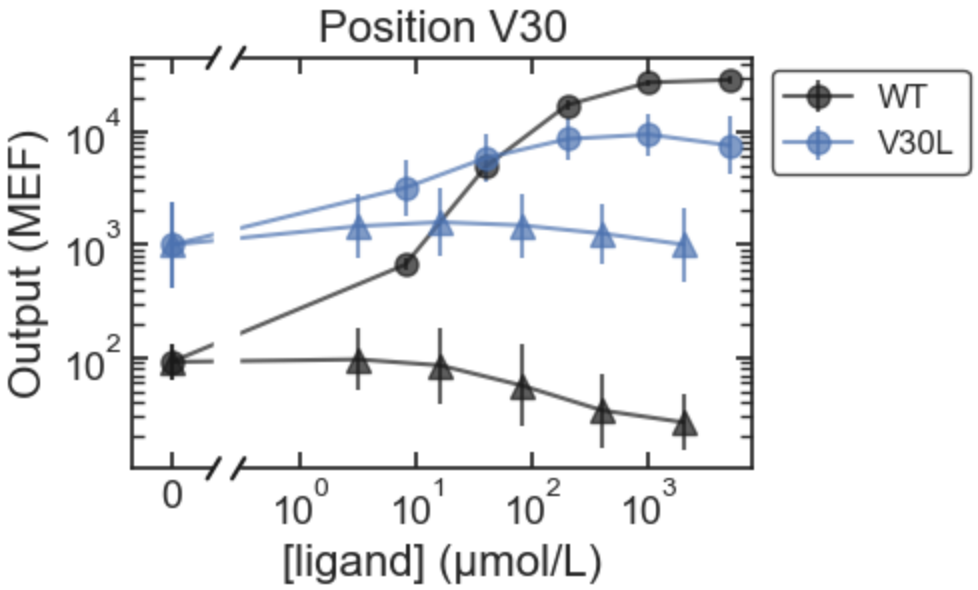

Position D8
Output (MEF)
[ligand] (μmol/L)
WT
D8E
D8H
D8Y

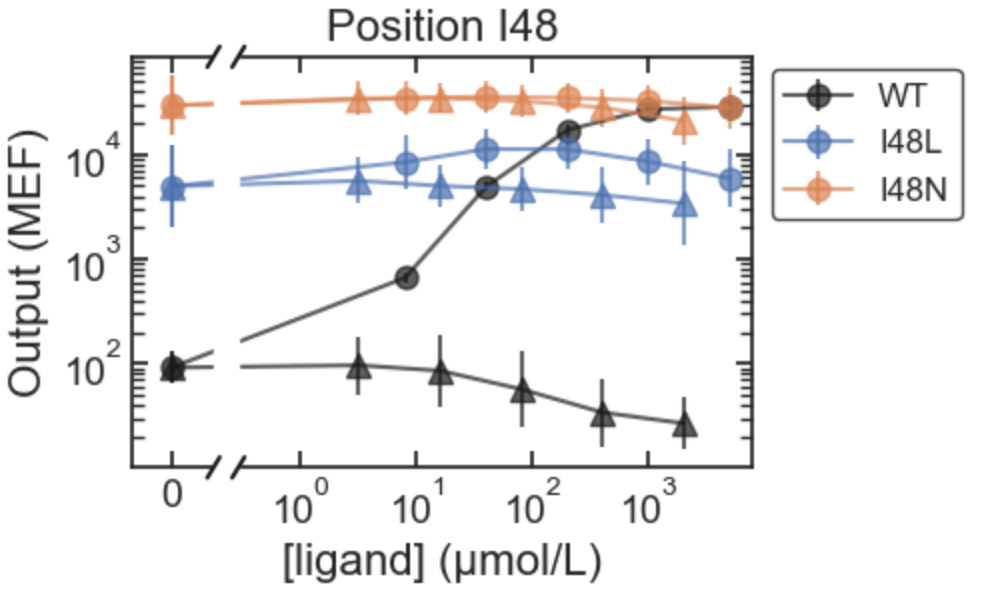

Position V9
Output (MEF)
[ligand] (μmol/L)
WT
V9F

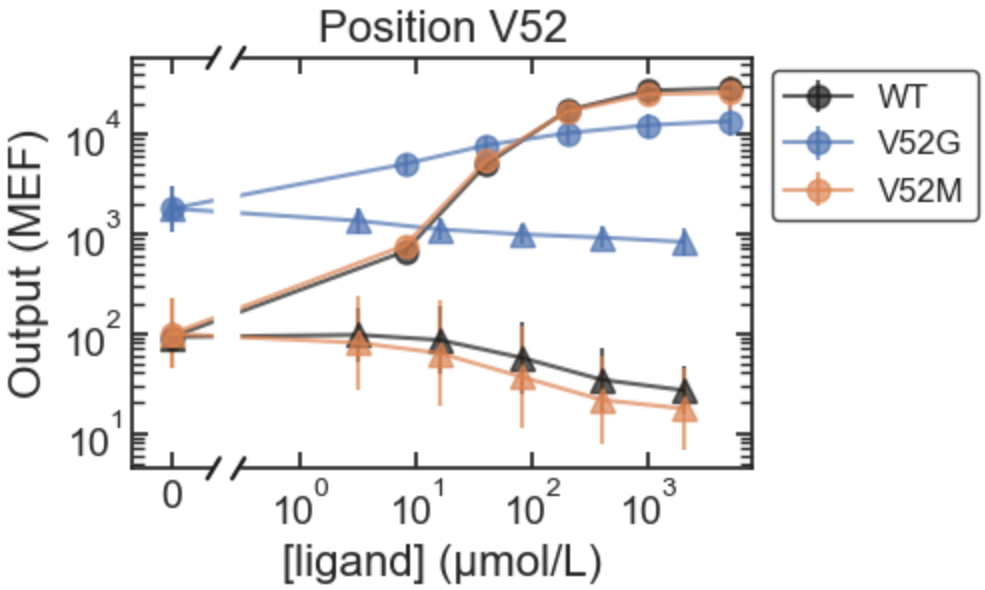

Position V23
Output (MEF)
[ligand] (μmol/L)
WT
V23A
V23E
V23M

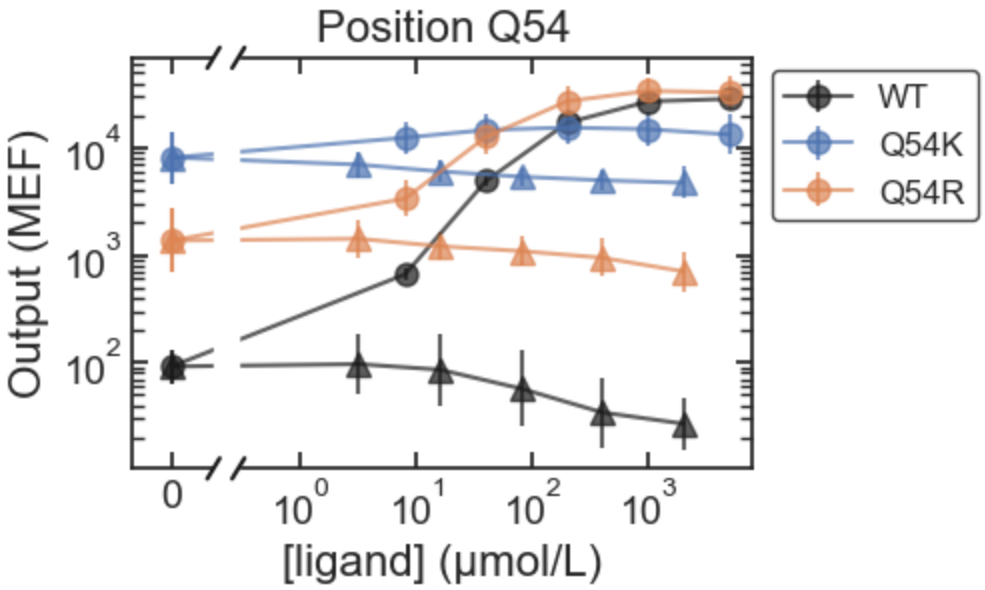

Position V24
Output (MEF)
[ligand] (μmol/L)
WT
V24A
V24L

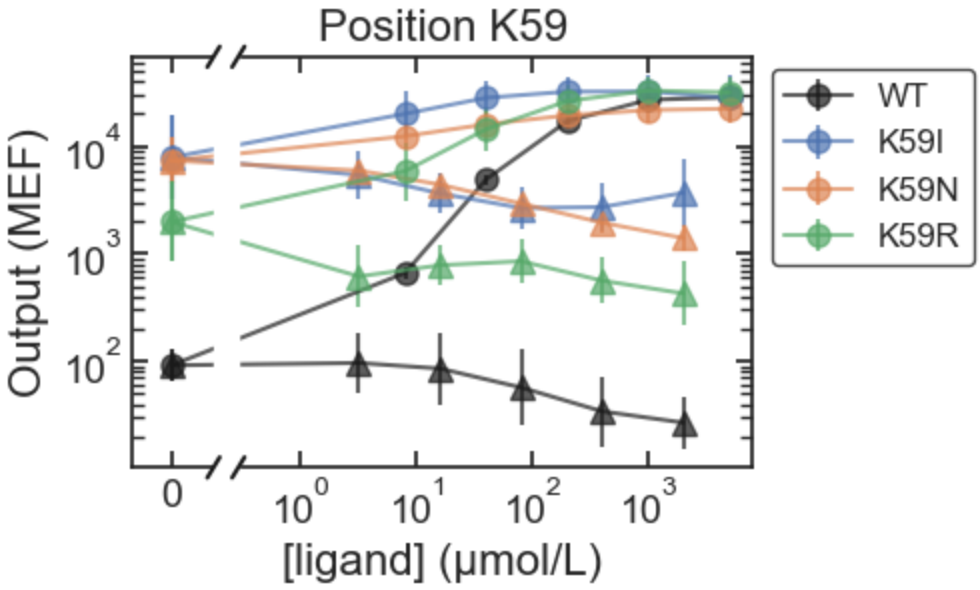

Position A27
Output (MEF)
[ligand] (μmol/L)
WT
A27D
A27T
A27V

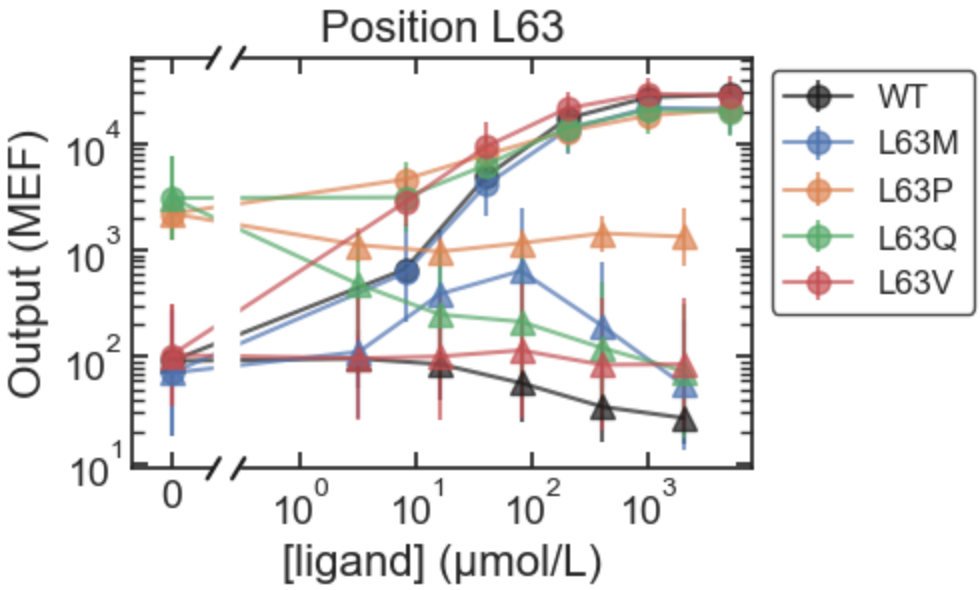

Position V30
Output (MEF)
[ligand] (μmol/L)
WT
V30L

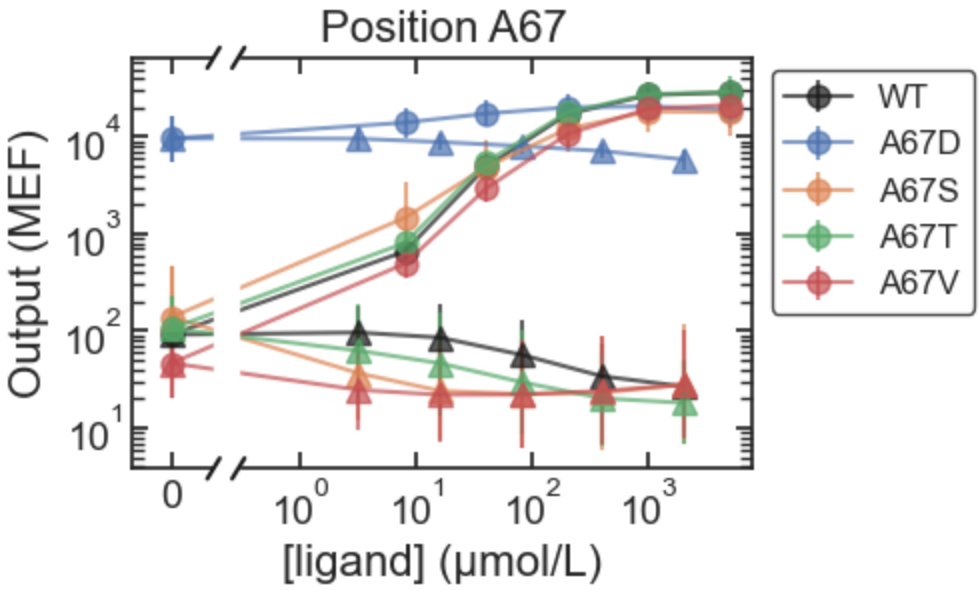

Position I48
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I48L
I48N

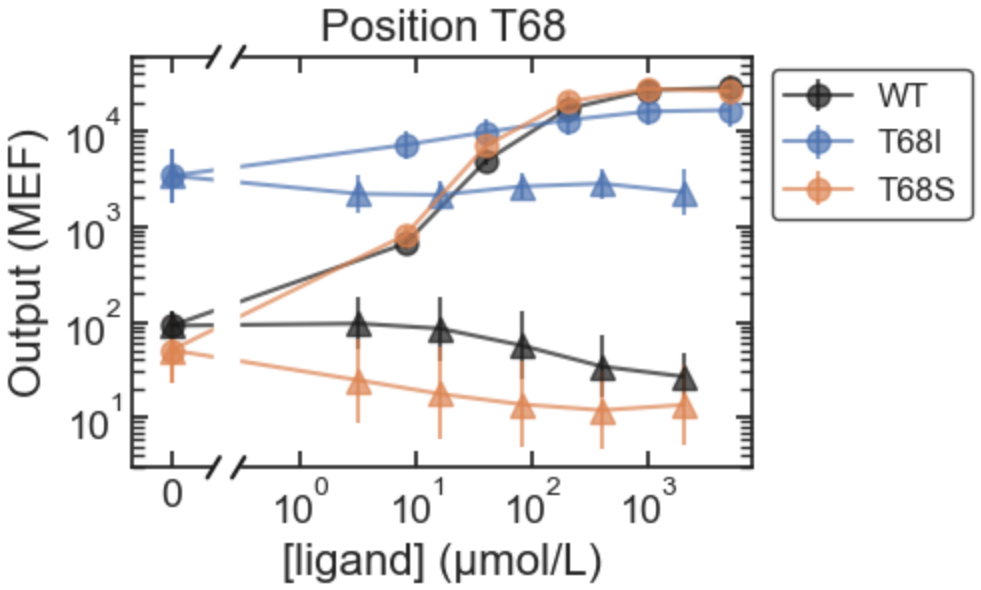

Position V52
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V52G
V52M

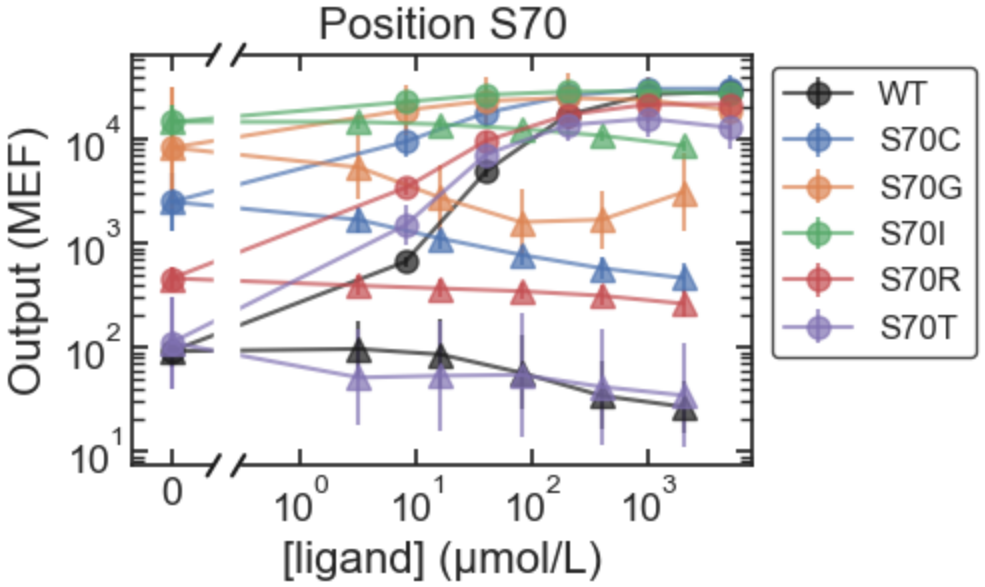

Position Q54
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q54K
Q54R

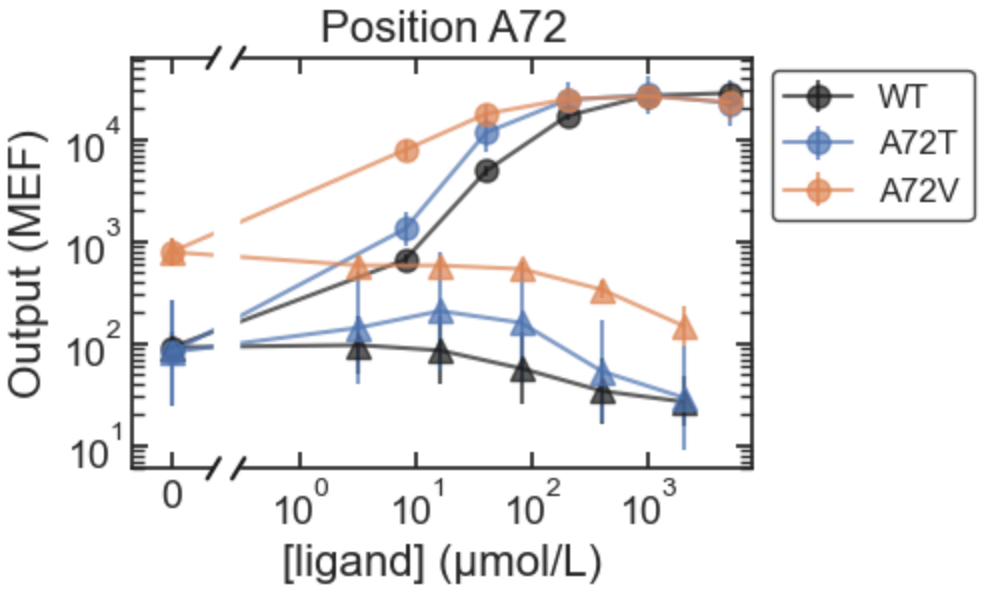

Position K59
Output (MEF)
[ligand] (μmol/L)
WT
K59I
K59N
K59R

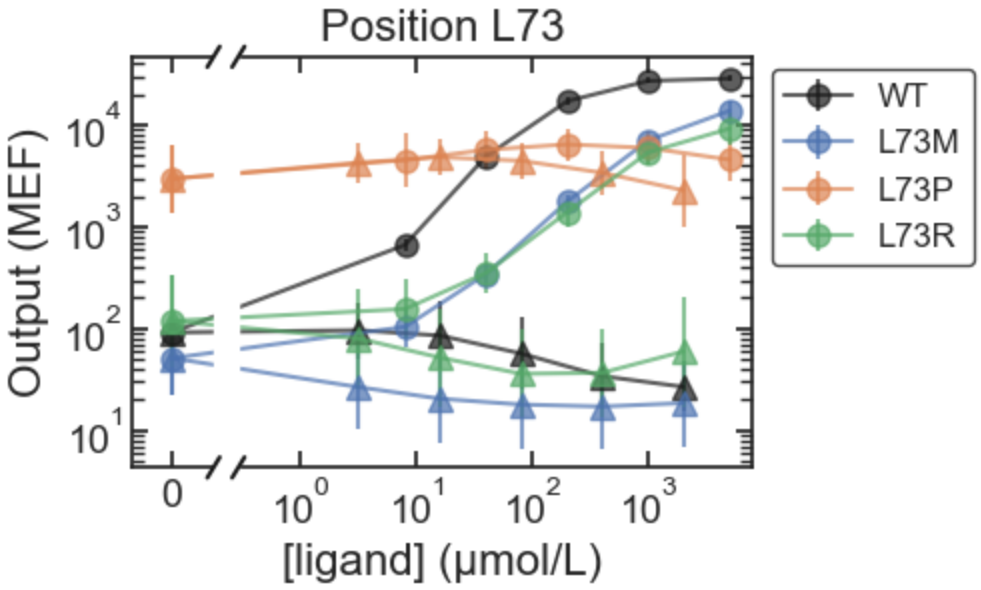

Position L63
Output (MEF)
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

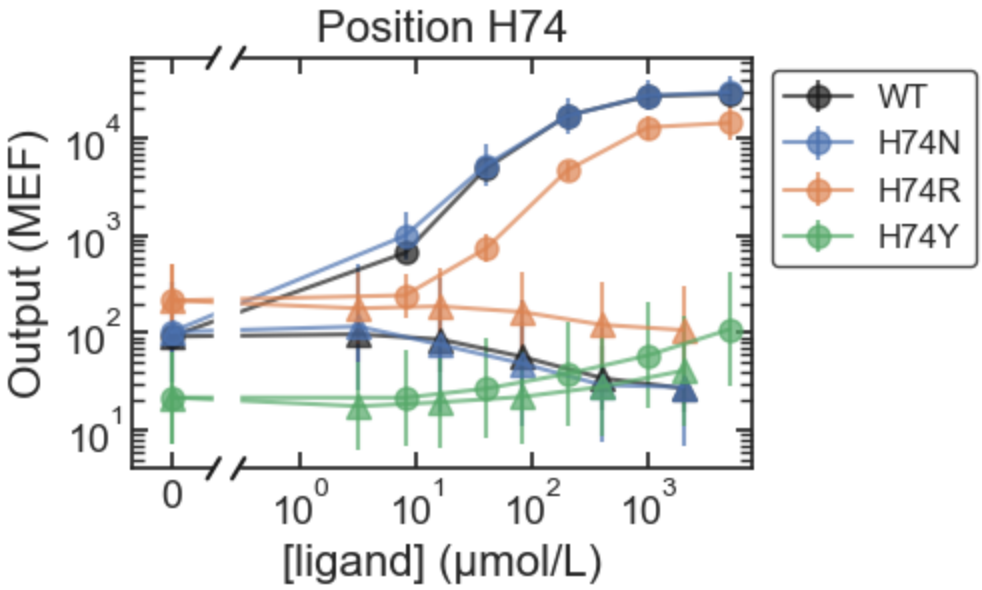

Position A67
Output (MEF)
[ligand] (μmol/L)
WT
A67D
A67S
A67T
A67V

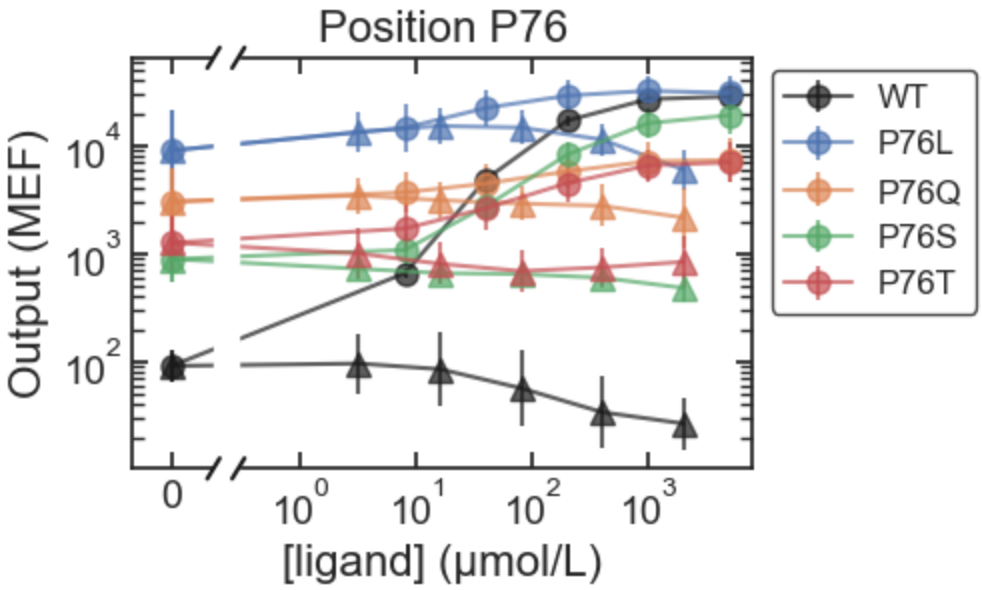

Position T68
Output (MEF)
[ligand] (μmol/L)
WT
T68I
T68S

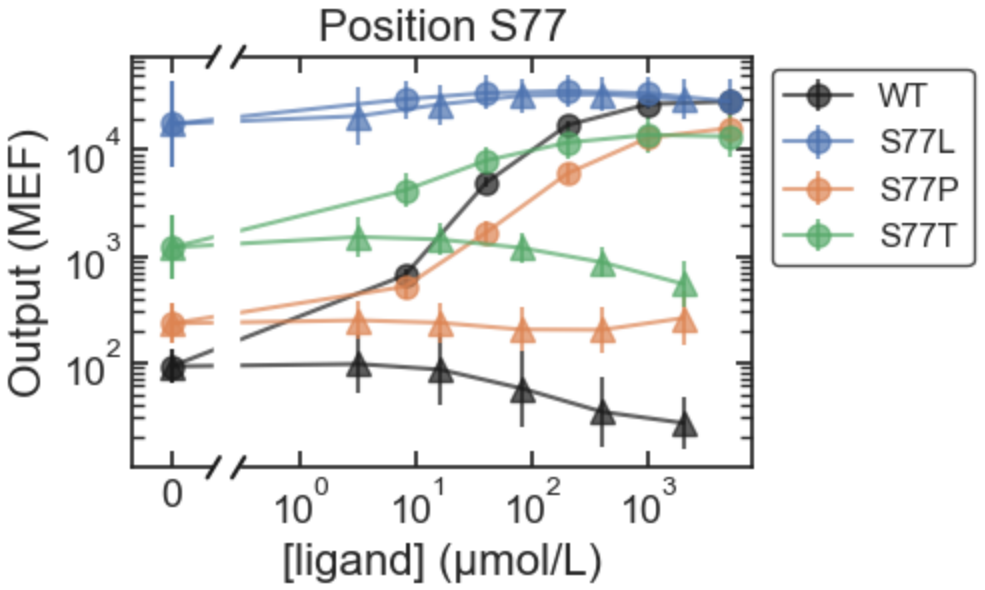

Position S70
Output (MEF)
[ligand] (μmol/L)
WT
S70C
S70G
S70I
S70R
S70T

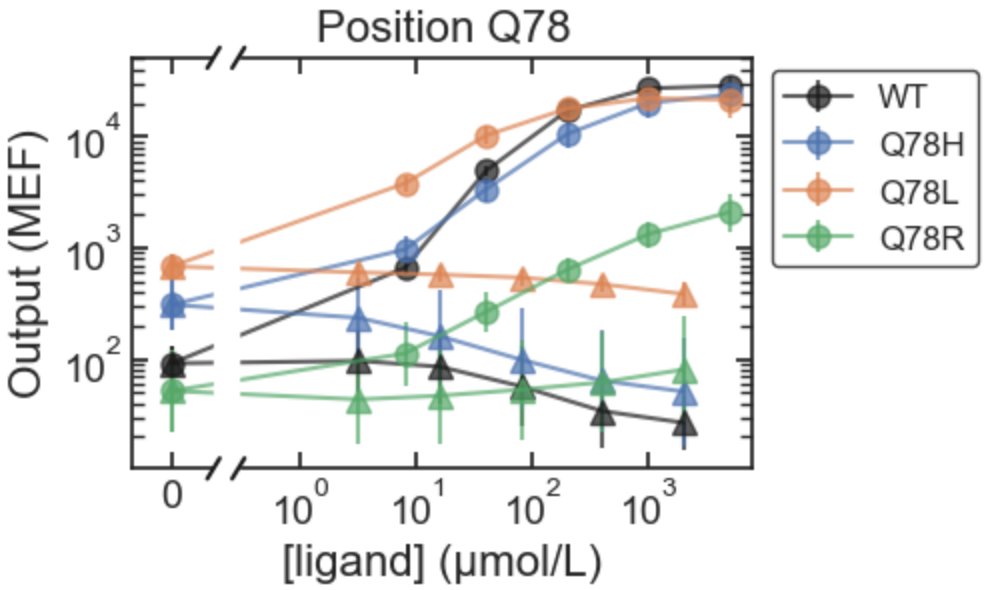

Position A72
Output (MEF)
[ligand] (μmol/L)
WT
A72T
A72V

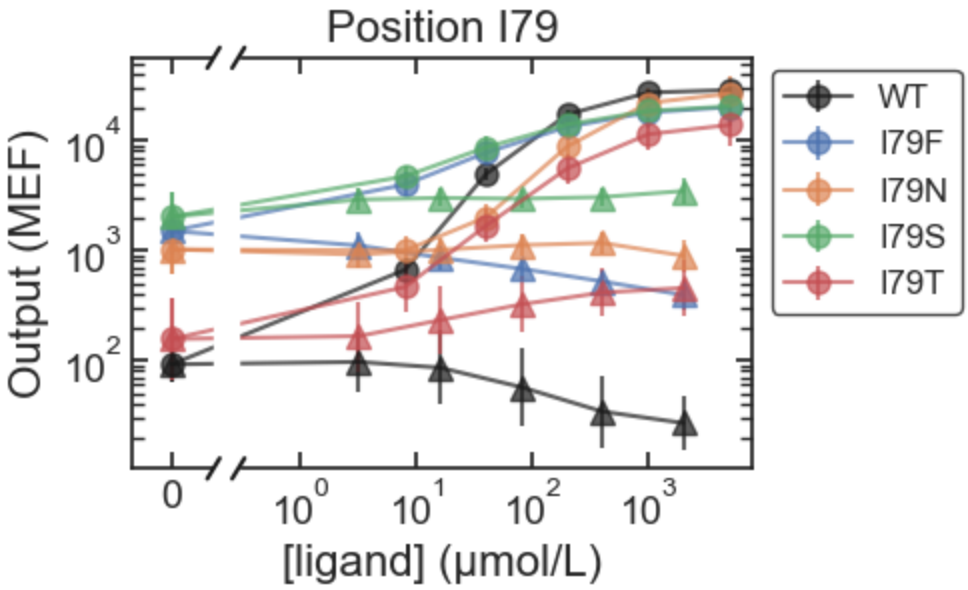

Position L73
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L73M
L73P
L73R

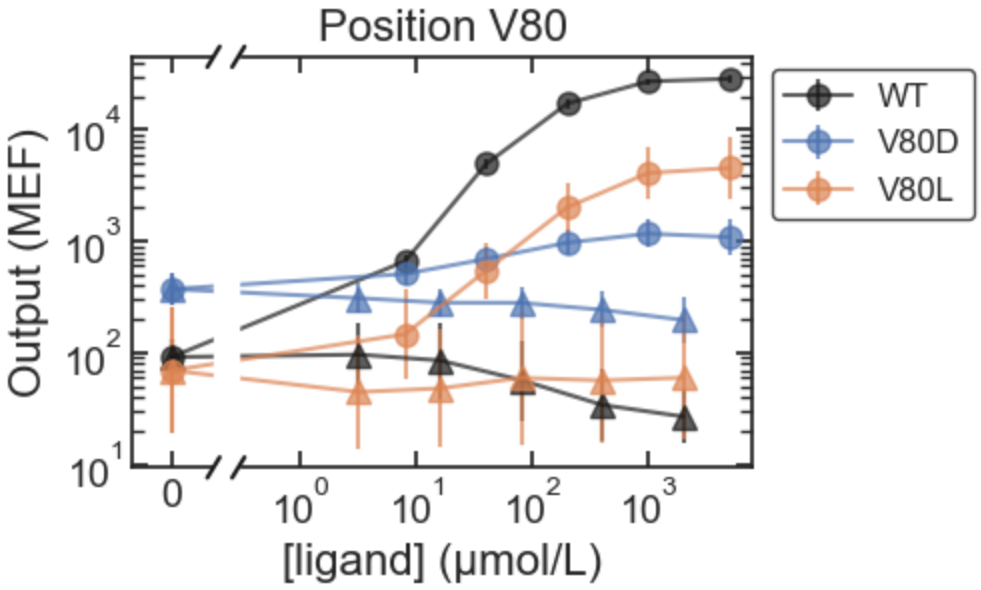

Position H74
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
H74N
H74R
H74Y

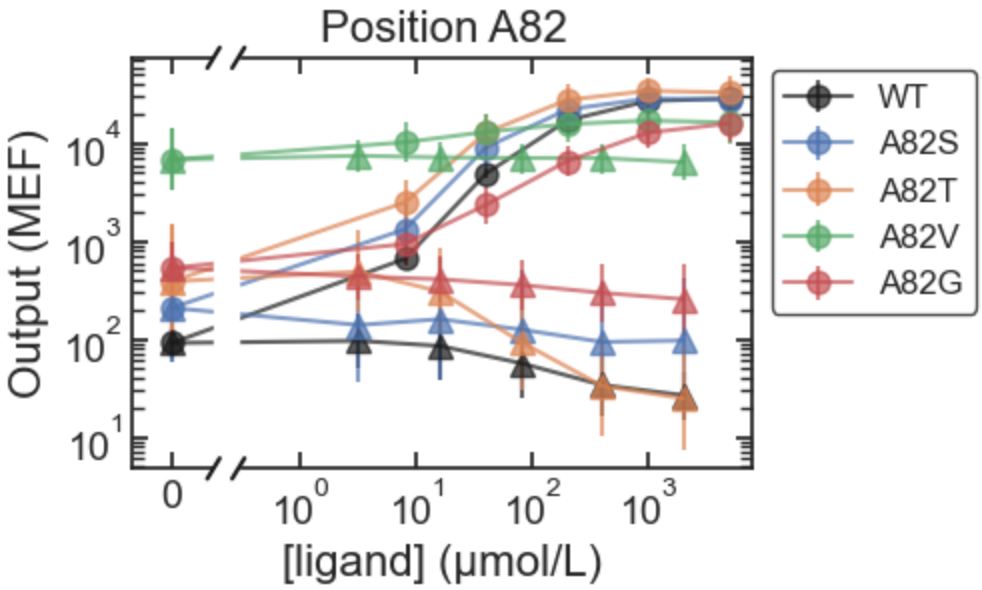

Position P76
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

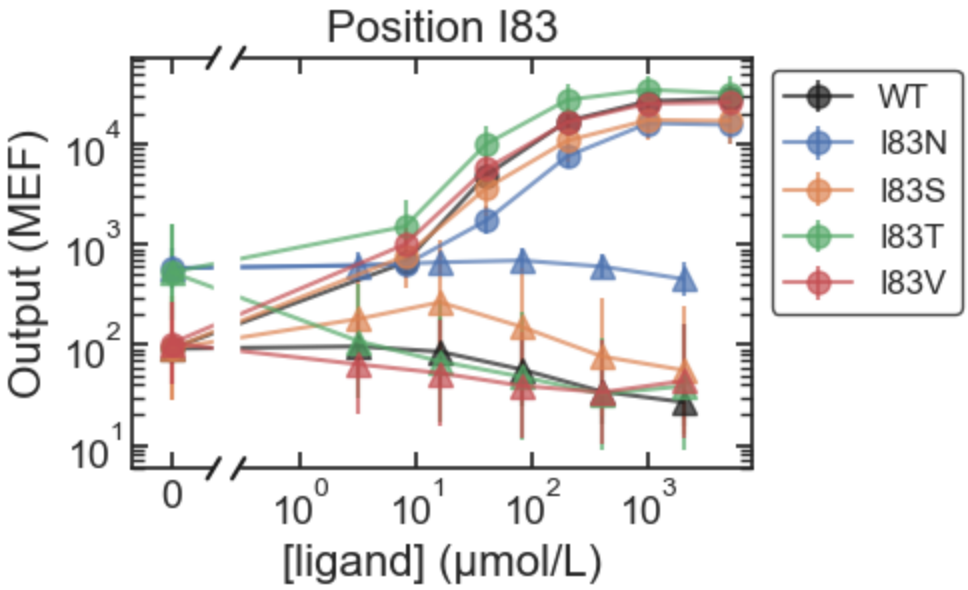

Position S77
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S77L
S77P
S77T

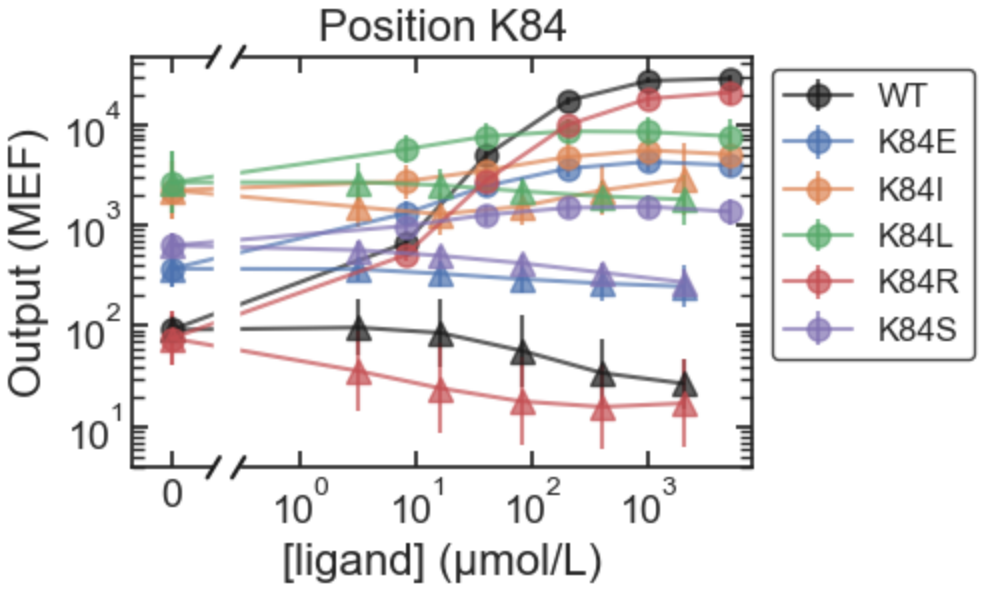

Position Q78
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q78H
Q78L
Q78R

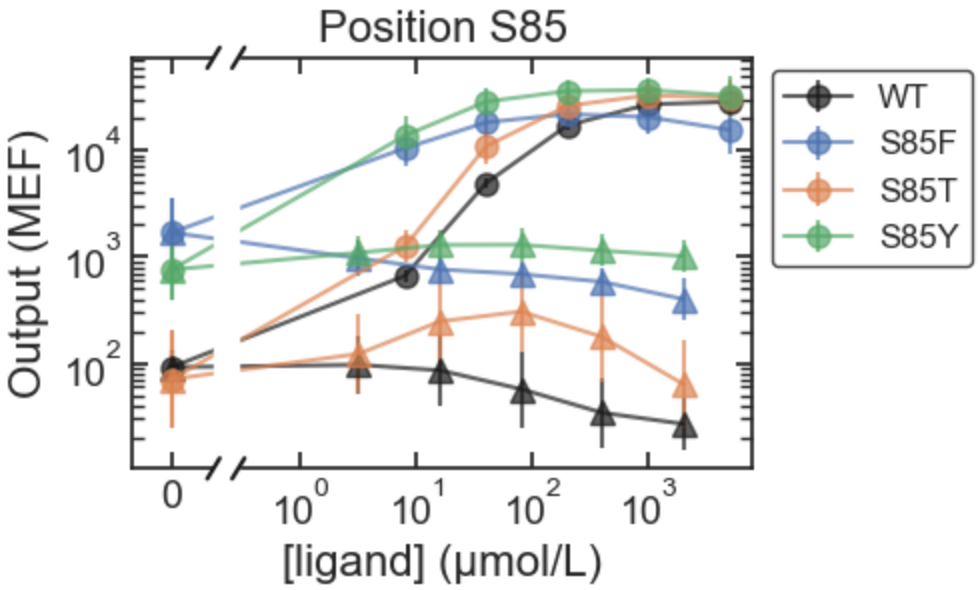

Position I79
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

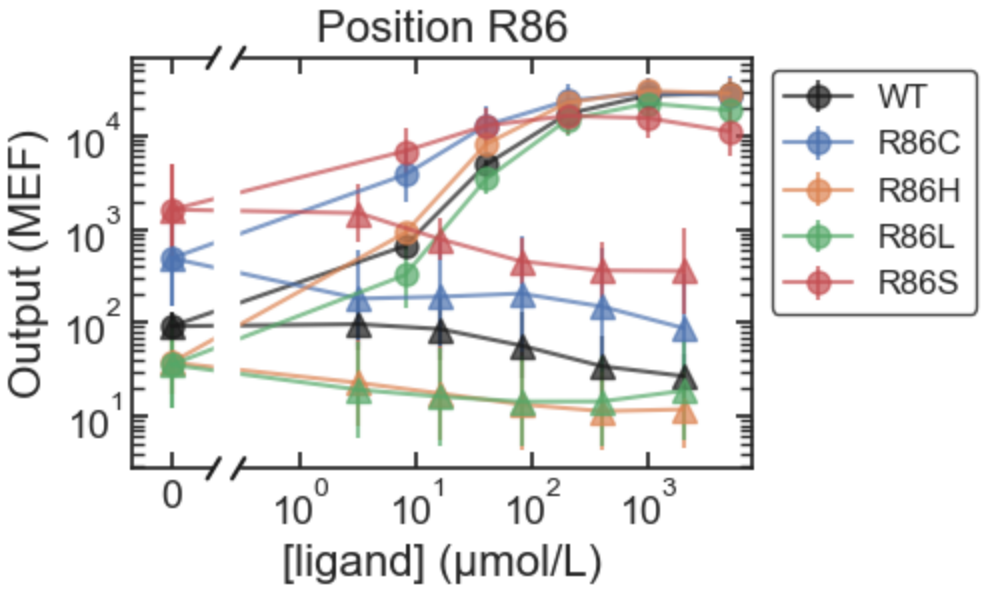

Position V80
Output (MEF)
[ligand] (μmol/L)
WT
V80D
V80L

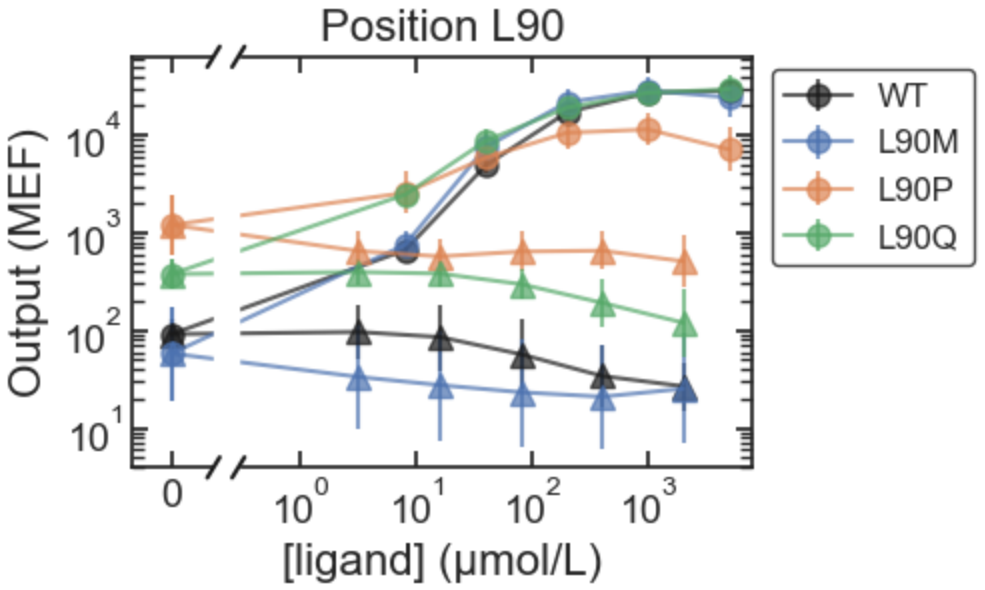

Position A82
Output (MEF)
[ligand] (μmol/L)
WT
A82S
A82T
A82V
A82G

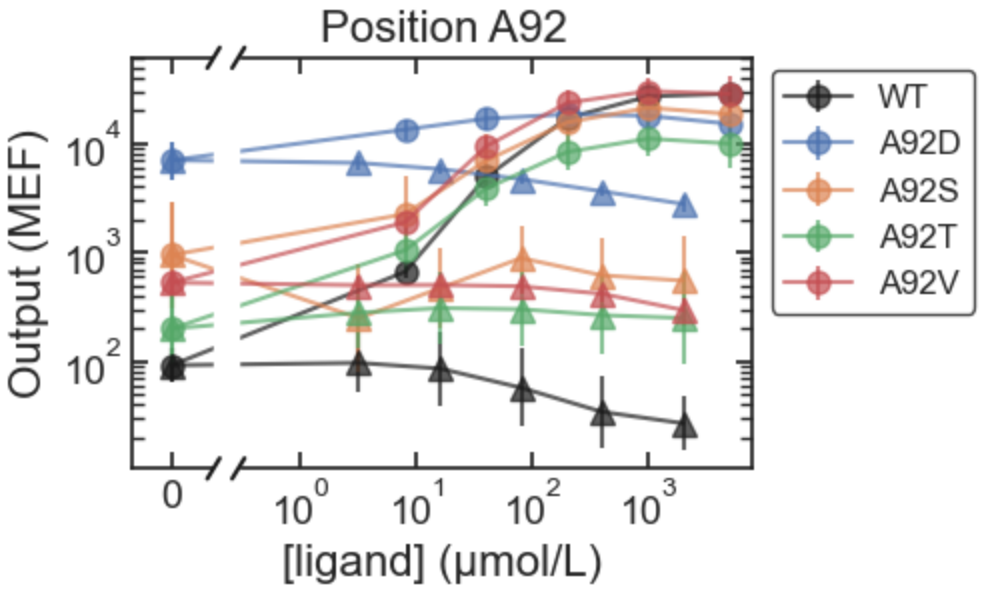

Position I83
Output (MEF)
[ligand] (μmol/L)
WT
I83N
I83S
I83T
I83V

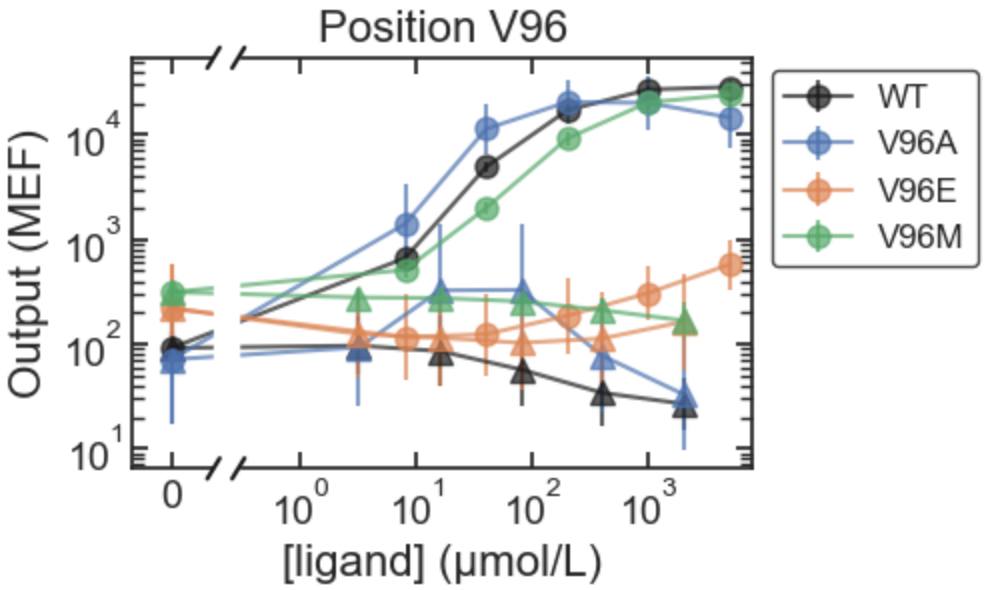

Position K84
Output (MEF)
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

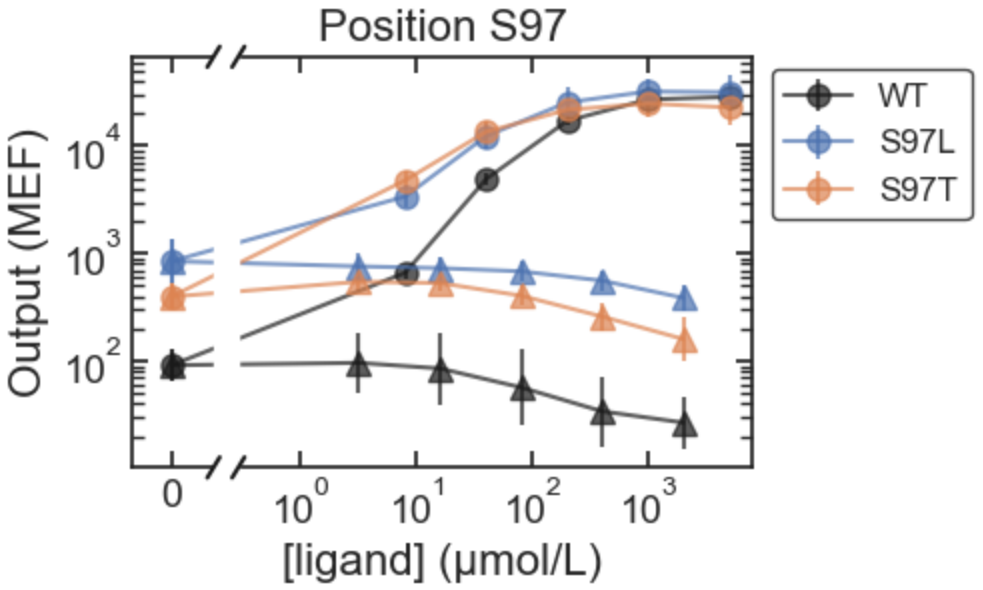

Position S85
Output (MEF)
[ligand] (μmol/L)
WT
S85F
S85T
S85Y

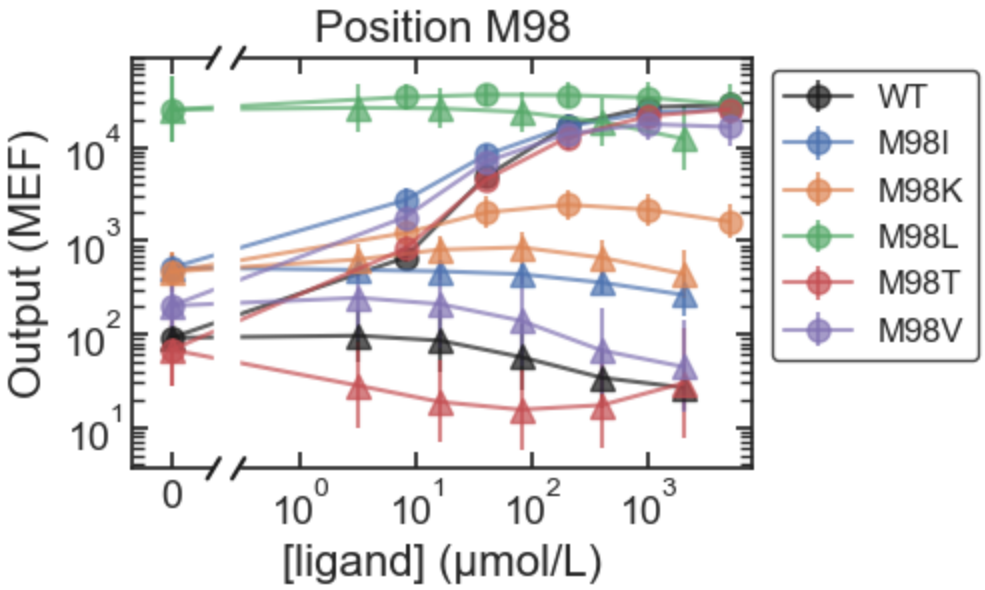

Position R86
Output (MEF)
[ligand] (μmol/L)
WT
R86C
R86H
R86L
R86S

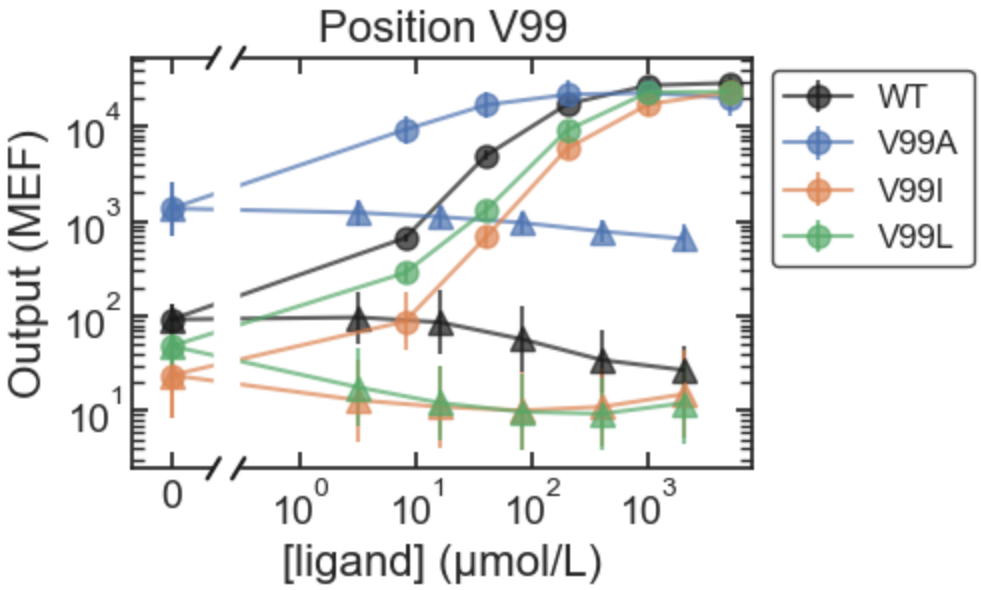

Position L90
Output (MEF)
[ligand] (μmol/L)
WT
L90M
L90P
L90Q

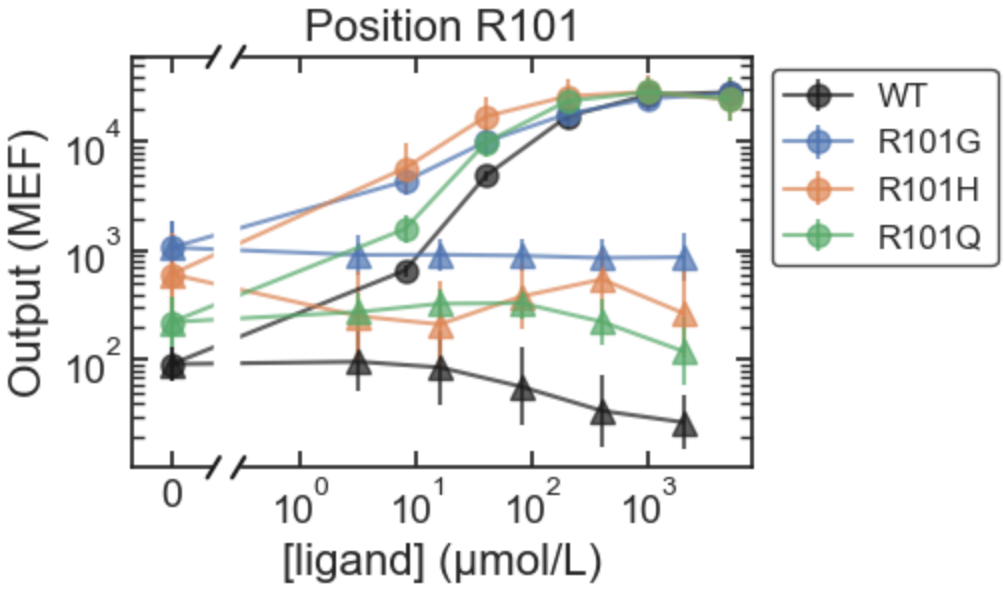

Position A92
Output (MEF)
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V

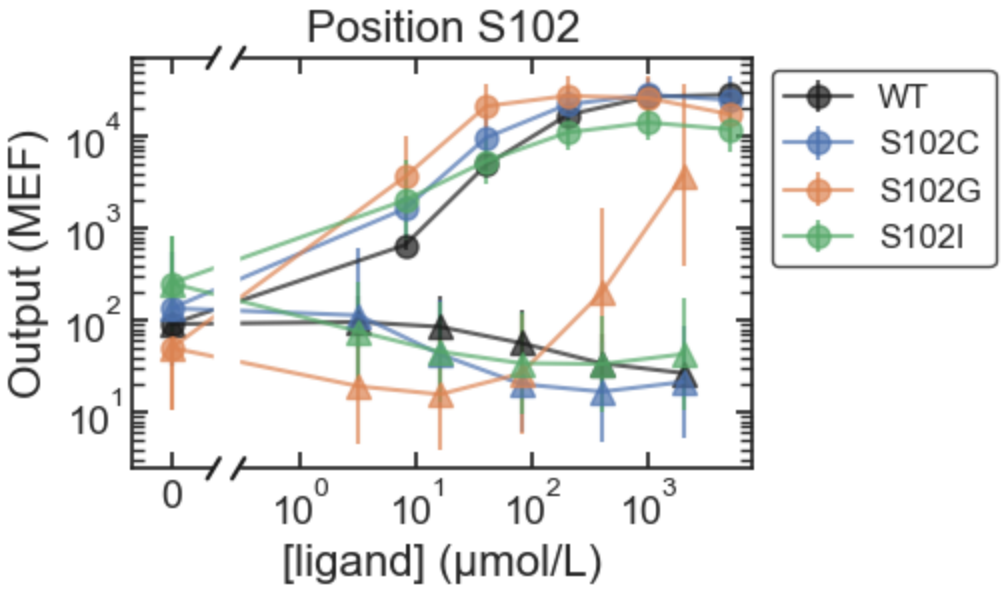

Position V96
Output (MEF)
[ligand] (μmol/L)
WT
V96A
V96E
V96M

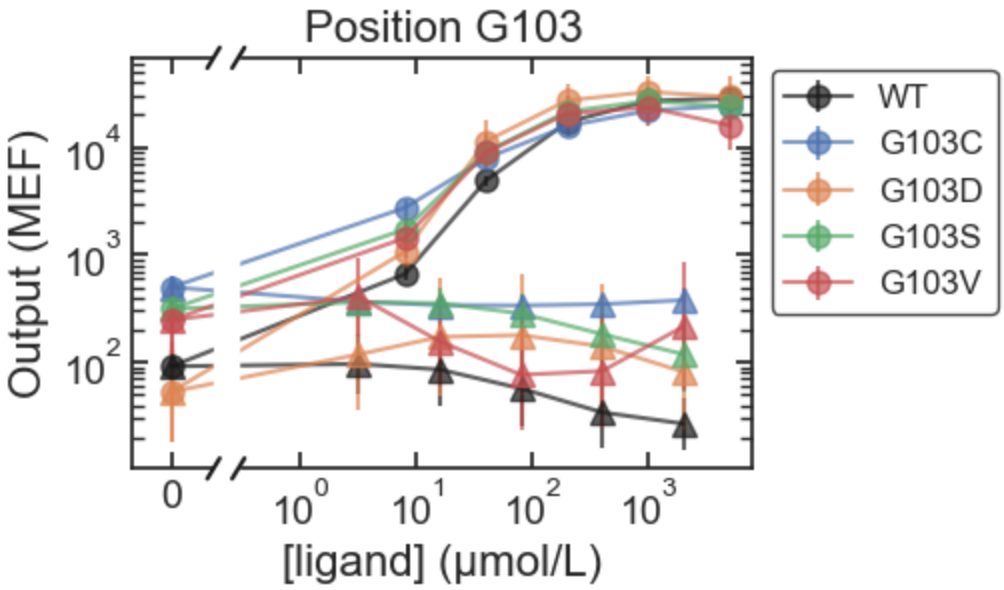

Position S97
Output (MEF)
[ligand] (μmol/L)
WT
S97L
S97T

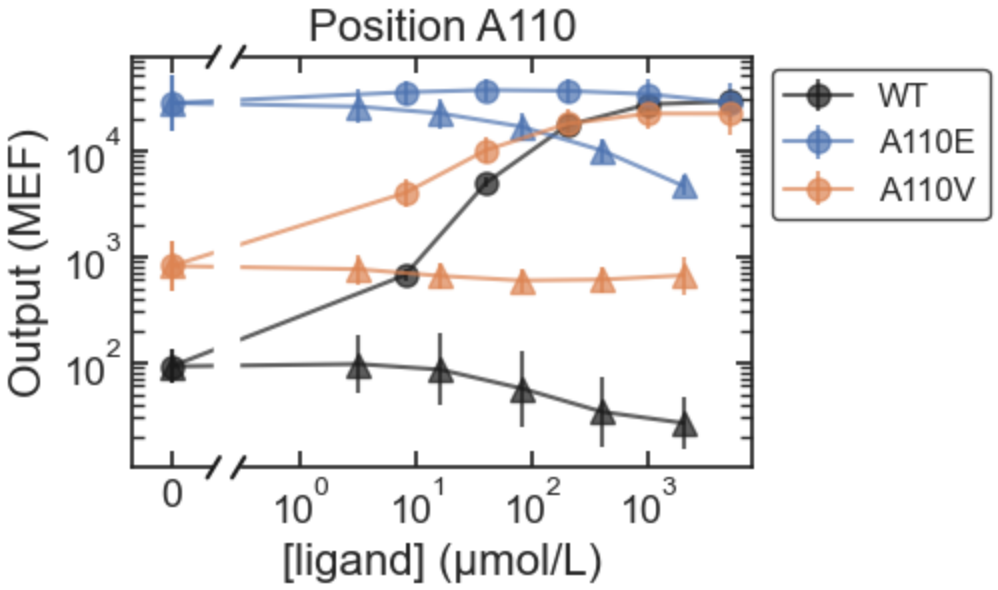

Position M98
Output (MEF)
[ligand] (μmol/L)
WT
M98I
M98K
M98L
M98T
M98V

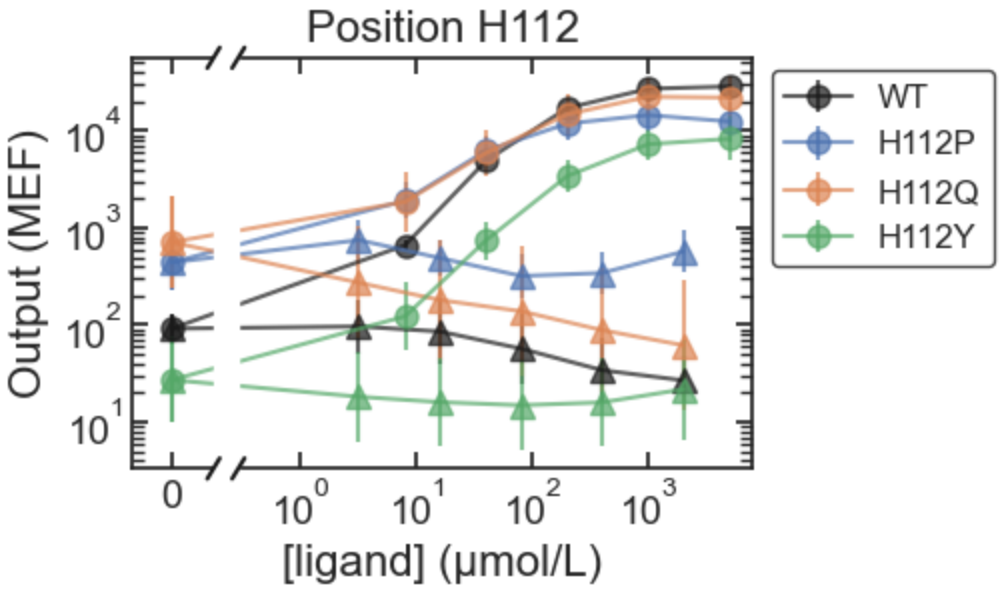

Position V99
Output (MEF)
[ligand] (μmol/L)
WT
V99A
V99I
V99L

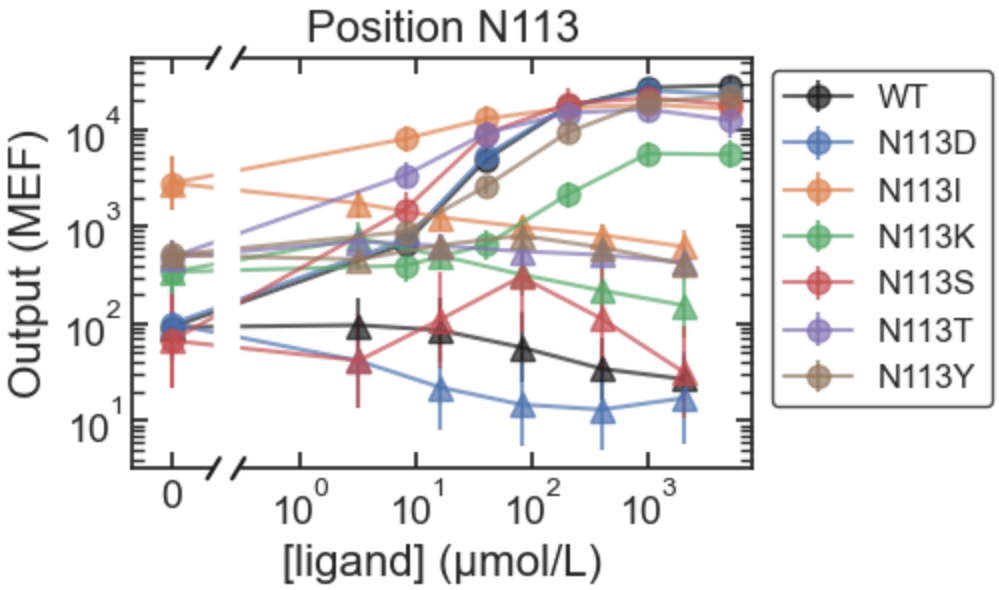

Position R101
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

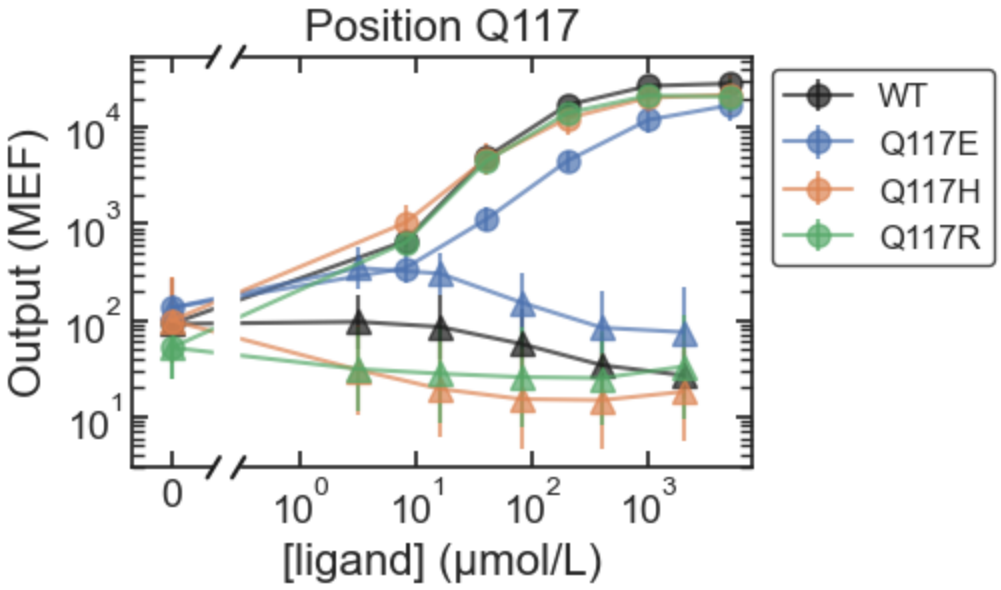

Position S102
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S102C
S102G
S102I

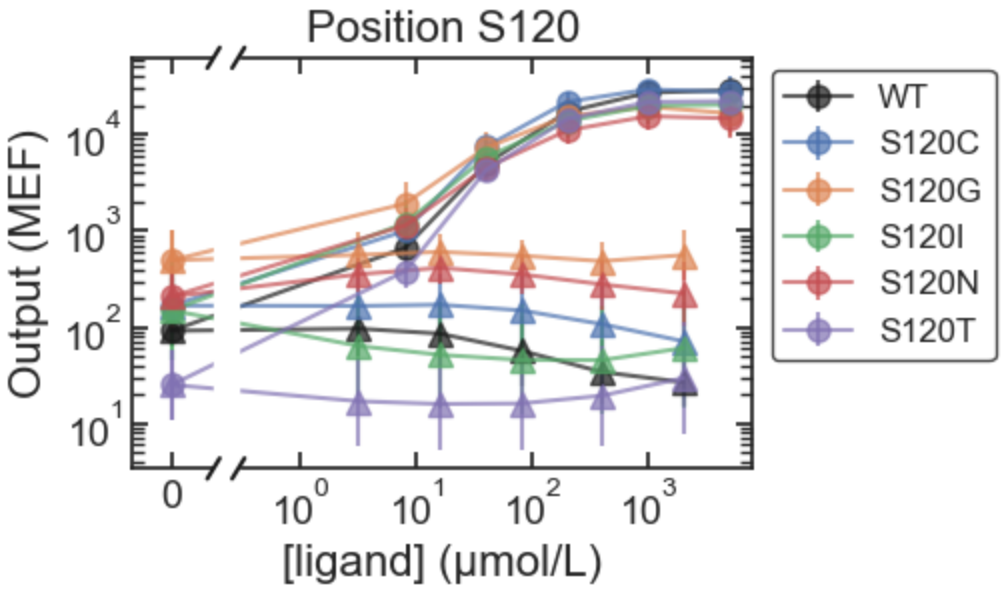

Position G103
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

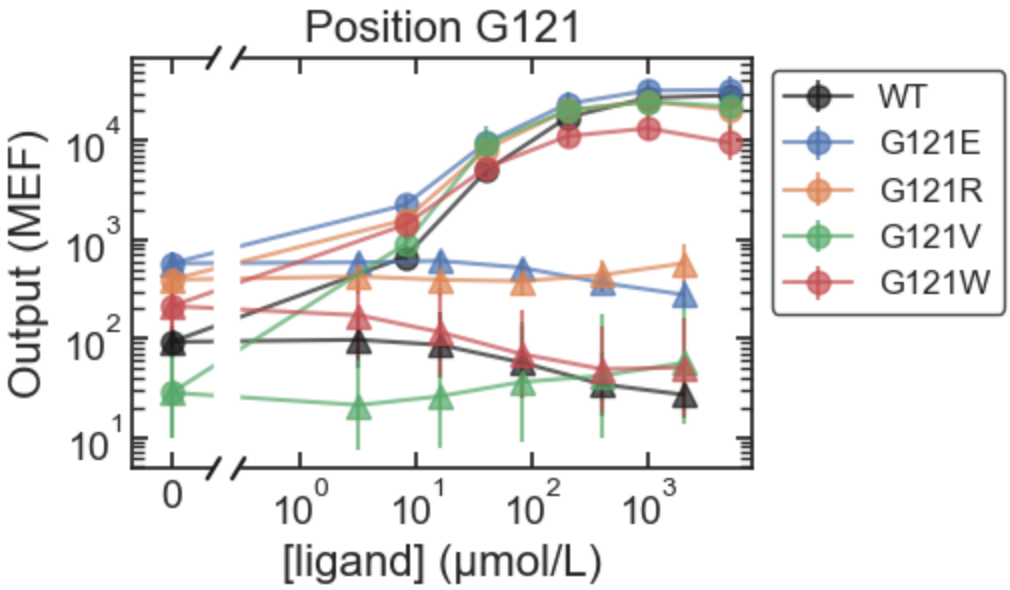

Position A110
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A110E
A110V

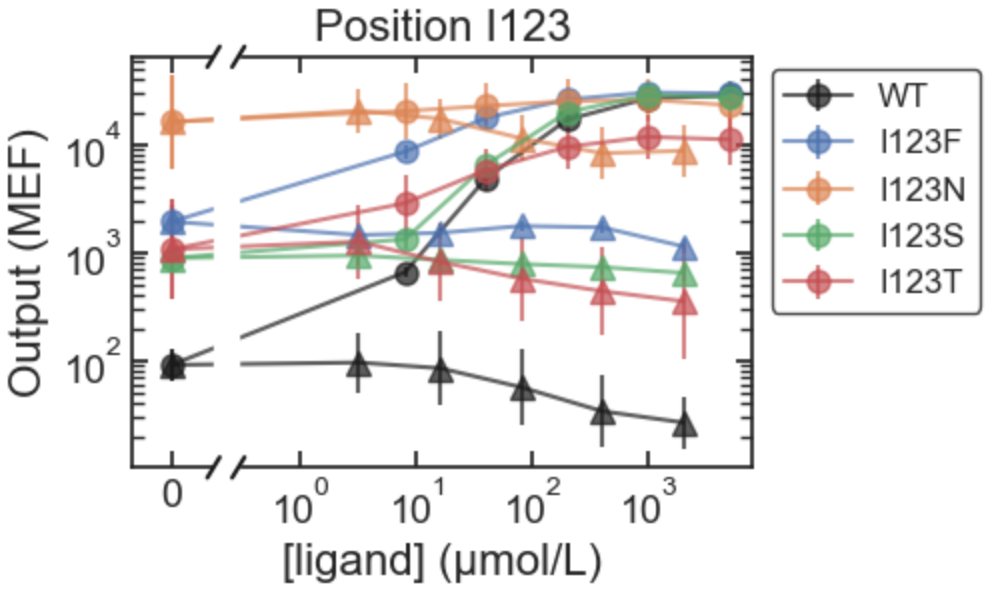

Position H112
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

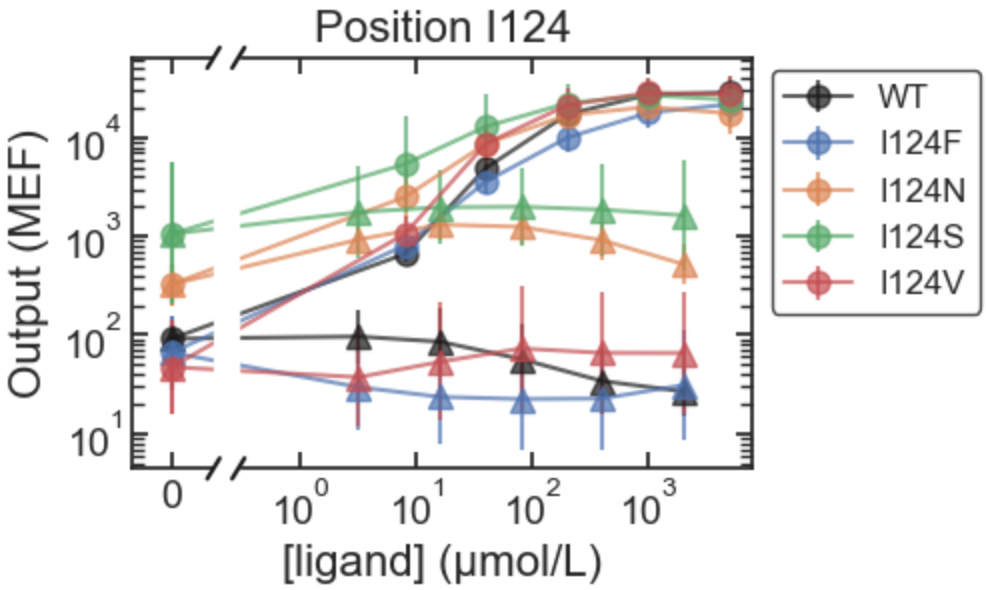

Position N113
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

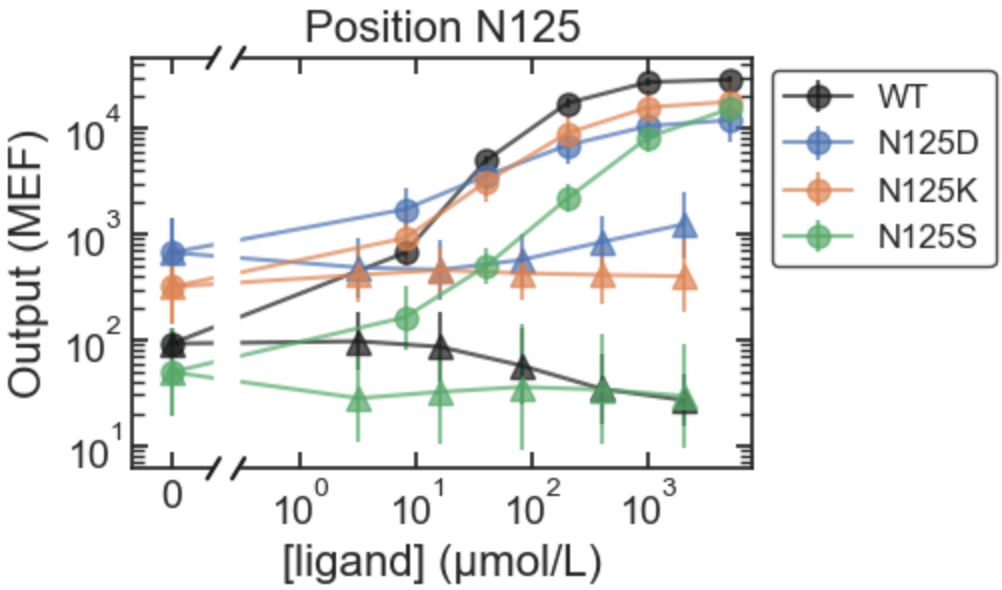

Position Q117
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q117E
Q117H
Q117R

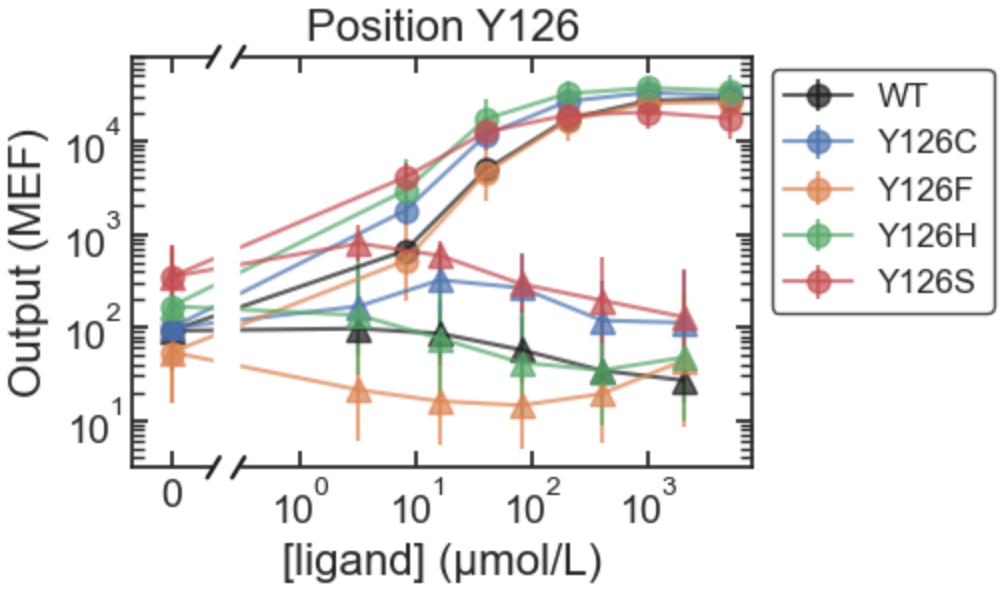

Position S120
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S120C
S120G
S120I
S120N
S120T

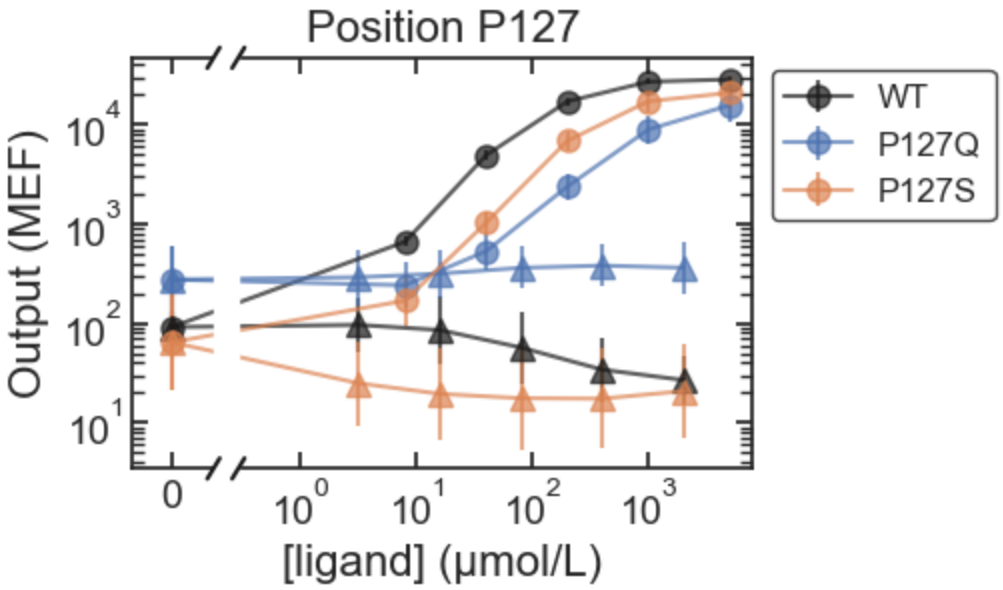

Position G121
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

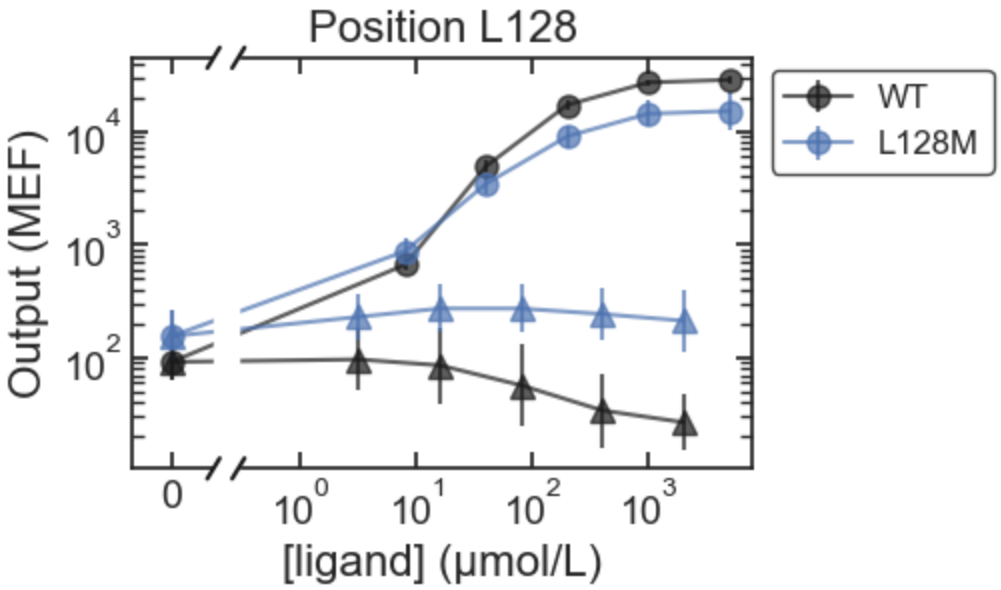

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

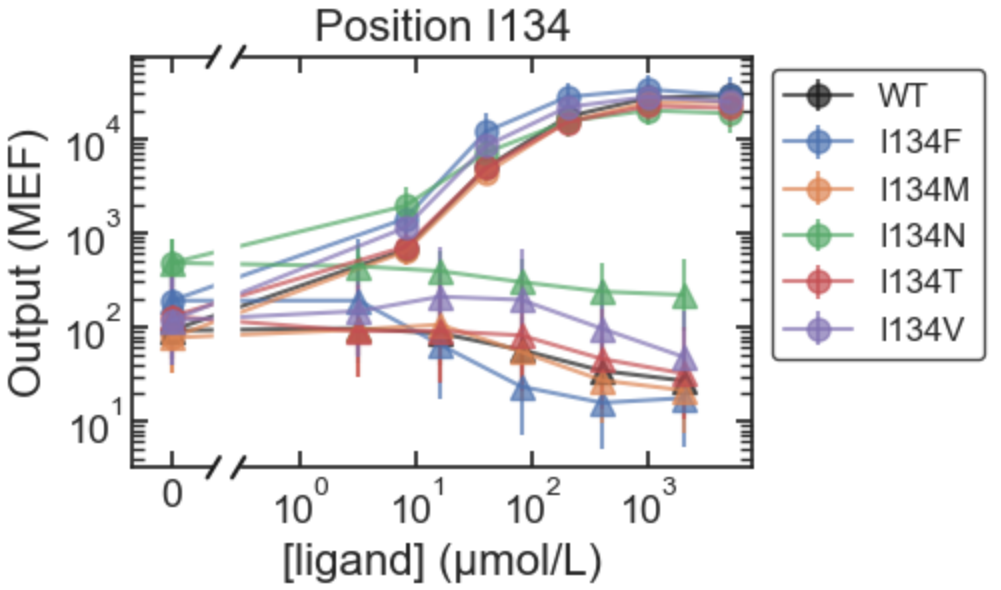

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V

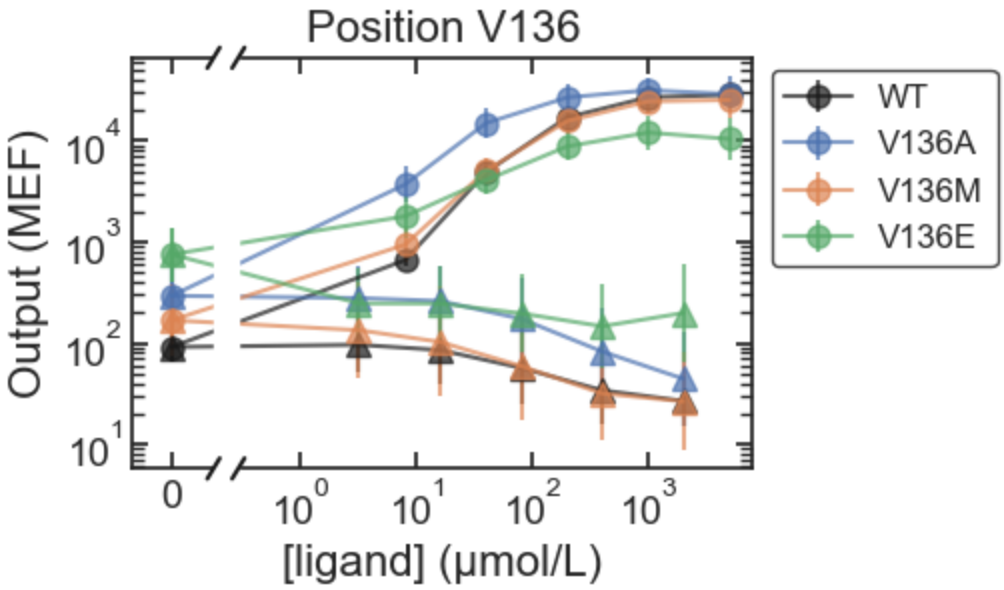

Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S

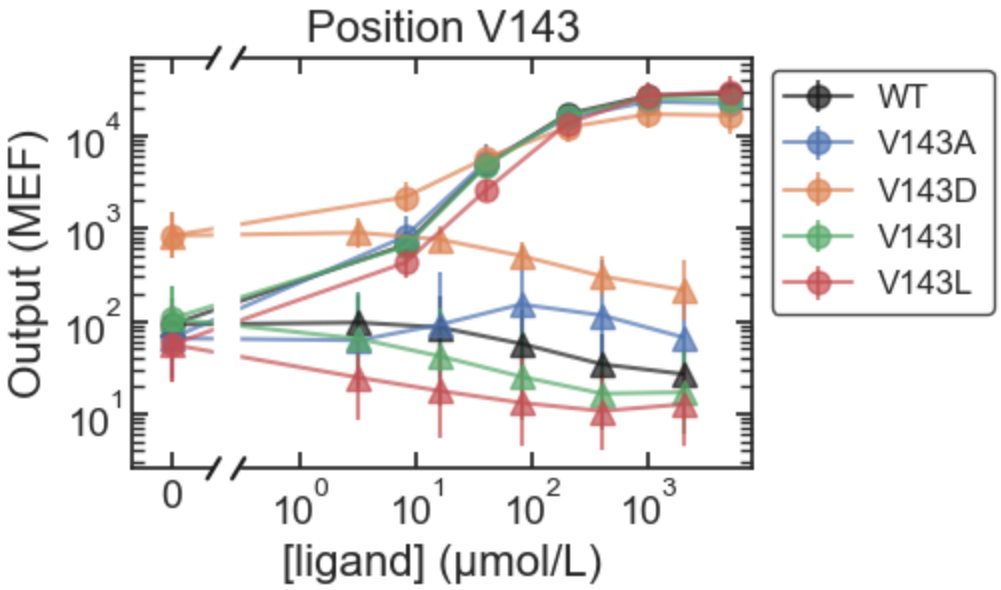

Position Y126
Output (MEF)
[ligand] (μmol/L)
WT
Y126C
Y126F
Y126H
Y126S

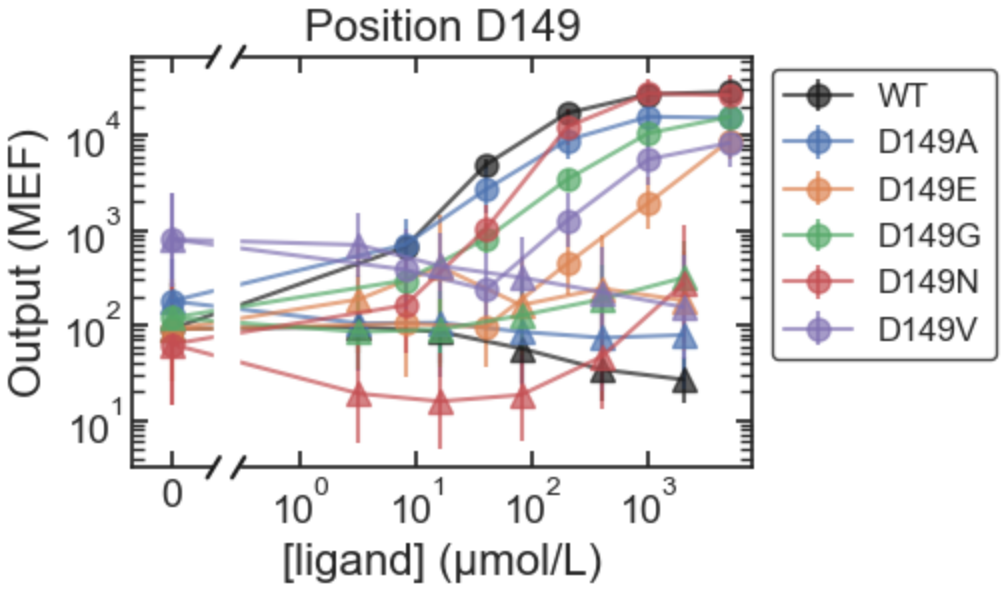

Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

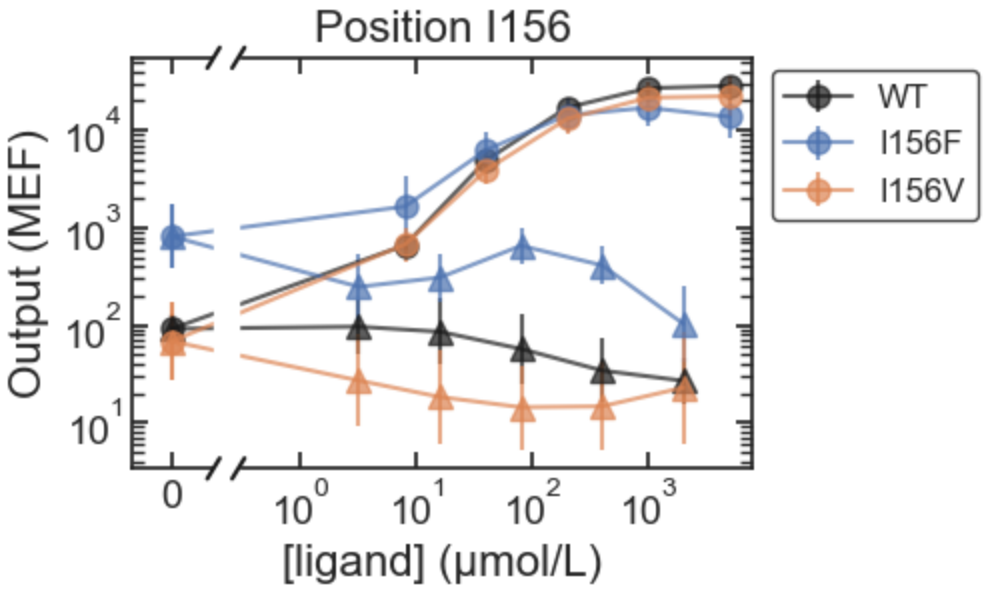

Position L128
Output (MEF)
[ligand] (μmol/L)
WT
L128M

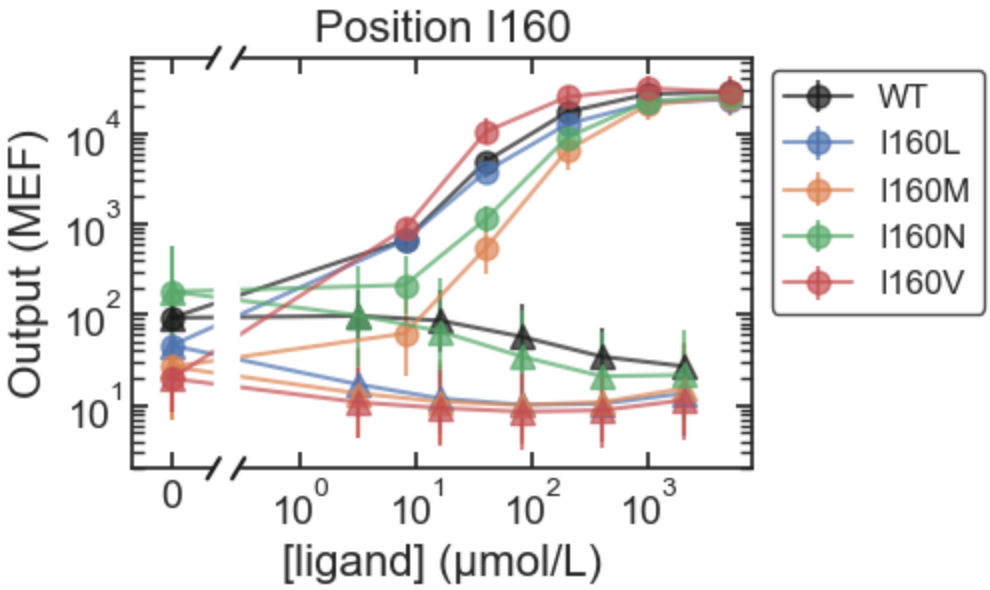

Position I134
Output (MEF)
[ligand] (µmol/L)
WT
I134F
I134M
I134N
I134T
I134V

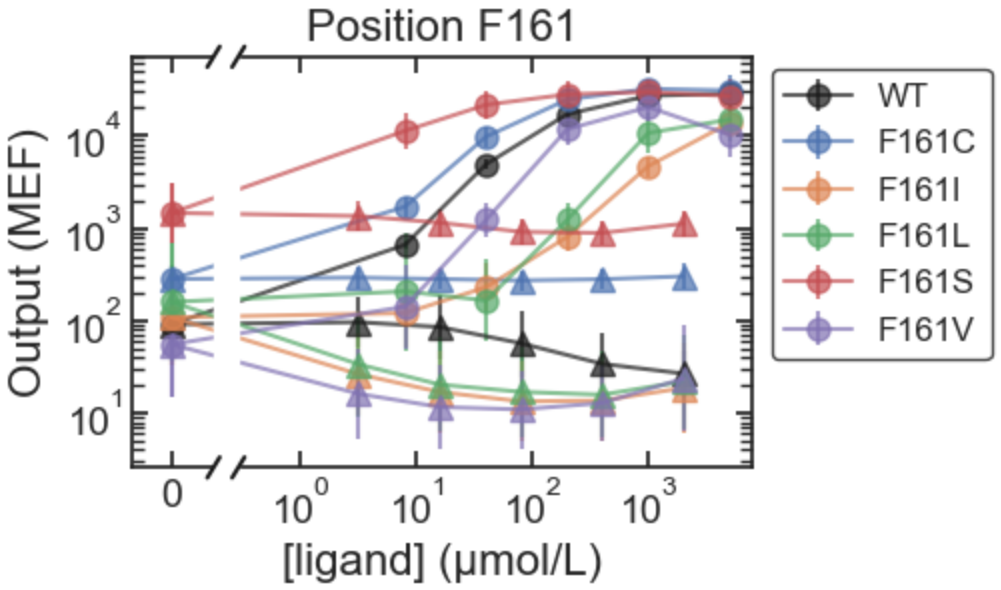

Position V136
Output (MEF)
[ligand] (µmol/L)
WT
V136A
V136M
V136E

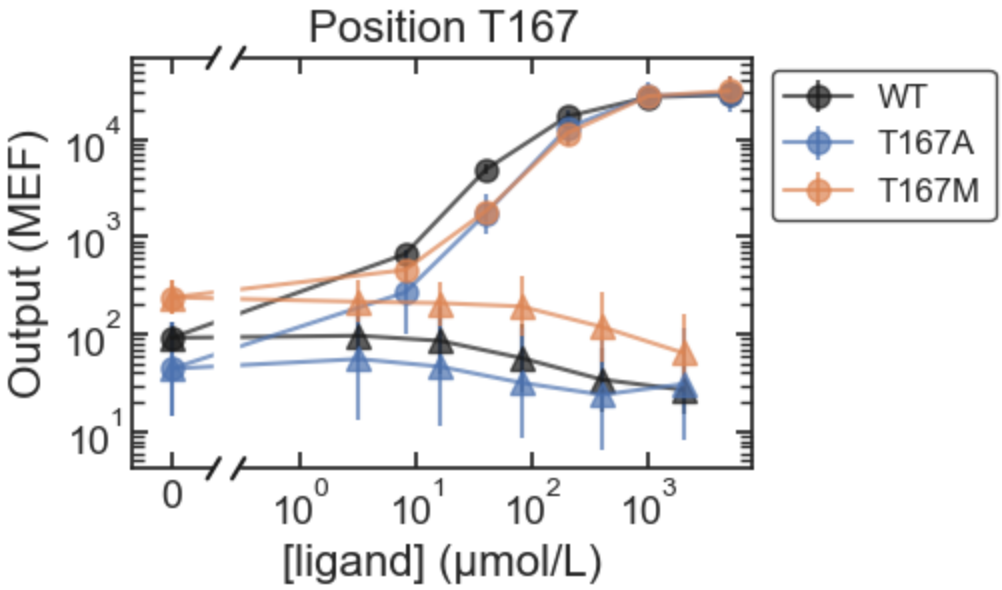

Position V143
Output (MEF)
[ligand] (µmol/L)
WT
V143A
V143D
V143I
V143L

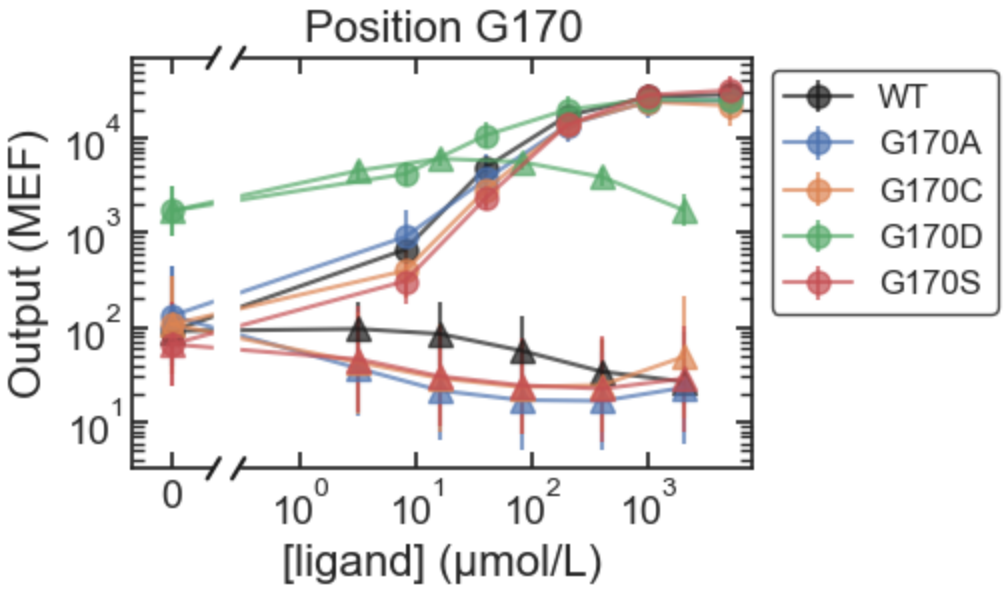

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

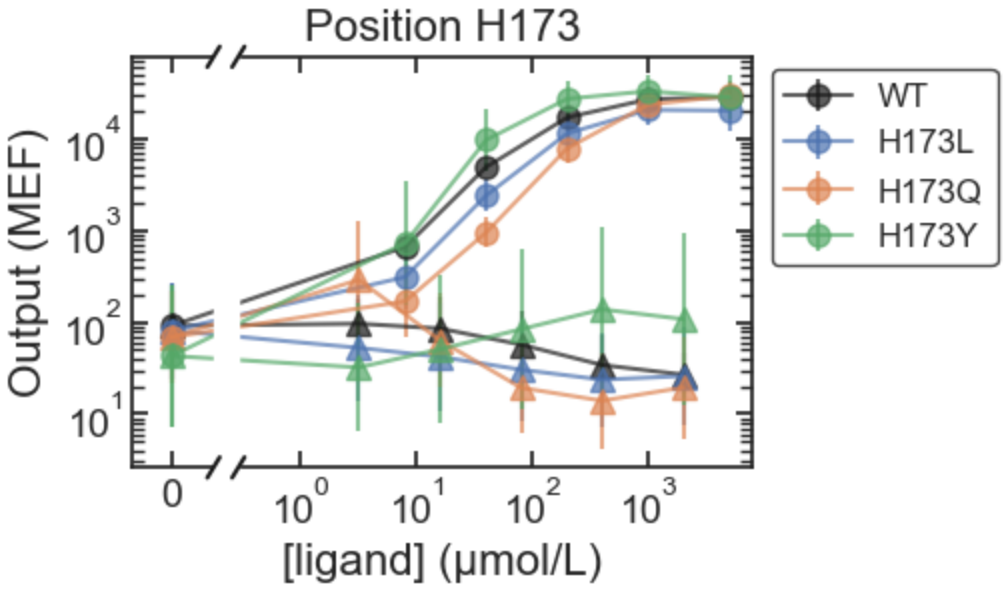

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

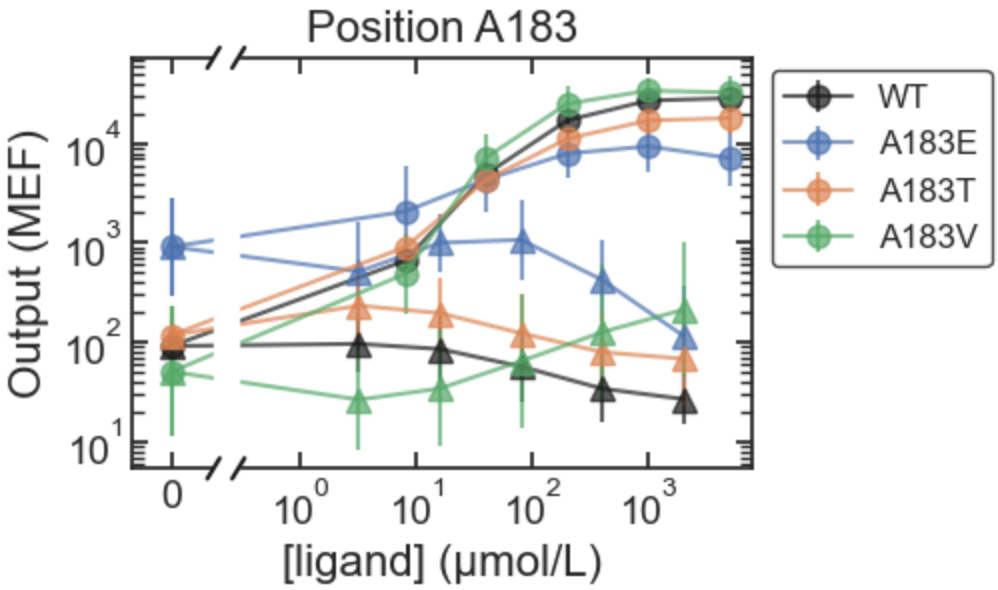

Position I160
Output (MEF)
[ligand] (μmol/L)
WT
I160L
I160M
I160N
I160V

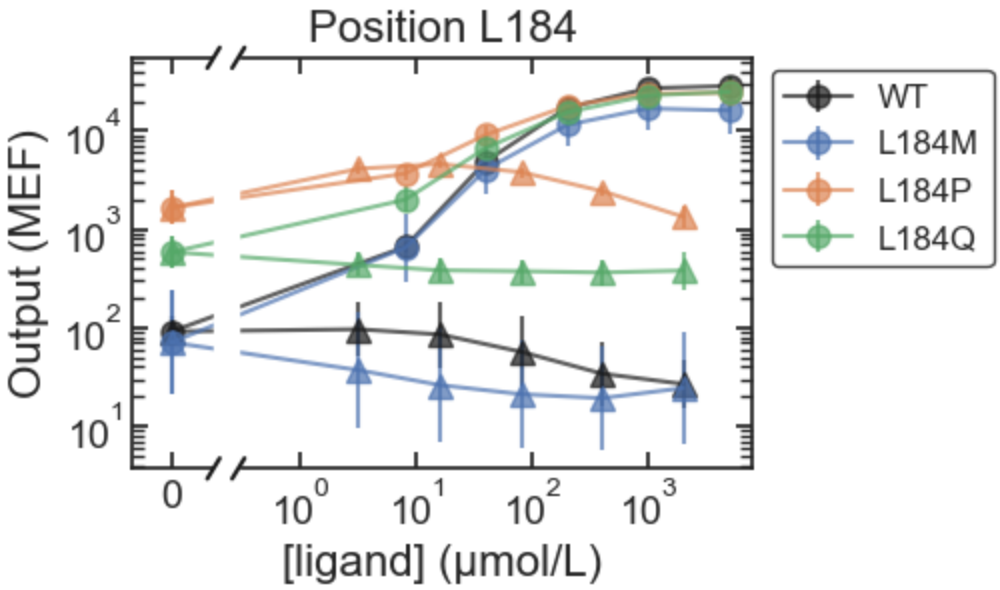

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

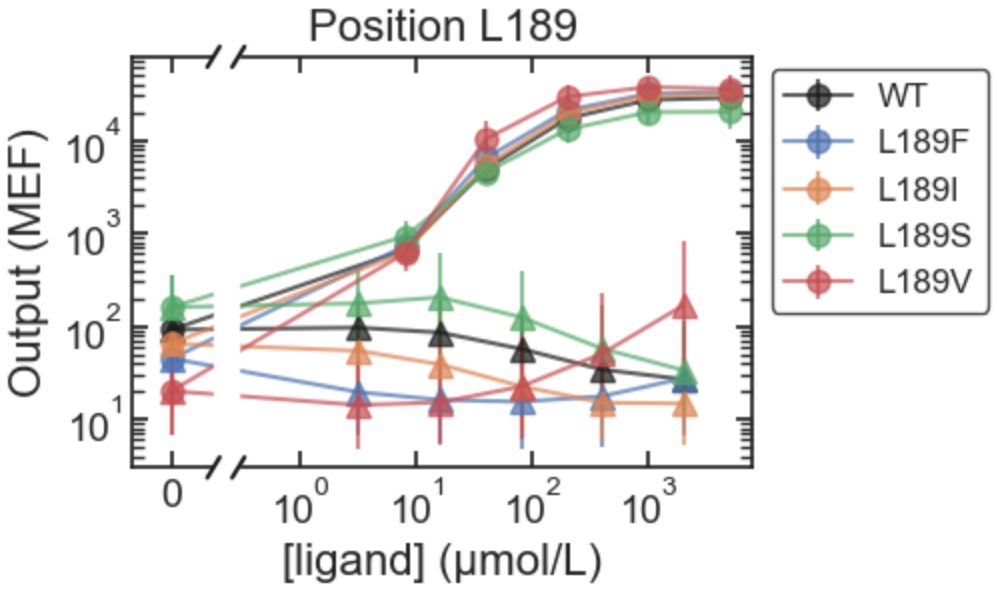

Position T167
Output (MEF)
[ligand] (μmol/L)
WT
T167A
T167M

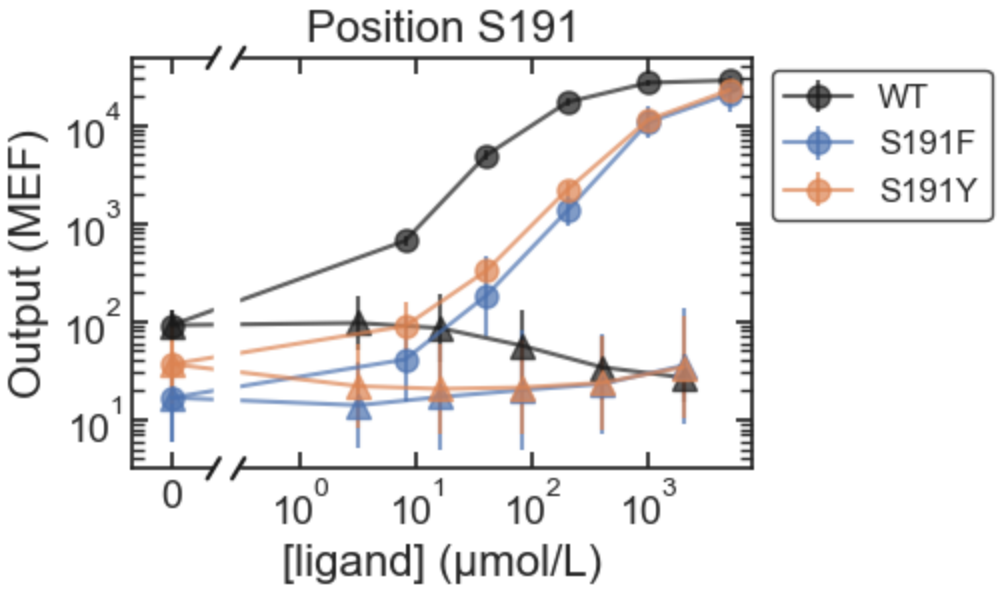

Position G170
Output (MEF)
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

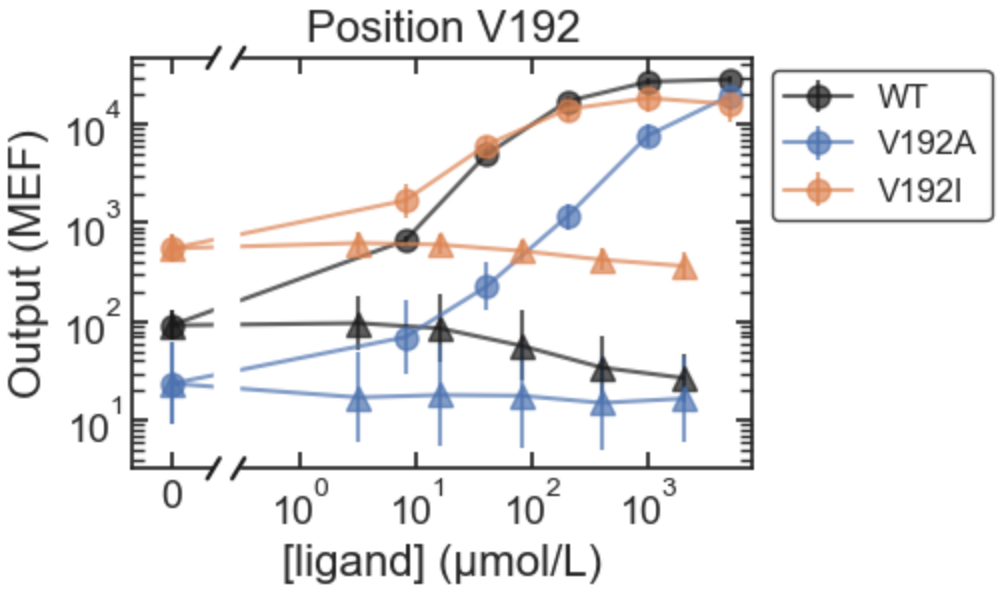

Position H173
Output (MEF)
[ligand] (μmol/L)
WT
H173L
H173Q
H173Y

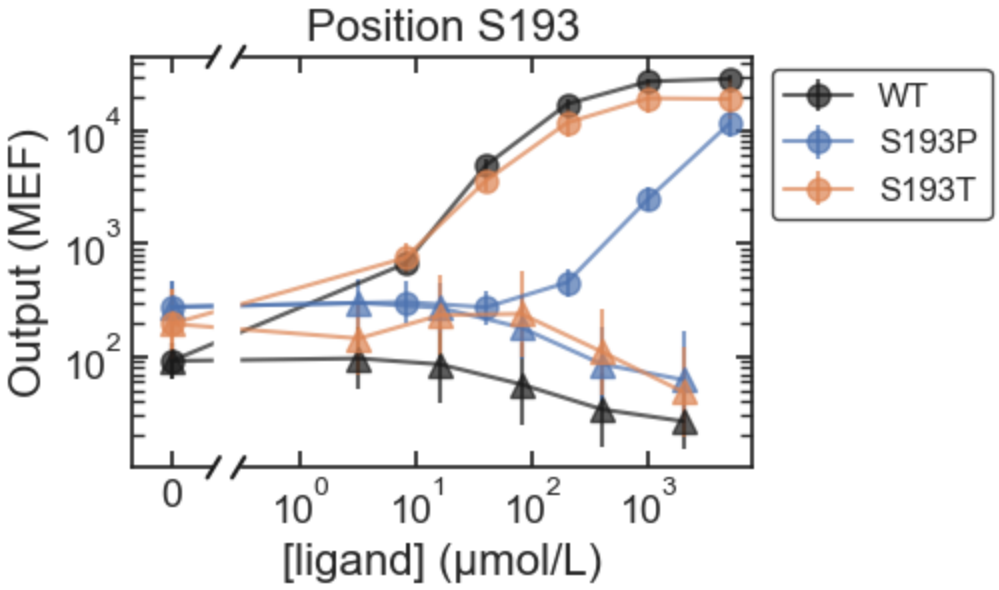

Position A183
Output (MEF)
[ligand] (μmol/L)
WT
A183E
A183T
A183V

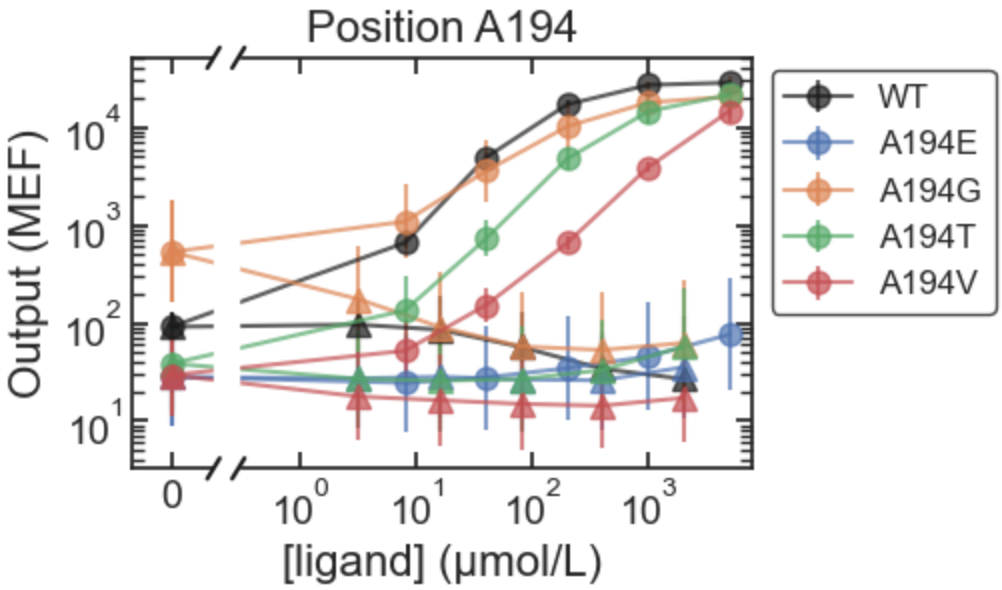

Position L184
Output (MEF)
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

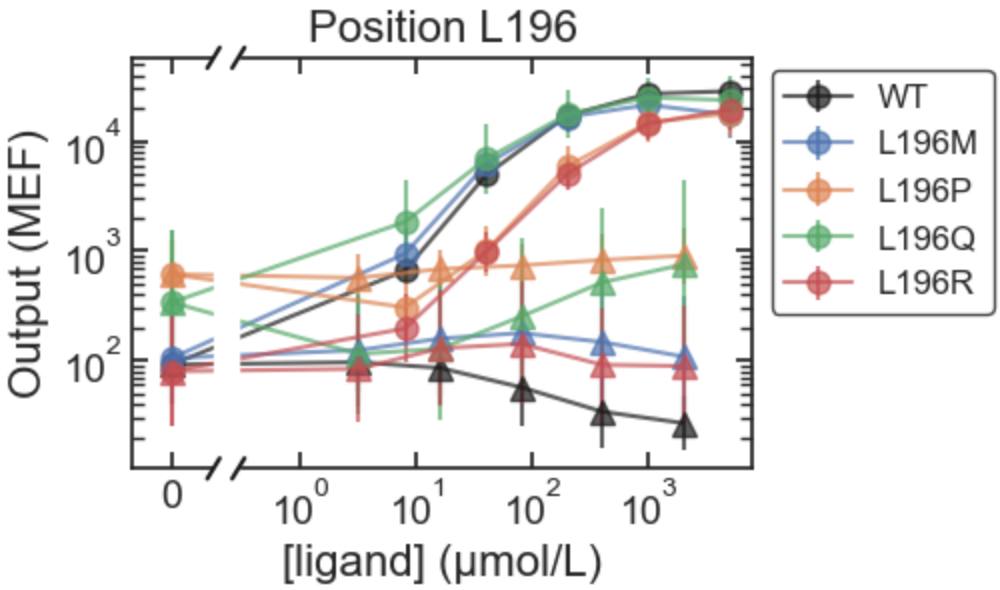

Position L189
Output (MEF)
[ligand] (μmol/L)
WT
L189F
L189I
L189S
L189V

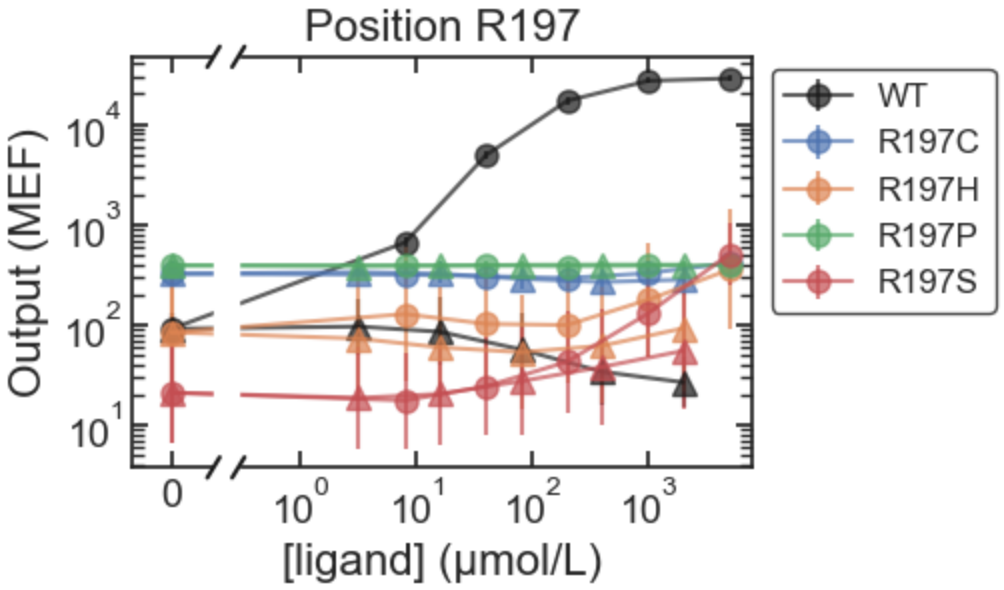

Position S191
Output (MEF)
[ligand] (μmol/L)
WT
S191F
S191Y

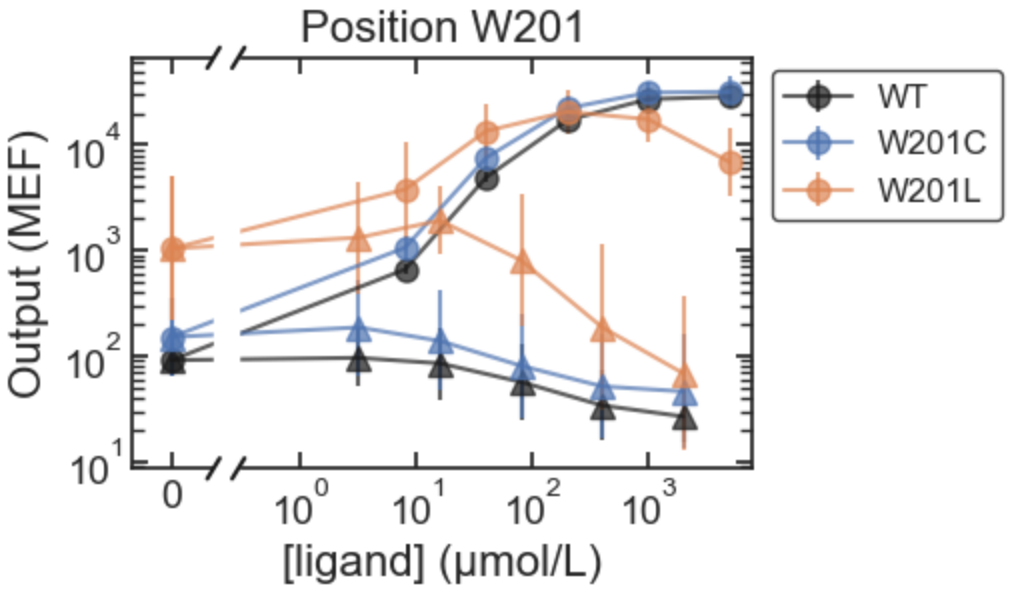

Position V192
Output (MEF)
[ligand] (μmol/L)
WT
V192A
V192I

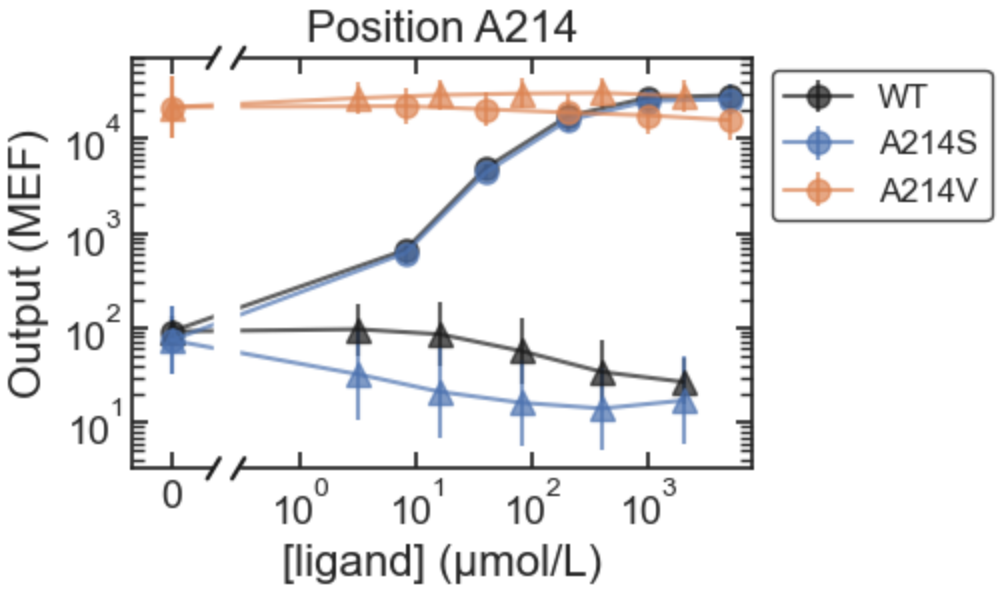

Position S193
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S193P
S193T

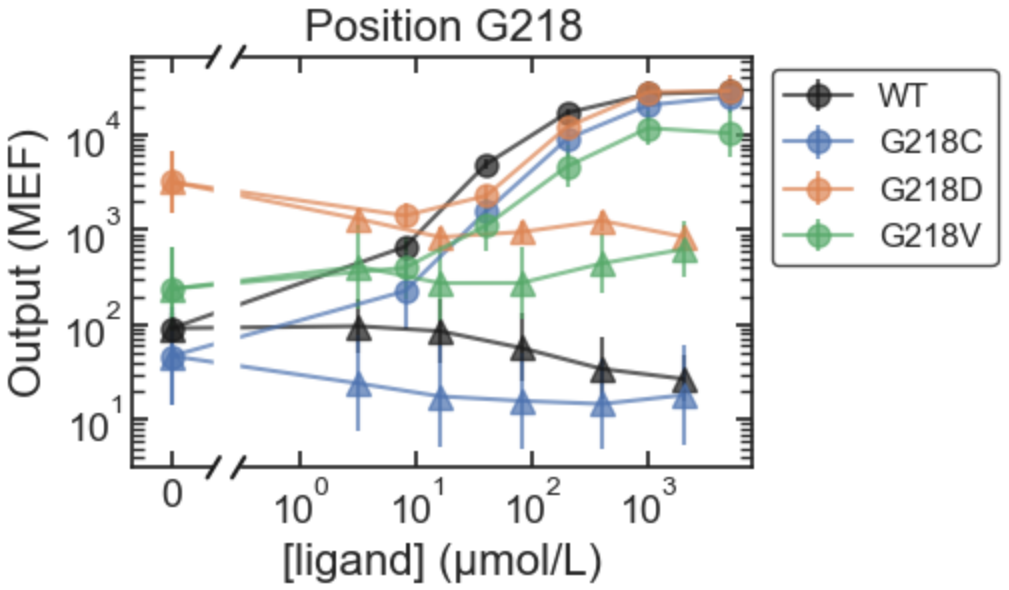

Position A194
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A194E
A194G
A194T
A194V

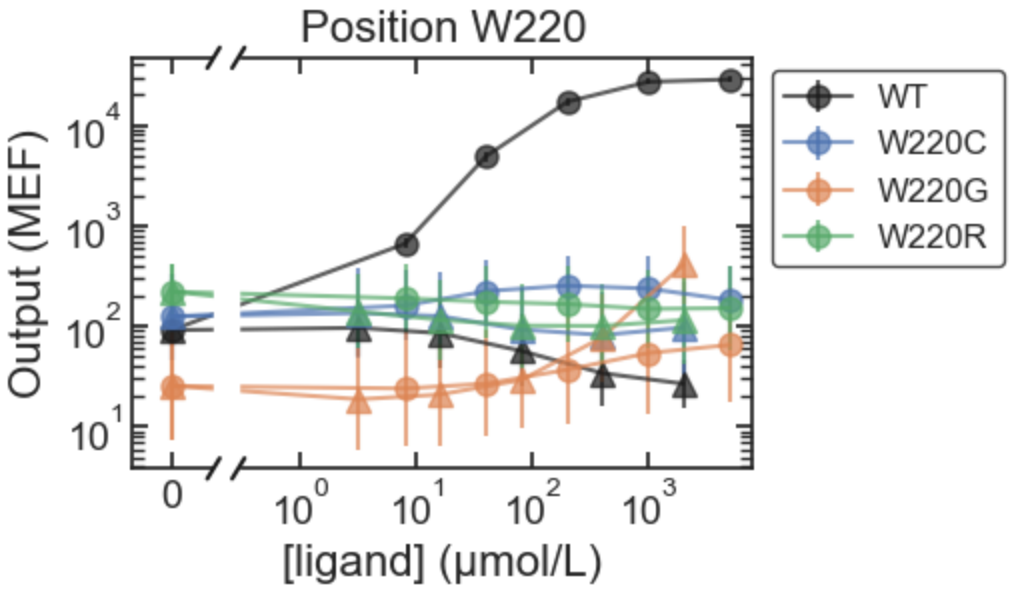

Position L196
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

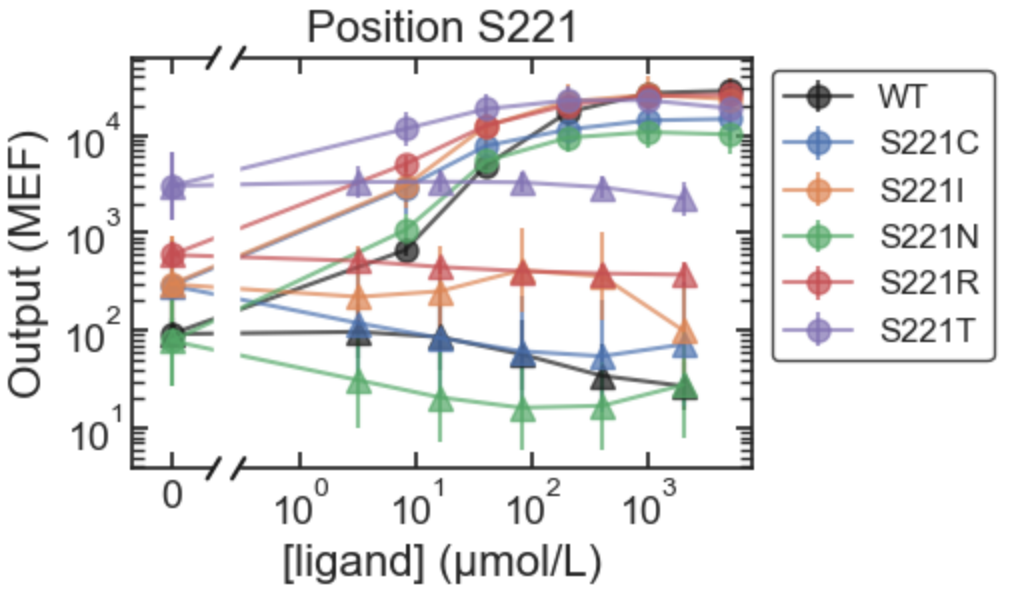

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

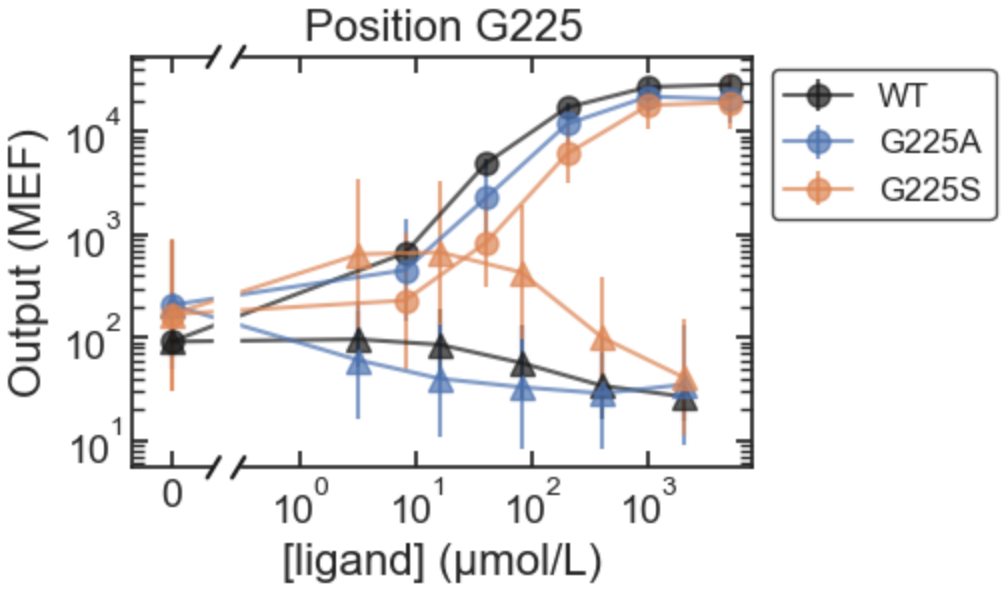

Position W201
Output (MEF)
[ligand] (μmol/L)
WT
W201C
W201L

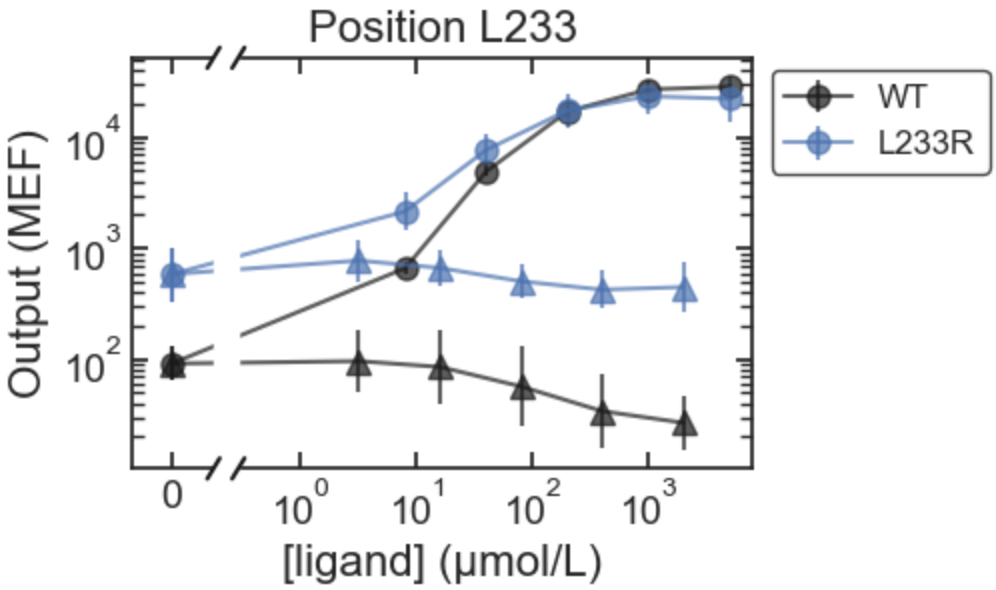

Position A214
Output (MEF)
[ligand] (μmol/L)
WT
A214S
A214V

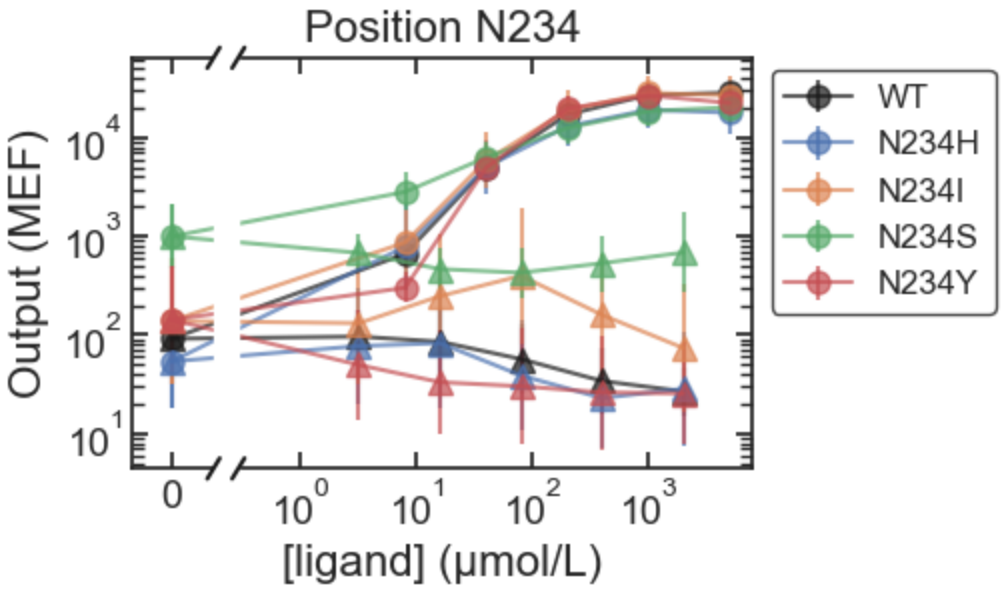

Position G218
Output (MEF)
[ligand] (μmol/L)
WT
G218C
G218D
G218V

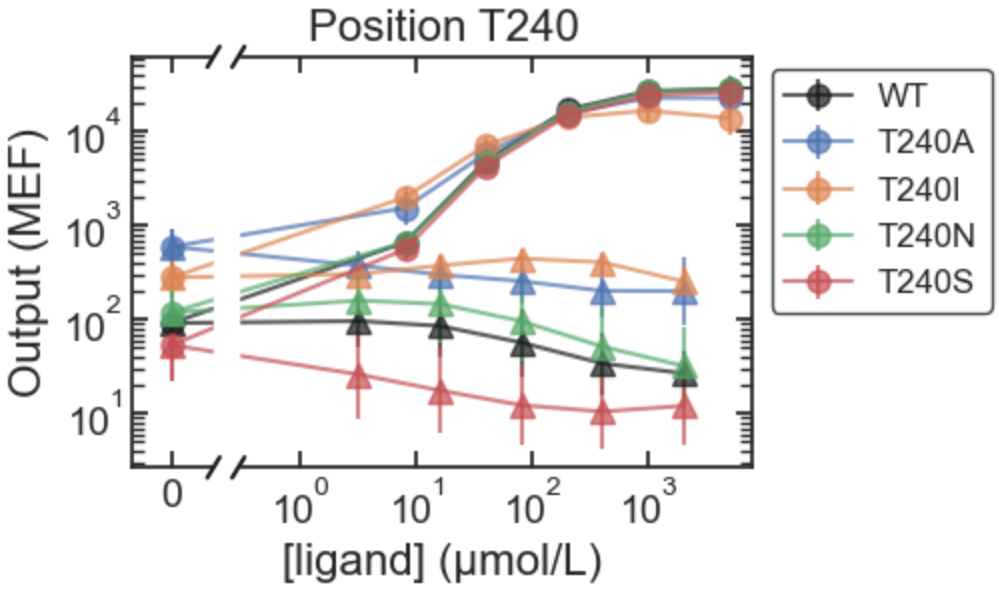

Position W220
Output (MEF)
[ligand] (μmol/L)
WT
W220C
W220G
W220R

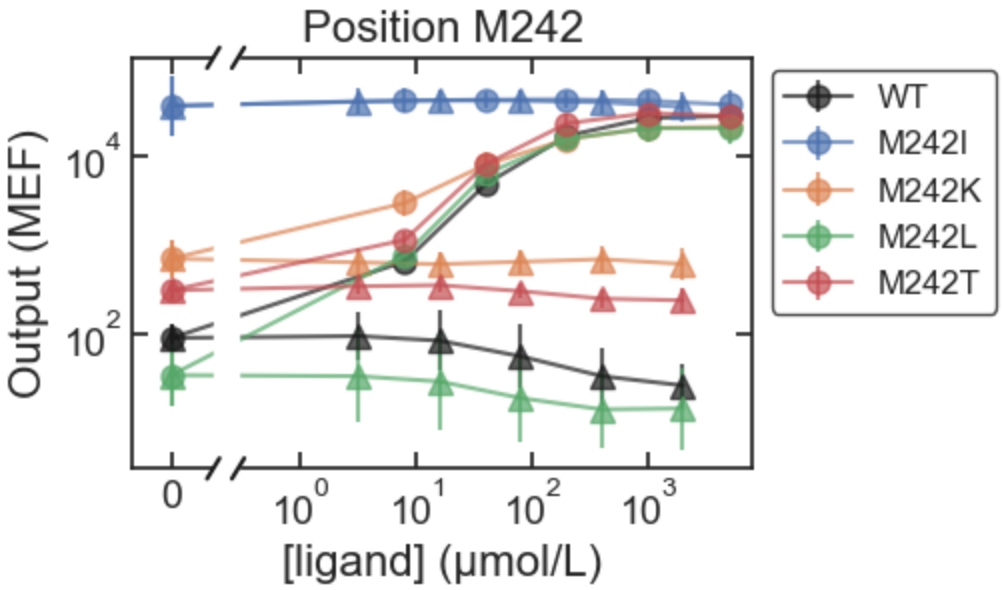

Position S221
Output (MEF)
[ligand] (μmol/L)
WT
S221C
S221I
S221N
S221R
S221T

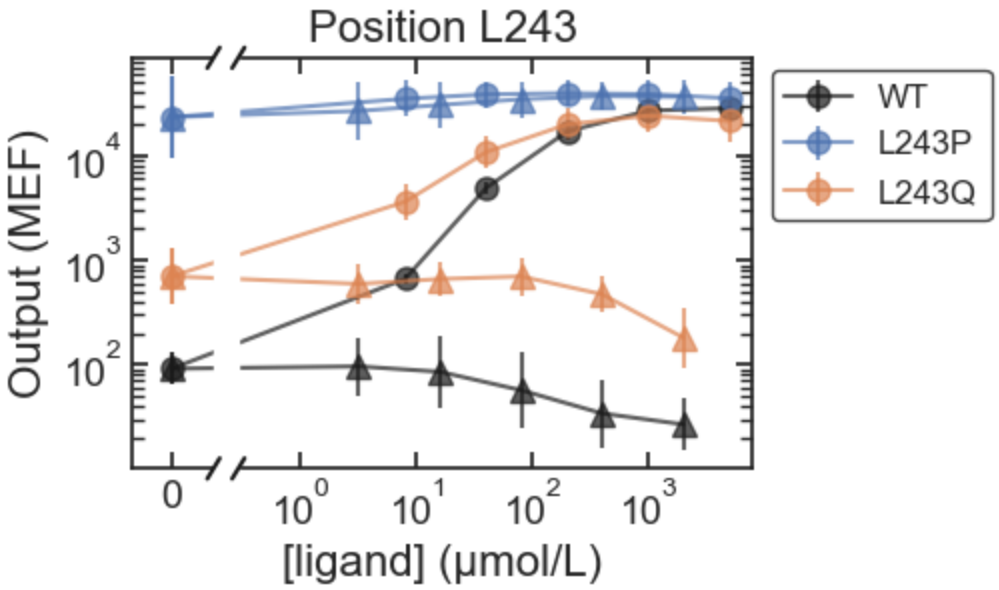

Position G225
Output (MEF)
[ligand] (μmol/L)
WT
G225A
G225S

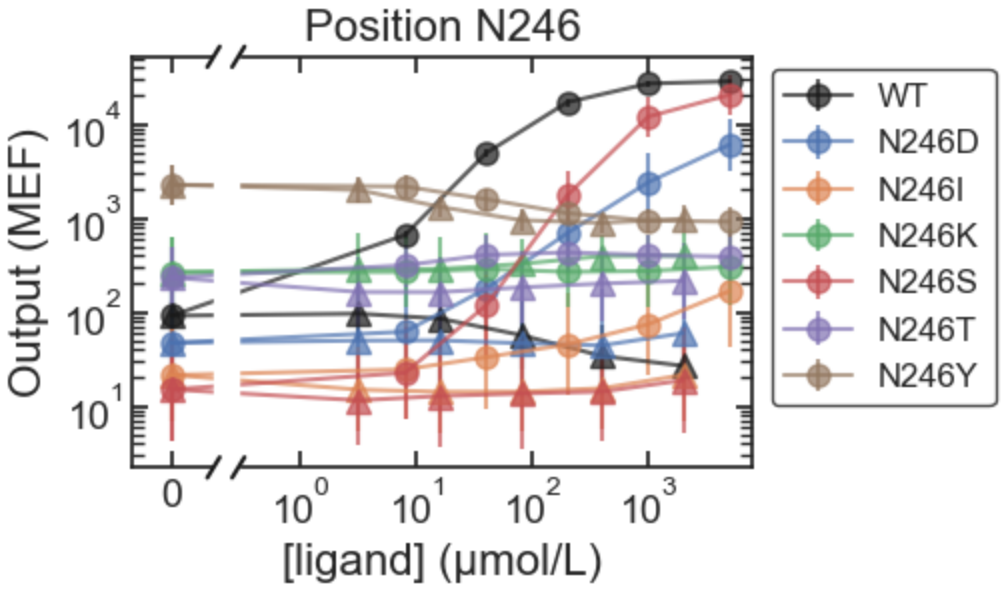

Position L233
Output (MEF)
[ligand] (μmol/L)
WT
L233R

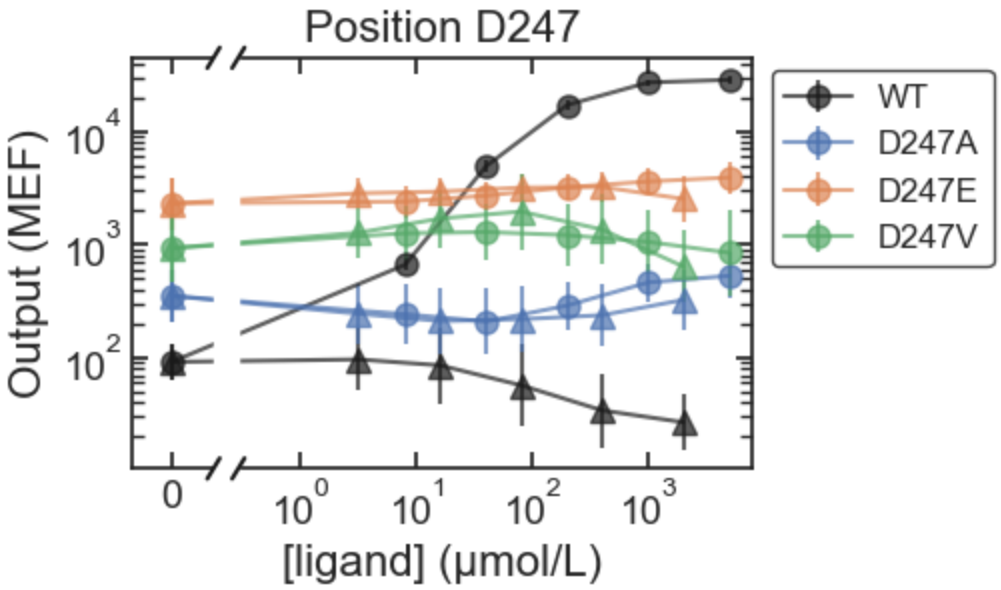

Position N234
Output (MEF)
[ligand] (μmol/L)
WT
N234H
N234I
N234S
N234Y

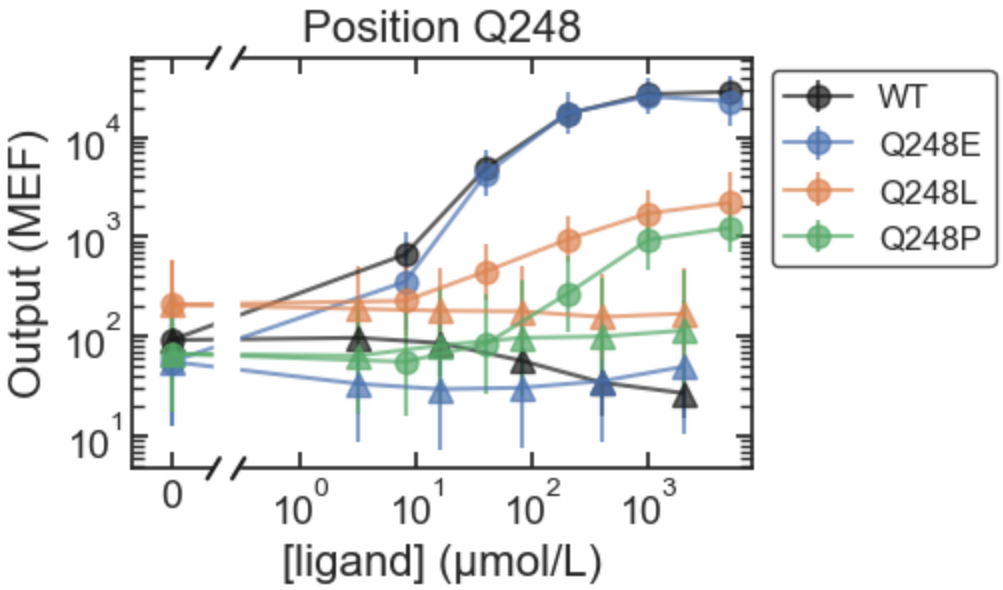

Position T240
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
T240A
T240I
T240N
T240S

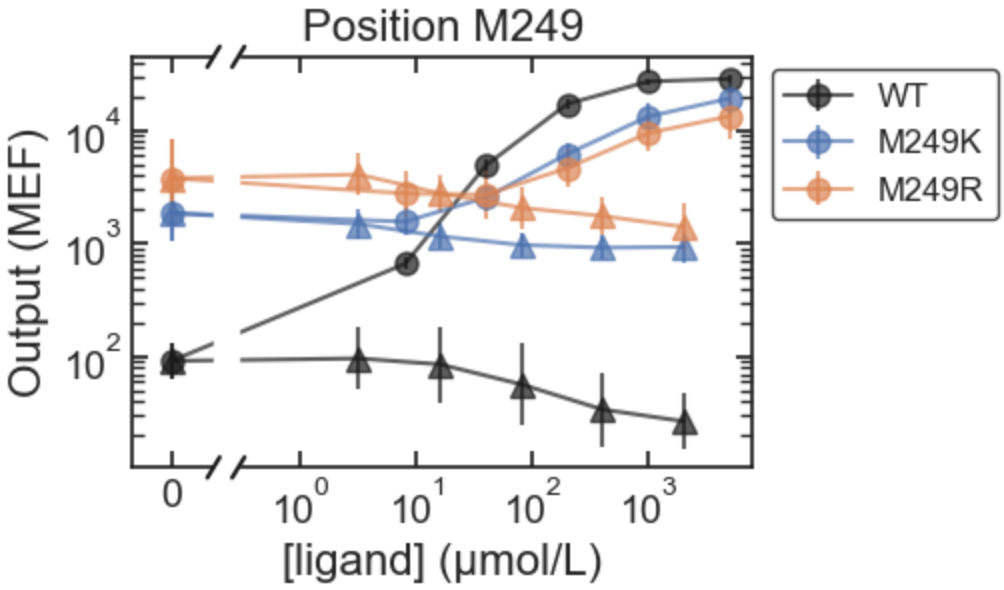

Position M242
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

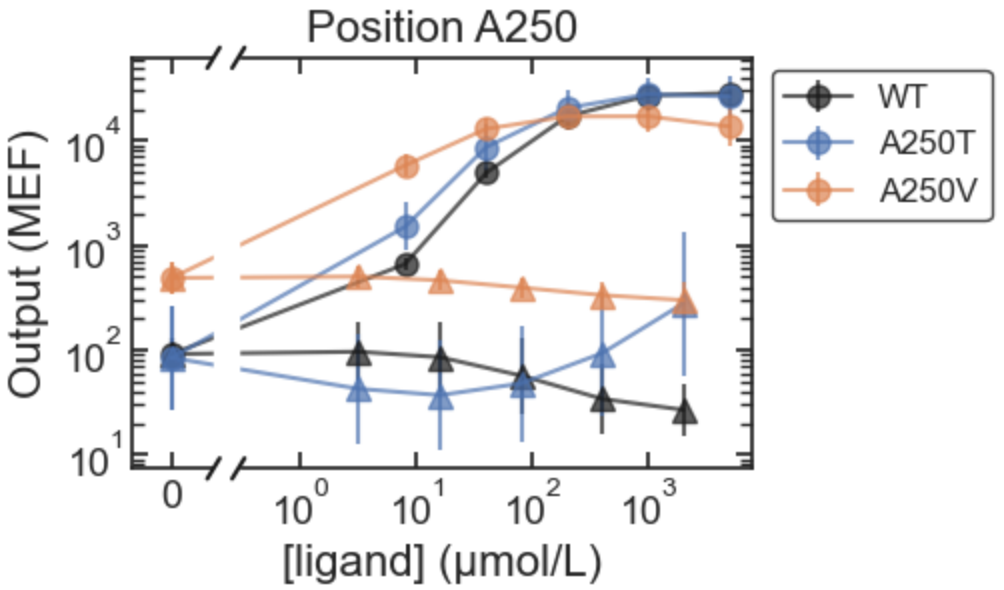

Position L243
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
L243P
L243Q

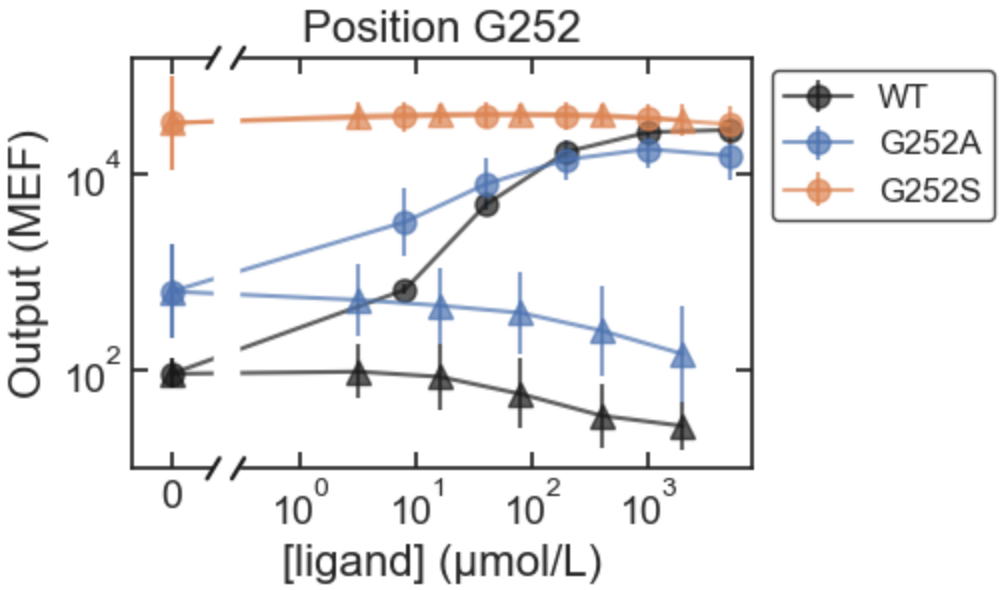

Position N246
Output (MEF)
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

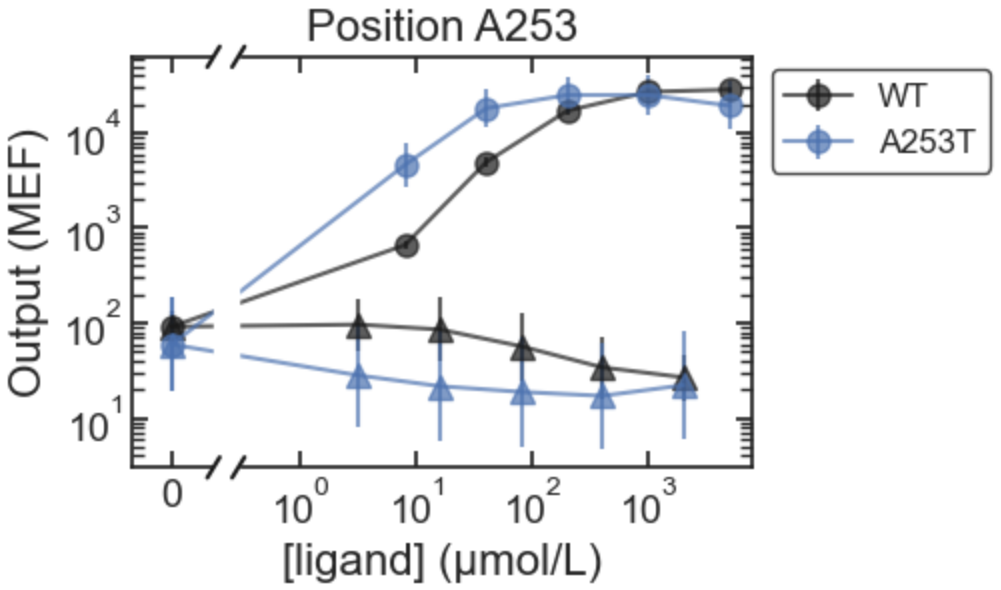

Position D247
Output (MEF)
[ligand] (μmol/L)
WT
D247A
D247E
D247V

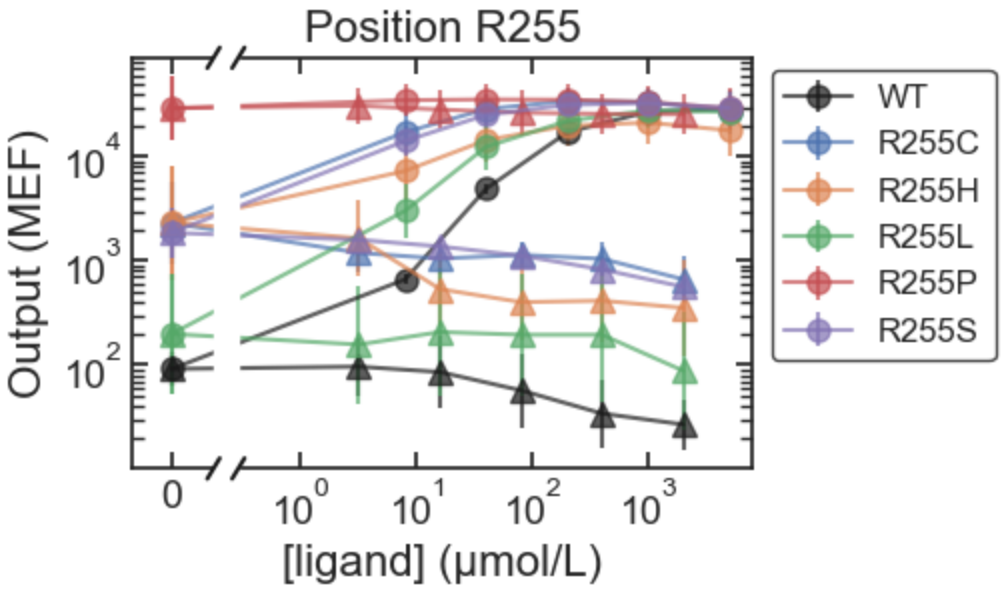

Position Q248
Output (MEF)
[ligand] (μmol/L)
WT
Q248E
Q248L
Q248P

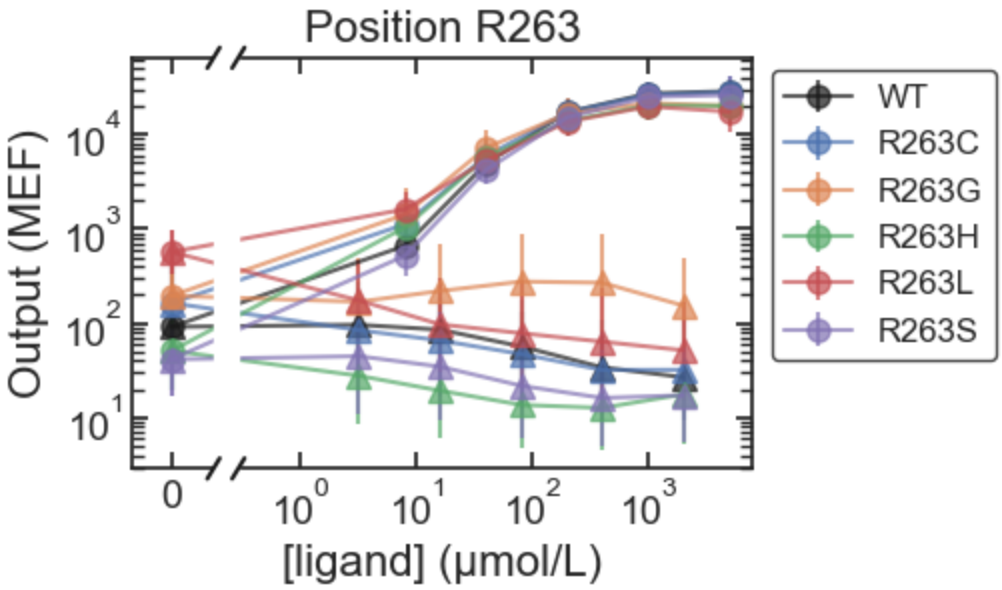

Position M249
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M249K
M249R

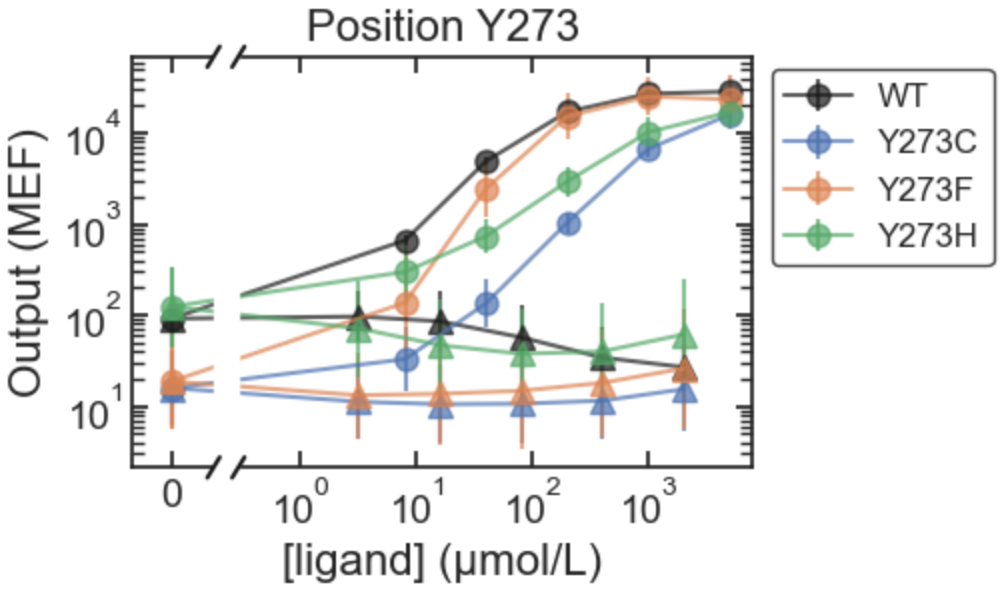

Position A250
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A250T
A250V

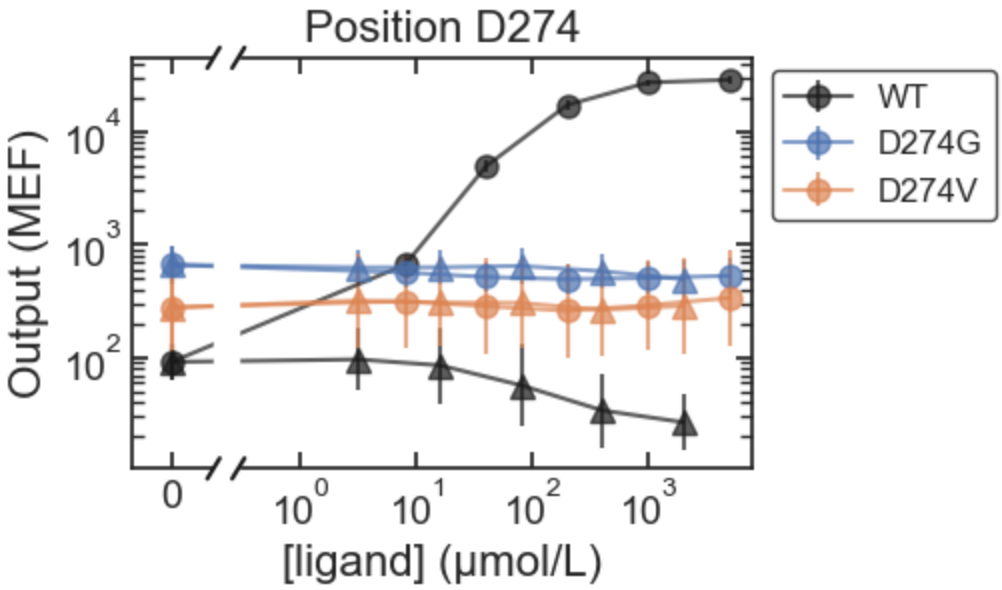

Position G252
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G252A
G252S

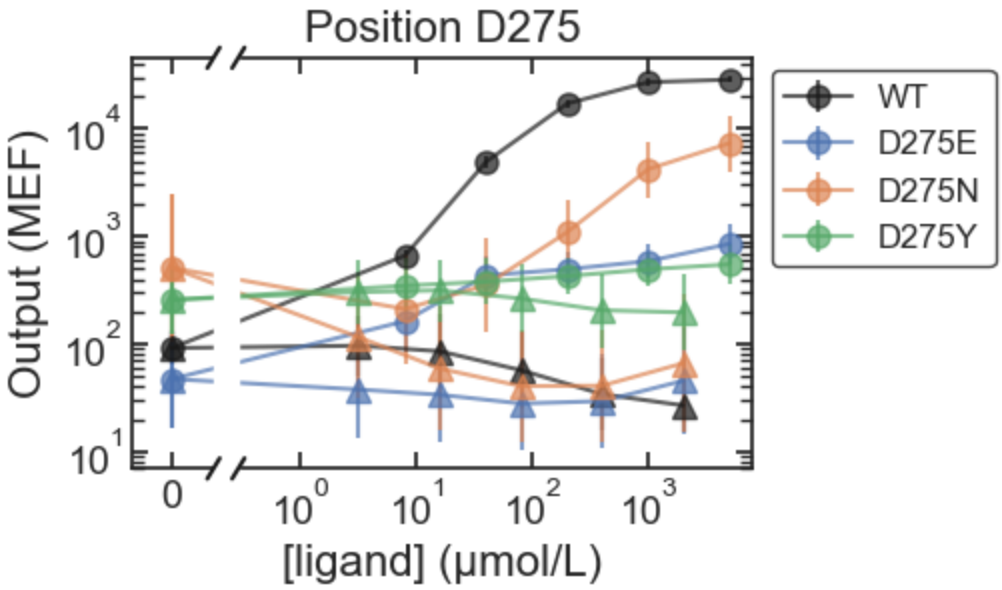

Position A253
Output (MEF)
[ligand] (μmol/L)
WT
A253T

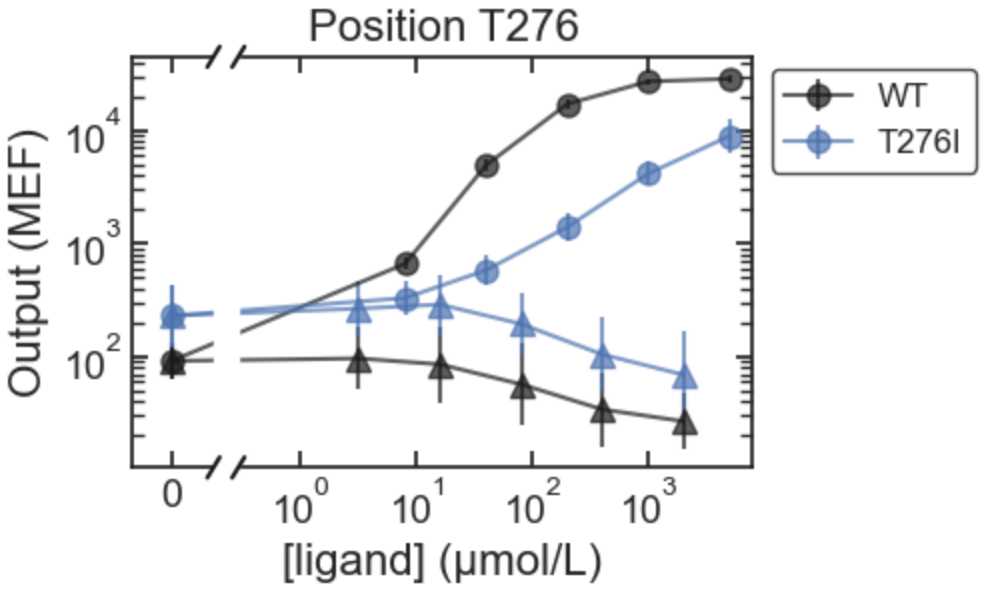

Position R255
Output (MEF)
[ligand] (μmol/L)
WT
R255C
R255H
R255L
R255P
R255S

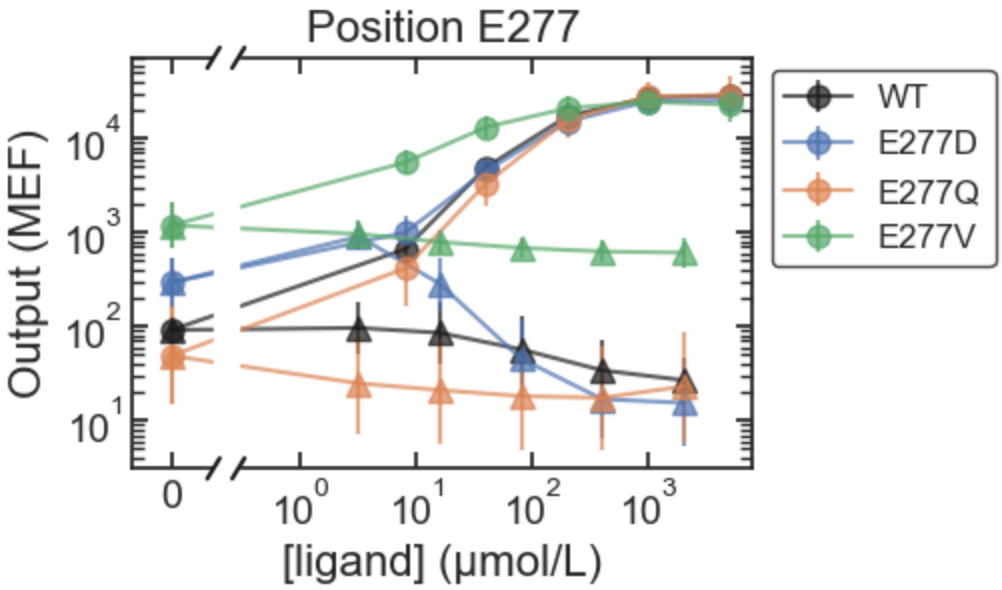

Position R263
Output (MEF)
[ligand] (μmol/L)
WT
R263C
R263G
R263H
R263L
R263S

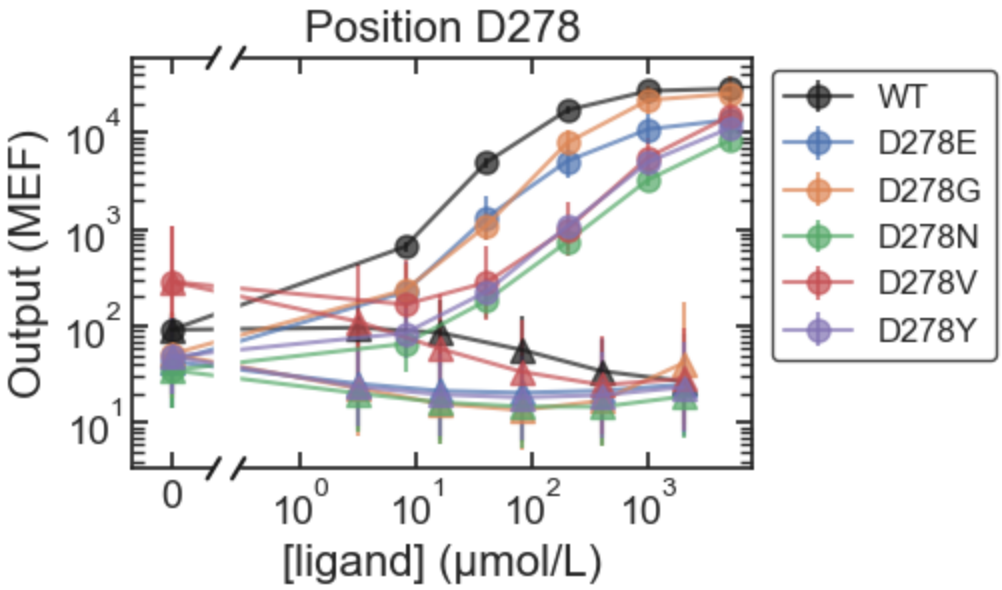

Position Y273
Output (MEF)
[ligand] (μmol/L)
WT
Y273C
Y273F
Y273H

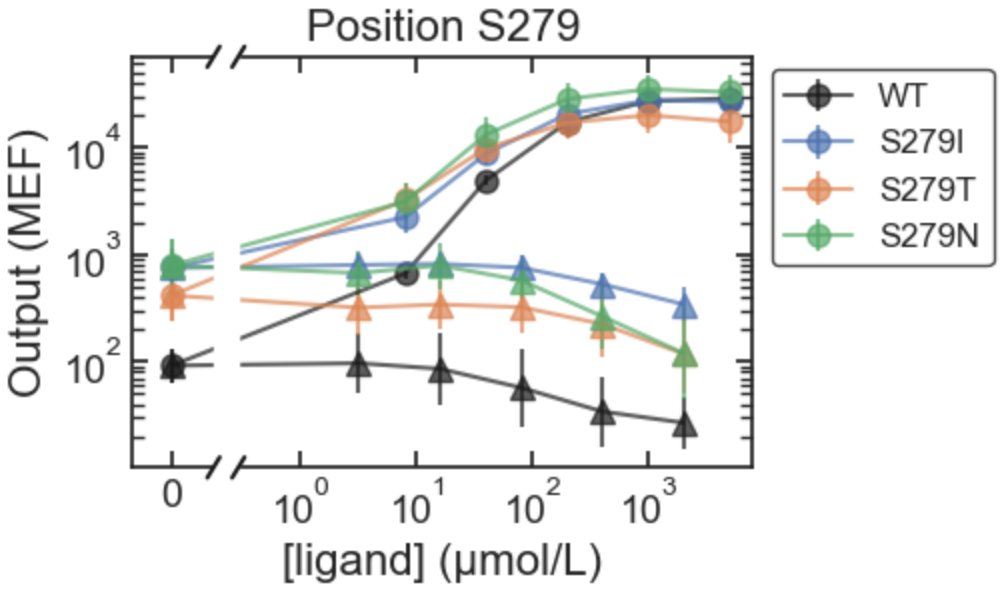

Position D274
Output (MEF)
[ligand] (μmol/L)
WT
D274G
D274V

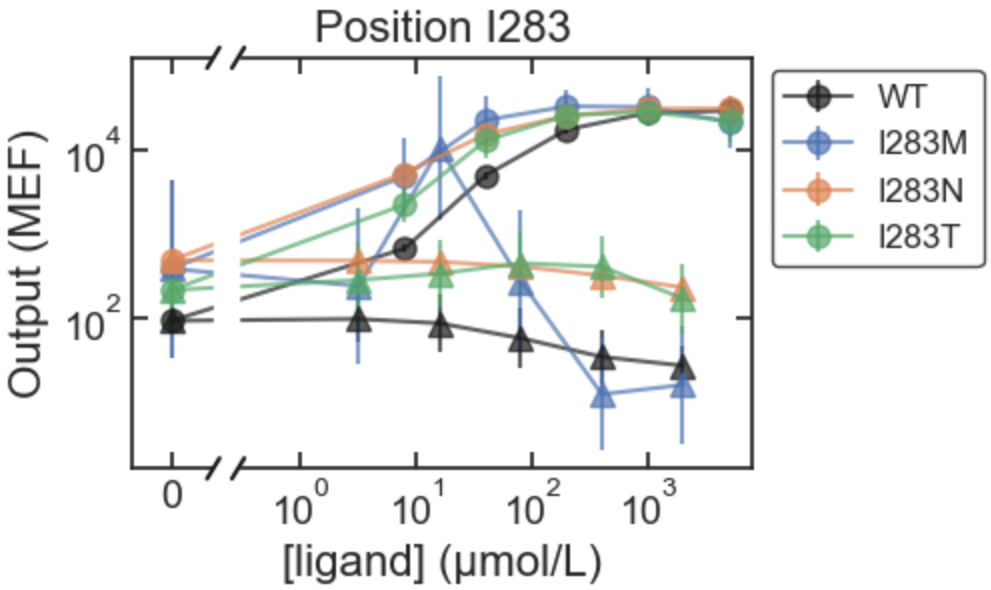

Position D275
Output (MEF)
[ligand] (μmol/L)
WT
D275E
D275N
D275Y

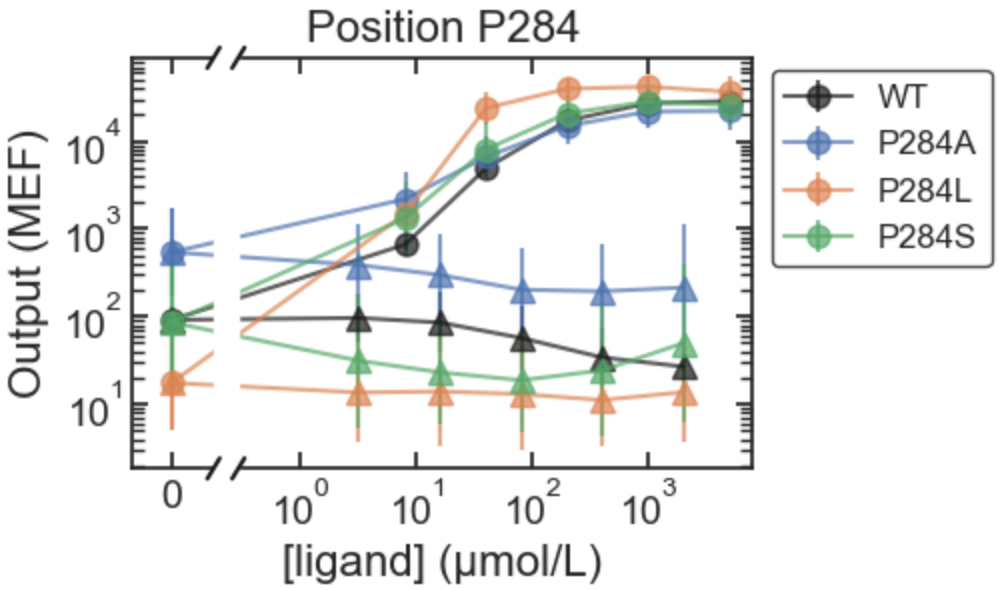

Position T276
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
T276I

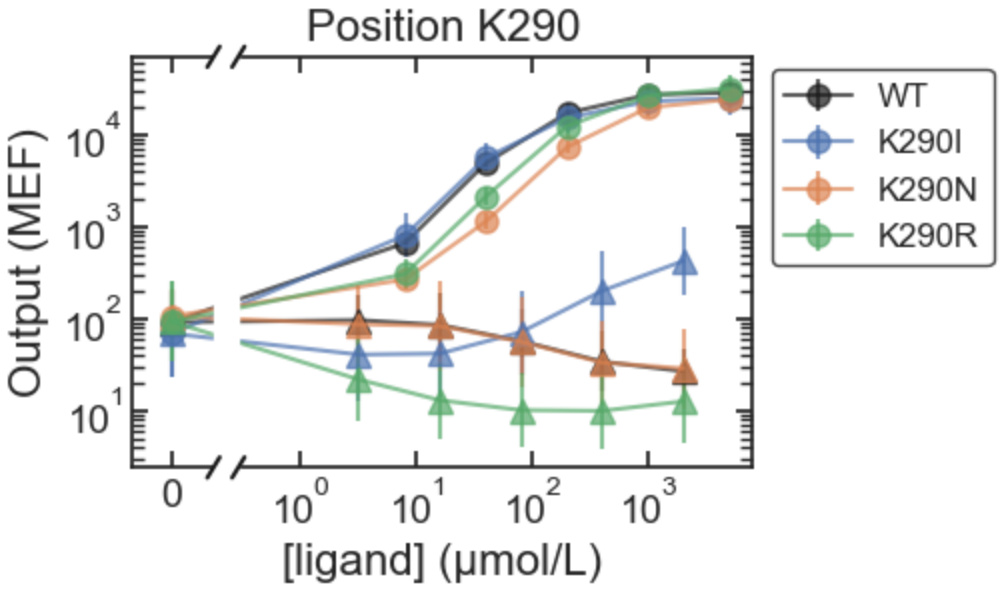

Position E277
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
E277D
E277Q
E277V

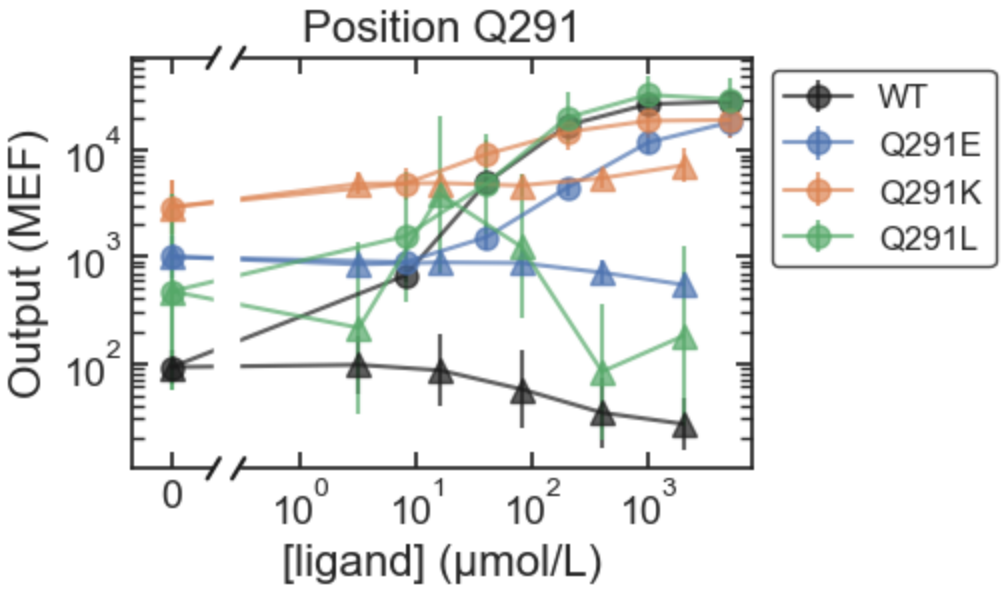

Position D278
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D278E
D278G
D278N
D278V
D278Y

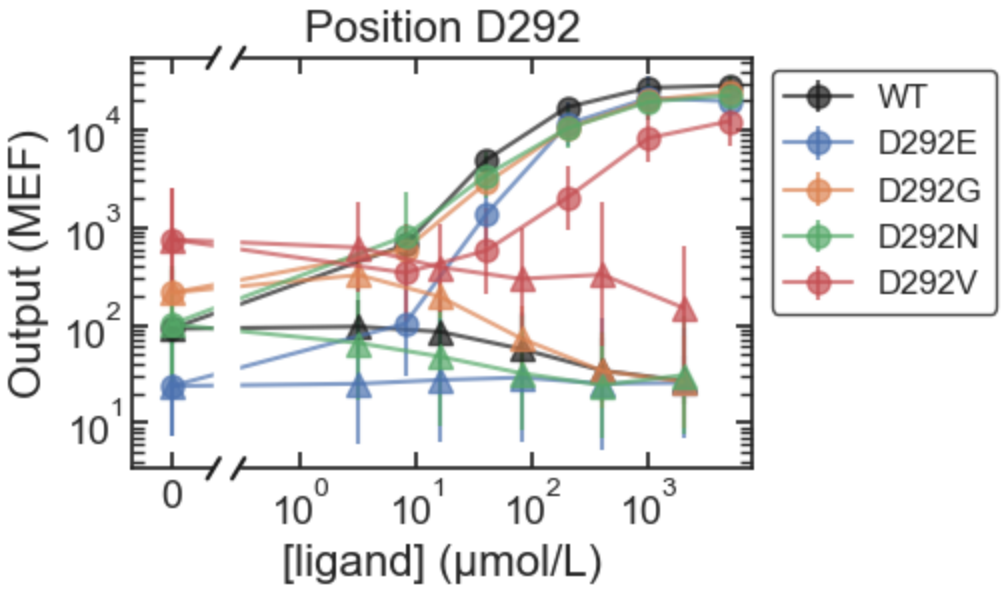

Position S279
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S279I
S279T
S279N

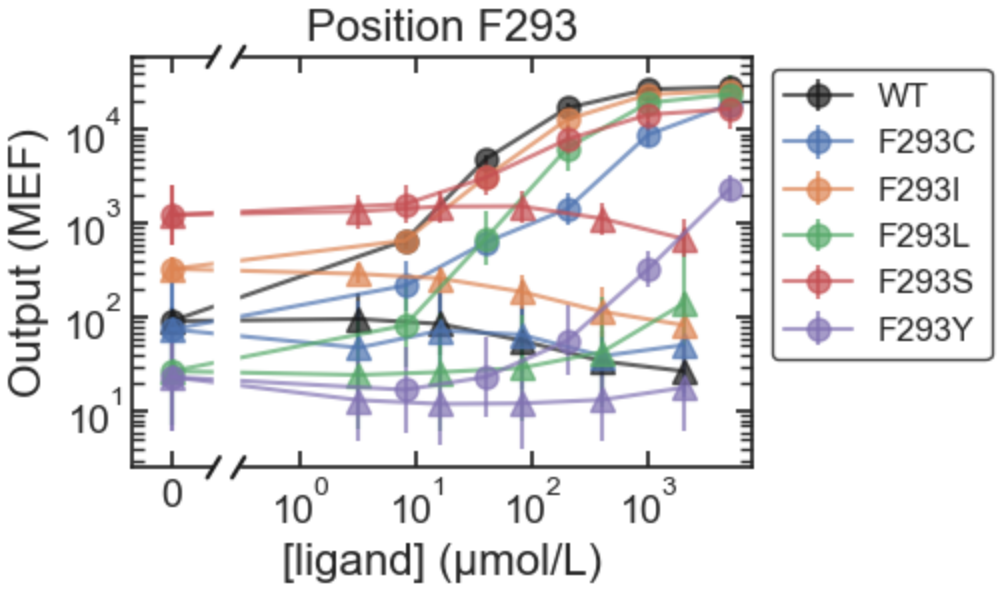

Position I283
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I283M
I283N
I283T

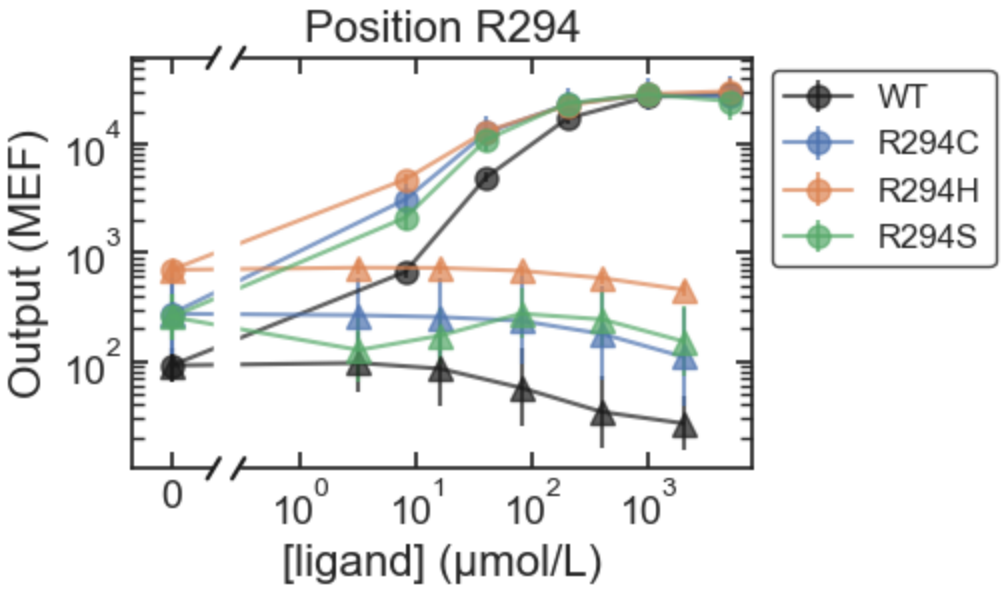

Position P284
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
P284A
P284L
P284S

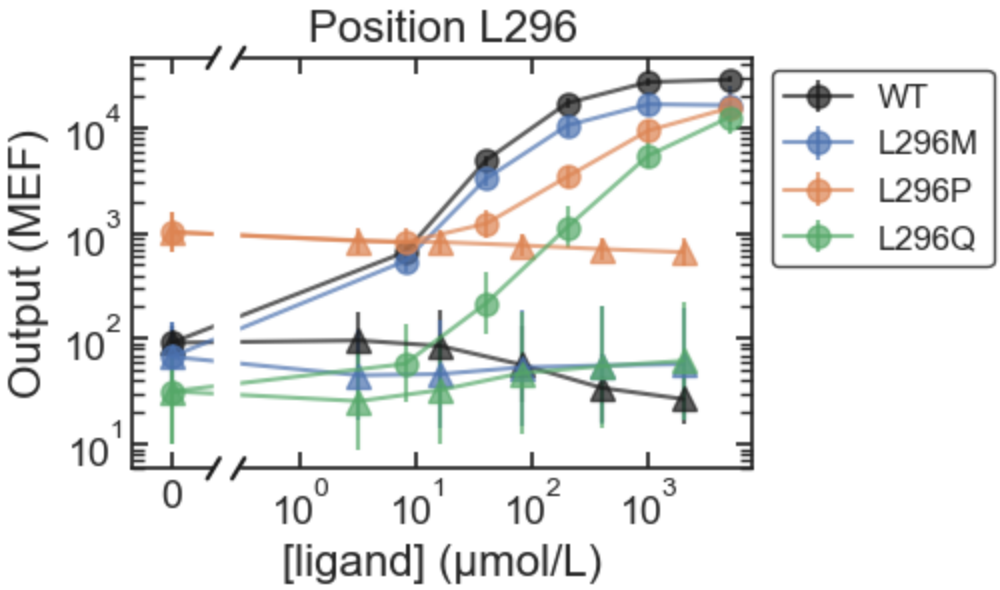

Position K290
Output (MEF)
[ligand] (μmol/L)
WT
K290I
K290N
K290R

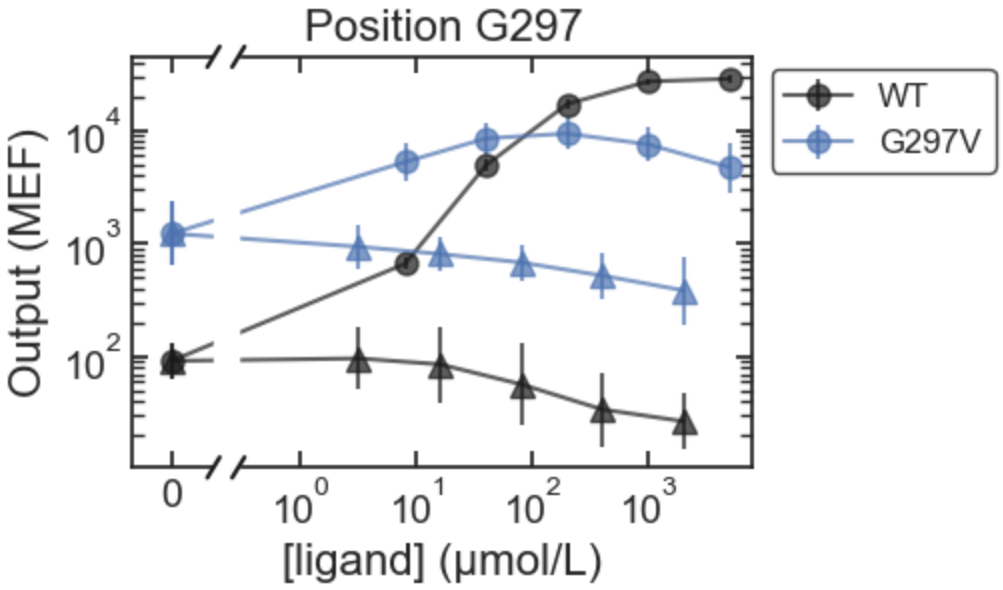

Position Q291
Output (MEF)
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L

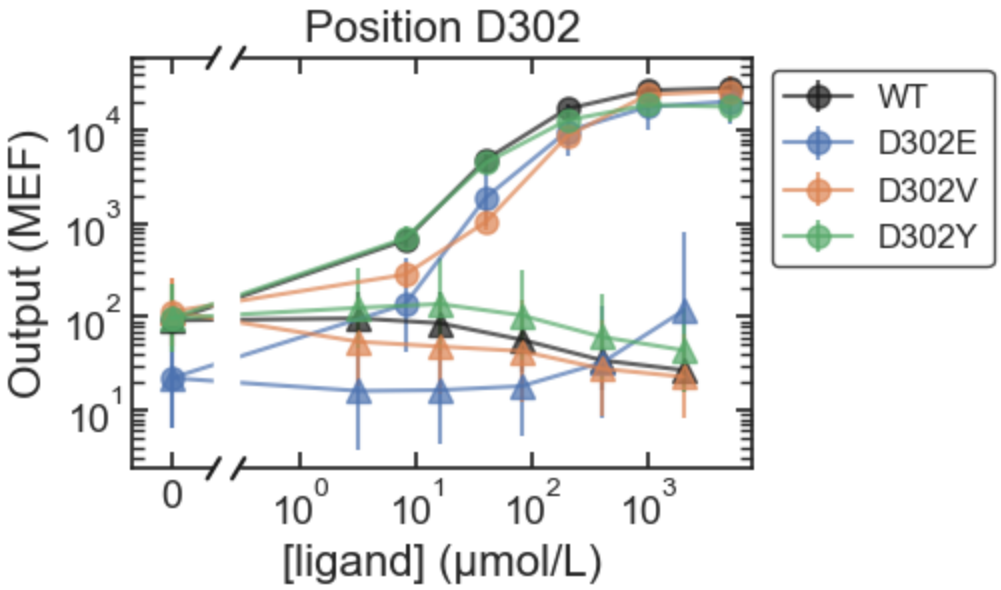

Position D292
Output (MEF)
[ligand] (μmol/L)
WT
D292E
D292G
D292N
D292V

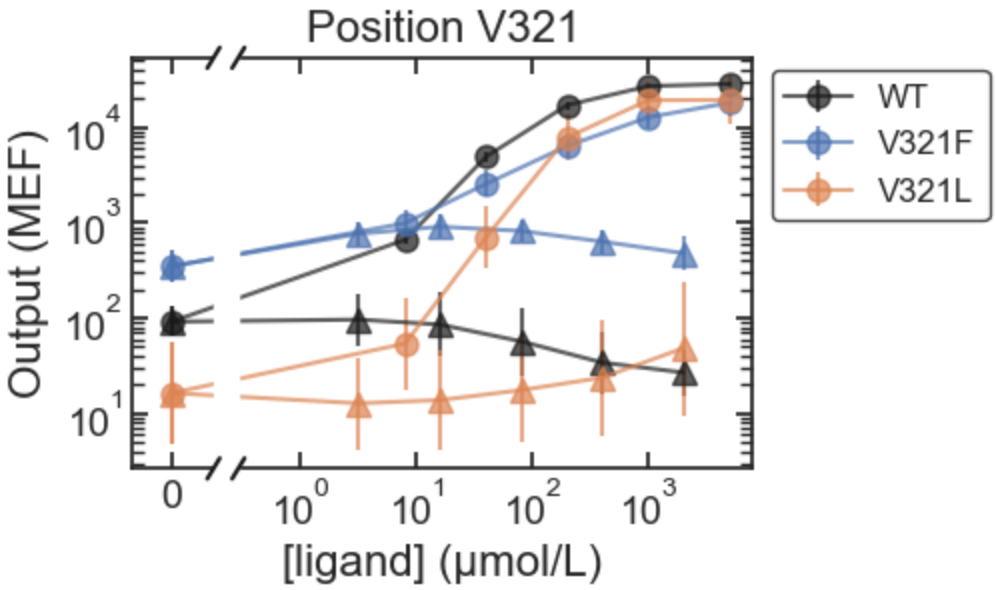

Position F293
Output (MEF)
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

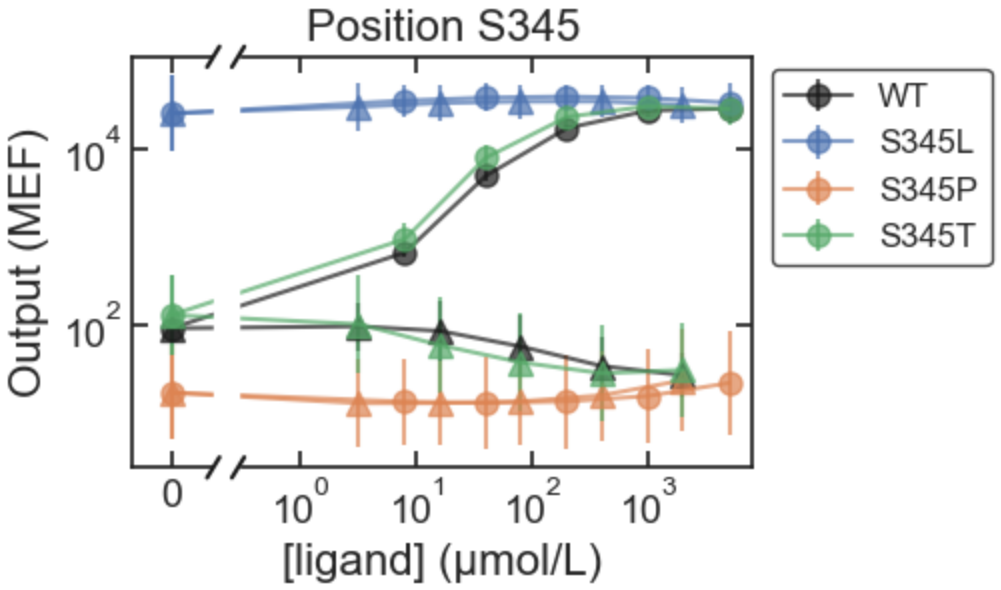

Position R294
Output (MEF)
[ligand] (μmol/L)
WT
R294C
R294H
R294S

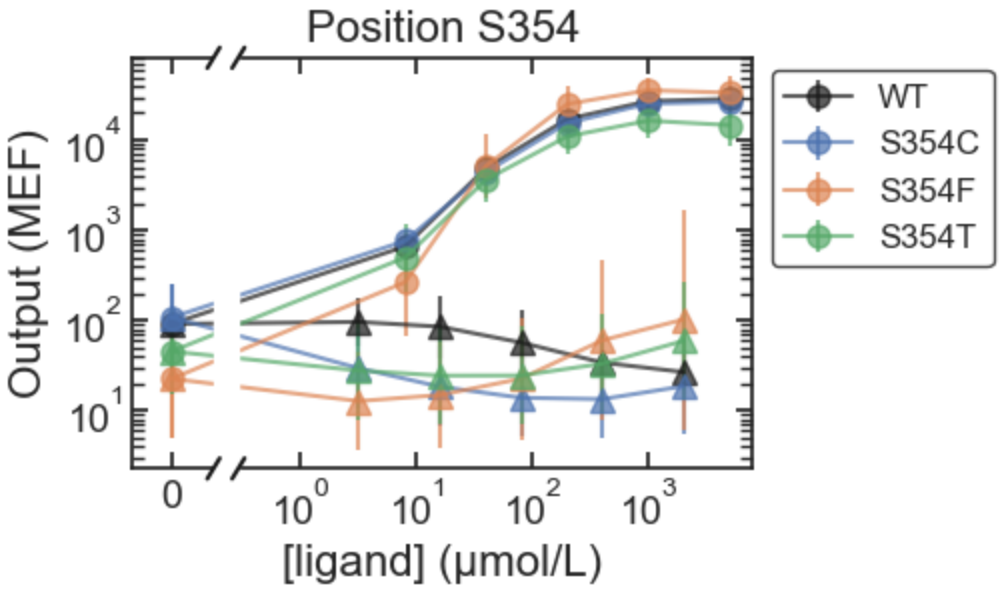

Position L296
Output (MEF)
[ligand] (μmol/L)
WT
L296M
L296P
L296Q

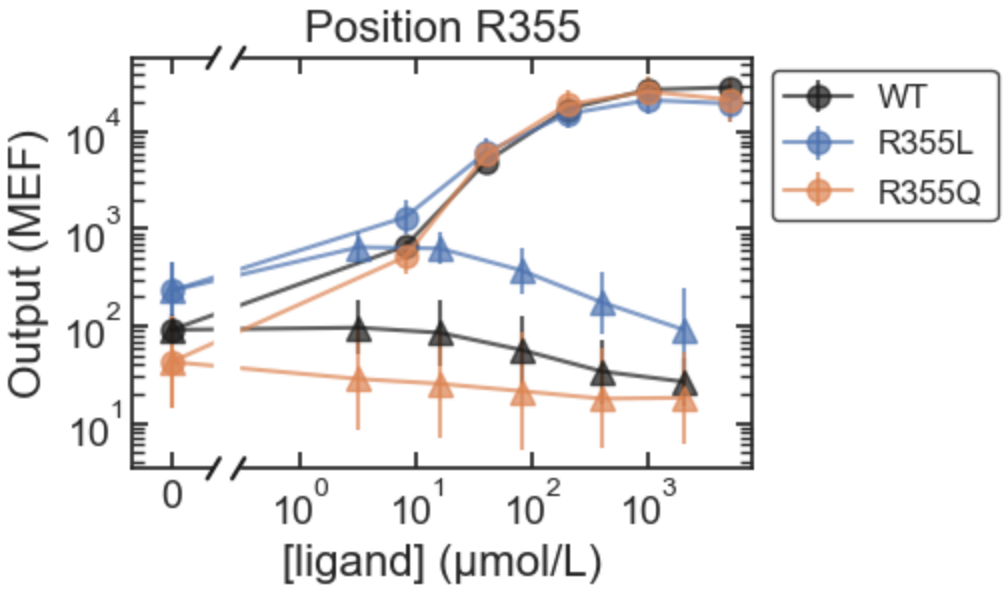

Position G297
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G297V

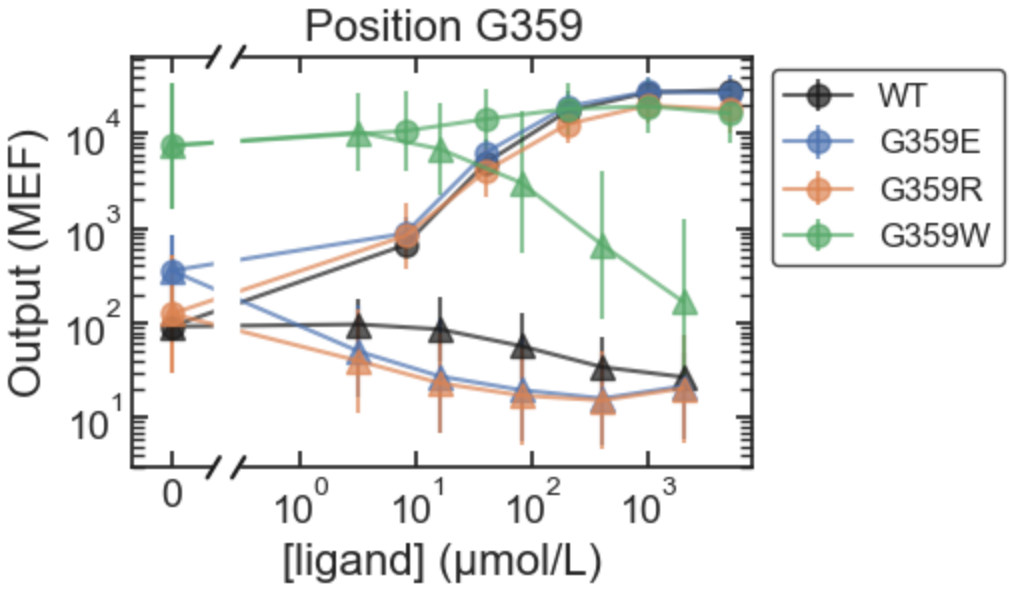

Position D302
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D302E
D302V
D302Y


Position V321
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V321F
V321L

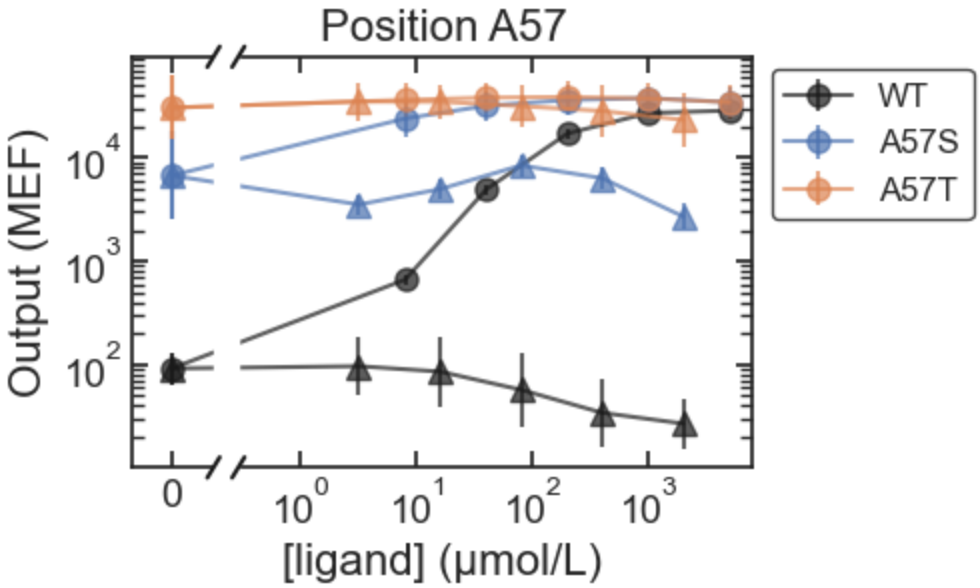

Position S345
Output (MEF)
[ligand] (μmol/L)
WT
S345L
S345P
S345T

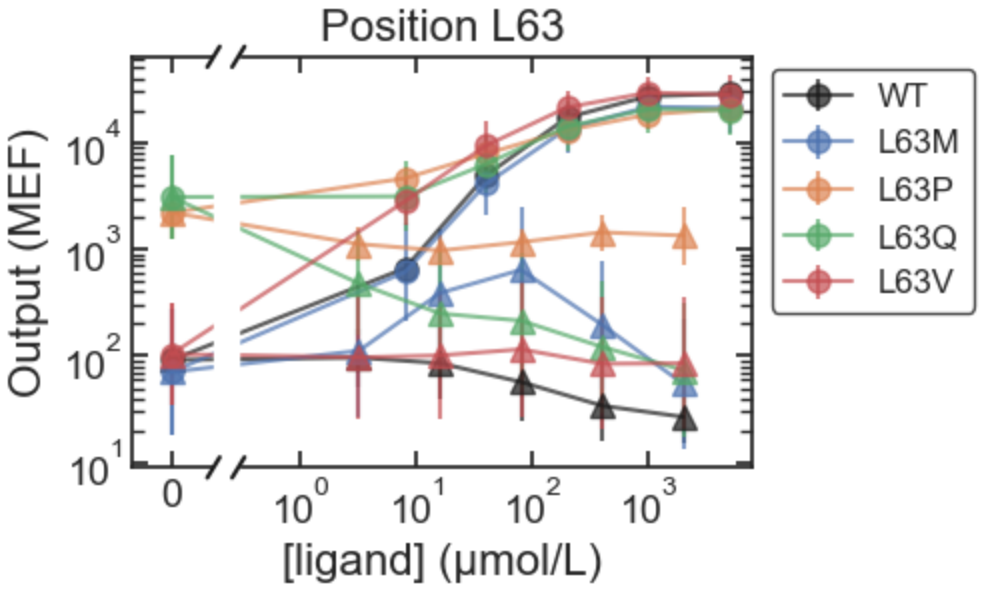

Position S354
Output (MEF)
[ligand] (μmol/L)
WT
S354C
S354F
S354T

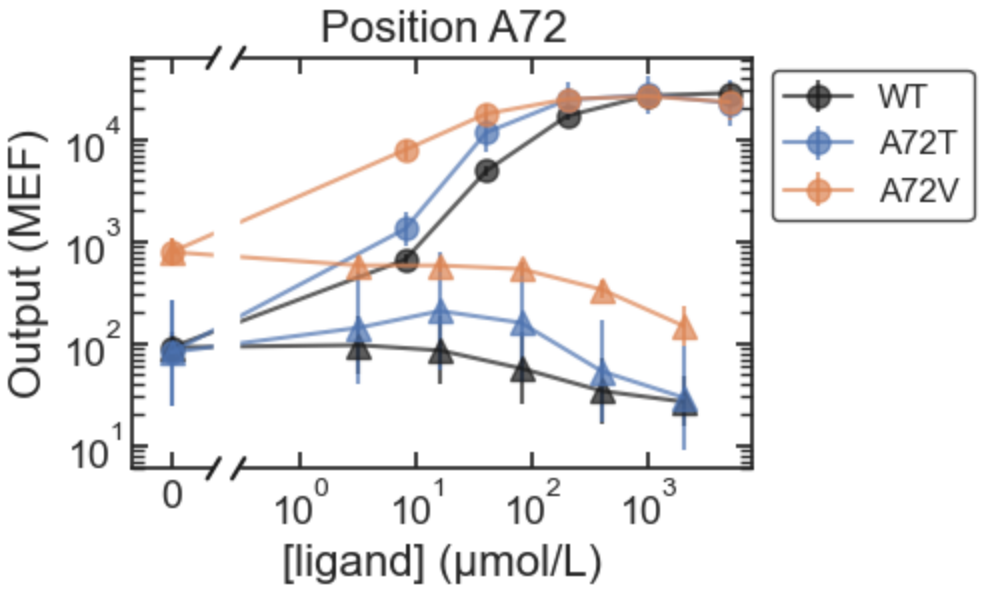

Position R355
Output (MEF)
[ligand] (μmol/L)
WT
R355L
R355Q

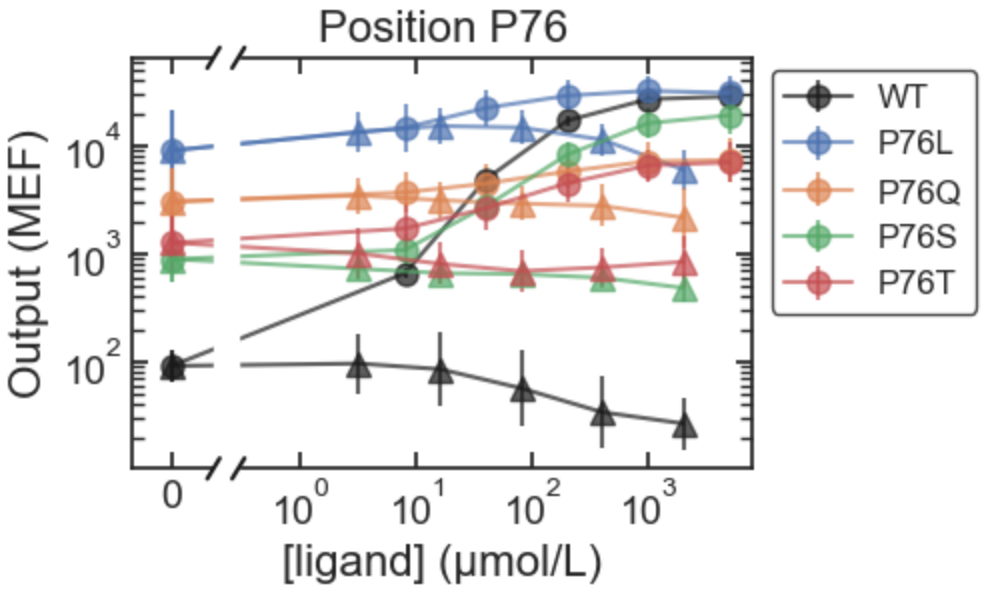

Position G359
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G359E
G359R
G359W

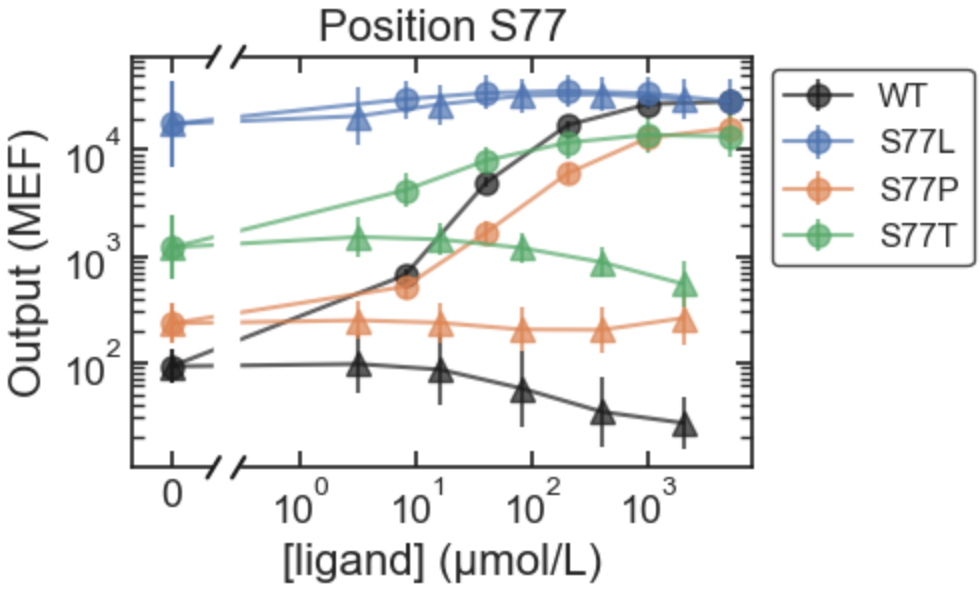
ONPF

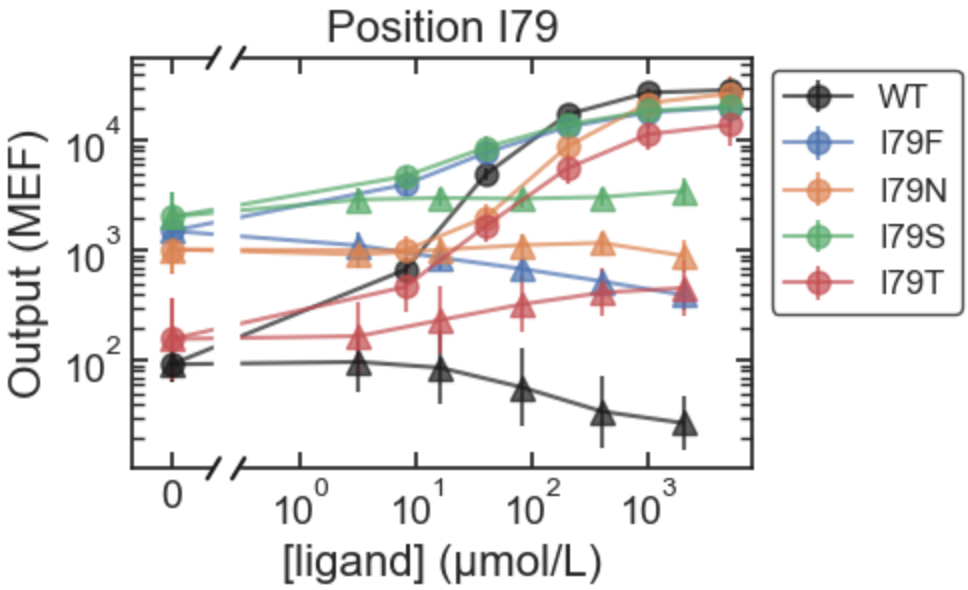

Position A57
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A57S
A57T

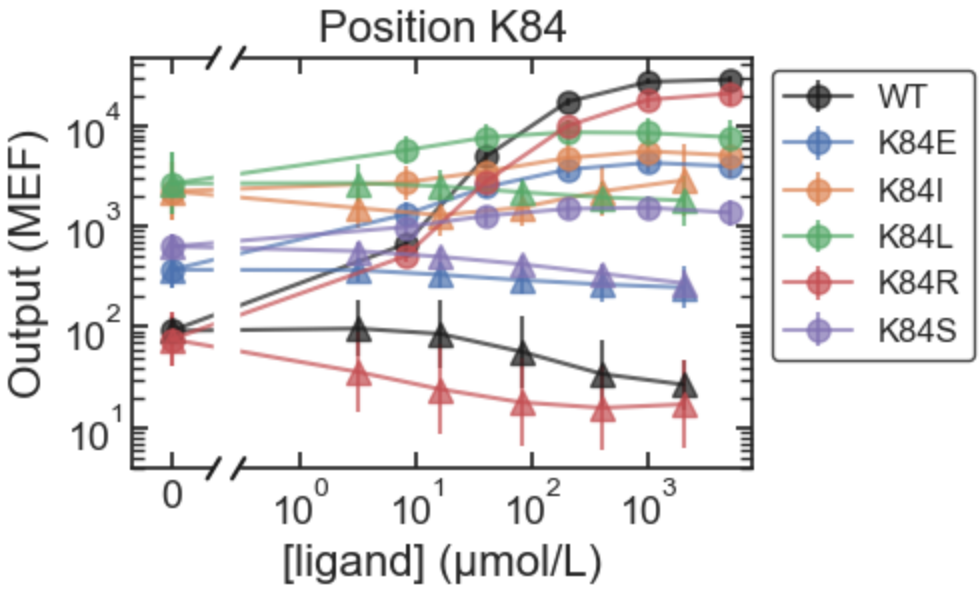

Position L63
Output (MEF)
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

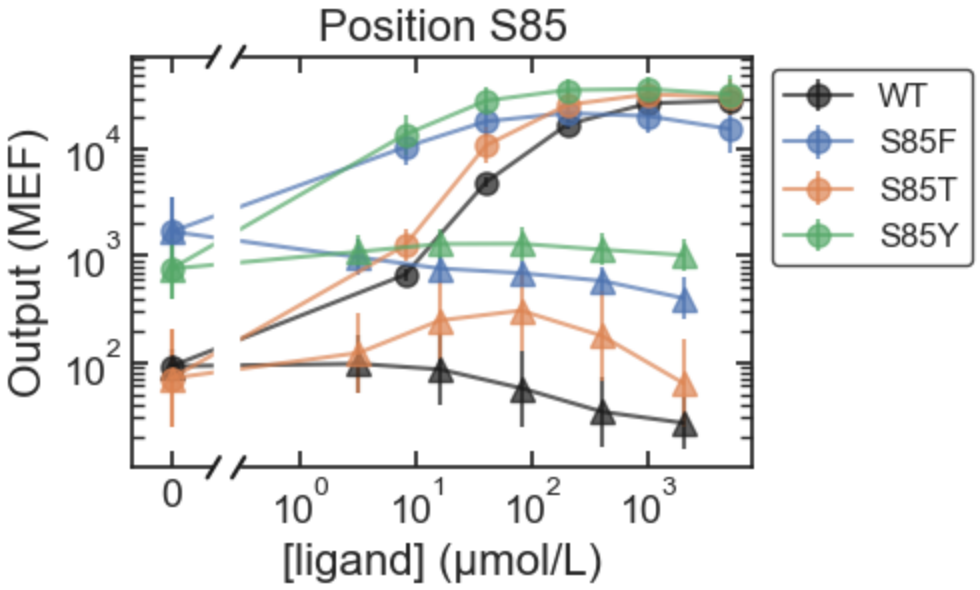

Position A72
Output (MEF)
[ligand] (μmol/L)
WT
A72T
A72V

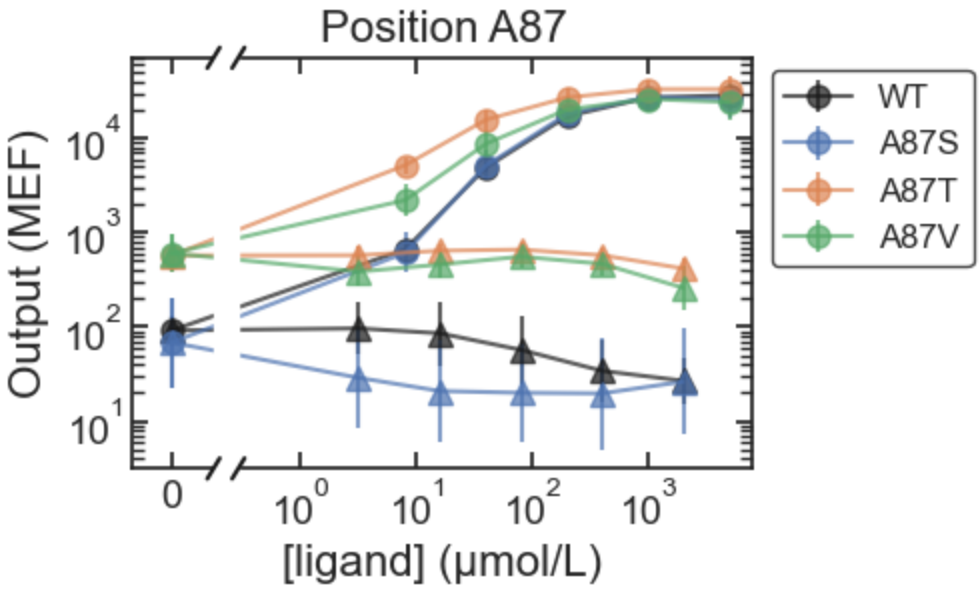

Position P76
Output (MEF)
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

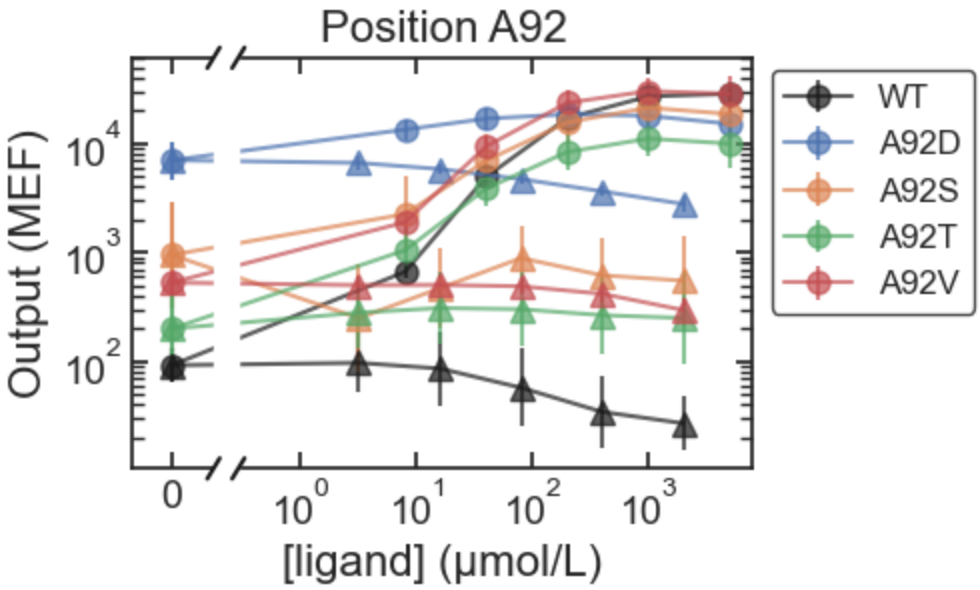

Position S77
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S77L
S77P
S77T

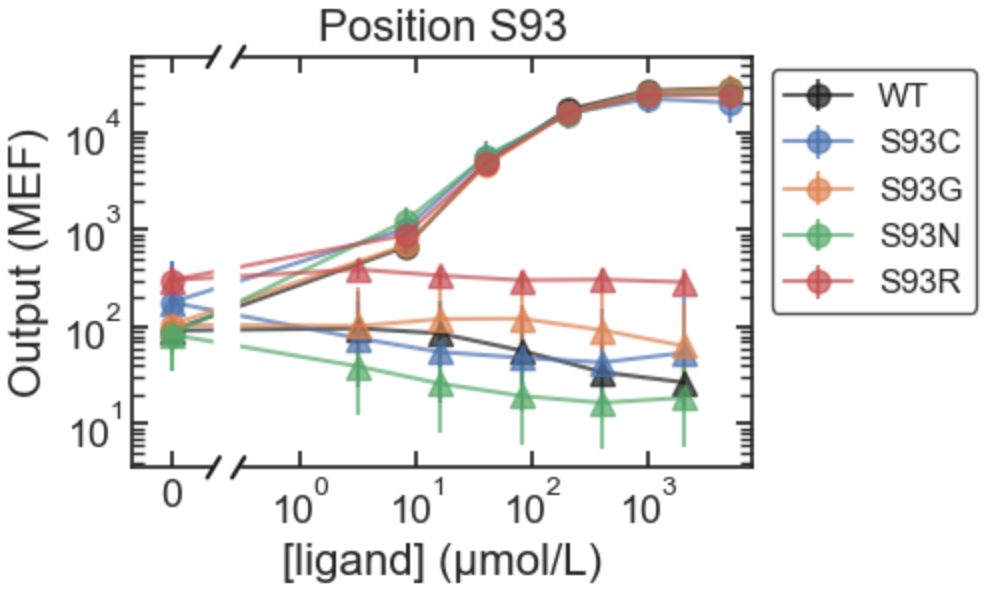

Position I79
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

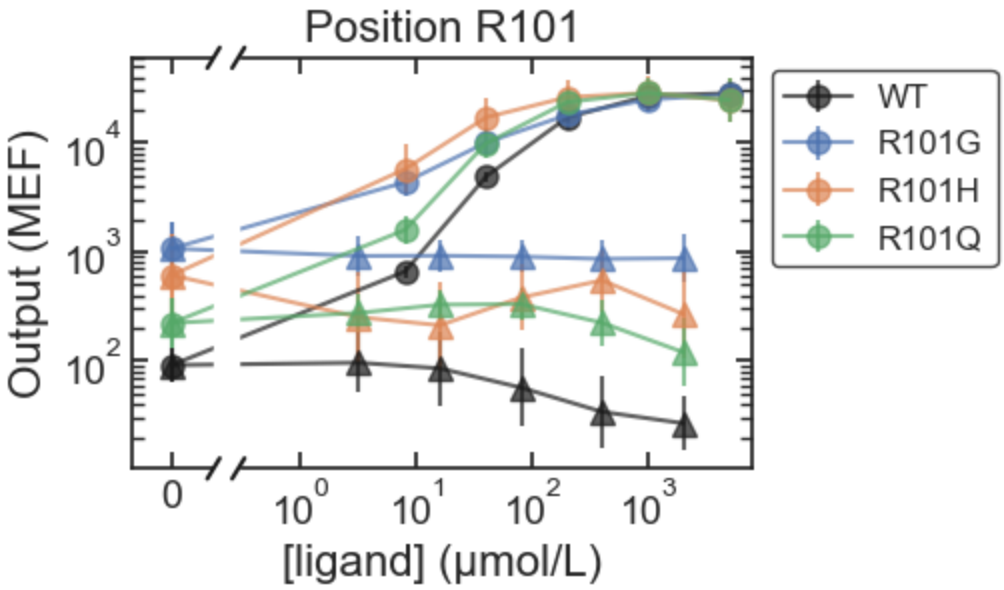

Position K84
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

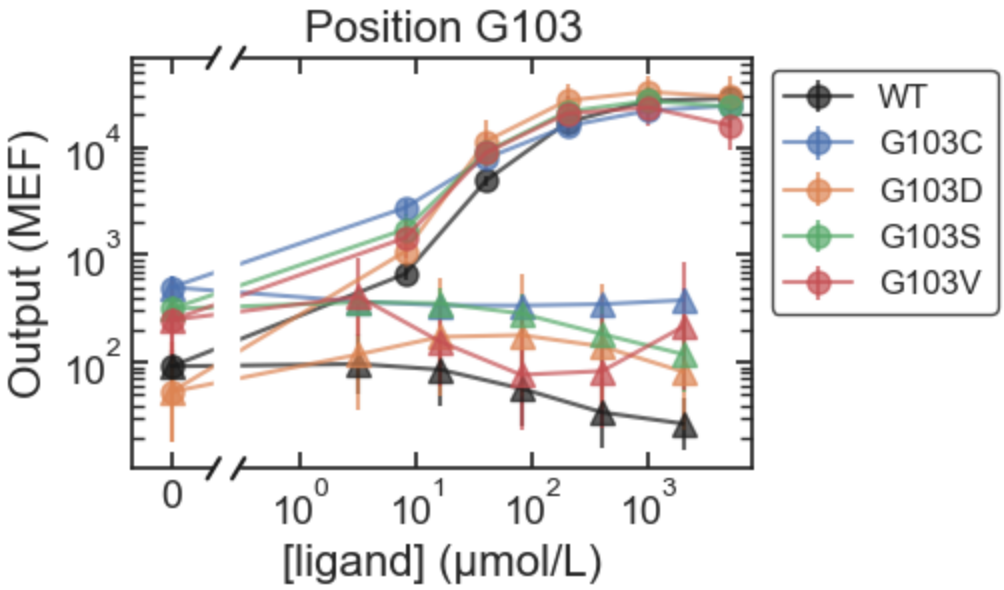

Position S85
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S85F
S85T
S85Y

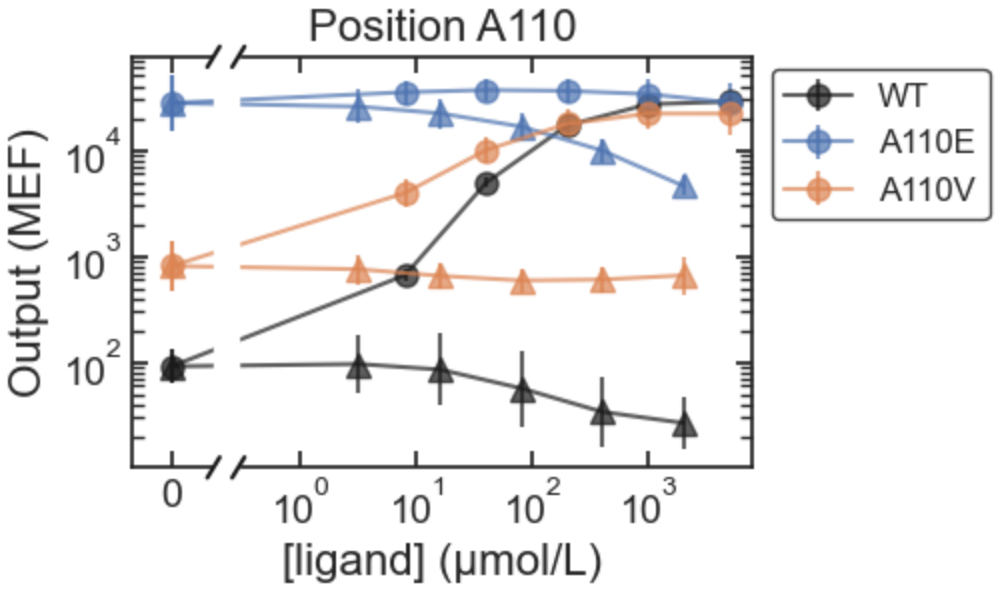

Position A87
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A87S
A87T
A87V

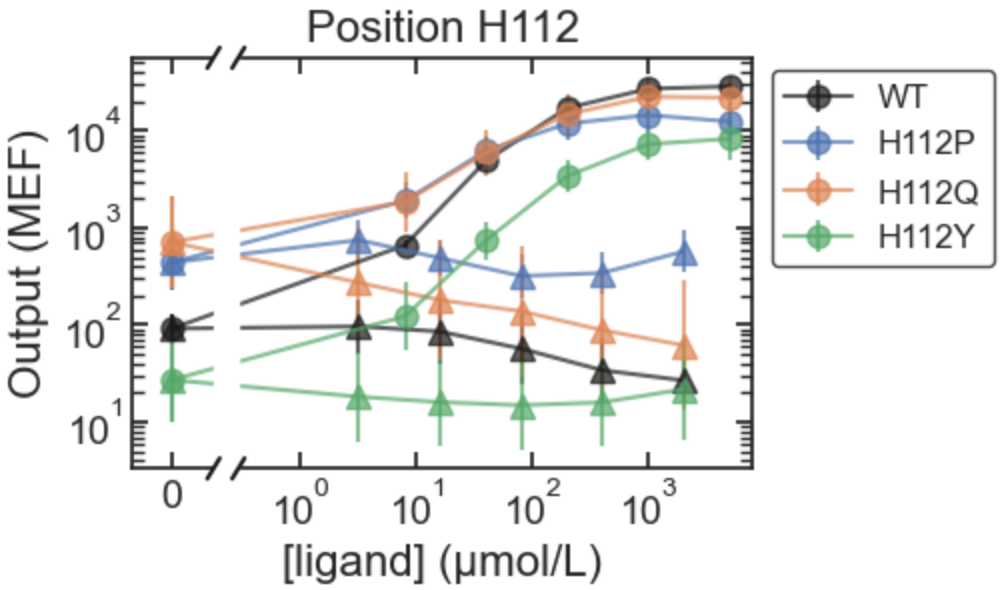

Position A92
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V

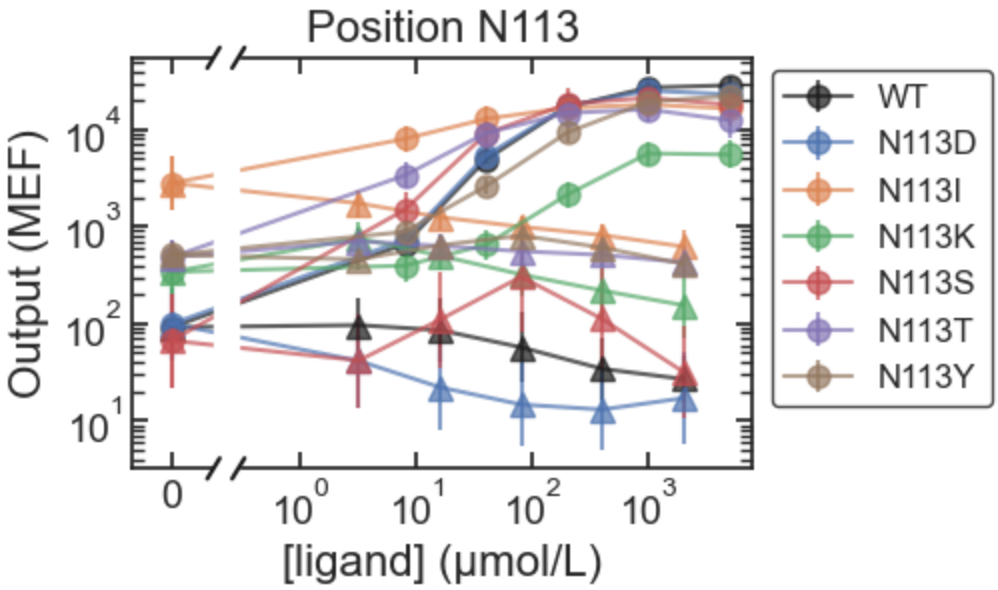

Position S93
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S93C
S93G
S93N
S93R

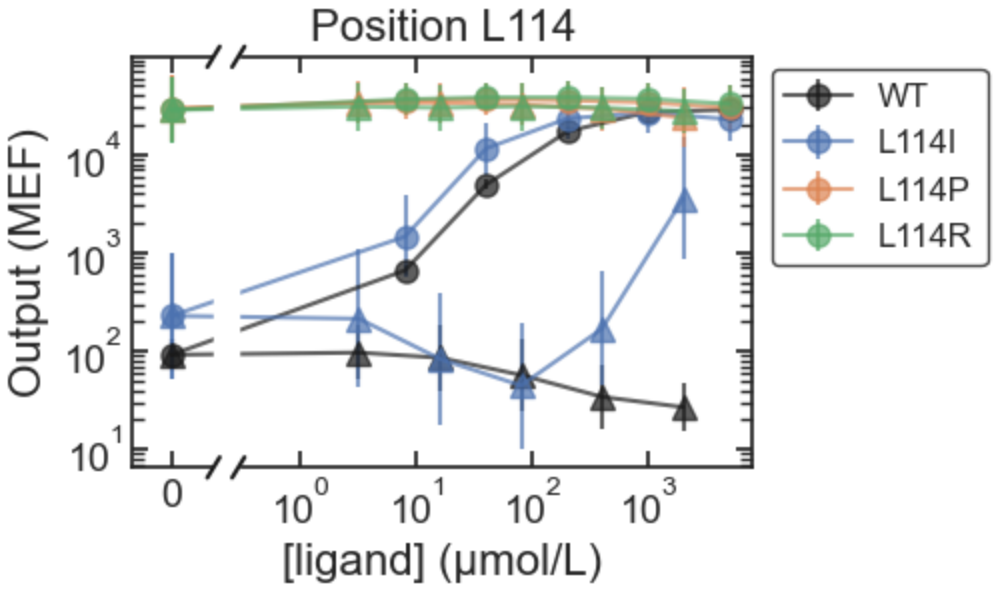

Position R101
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

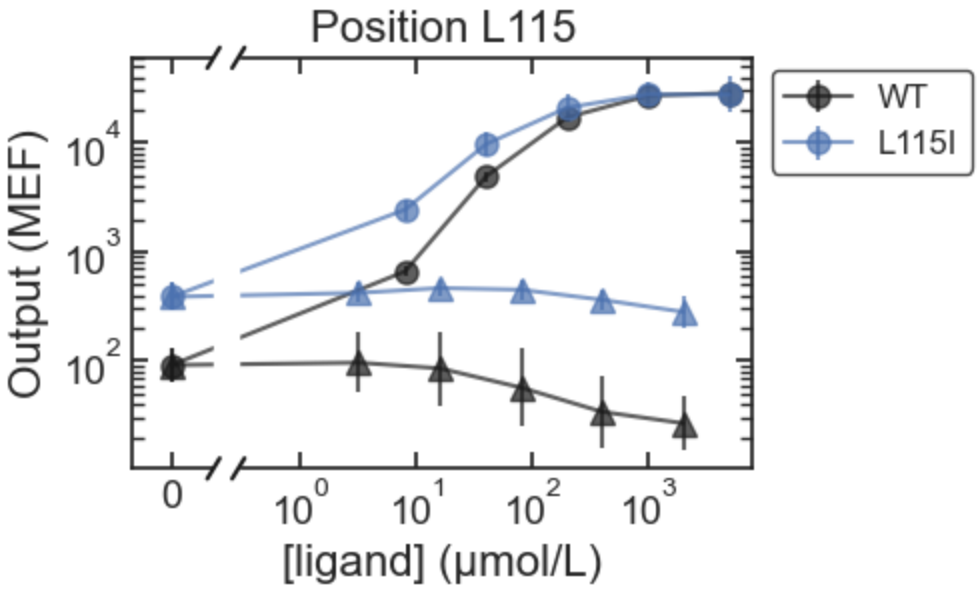

Position G103
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

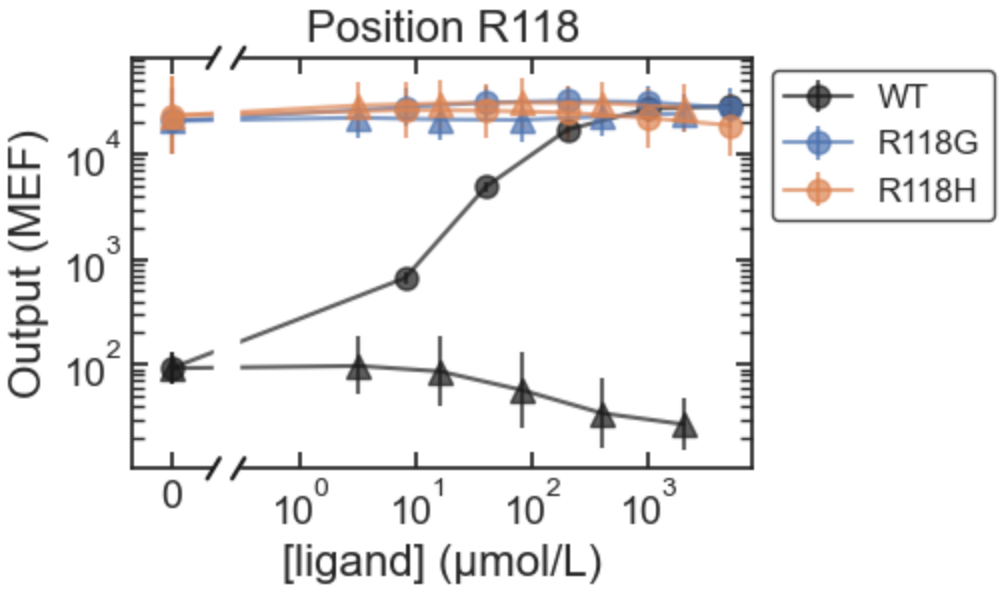

Position A110
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A110E
A110V

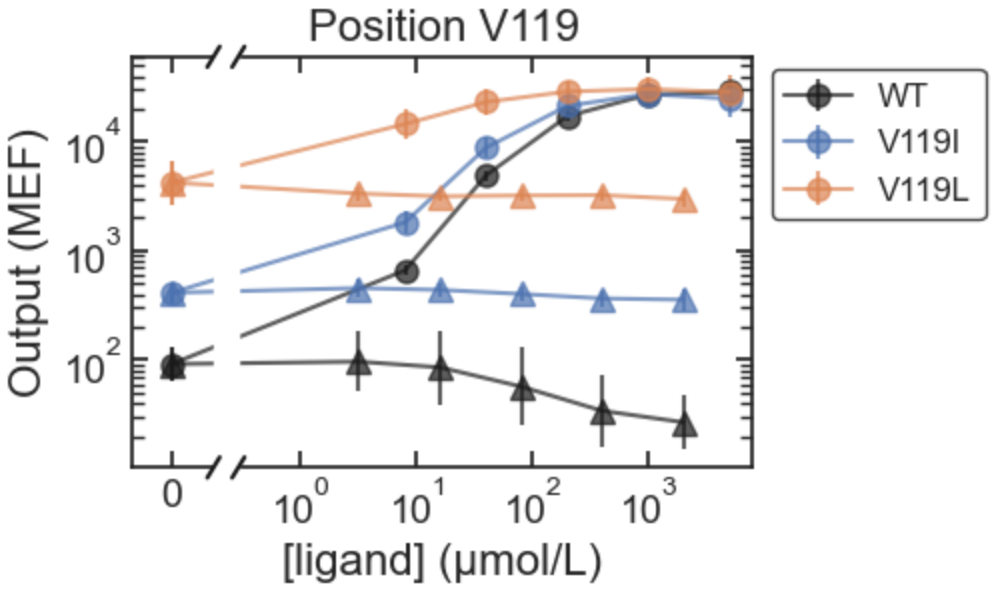

Position H112
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

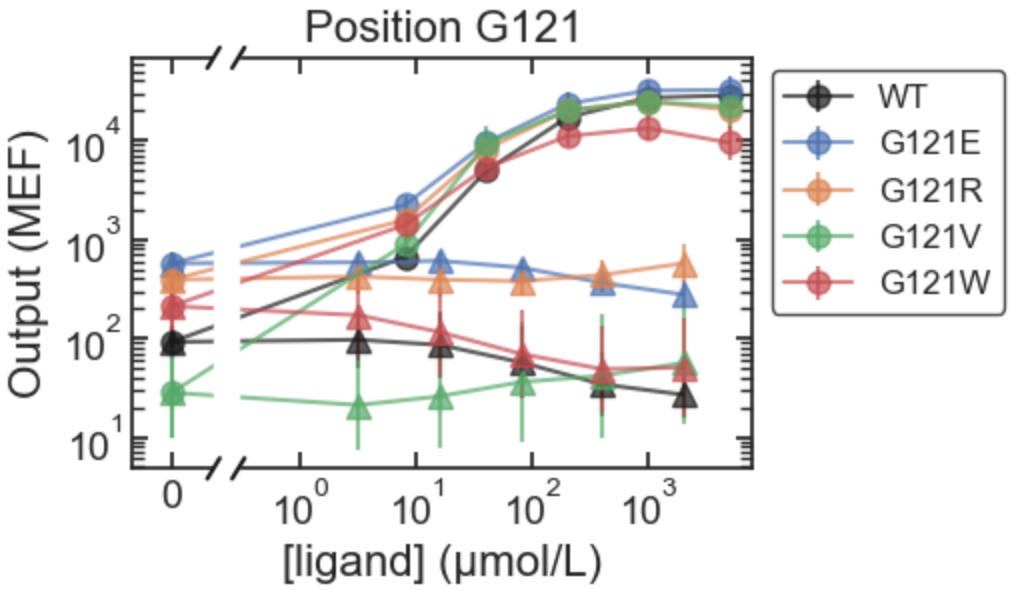

Position N113
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

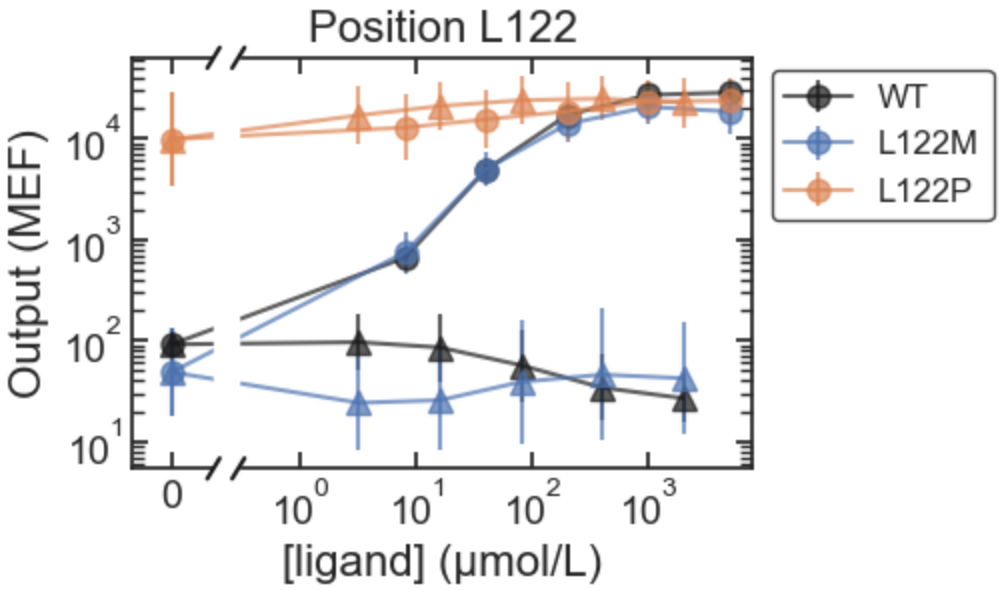

Position L114
Output (MEF)
[ligand] (µmol/L)
WT
L114I
L114P
L114R

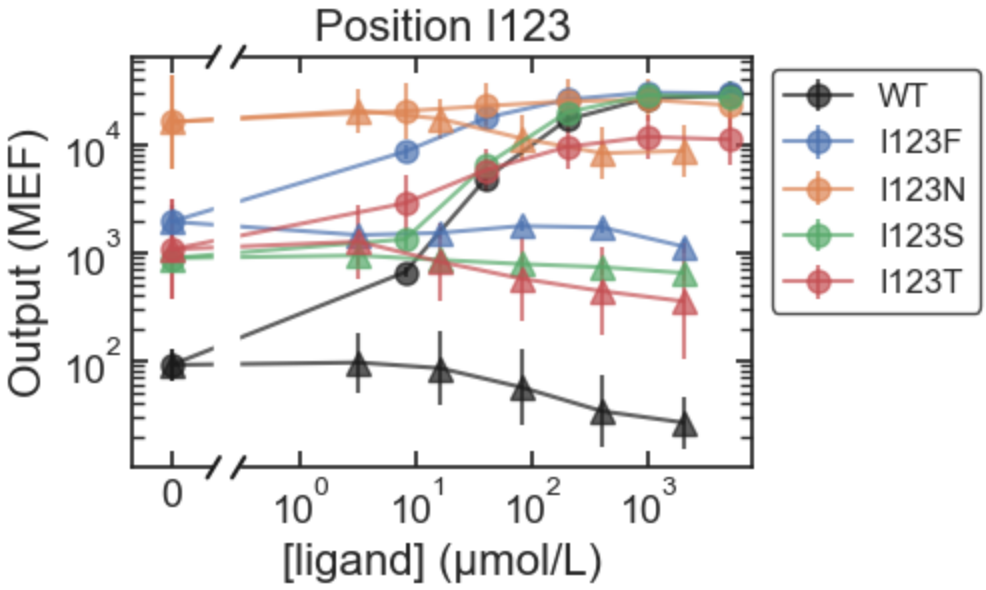

Position L115
Output (MEF)
[ligand] (µmol/L)
WT
L115I

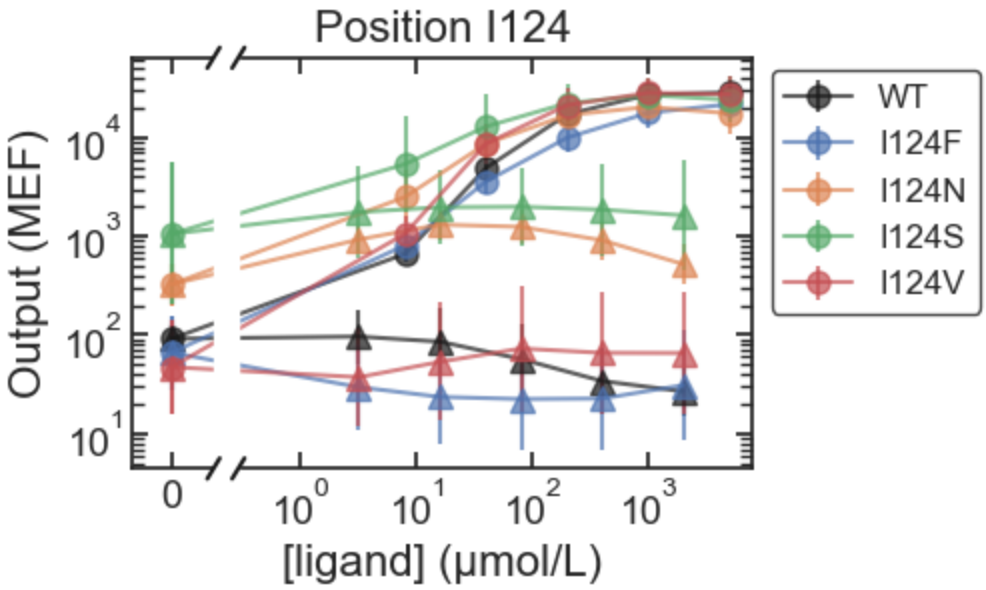

Position R118
Output (MEF)
[ligand] (µmol/L)
WT
R118G
R118H

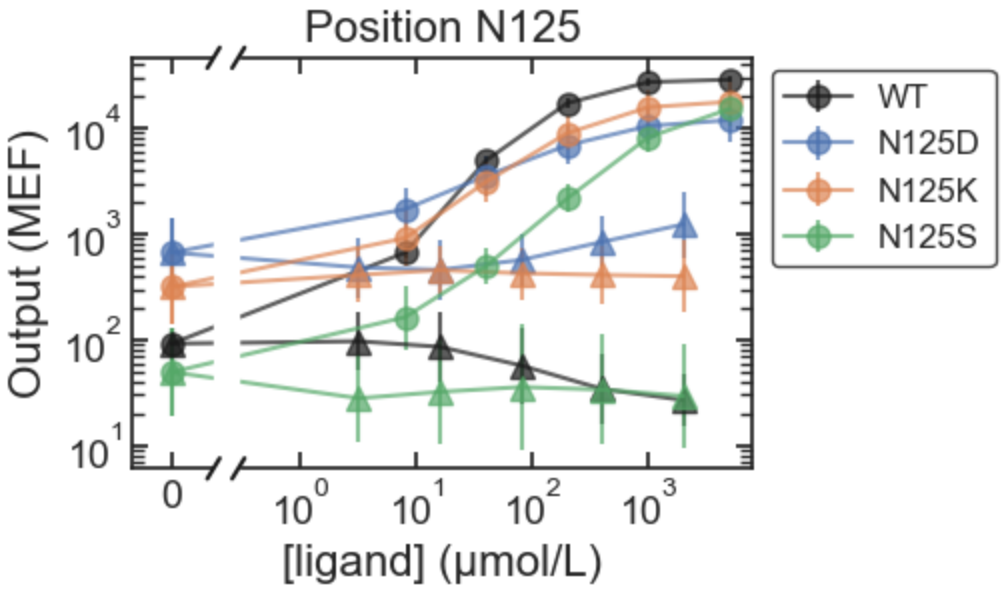

Position V119
Output (MEF)
[ligand] (μmol/L)
WT
V119I
V119L

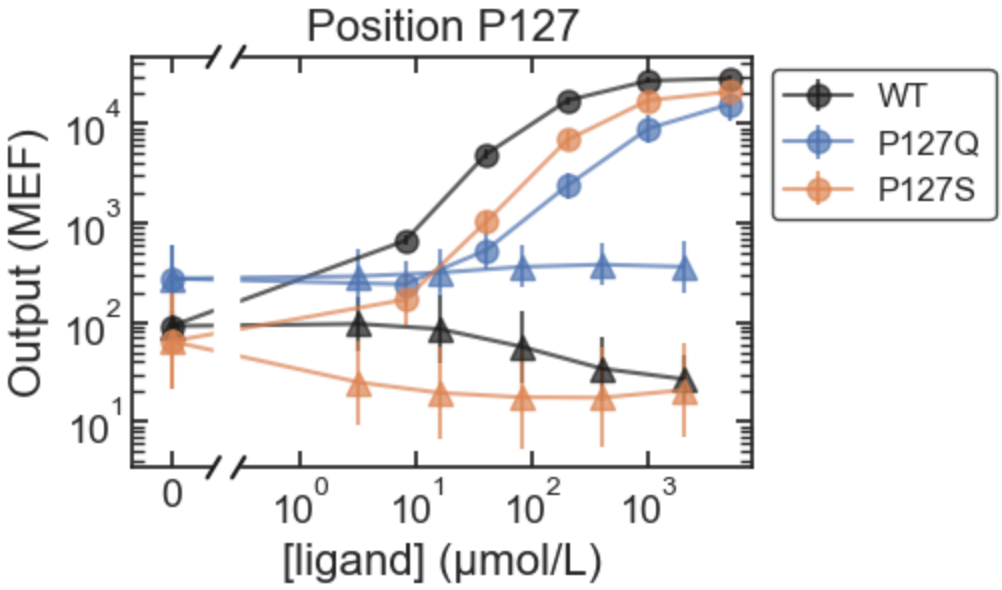

Position G121
Output (MEF)
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

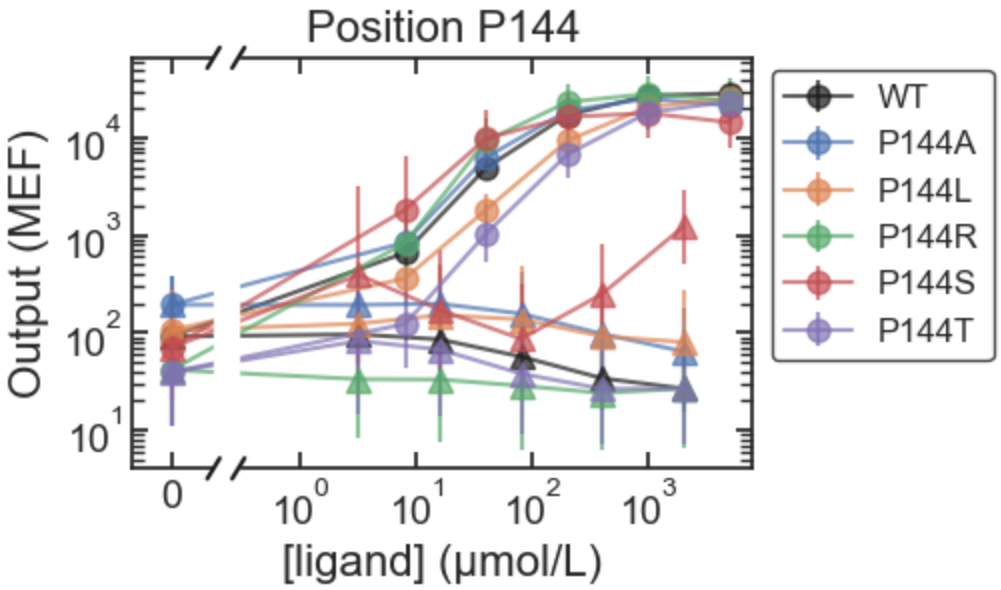

Position L122
Output (MEF)
[ligand] (μmol/L)
WT
L122M
L122P

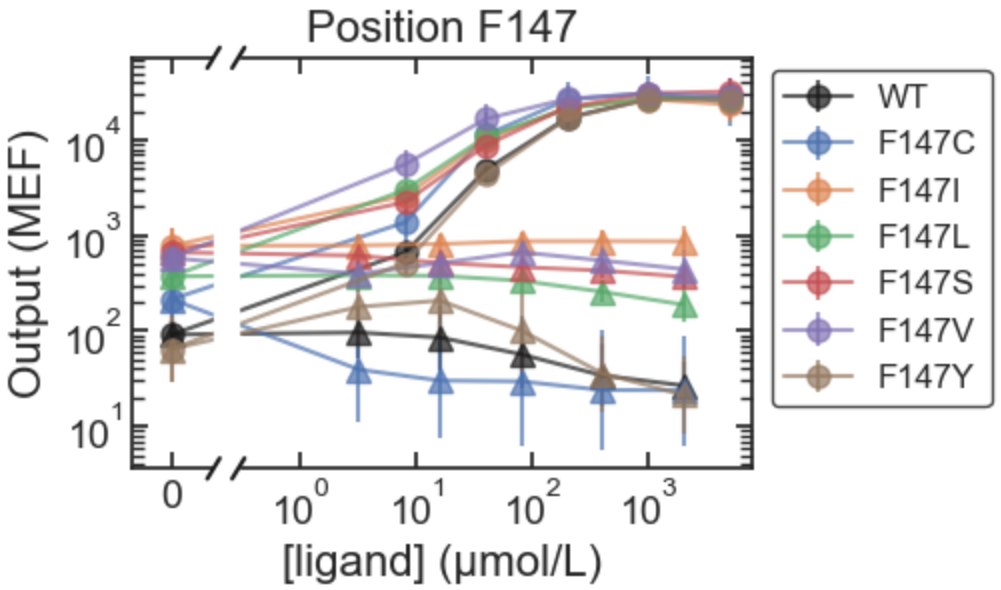

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

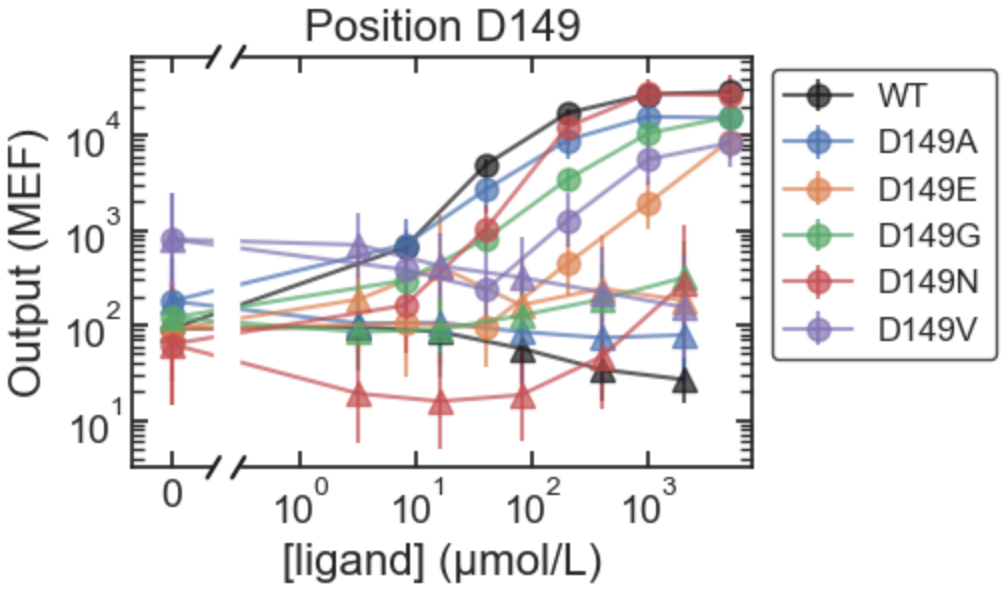

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V

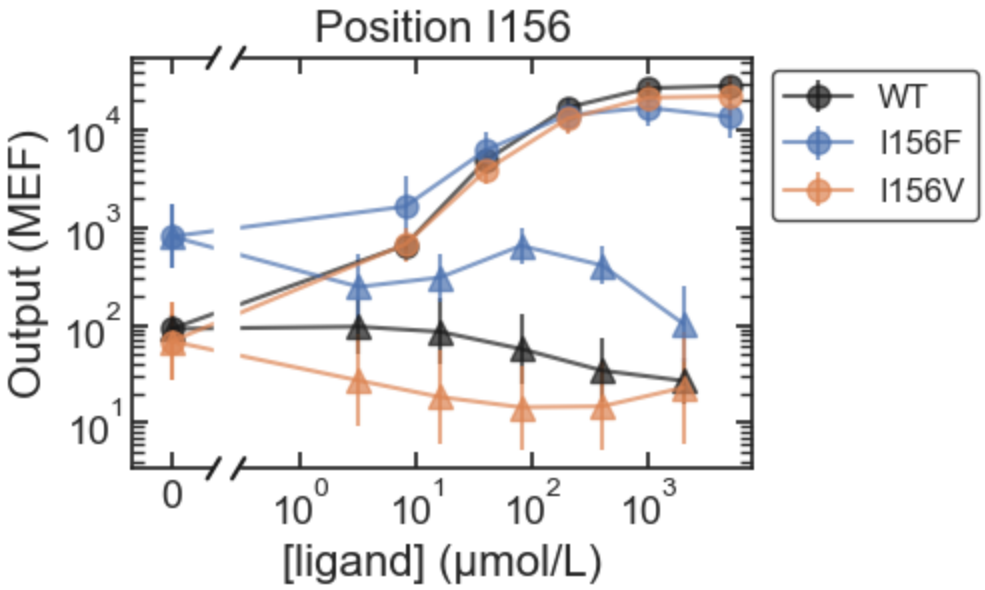

Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S

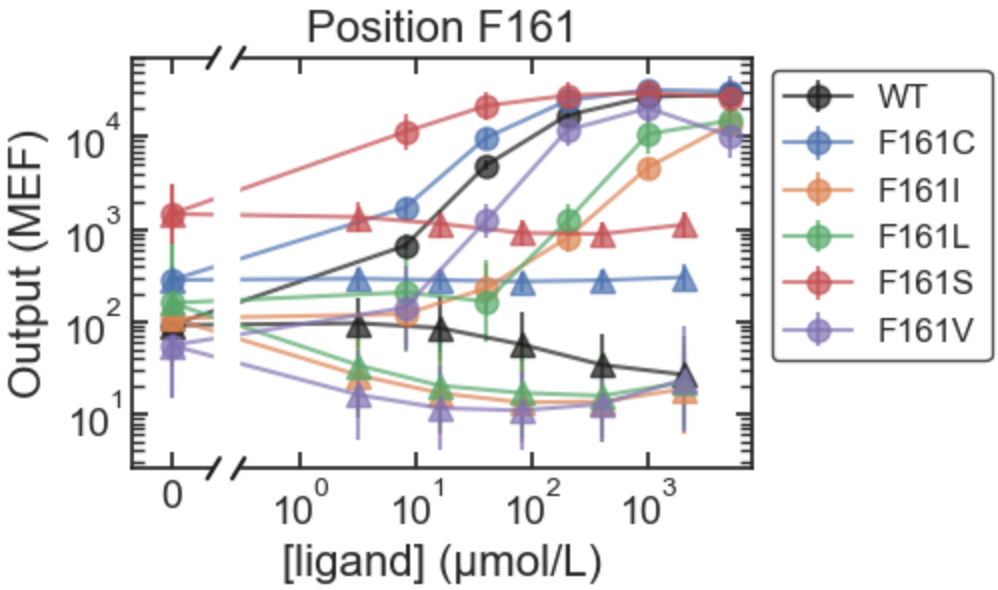

Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

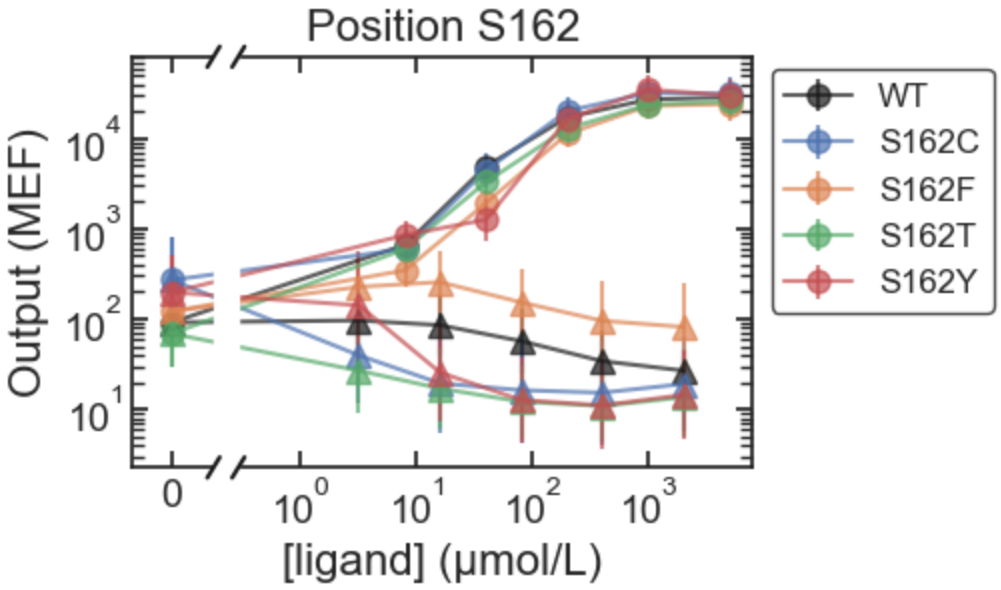

Position P144
Output (MEF)
[ligand] (μmol/L)
WT
P144A
P144L
P144R
P144S
P144T

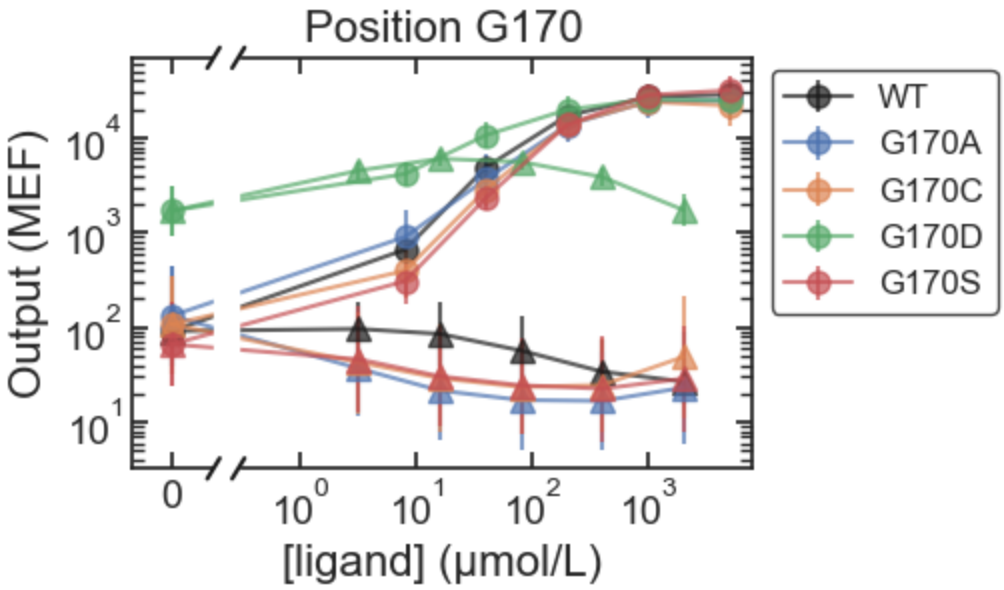

Position F147
Output (MEF)
[ligand] (μmol/L)
WT
F147C
F147I
F147L
F147S
F147V
F147Y

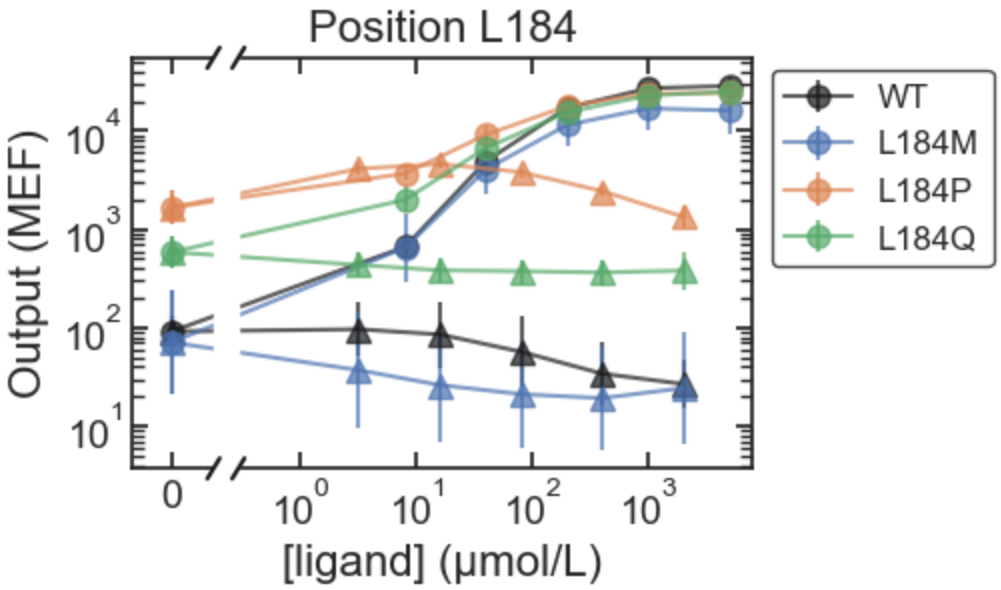

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

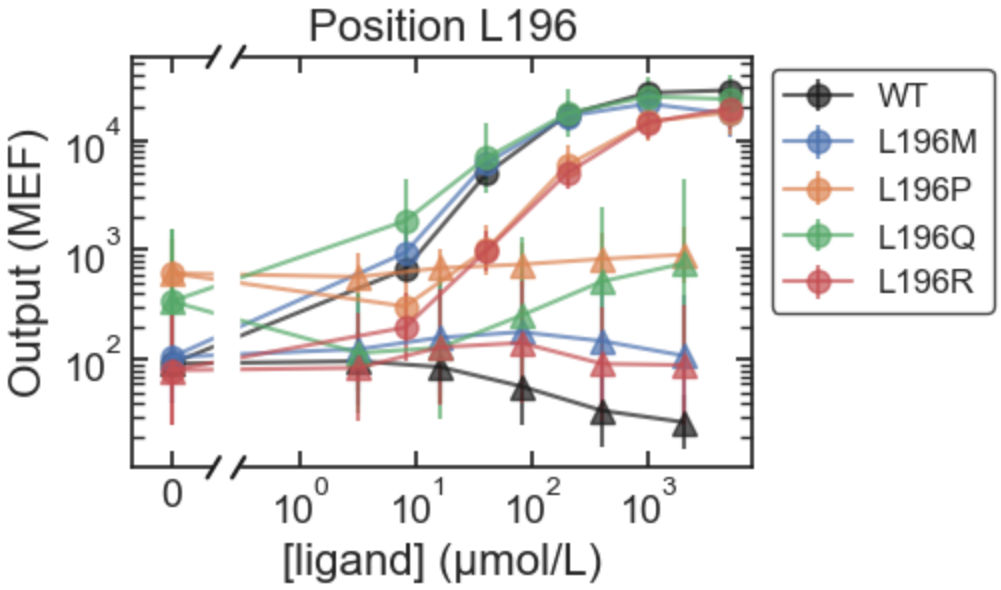

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

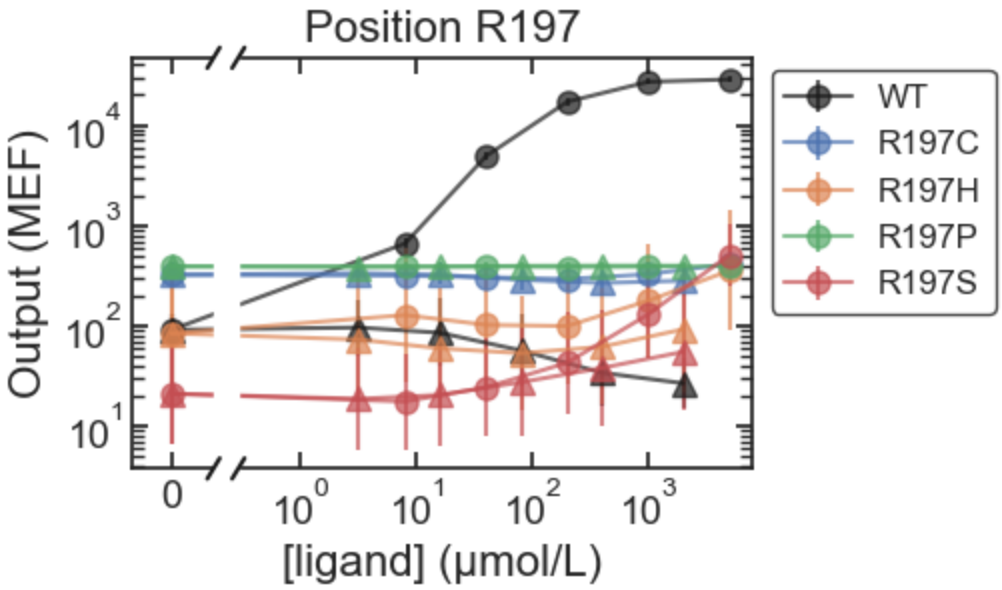

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

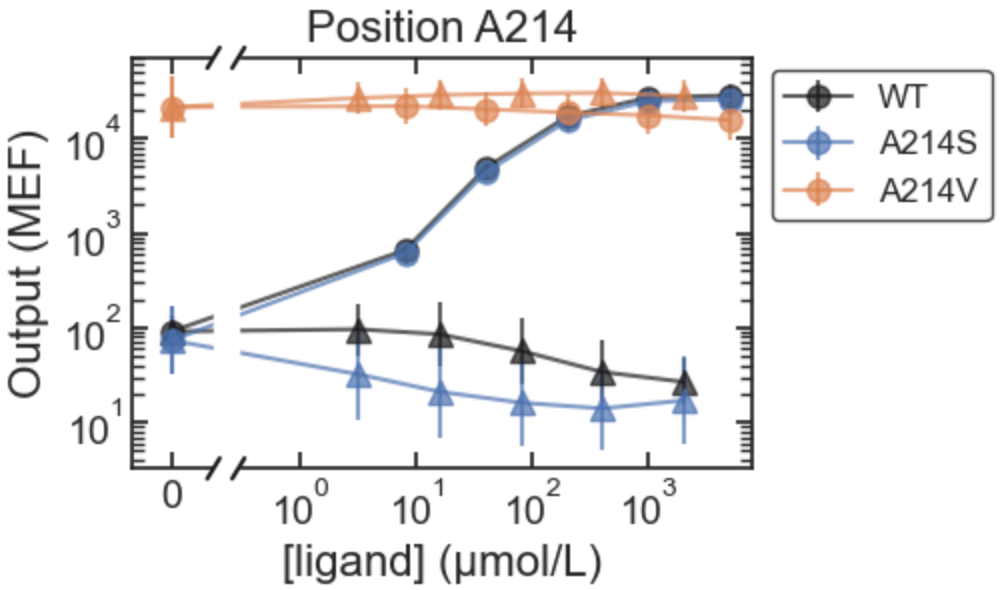

Position S162
Output (MEF)
[ligand] (μmol/L)
WT
S162C
S162F
S162T
S162Y

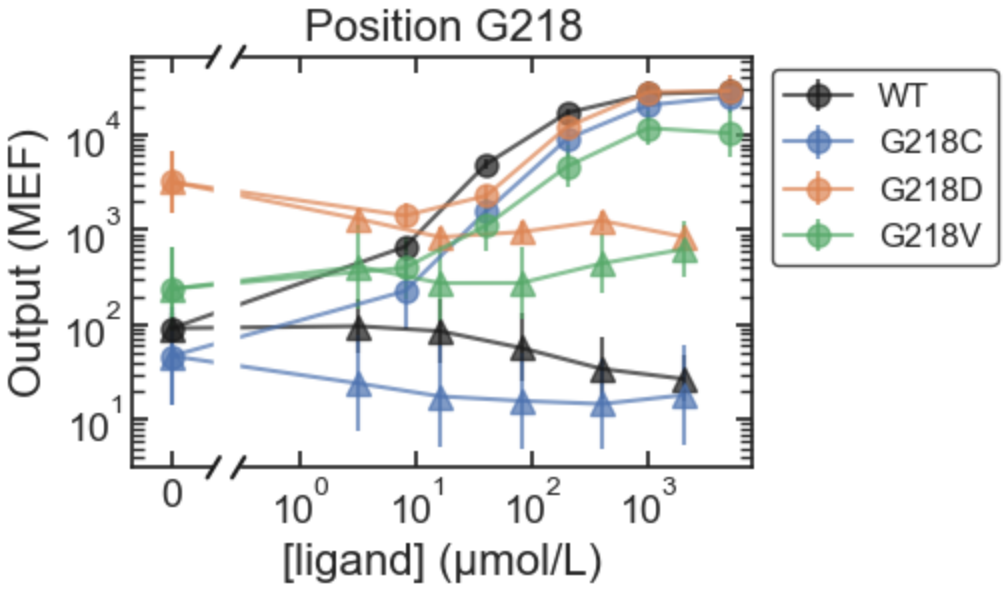

Position G170
Output (MEF)
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

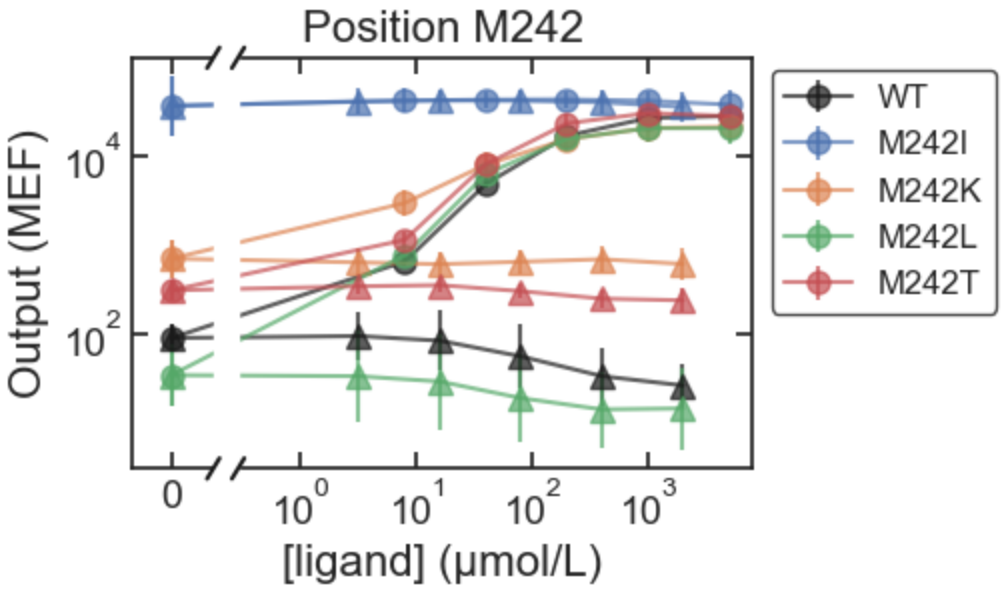

Position L184
Output (MEF)
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

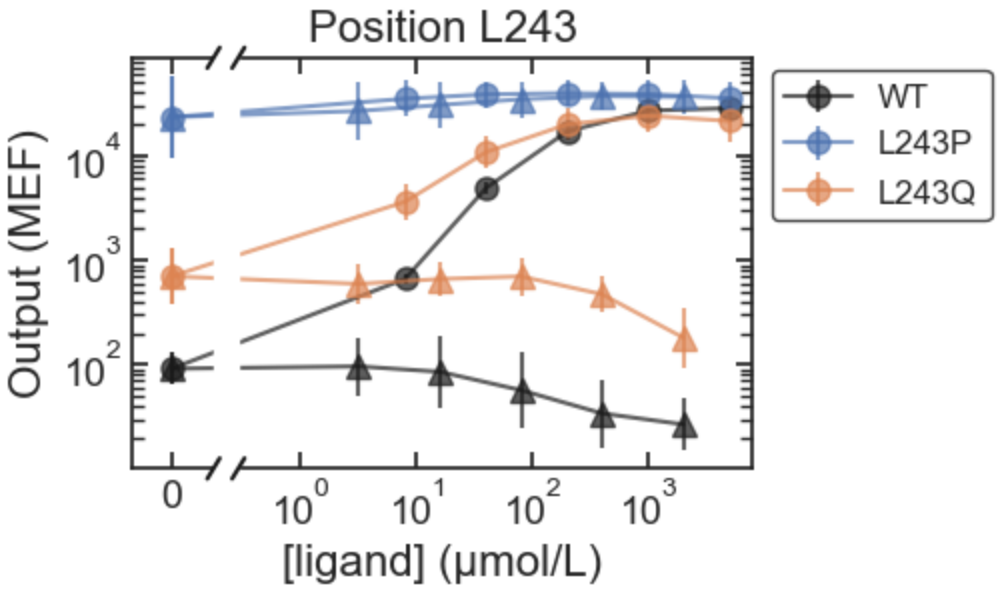

Position L196
Output (MEF)
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

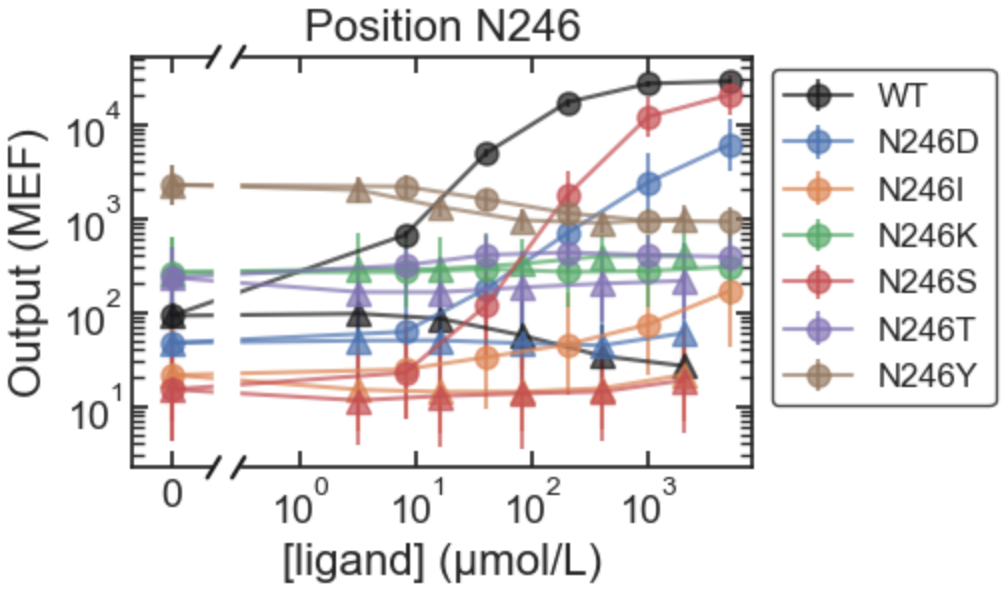

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

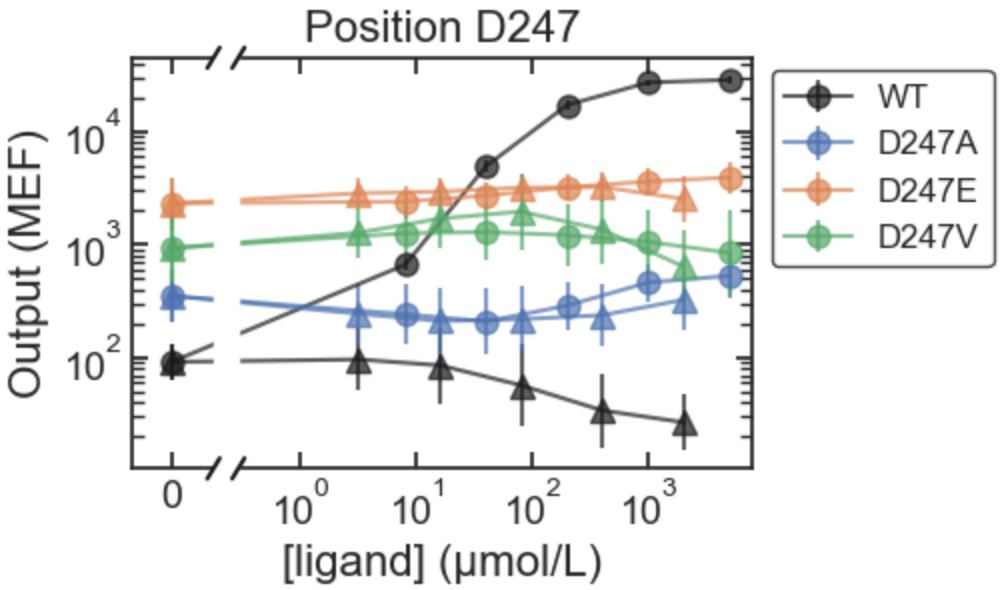

Position A214
Output (MEF)
[ligand] (μmol/L)
WT
A214S
A214V

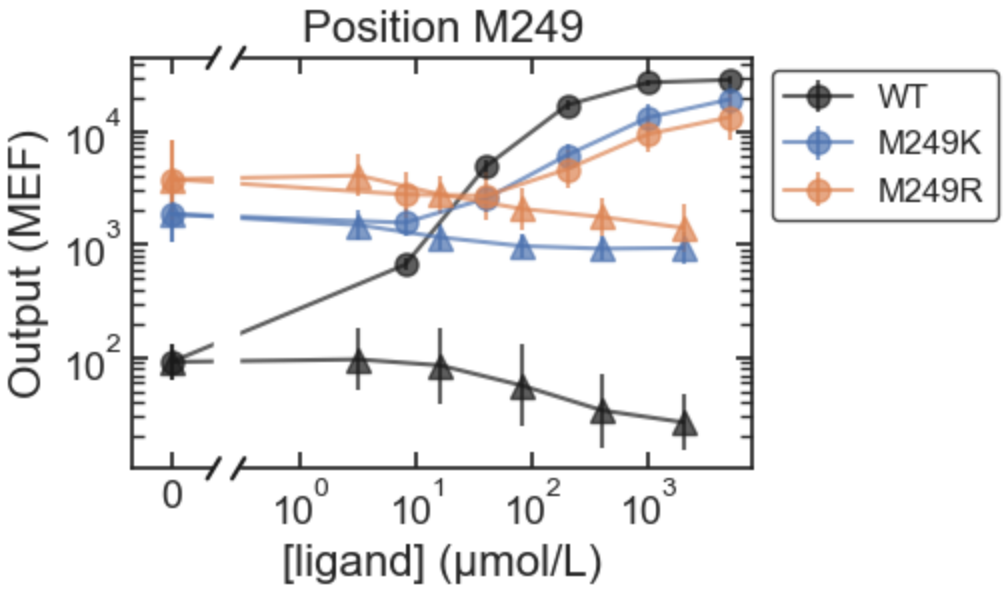

Position G218
Output (MEF)
[ligand] (μmol/L)
WT
G218C
G218D
G218V

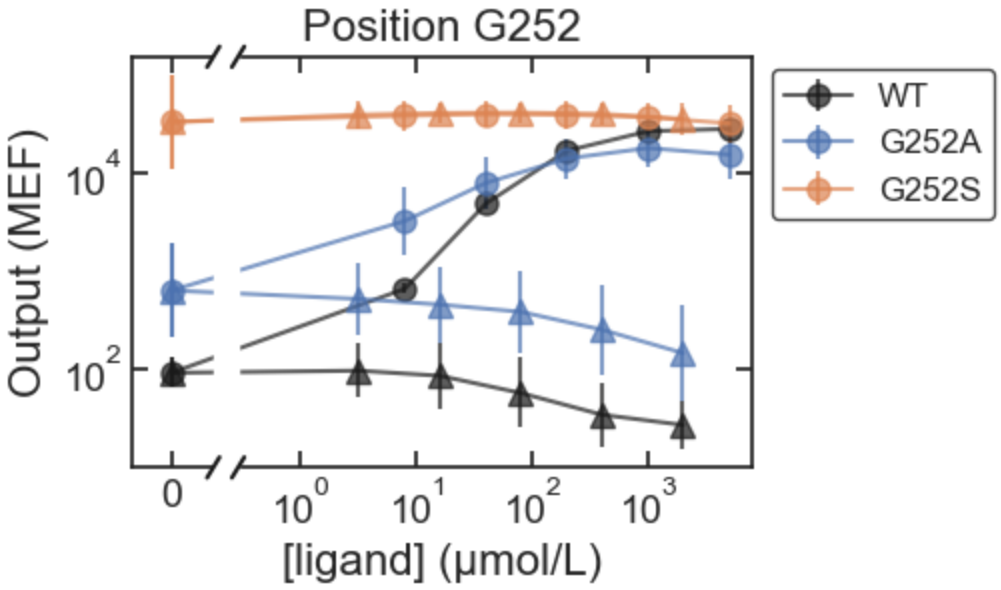

Position M242
Output (MEF)
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

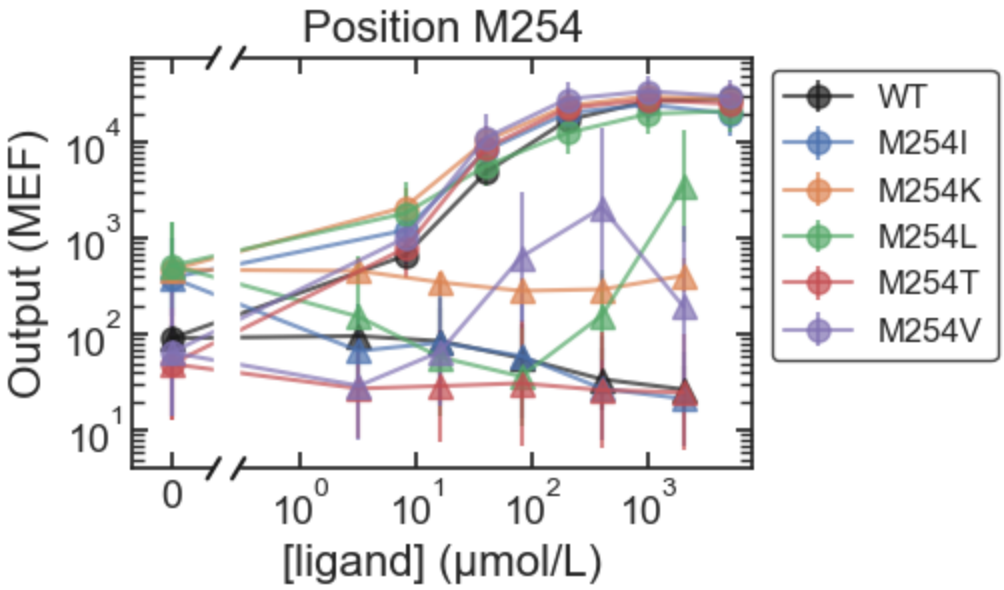

Position L243
Output (MEF)
[ligand] (μmol/L)
WT
L243P
L243Q

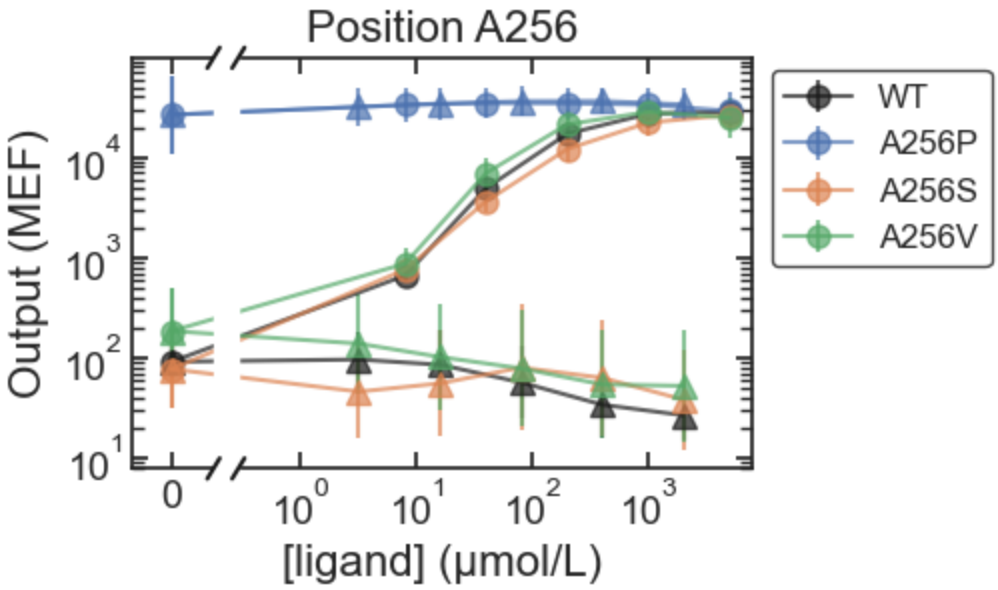

Position N246
Output (MEF)
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

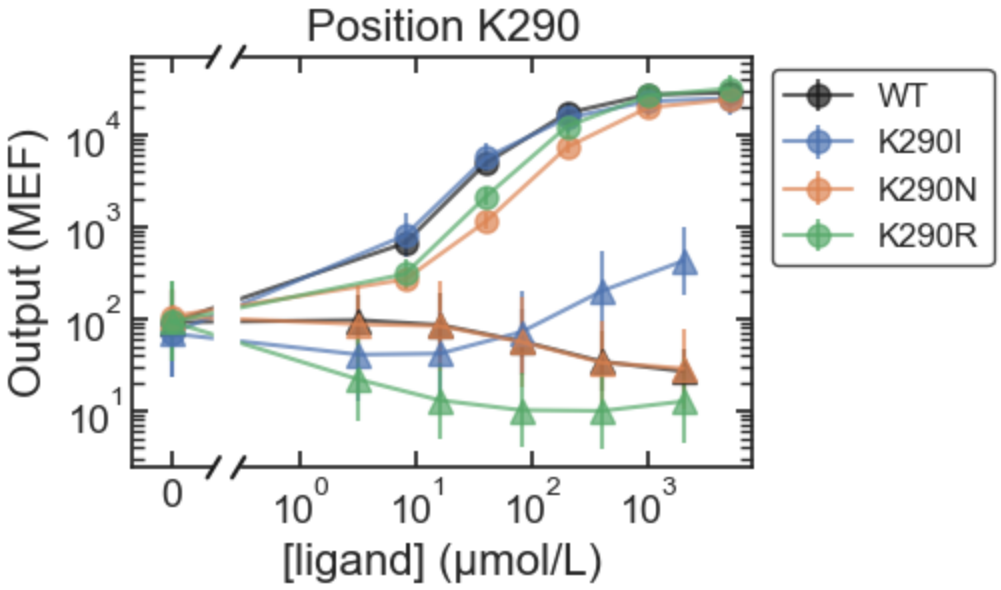

Position D247
Output (MEF)
[ligand] (μmol/L)
WT
D247A
D247E
D247V

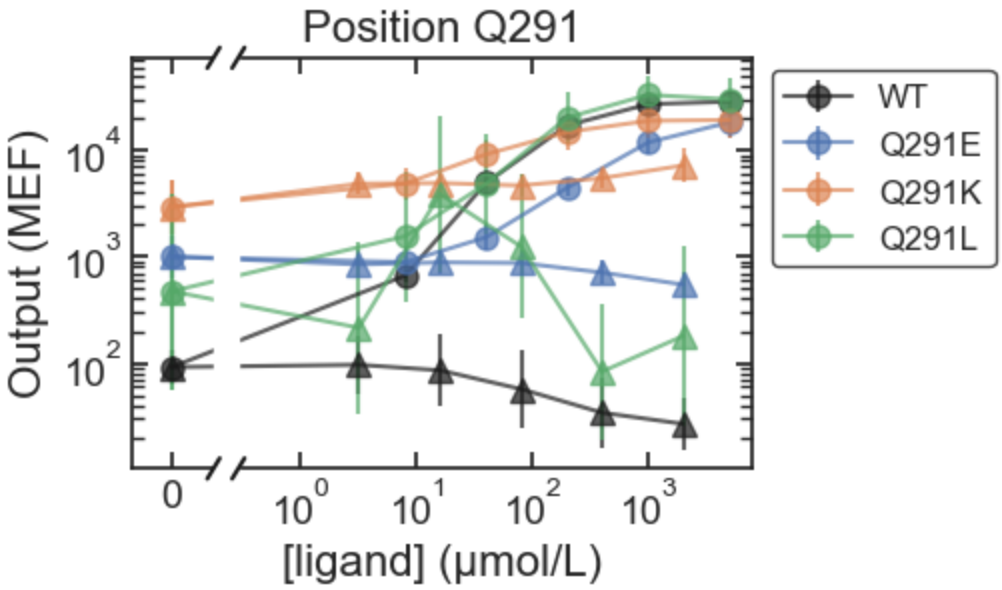

Position M249
Output (MEF)
[ligand] (μmol/L)
WT
M249K
M249R

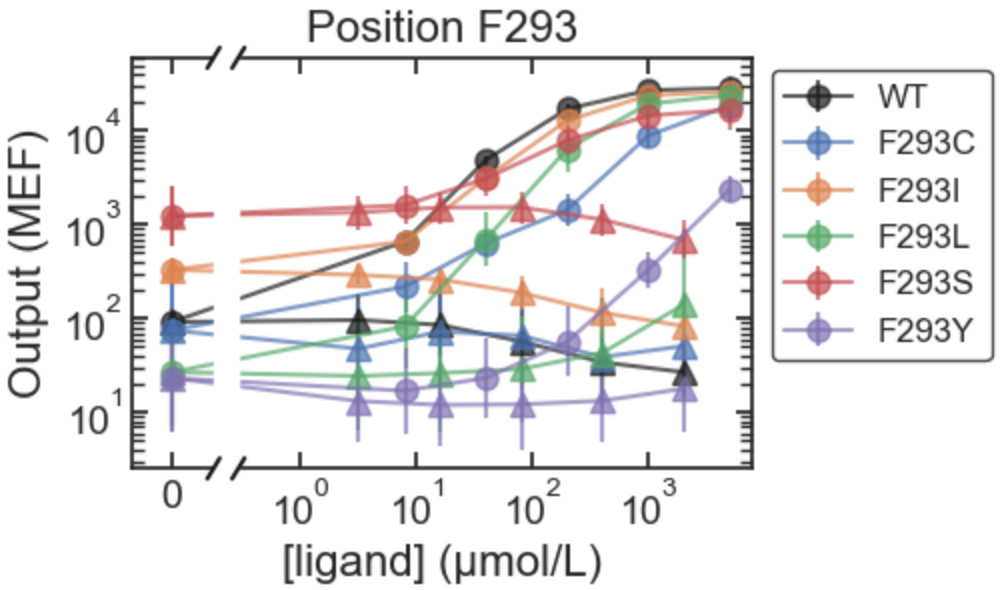

Position G252
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G252A
G252S

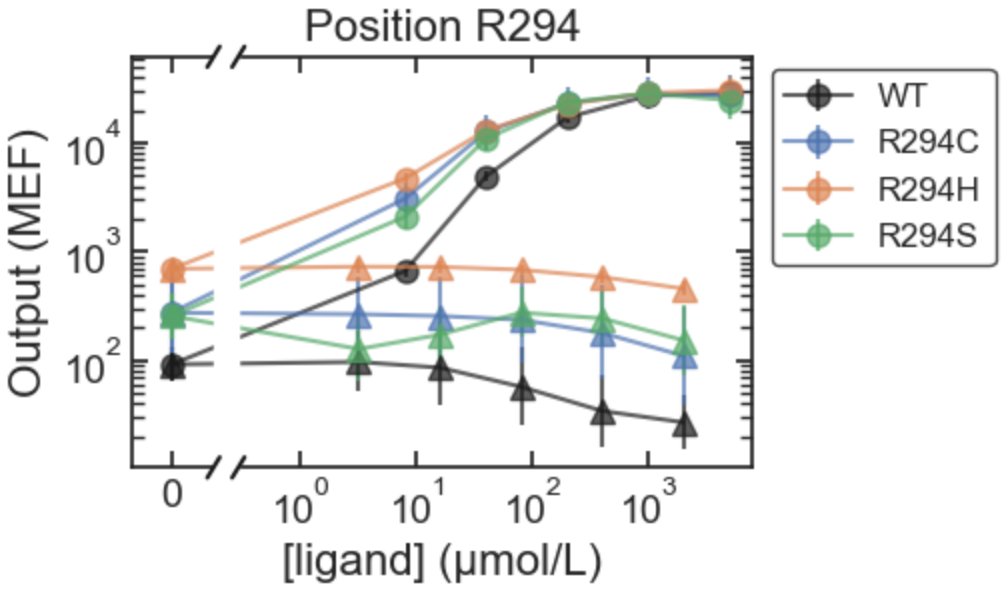

Position M254
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M254I
M254K
M254L
M254T
M254V

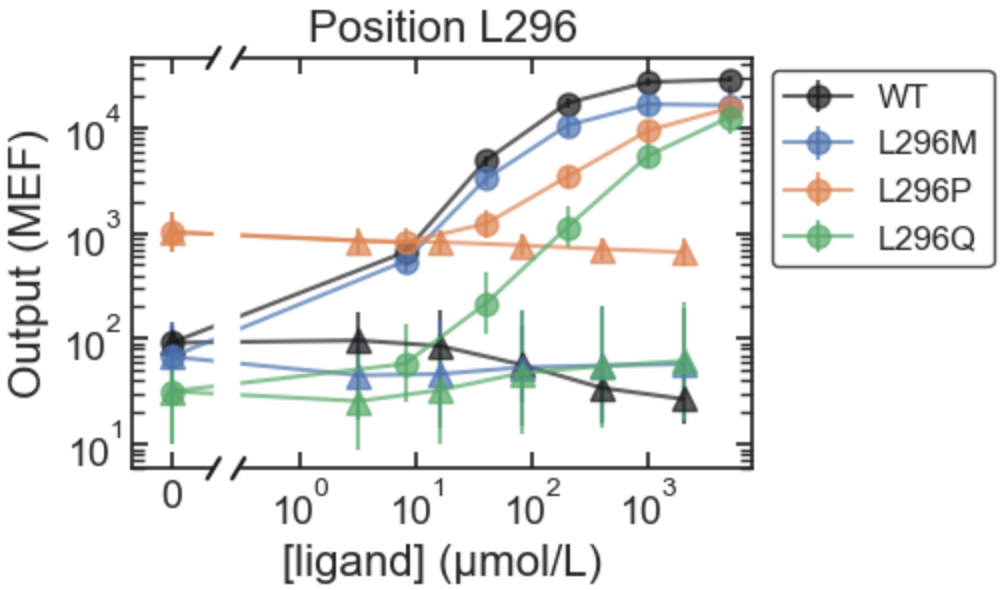

Position A256
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A256P
A256S
A256V

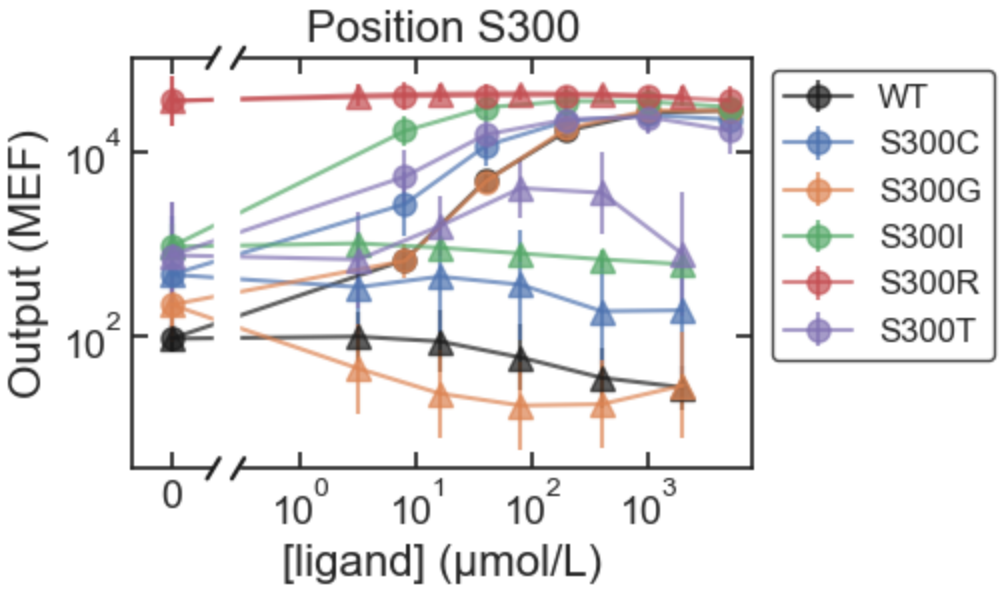

Position K290
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K290I
K290N
K290R

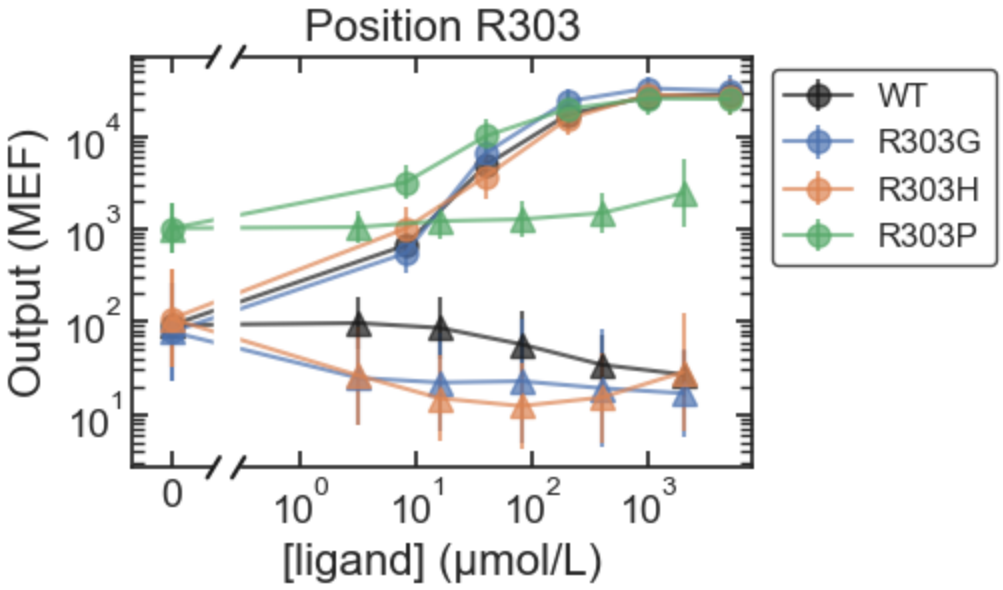

Position Q291
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L

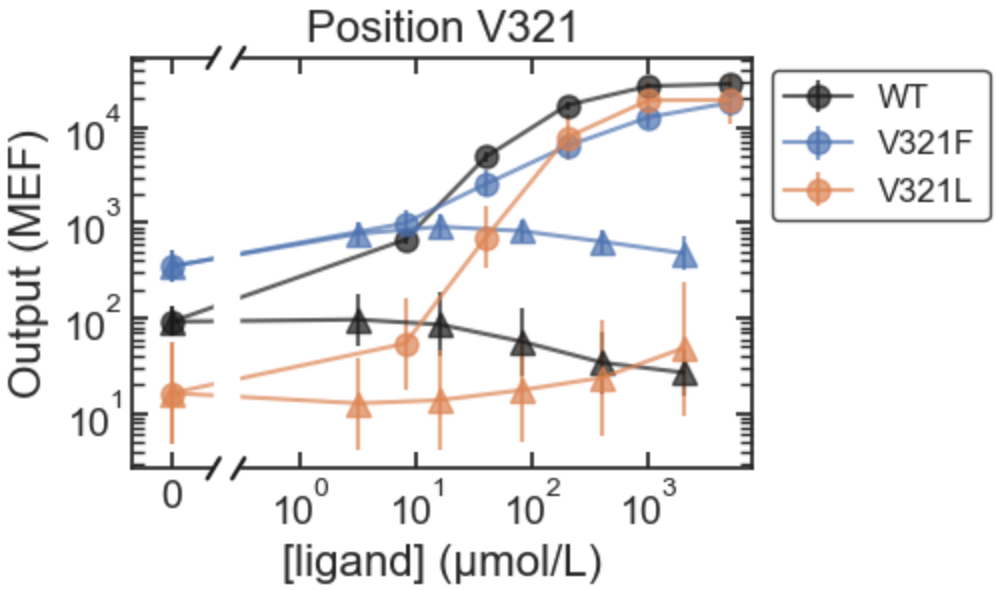

Position F293
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

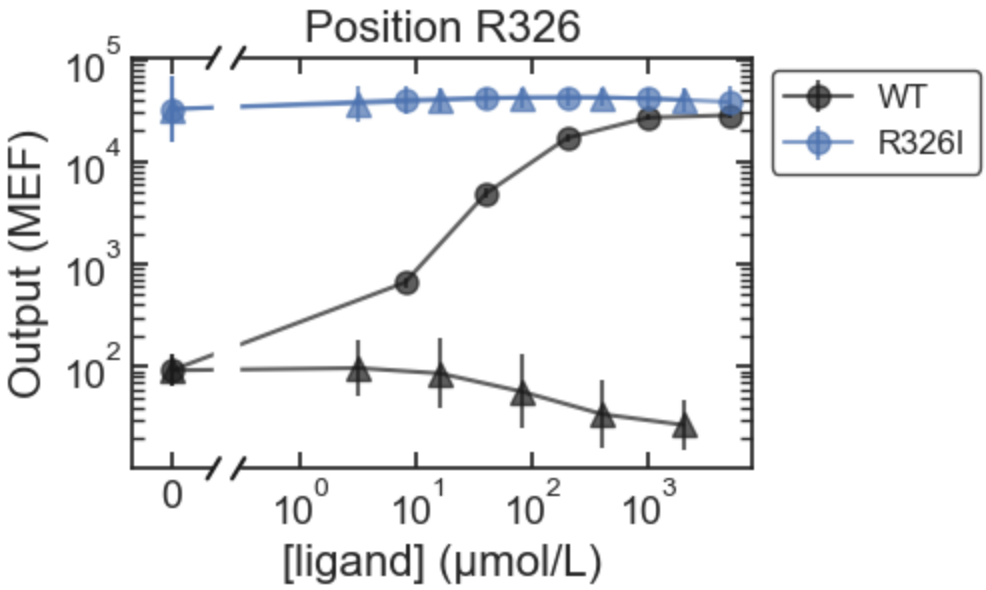

Position R294
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R294C
R294H
R294S

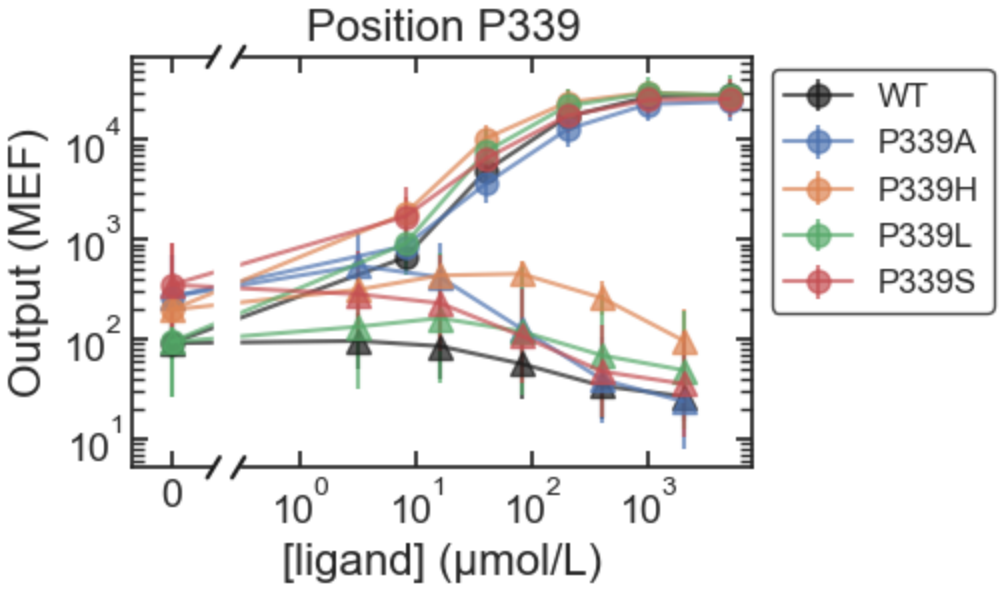

Position L296
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L296M
L296P
L296Q

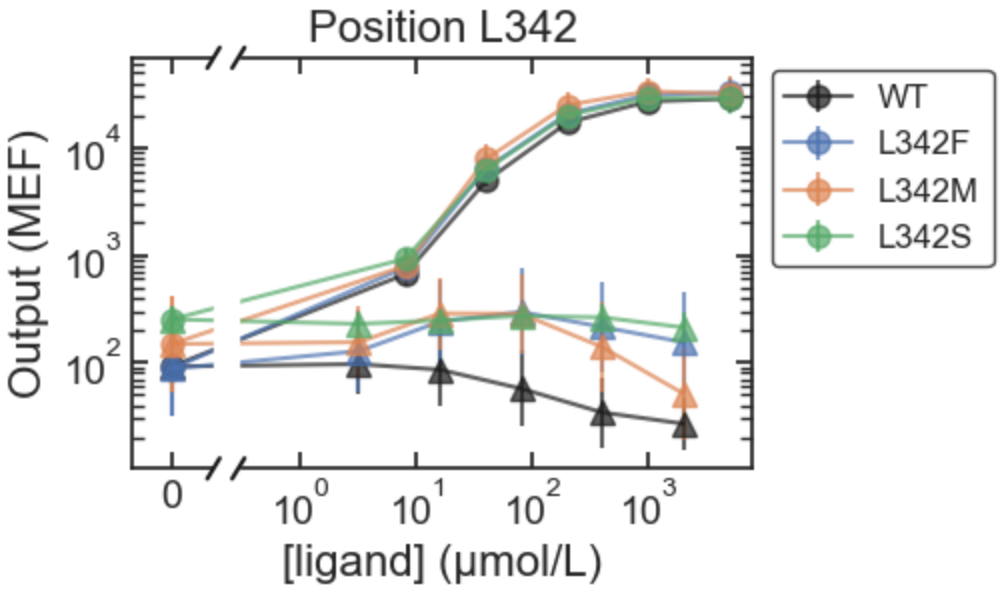

Position S300
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S300C
S300G
S300I
S300R
S300T

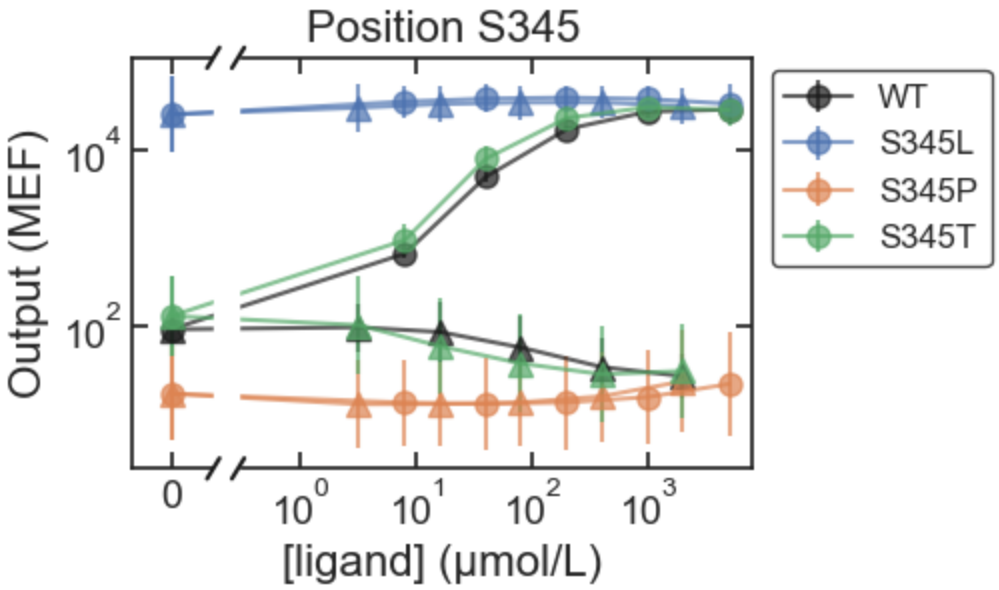

Position R303
Output (MEF)
[ligand] (μmol/L)
WT
R303G
R303H
R303P


Position V321
Output (MEF)
[ligand] (μmol/L)
WT
V321F
V321L

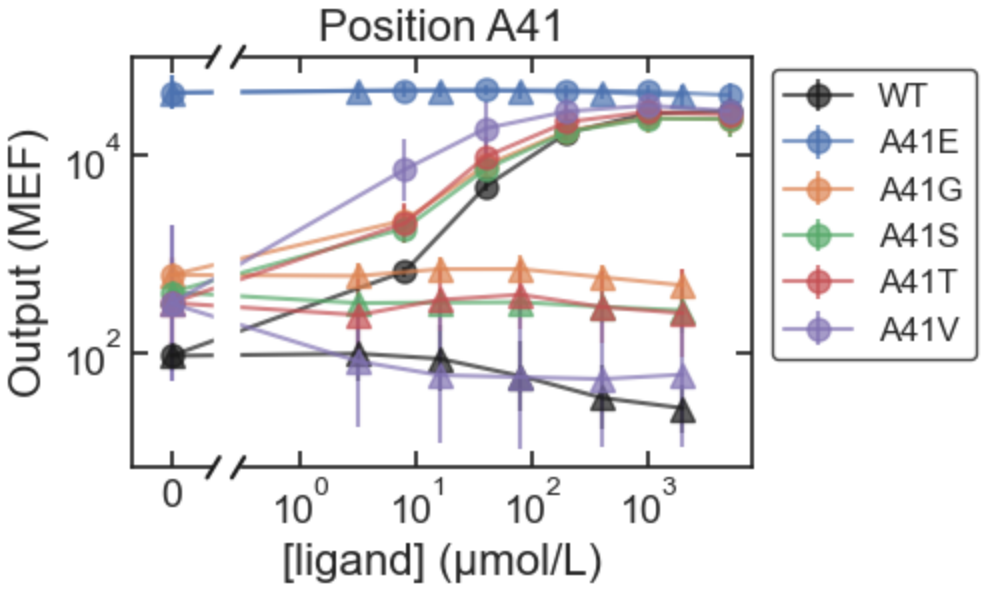

Position R326
Output (MEF)
[ligand] (μmol/L)
WT
R326I

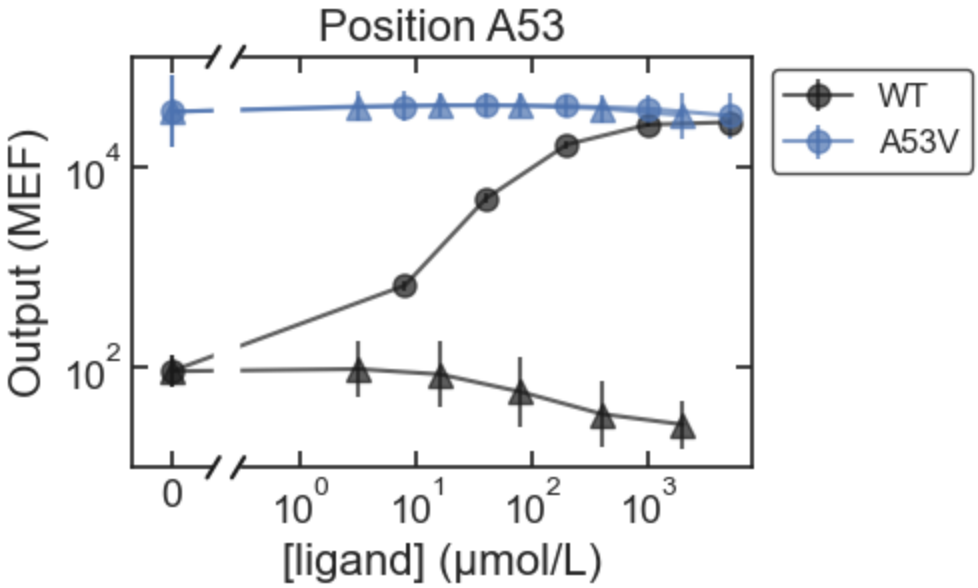

Position P339
Output (MEF)
[ligand] (µmol/L)
WT
P339A
P339H
P339L
P339S

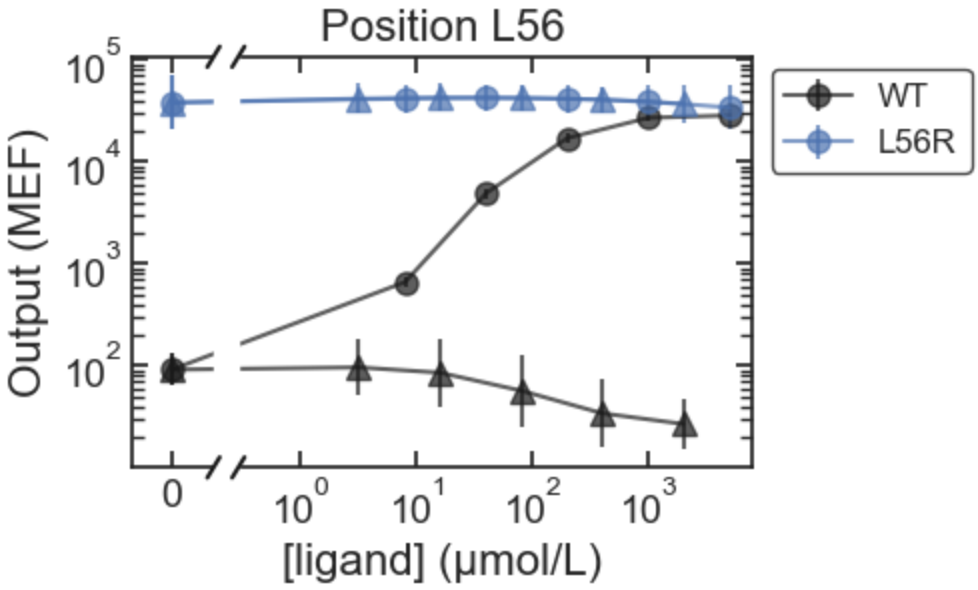

Position L342
Output (MEF)
[ligand] (µmol/L)
WT
L342F
L342M
L342S

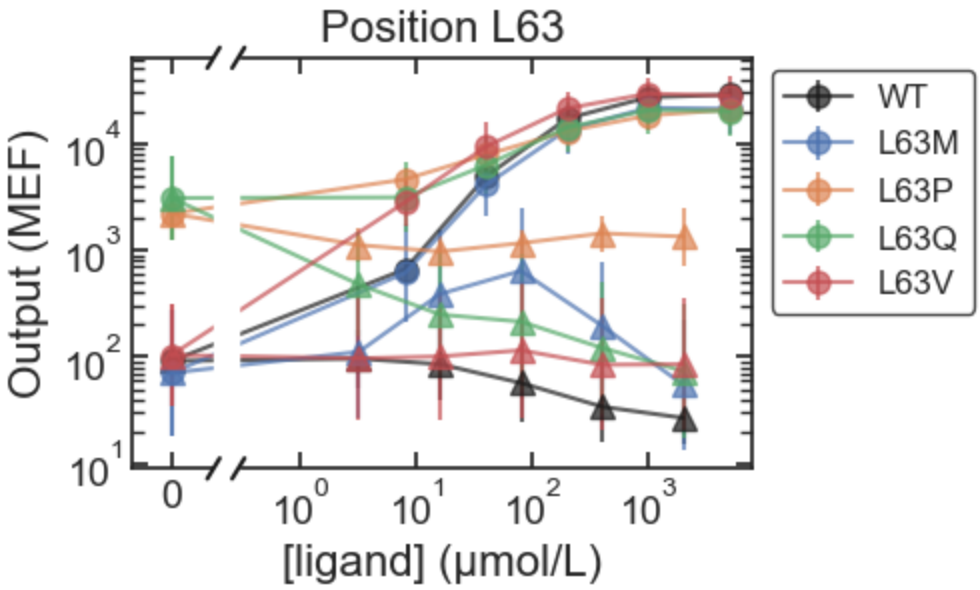

Position S345
Output (MEF)
[ligand] (µmol/L)
WT
S345L
S345P
S345T

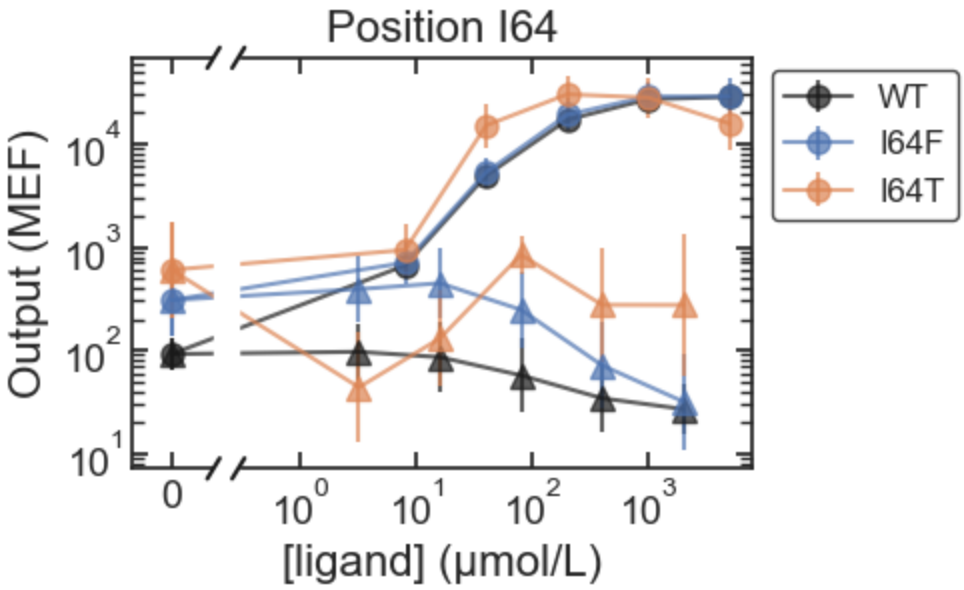
MWC

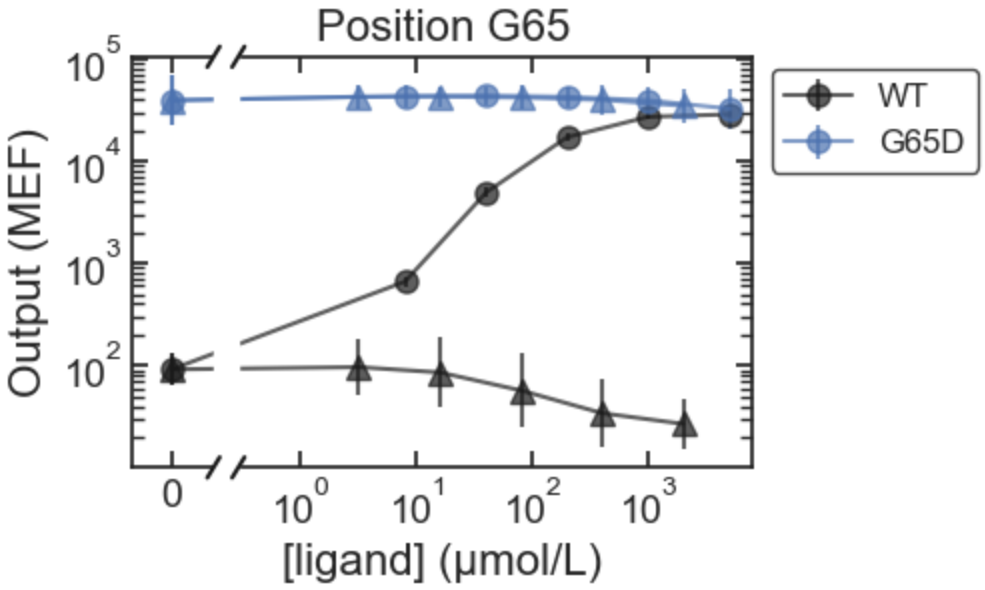

Position A41
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A41E
A41G
A41S
A41T
A41V

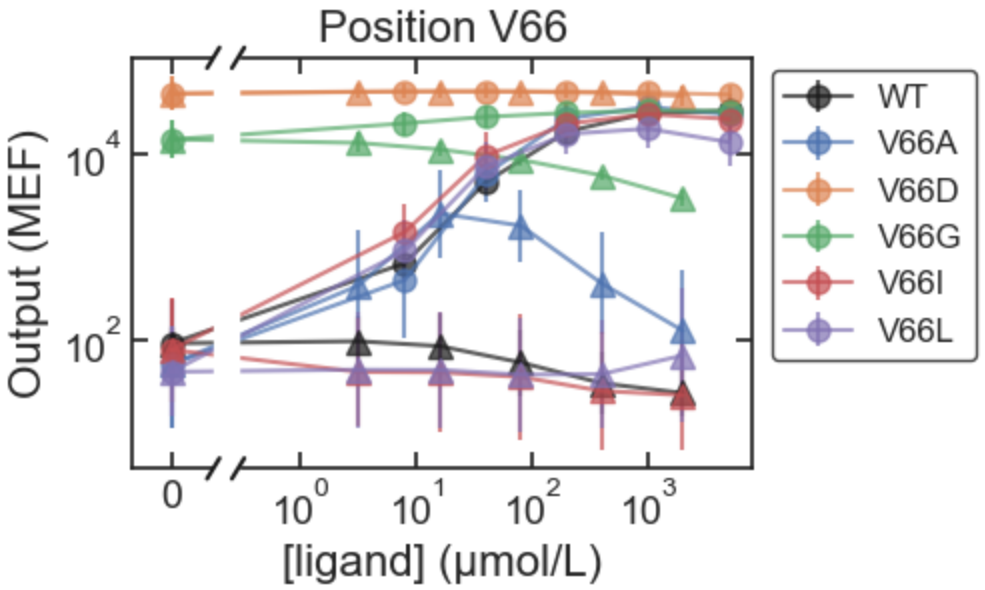

Position A53
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A53V

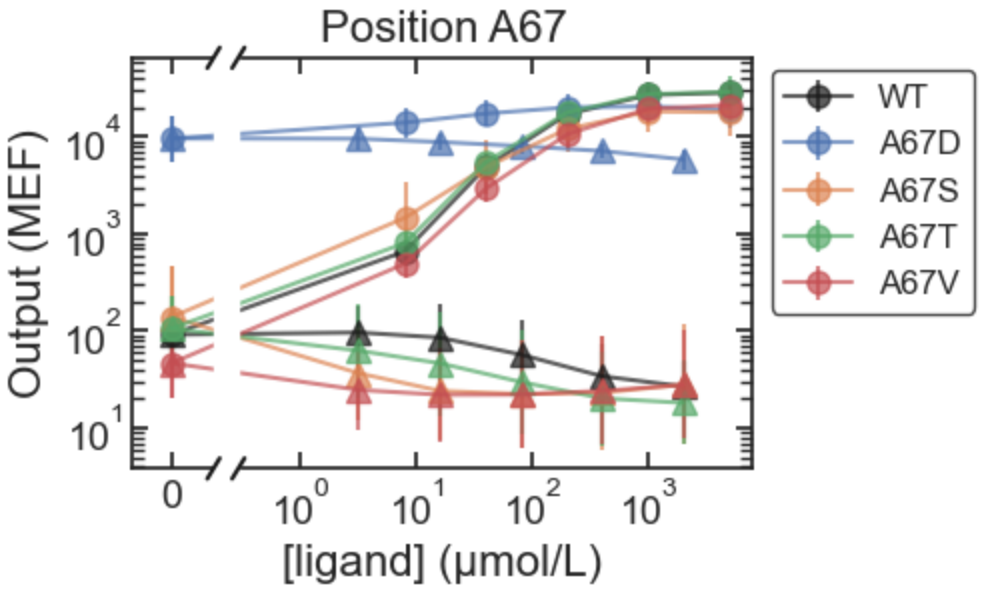

Position L56
Output (MEF)
[ligand] (μmol/L)
WT
L56R

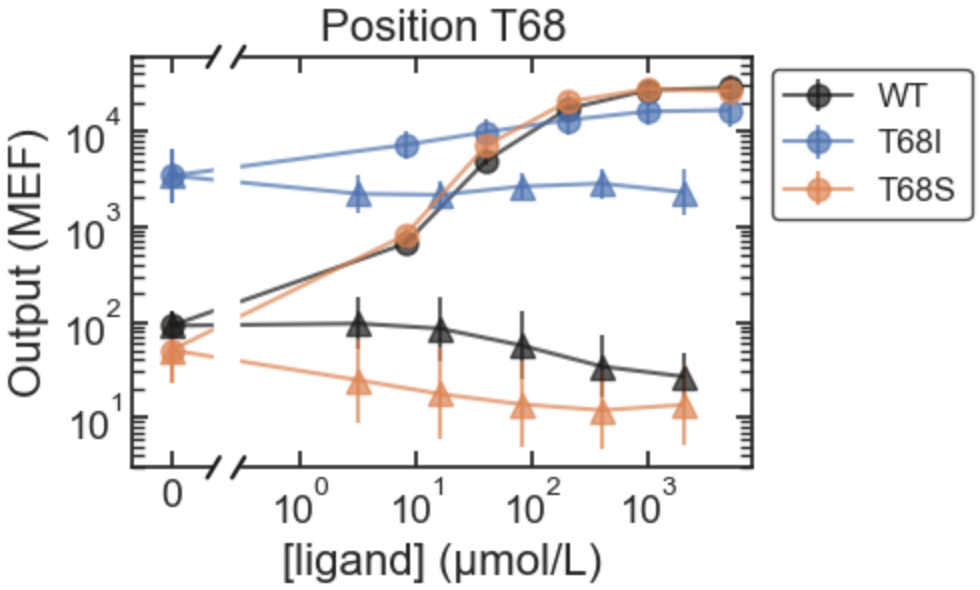

Position L63
Output (MEF)
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

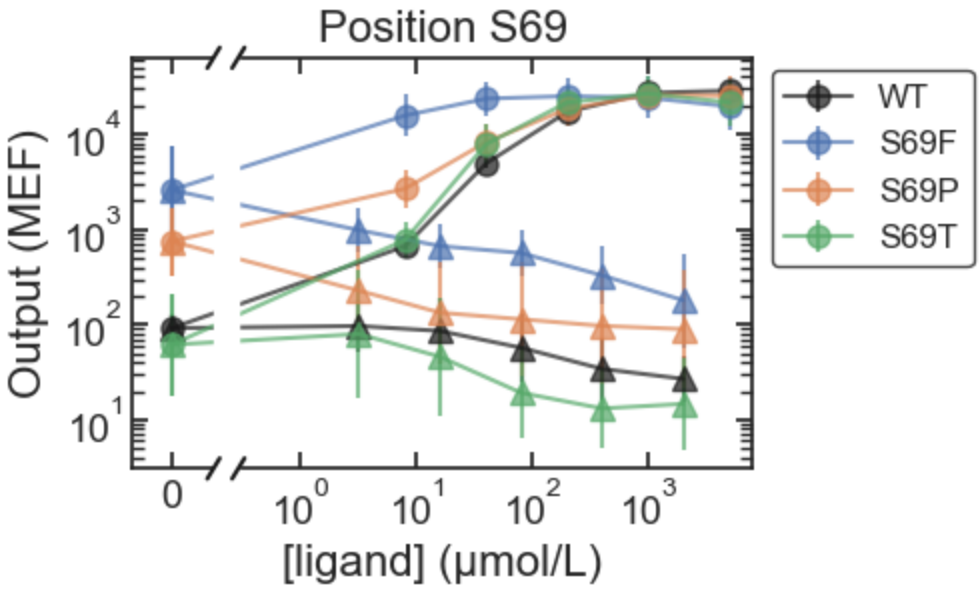

Position I64
Output (MEF)
[ligand] (μmol/L)
WT
I64F
I64T

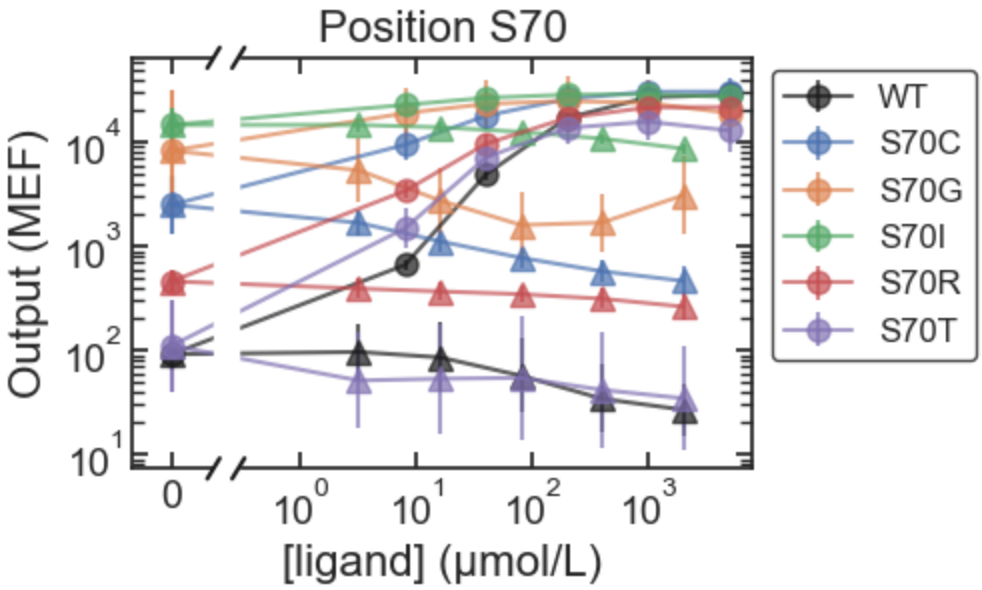

Position G65
Output (MEF)
10^5
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G65D

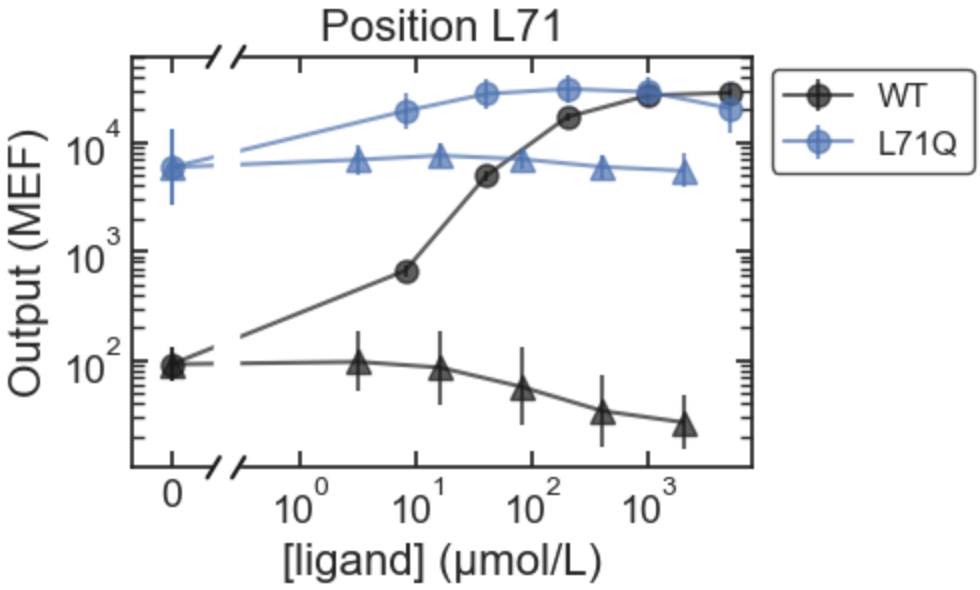

Position V66
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V66A
V66D
V66G
V66I
V66L

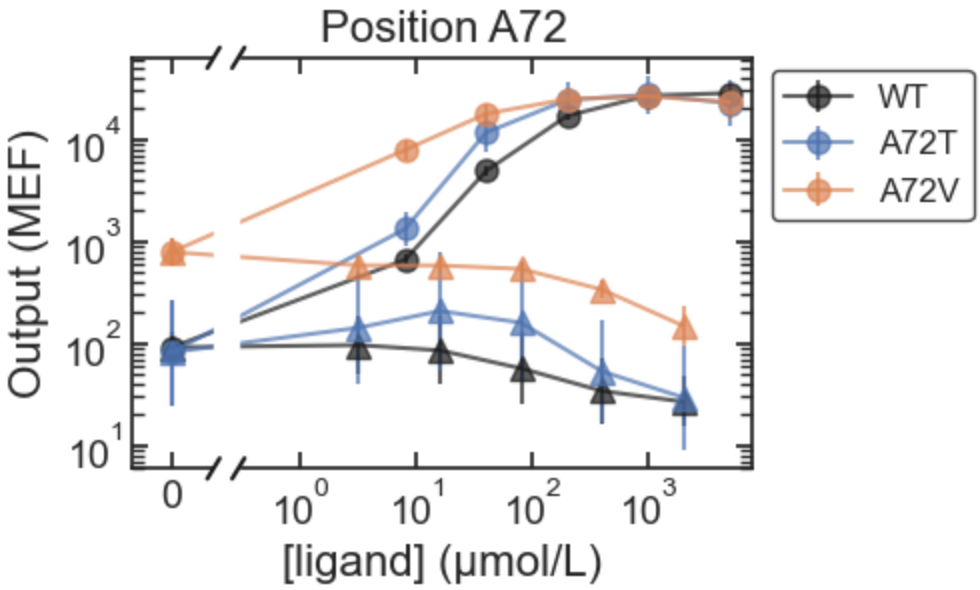

Position A67
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A67D
A67S
A67T
A67V

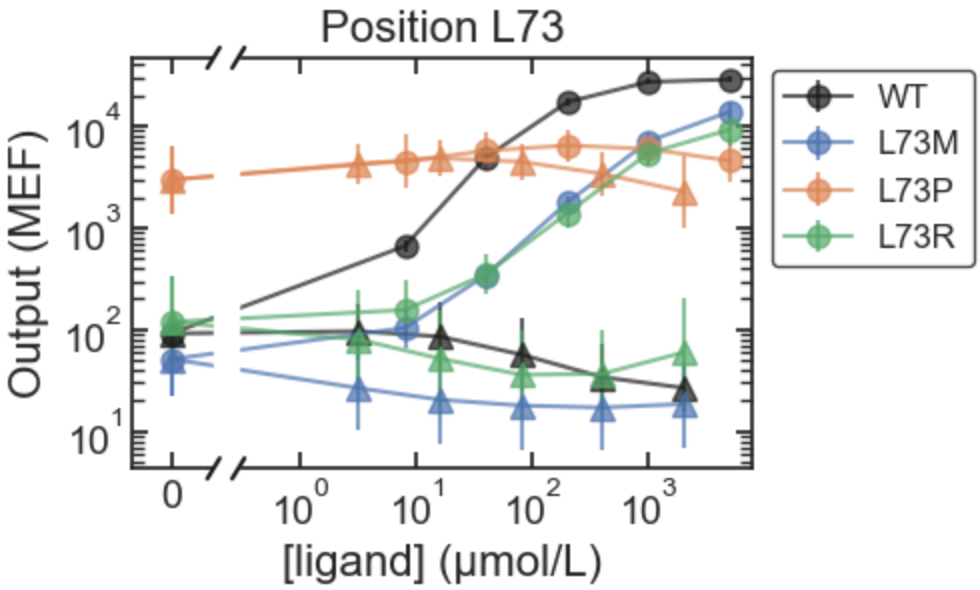

Position T68
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
T68I
T68S

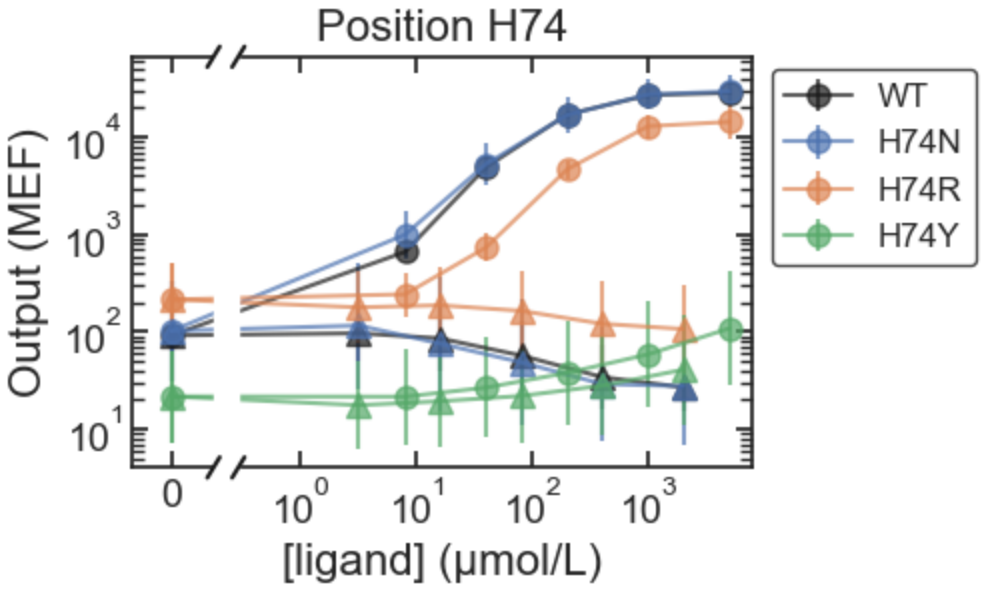

Position S69
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S69F
S69P
S69T

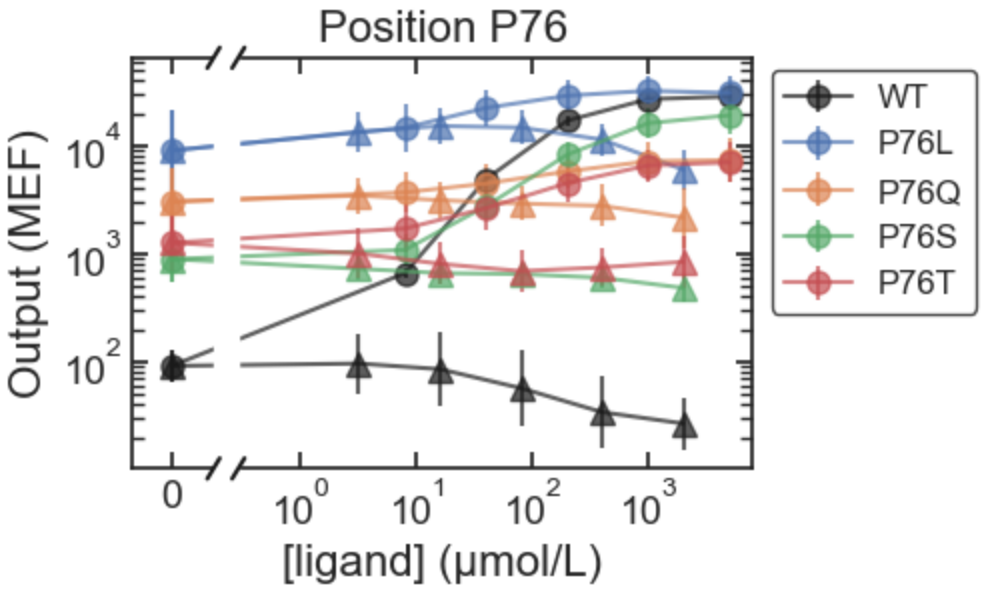

Position S70
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S70C
S70G
S70I
S70R
S70T

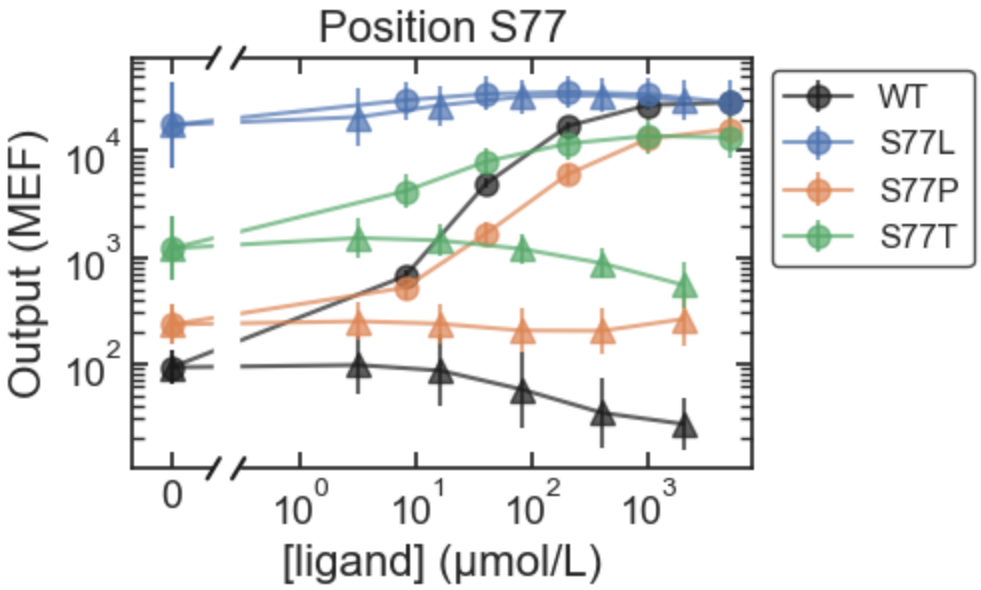

Position L71
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L71Q

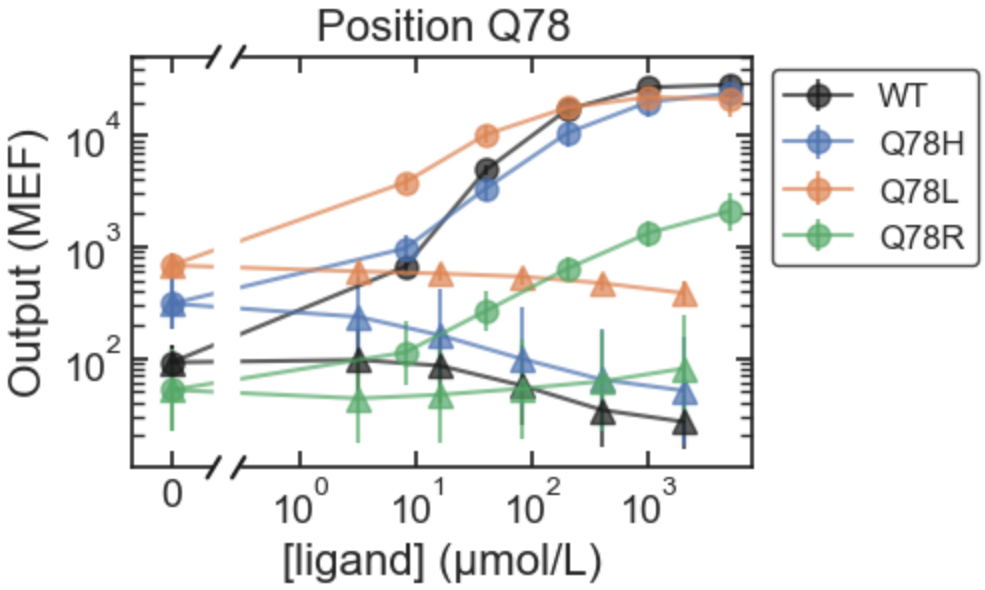

Position A72
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A72T
A72V

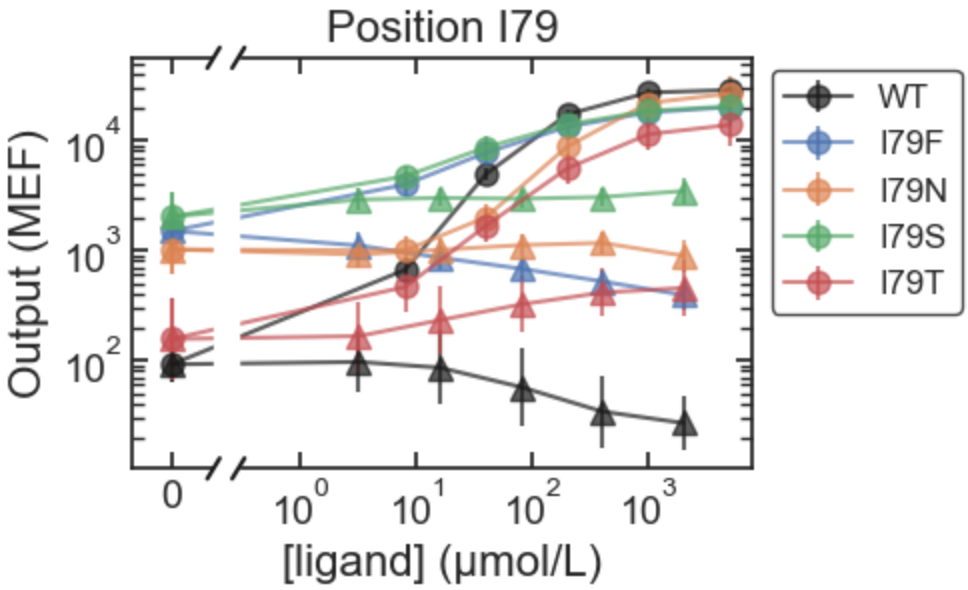

Position L73
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L73M
L73P
L73R

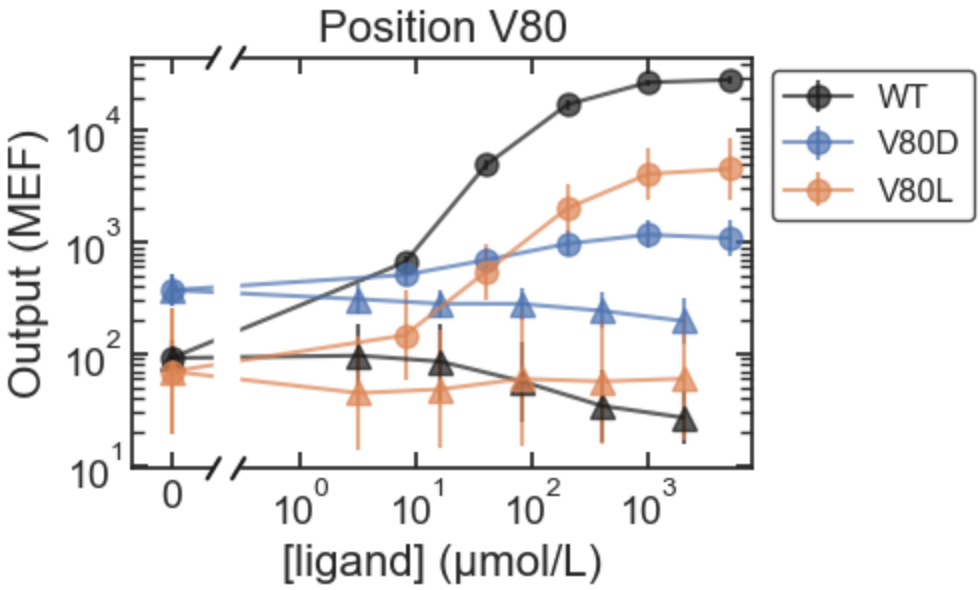

Position H74
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
H74N
H74R
H74Y

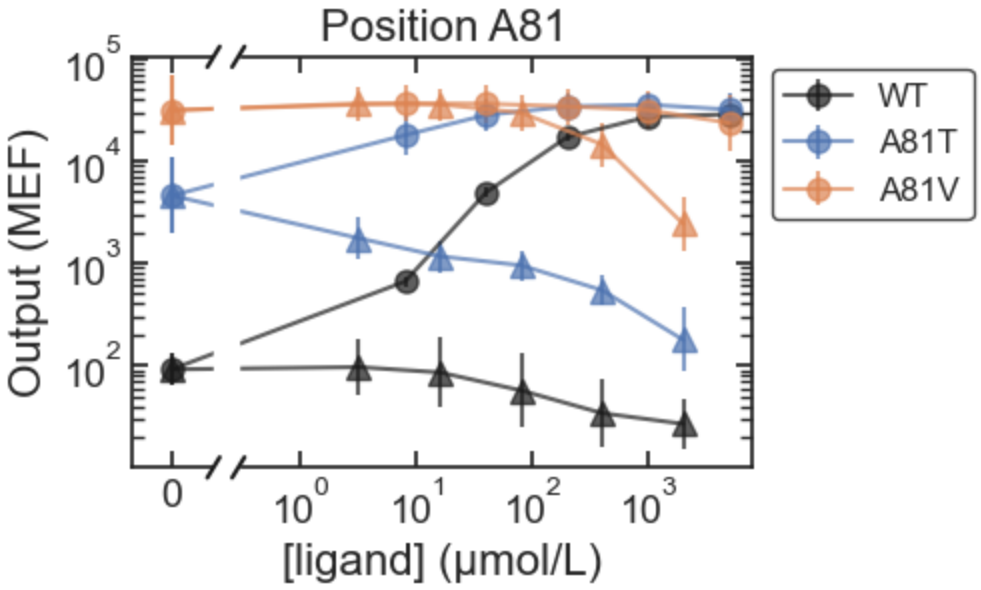

Position P76
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

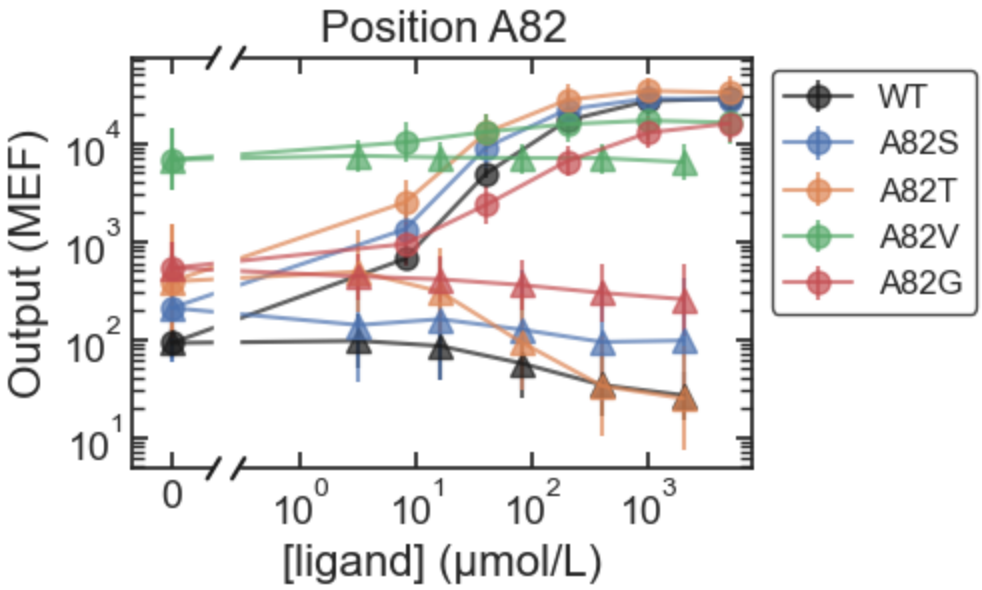

Position S77
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
S77L
S77P
S77T

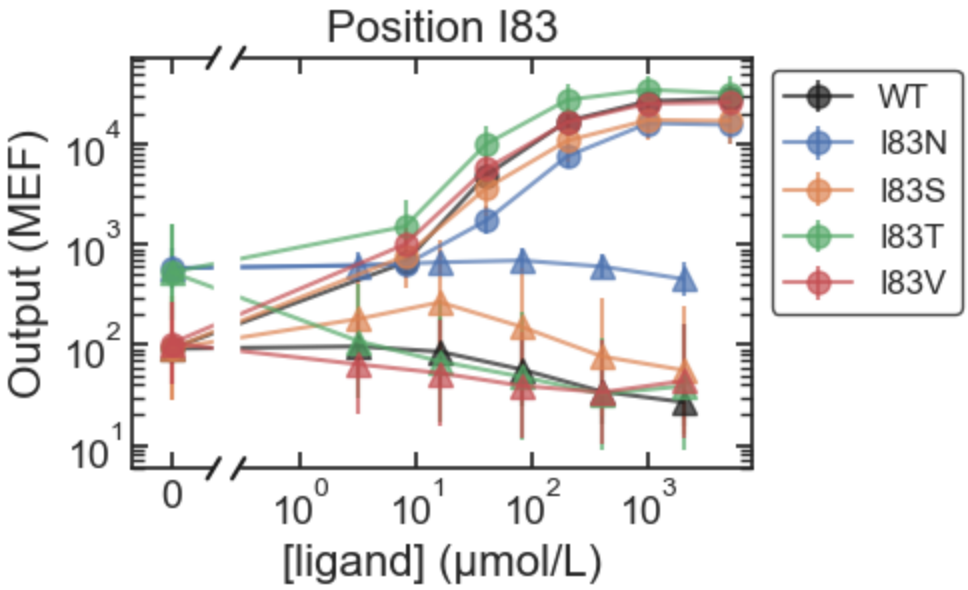

Position Q78
Output (MEF)
[ligand] (μmol/L)
WT
Q78H
Q78L
Q78R

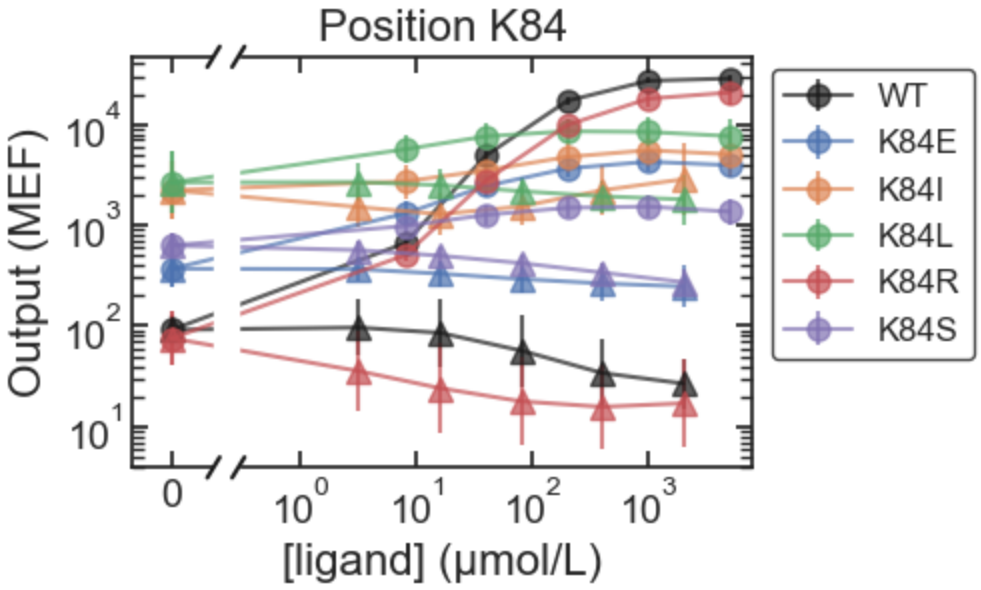

Position I79
Output (MEF)
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

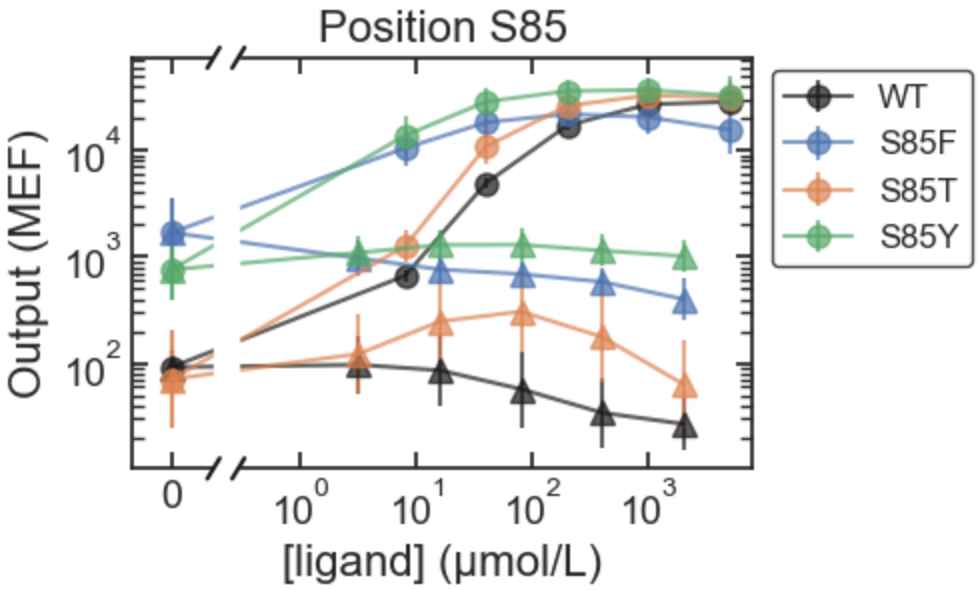

Position V80
Output (MEF)
[ligand] (μmol/L)
WT
V80D
V80L

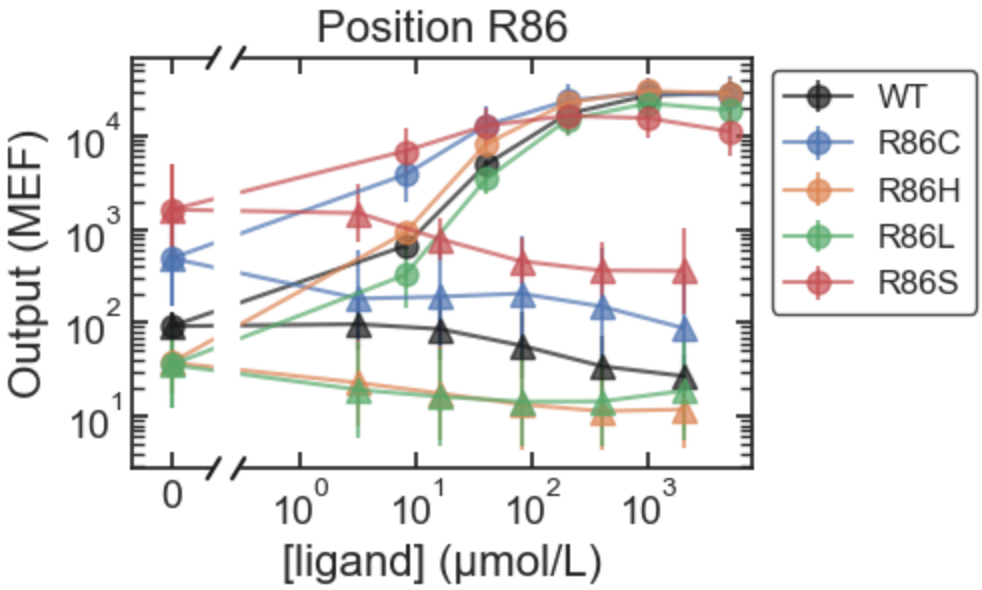

Position A81
Output (MEF)
[ligand] (μmol/L)
WT
A81T
A81V

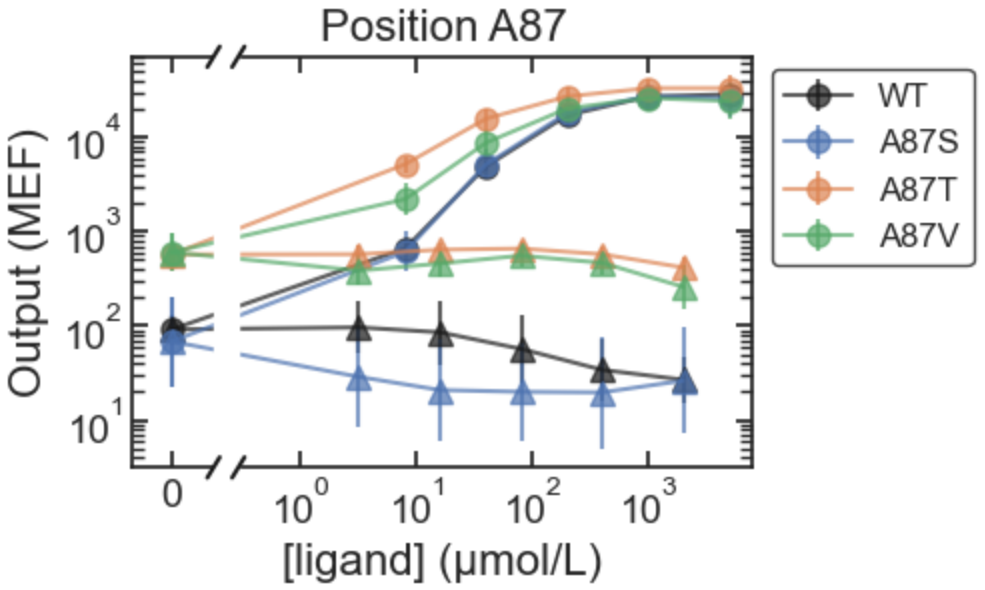

Position A82
Output (MEF)
[ligand] (μmol/L)
WT
A82S
A82T
A82V
A82G

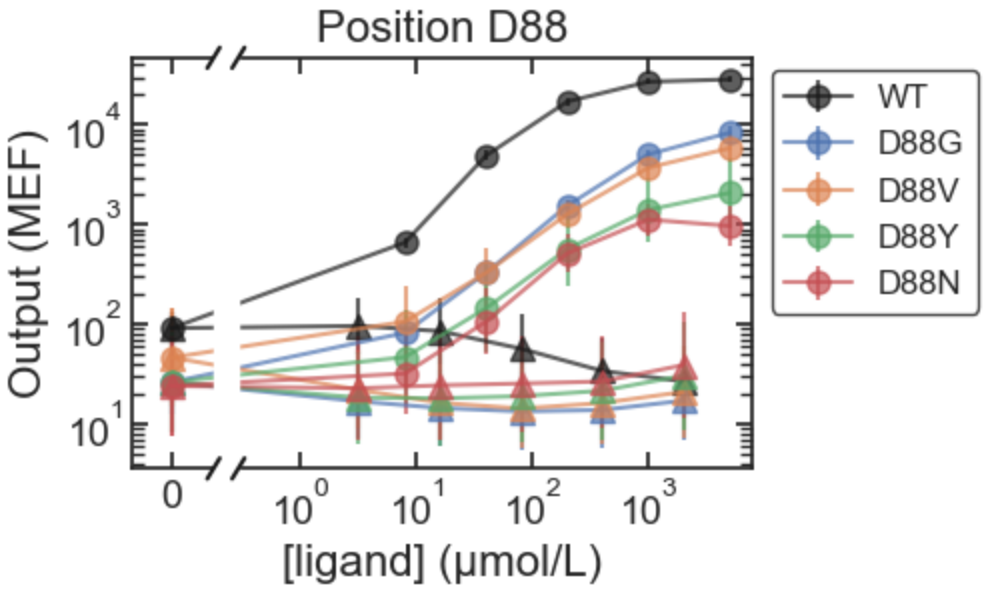

Position I83
Output (MEF)
[ligand] (μmol/L)
WT
I83N
I83S
I83T
I83V

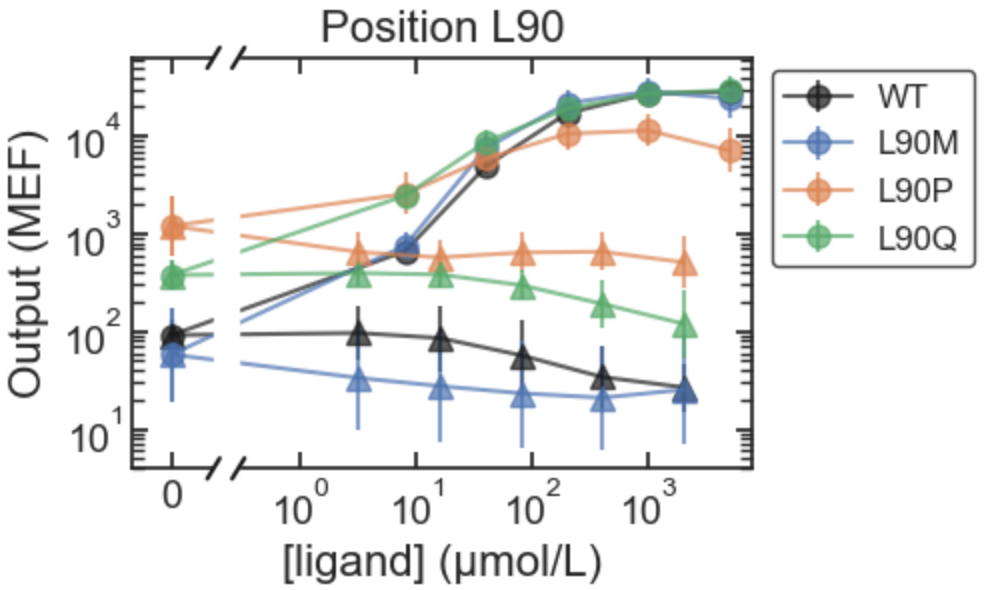

Position K84
Output (MEF)
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

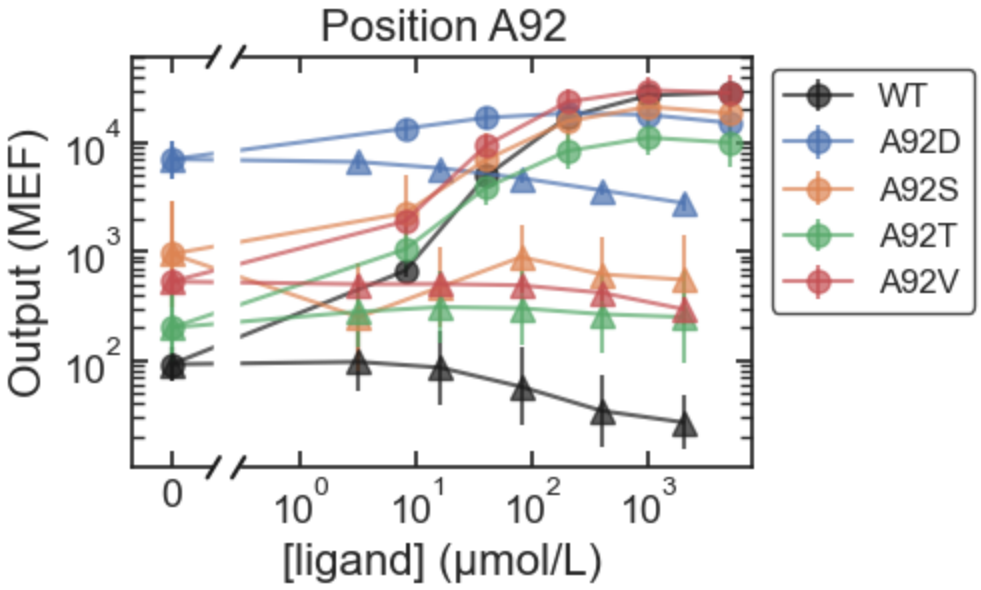

Position S85
Output (MEF)
[ligand] (μmol/L)
WT
S85F
S85T
S85Y

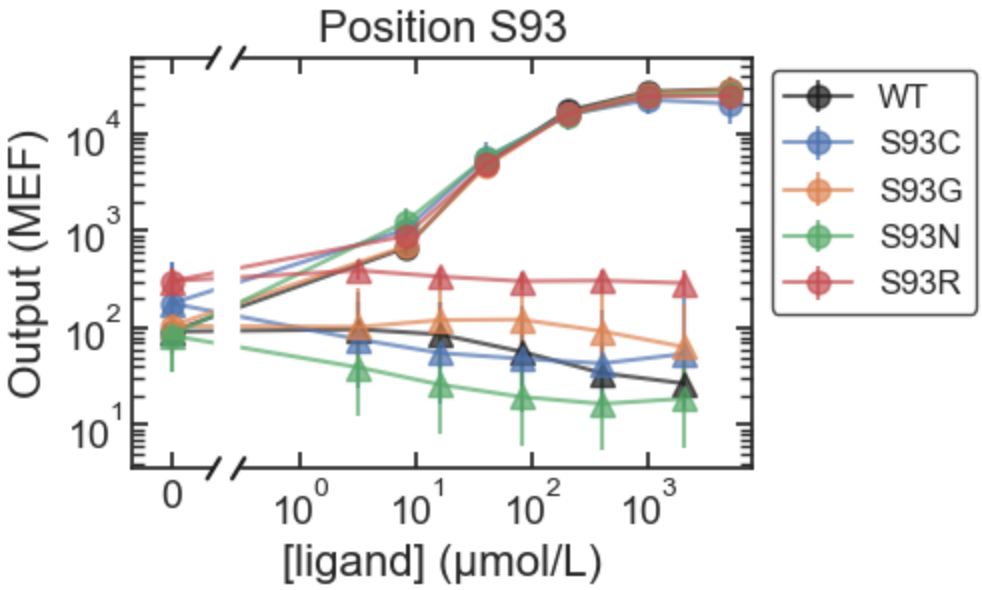

Position R86
Output (MEF)
[ligand] (μmol/L)
WT
R86C
R86H
R86L
R86S

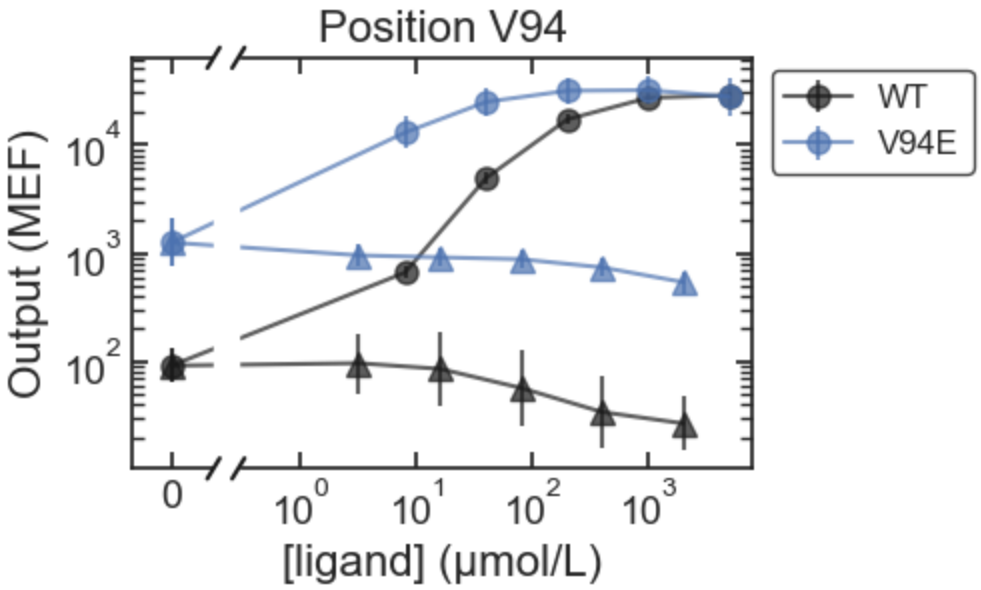

Position A87
Output (MEF)
[ligand] (μmol/L)
WT
A87S
A87T
A87V

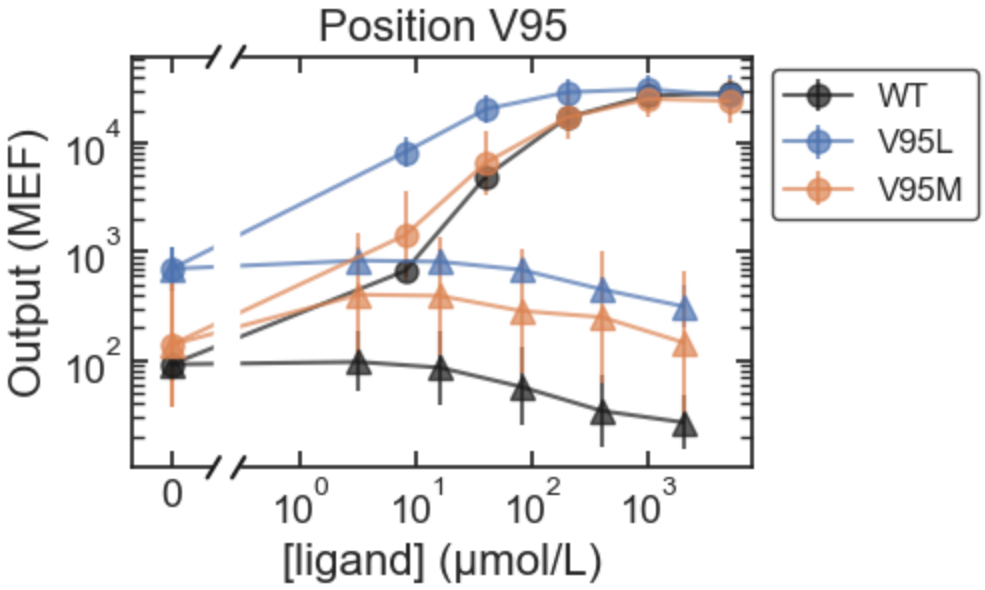

Position D88
Output (MEF)
[ligand] (μmol/L)
WT
D88G
D88V
D88Y
D88N

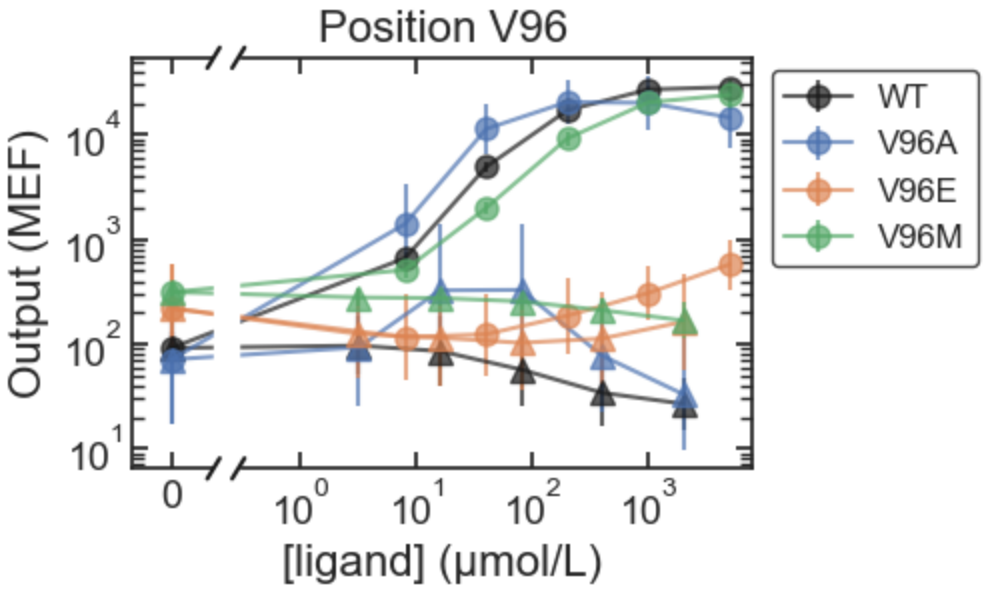

Position L90
Output (MEF)
[ligand] (μmol/L)
WT
L90M
L90P
L90Q

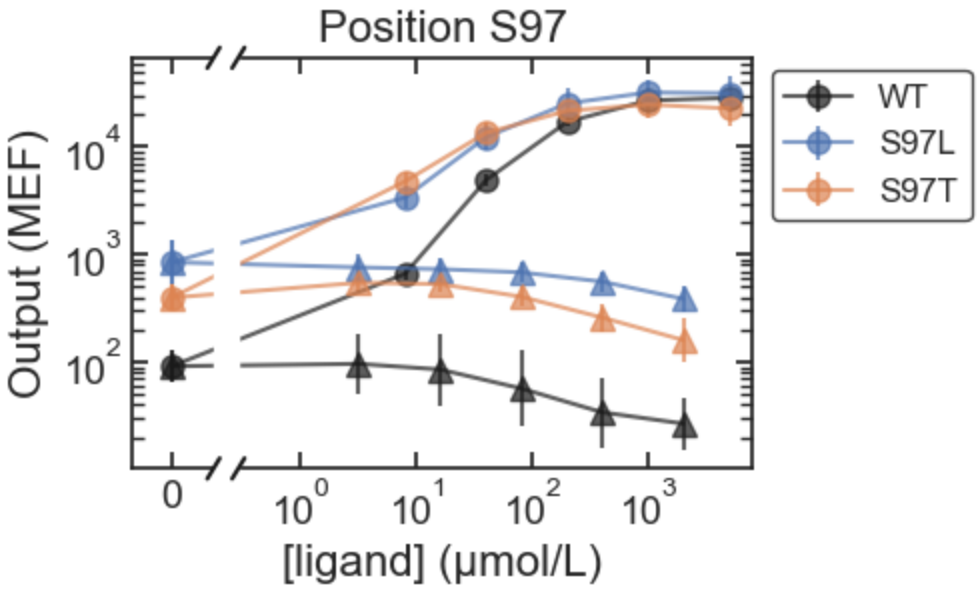

Position A92
Output (MEF)
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V

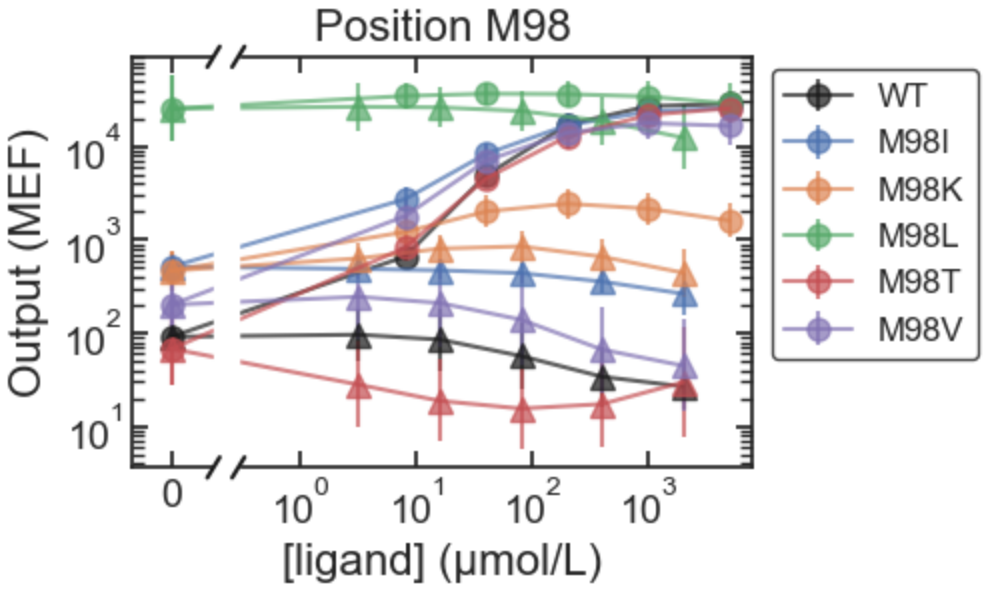

Position S93
Output (MEF)
[ligand] (μmol/L)
WT
S93C
S93G
S93N
S93R

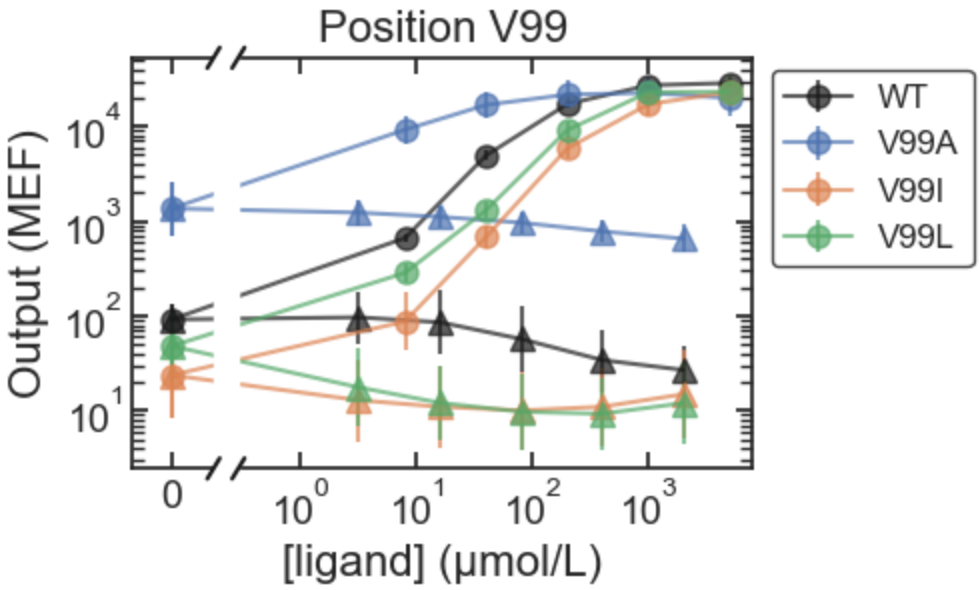

Position V94
Output (MEF)
[ligand] (μmol/L)
WT
V94E

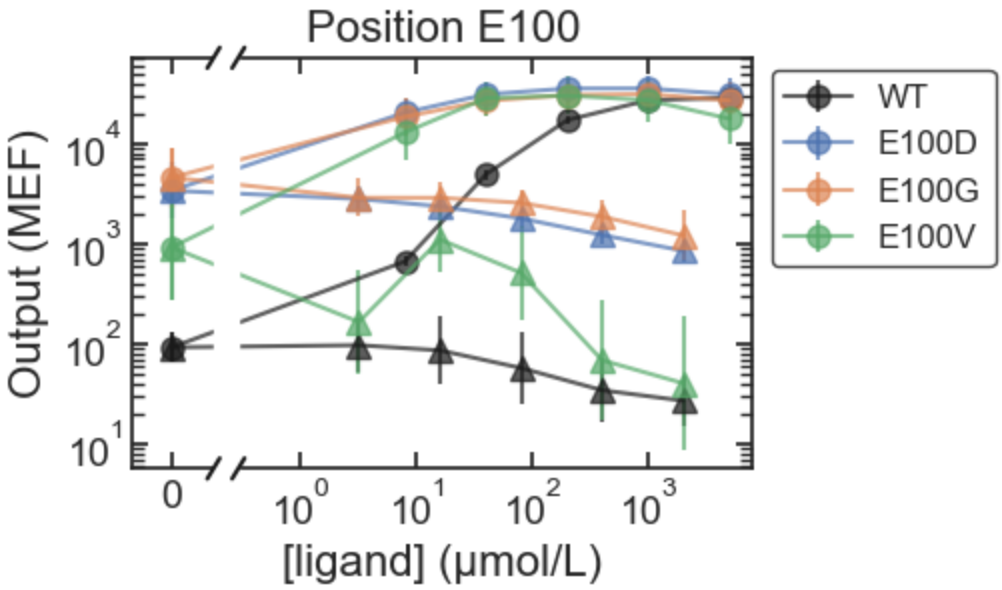

Position V95
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V95L
V95M

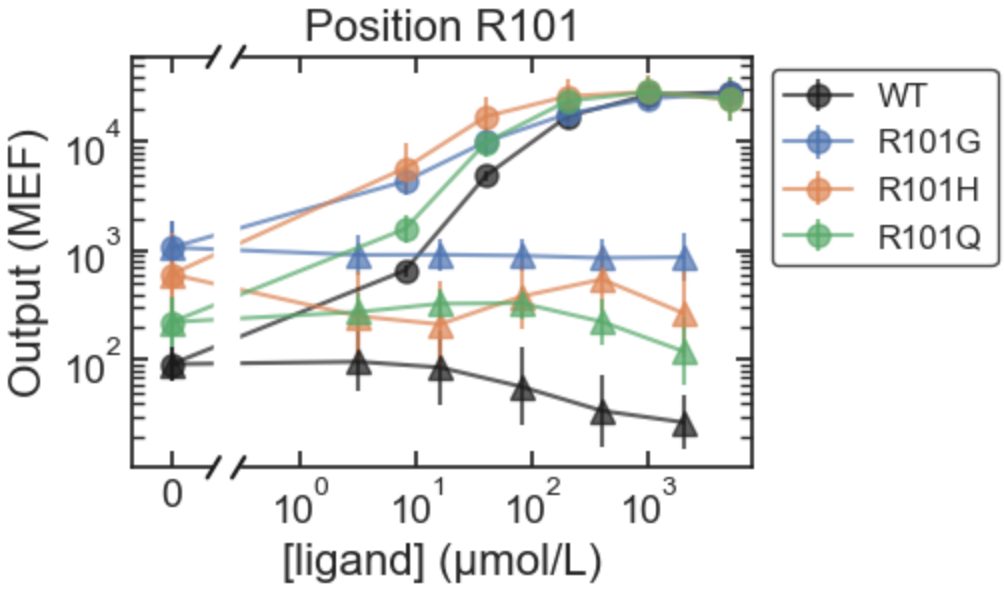

Position V96
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V96A
V96E
V96M

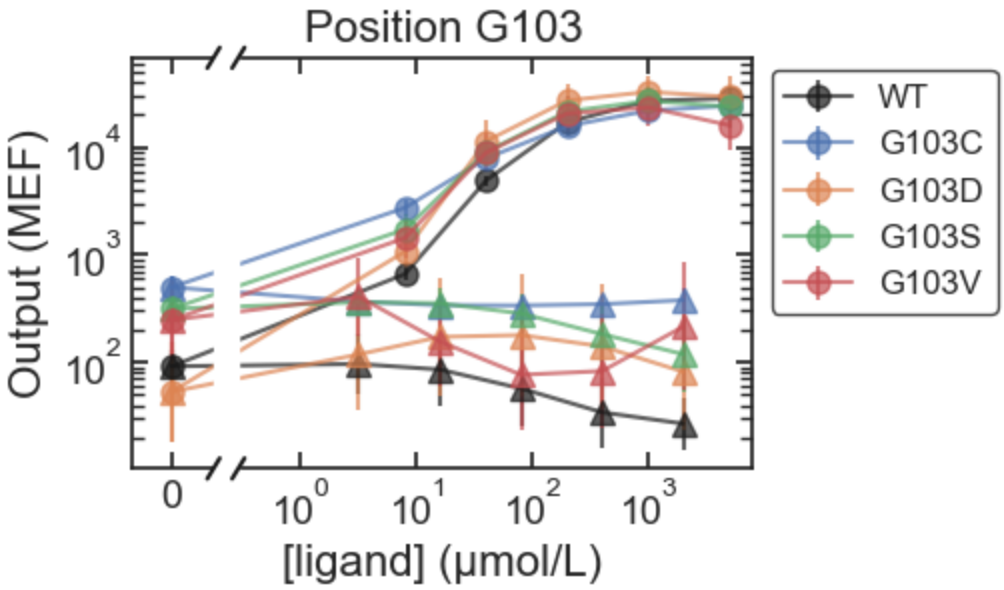

Position S97
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S97L
S97T

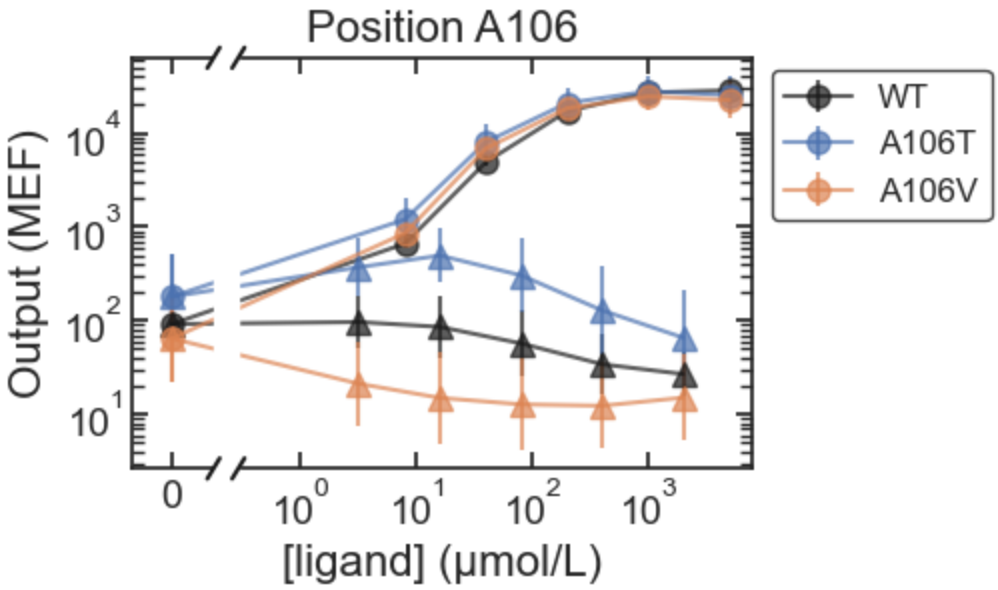

Position M98
Output (MEF)
[ligand] (μmol/L)
WT
M98I
M98K
M98L
M98T
M98V

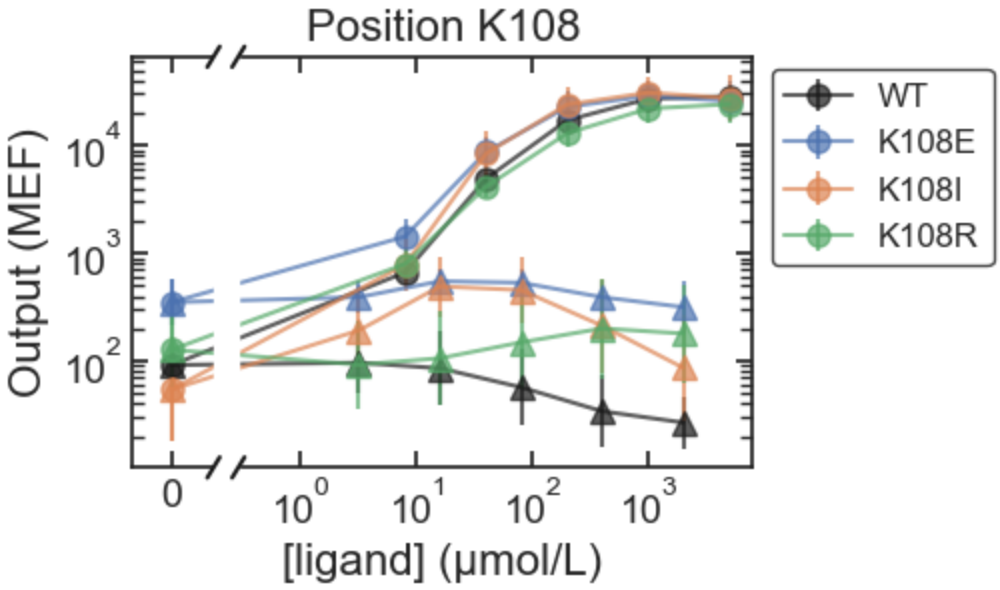

Position V99
Output (MEF)
[ligand] (μmol/L)
WT
V99A
V99I
V99L

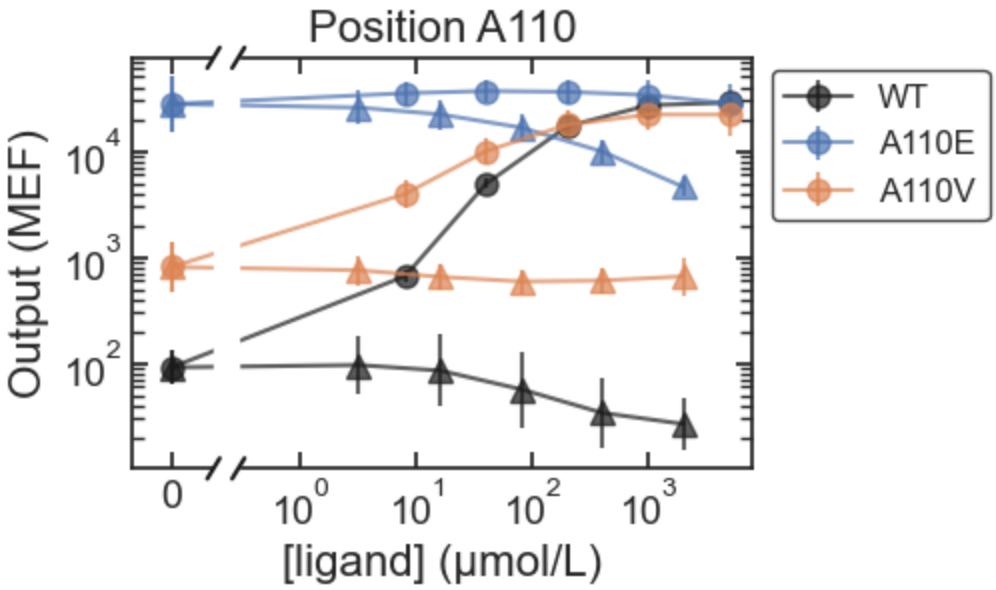

Position E100
Output (MEF)
[ligand] (μmol/L)
WT
E100D
E100G
E100V

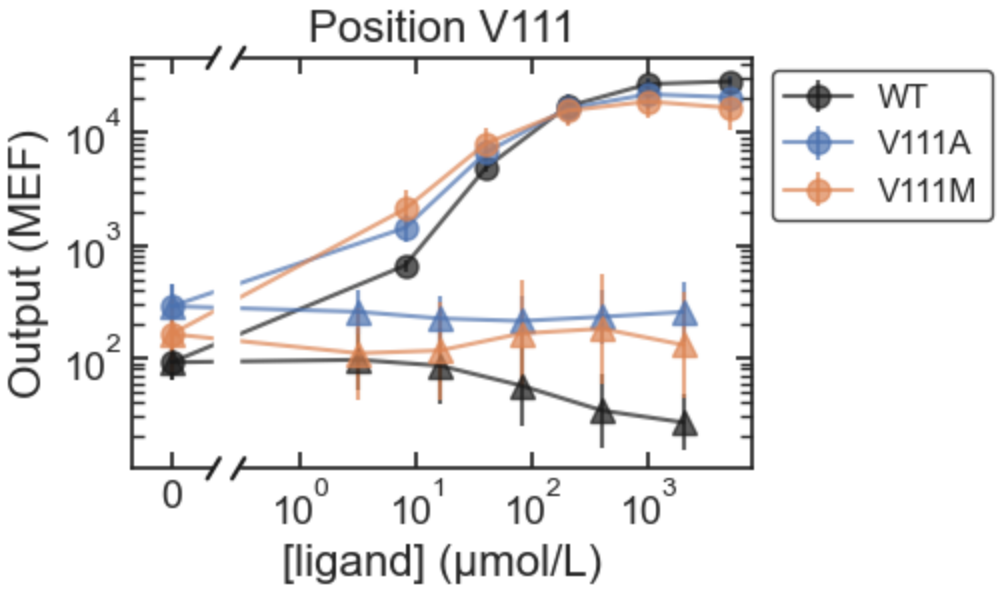

Position R101
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

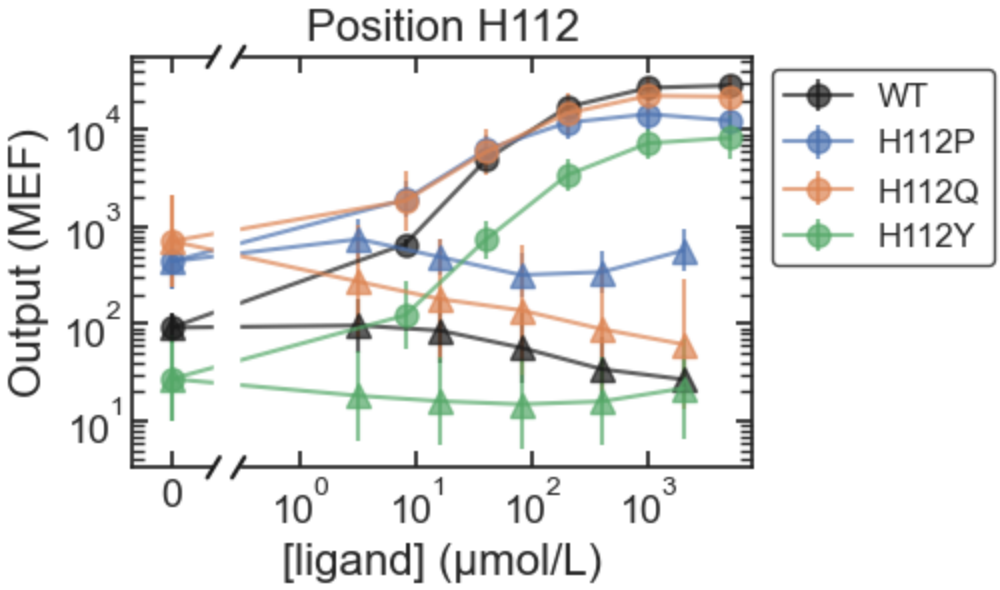

Position G103
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

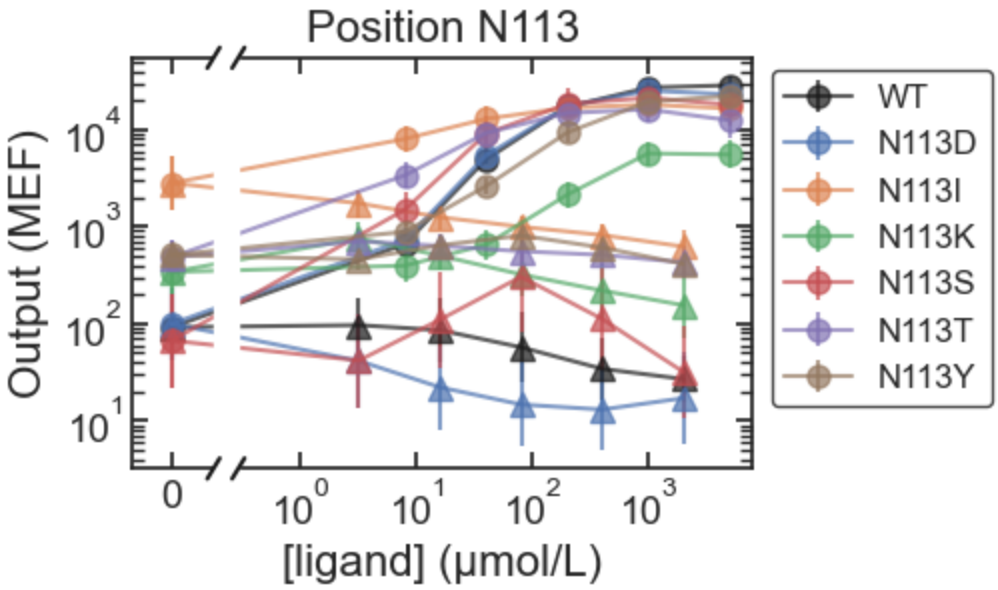

Position A106
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A106T
A106V

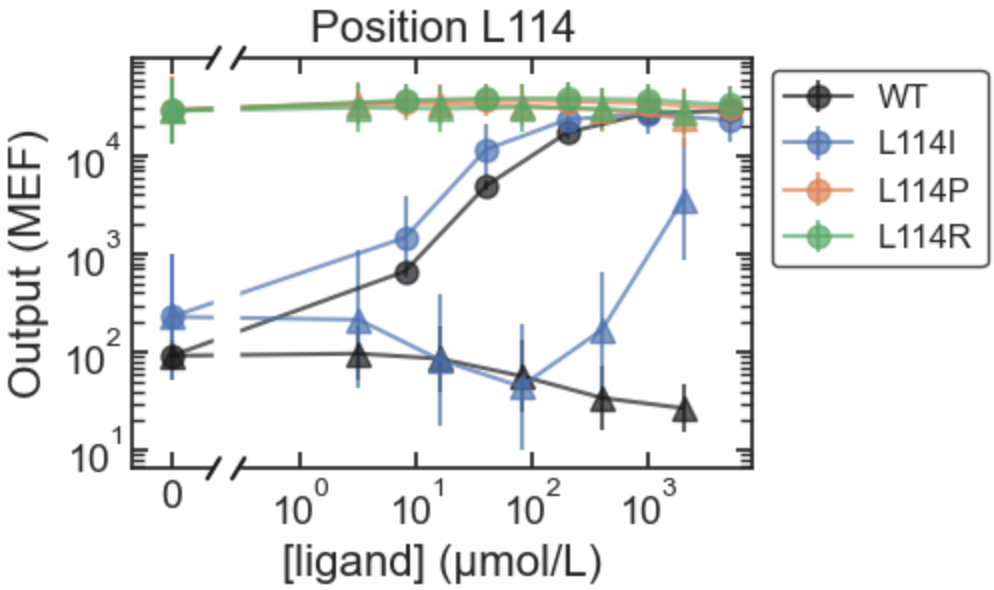

Position K108
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K108E
K108I
K108R

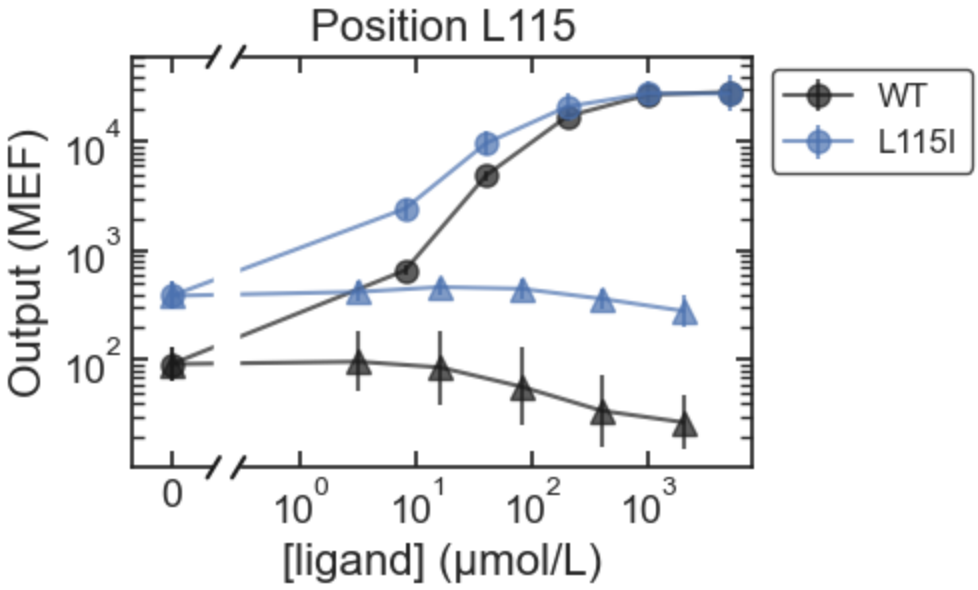

Position A110
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A110E
A110V

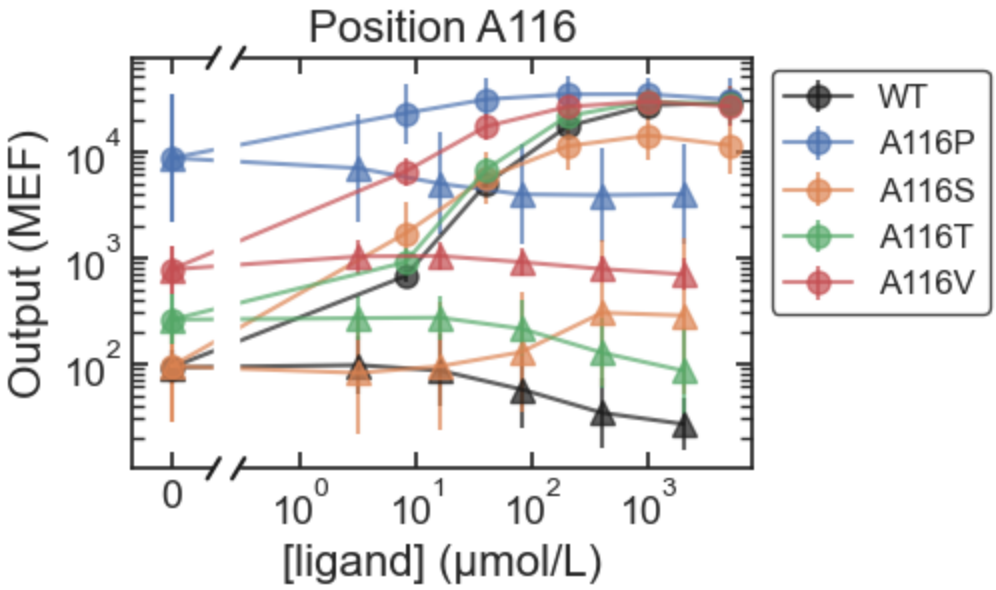

Position V111
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V111A
V111M

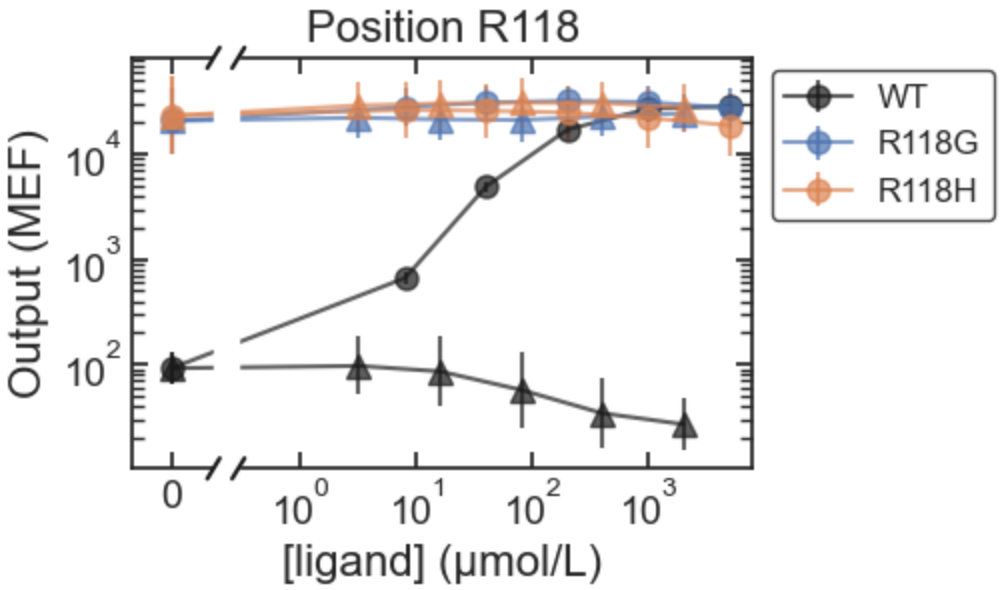

Position H112
Output (MEF)
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

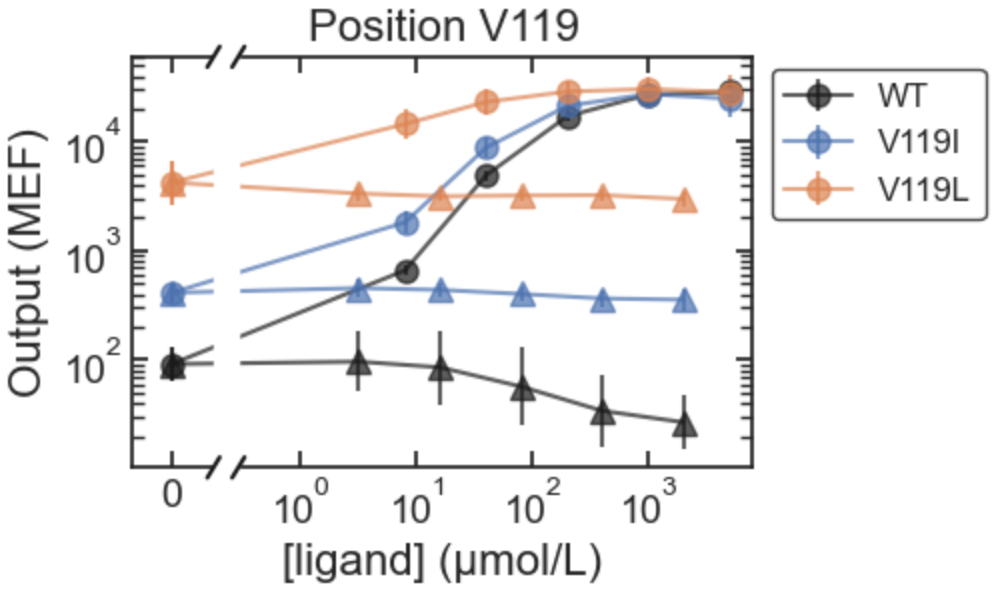

Position N113
Output (MEF)
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

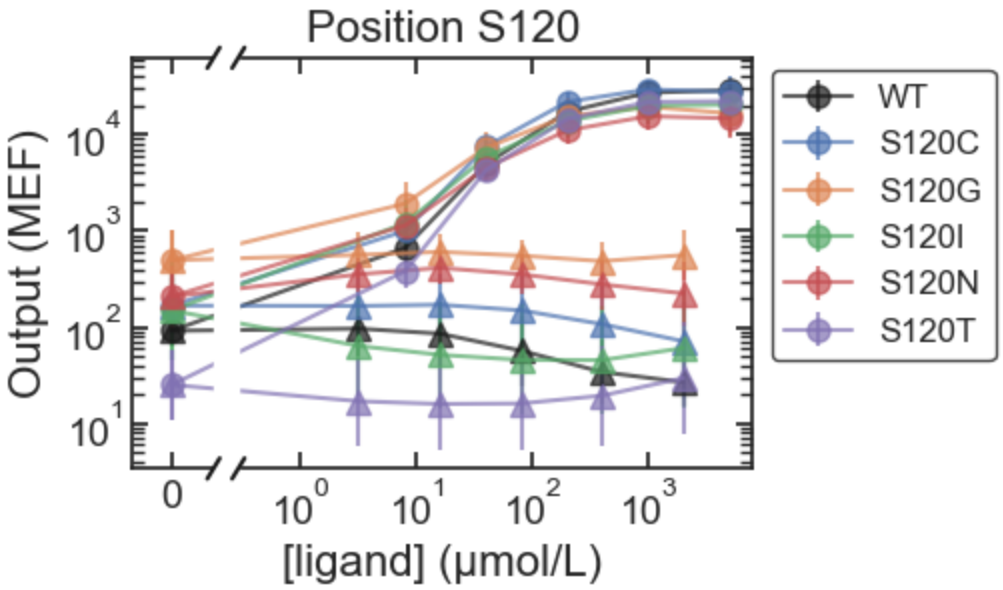

Position L114
Output (MEF)
[ligand] (μmol/L)
WT
L114I
L114P
L114R

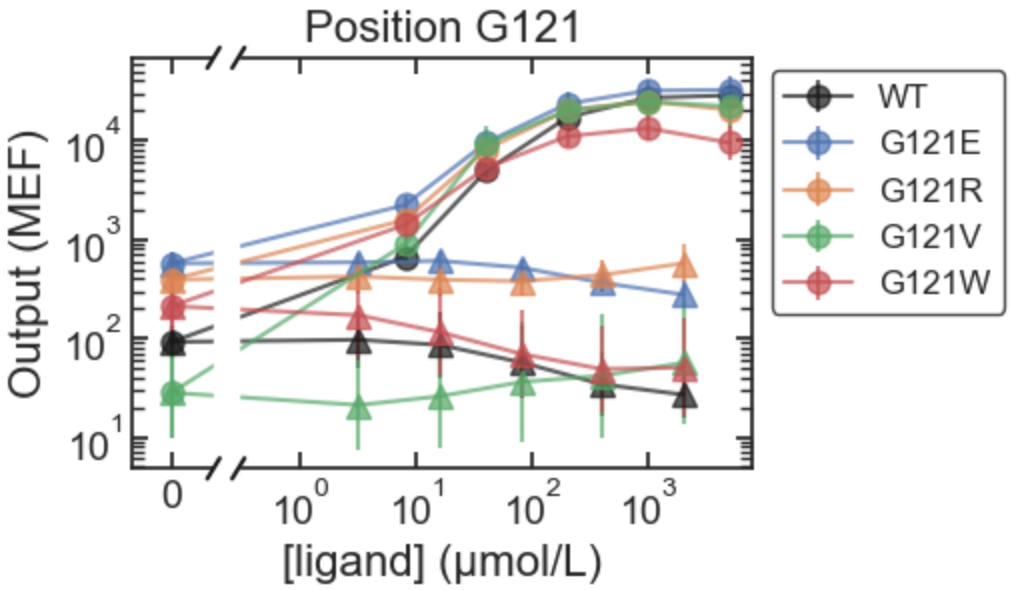

Position L115
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L115I

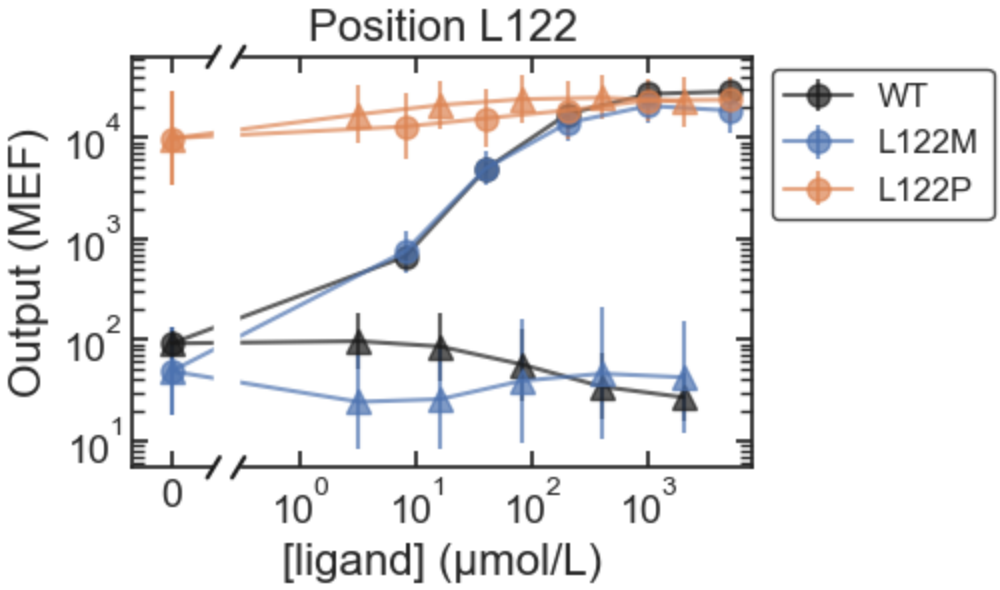

Position A116
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A116P
A116S
A116T
A116V

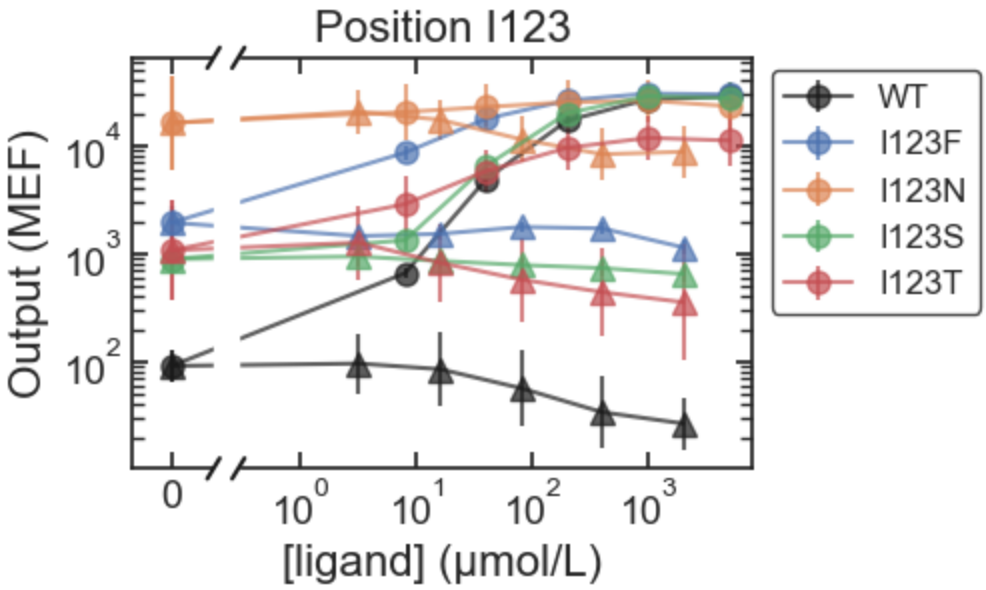

Position R118
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R118G
R118H

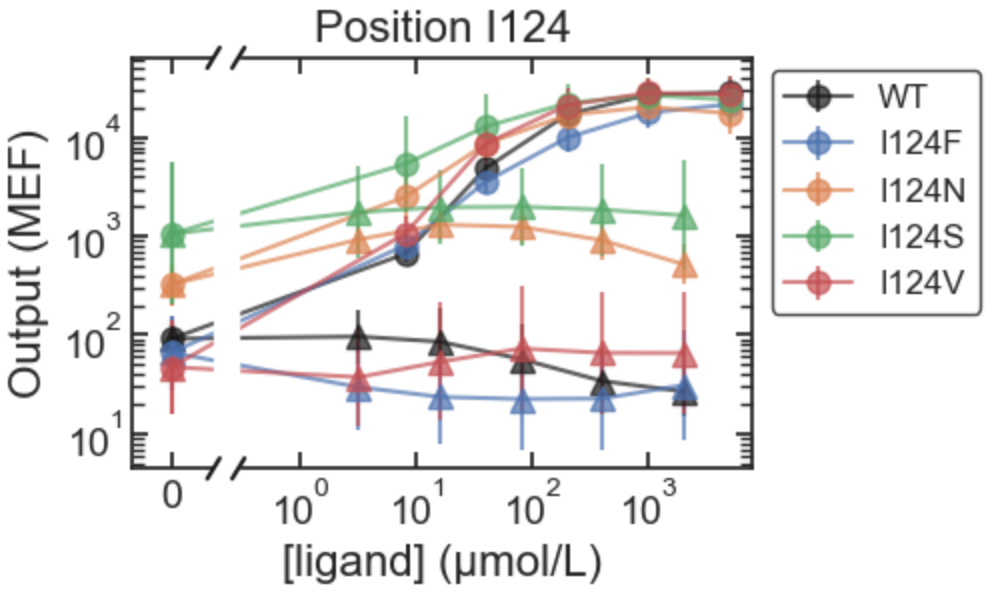

Position V119
Output (MEF)
[ligand] (μmol/L)
WT
V119I
V119L

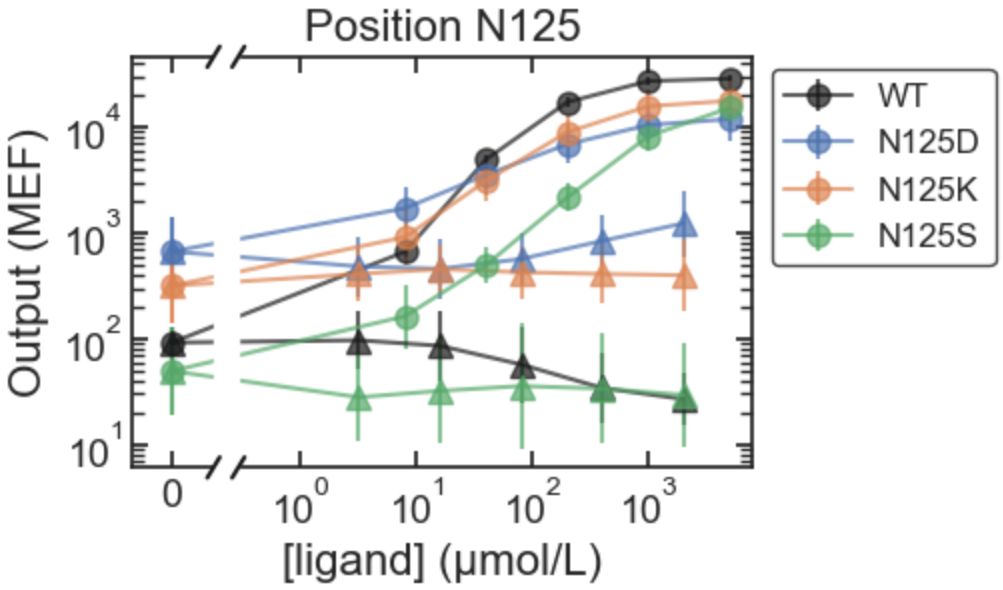

Position S120
Output (MEF)
[ligand] (μmol/L)
WT
S120C
S120G
S120I
S120N
S120T

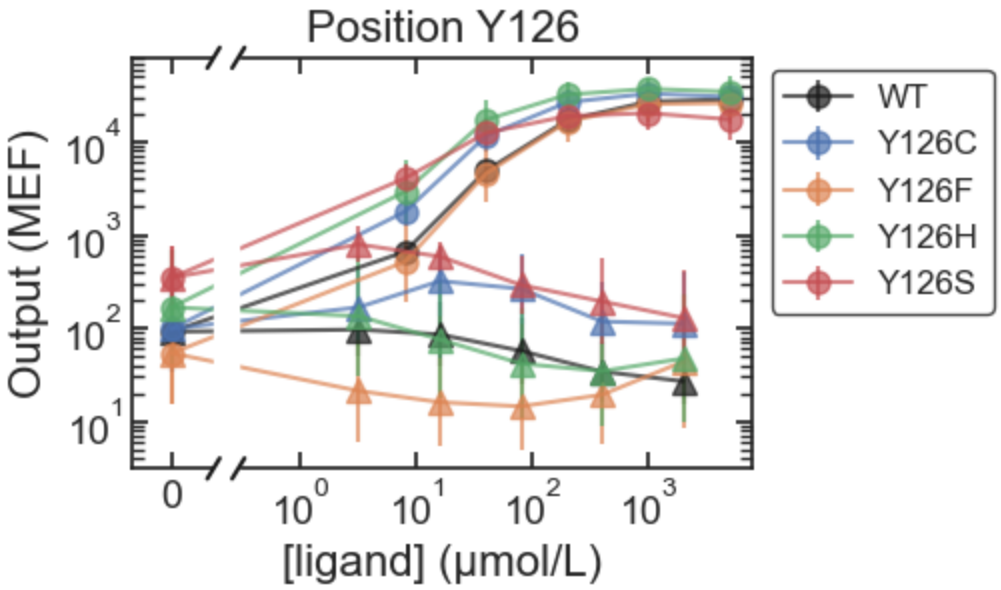

Position G121
Output (MEF)
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

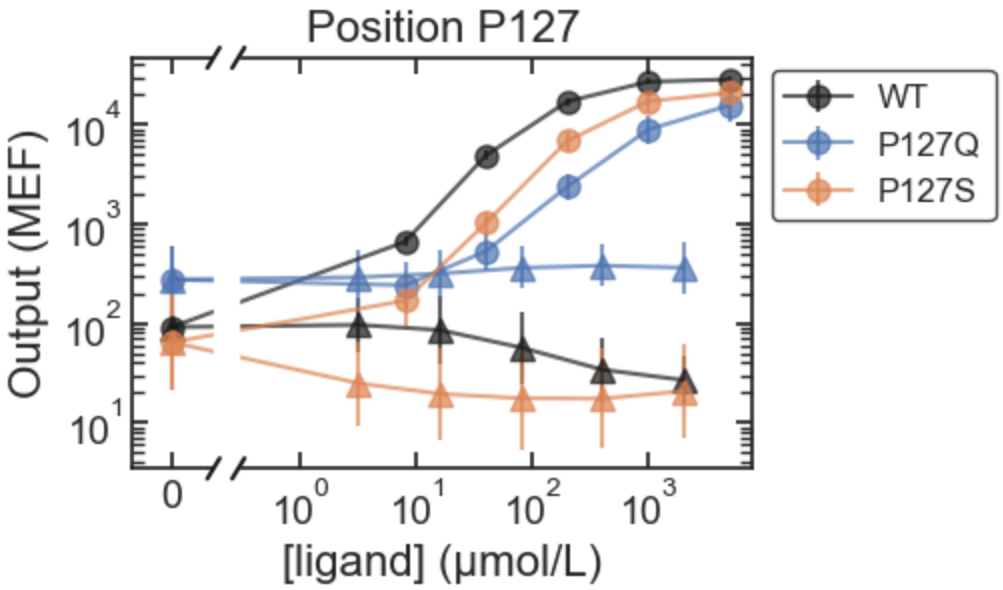

Position L122
Output (MEF)
[ligand] (μmol/L)
WT
L122M
L122P

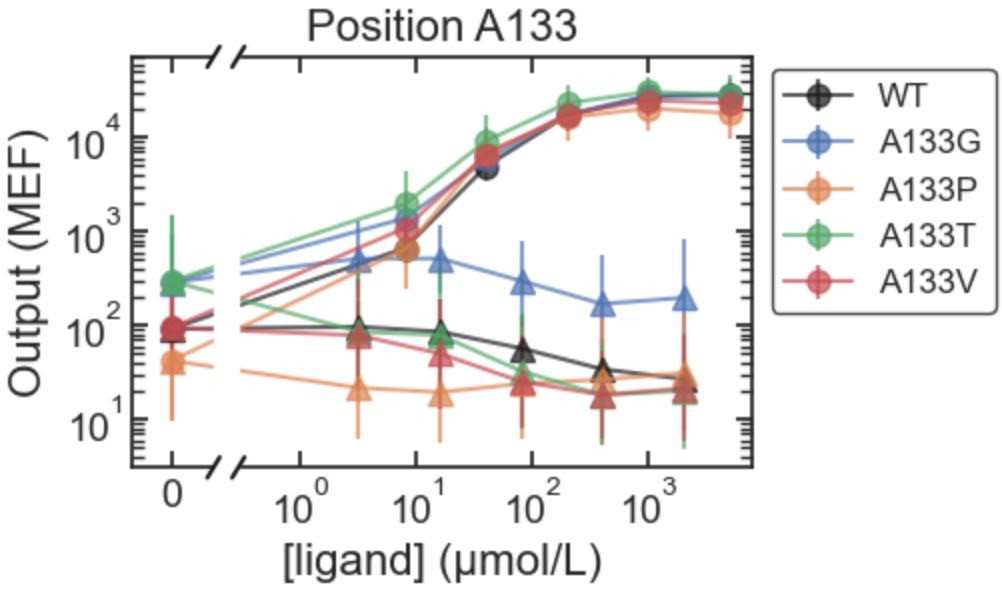

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

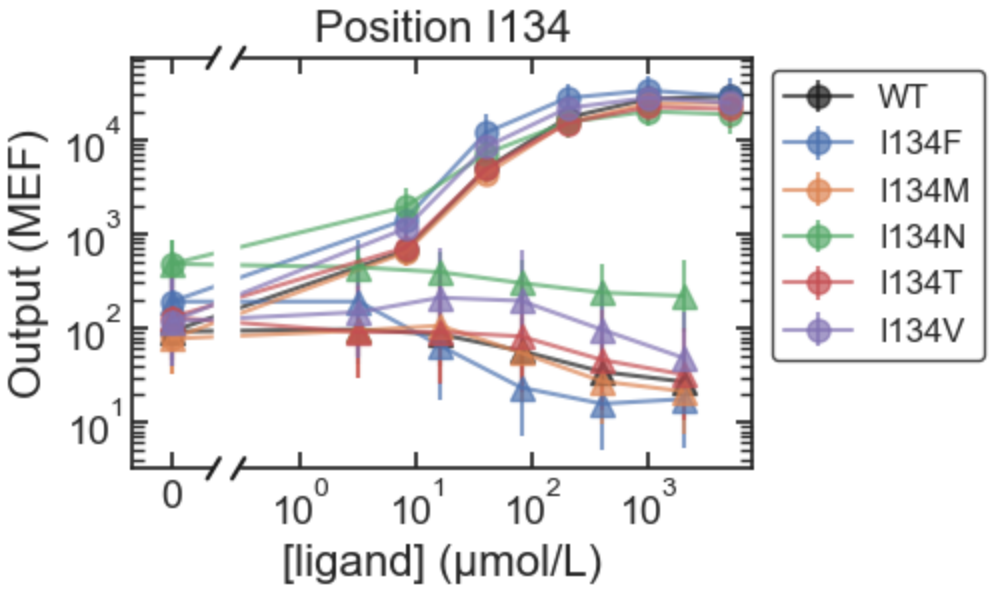

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V

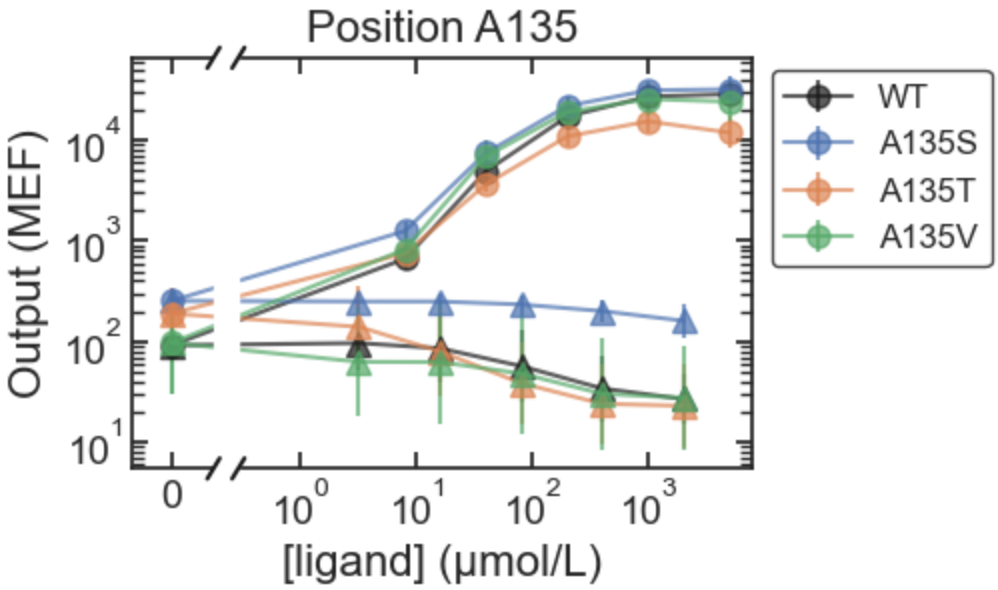

Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S

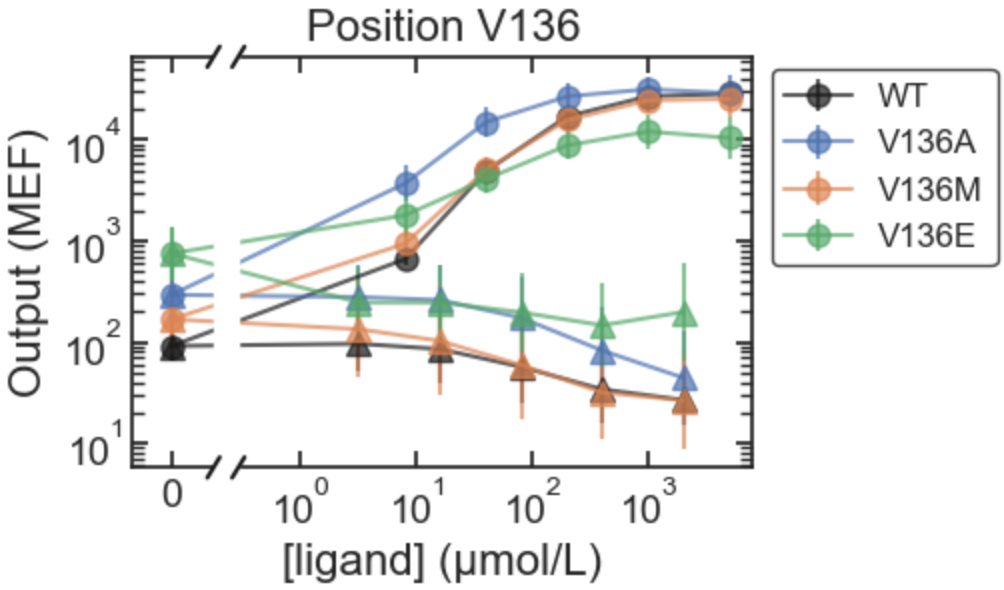

Position Y126
Output (MEF)
[ligand] (μmol/L)
WT
Y126C
Y126F
Y126H
Y126S

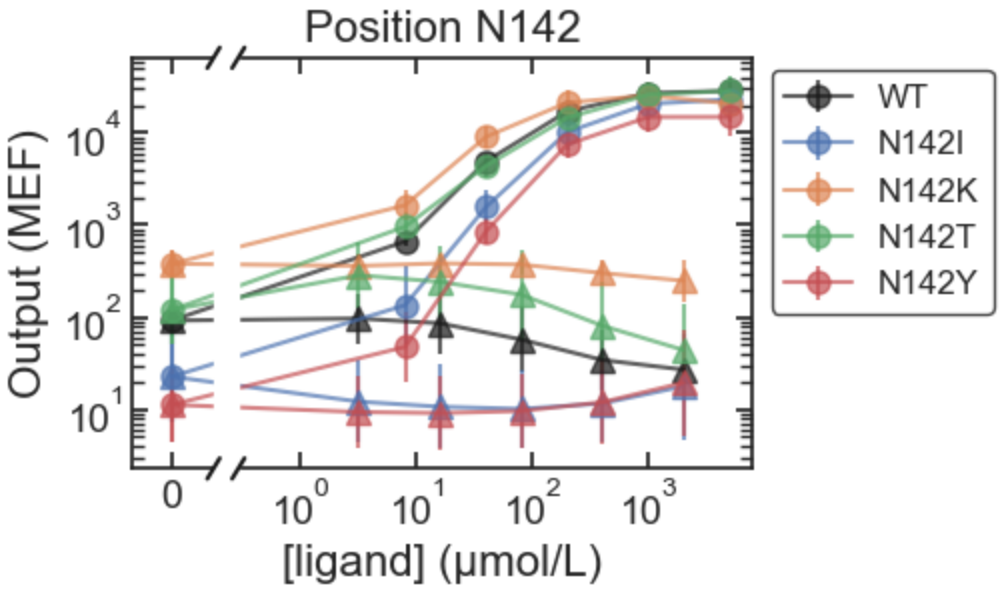

Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

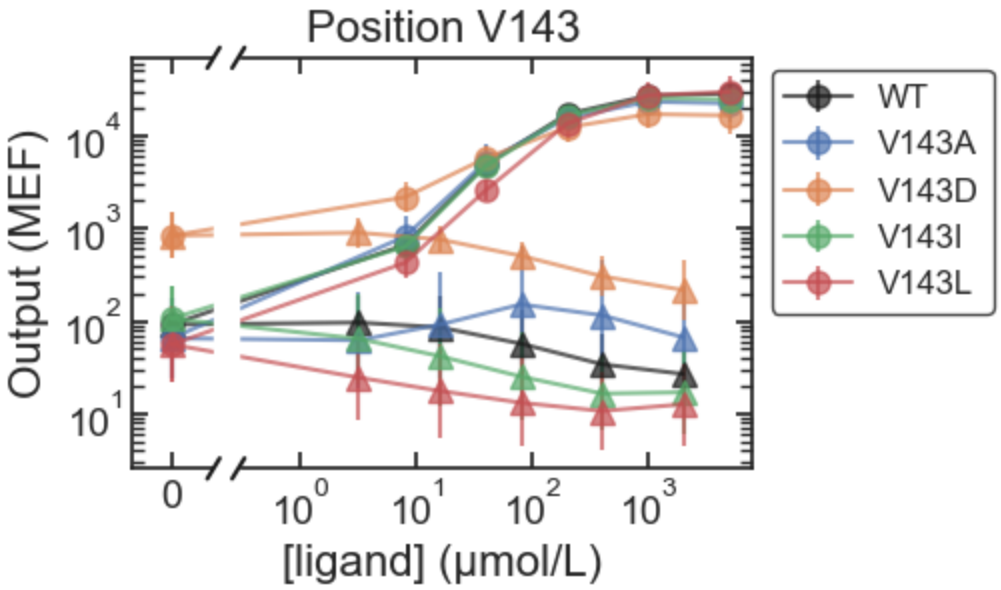

Position A133
Output (MEF)
[ligand] (μmol/L)
WT
A133G
A133P
A133T
A133V

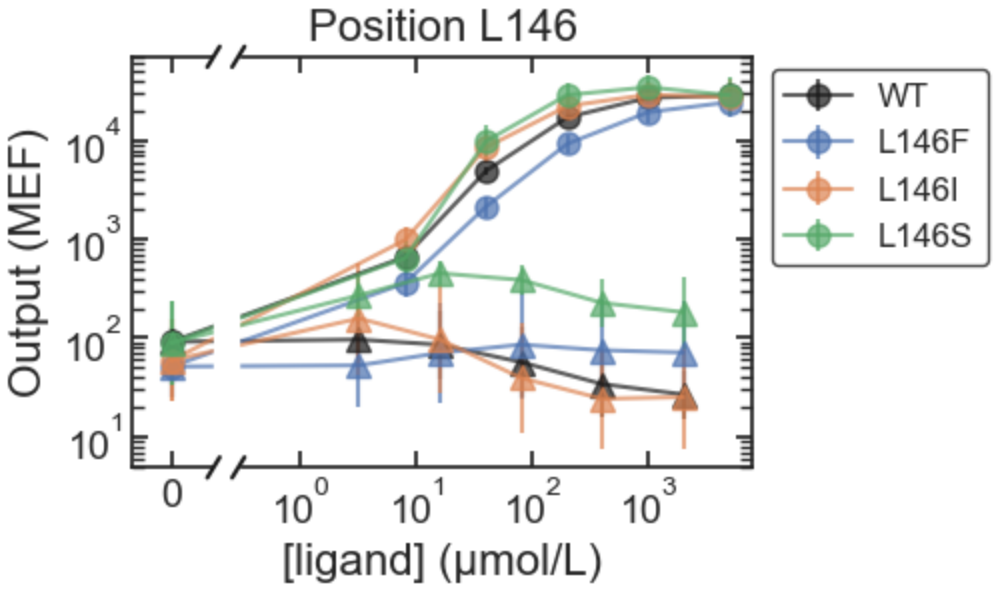

Position I134
Output (MEF)
[ligand] (μmol/L)
WT
I134F
I134M
I134N
I134T
I134V

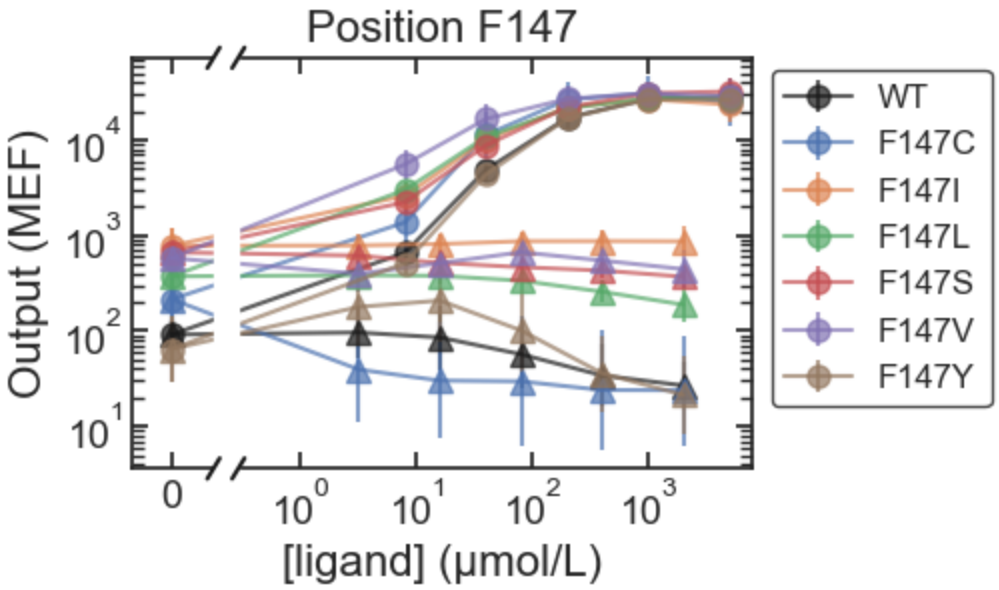

Position A135
Output (MEF)
[ligand] (μmol/L)
WT
A135S
A135T
A135V

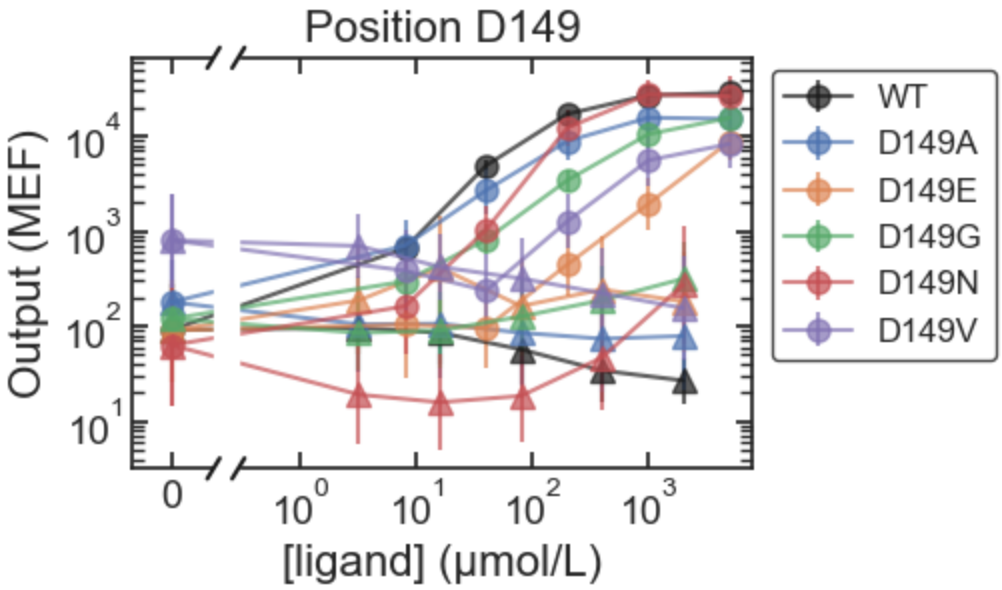

Position V136
Output (MEF)
[ligand] (μmol/L)
WT
V136A
V136M
V136E

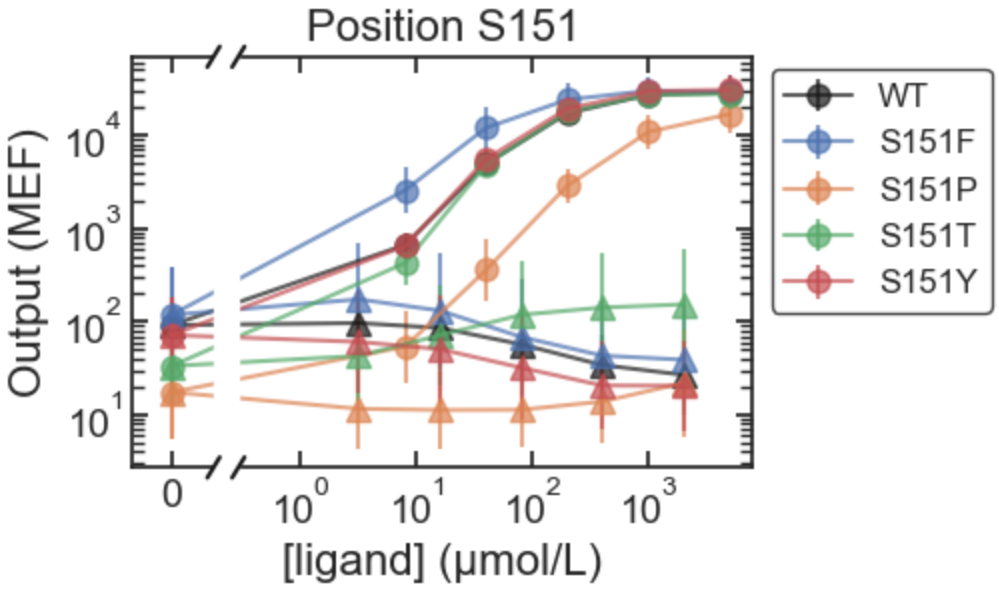

Position N142
Output (MEF)
[ligand] (μmol/L)
WT
N142I
N142K
N142T
N142Y

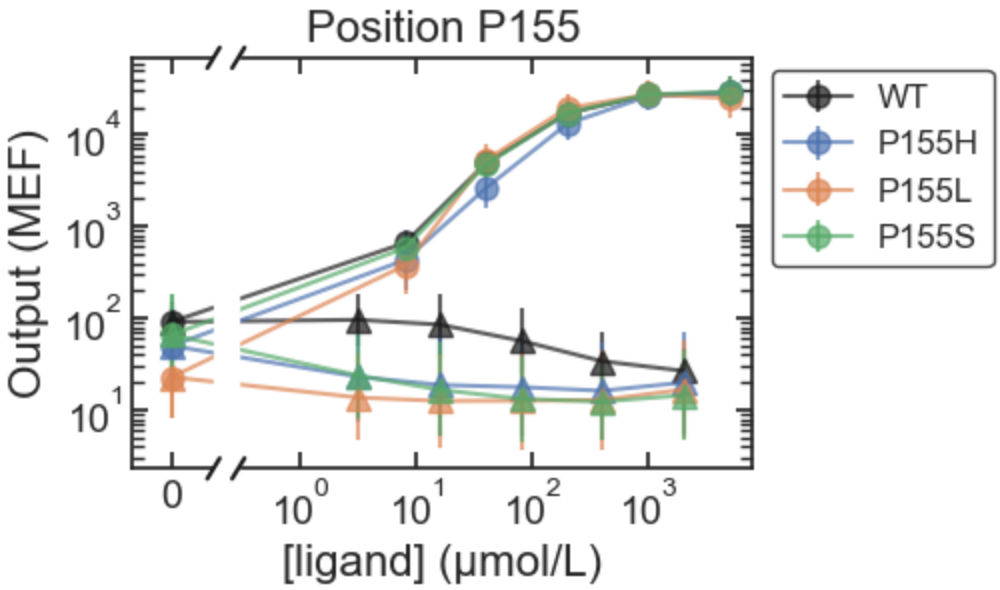

Position V143
Output (MEF)
[ligand] (μmol/L)
WT
V143A
V143D
V143I
V143L

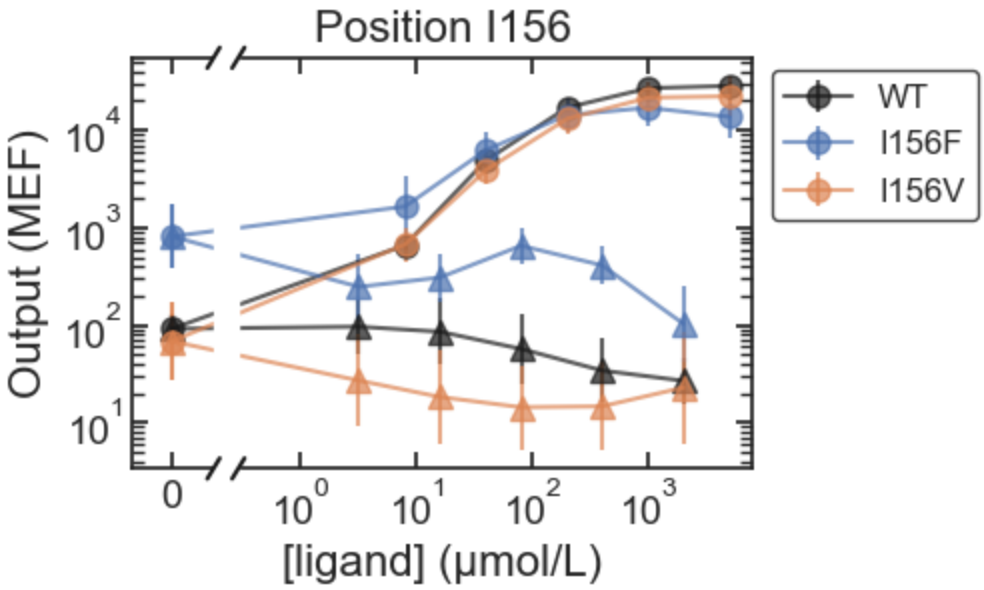

Position L146
Output (MEF)
[ligand] (μmol/L)
WT
L146F
L146I
L146S

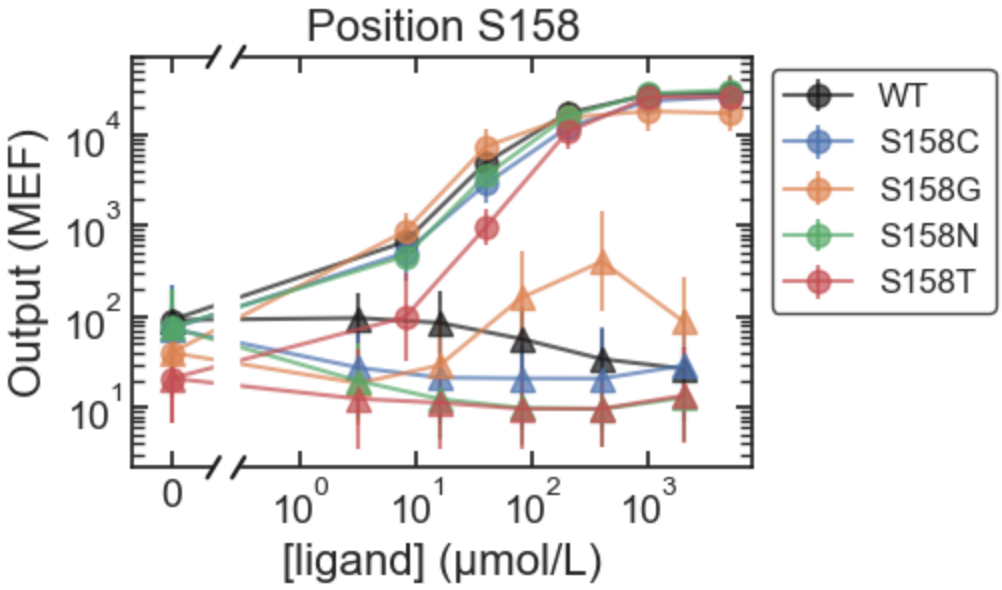

Position F147
Output (MEF)
[ligand] (μmol/L)
WT
F147C
F147I
F147L
F147S
F147V
F147Y

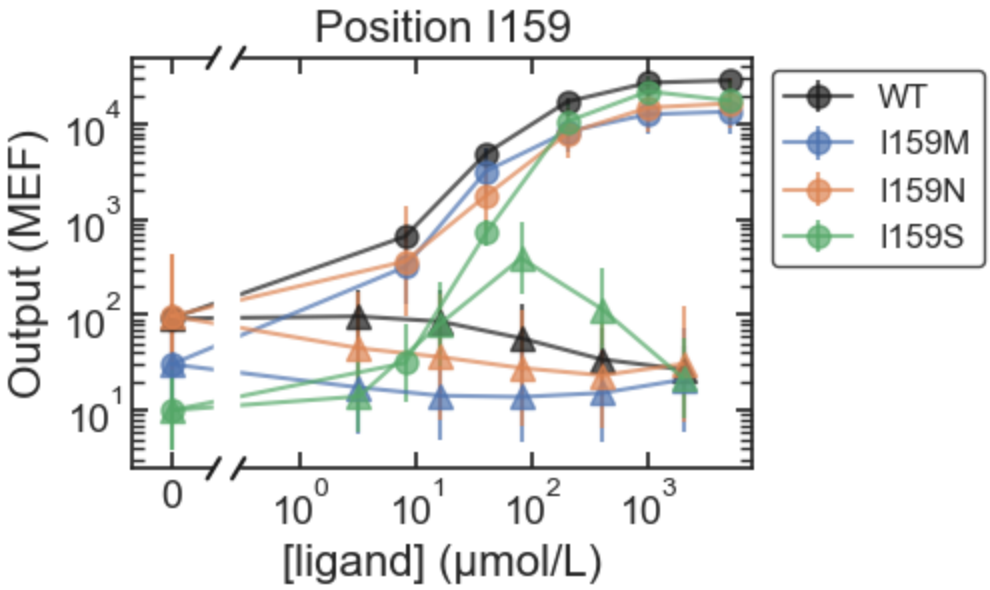

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

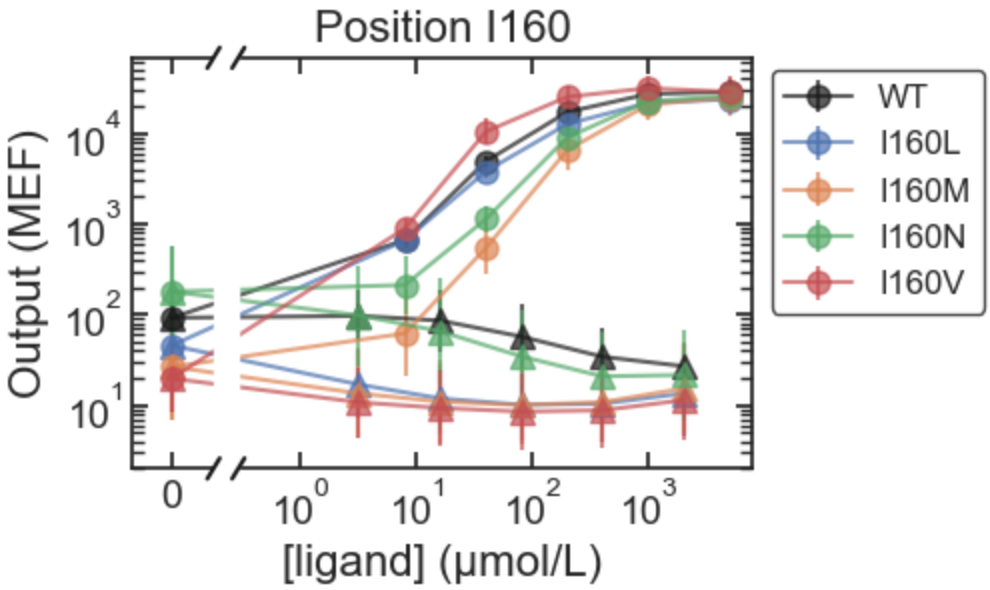

Position S151
Output (MEF)
[ligand] (μmol/L)
WT
S151F
S151P
S151T
S151Y

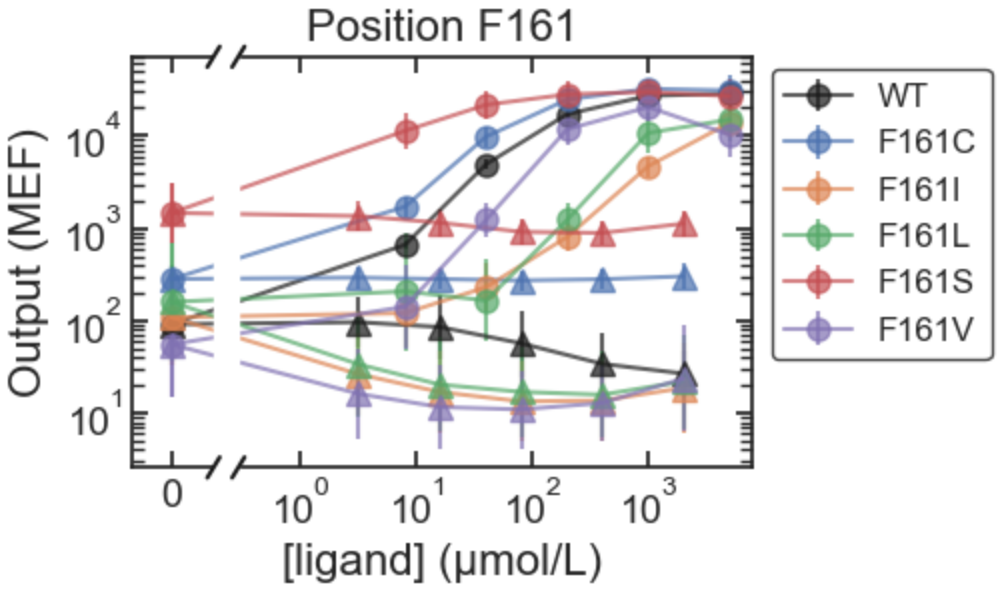

Position P155
Output (MEF)
[ligand] (μmol/L)
WT
P155H
P155L
P155S

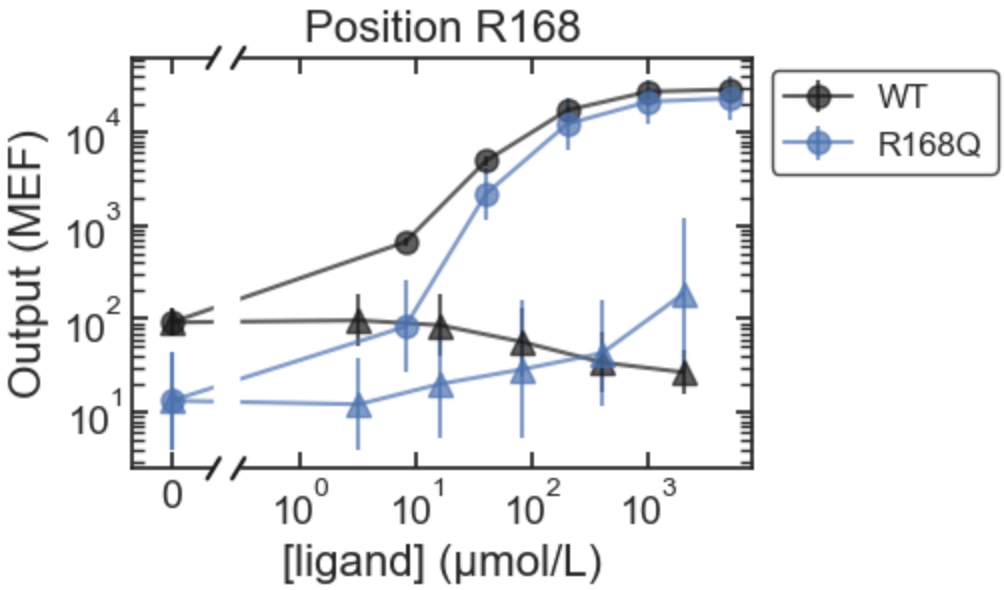

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

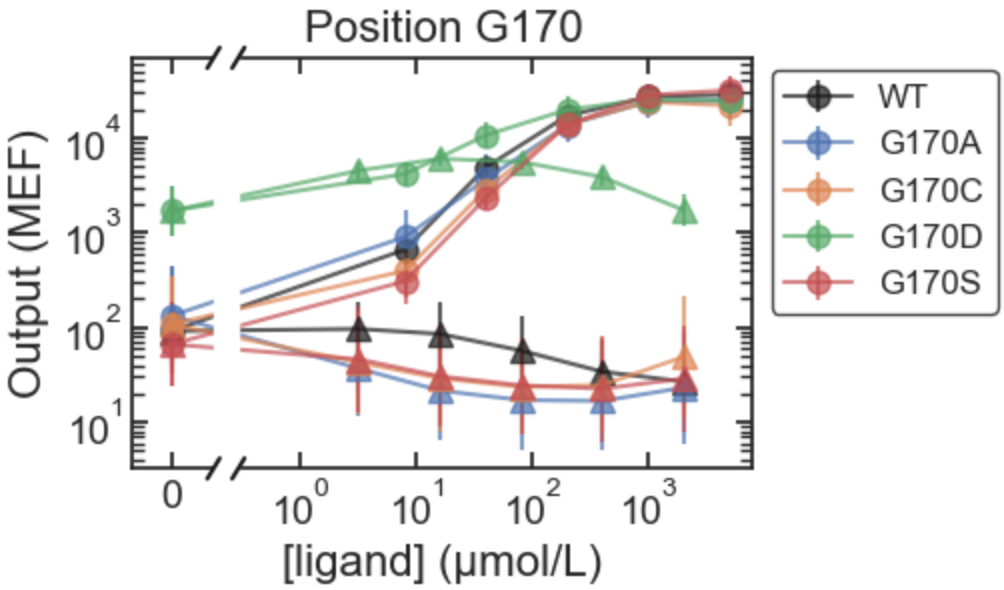

Position S158
Output (MEF)
[ligand] (μmol/L)
WT
S158C
S158G
S158N
S158T

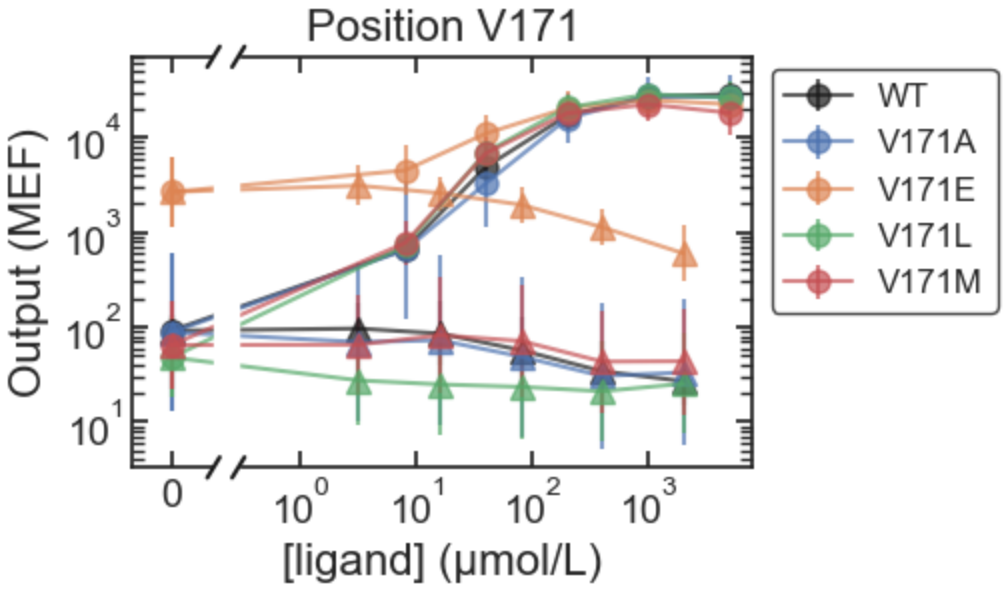

Position I159
Output (MEF)
[ligand] (μmol/L)
WT
I159M
I159N
I159S

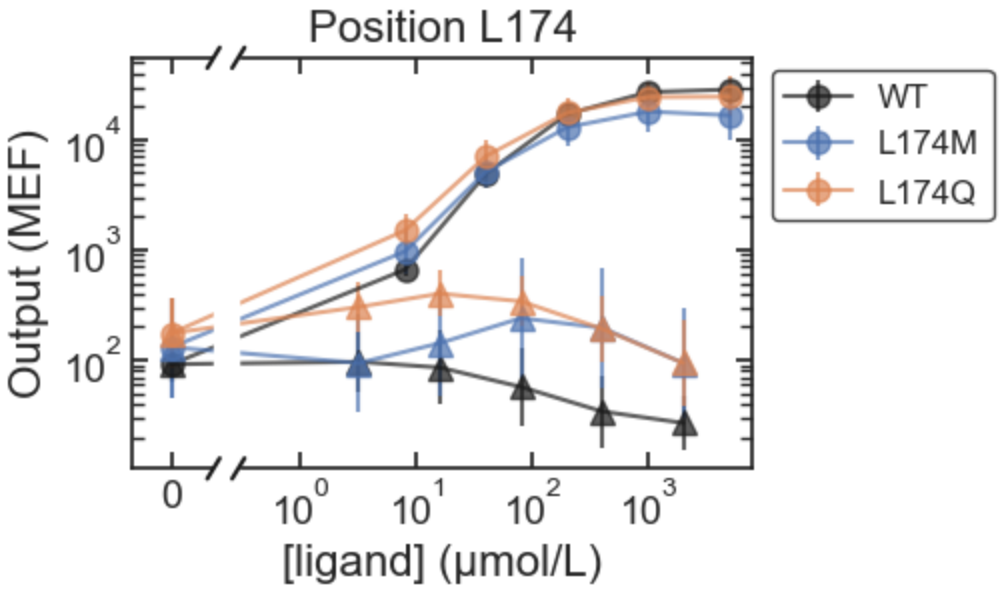

Position I160
Output (MEF)
[ligand] (μmol/L)
WT
I160L
I160M
I160N
I160V

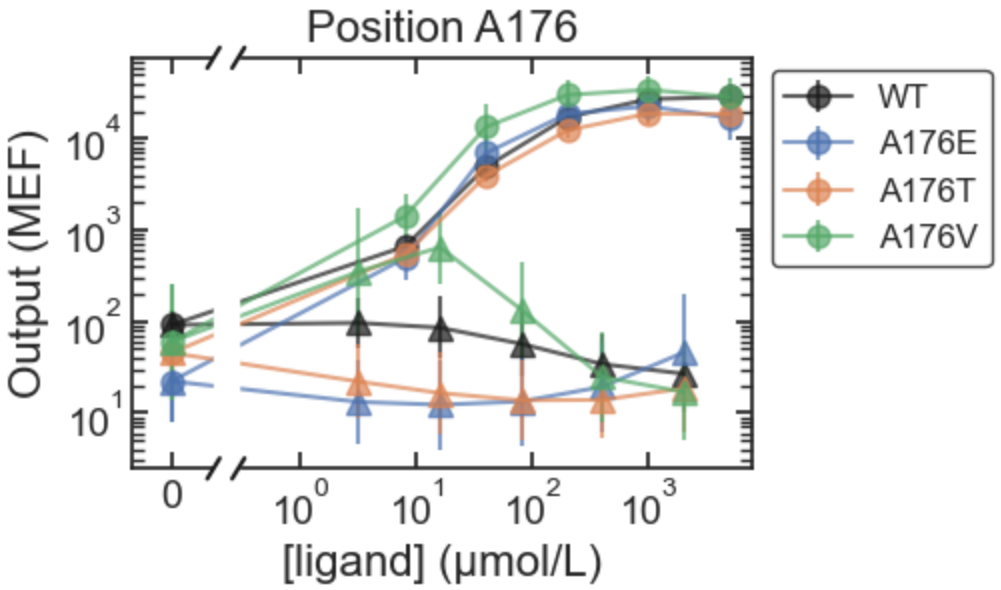

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

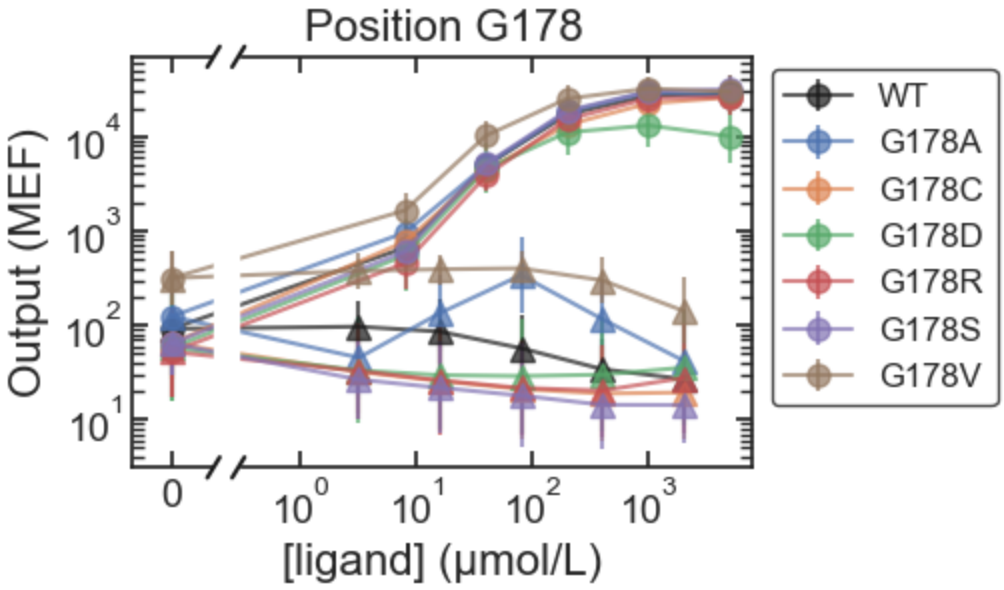

Position R168
Output (MEF)
[ligand] (μmol/L)
WT
R168Q

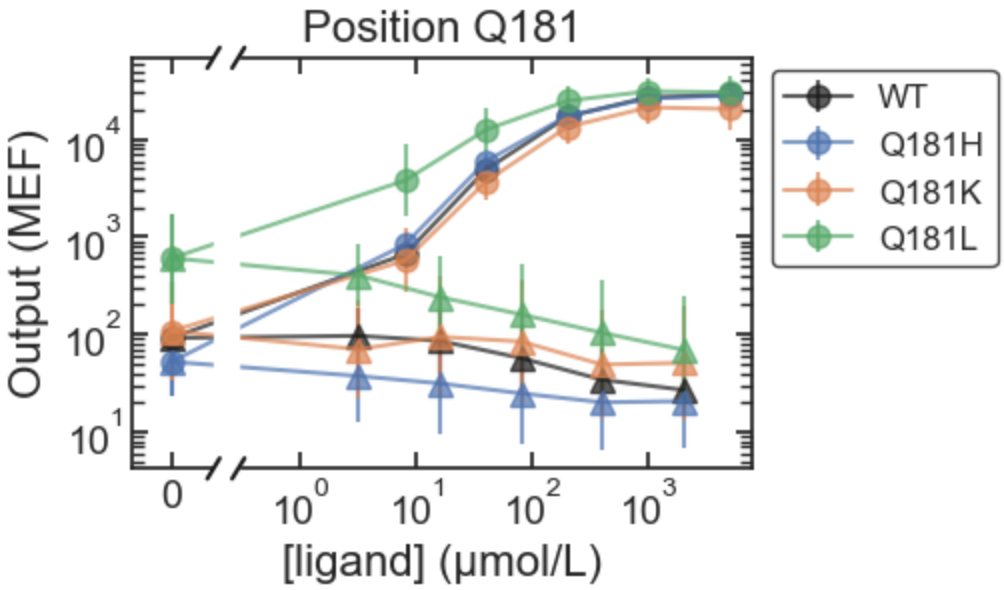

Position G170
Output (MEF)
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

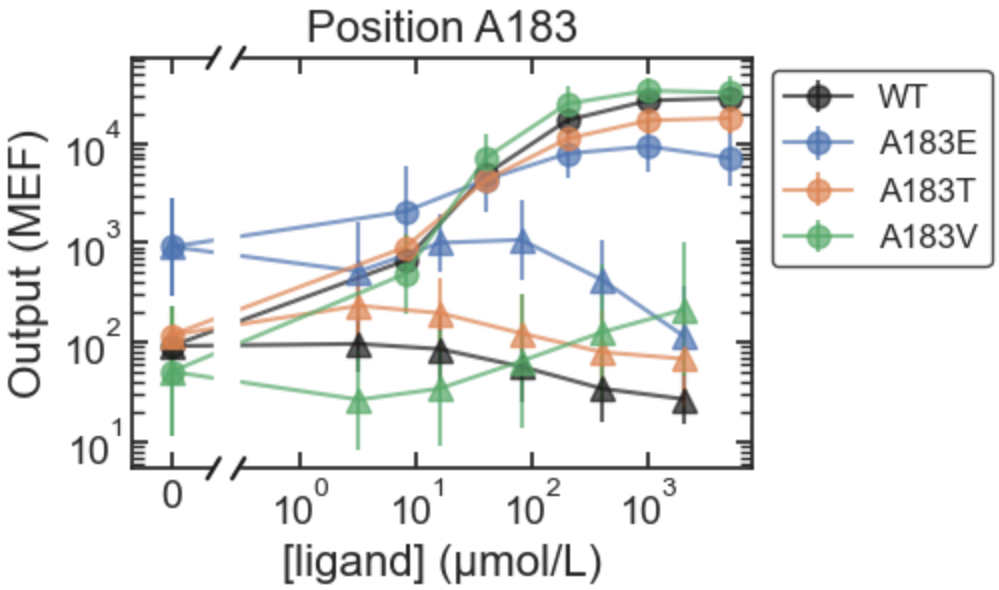

Position V171
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V171A
V171E
V171L
V171M

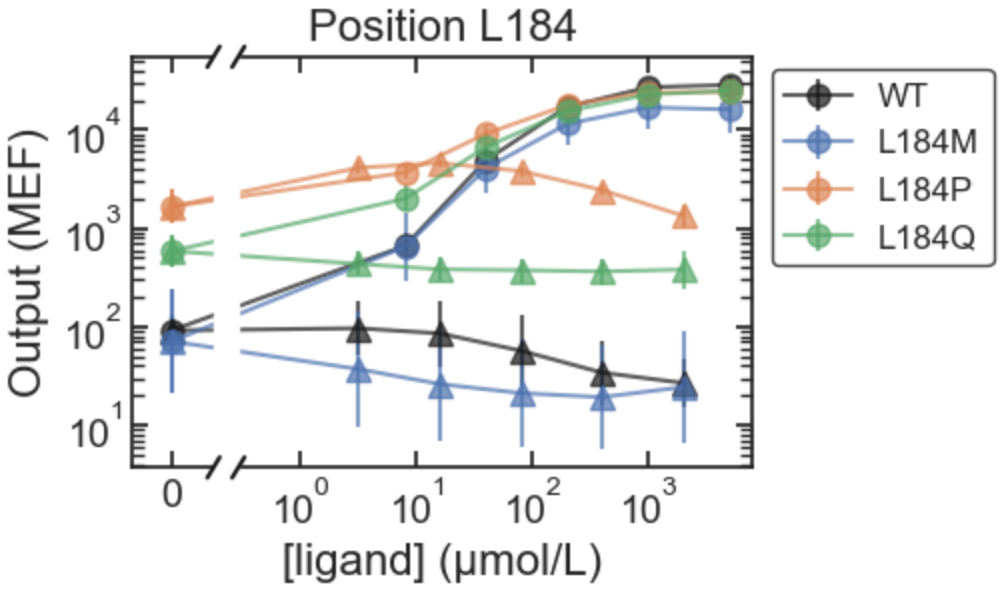

Position L174
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L174M
L174Q

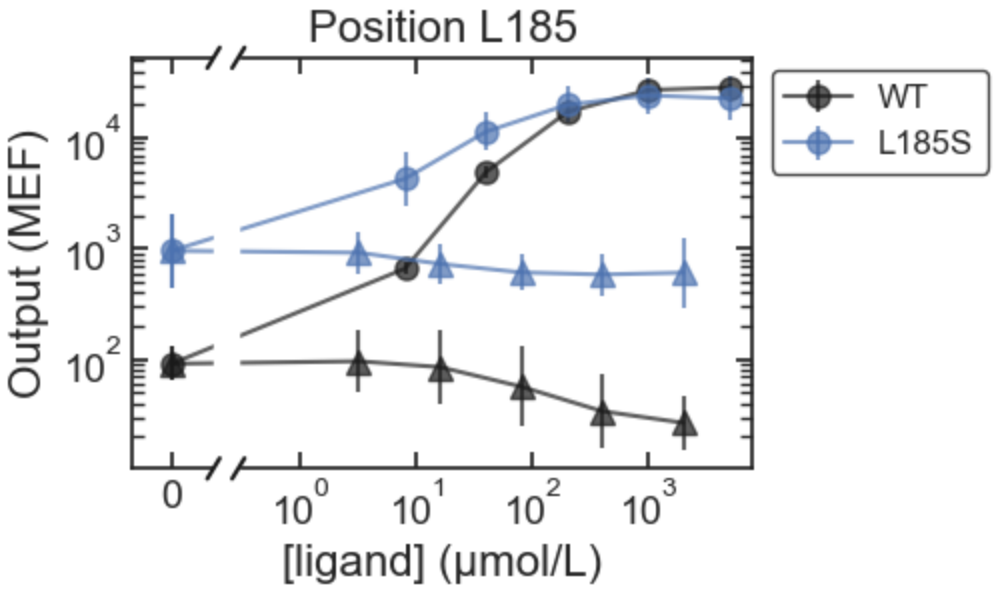

Position A176
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A176E
A176T
A176V

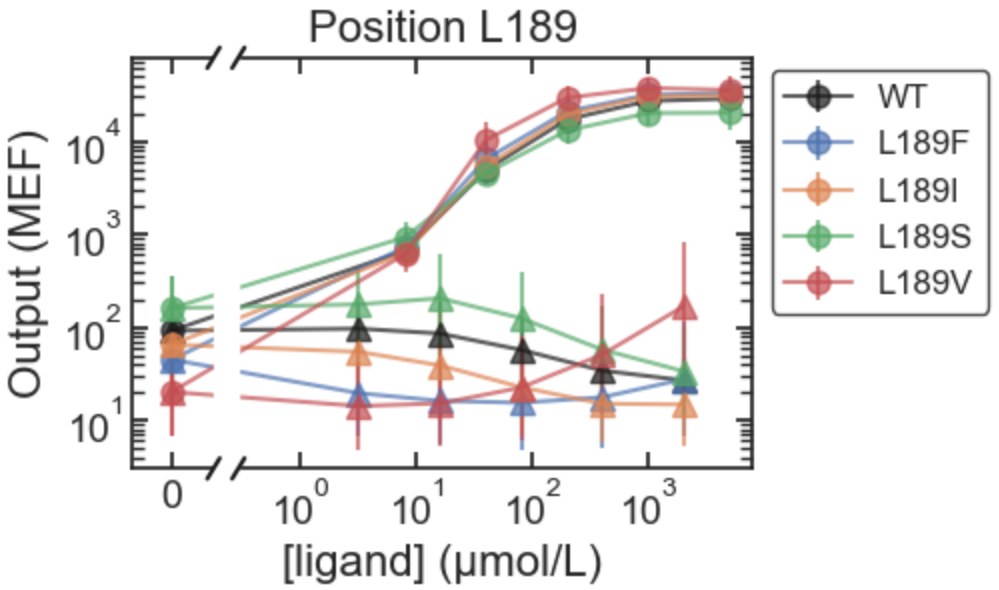

Position G178
Output (MEF)
[ligand] (μmol/L)
WT
G178A
G178C
G178D
G178R
G178S
G178V

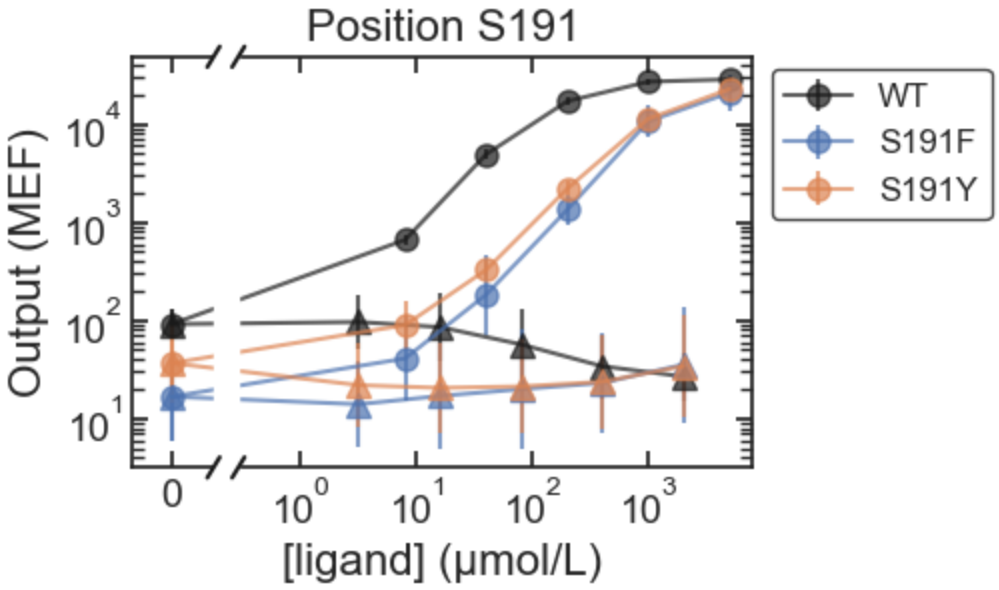

Position Q181
Output (MEF)
[ligand] (μmol/L)
WT
Q181H
Q181K
Q181L

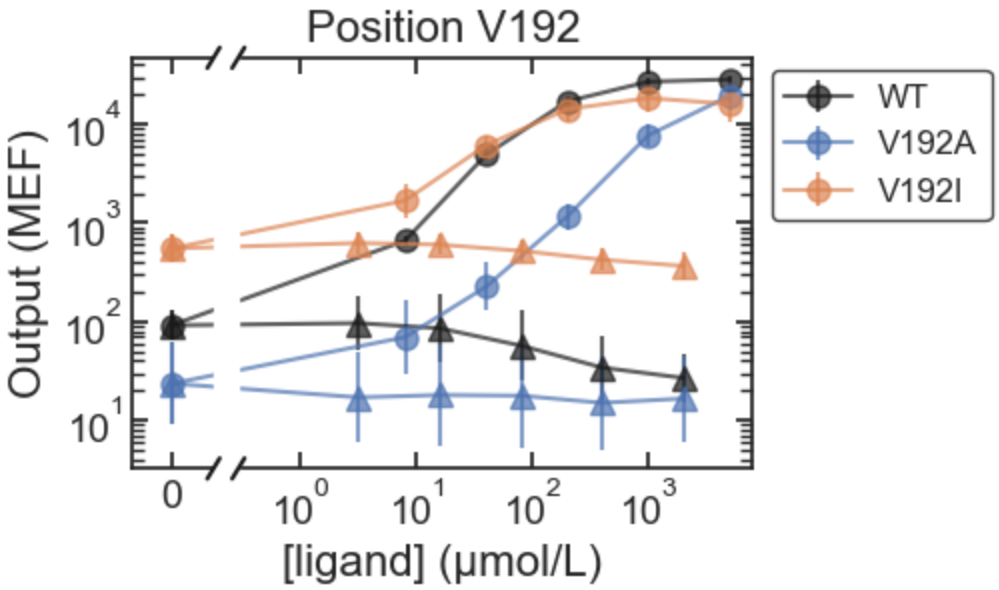

Position A183
Output (MEF)
[ligand] (μmol/L)
WT
A183E
A183T
A183V

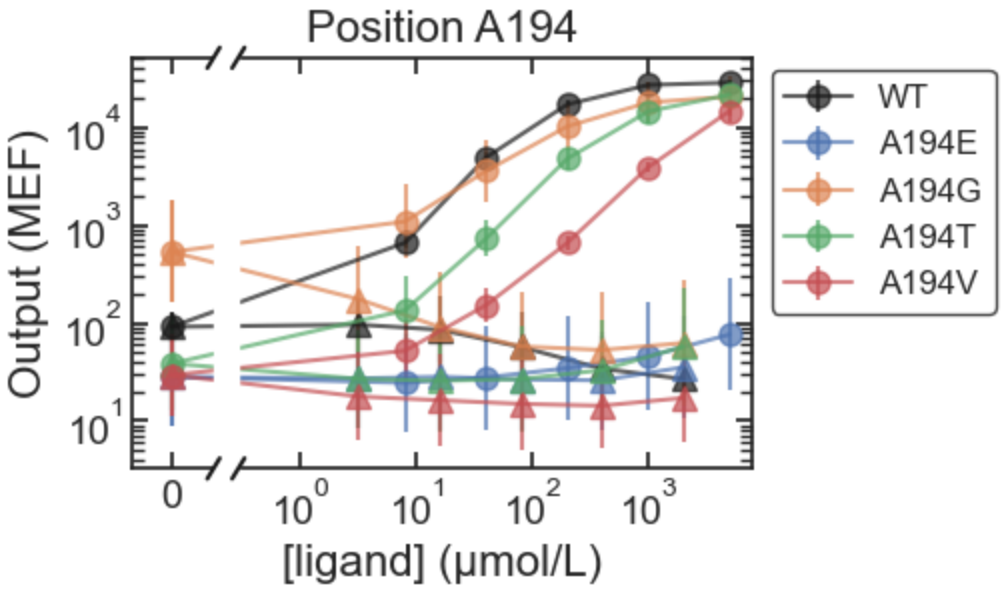

Position L184
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

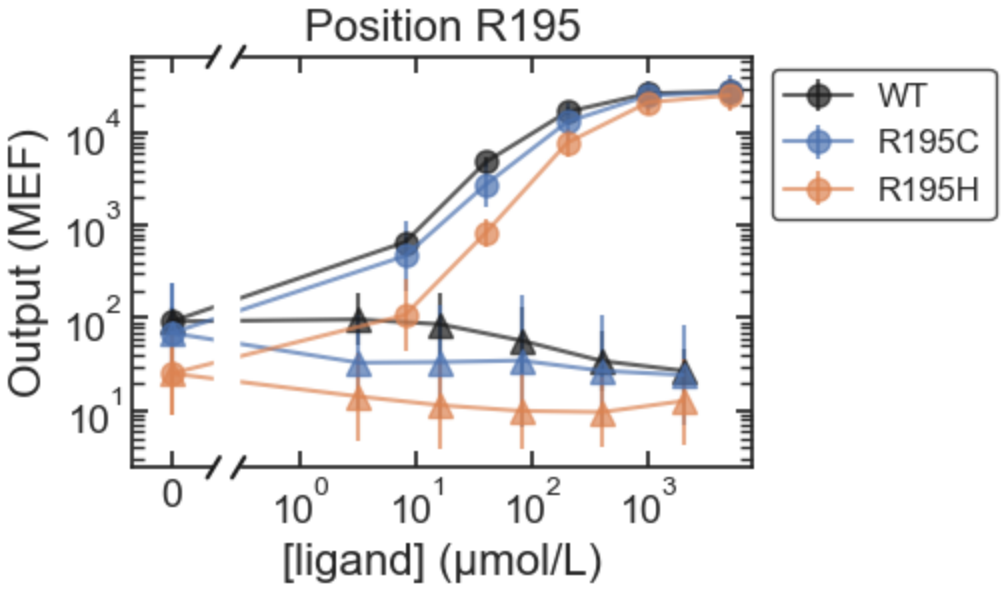

Position L185
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L185S

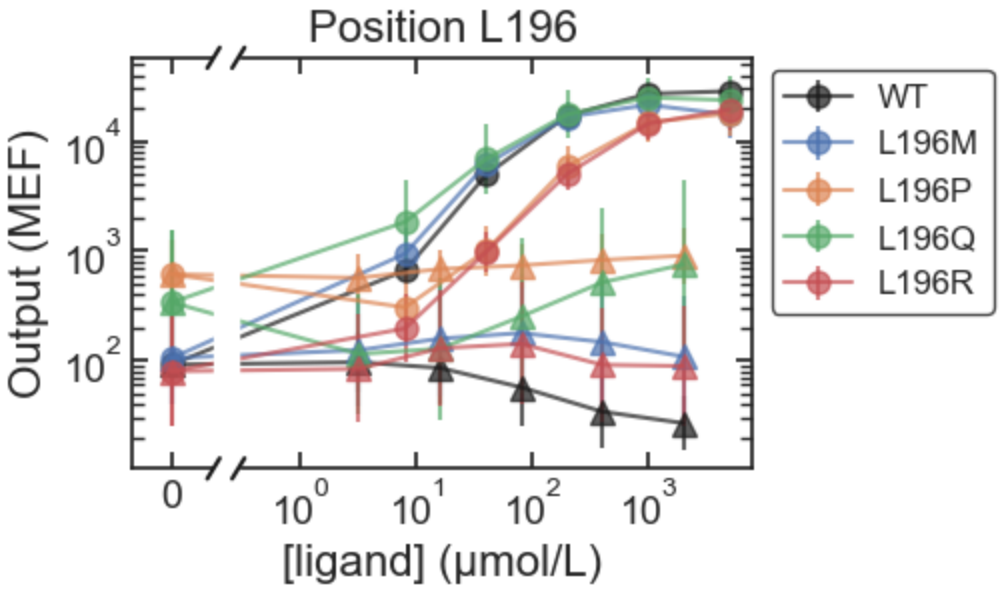

Position L189
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L189F
L189I
L189S
L189V

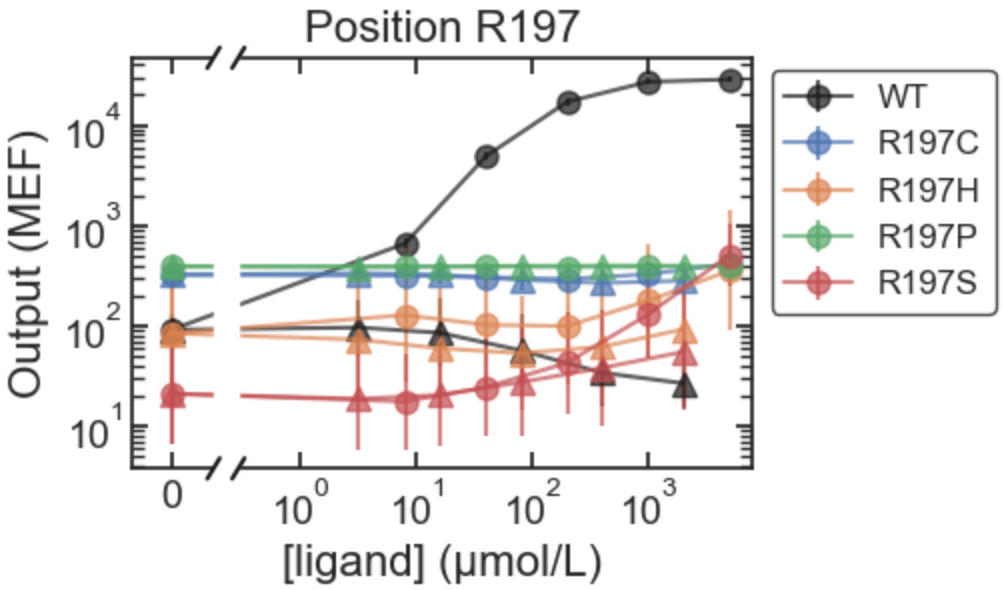

Position S191
Output (MEF)
[ligand] (μmol/L)
WT
S191F
S191Y

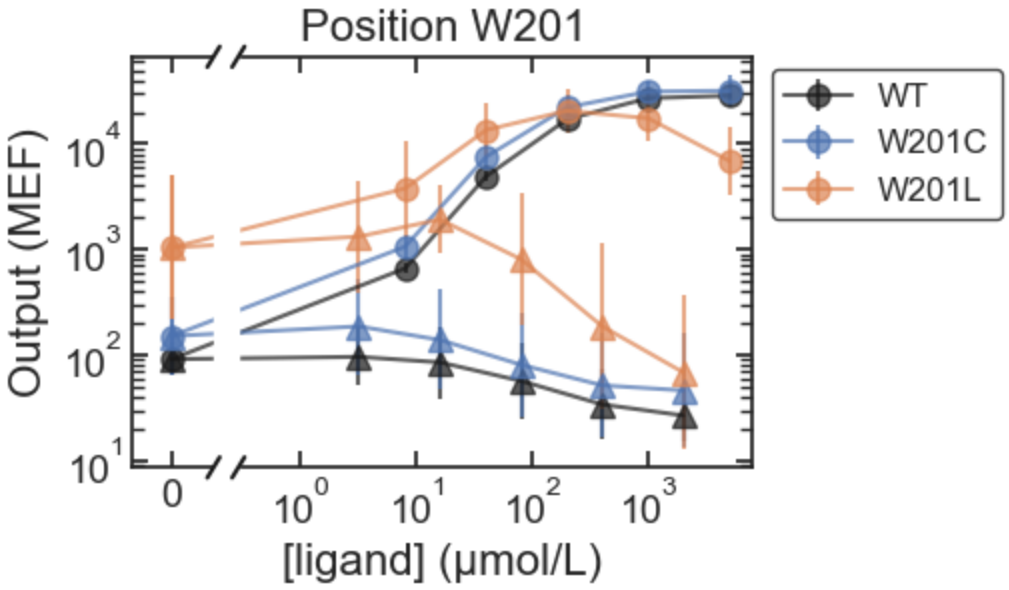

Position V192
Output (MEF)
[ligand] (μmol/L)
WT
V192A
V192I

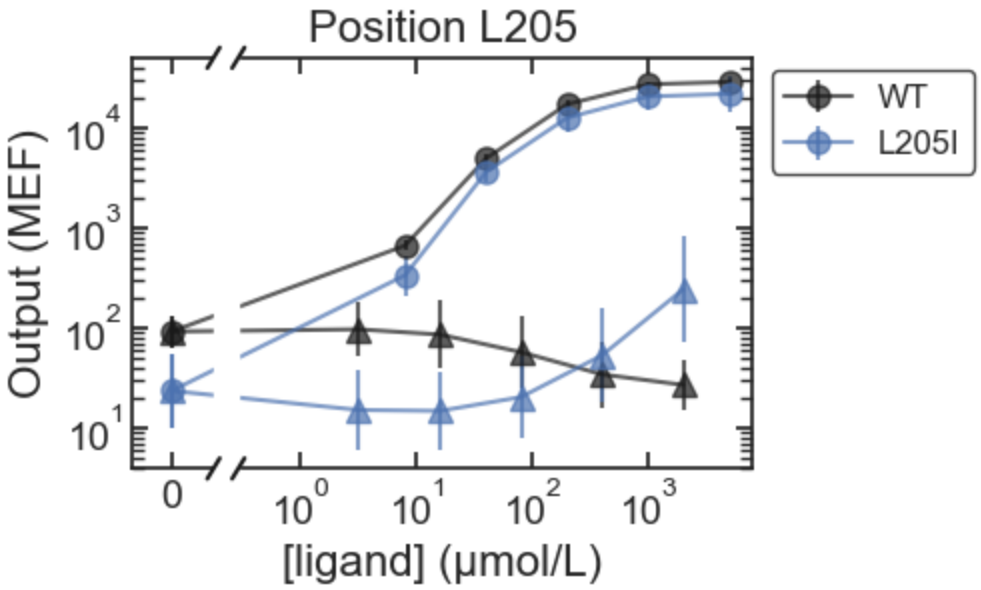

Position A194
Output (MEF)
[ligand] (μmol/L)
WT
A194E
A194G
A194T
A194V

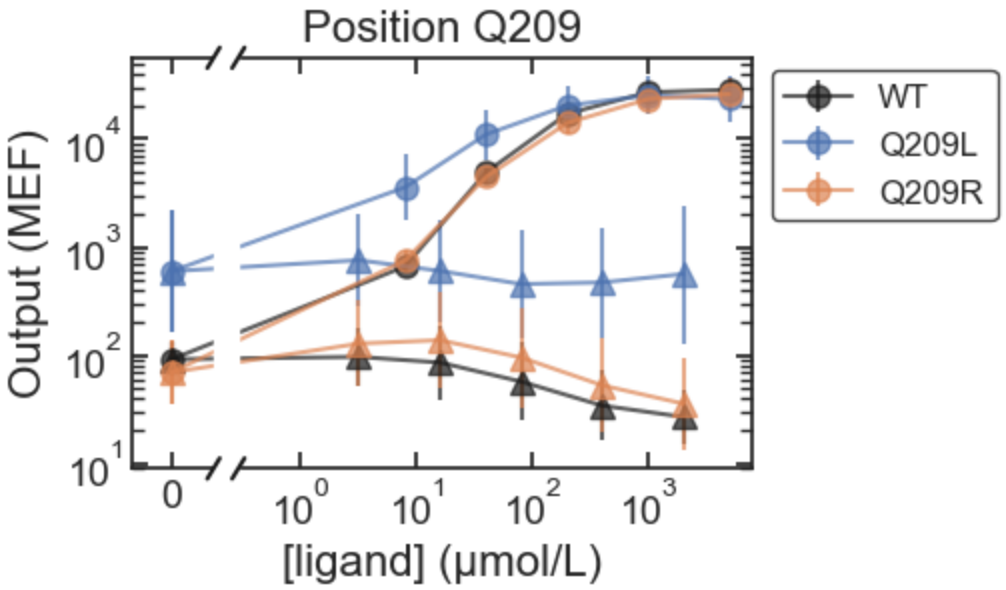

Position R195
Output (MEF)
[ligand] (μmol/L)
WT
R195C
R195H

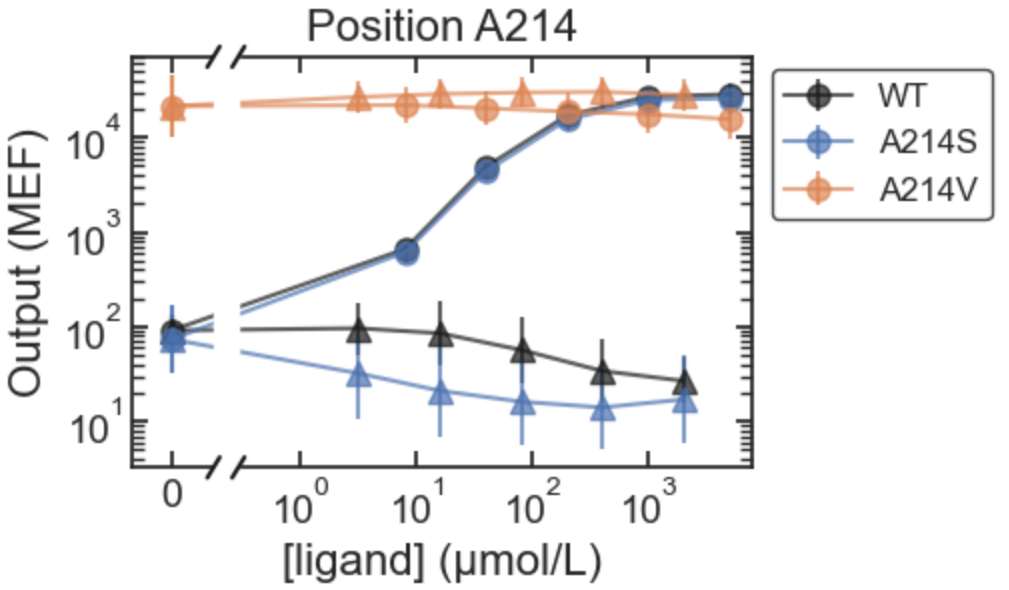

Position L196
Output (MEF)
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

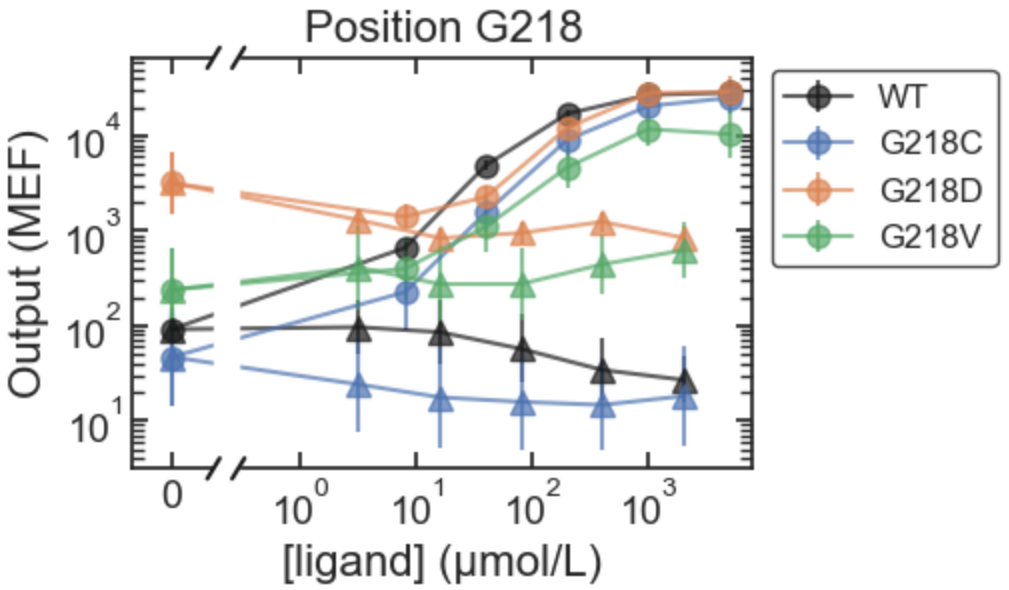

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

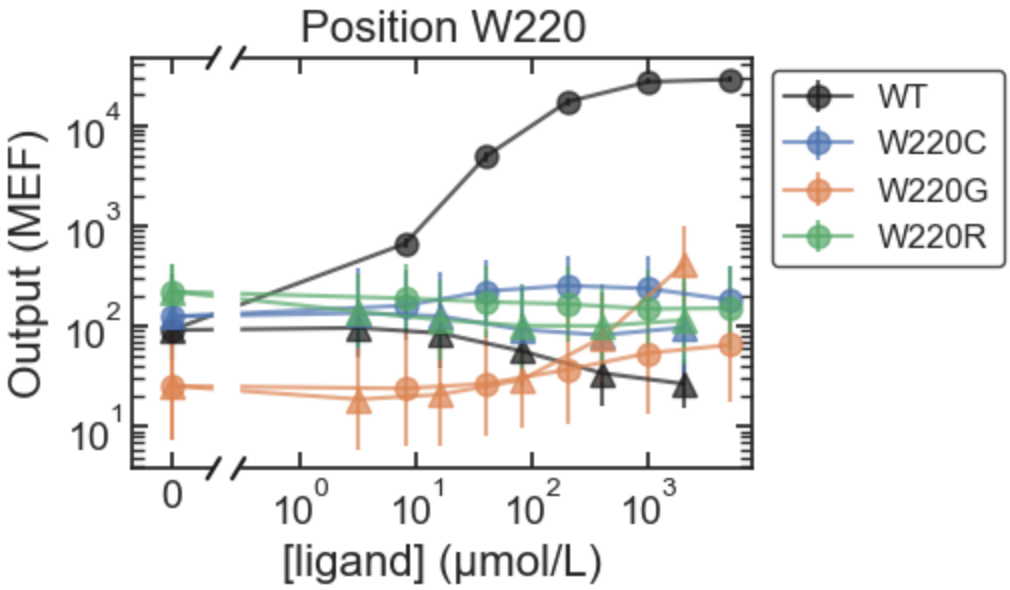

Position W201
Output (MEF)
[ligand] (μmol/L)
WT
W201C
W201L

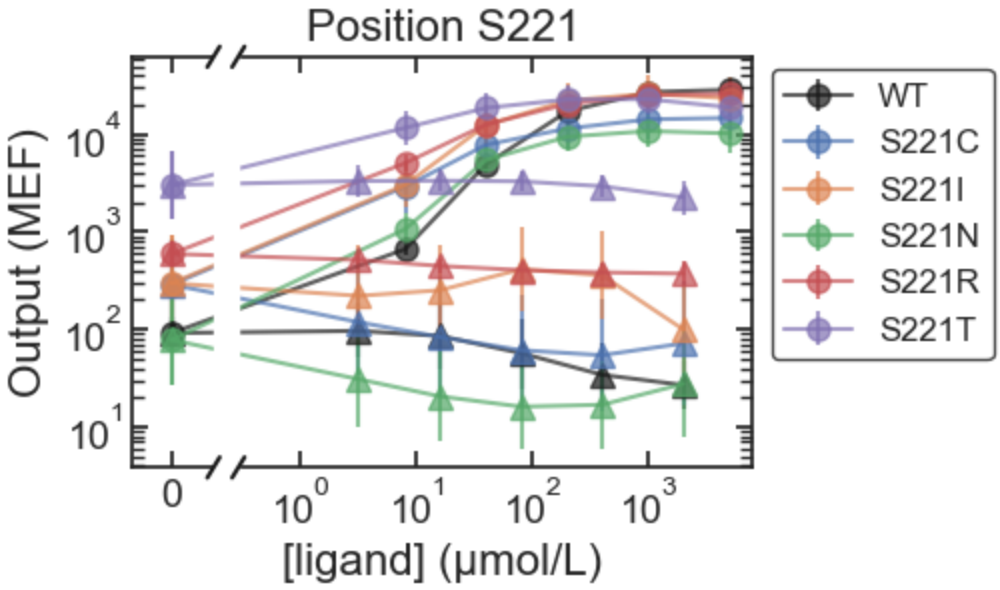

Position L205
Output (MEF)
[ligand] (μmol/L)
WT
L205I

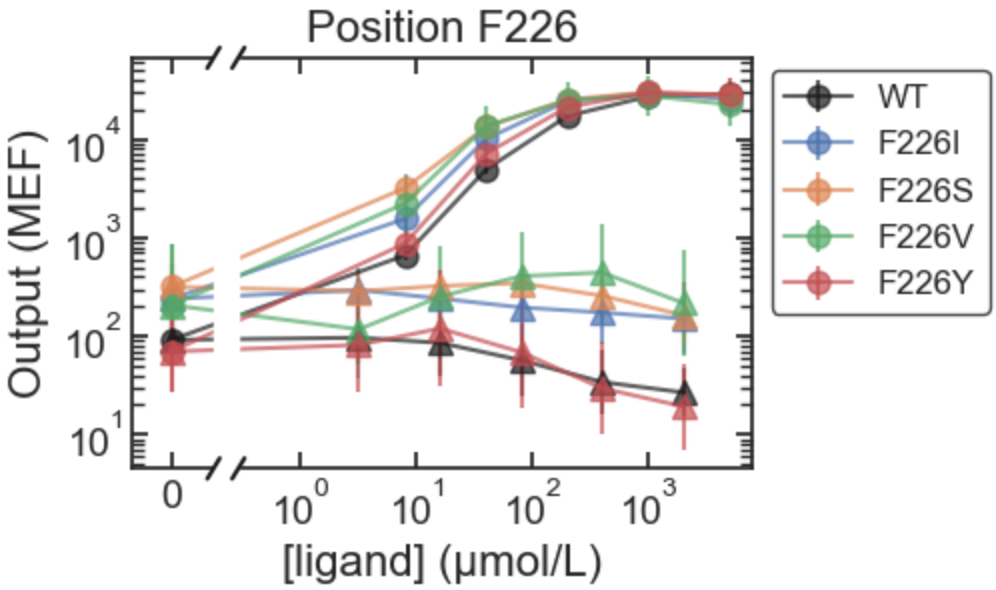

Position Q209
Output (MEF)
[ligand] (μmol/L)
WT
Q209L
Q209R

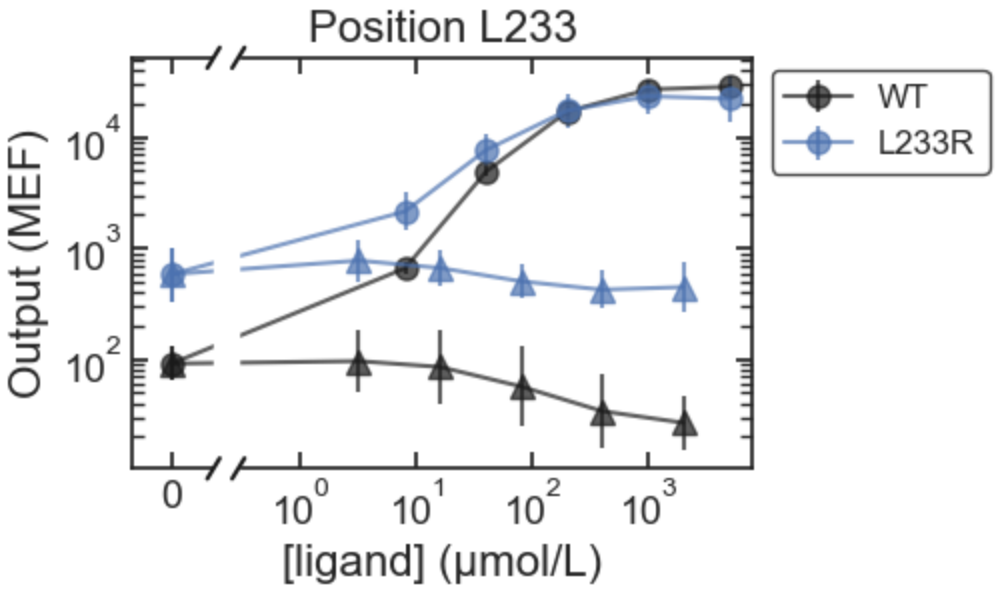

Position A214
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A214S
A214V

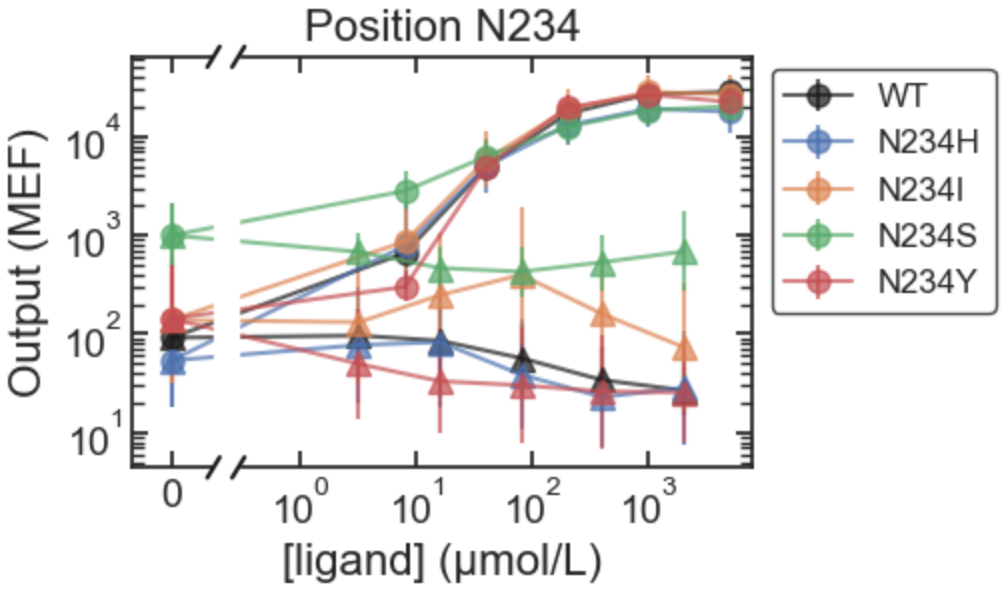

Position G218
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G218C
G218D
G218V

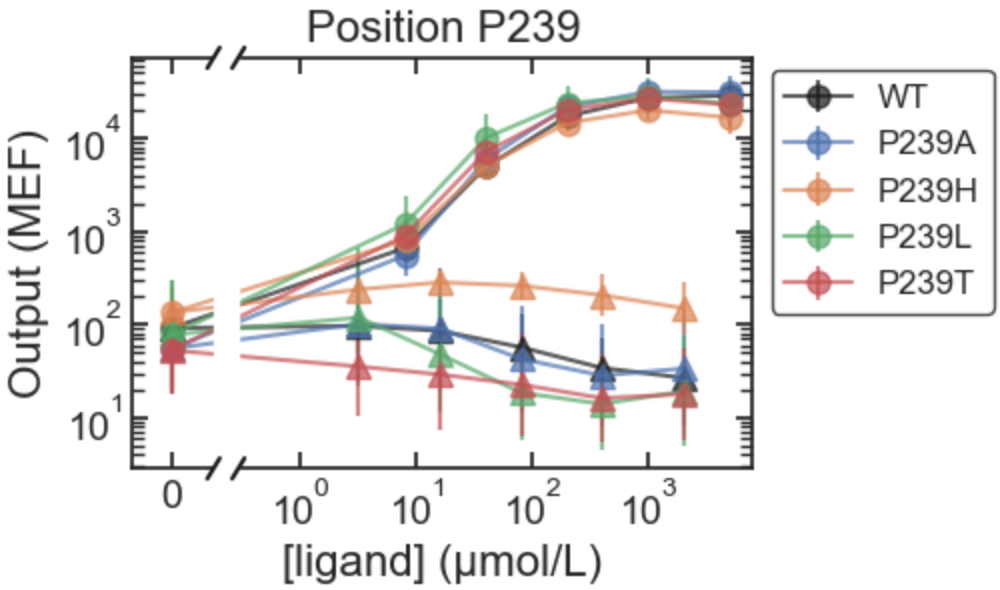

Position W220
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
W220C
W220G
W220R

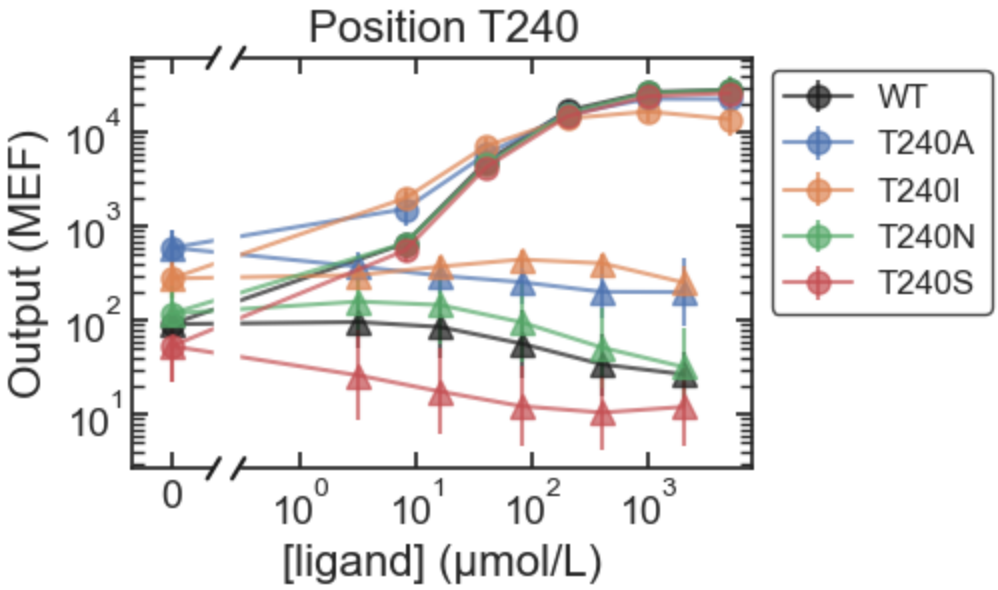

Position S221
Output (MEF)
[ligand] (μmol/L)
WT
S221C
S221I
S221N
S221R
S221T

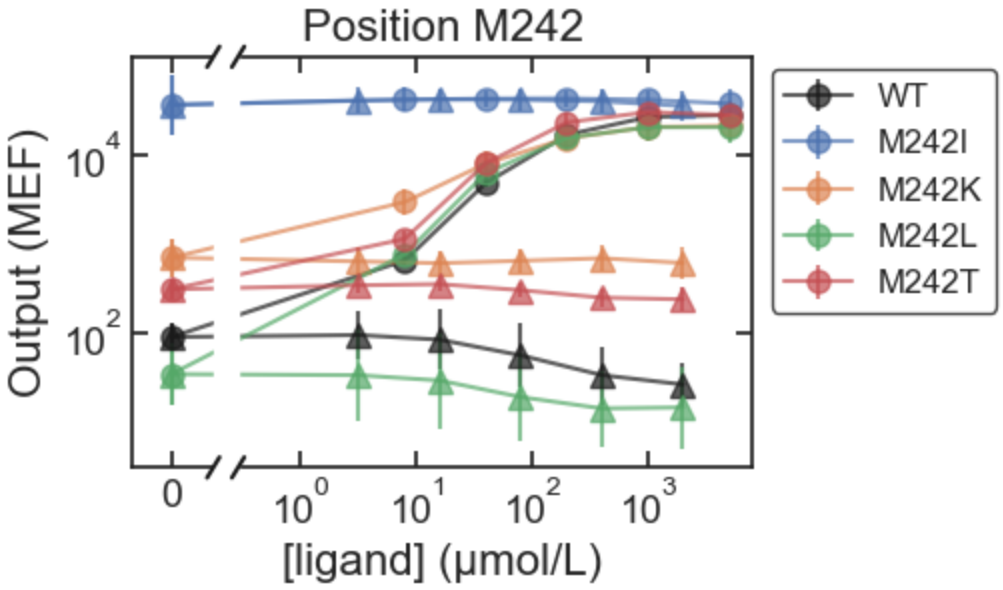

Position F226
Output (MEF)
[ligand] (μmol/L)
WT
F226I
F226S
F226V
F226Y

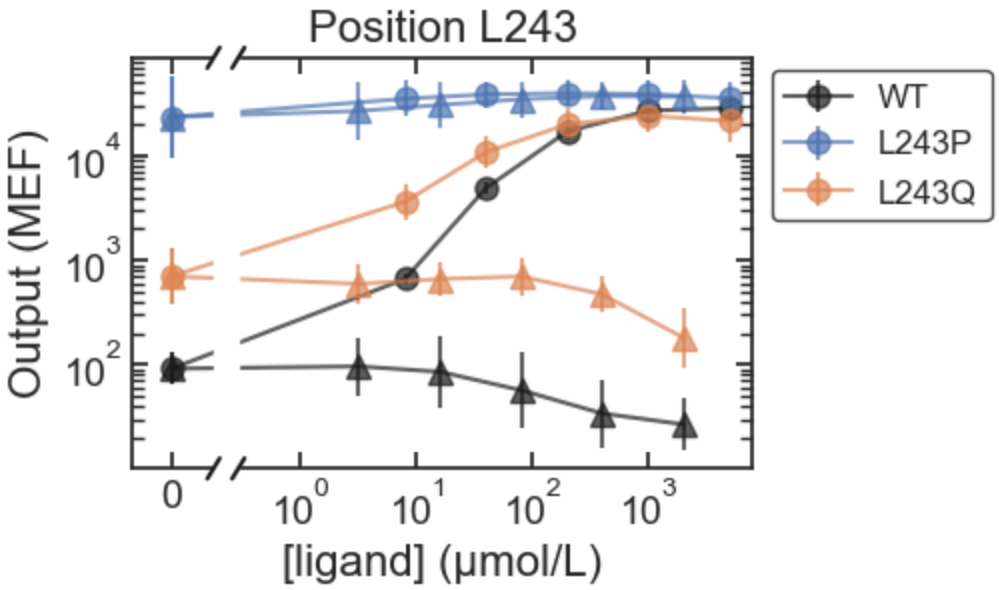

Position L233
Output (MEF)
[ligand] (μmol/L)
WT
L233R

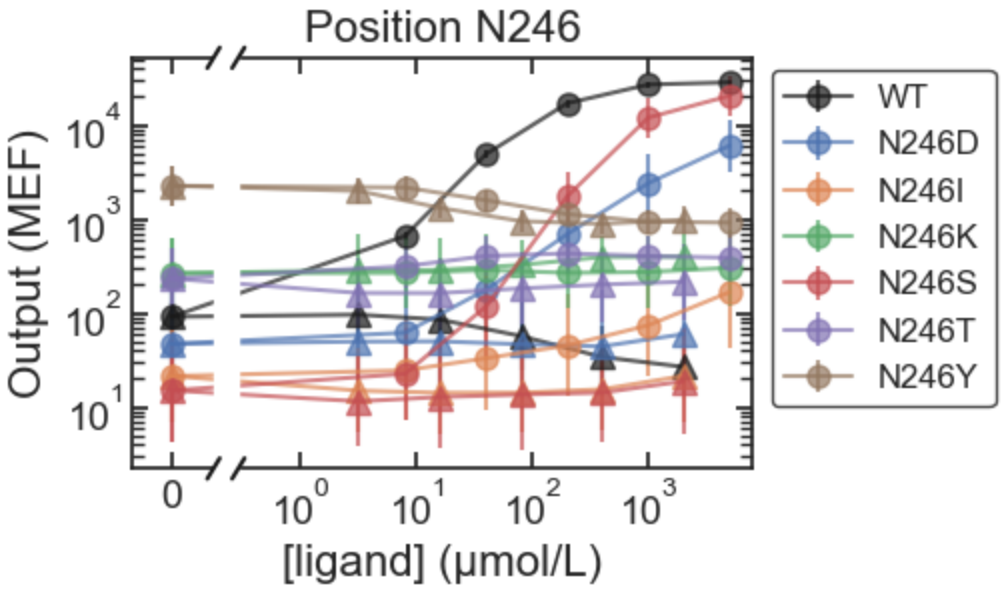

Position N234
Output (MEF)
[ligand] (μmol/L)
WT
N234H
N234I
N234S
N234Y

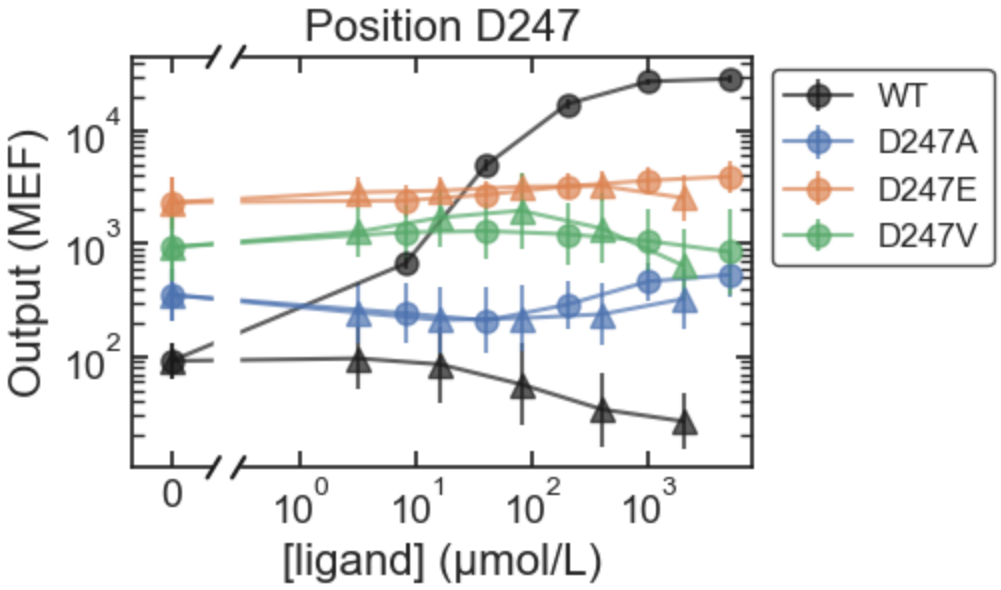

Position P239
Output (MEF)
[ligand] (μmol/L)
WT
P239A
P239H
P239L
P239T

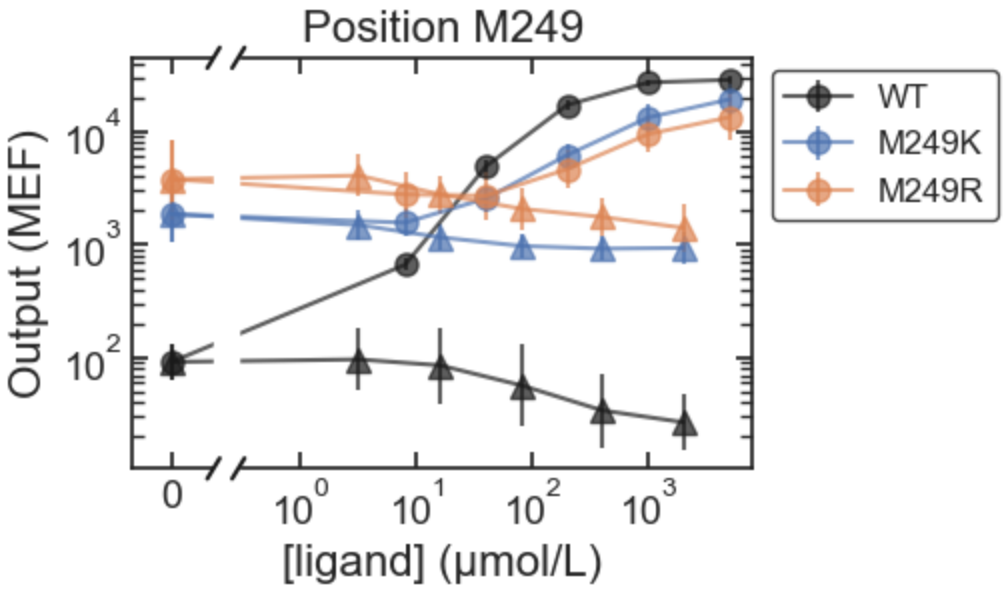

Position T240
Output (MEF)
[ligand] (μmol/L)
WT
T240A
T240I
T240N
T240S

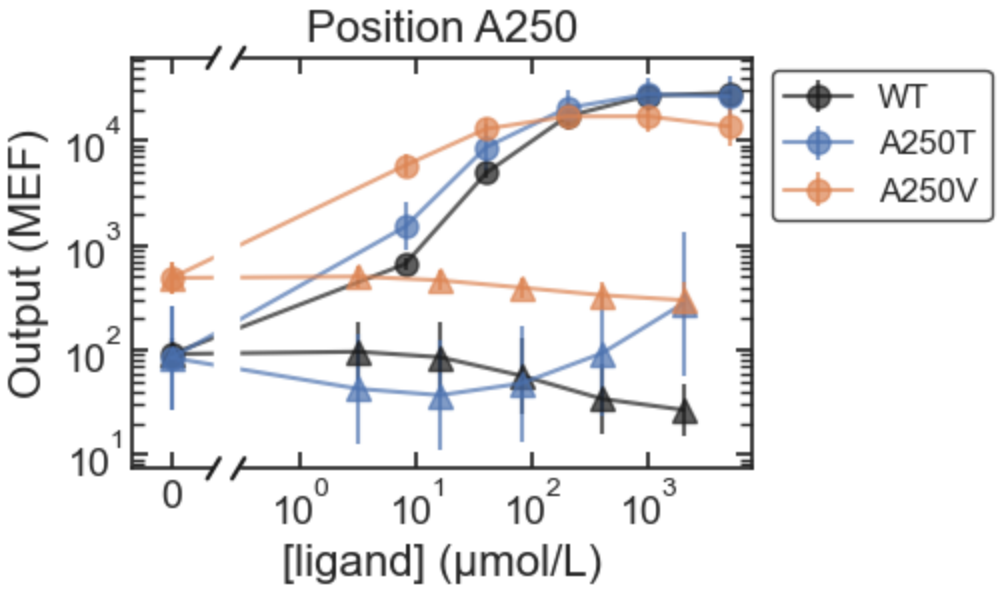

Position M242
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

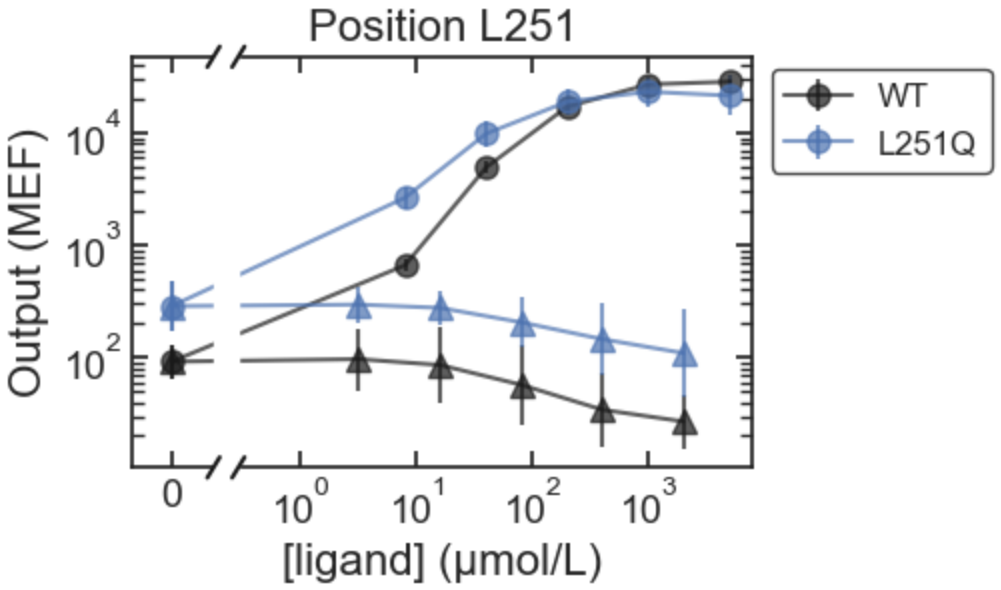

Position L243
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L243P
L243Q

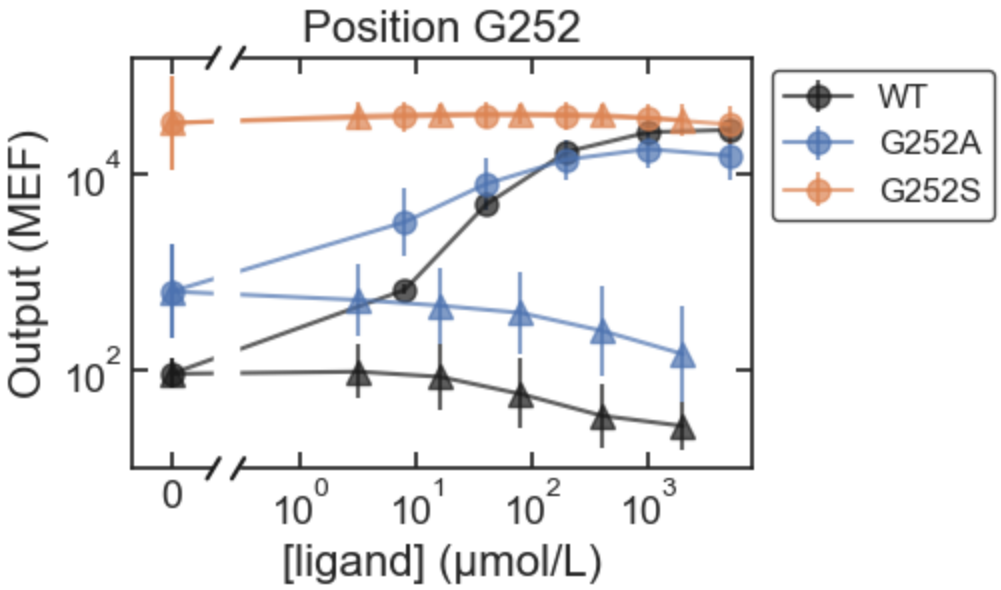

Position N246
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

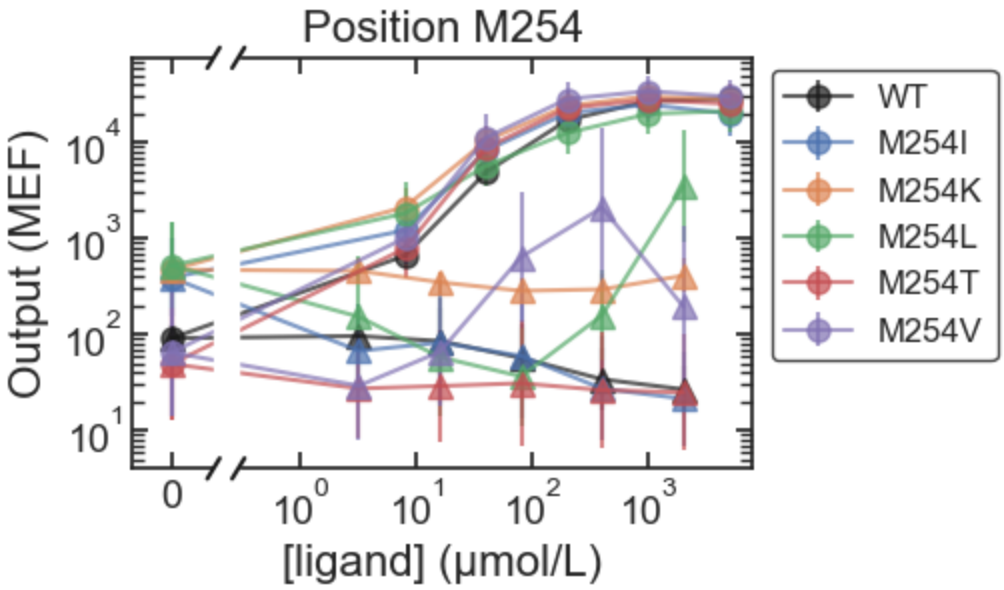

Position D247
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D247A
D247E
D247V

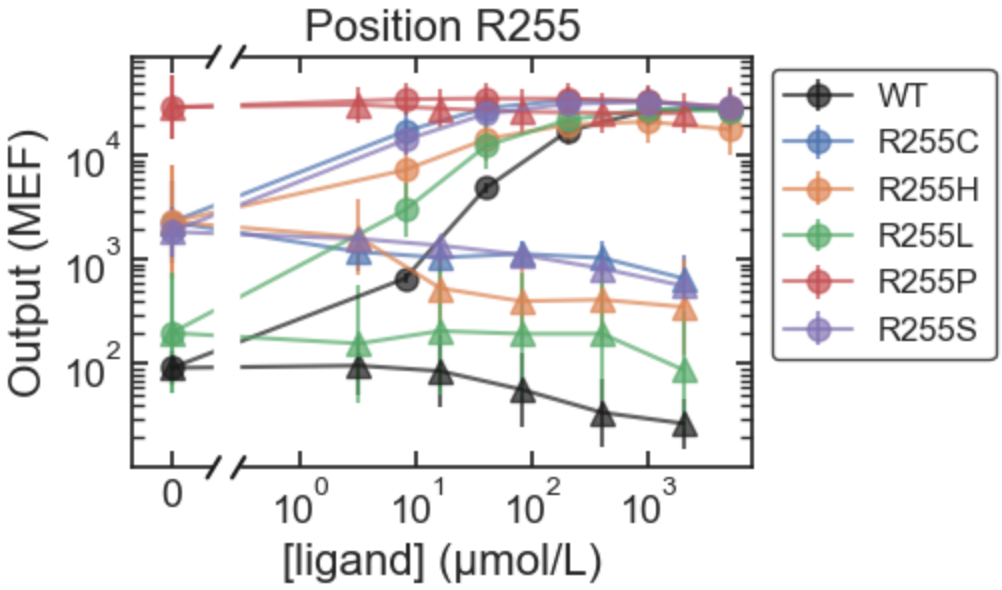

Position M249
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M249K
M249R

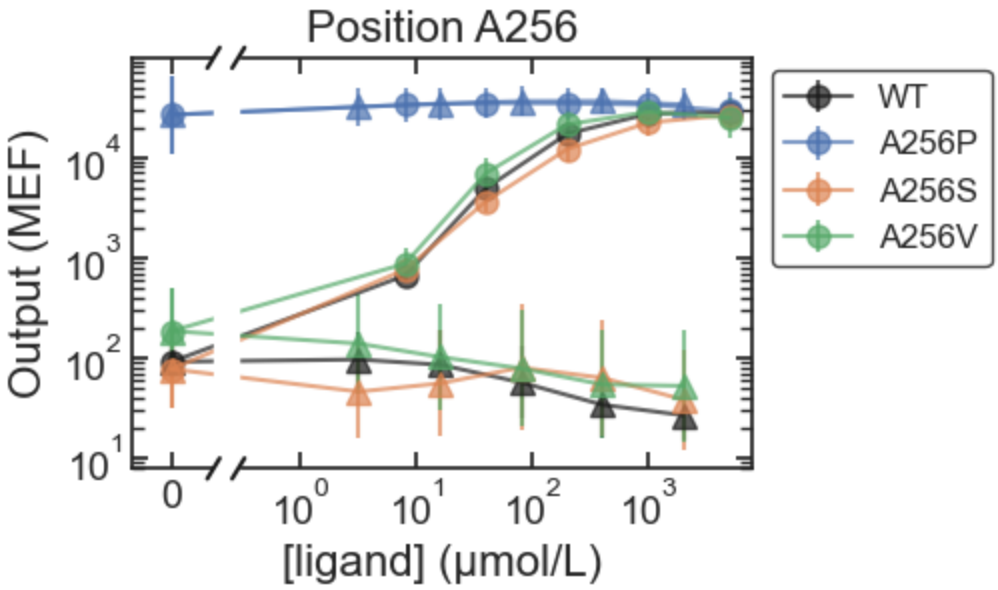

Position A250
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A250T
A250V

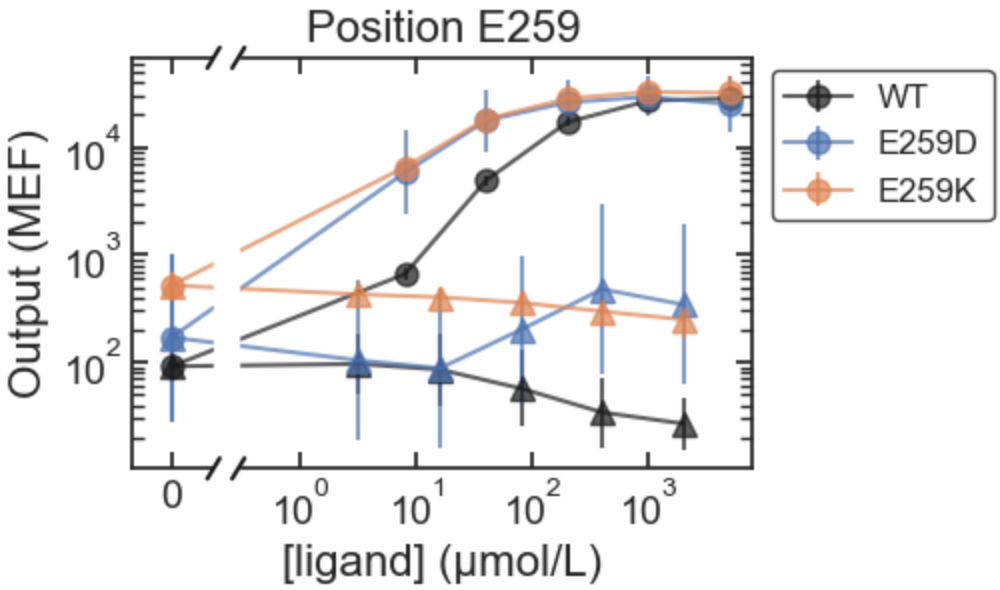

Position L251
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L251Q

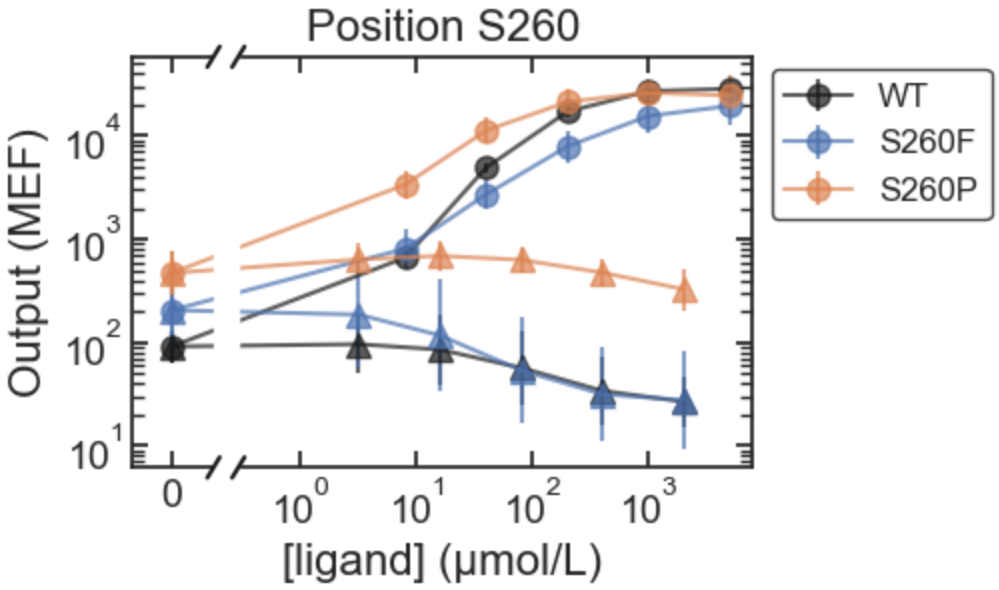

Position G252
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G252A
G252S

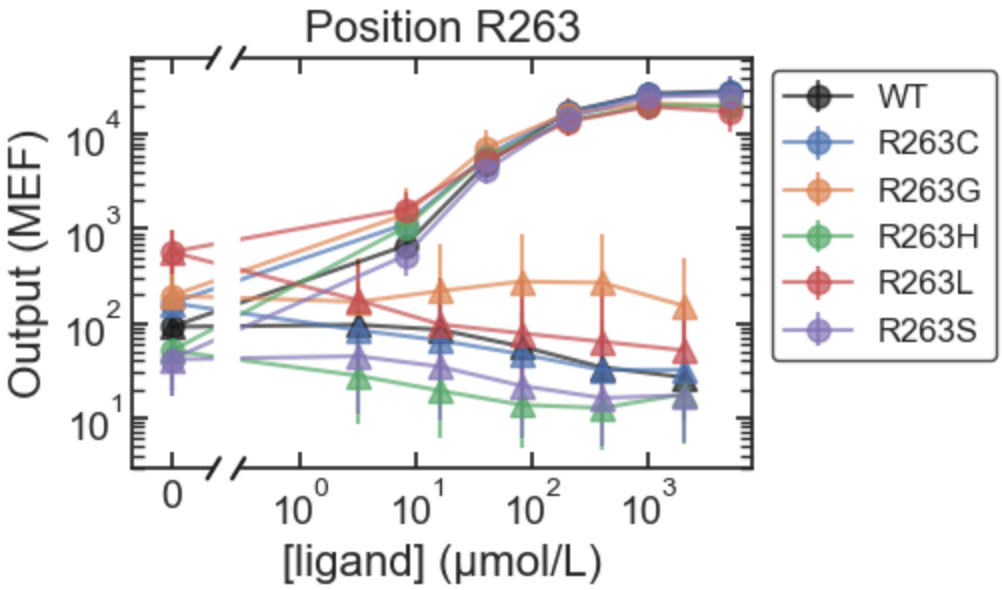

Position M254
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M254I
M254K
M254L
M254T
M254V

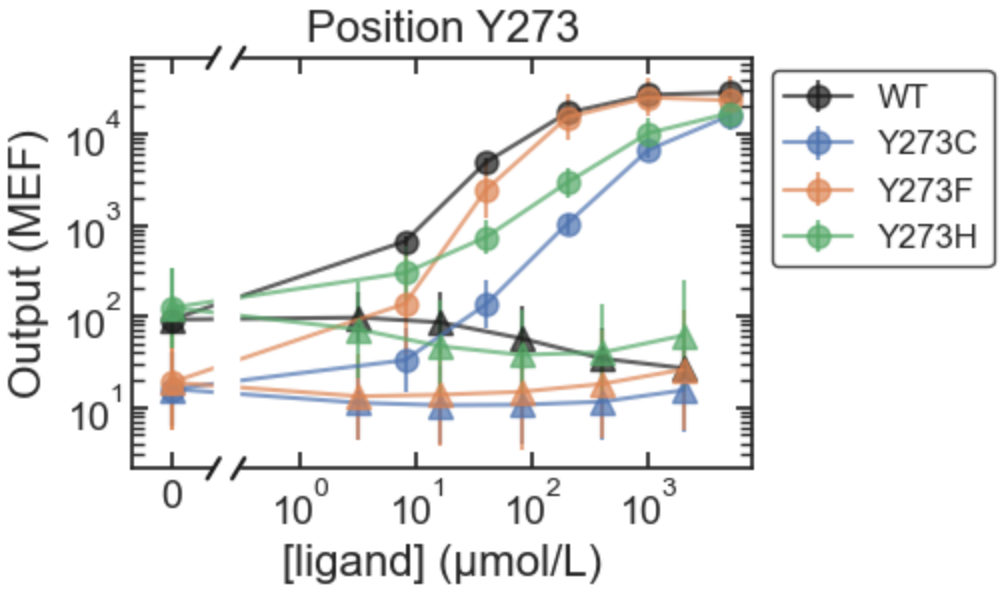

Position R255
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R255C
R255H
R255L
R255P
R255S

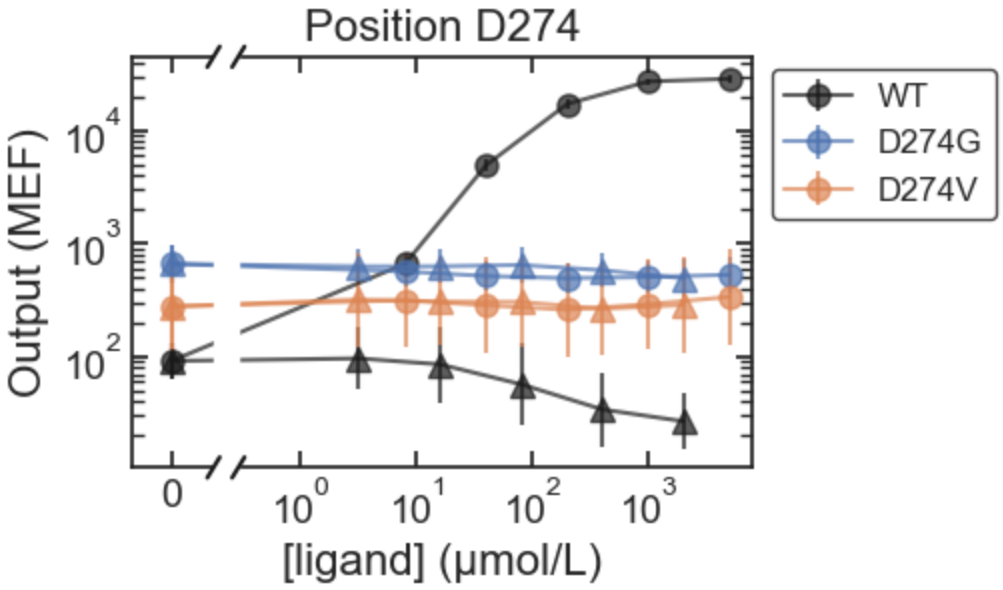

Position A256
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A256P
A256S
A256V

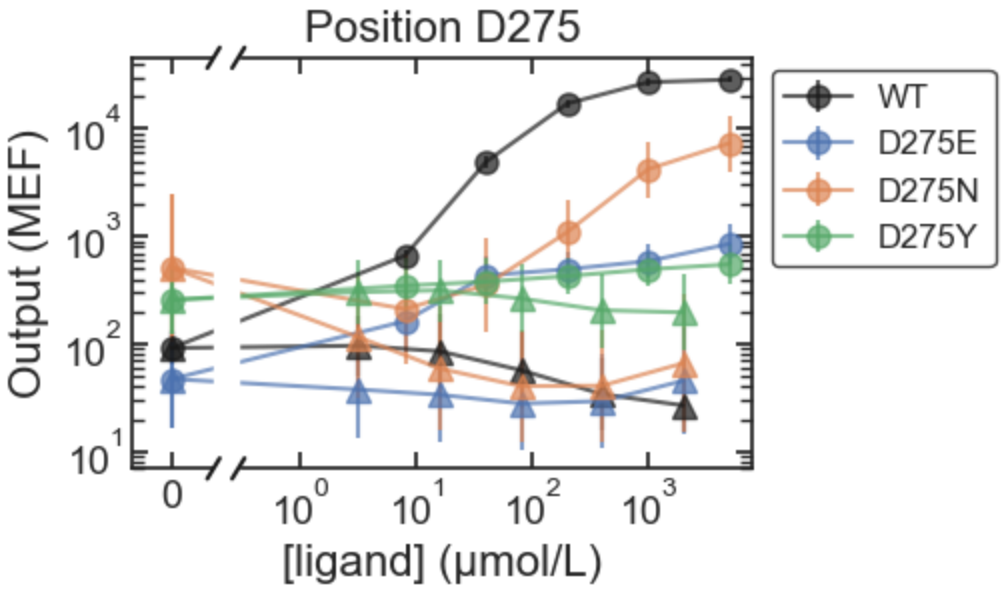

Position E259
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
E259D
E259K

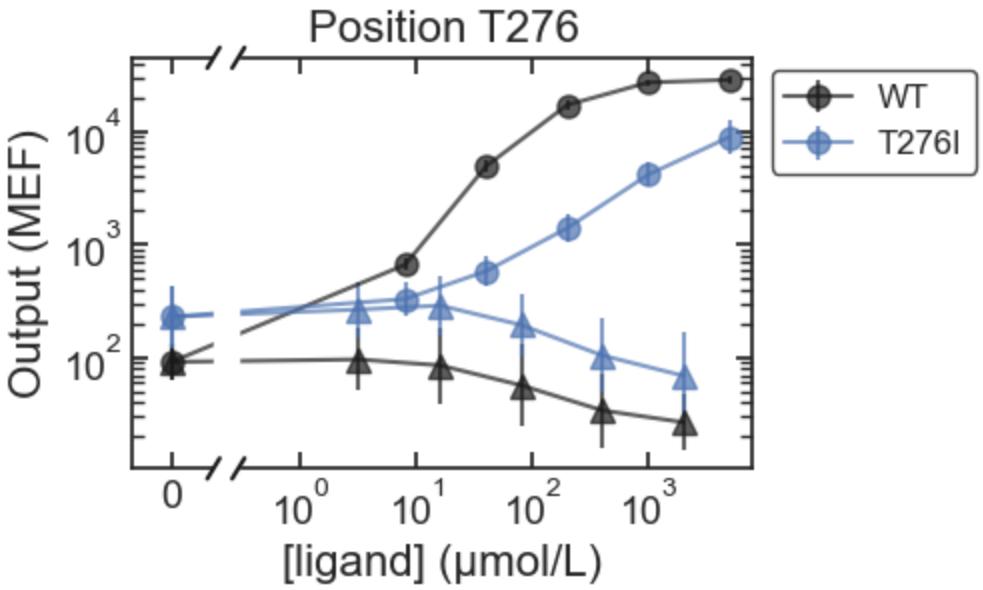

Position S260
Output (MEF)
[ligand] (μmol/L)
WT
S260F
S260P

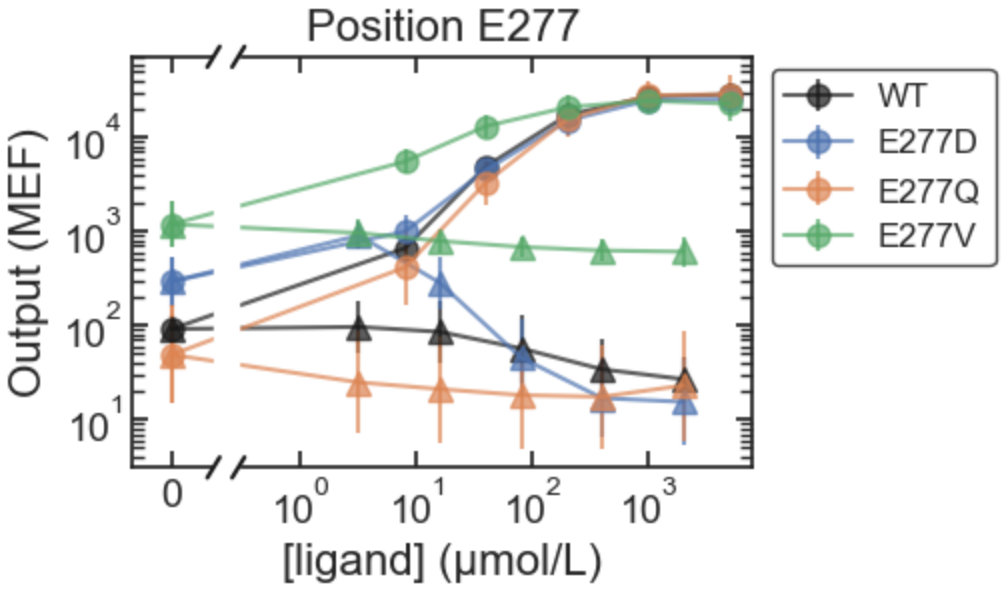

Position R263
Output (MEF)
[ligand] (μmol/L)
WT
R263C
R263G
R263H
R263L
R263S

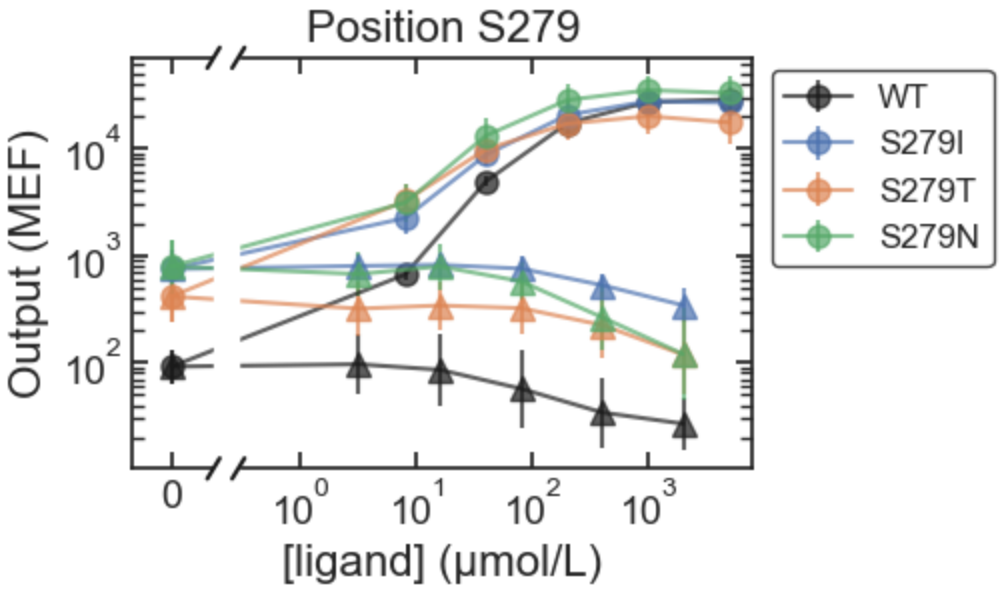

Position Y273
Output (MEF)
[ligand] (μmol/L)
WT
Y273C
Y273F
Y273H

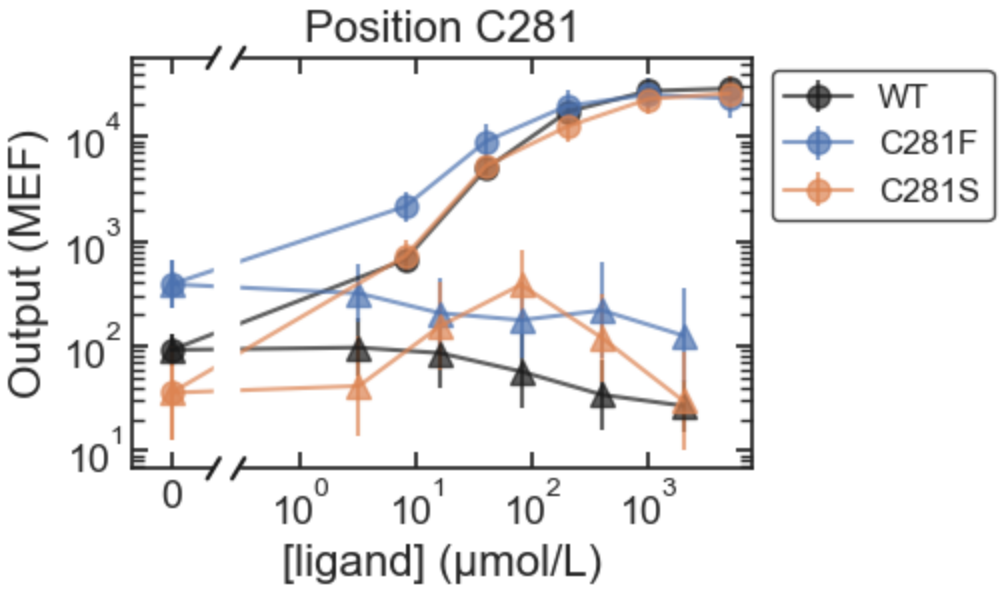

Position D274
Output (MEF)
[ligand] (μmol/L)
WT
D274G
D274V

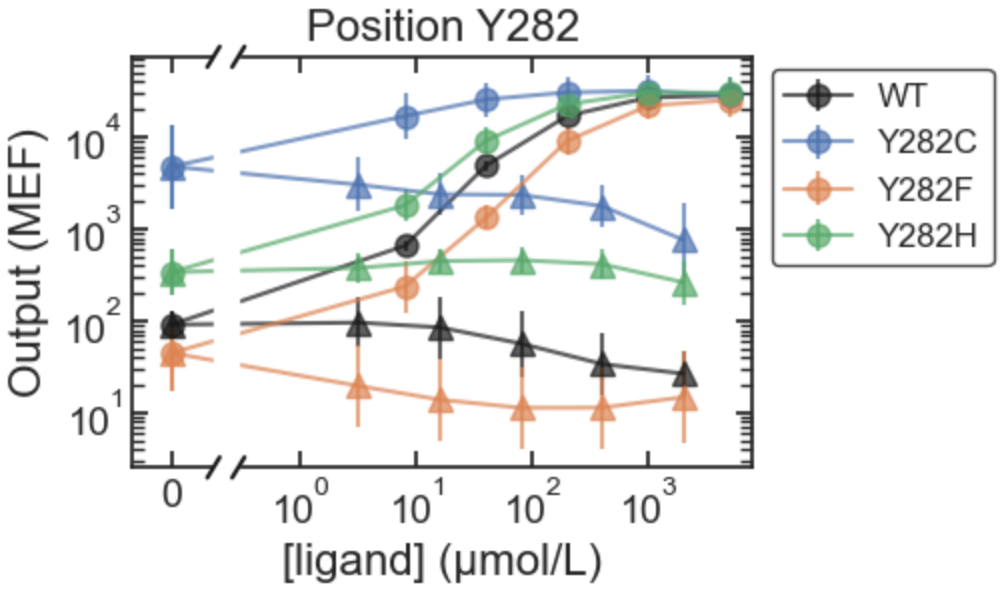

Position D275
Output (MEF)
[ligand] (μmol/L)
WT
D275E
D275N
D275Y

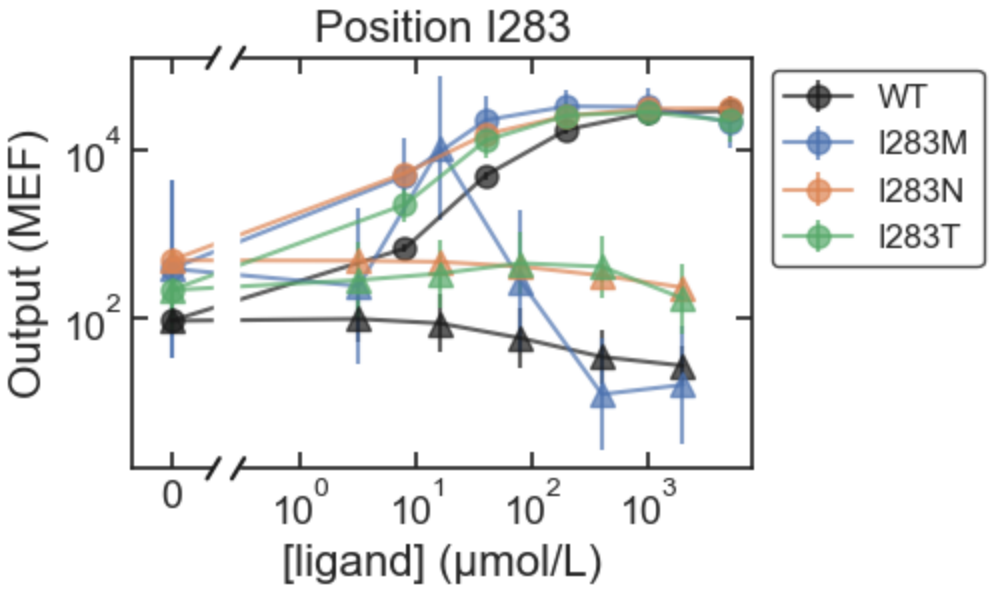

Position T276
Output (MEF)
[ligand] (μmol/L)
WT
T276I

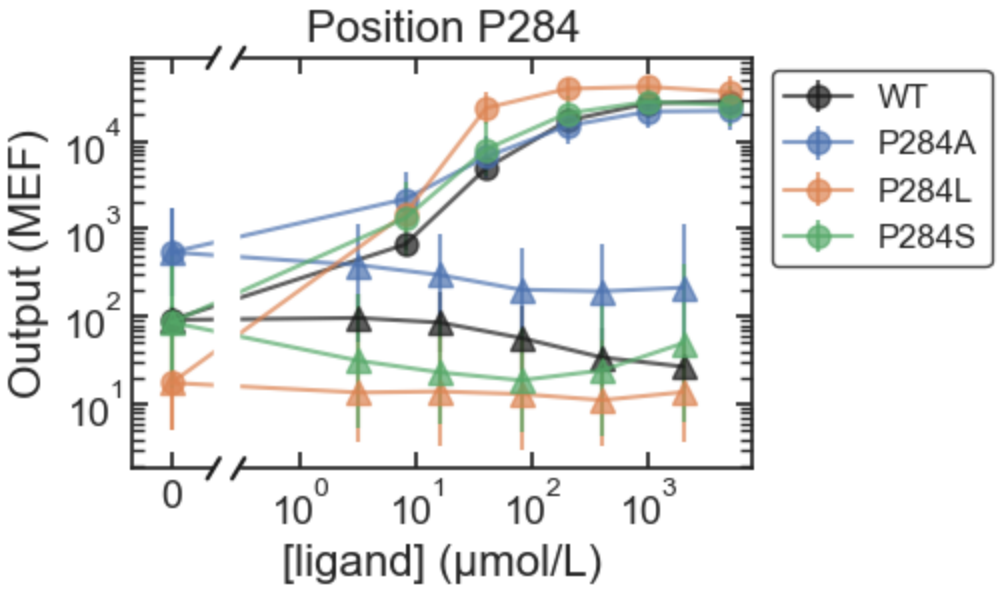

Position E277
Output (MEF)
[ligand] (μmol/L)
WT
E277D
E277Q
E277V

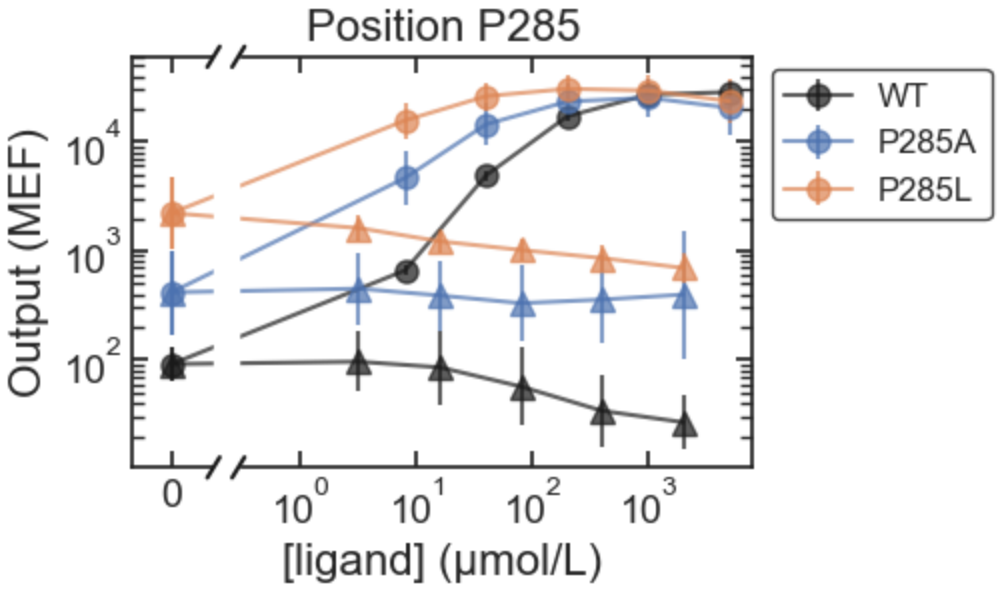

Position S279
Output (MEF)
[ligand] (μmol/L)
WT
S279I
S279T
S279N

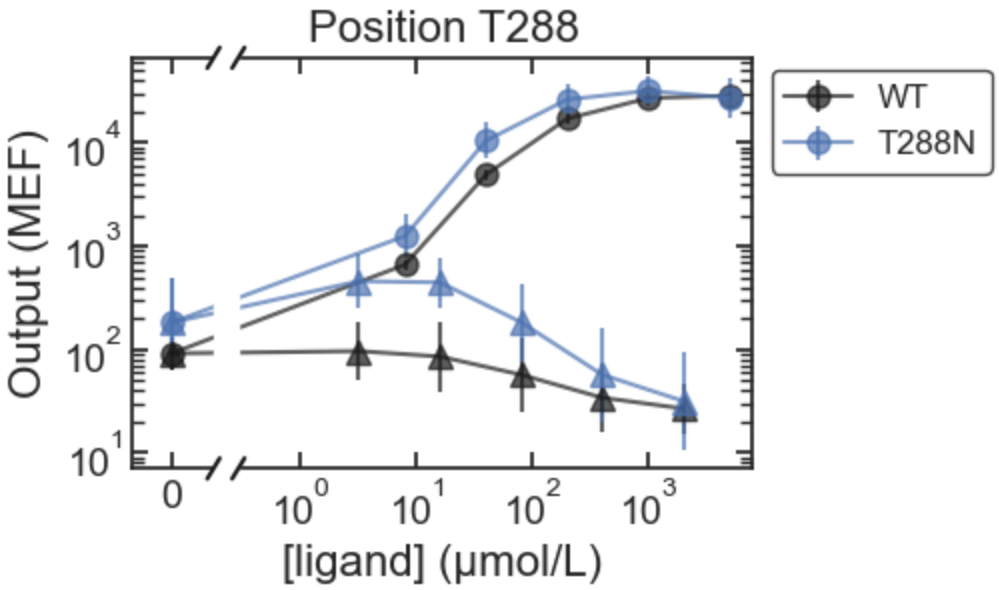

Position C281
Output (MEF)
[ligand] (μmol/L)
WT
C281F
C281S

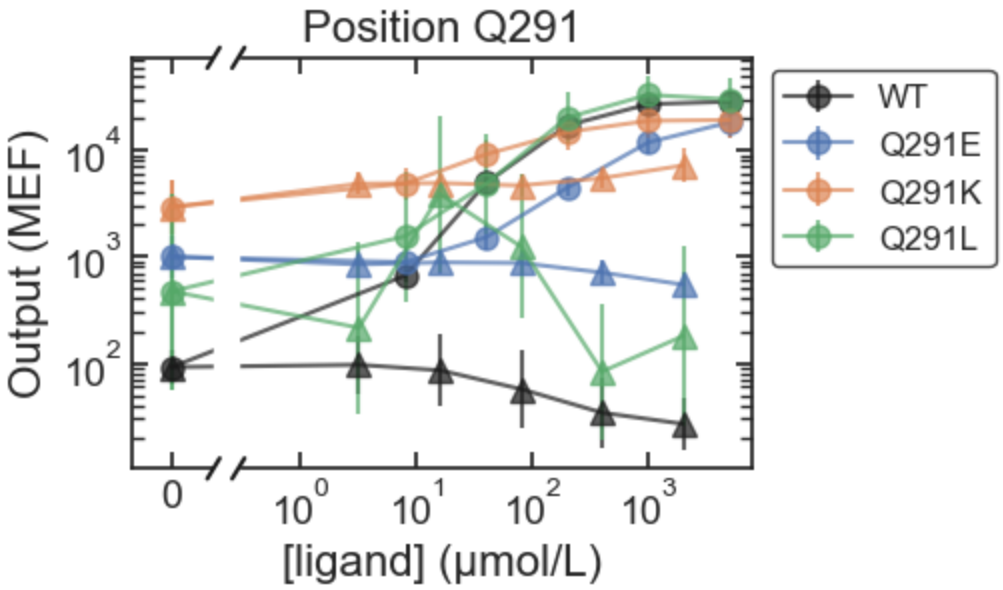

Position Y282
Output (MEF)
[ligand] (µmol/L)
WT
Y282C
Y282F
Y282H

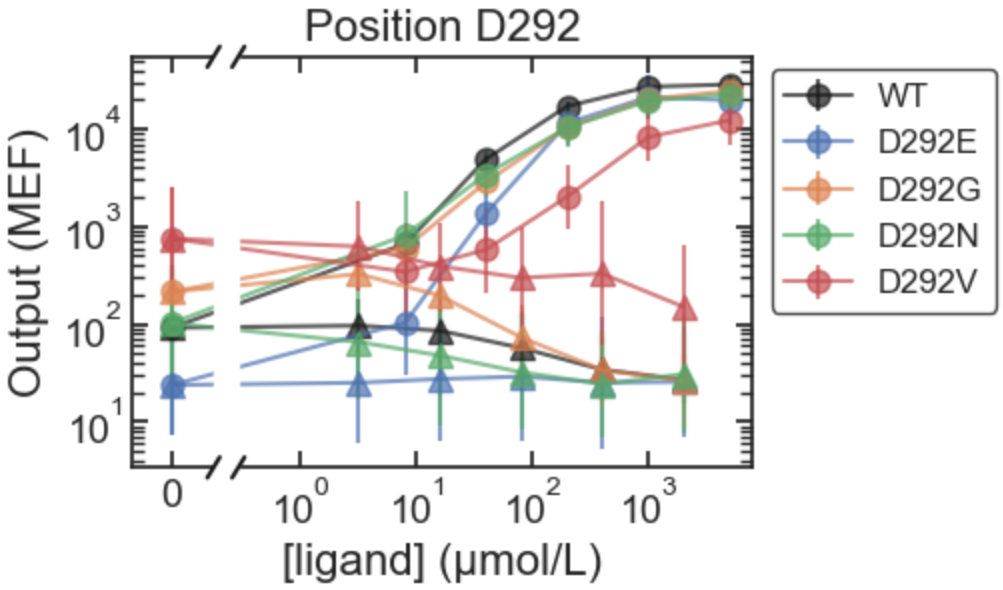

Position I283
Output (MEF)
[ligand] (µmol/L)
WT
I283M
I283N
I283T

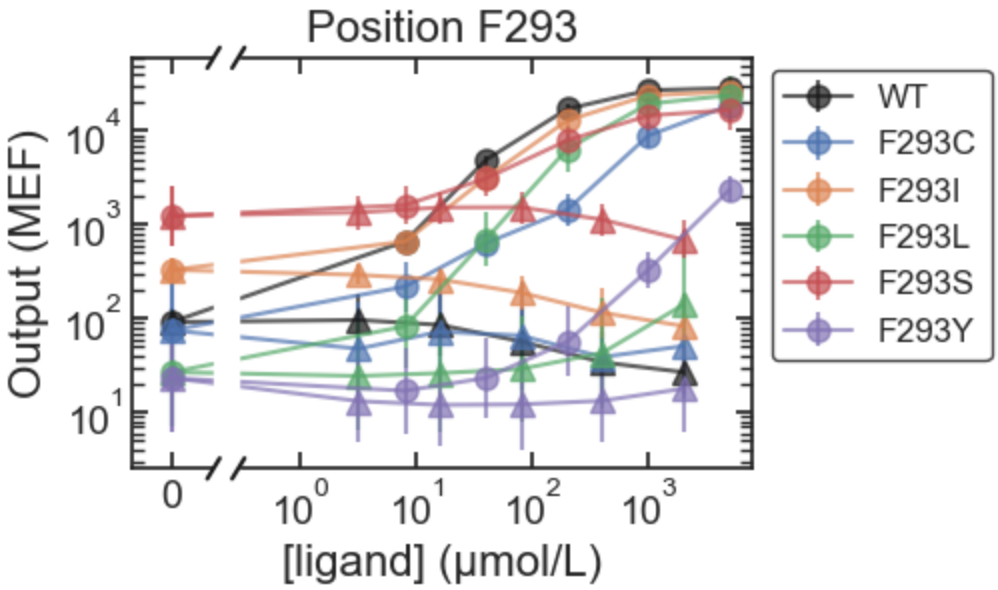

Position P284
Output (MEF)
[ligand] (µmol/L)
WT
P284A
P284L
P284S

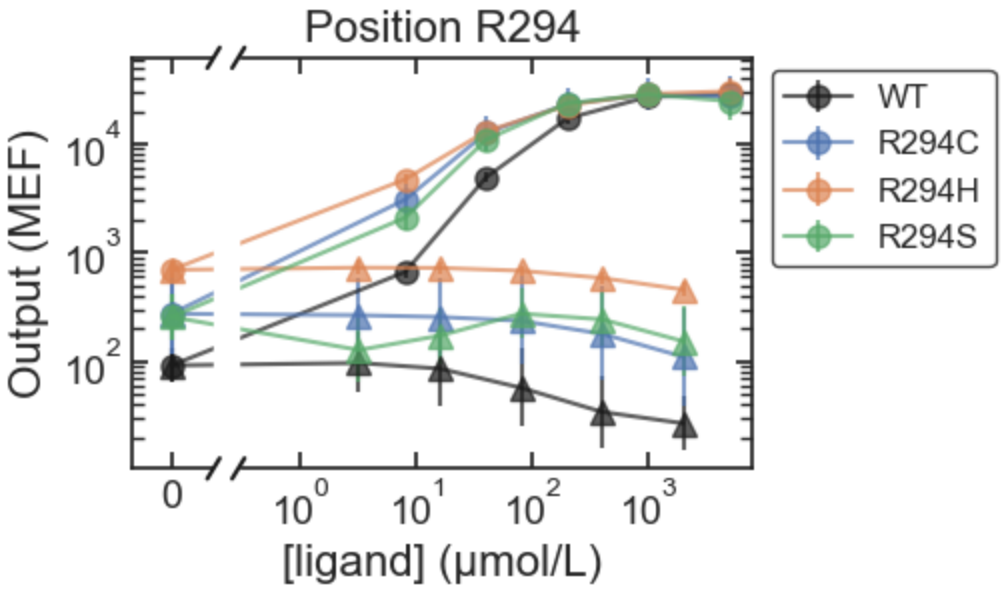

Position P285
Output (MEF)
[ligand] (μmol/L)
WT
P285A
P285L

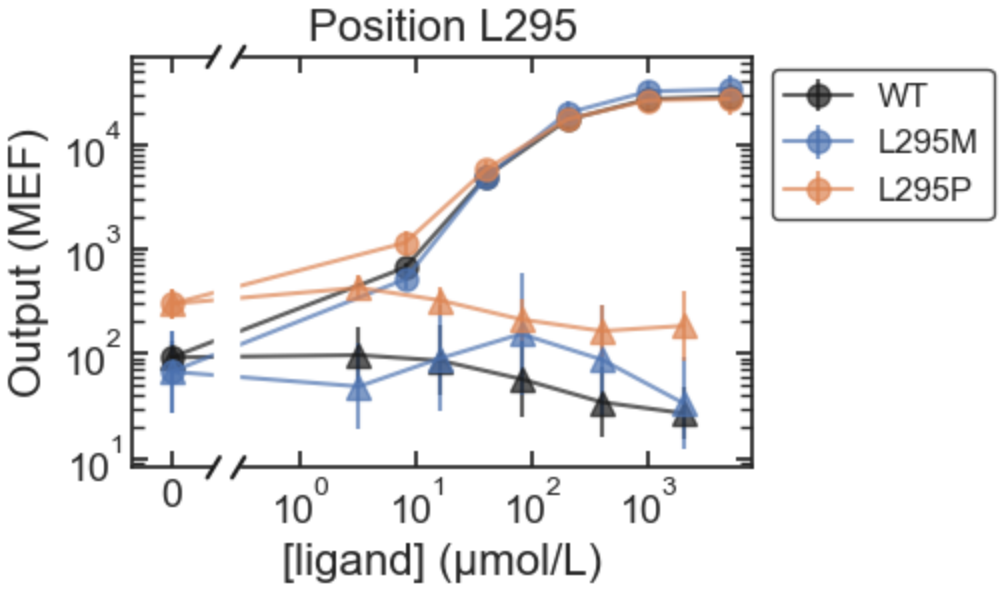

Position T288
Output (MEF)
[ligand] (μmol/L)
WT
T288N

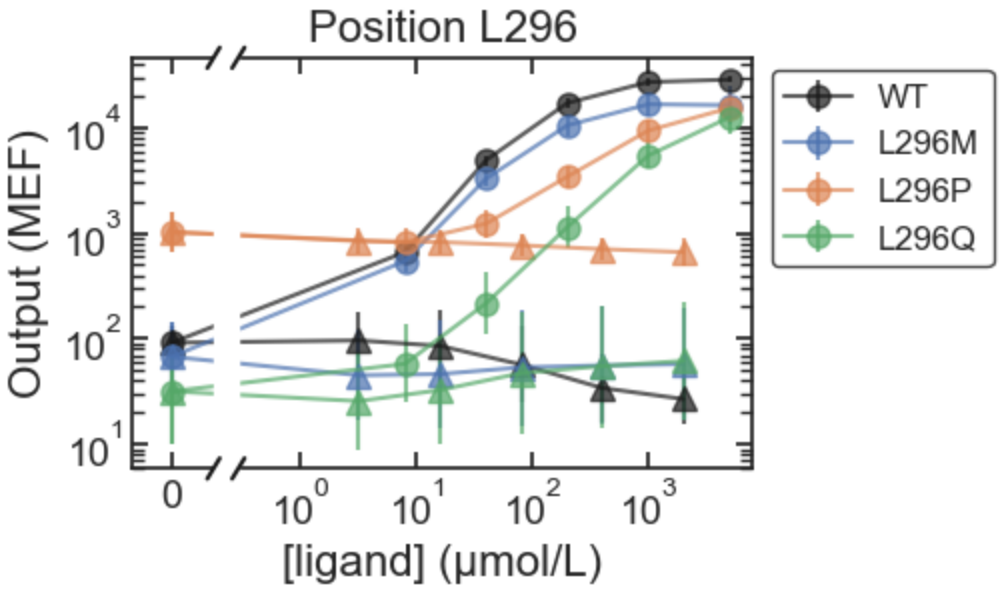

Position Q291
Output (MEF)
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L

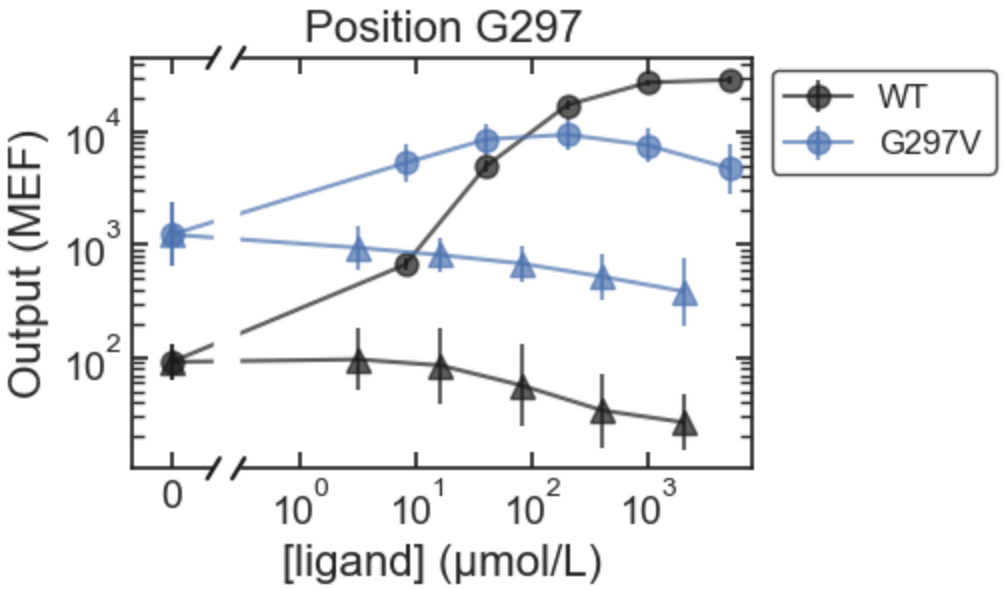

Position D292
Output (MEF)
[ligand] (μmol/L)
WT
D292E
D292G
D292N
D292V

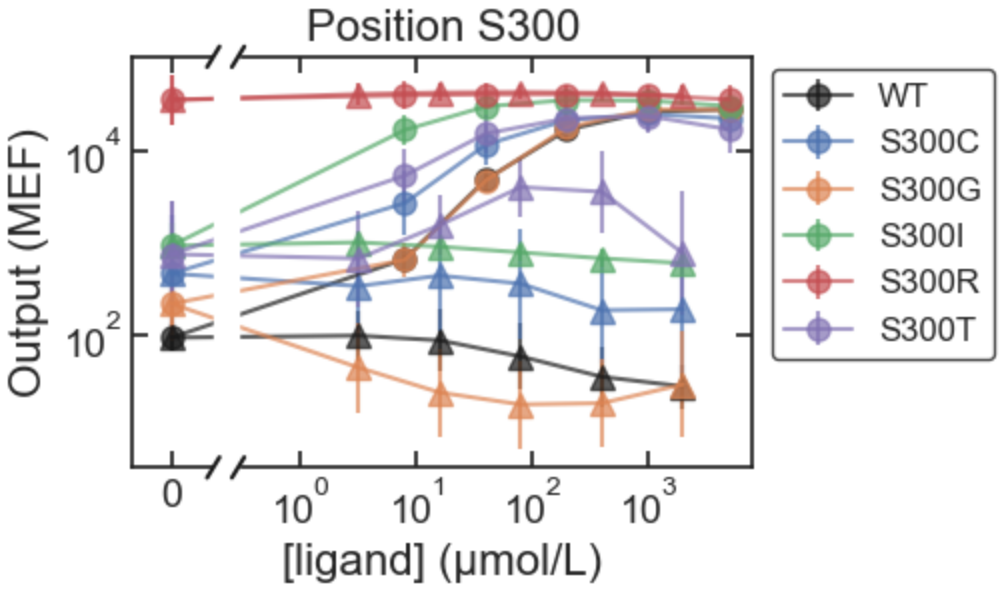

Position F293
Output (MEF)
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

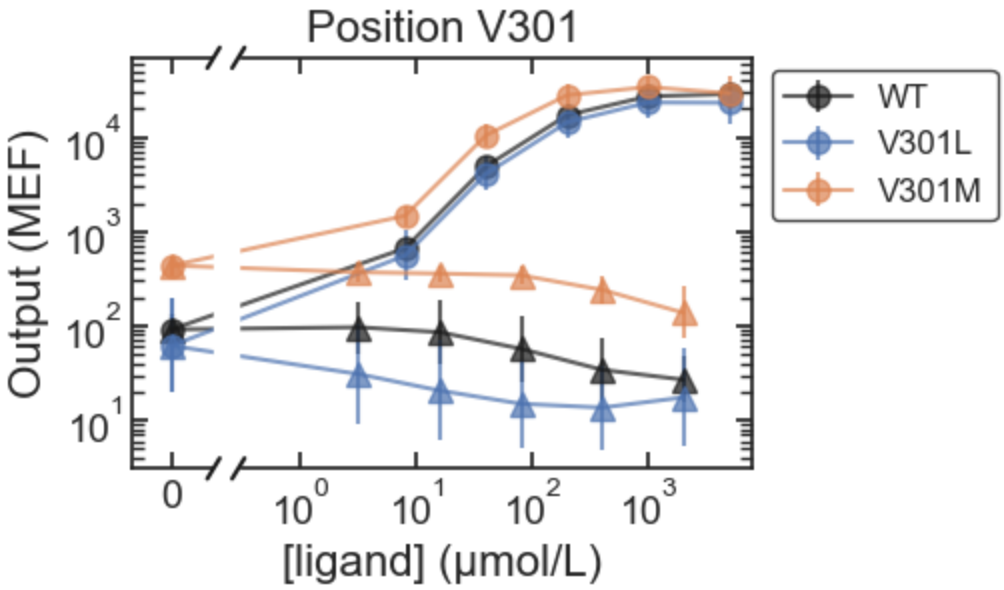

Position R294
Output (MEF)
[ligand] (μmol/L)
WT
R294C
R294H
R294S

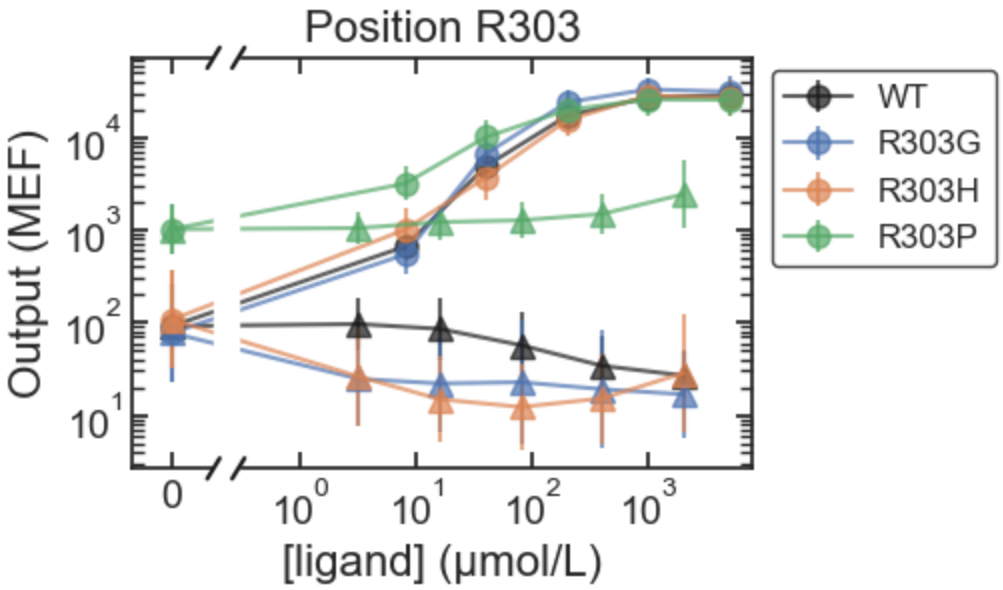

Position L295
Output (MEF)
[ligand] (μmol/L)
WT
L295M
L295P

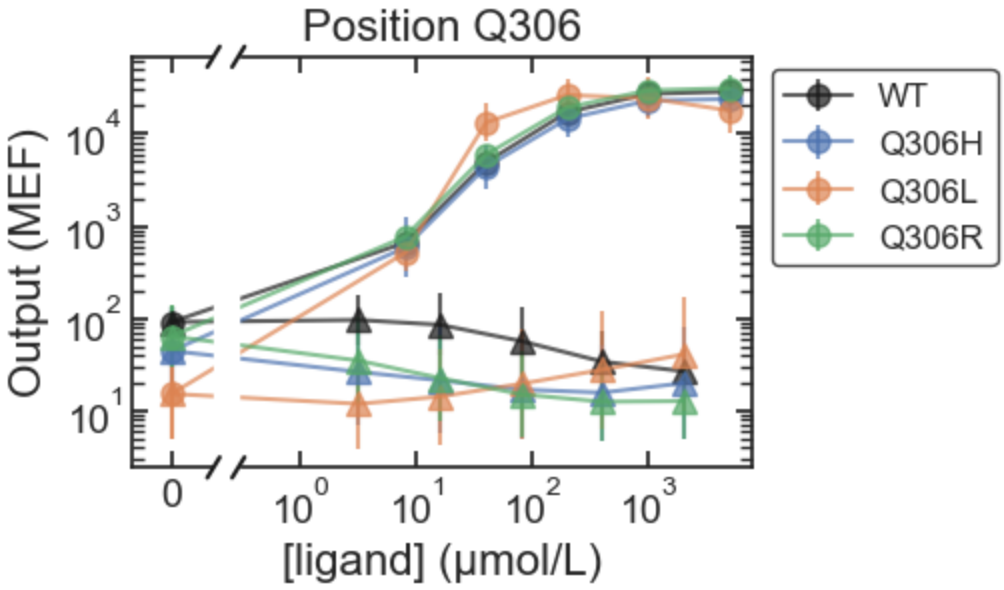

Position L296
Output (MEF)
[ligand] (μmol/L)
WT
L296M
L296P
L296Q

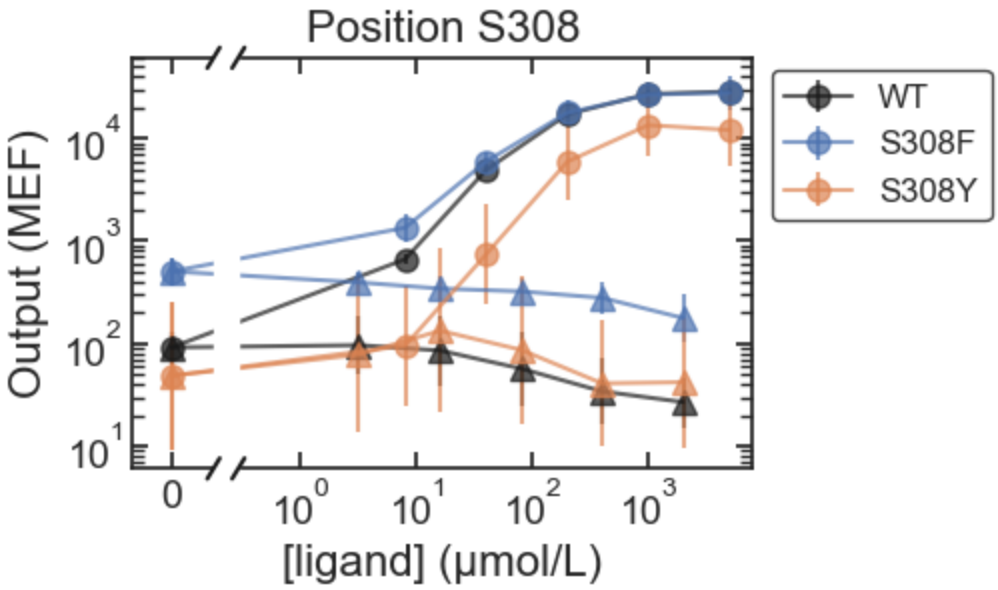

Position G297
Output (MEF)
[ligand] (μmol/L)
WT
G297V

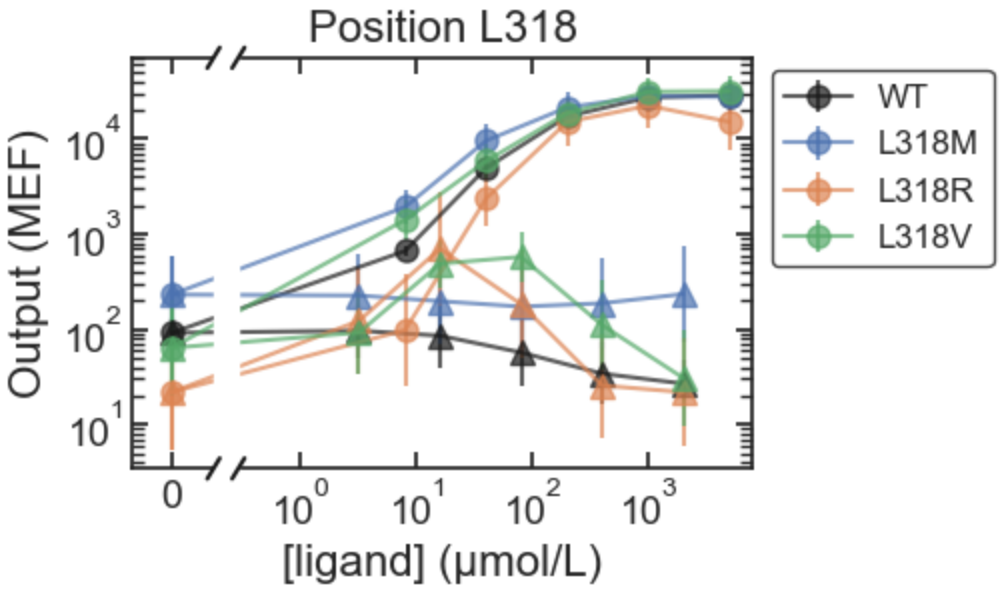

Position S300
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S300C
S300G
S300I
S300R
S300T

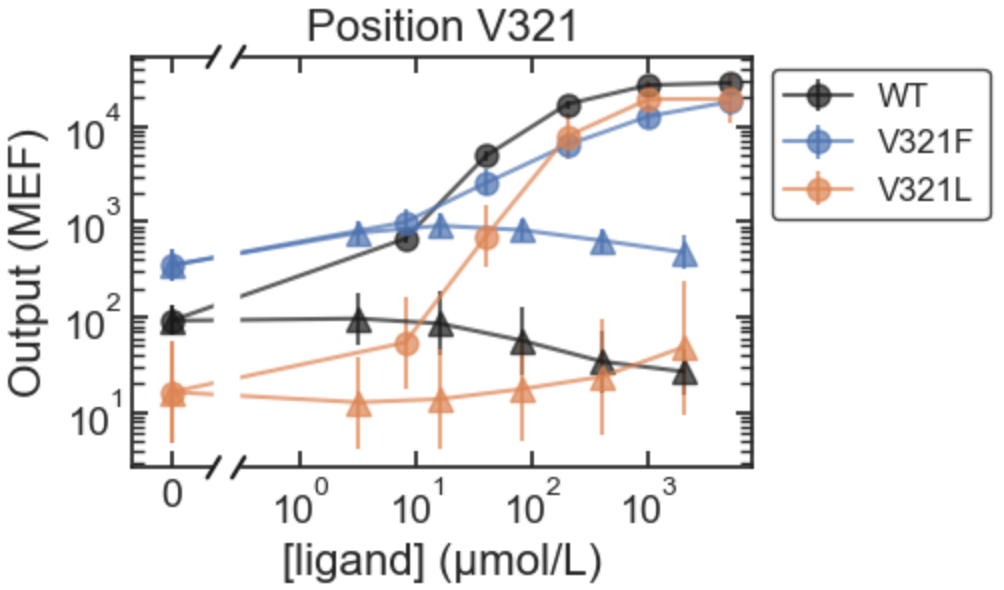

Position V301
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V301L
V301M

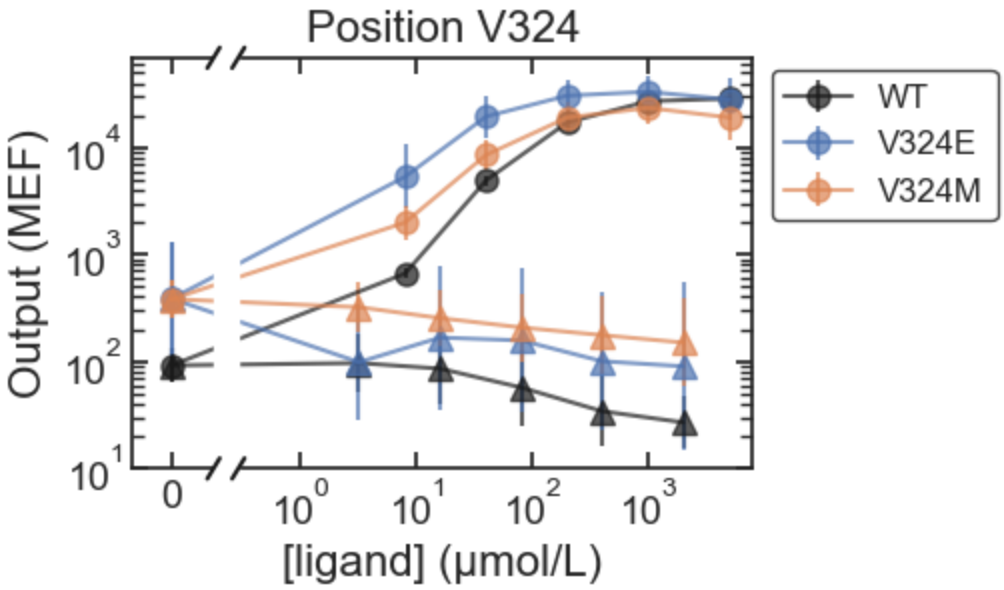

Position R303
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R303G
R303H
R303P

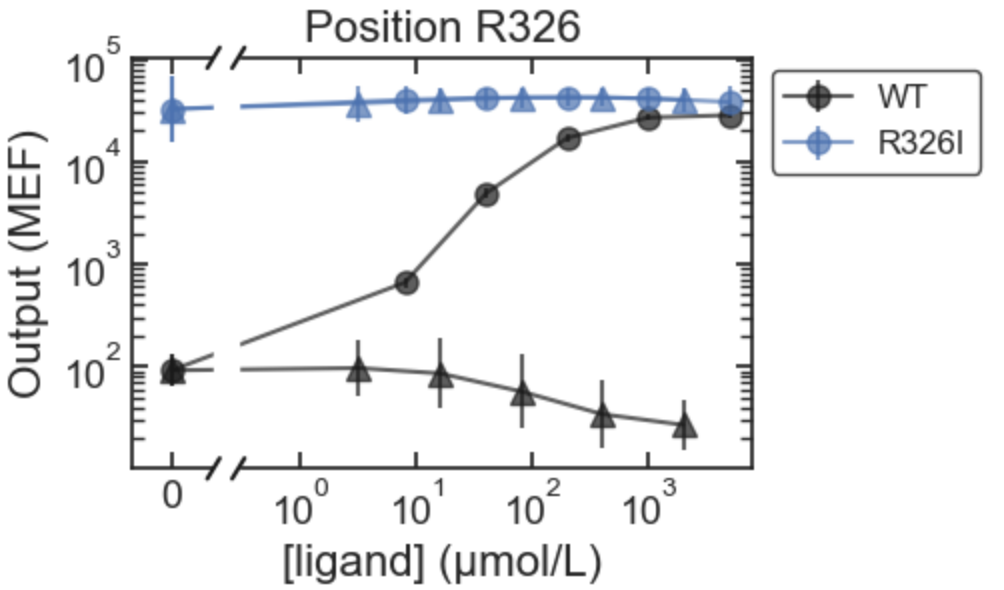

Position Q306
Output (MEF)
[ligand] (µmol/L)
WT
Q306H
Q306L
Q306R

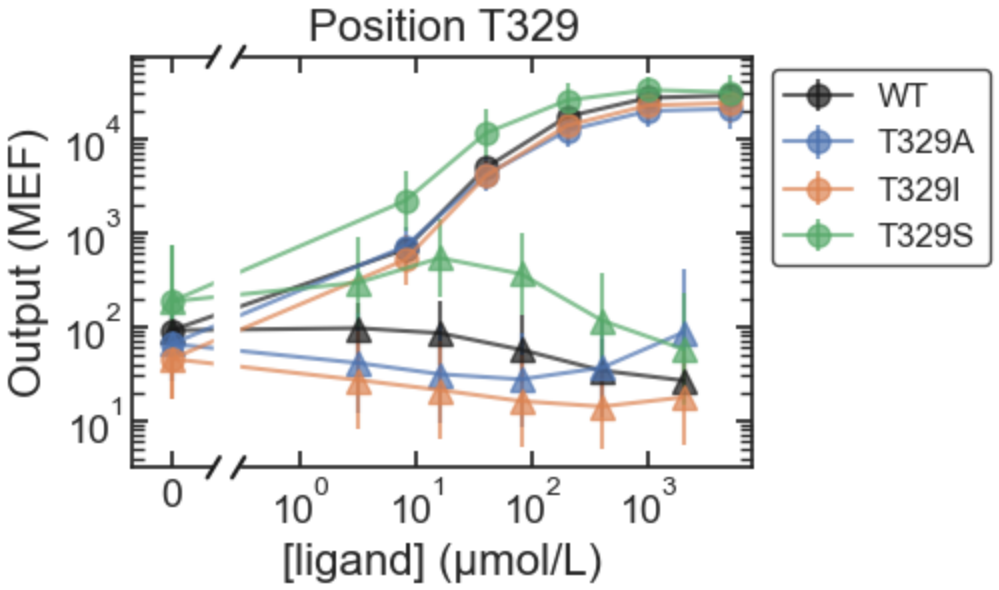

Position S308
Output (MEF)
[ligand] (µmol/L)
WT
S308F
S308Y

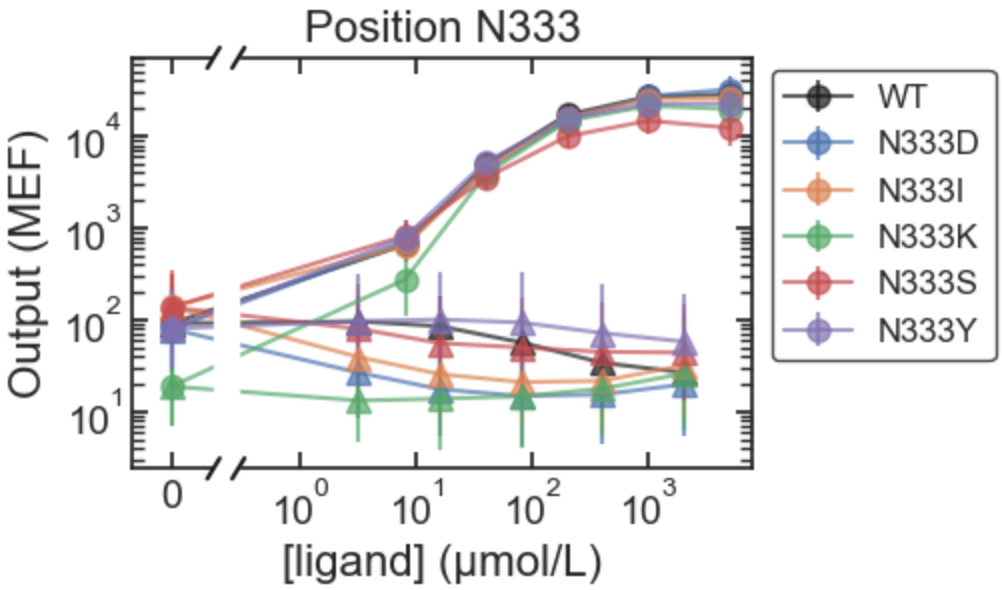

Position L318
Output (MEF)
[ligand] (µmol/L)
WT
L318M
L318R
L318V

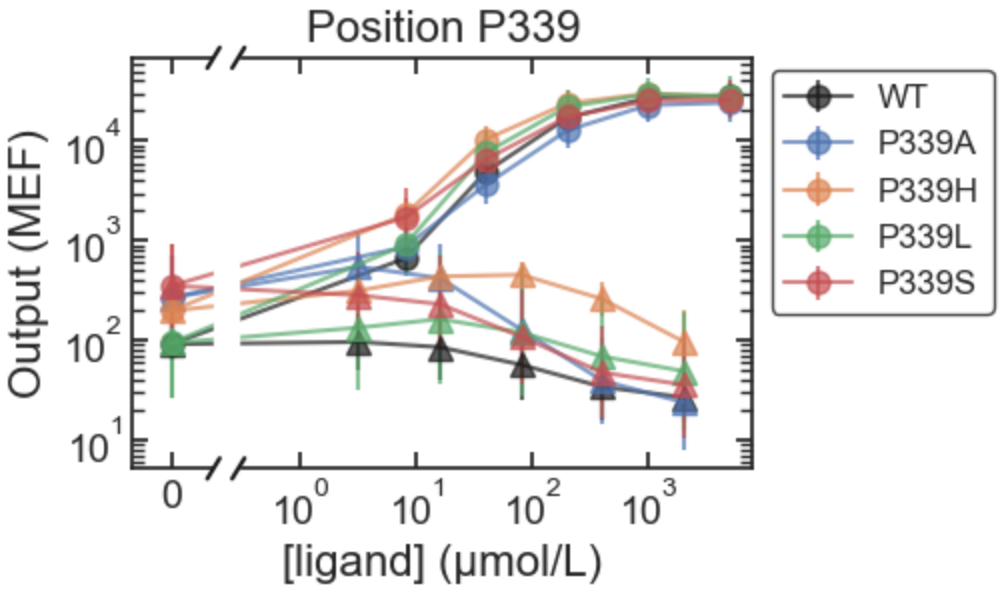

Position V321
Output (MEF)
[ligand] (μmol/L)
WT
V321F
V321L

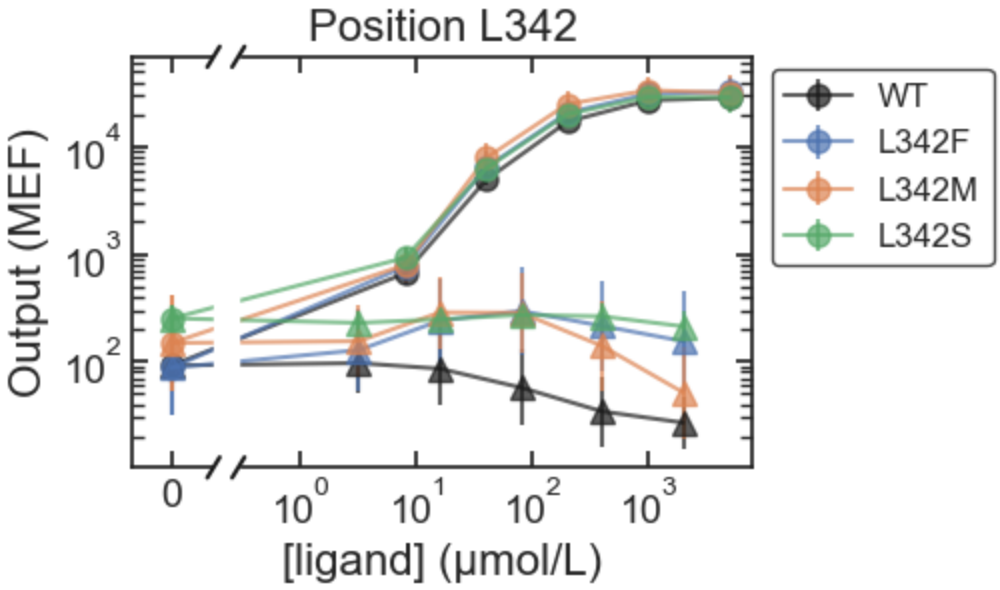

Position V324
Output (MEF)
[ligand] (μmol/L)
WT
V324E
V324M

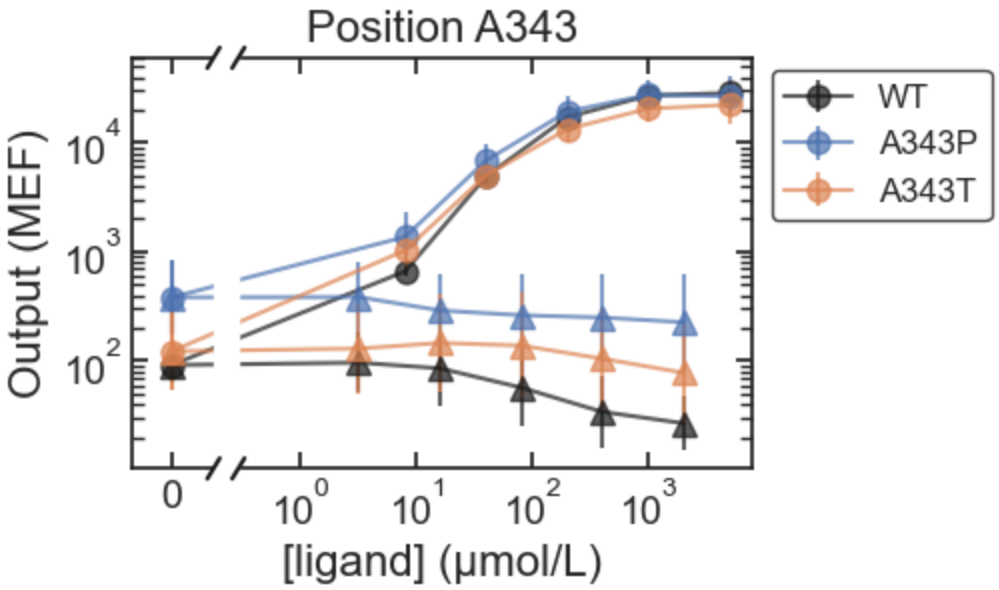

Position R326
Output (MEF)
[ligand] (μmol/L)
WT
R326I

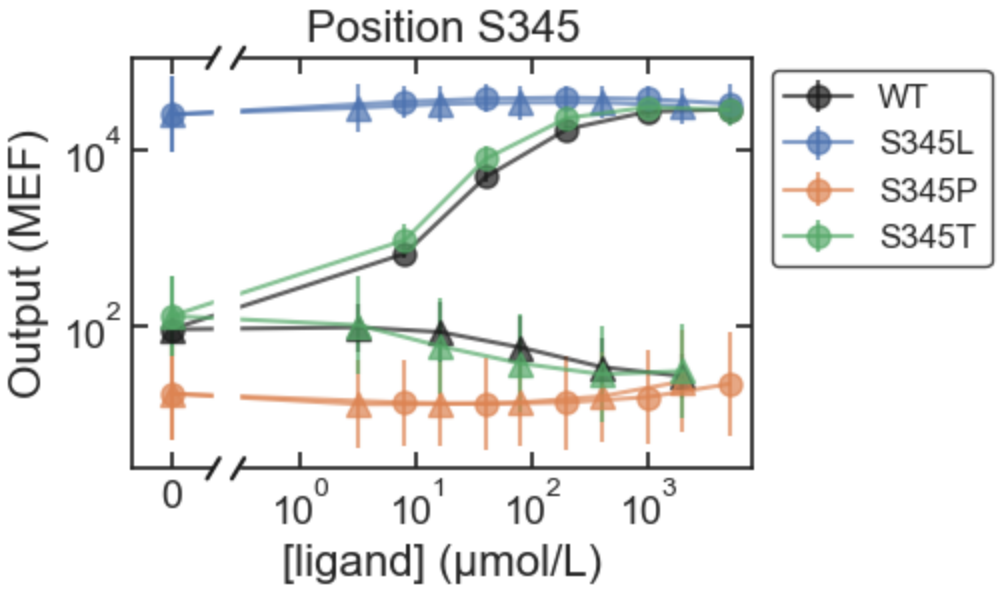

Position T329
Output (MEF)
[ligand] (μmol/L)
WT
T329A
T329I
T329S

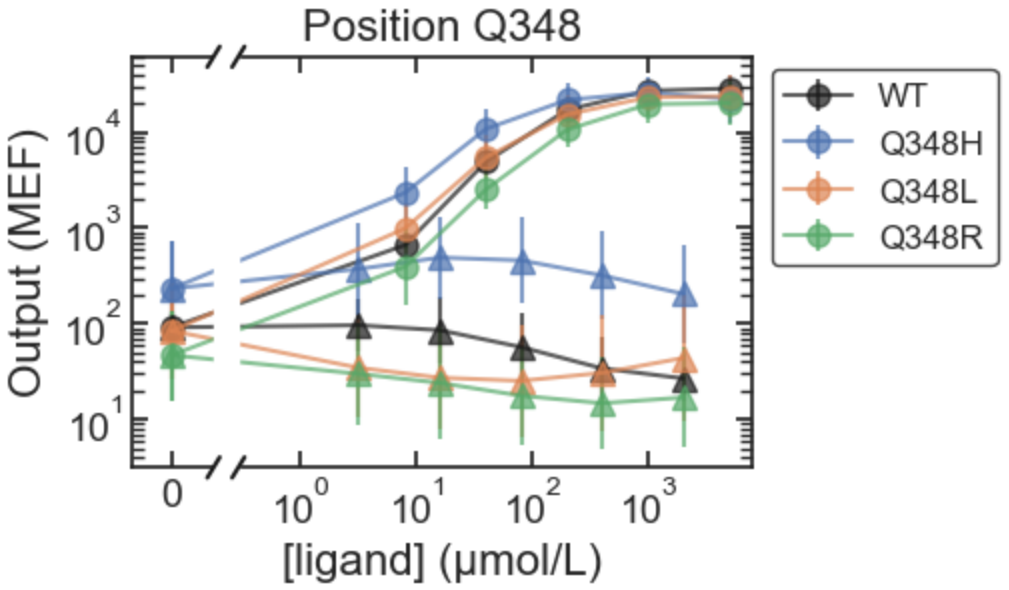

Position N333
Output (MEF)
[ligand] (μmol/L)
WT
N333D
N333I
N333K
N333S
N333Y

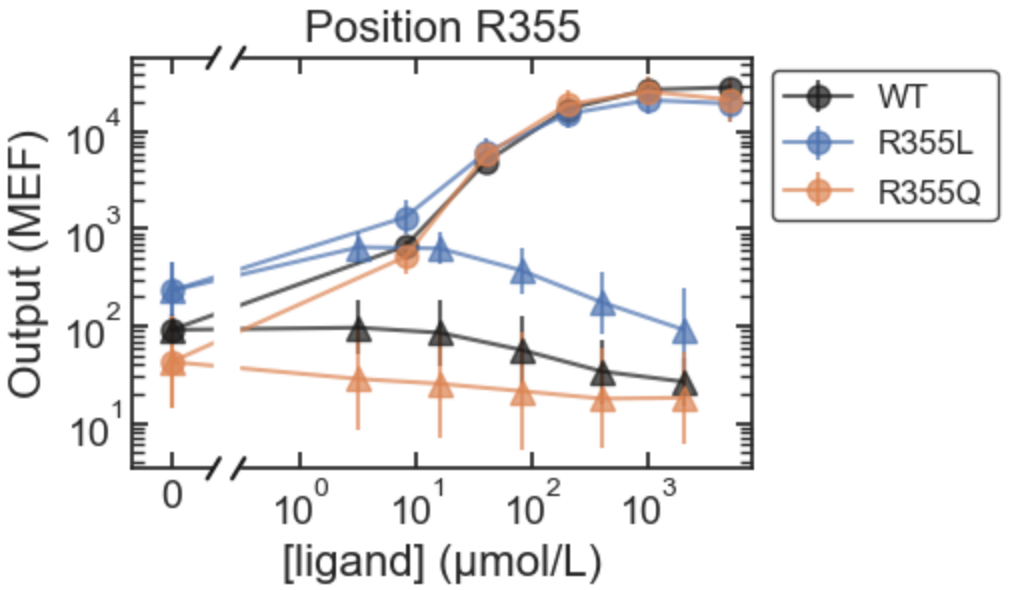

Position P339
Output (MEF)
[ligand] (μmol/L)
WT
P339A
P339H
P339L
P339S

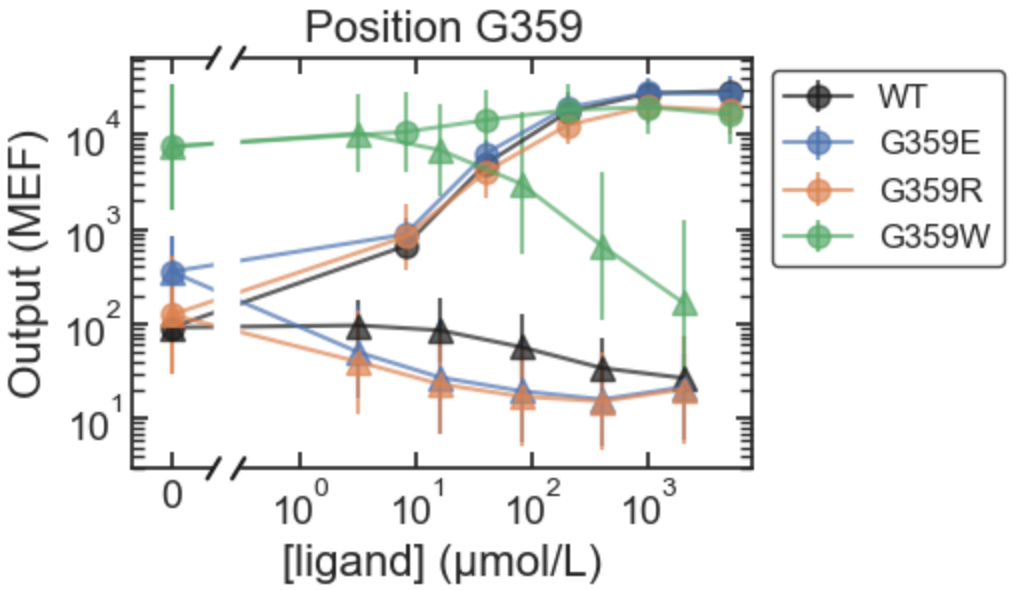

Position L342
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L342F
L342M
L342S


Position A343
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A343P
A343T

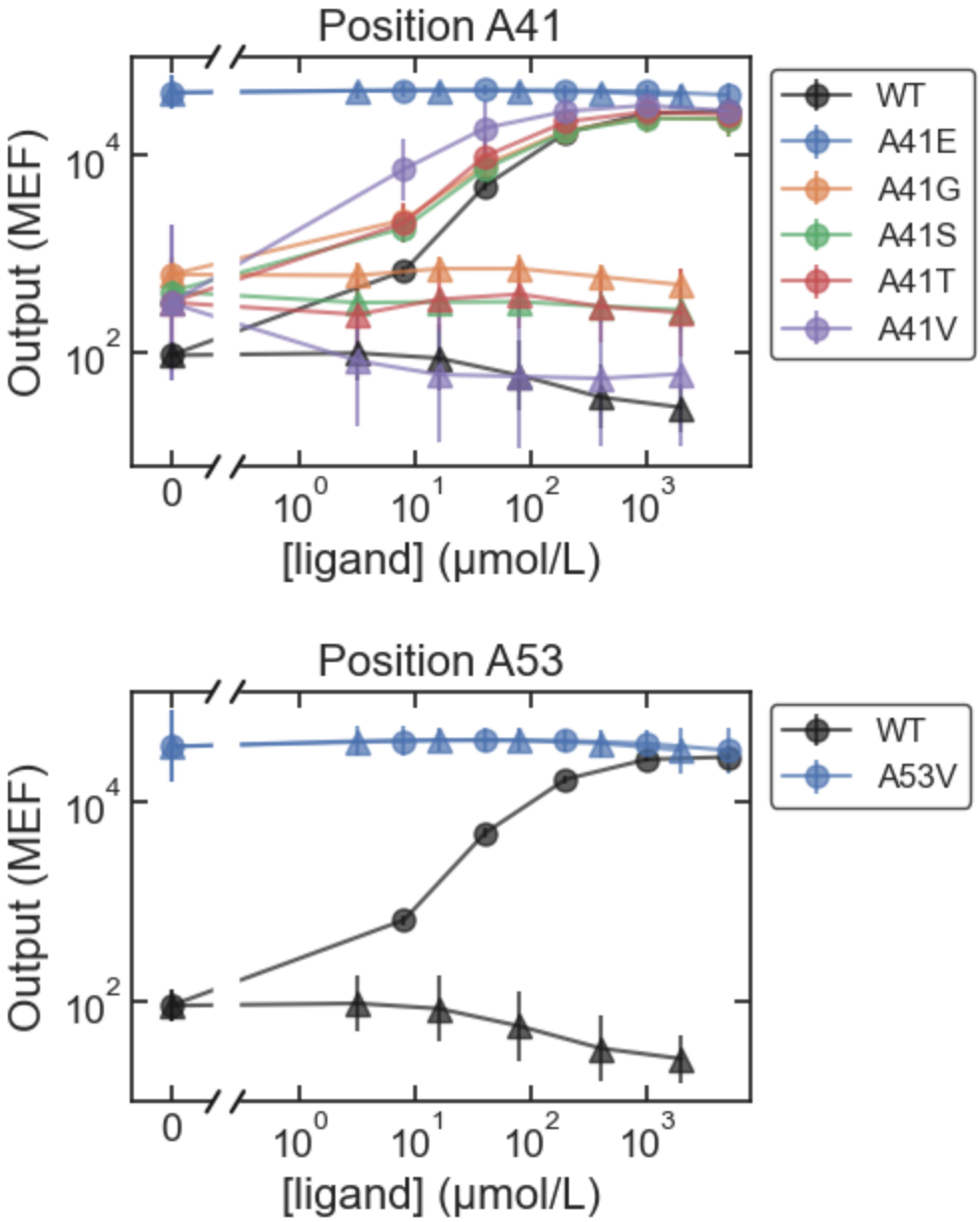

Position S345
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S345L
S345P
S345T

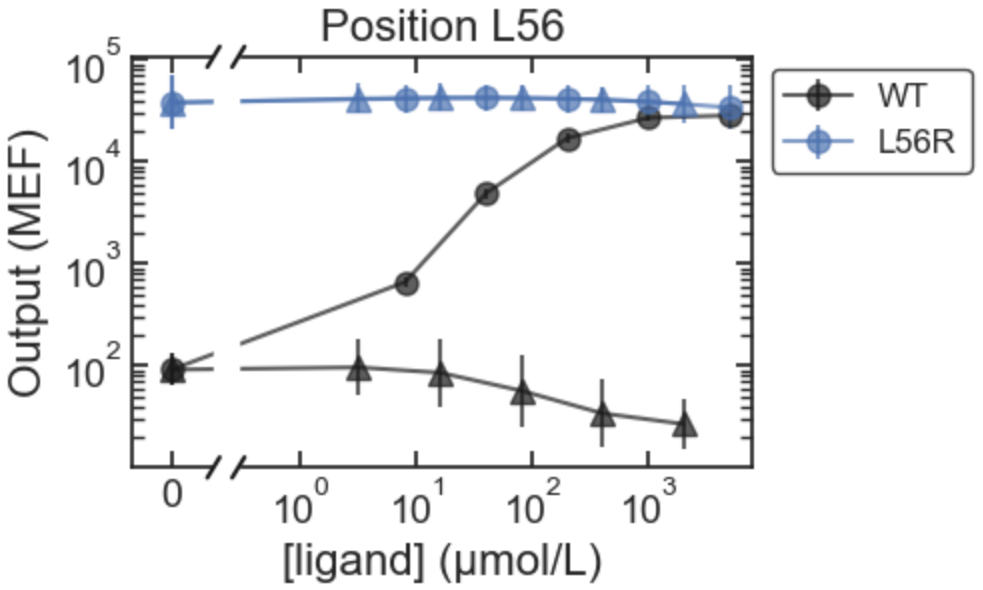

Position Q348
Output (MEF)
[ligand] (μmol/L)
WT
Q348H
Q348L
Q348R

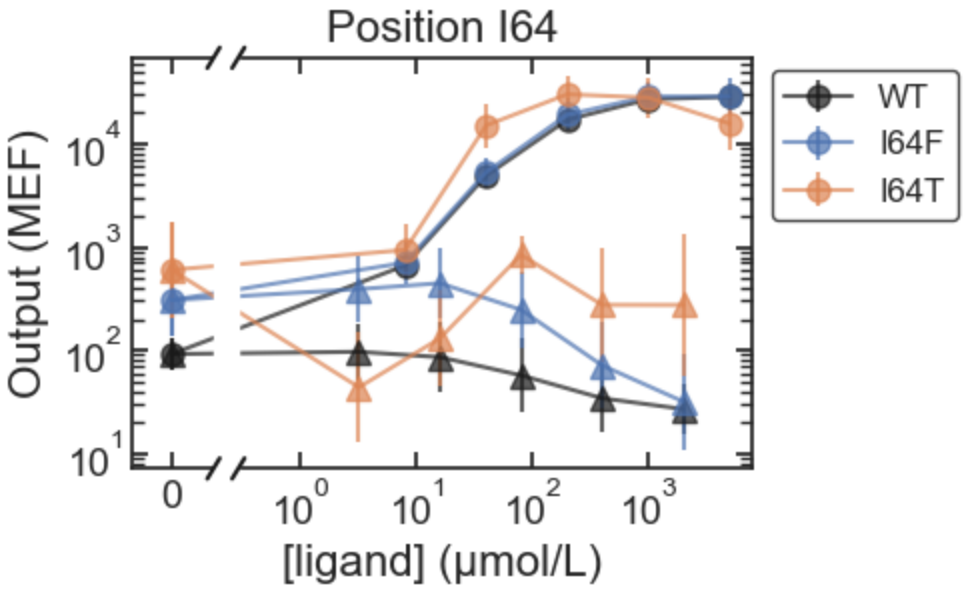

Position R355
Output (MEF)
[ligand] (μmol/L)
WT
R355L
R355Q

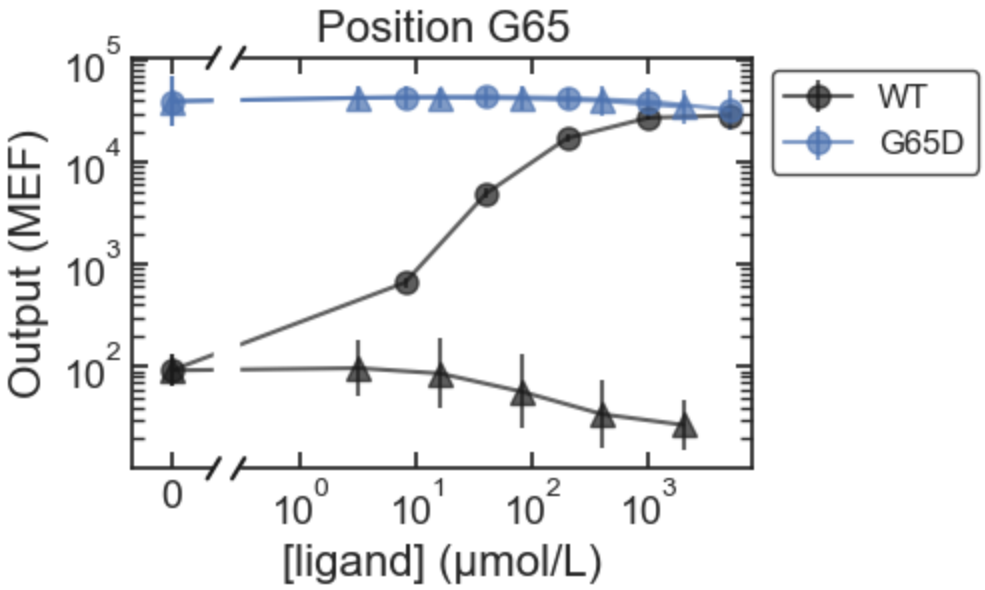

Position G359
Output (MEF)
[ligand] (μmol/L)
WT
G359E
G359R
G359W

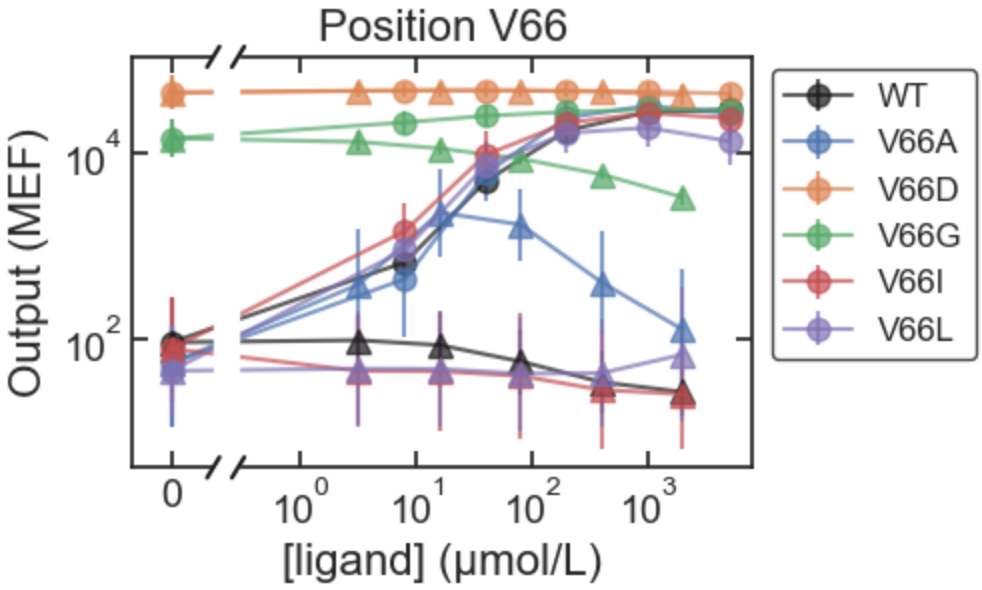
MWC_only

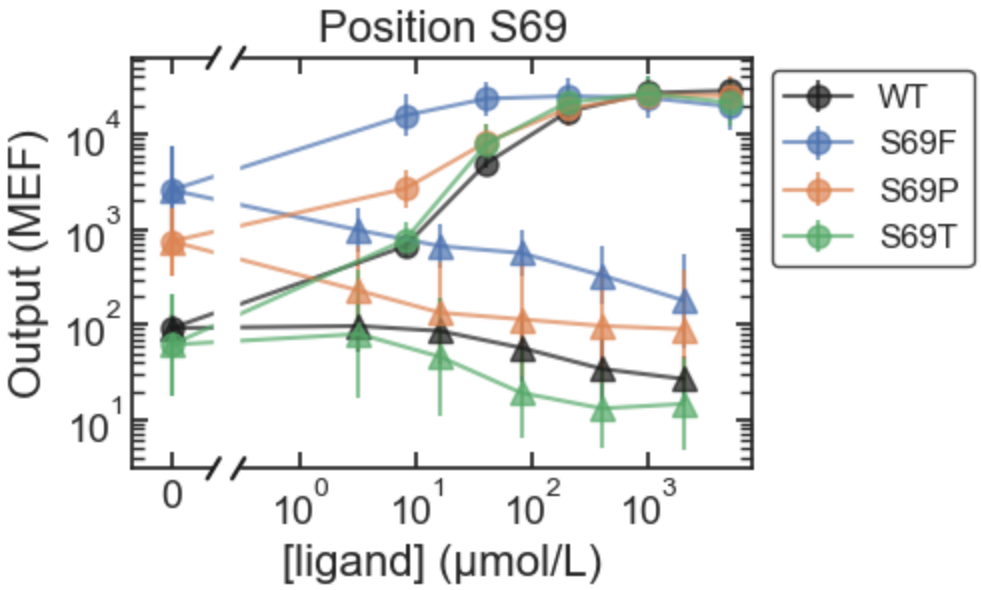

Position A41
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A41E
A41G
A41S
A41T
A41V
Position A53
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A53V

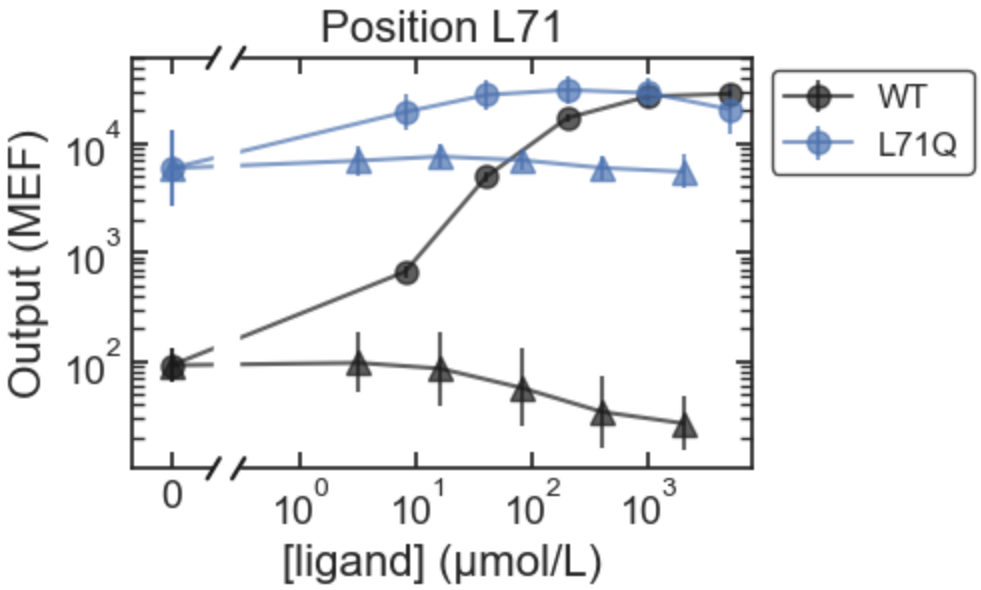

Position L56
Output (MEF)
[ligand] (μmol/L)
WT
L56R

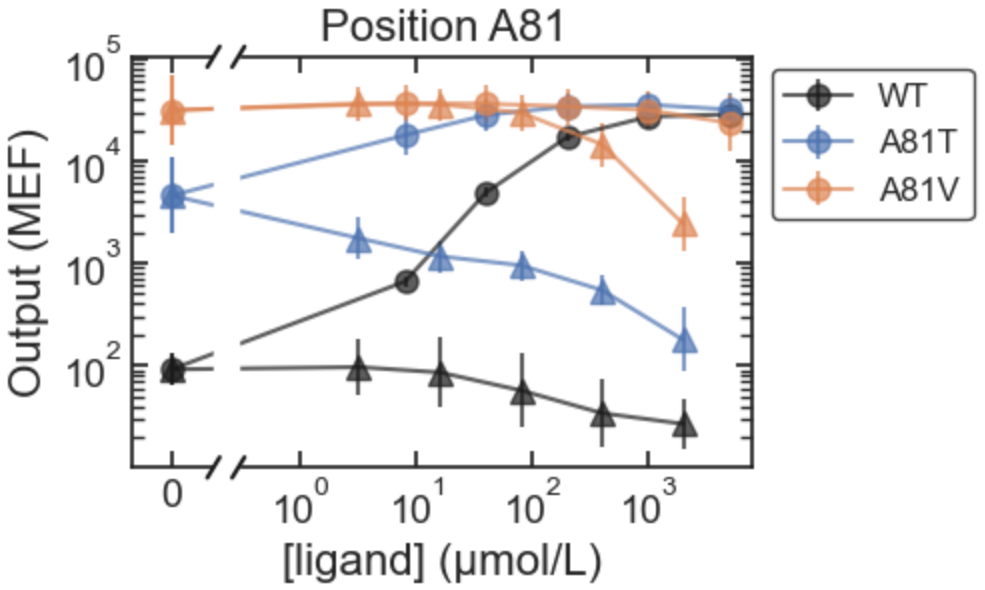

Position I64
Output (MEF)
[ligand] (μmol/L)
WT
I64F
I64T

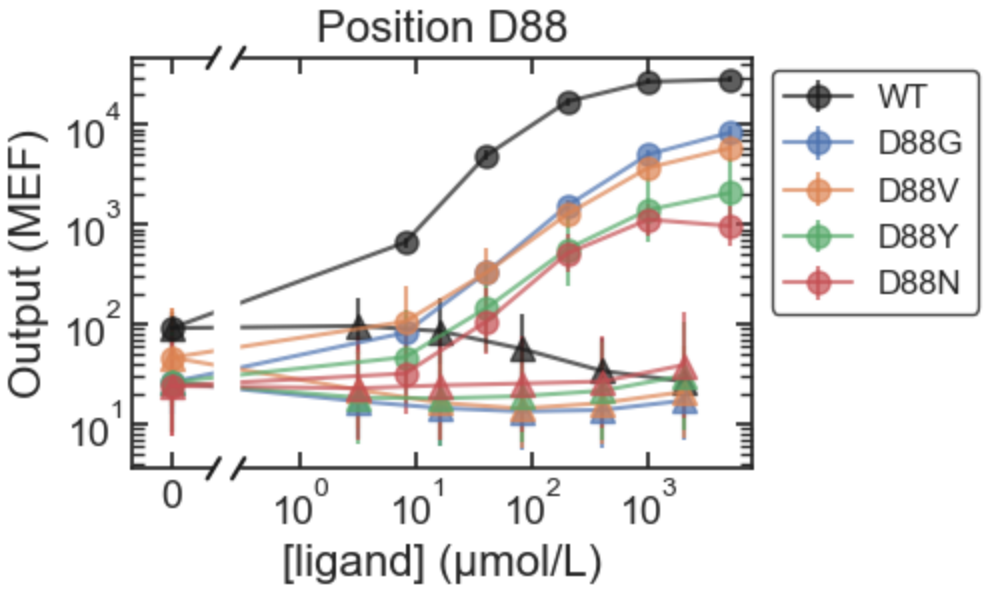

Position G65
Output (MEF)
[ligand] (μmol/L)
WT
G65D

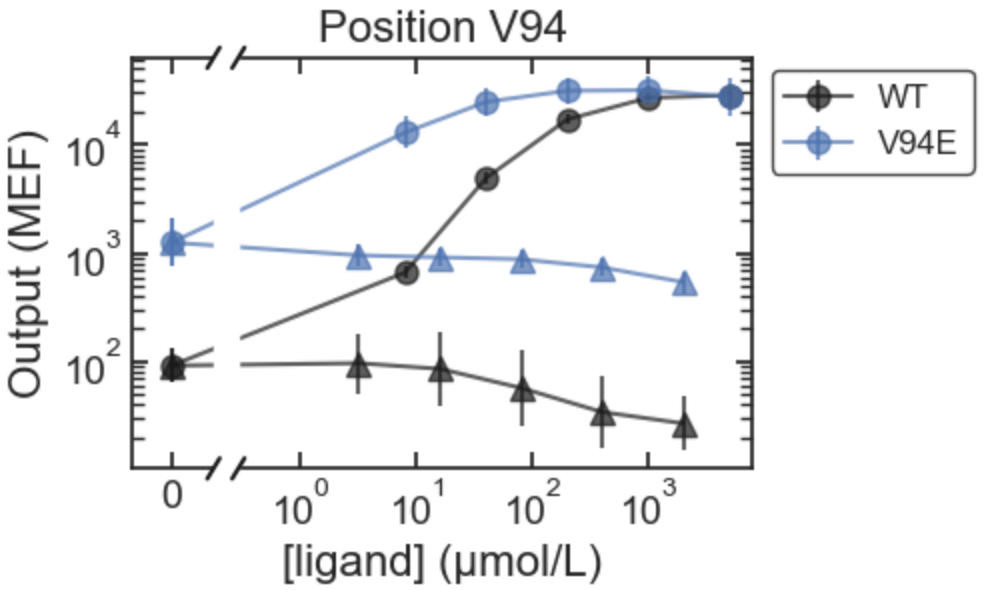

Position V66
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V66A
V66D
V66G
V66I
V66L

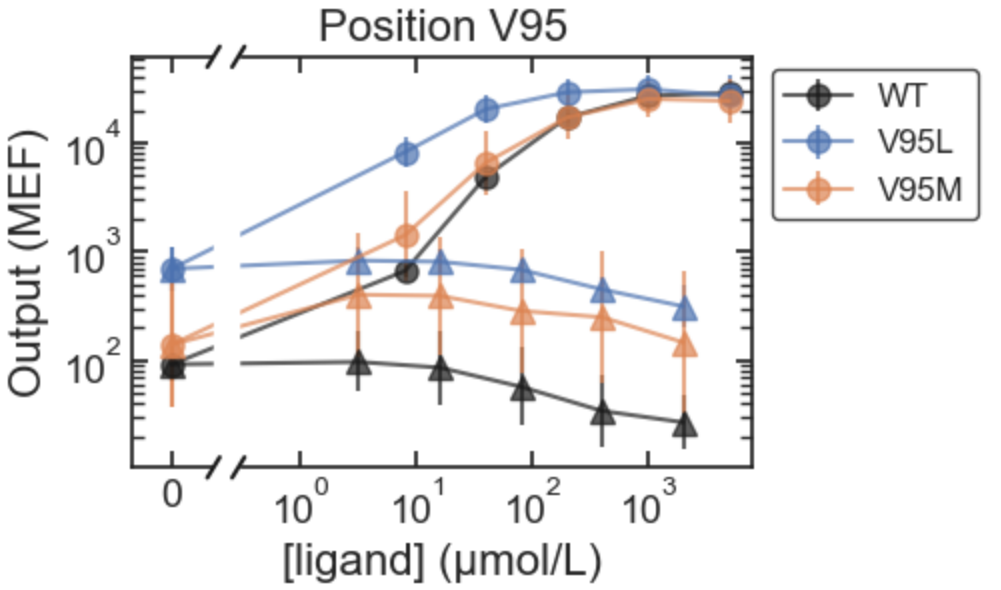

Position S69
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S69F
S69P
S69T

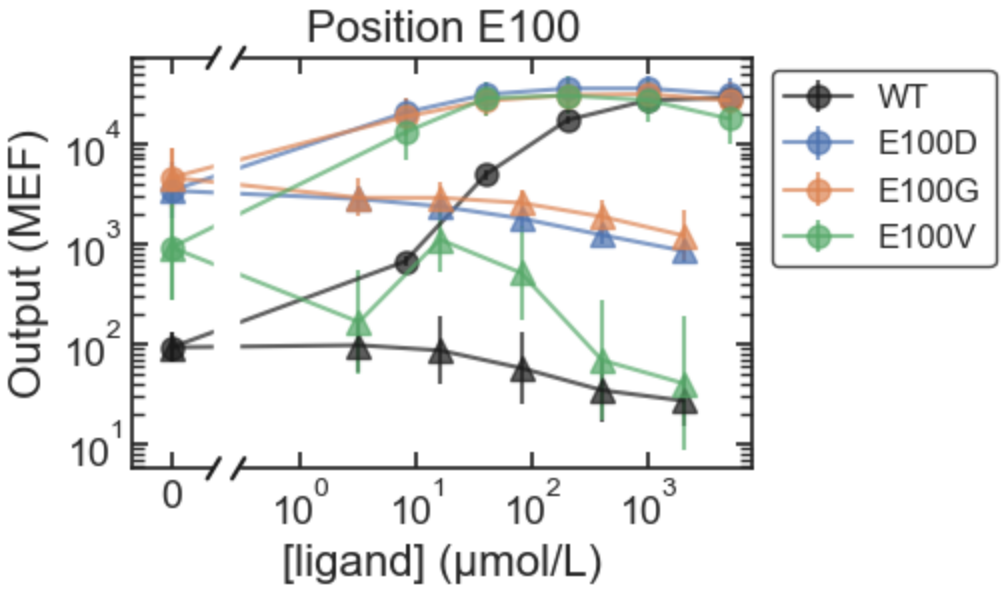

Position L71
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L71Q

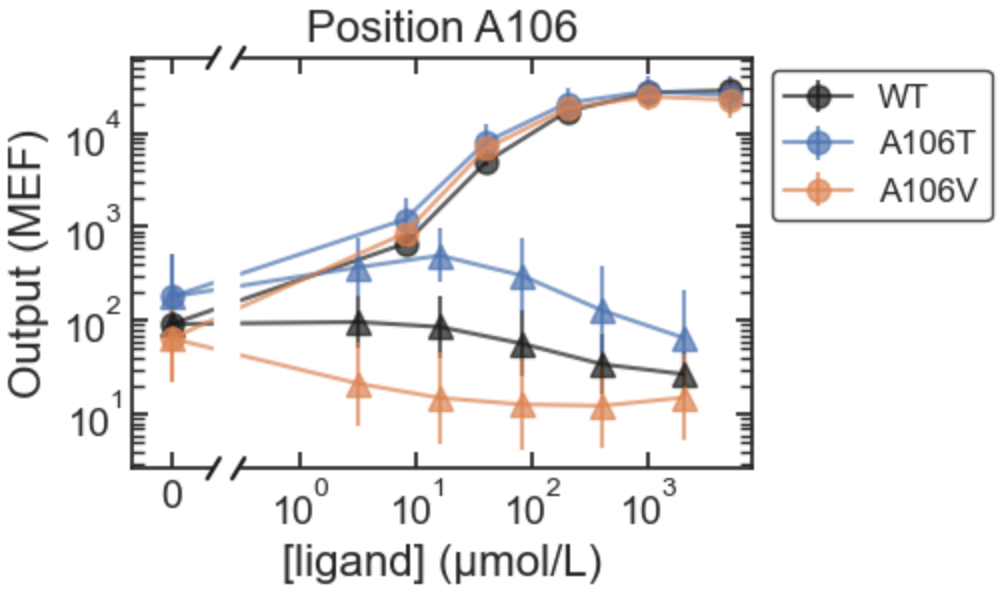

Position A81
Output (MEF)
[ligand] (μmol/L)
WT
A81T
A81V

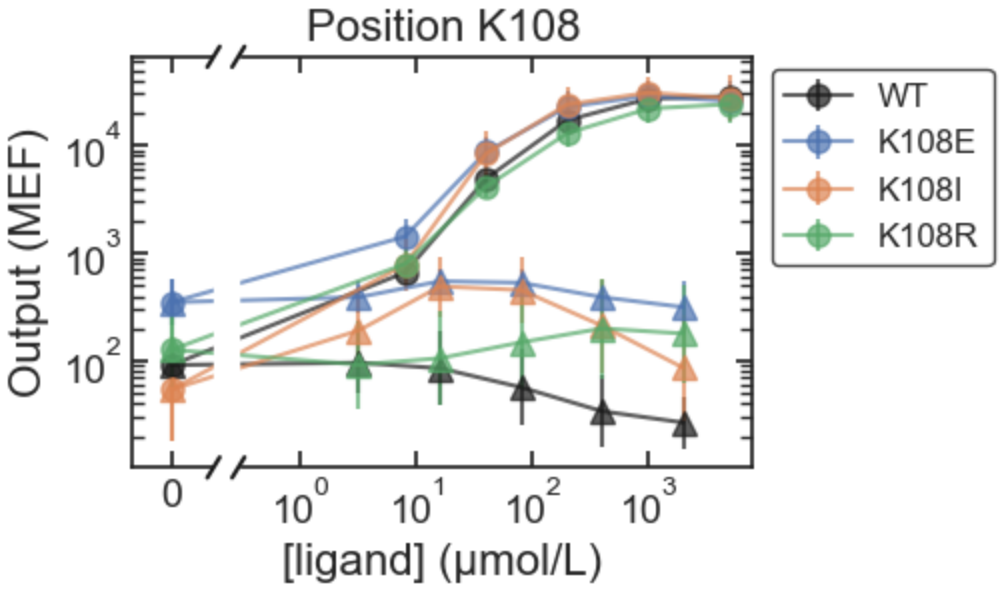

Position D88
Output (MEF)
[ligand] (μmol/L)
WT
D88G
D88V
D88Y
D88N

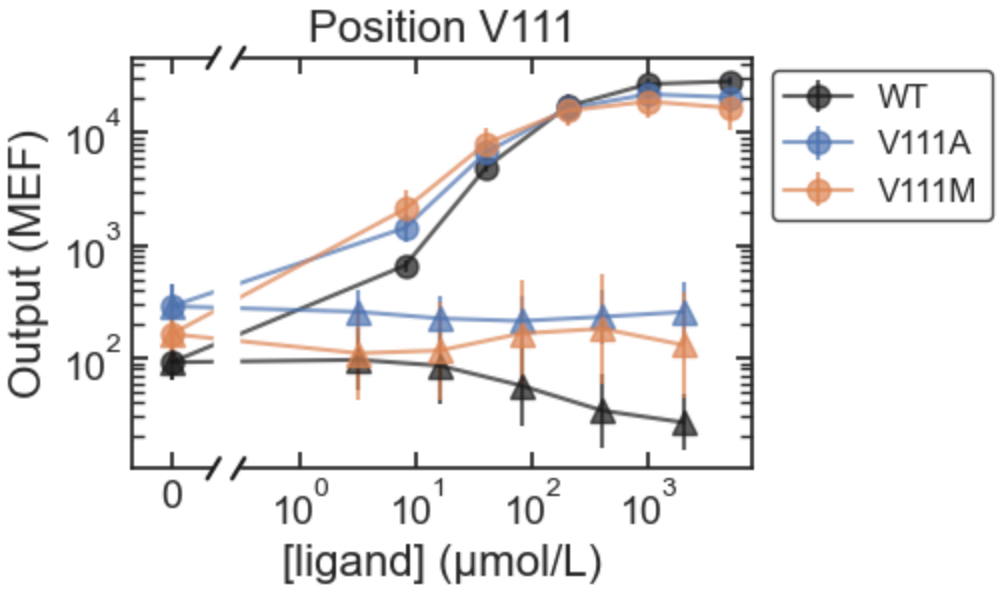

Position V94
Output (MEF)
[ligand] (μmol/L)
WT
V94E

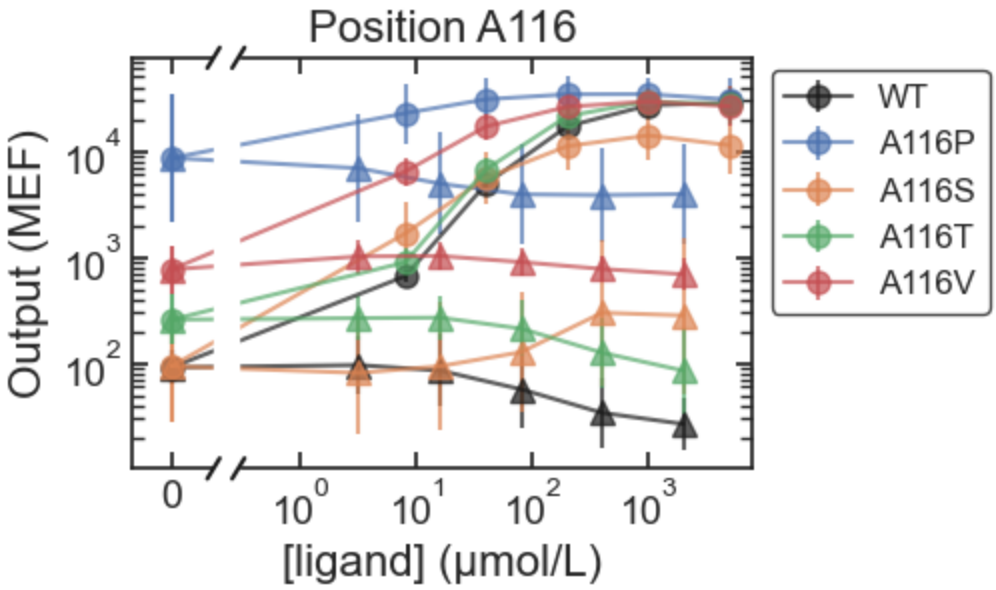

Position V95
Output (MEF)
[ligand] (μmol/L)
WT
V95L
V95M

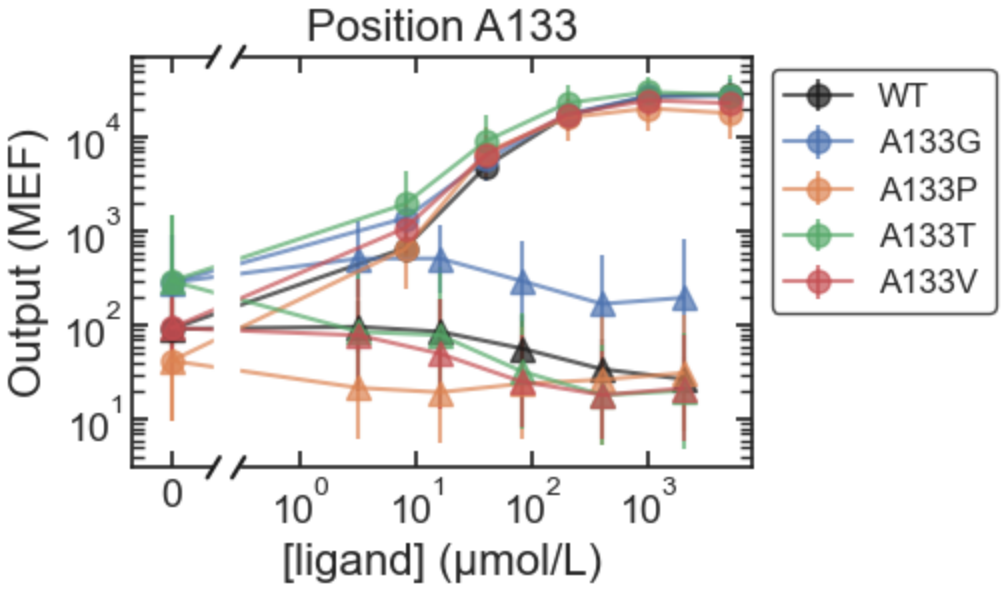

Position E100
Output (MEF)
[ligand] (μmol/L)
WT
E100D
E100G
E100V

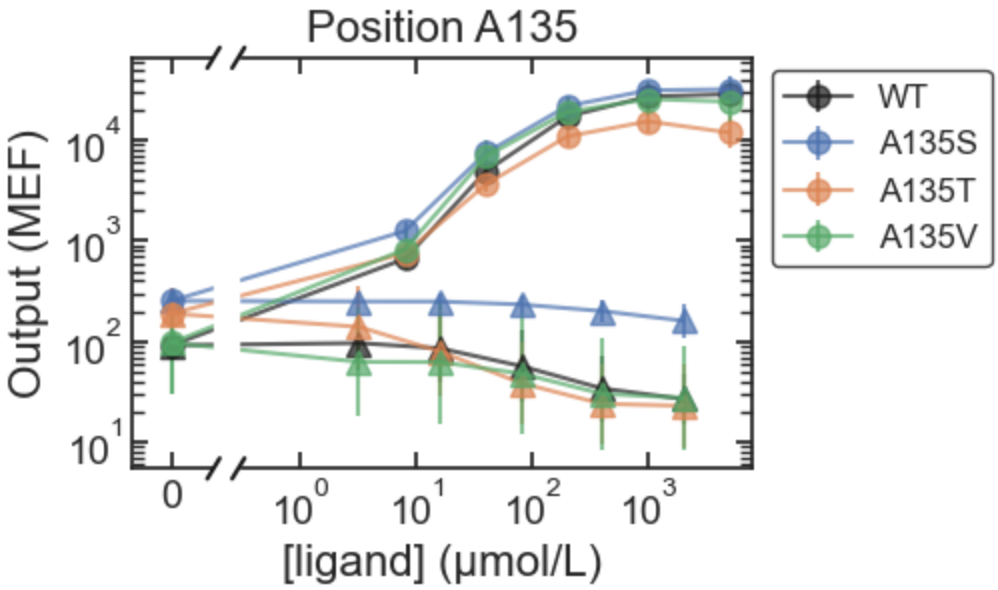

Position A106
Output (MEF)
[ligand] (μmol/L)
WT
A106T
A106V

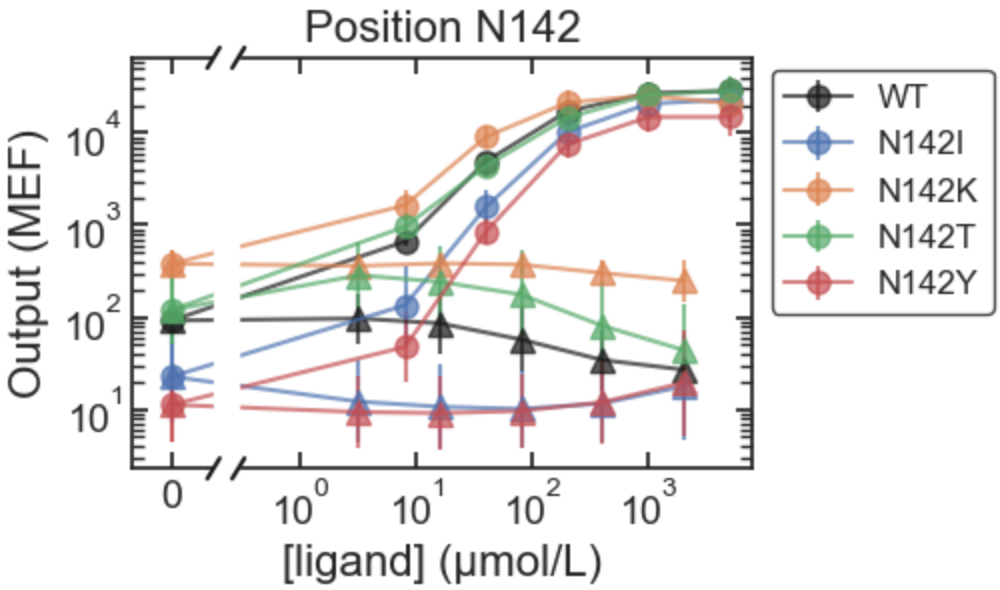

Position K108
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K108E
K108I
K108R

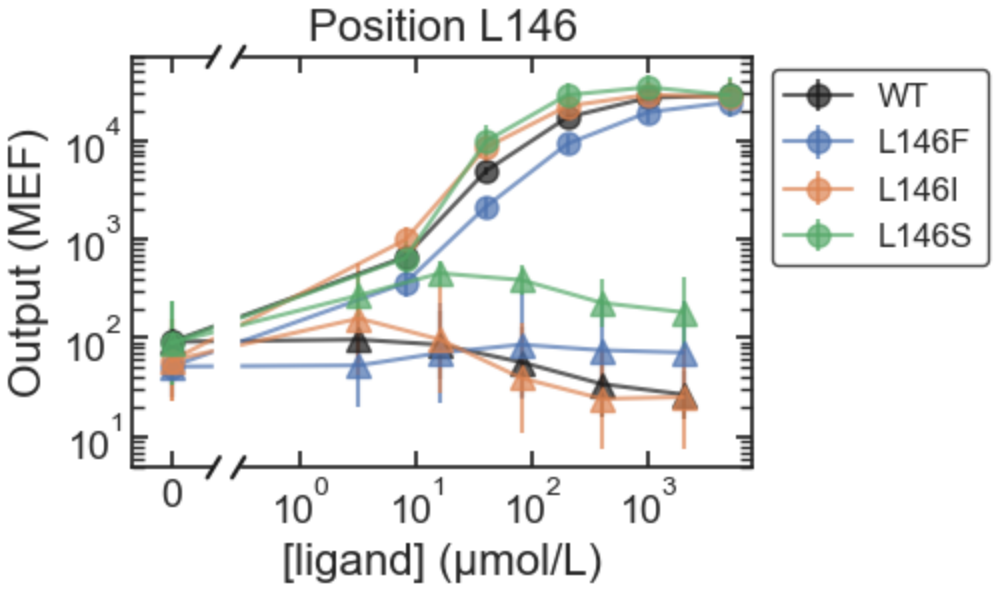

Position V111
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V111A
V111M

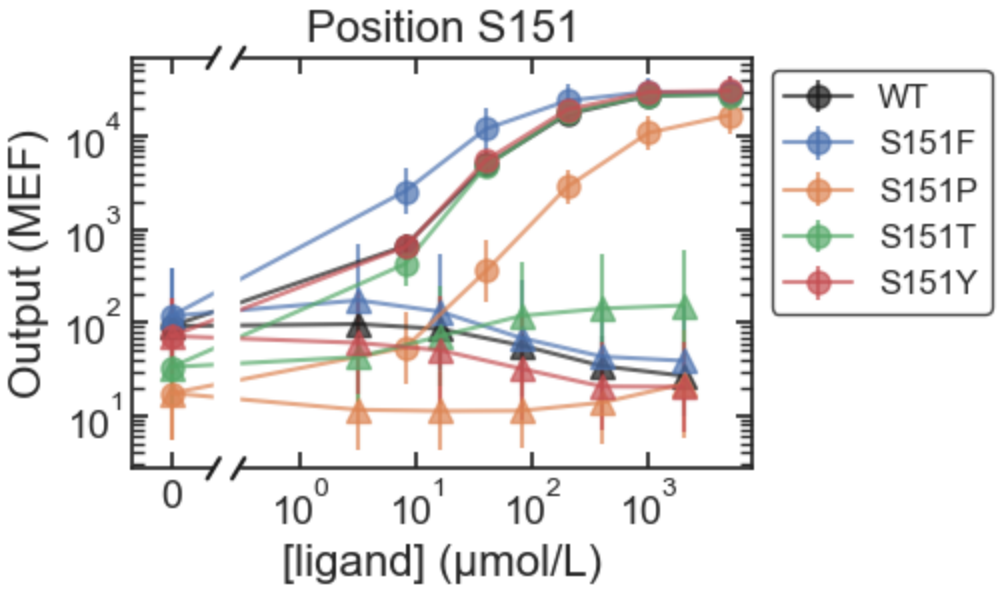

Position A116
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A116P
A116S
A116T
A116V

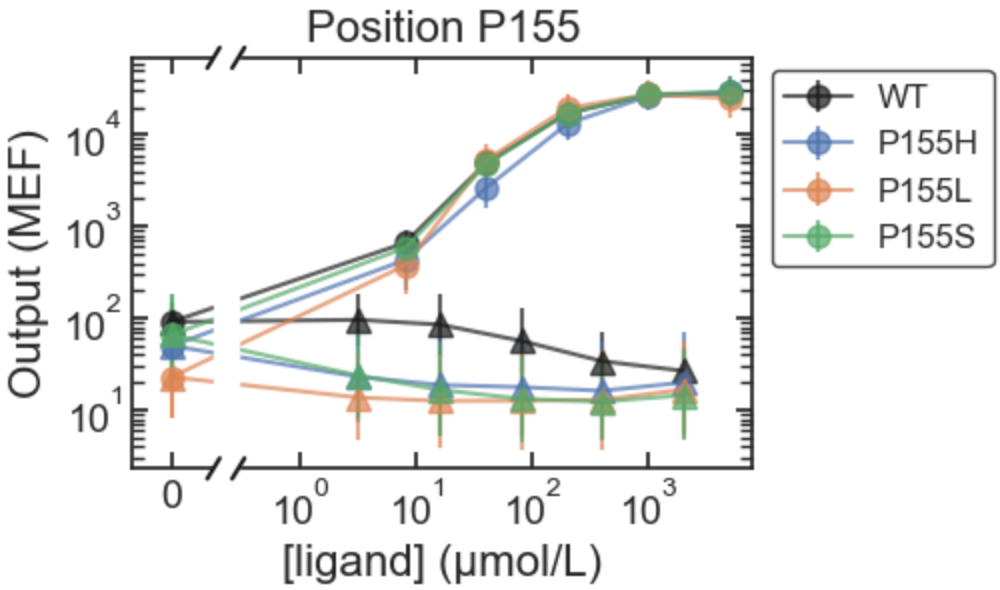

Position A133
Output (MEF)
[ligand] (μmol/L)
WT
A133G
A133P
A133T
A133V

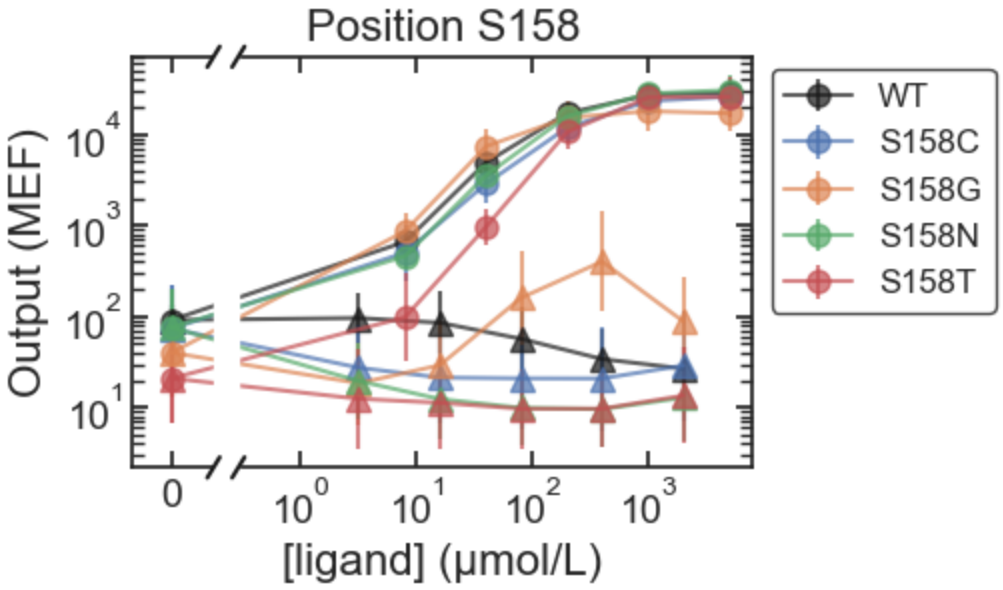

Position A135
Output (MEF)
[ligand] (μmol/L)
WT
A135S
A135T
A135V

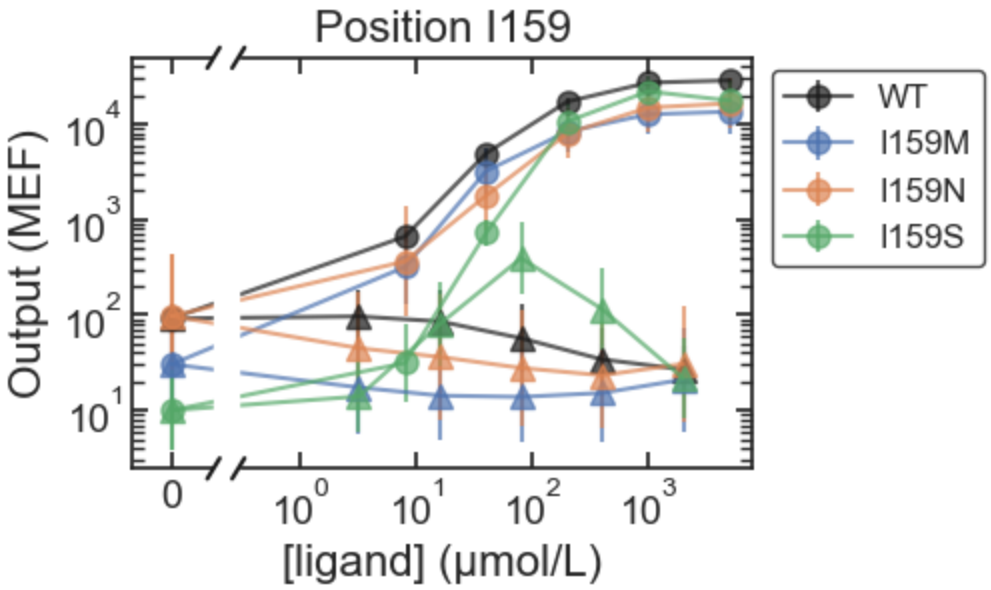

Position N142
Output (MEF)
[ligand] (μmol/L)
WT
N142I
N142K
N142T
N142Y

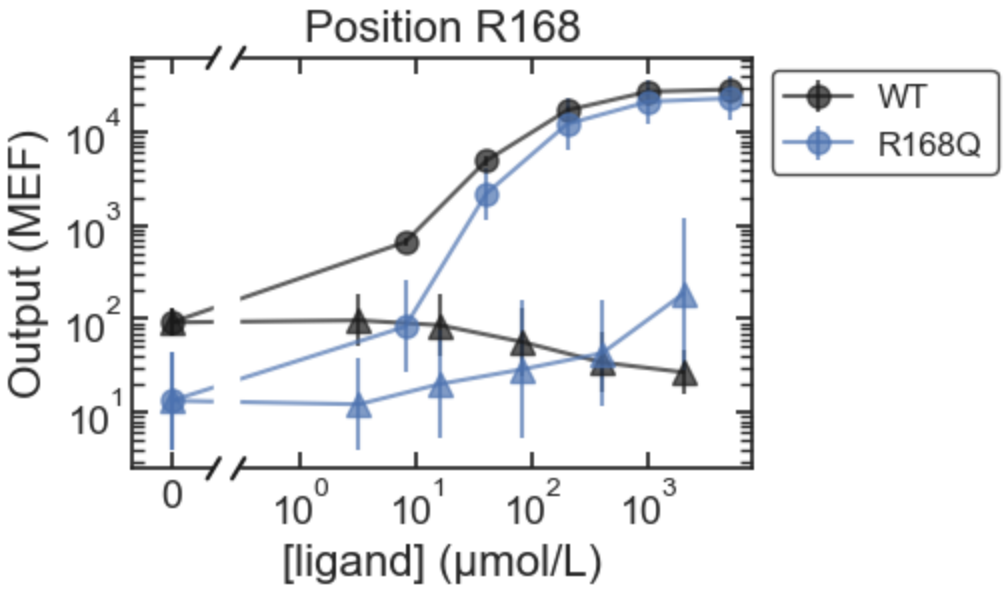

Position L146
Output (MEF)
[ligand] (μmol/L)
WT
L146F
L146I
L146S

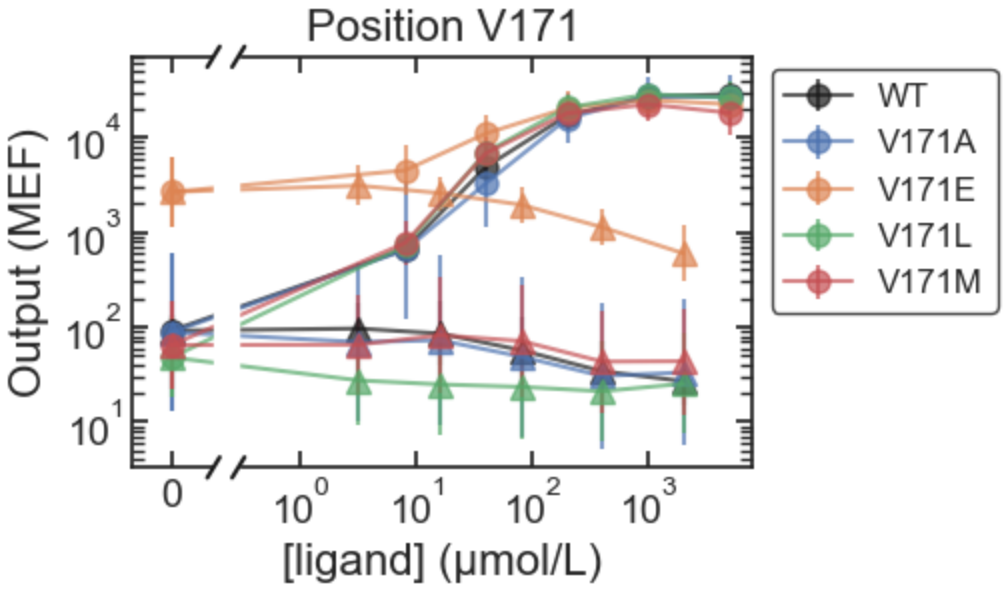

Position S151
Output (MEF)
[ligand] (μmol/L)
WT
S151F
S151P
S151T
S151Y

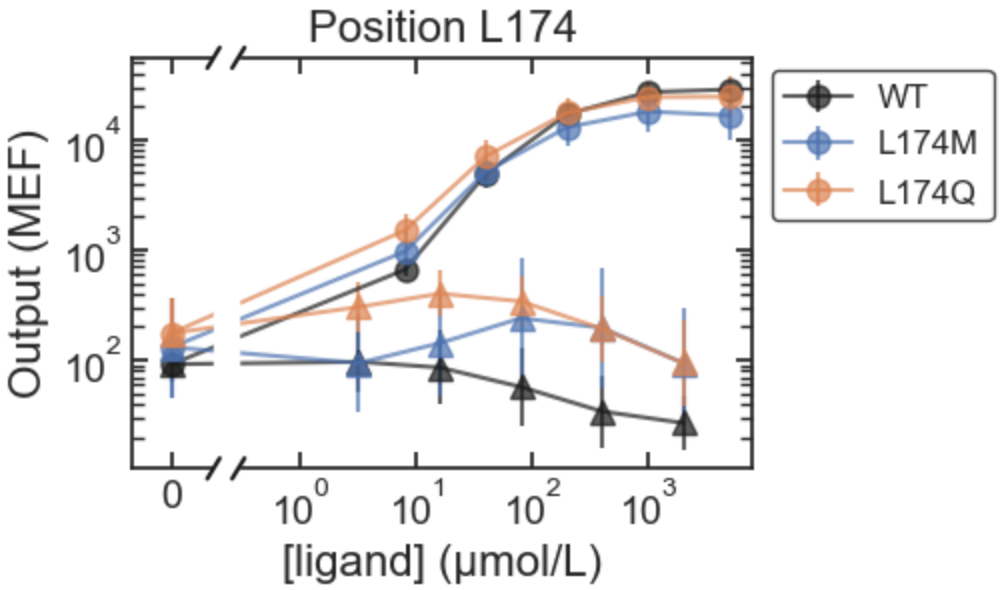

Position P155
Output (MEF)
[ligand] (μmol/L)
WT
P155H
P155L
P155S

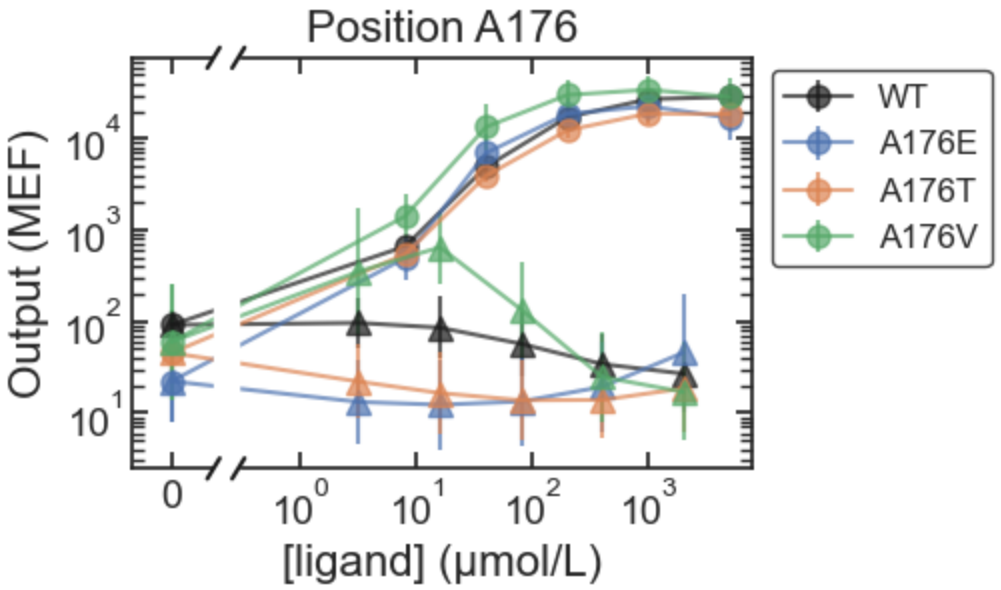

Position S158
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S158C
S158G
S158N
S158T

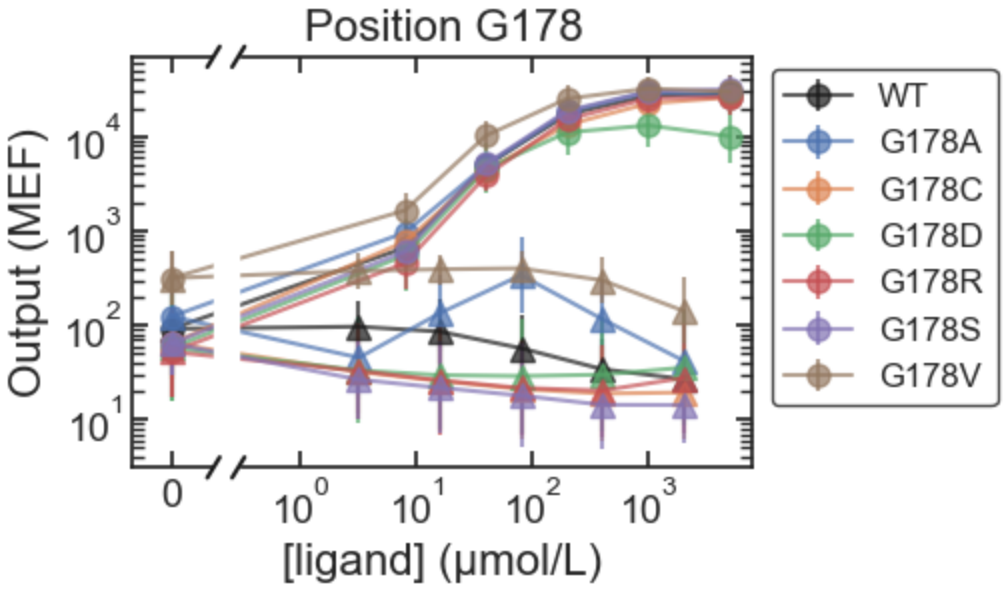

Position I159
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I159M
I159N
I159S

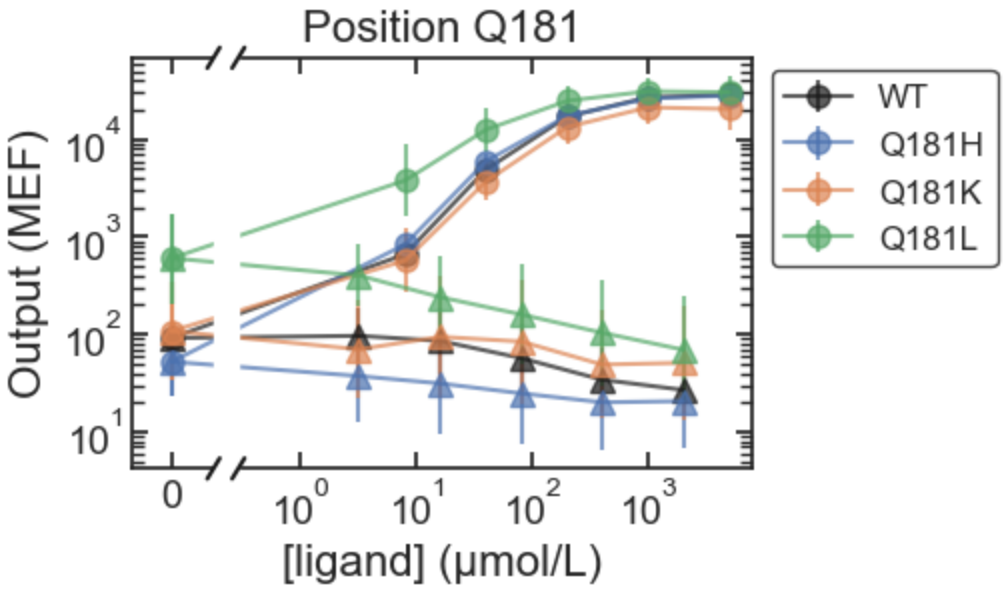

Position R168
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R168Q

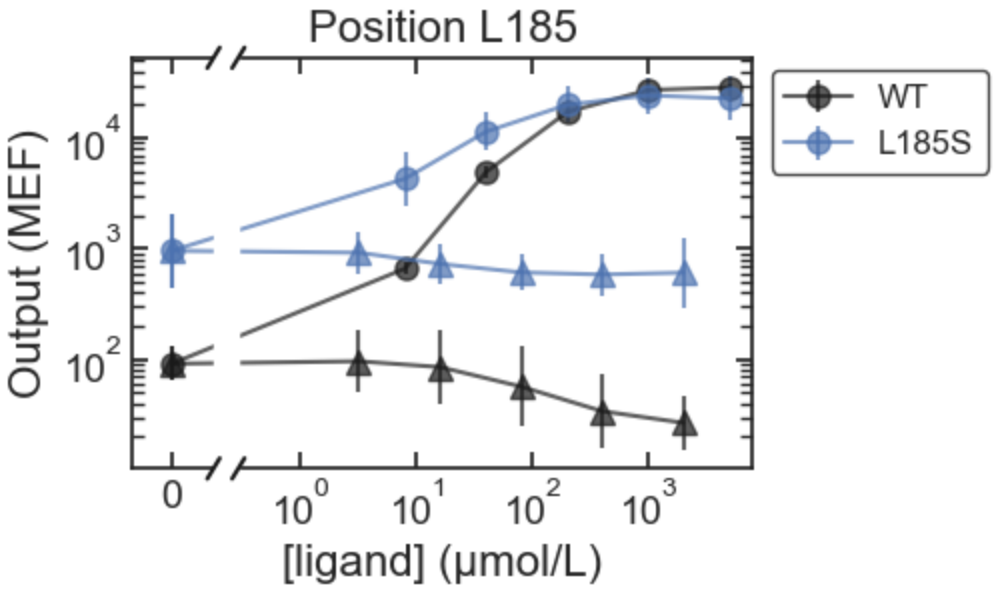

Position V171
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
V171A
V171E
V171L
V171M

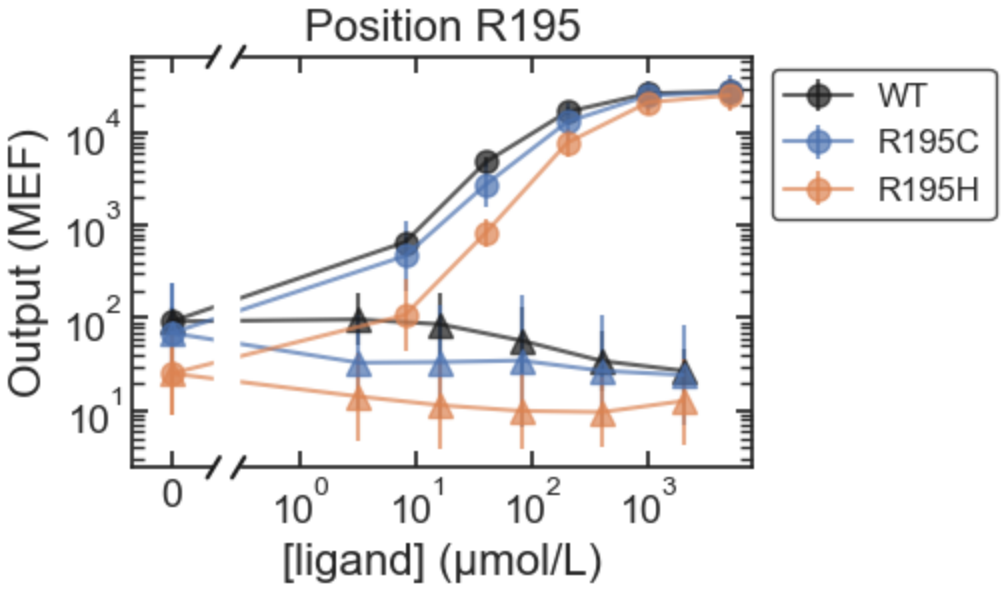

Position L174
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
L174M
L174Q

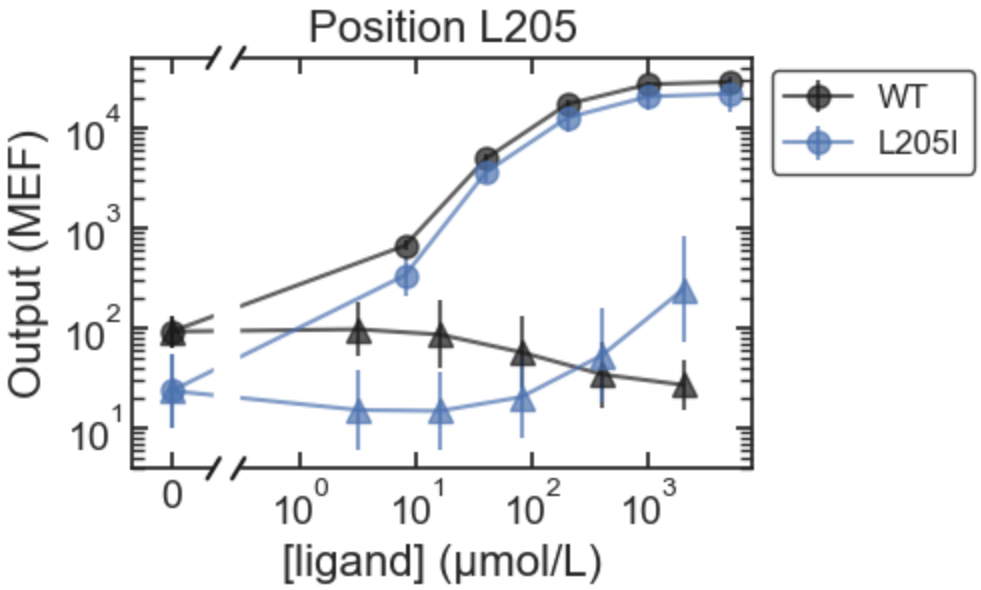

Position A176
Output (MEF)
0
10^0
10^1
10^2
10^3
10^4
[ligand] (μmol/L)
WT
A176E
A176T
A176V

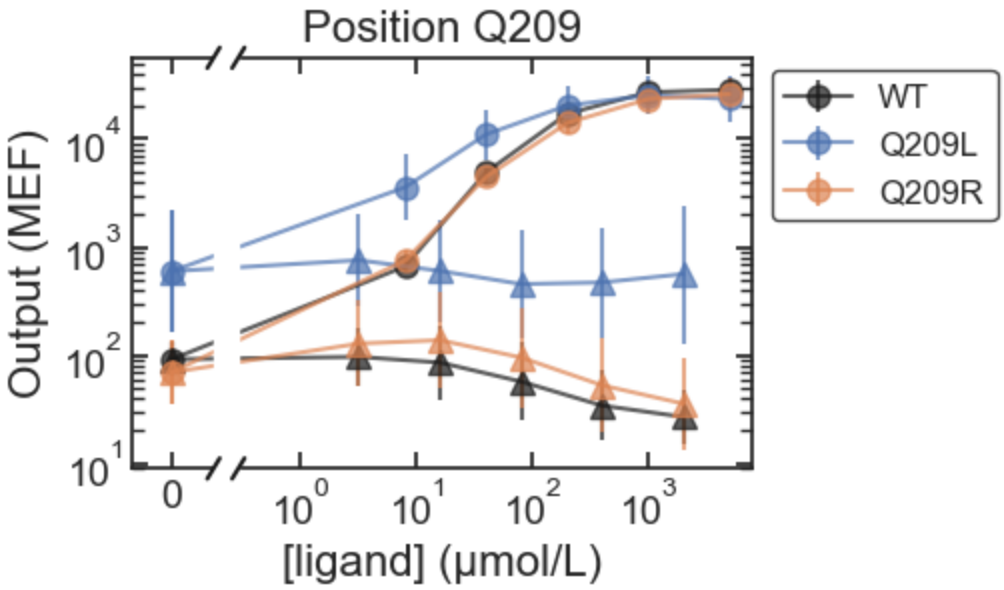

Position G178
Output (MEF)
[ligand] (μmol/L)
WT
G178A
G178C
G178D
G178R
G178S
G178V

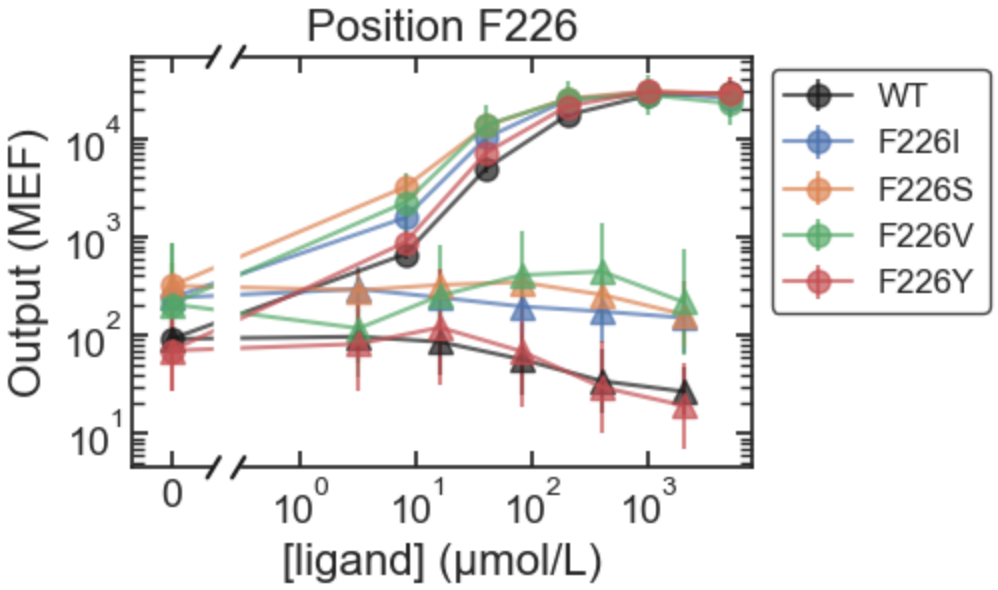

Position Q181
Output (MEF)
[ligand] (μmol/L)
WT
Q181H
Q181K
Q181L

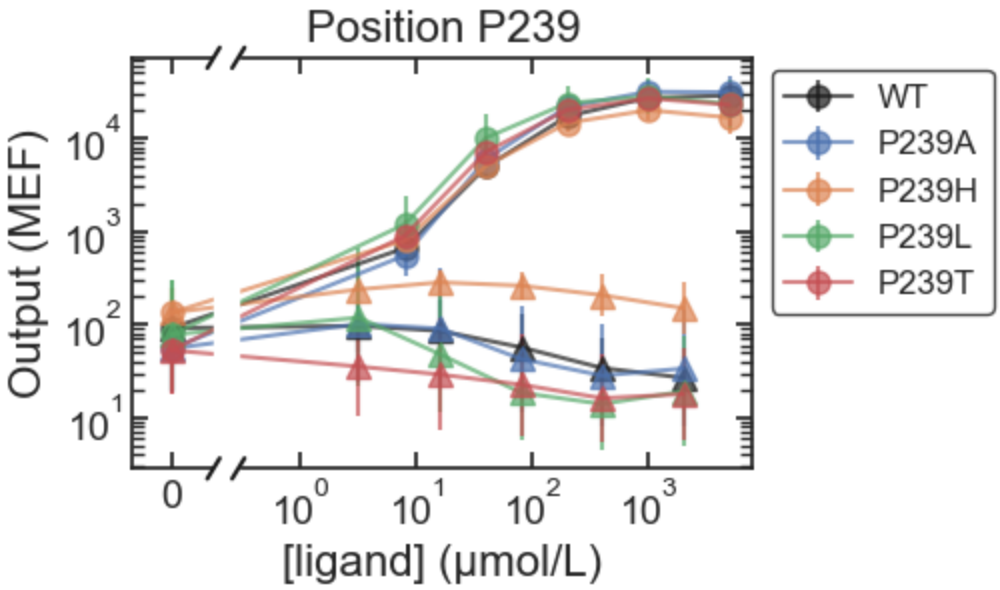

Position L185
Output (MEF)
[ligand] (μmol/L)
WT
L185S

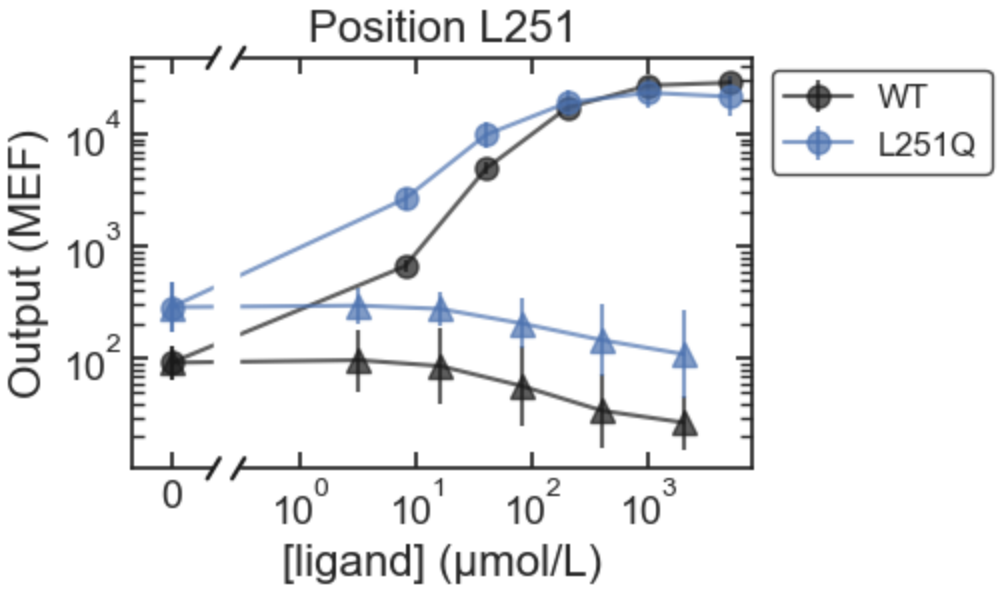

Position R195
Output (MEF)
[ligand] (μmol/L)
WT
R195C
R195H

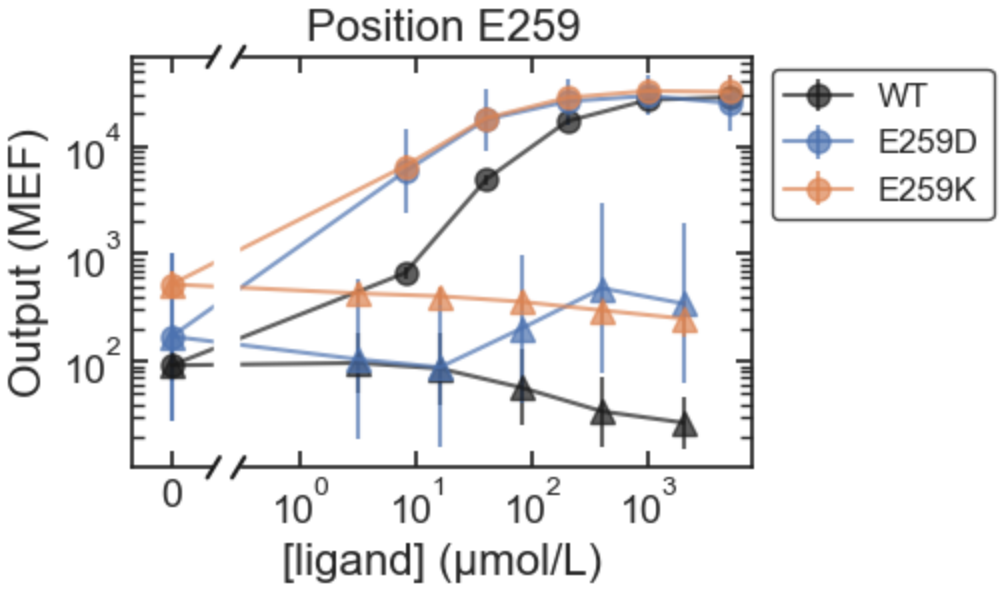

Position L205
Output (MEF)
[ligand] (μmol/L)
WT
L205I

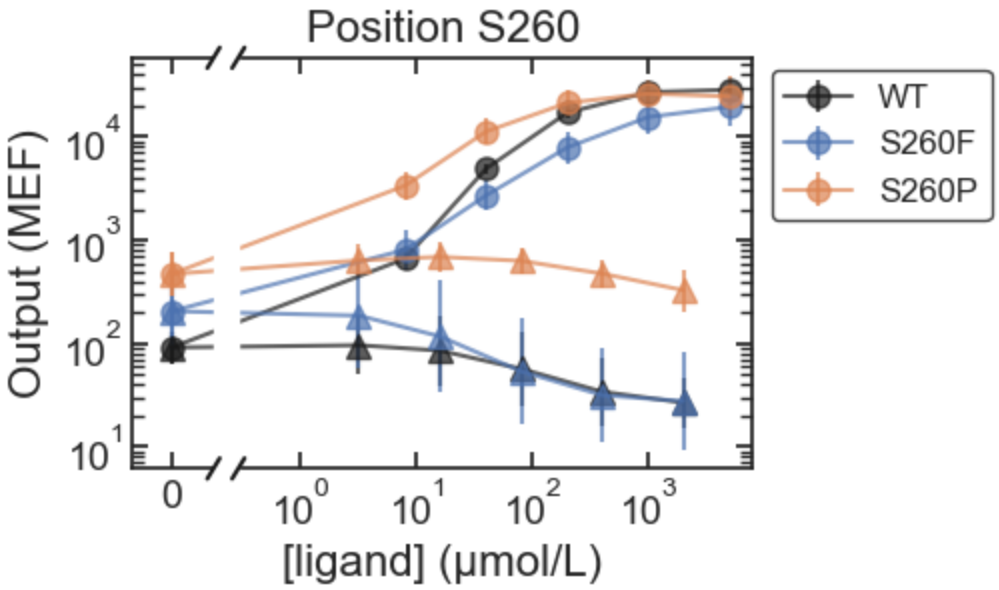

Position Q209
Output (MEF)
[ligand] (μmol/L)
WT
Q209L
Q209R

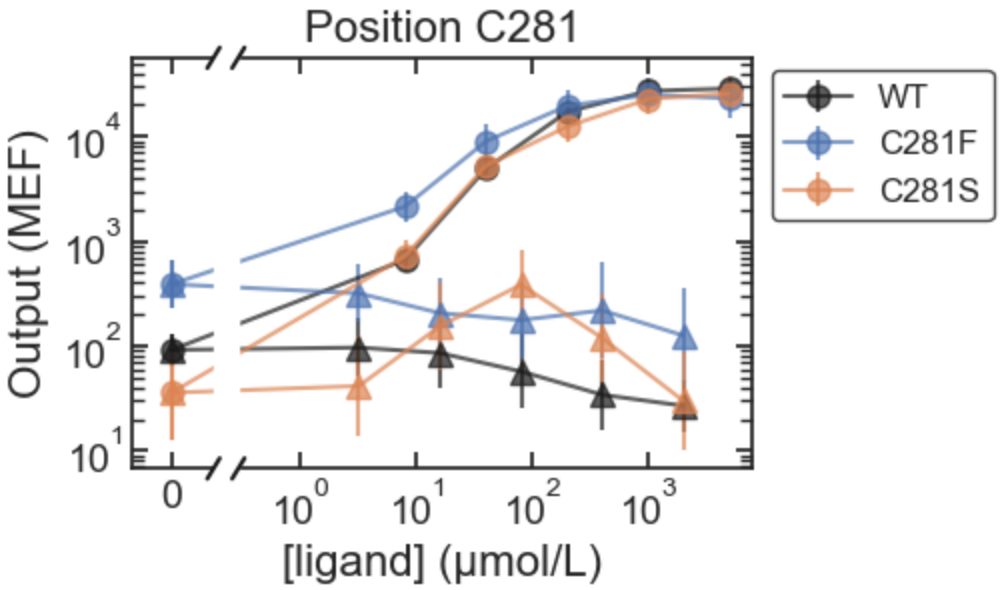

Position F226
Output (MEF)
[ligand] (μmol/L)
WT
F226I
F226S
F226V
F226Y

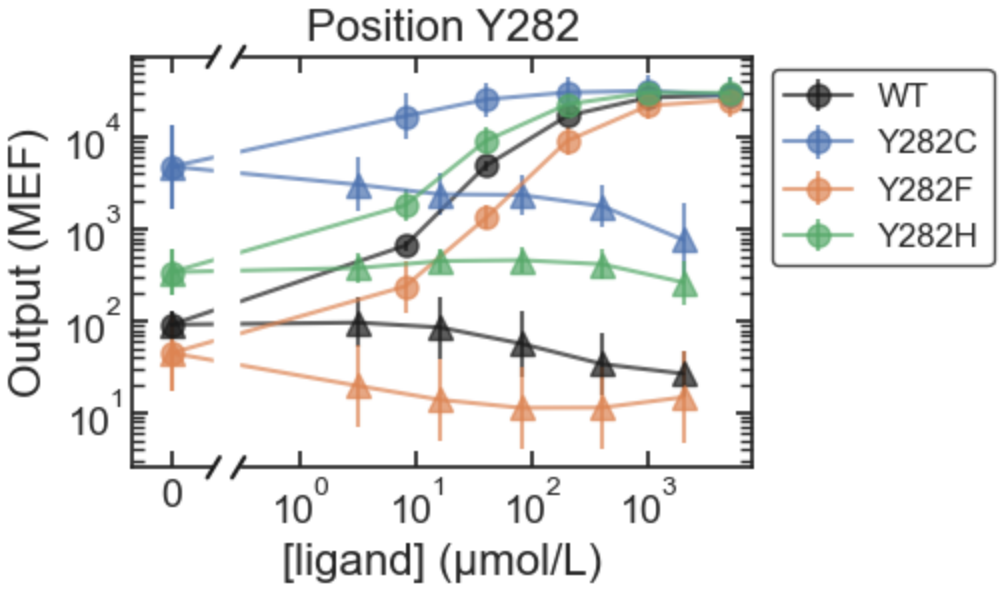

Position P239
Output (MEF)
[ligand] (μmol/L)
WT
P239A
P239H
P239L
P239T

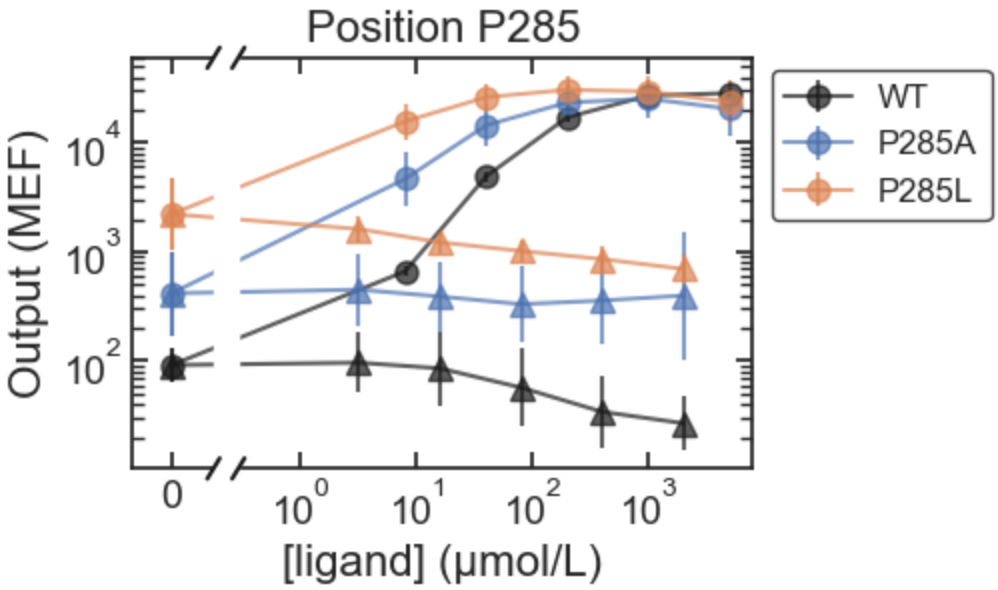

Position L251
Output (MEF)
[ligand] (μmol/L)
WT
L251Q

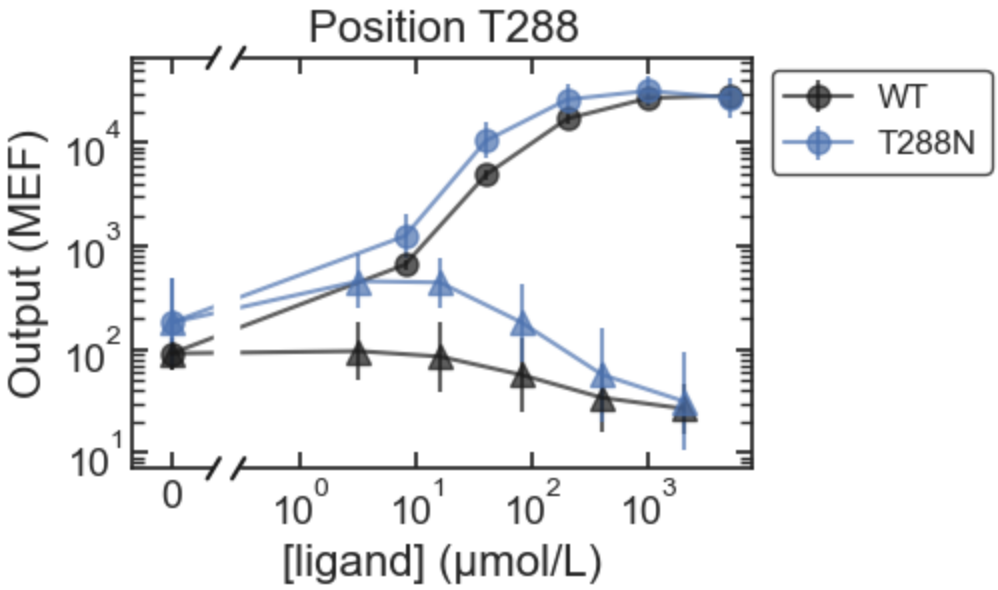

Position E259
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
E259D
E259K

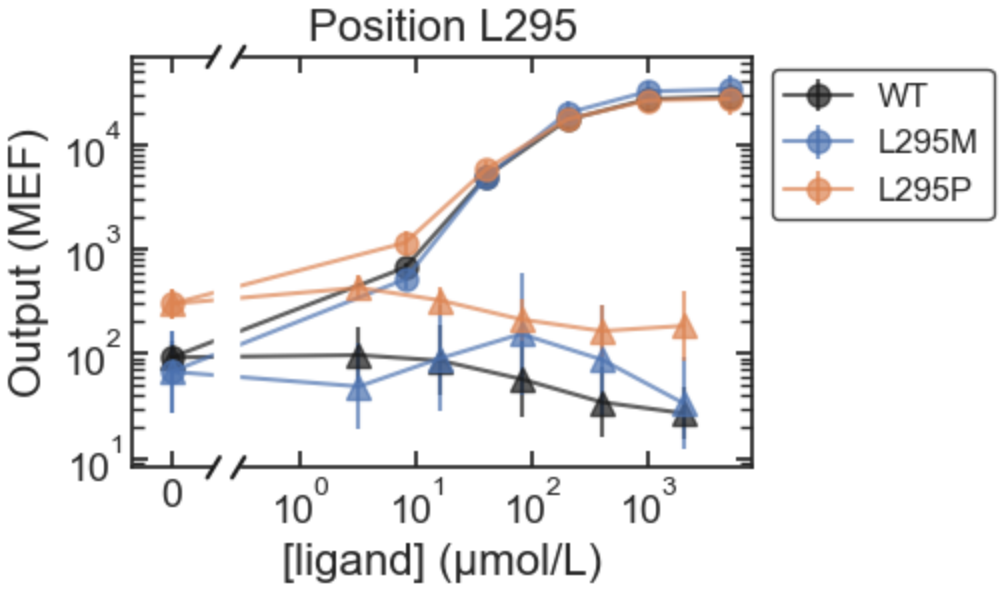

Position S260
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S260F
S260P

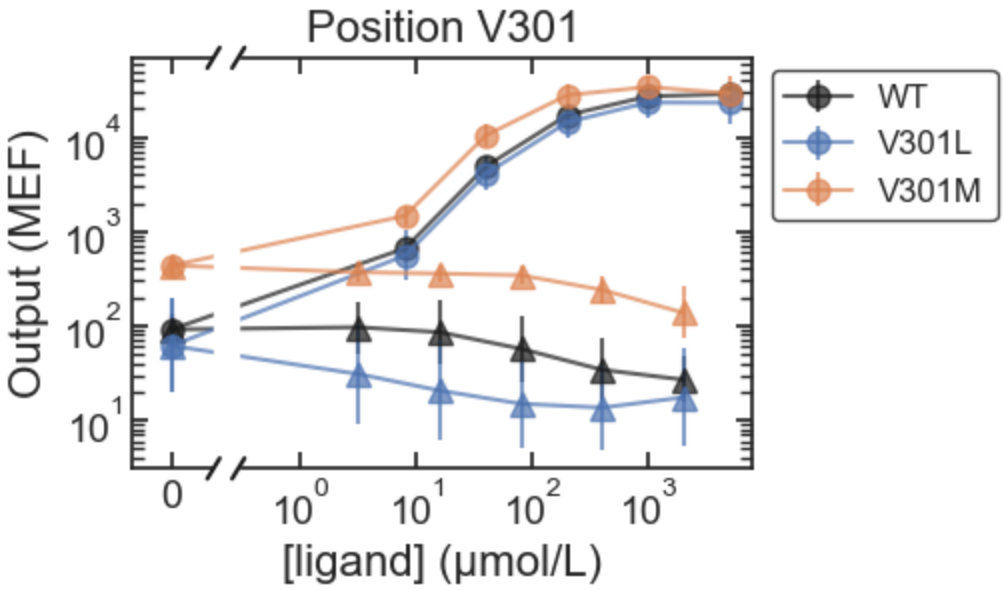

Position C281
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
C281F
C281S

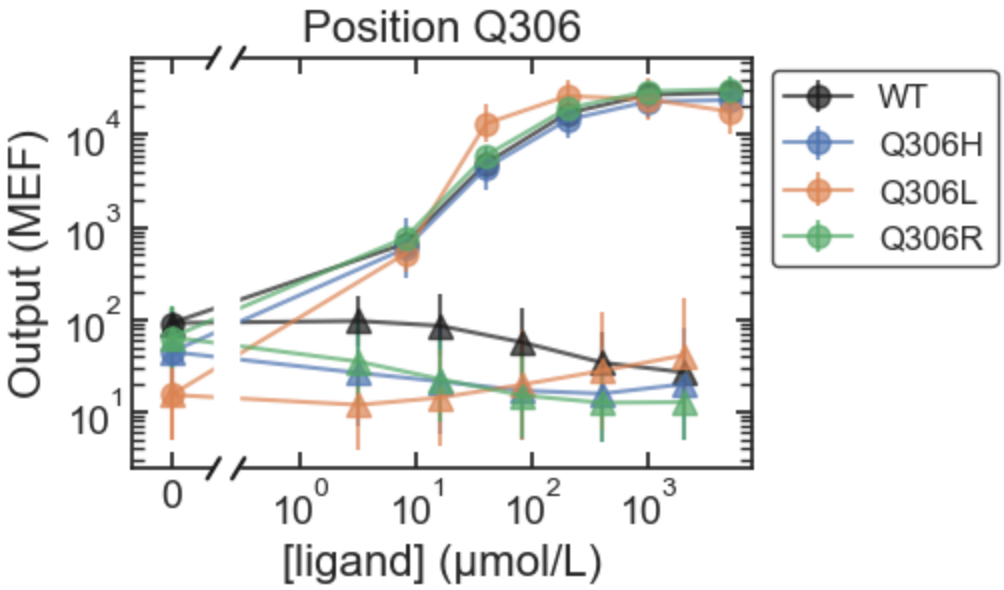

Position Y282
Output (MEF)
[ligand] (µmol/L)
WT
Y282C
Y282F
Y282H

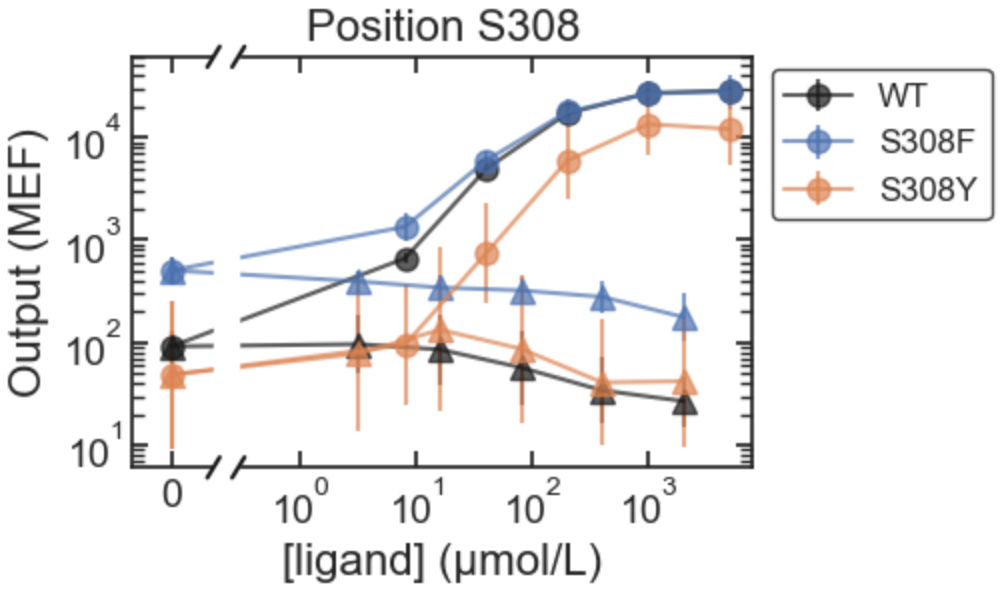

Position P285
Output (MEF)
[ligand] (µmol/L)
WT
P285A
P285L

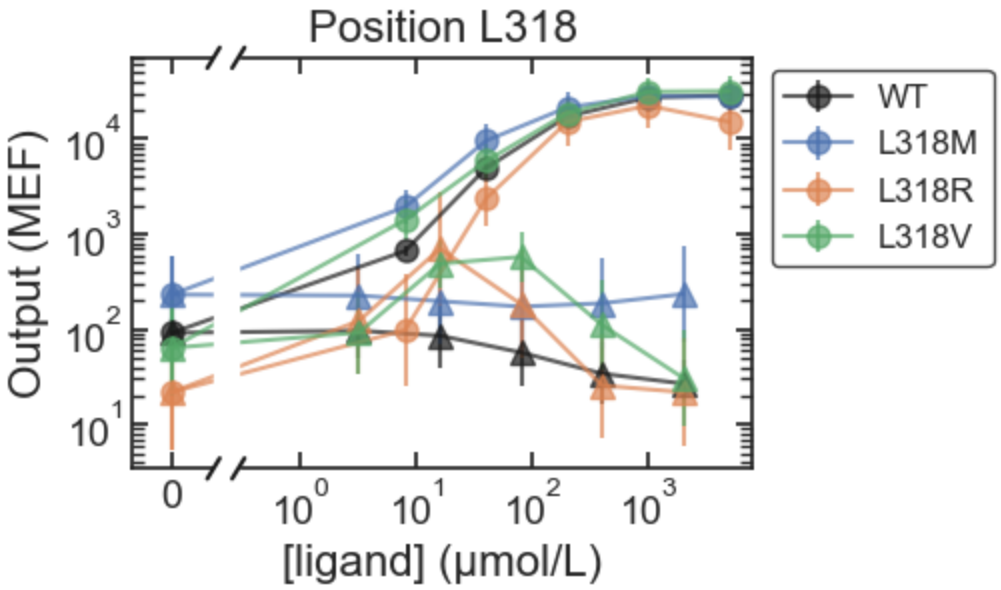

Position T288
Output (MEF)
[ligand] (µmol/L)
WT
T288N

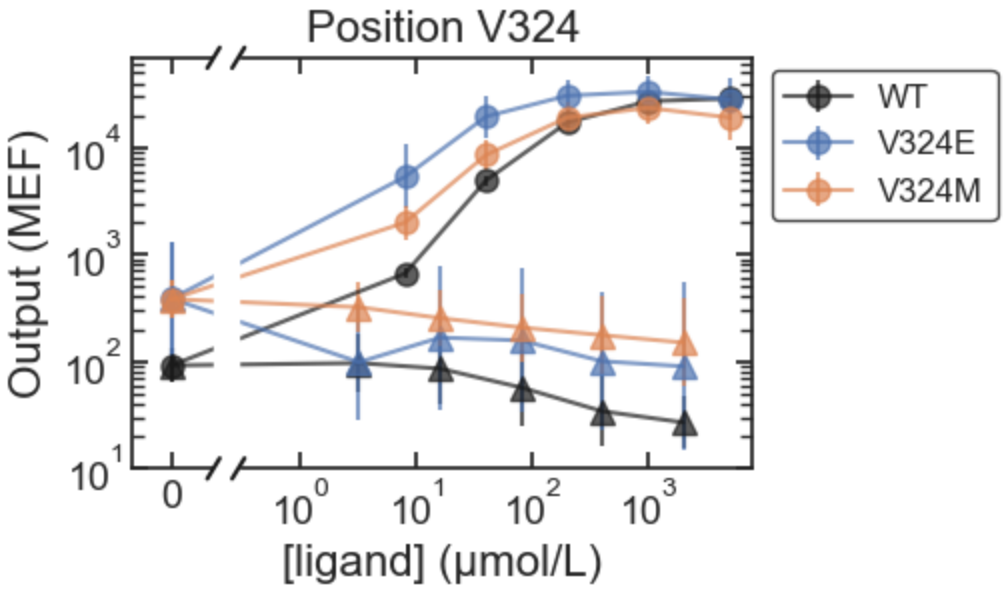

Position L295
Output (MEF)
[ligand] (μmol/L)
WT
L295M
L295P

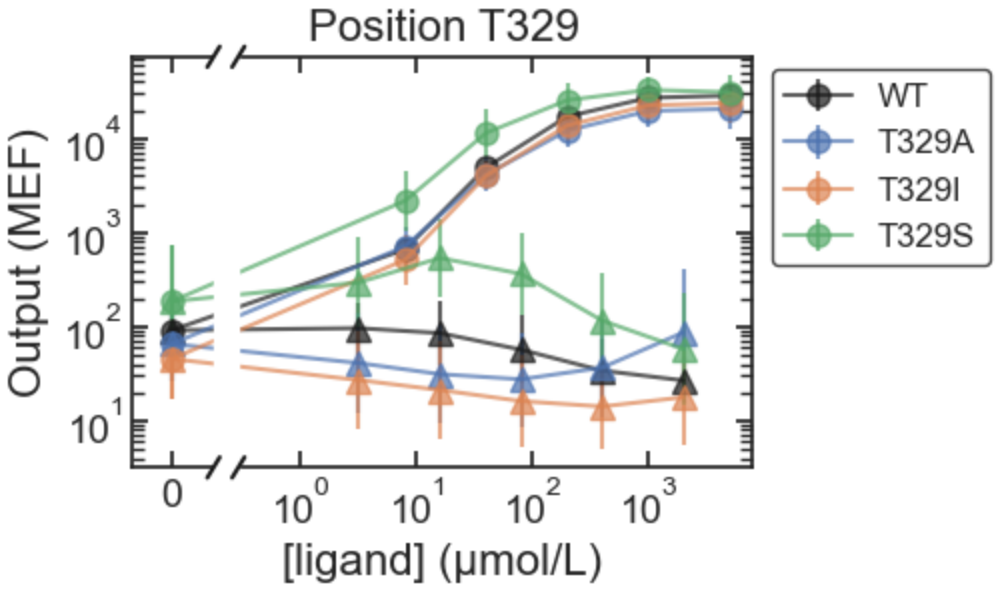

Position V301
Output (MEF)
[ligand] (μmol/L)
WT
V301L
V301M

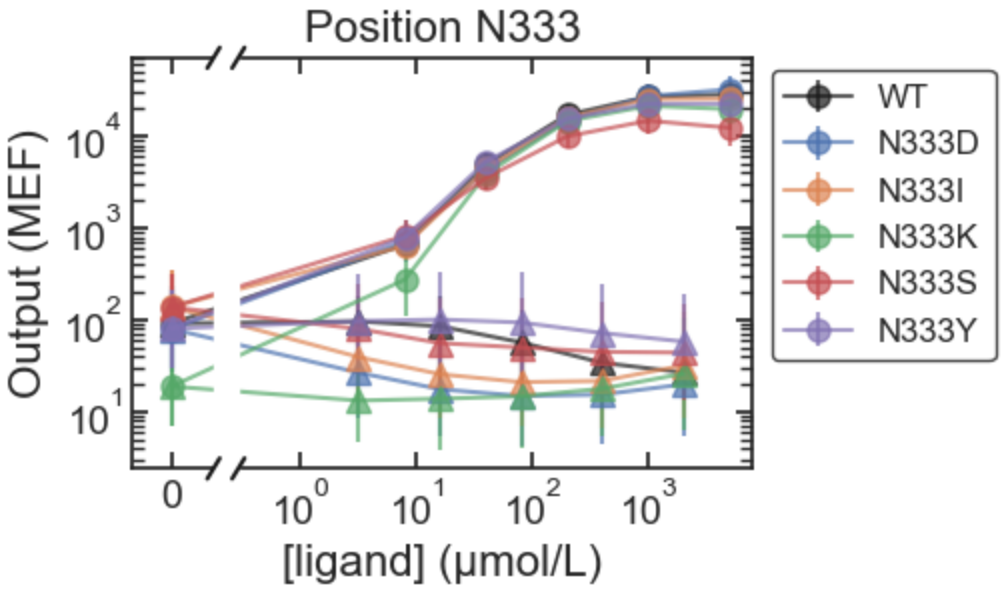

Position Q306
Output (MEF)
[ligand] (μmol/L)
WT
Q306H
Q306L
Q306R

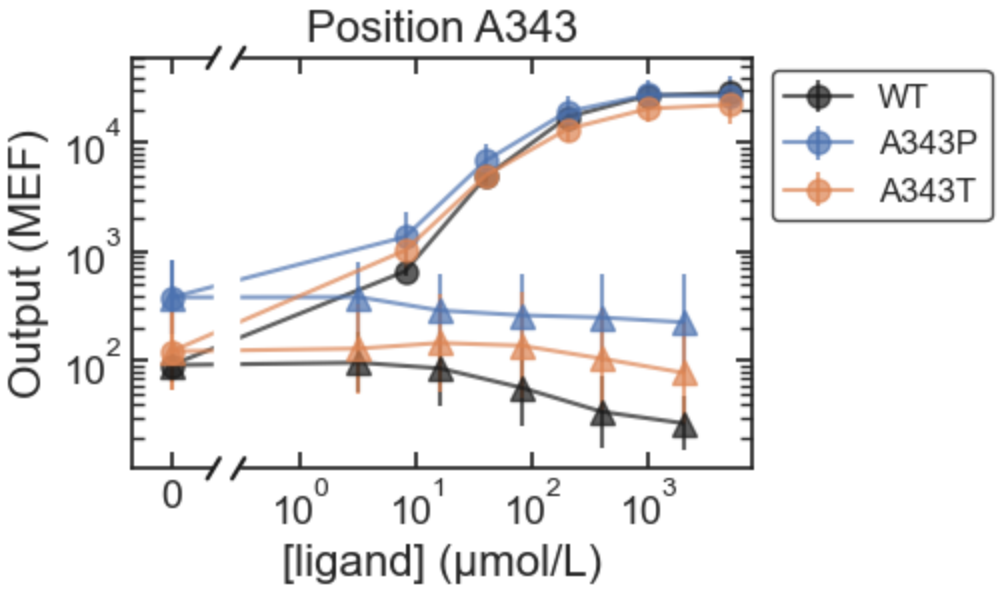

Position S308
Output (MEF)
[ligand] (μmol/L)
WT
S308F
S308Y

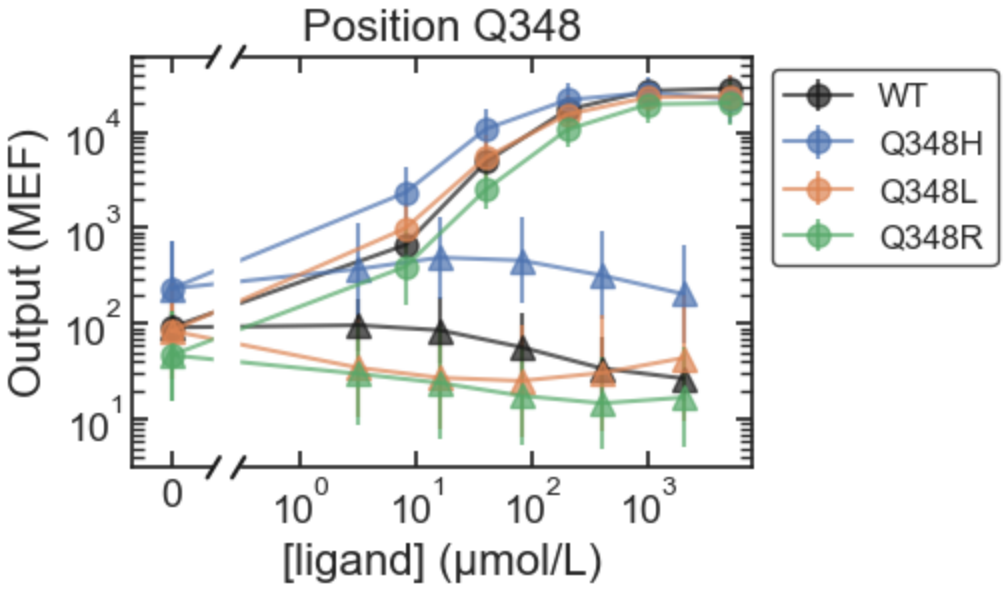

Position L318
Output (MEF)
[ligand] (μmol/L)
WT
L318M
L318R
L318V


Position V324
Output (MEF)
[ligand] (μmol/L)
WT
V324E
V324M

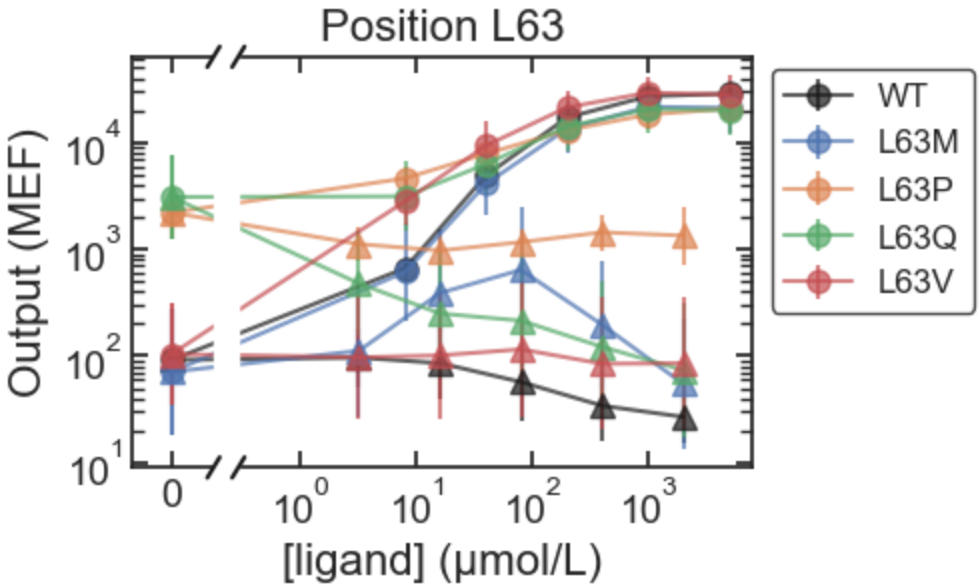

Position T329
Output (MEF)
[ligand] (μmol/L)
WT
T329A
T329I
T329S

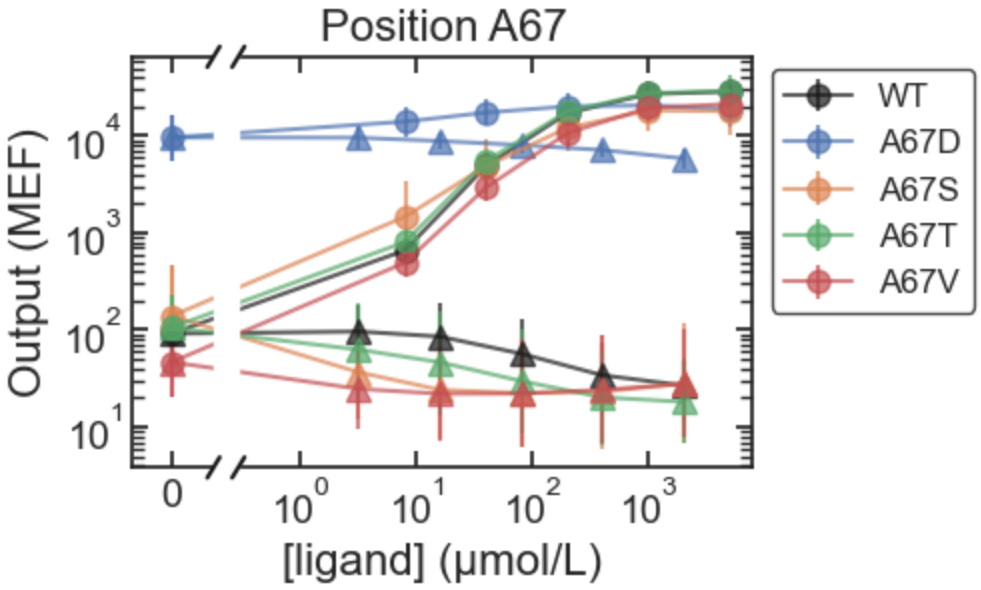

Position N333
Output (MEF)
[ligand] (μmol/L)
WT
N333D
N333I
N333K
N333S
N333Y

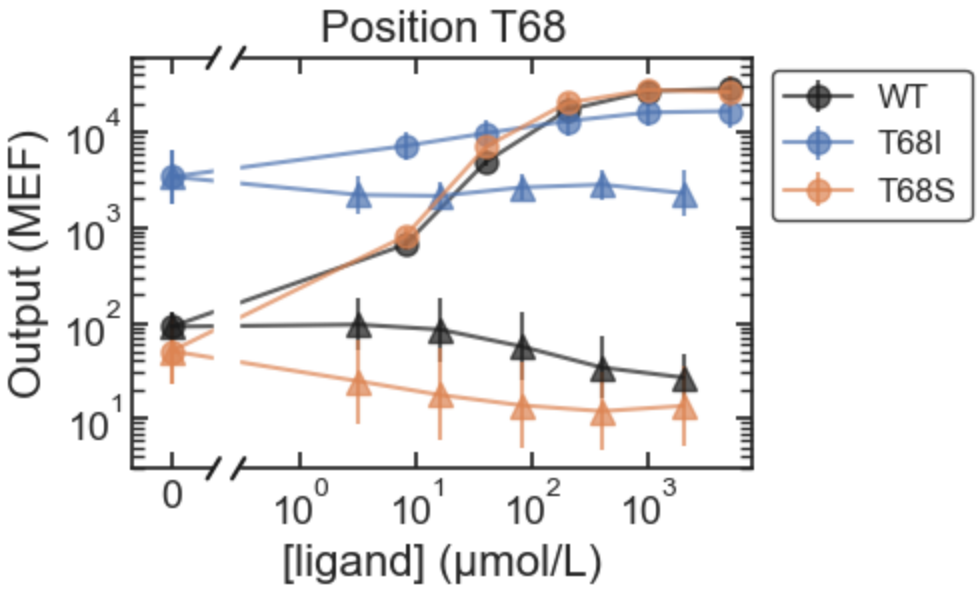

Position A343
Output (MEF)
[ligand] (μmol/L)
WT
A343P
A343T

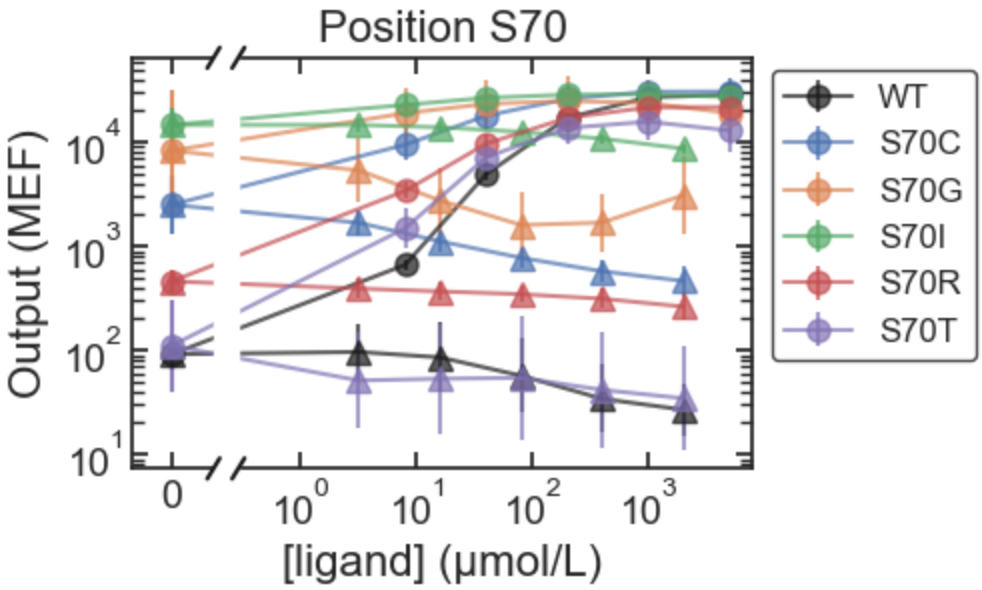

Position Q348
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q348H
Q348L
Q348R

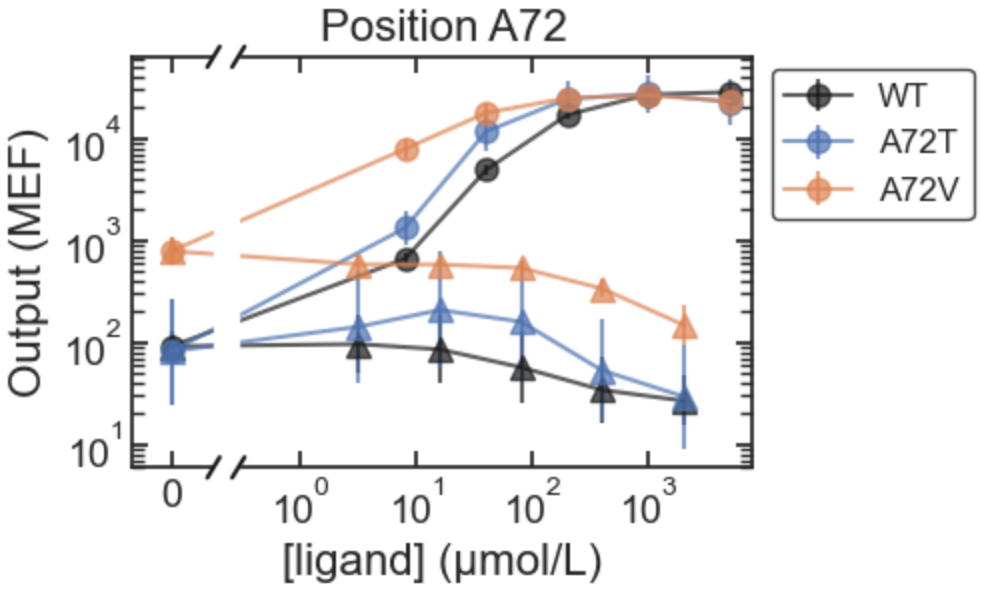
MWC_IPTG

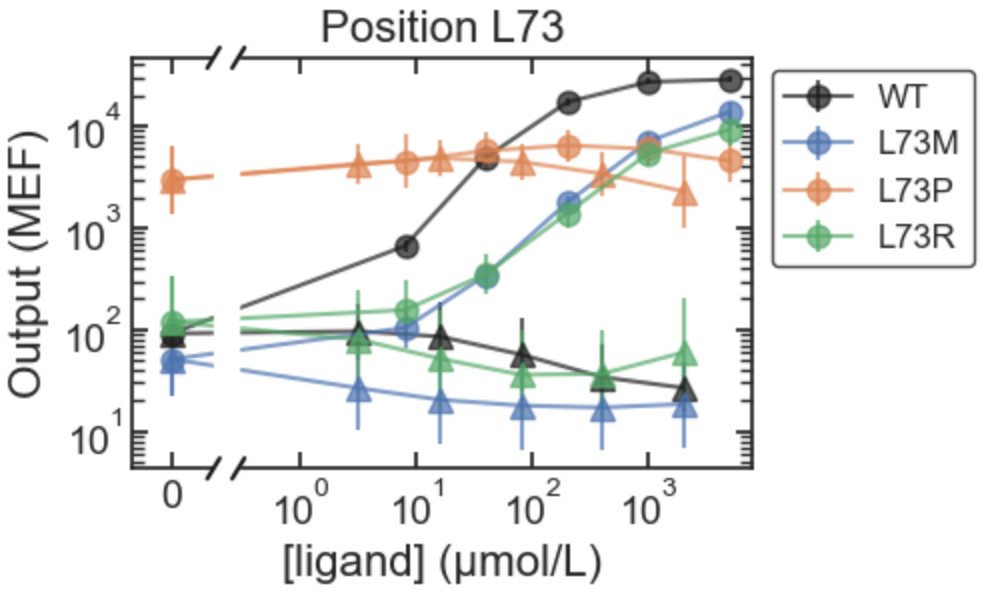

Position L63
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

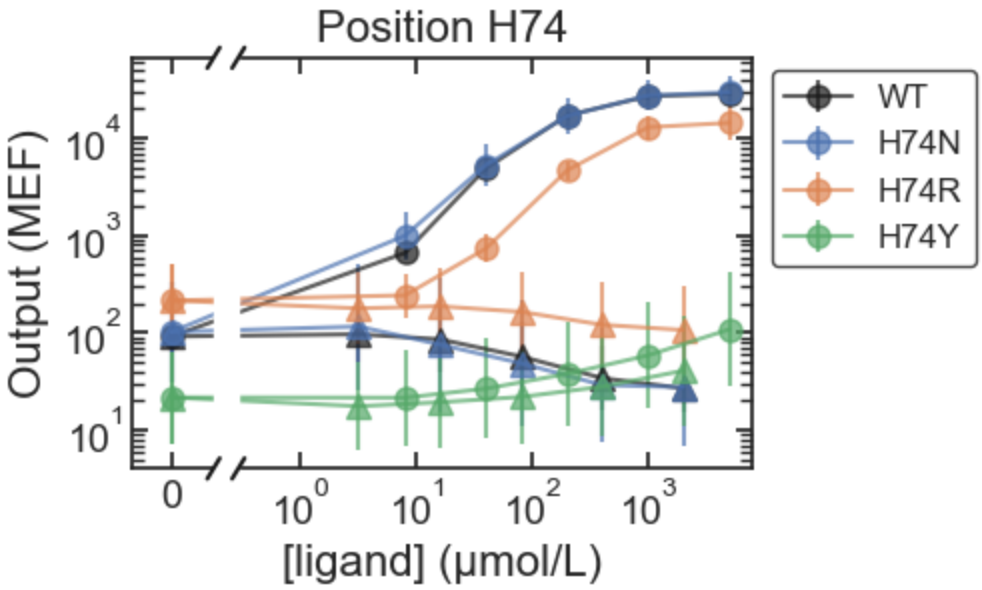

Position A67
Output (MEF)
[ligand] (μmol/L)
WT
A67D
A67S
A67T
A67V

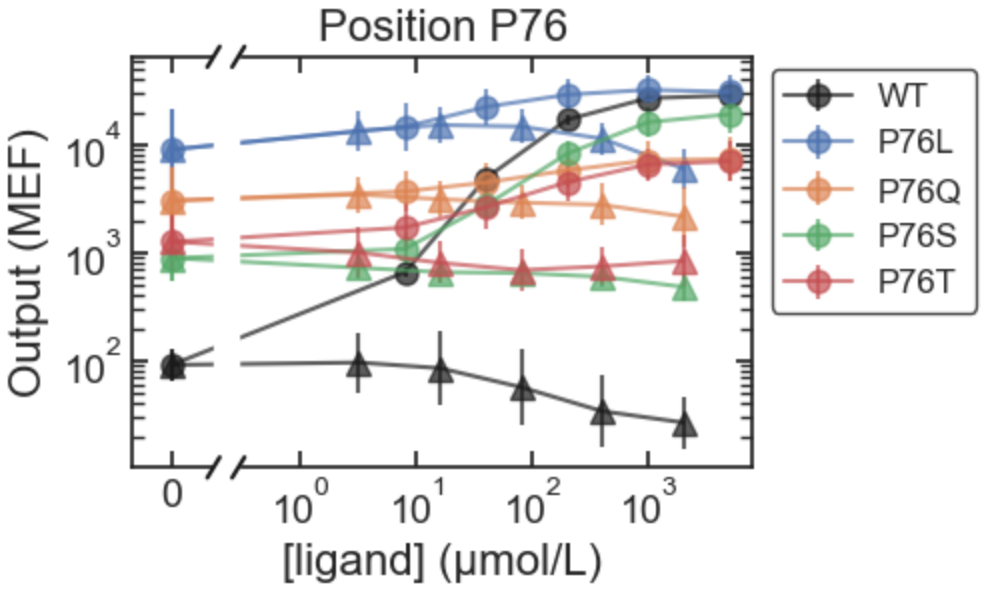

Position T68
Output (MEF)
[ligand] (μmol/L)
WT
T68I
T68S

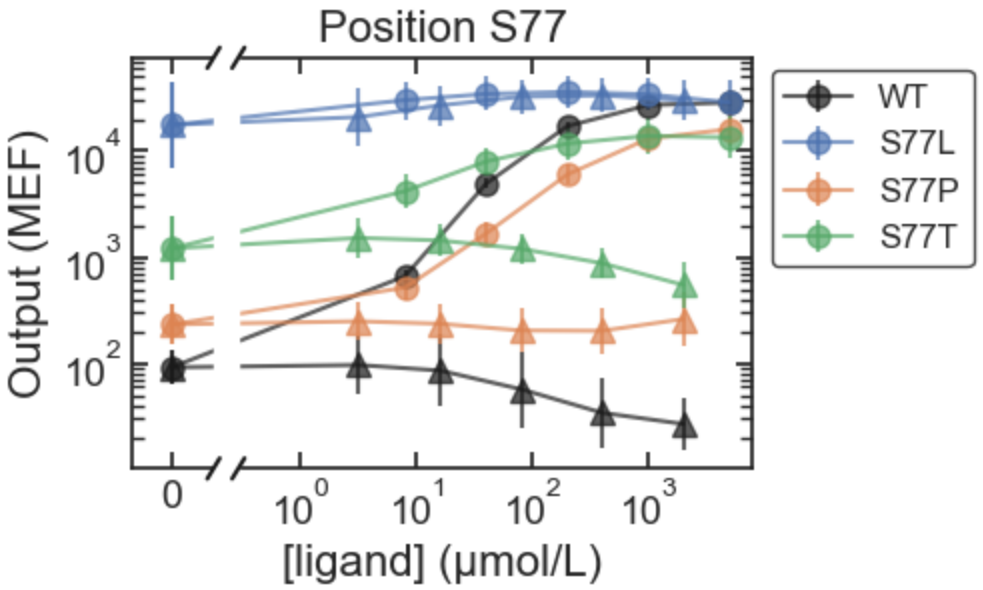

Position S70
Output (MEF)
[ligand] (μmol/L)
WT
S70C
S70G
S70I
S70R
S70T

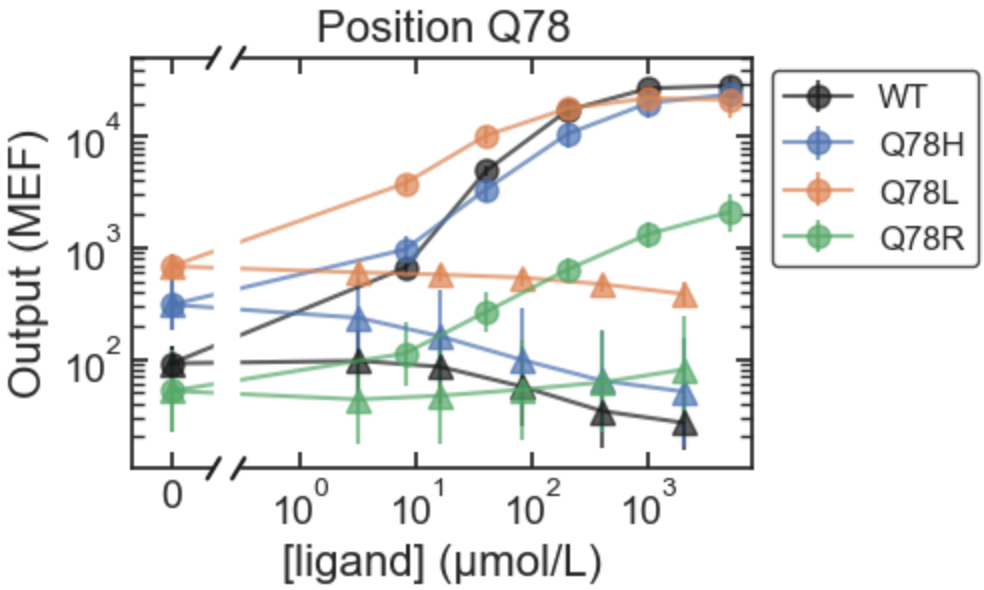

Position A72
Output (MEF)
[ligand] (μmol/L)
WT
A72T
A72V

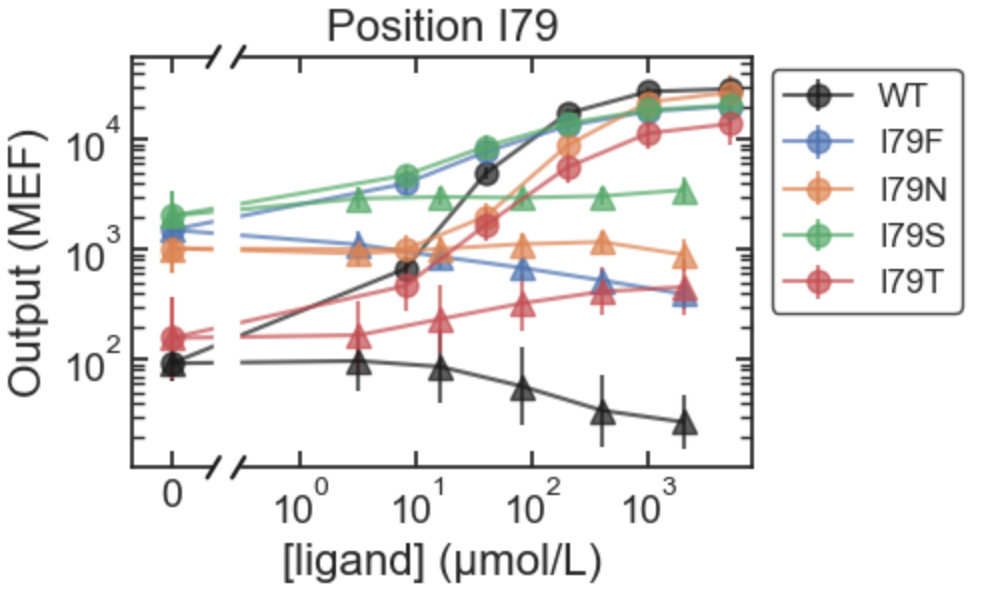

Position L73
Output (MEF)
[ligand] (μmol/L)
WT
L73M
L73P
L73R

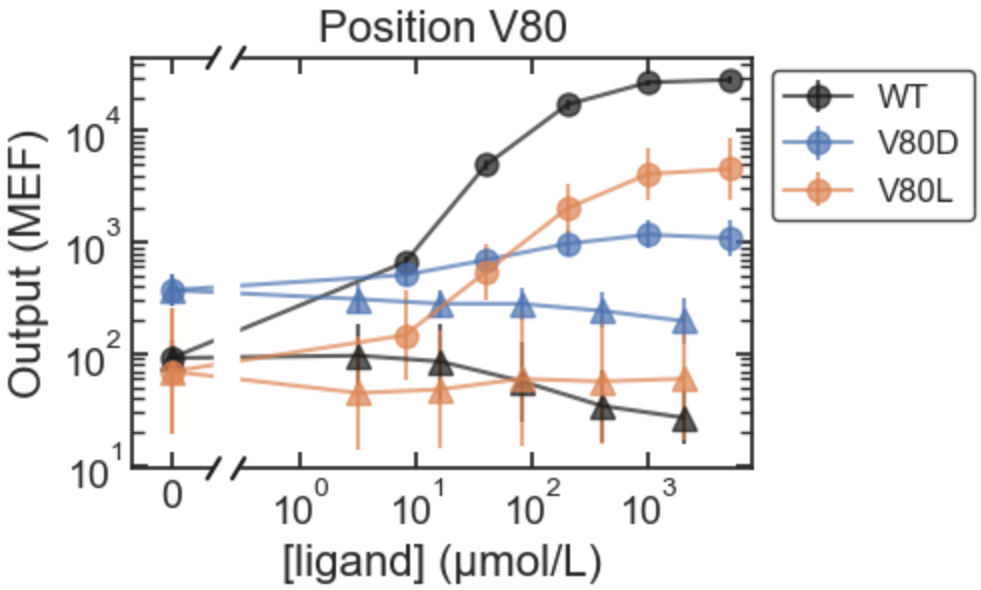

Position H74
Output (MEF)
[ligand] (μmol/L)
WT
H74N
H74R
H74Y

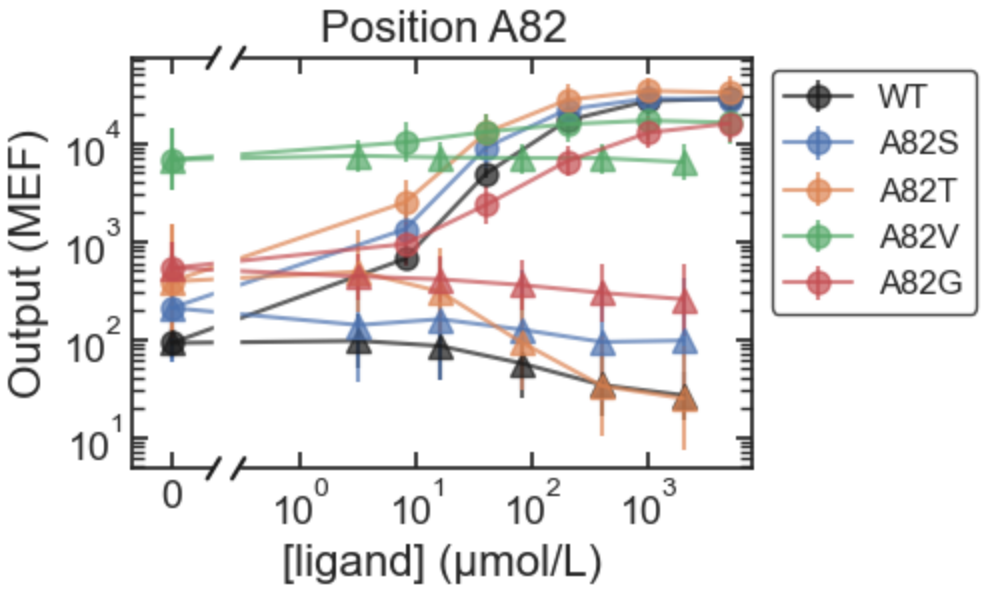

Position P76
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

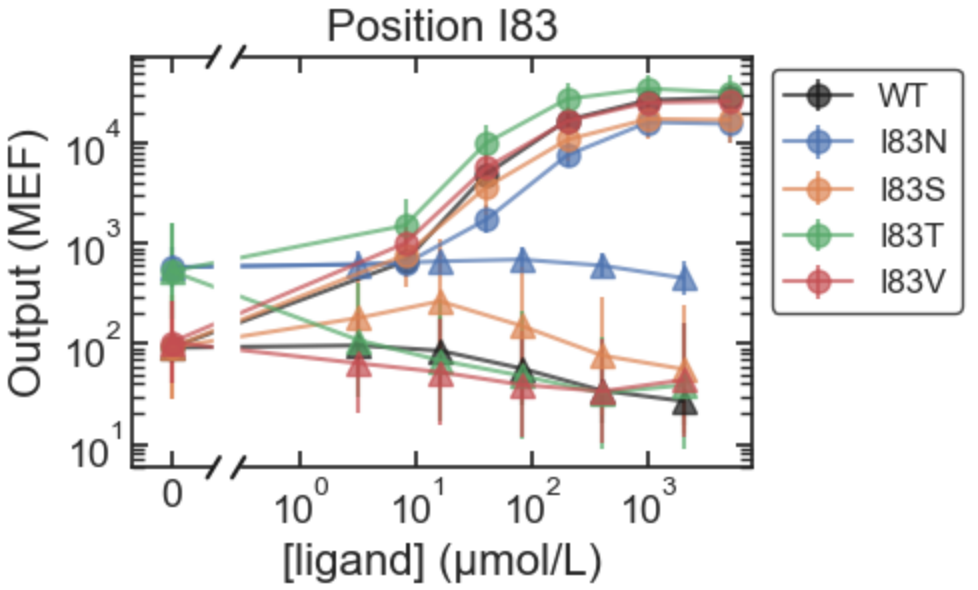

Position S77
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S77L
S77P
S77T

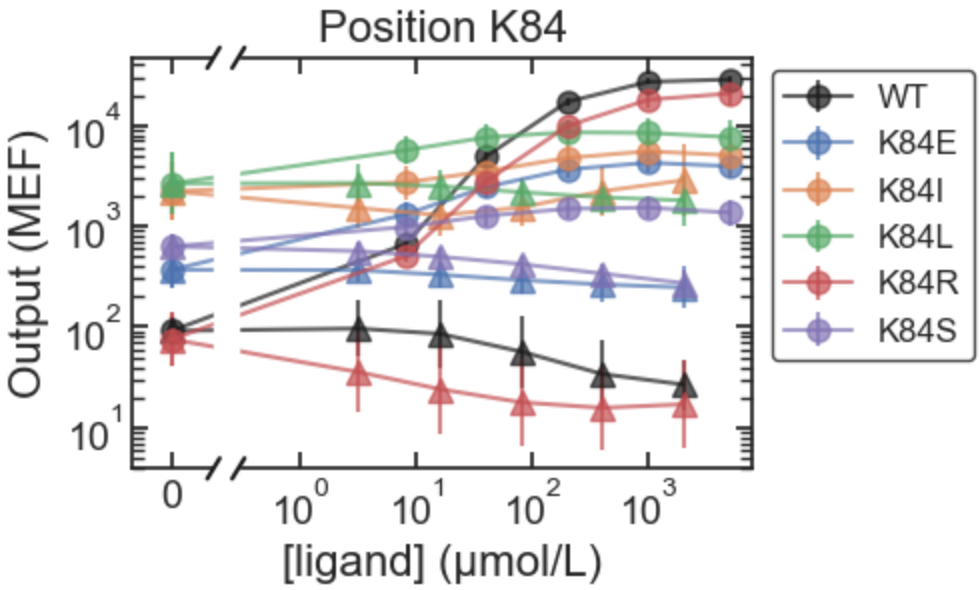

Position Q78
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q78H
Q78L
Q78R

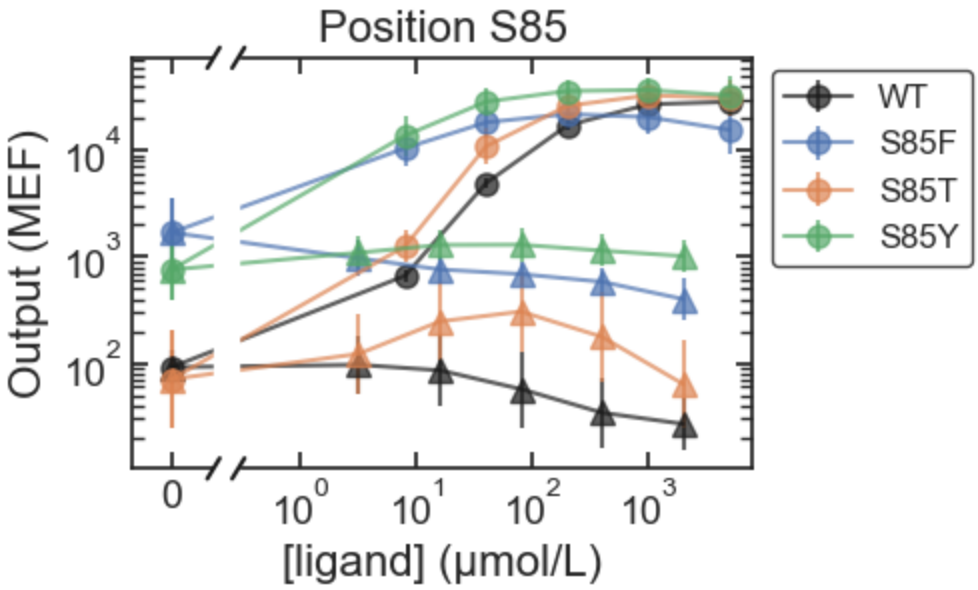

Position I79
Output (MEF)
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

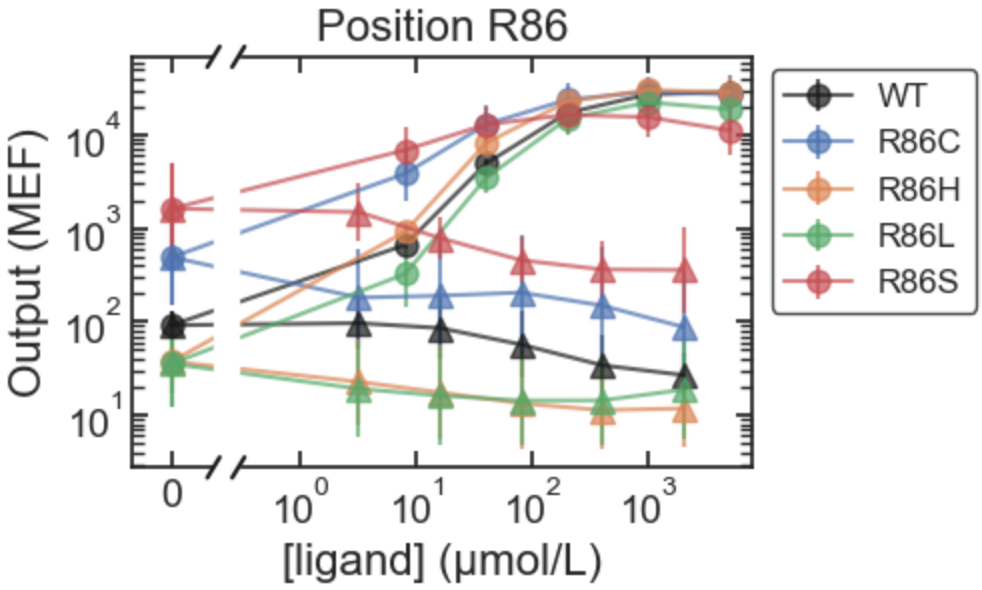

Position V80
Output (MEF)
[ligand] (μmol/L)
WT
V80D
V80L

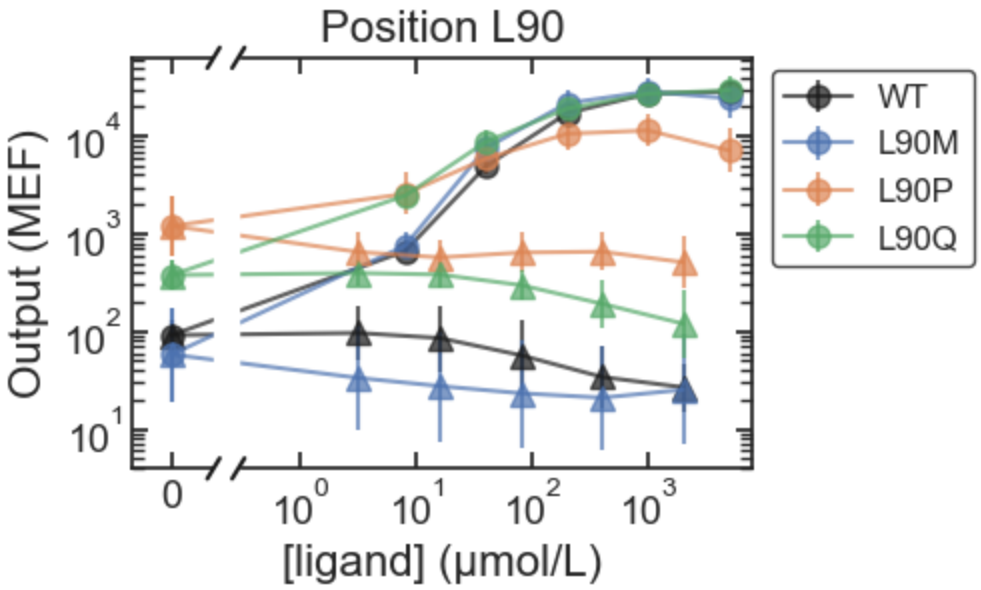

Position A82
Output (MEF)
[ligand] (μmol/L)
WT
A82S
A82T
A82V
A82G

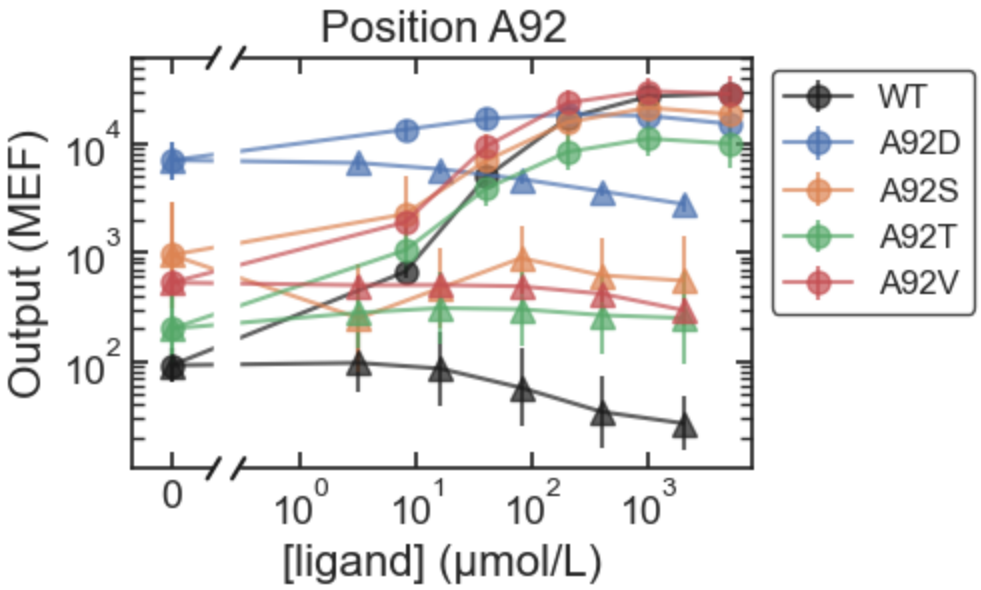

Position I83
Output (MEF)
[ligand] (μmol/L)
WT
I83N
I83S
I83T
I83V

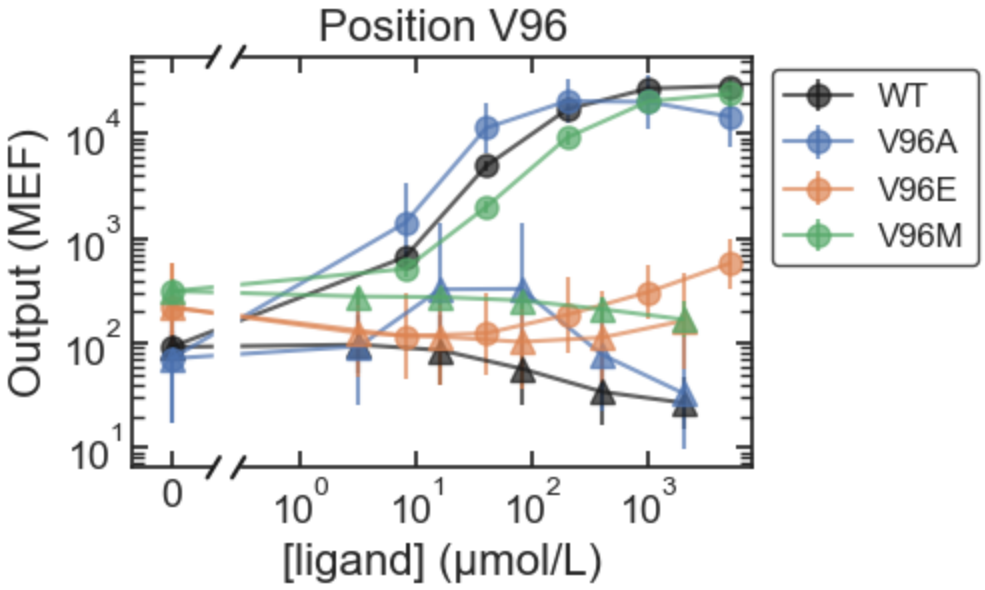

Position K84
Output (MEF)
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

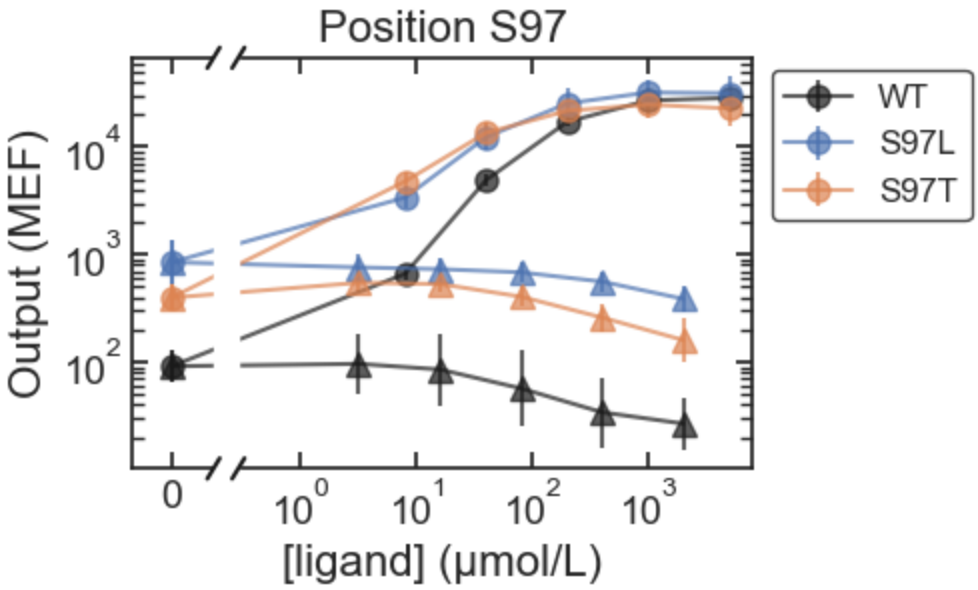

Position S85
Output (MEF)
[ligand] (μmol/L)
WT
S85F
S85T
S85Y

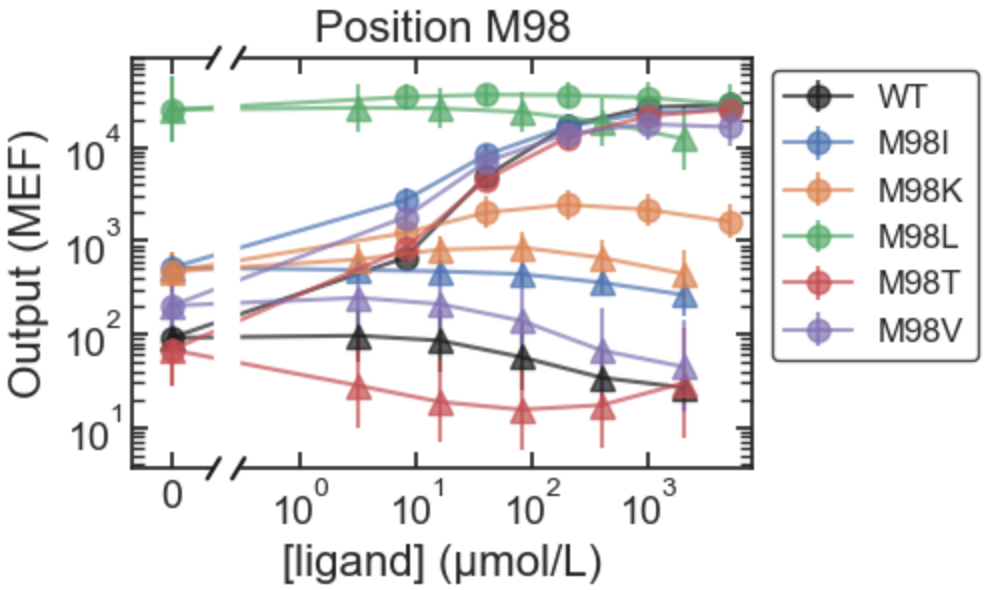

Position R86
Output (MEF)
[ligand] (μmol/L)
WT
R86C
R86H
R86L
R86S

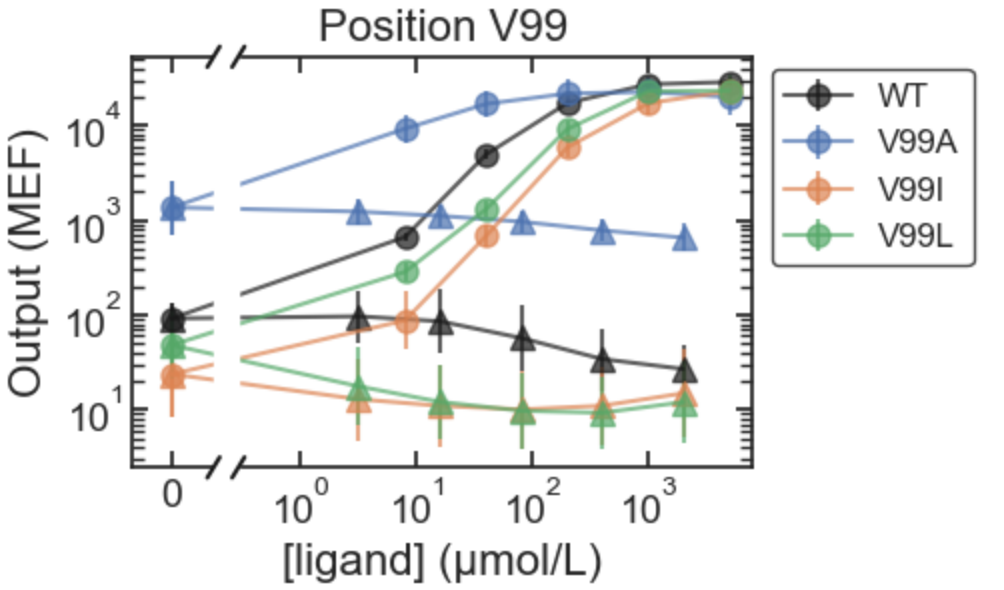

Position L90
Output (MEF)
[ligand] (μmol/L)
WT
L90M
L90P
L90Q

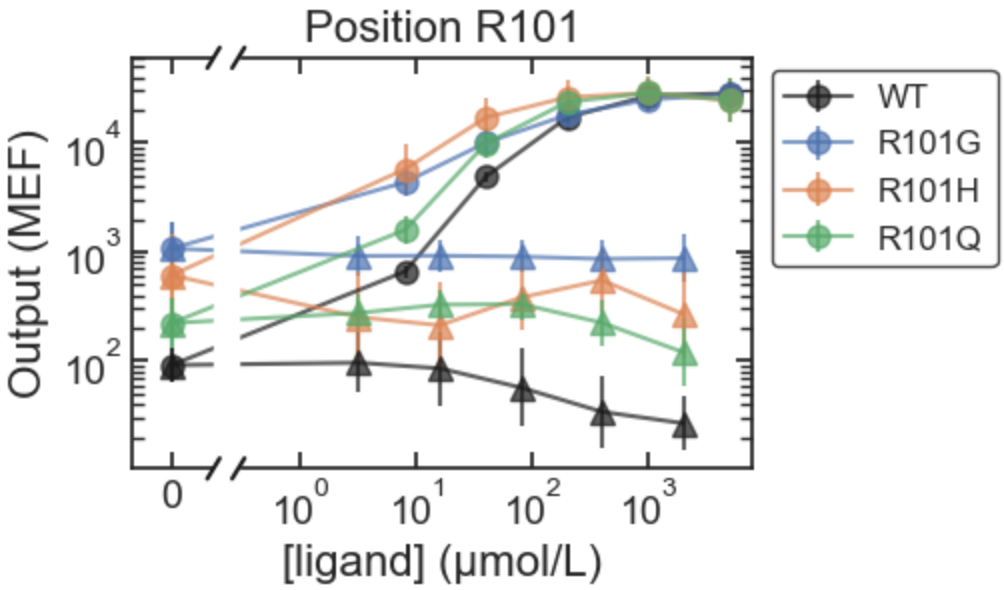

Position A92
Output (MEF)
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V

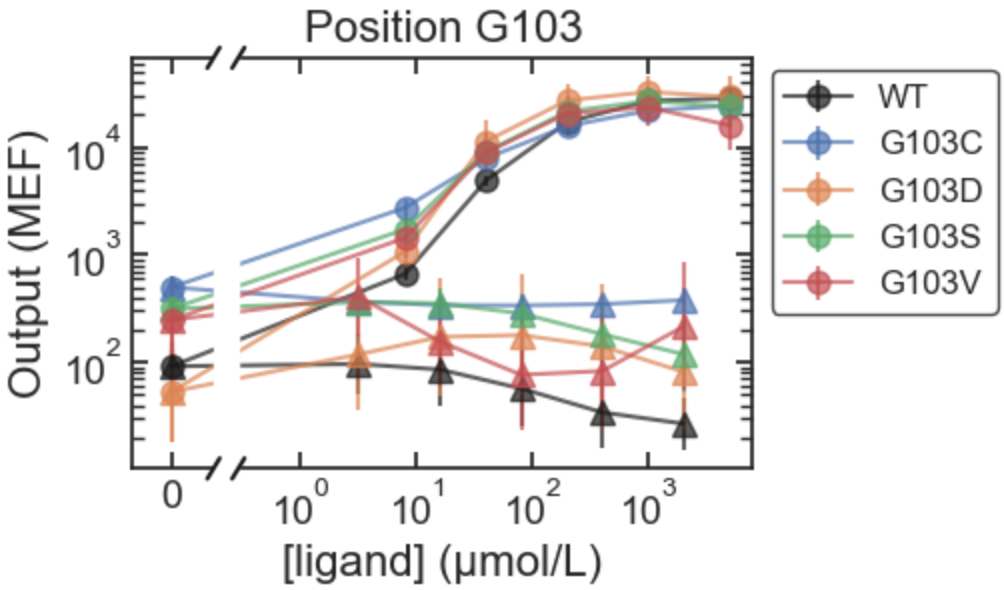

Position V96
Output (MEF)
[ligand] (μmol/L)
WT
V96A
V96E
V96M

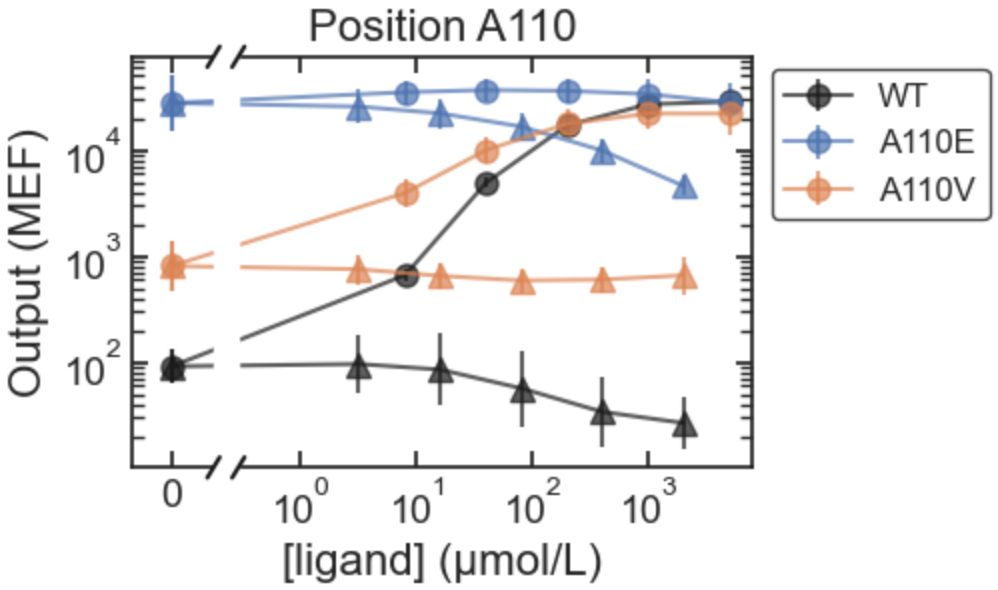

Position S97
Output (MEF)
[ligand] (μmol/L)
WT
S97L
S97T

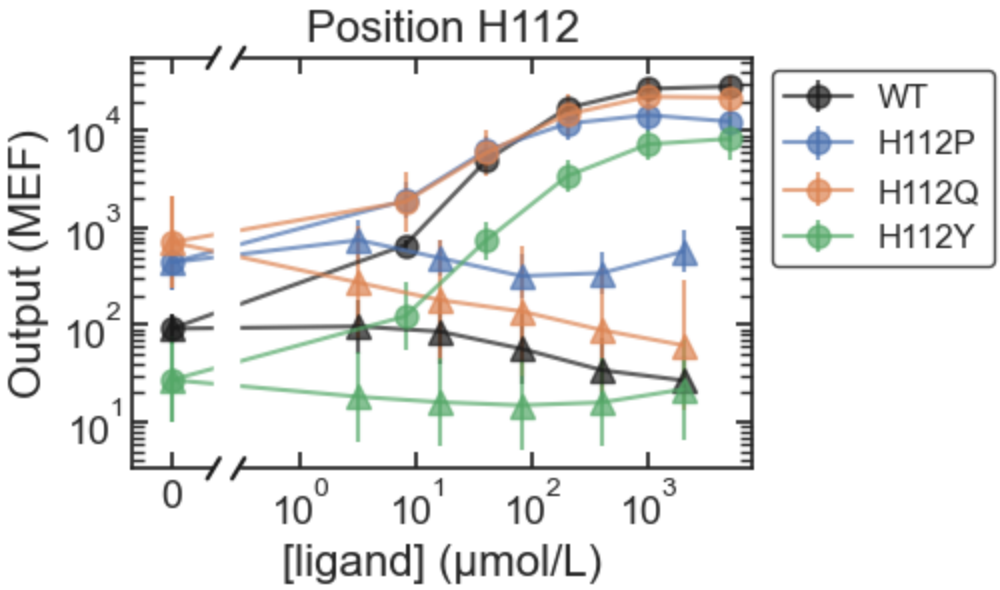

Position M98
Output (MEF)
[ligand] (μmol/L)
WT
M98I
M98K
M98L
M98T
M98V

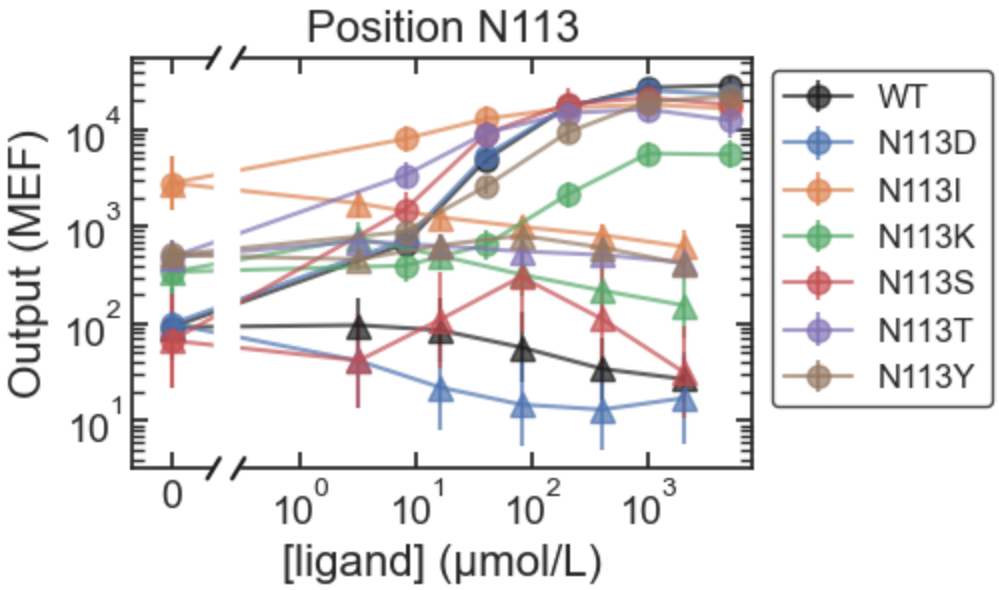

Position V99
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V99A
V99I
V99L

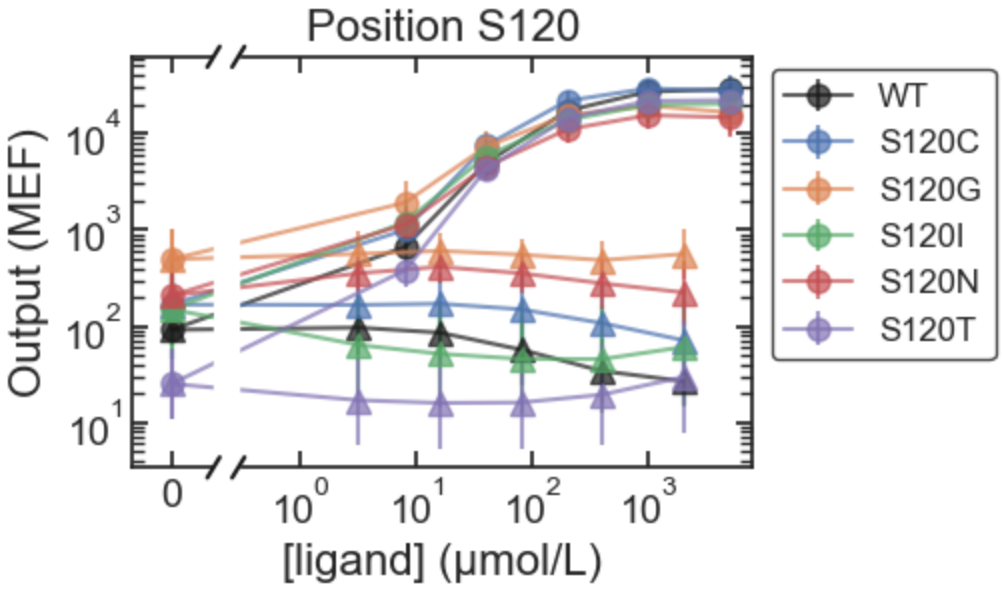

Position R101
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

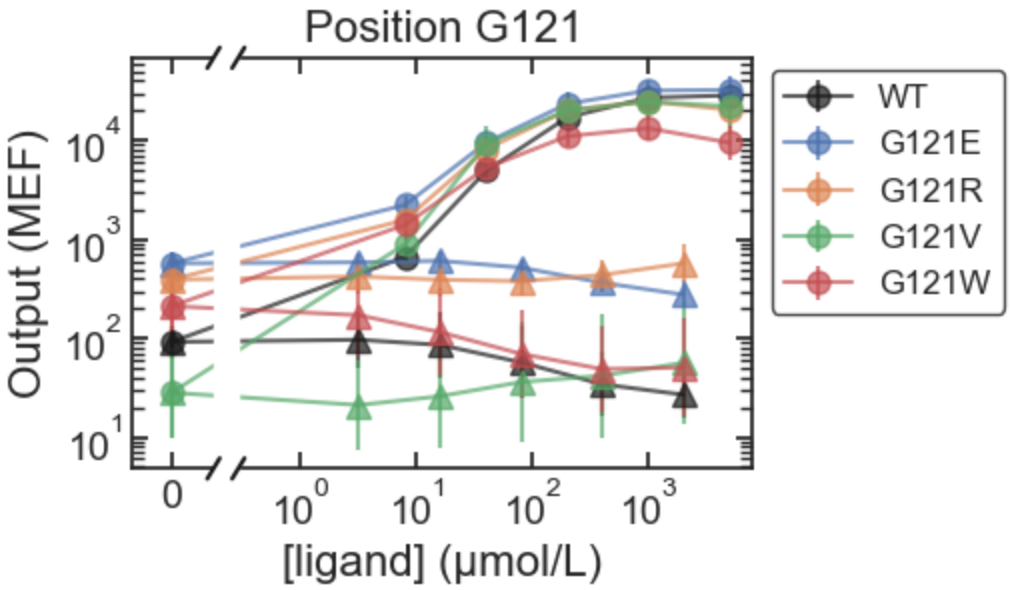

Position G103
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

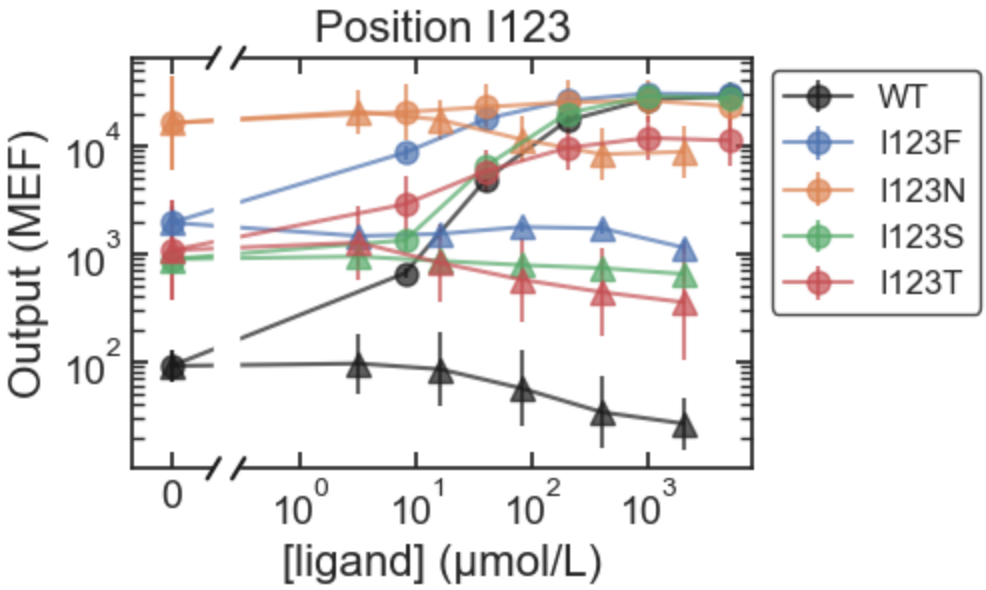

Position A110
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A110E
A110V

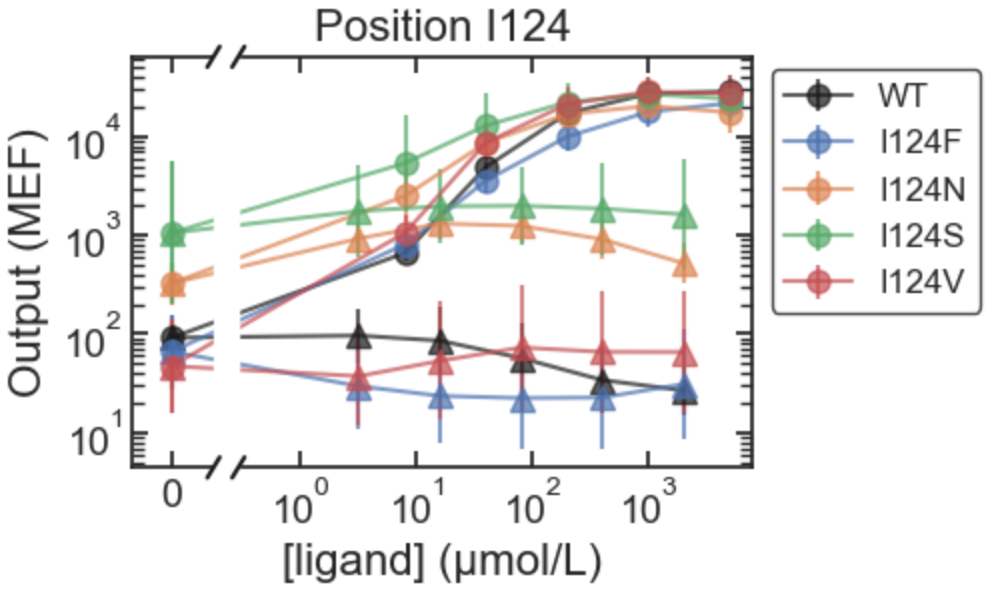

Position H112
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

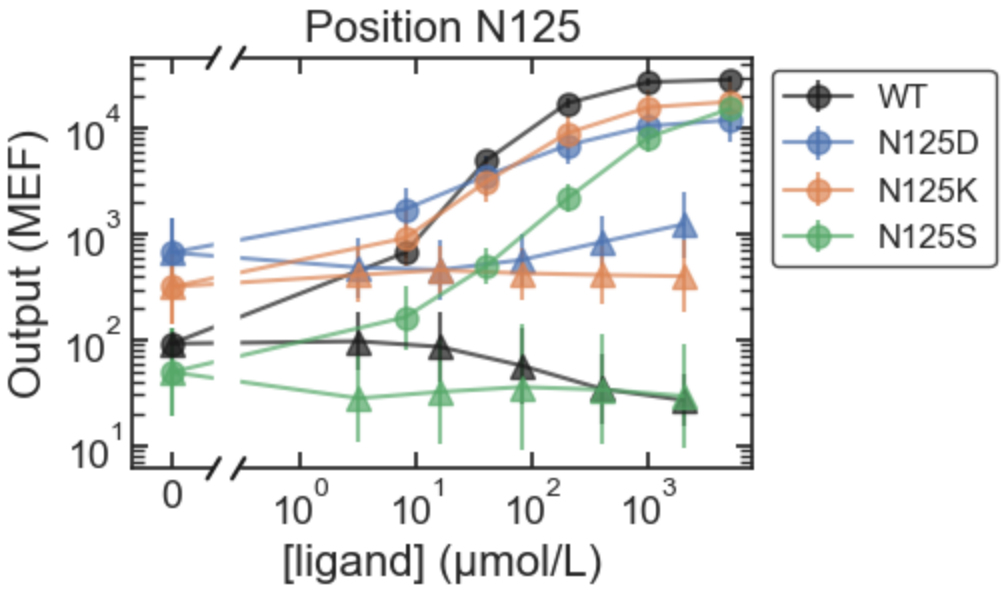

Position N113
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

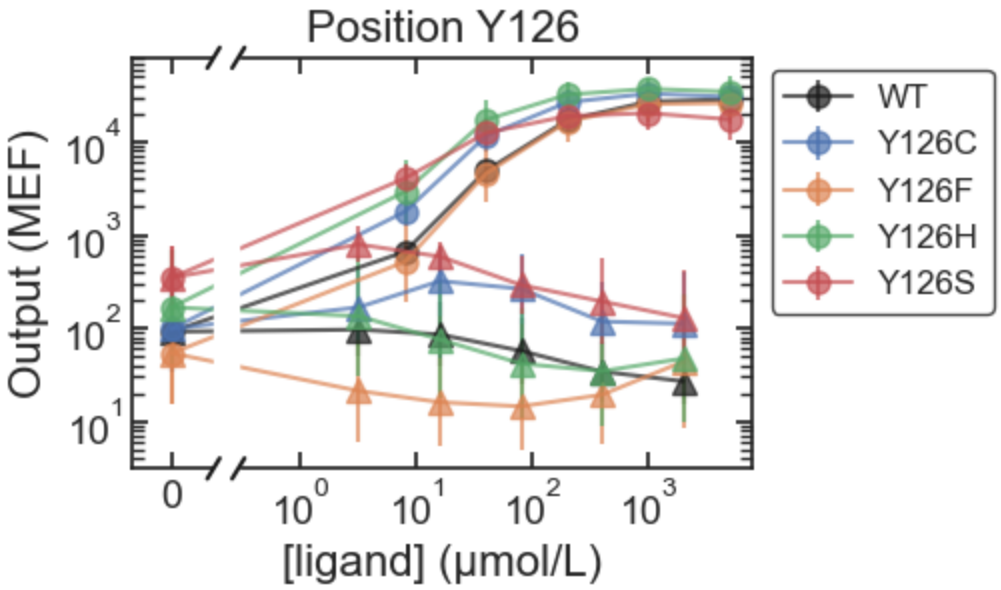

Position S120
Output (MEF)
[ligand] (μmol/L)
WT
S120C
S120G
S120I
S120N
S120T

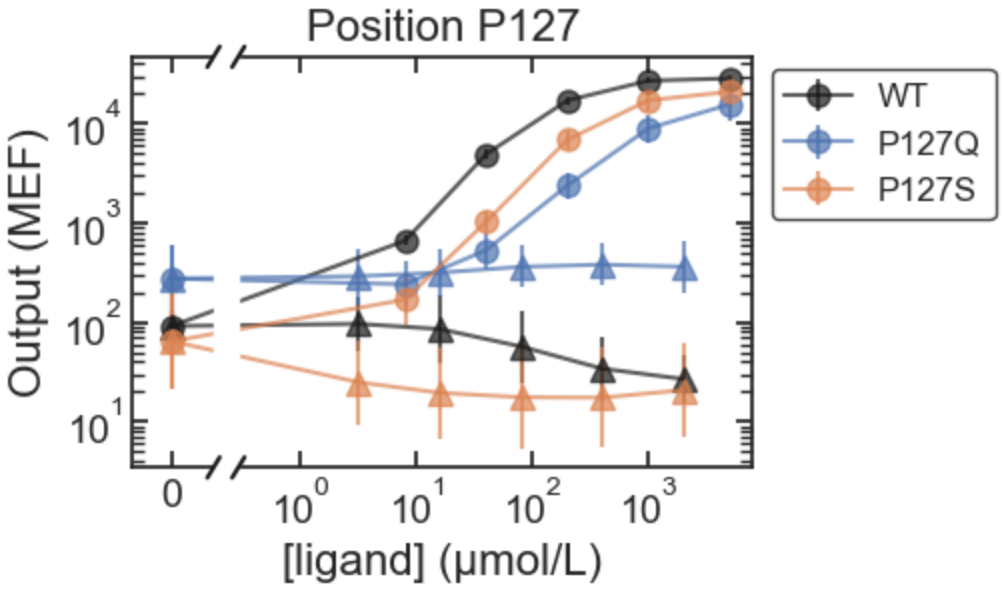

Position G121
Output (MEF)
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

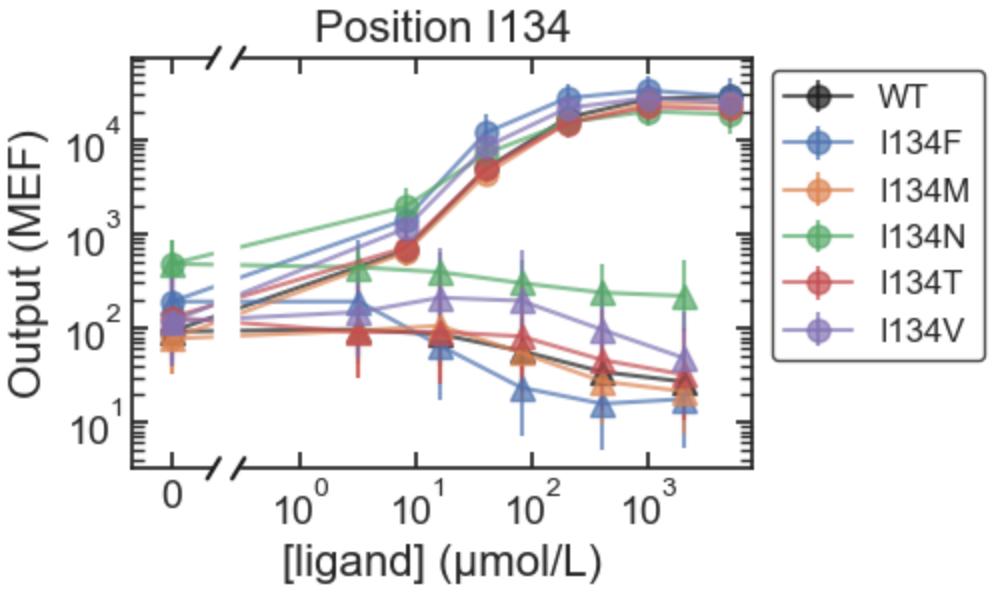

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

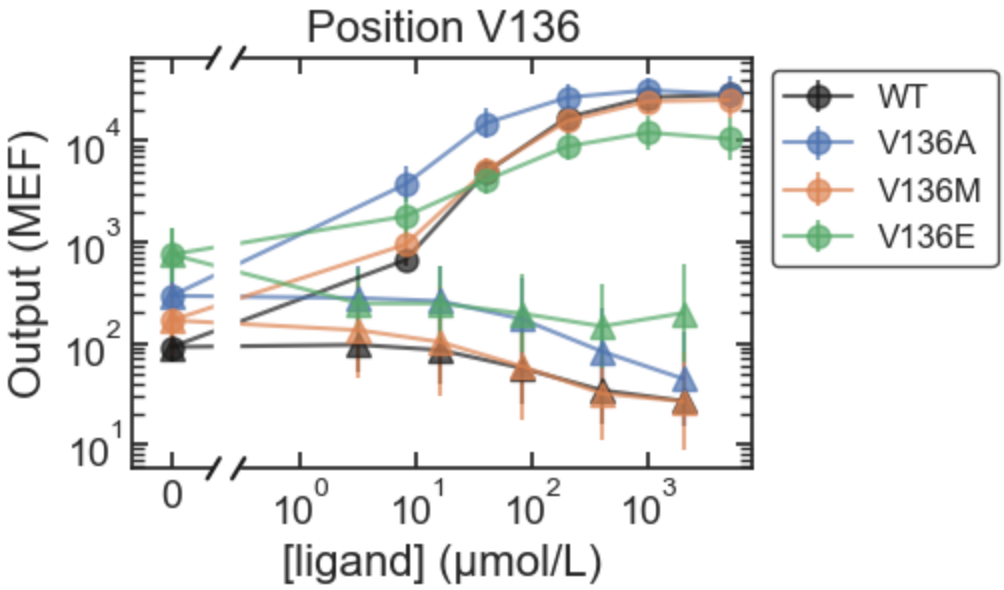

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V

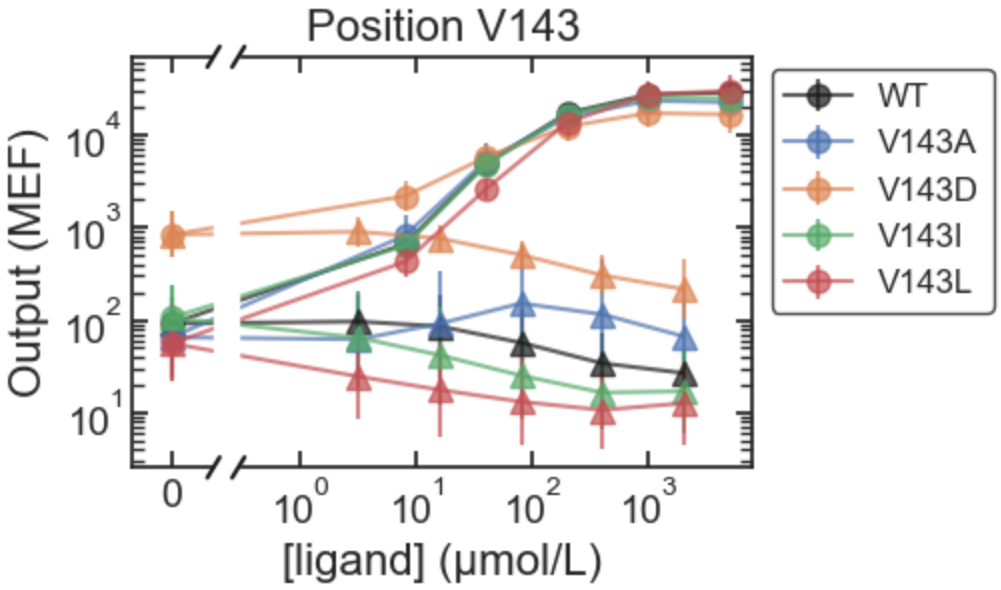

Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S

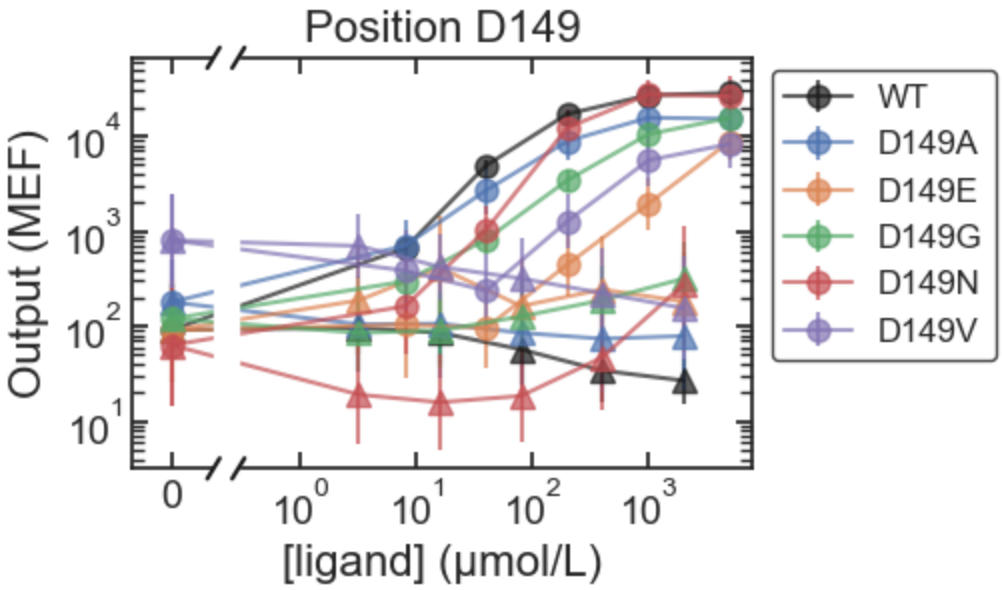

Position Y126
Output (MEF)
[ligand] (μmol/L)
WT
Y126C
Y126F
Y126H
Y126S

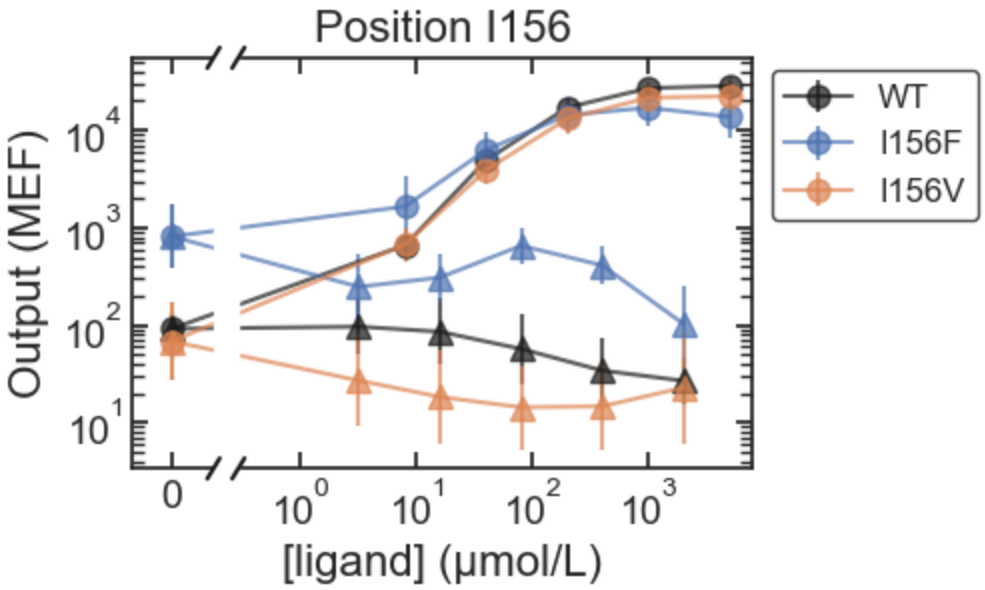

Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

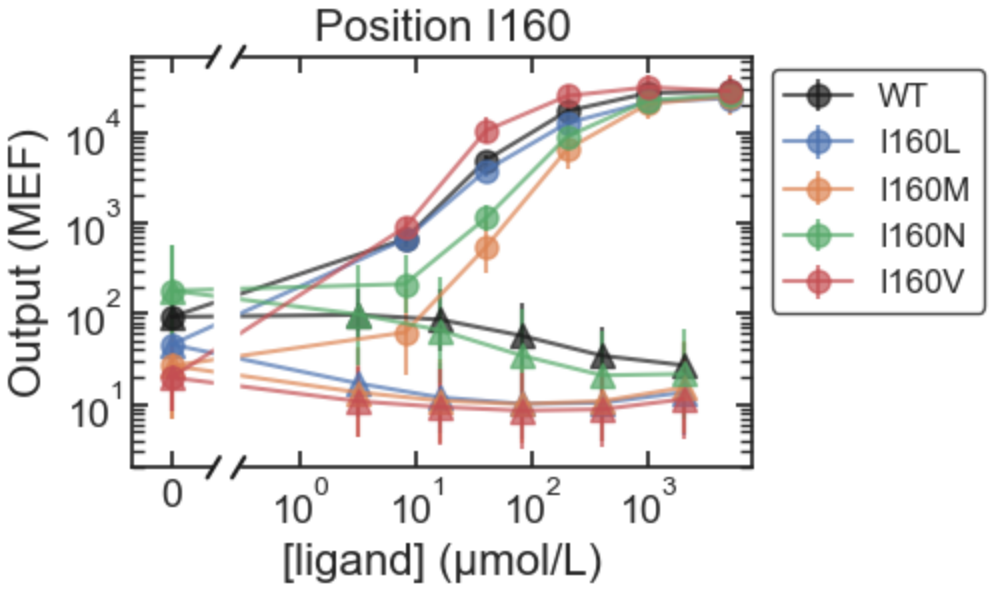

Position I134
Output (MEF)
[ligand] (μmol/L)
WT
I134F
I134M
I134N
I134T
I134V

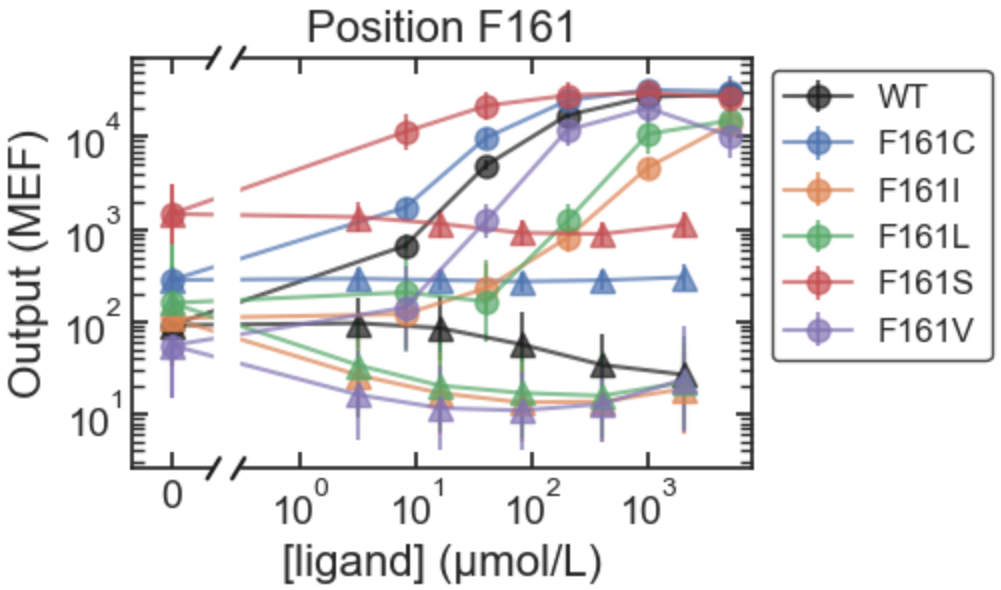

Position V136
Output (MEF)
[ligand] (μmol/L)
WT
V136A
V136M
V136E

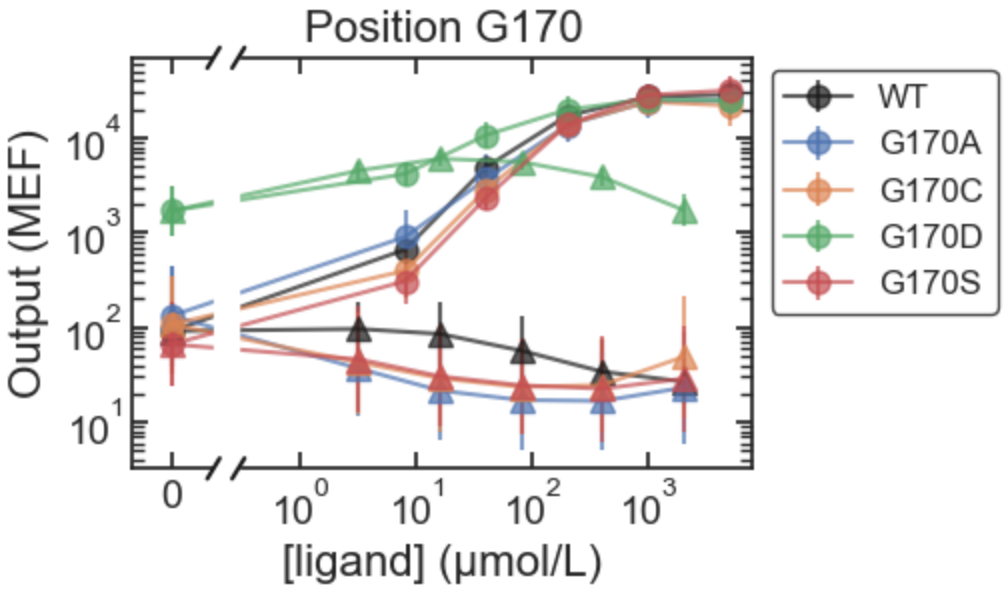

Position V143
Output (MEF)
[ligand] (μmol/L)
WT
V143A
V143D
V143I
V143L

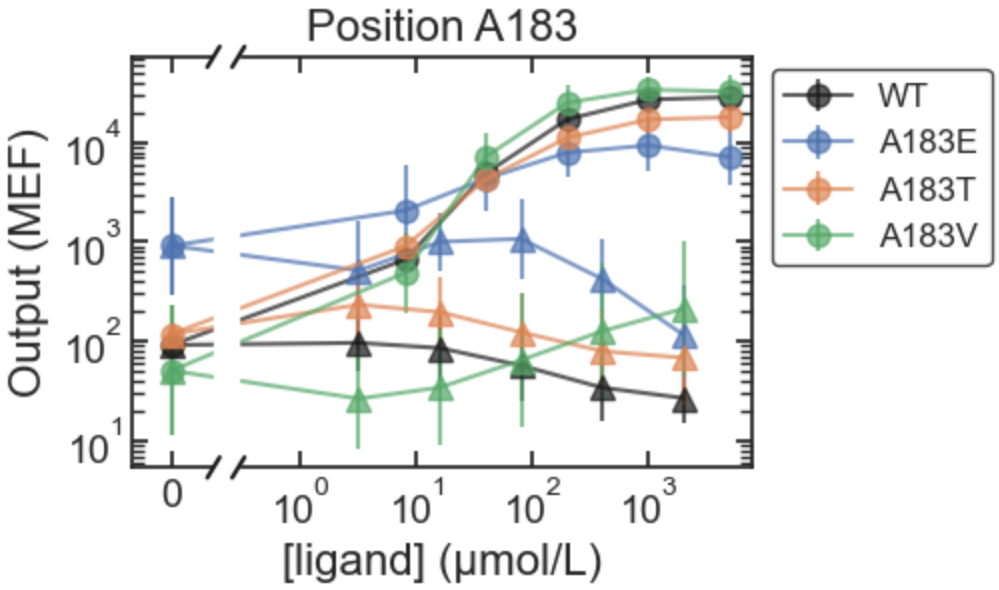

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

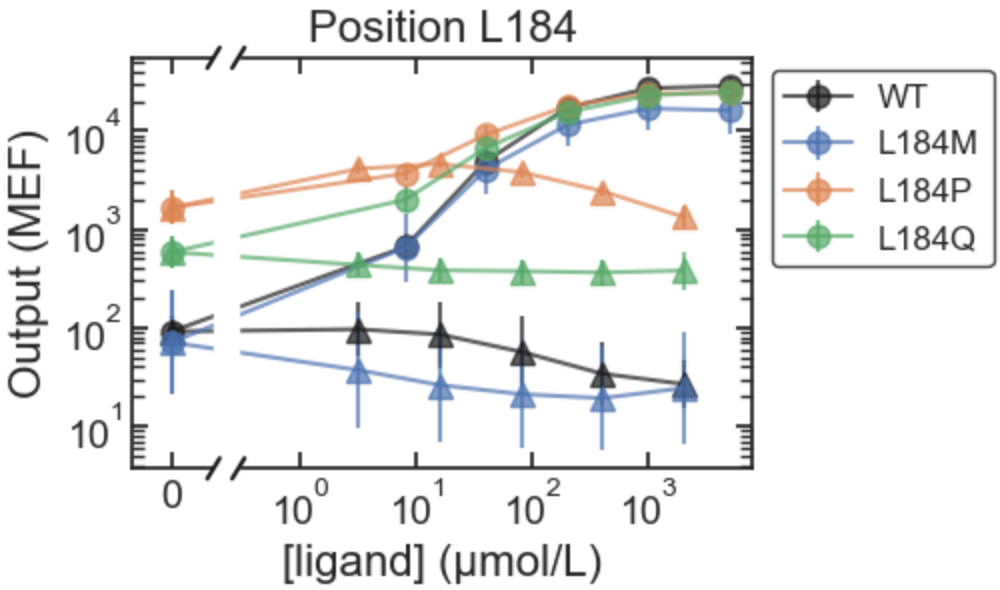

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

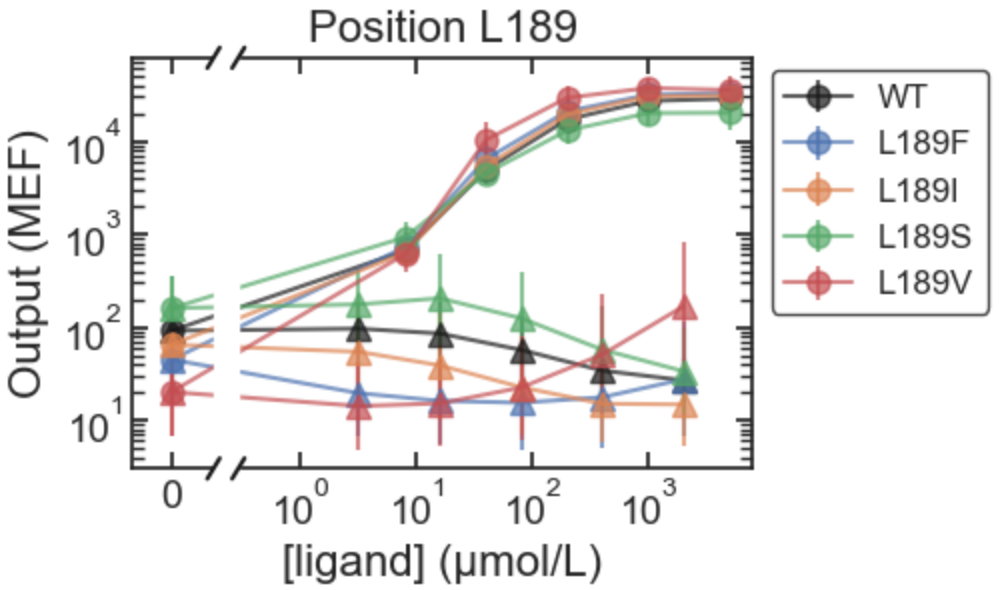

Position I160
Output (MEF)
[ligand] (μmol/L)
WT
I160L
I160M
I160N
I160V

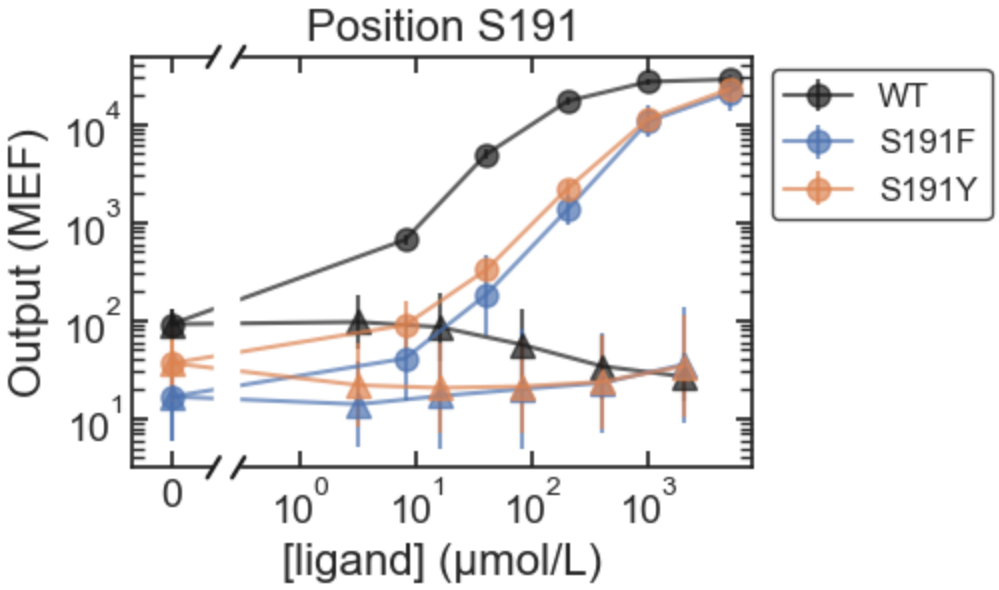

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

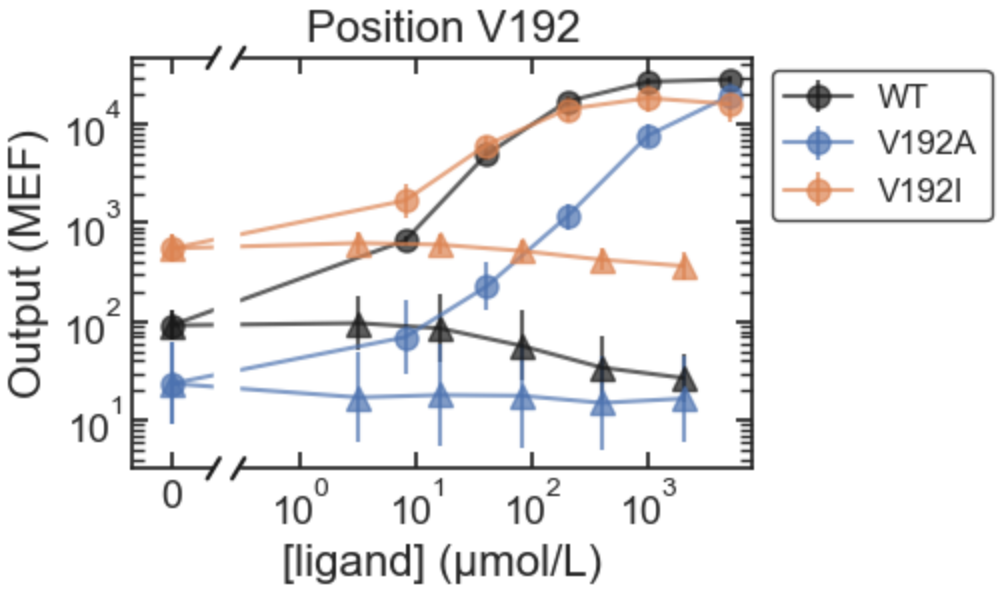

Position G170
Output (MEF)
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

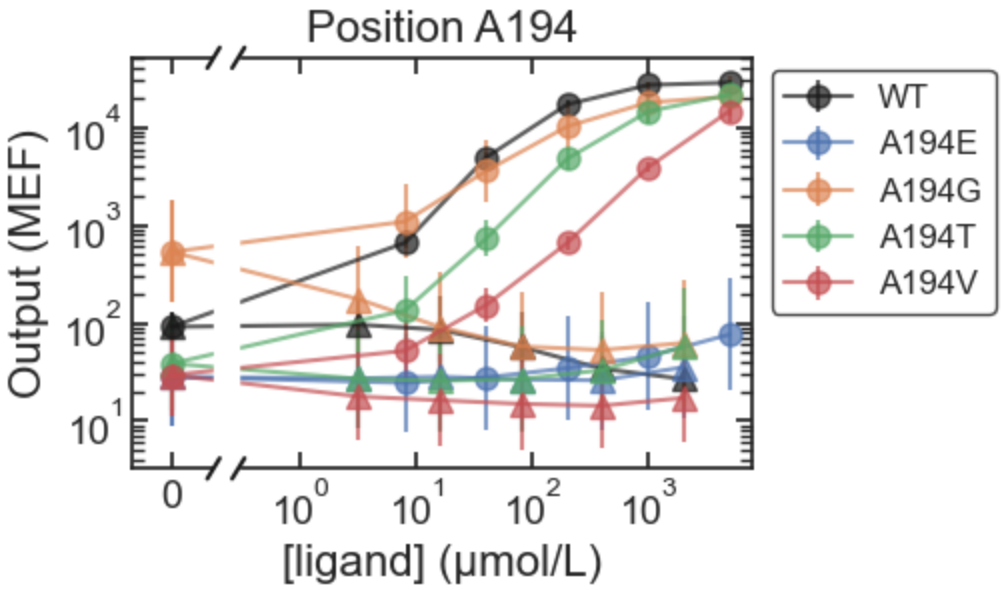

Position A183
Output (MEF)
[ligand] (μmol/L)
WT
A183E
A183T
A183V

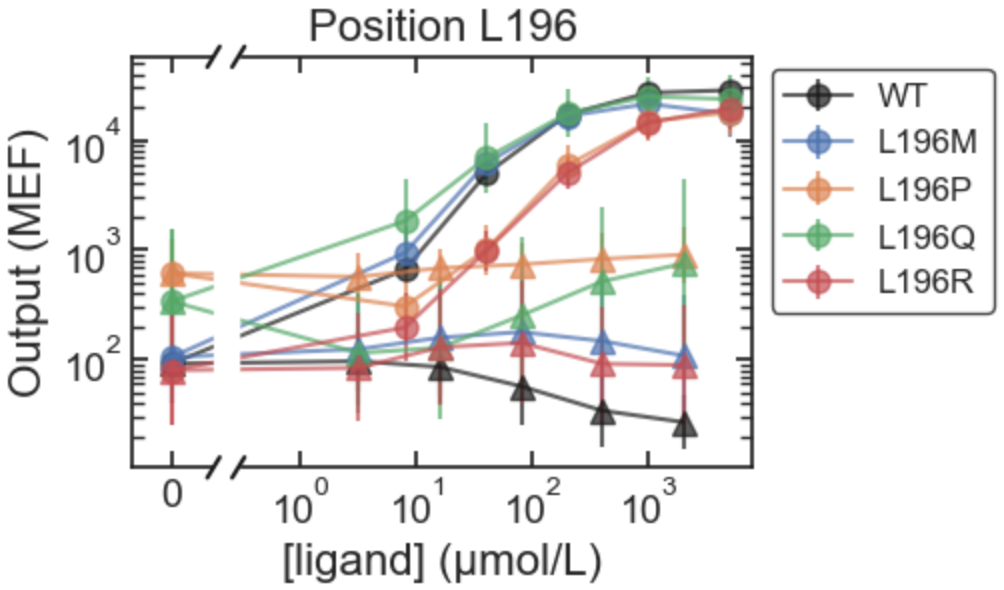

Position L184
Output (MEF)
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

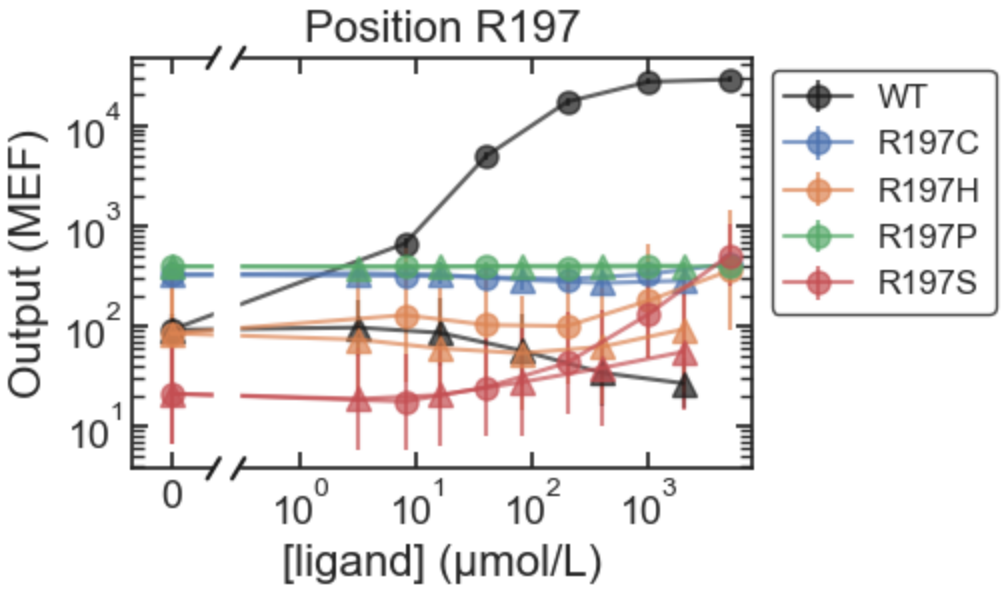

Position L189
Output (MEF)
[ligand] (μmol/L)
WT
L189F
L189I
L189S
L189V

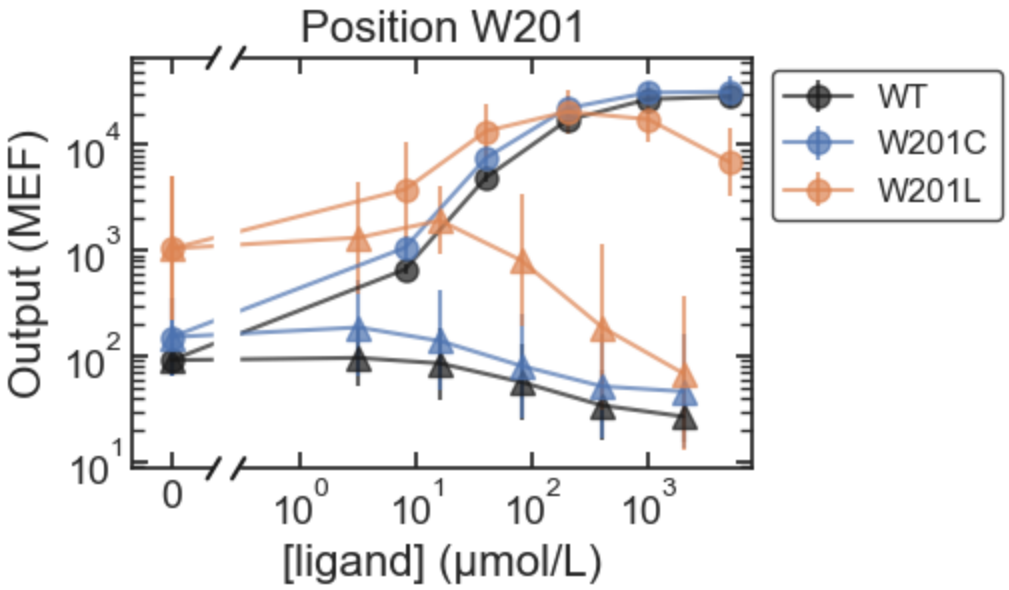

Position S191
Output (MEF)
[ligand] (μmol/L)
WT
S191F
S191Y

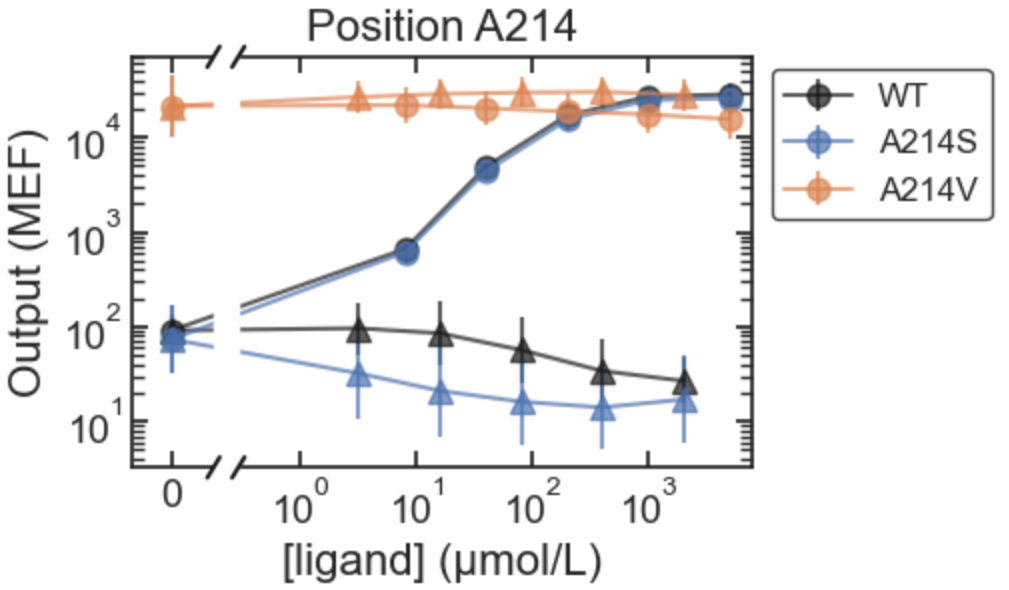

Position V192
Output (MEF)
[ligand] (μmol/L)
WT
V192A
V192I

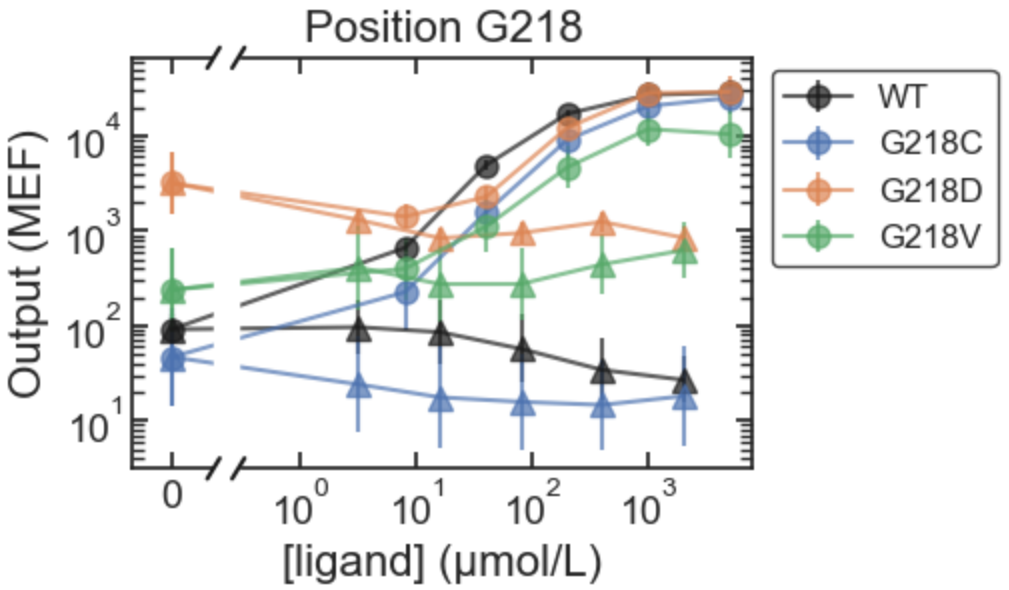

Position A194
Output (MEF)
[ligand] (μmol/L)
WT
A194E
A194G
A194T
A194V

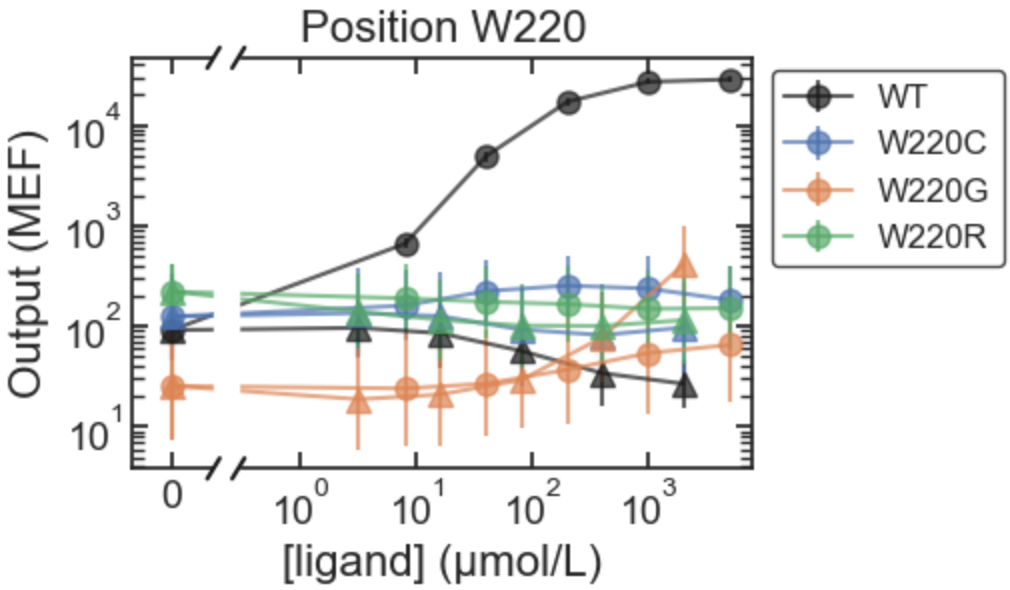

Position L196
Output (MEF)
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

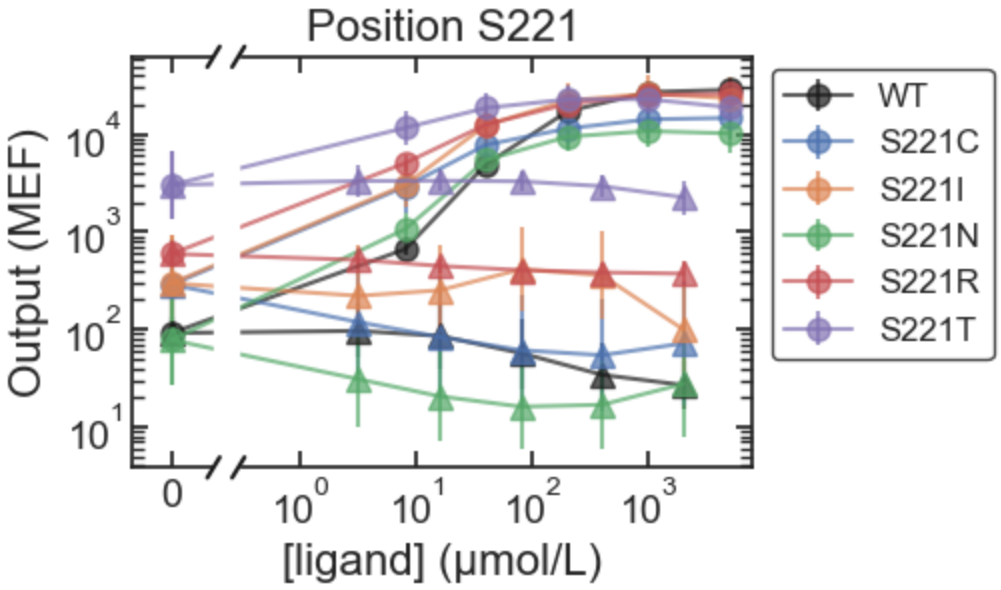

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

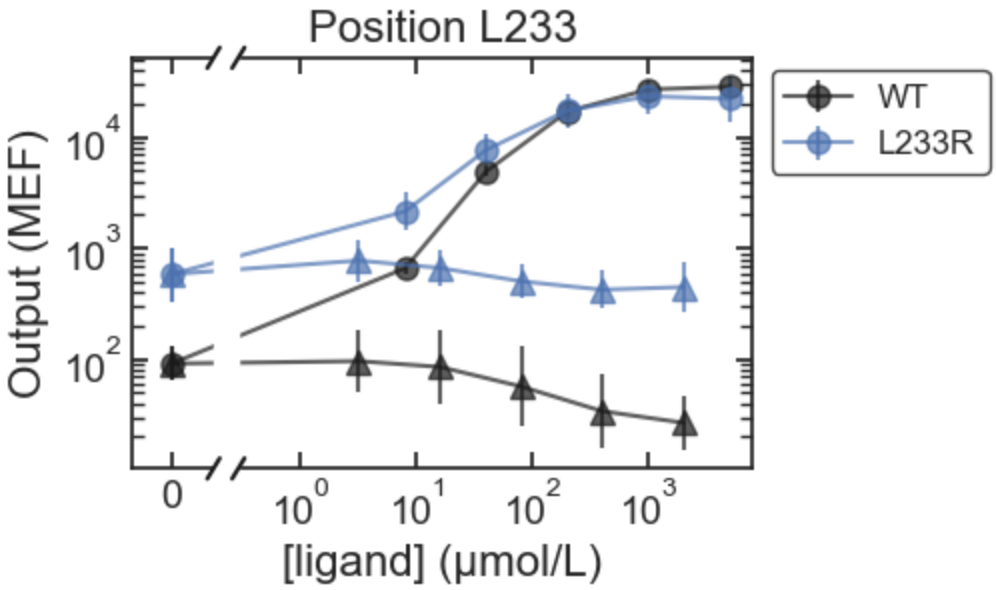

Position W201
Output (MEF)
[ligand] (μmol/L)
WT
W201C
W201L

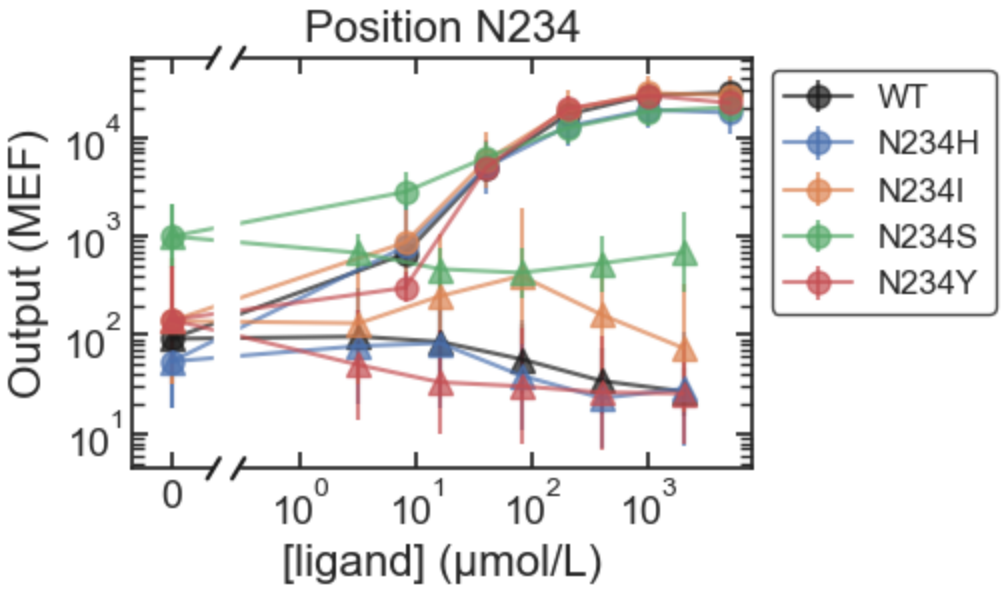

Position A214
Output (MEF)
[ligand] (μmol/L)
WT
A214S
A214V

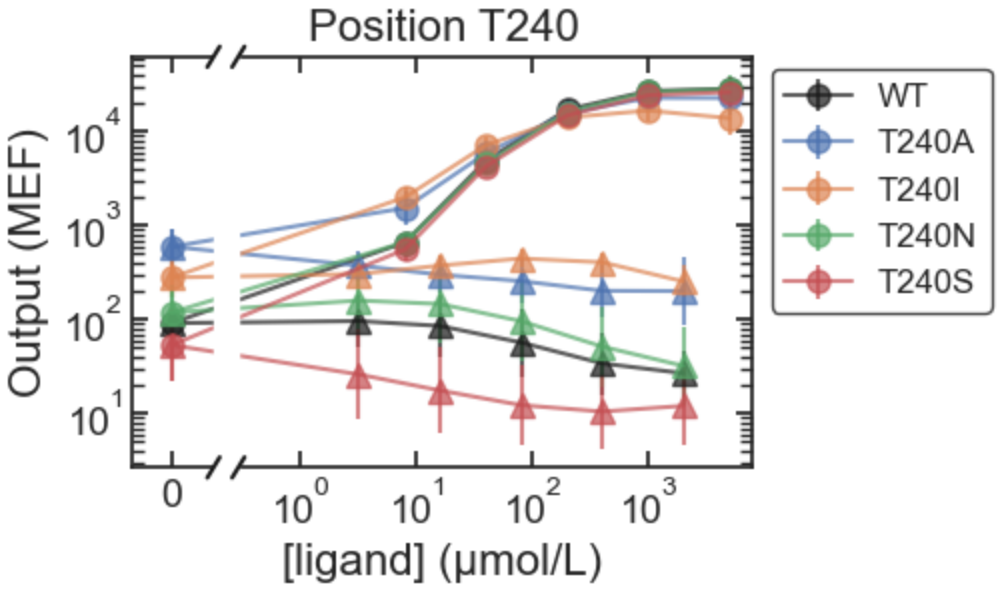

Position G218
Output (MEF)
[ligand] (μmol/L)
WT
G218C
G218D
G218V

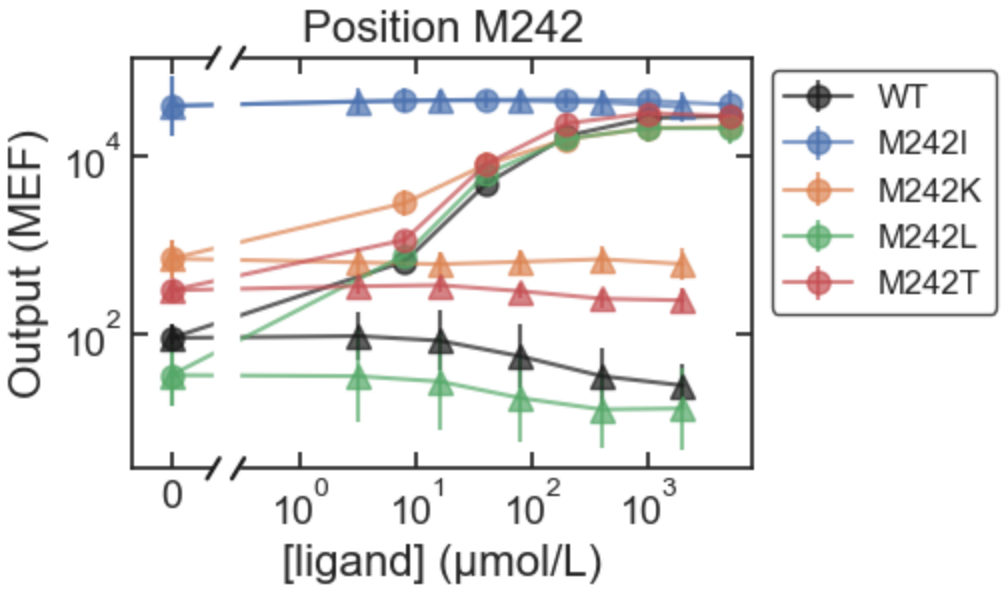

Position W220
Output (MEF)
[ligand] (μmol/L)
WT
W220C
W220G
W220R

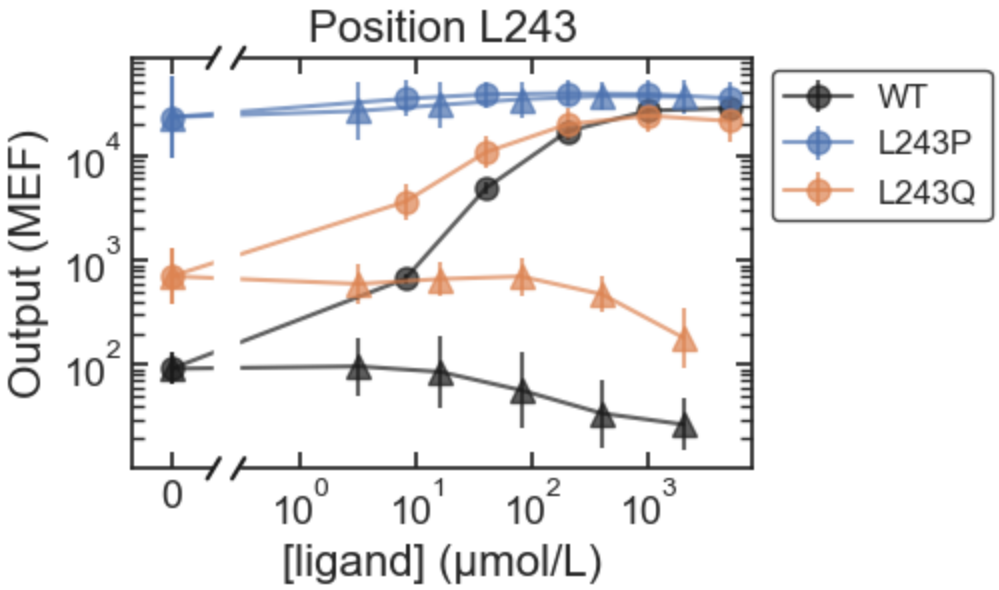

Position S221
Output (MEF)
[ligand] (μmol/L)
WT
S221C
S221I
S221N
S221R
S221T

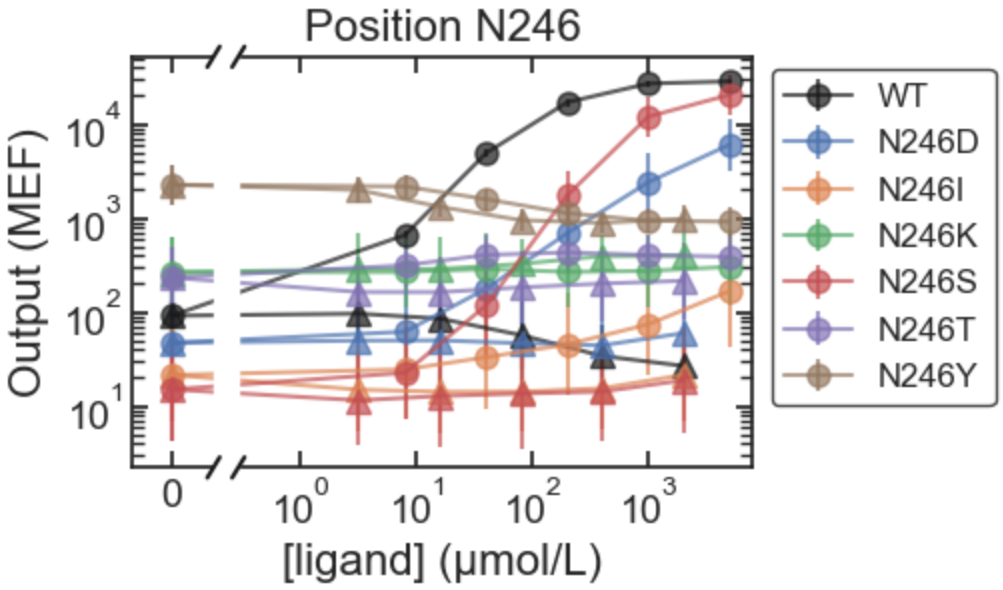

Position L233
Output (MEF)
[ligand] (μmol/L)
WT
L233R

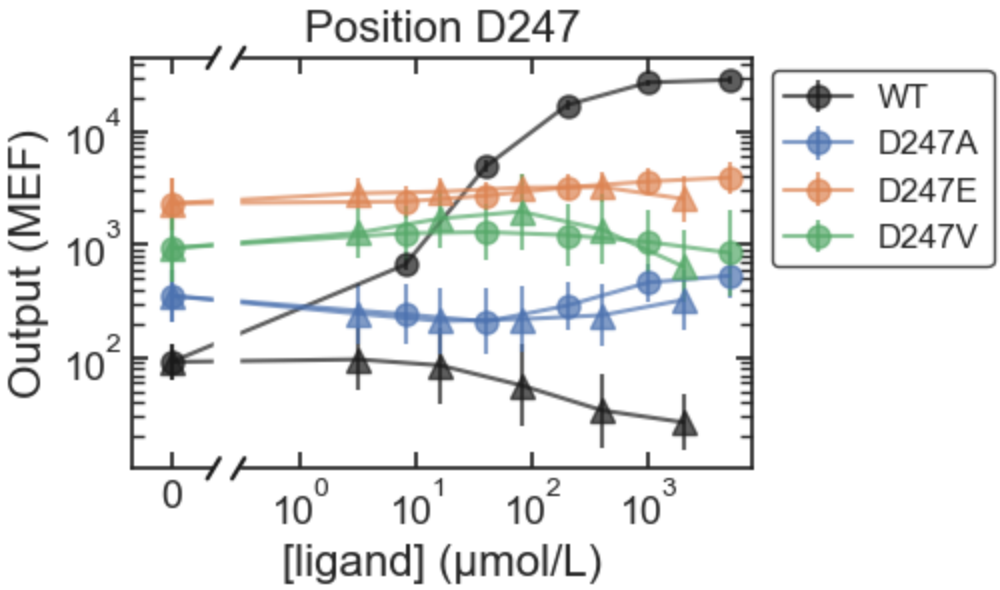

Position N234
Output (MEF)
[ligand] (μmol/L)
WT
N234H
N234I
N234S
N234Y

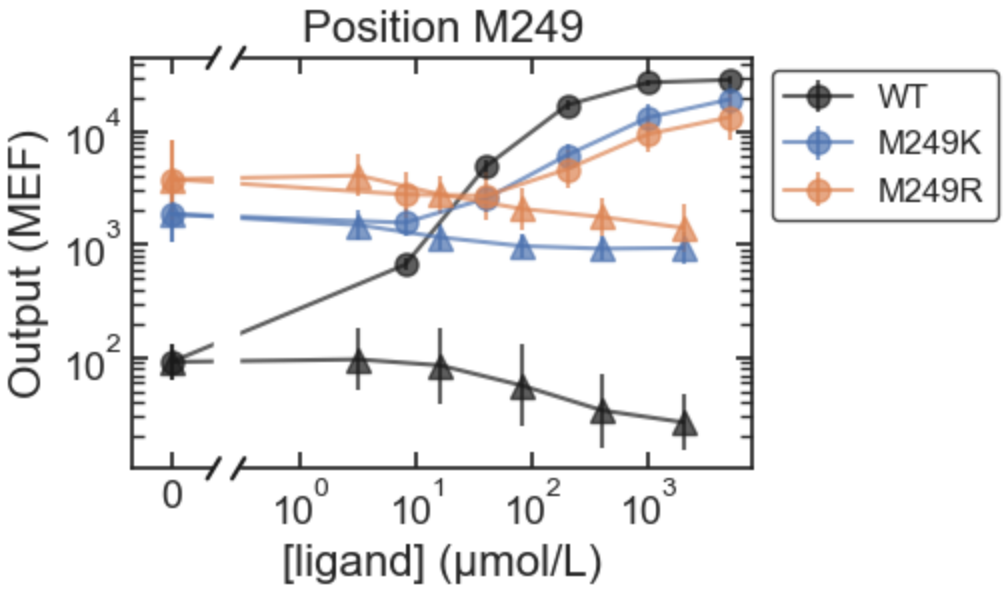

Position T240
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
T240A
T240I
T240N
T240S

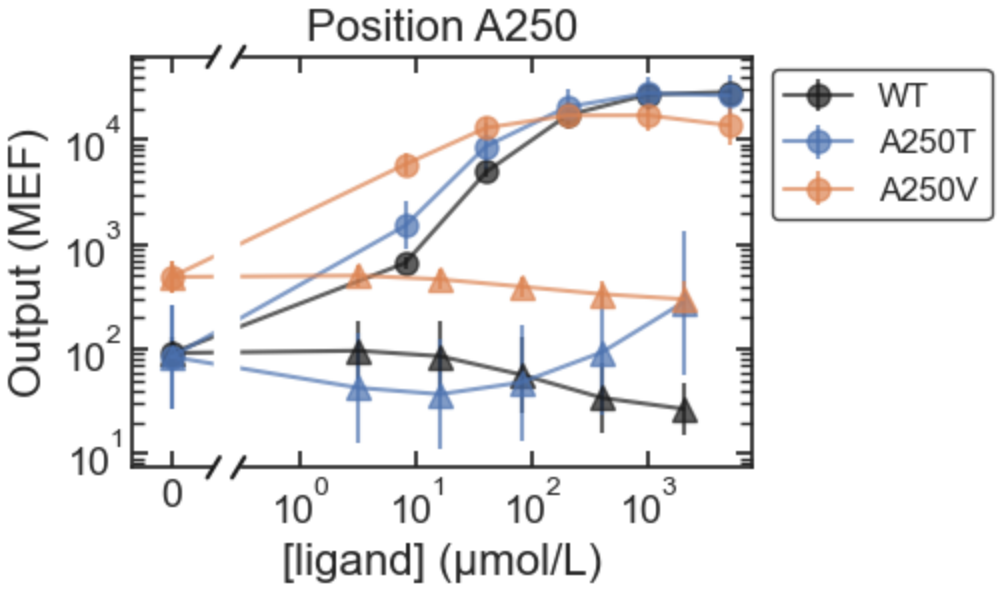

Position M242
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

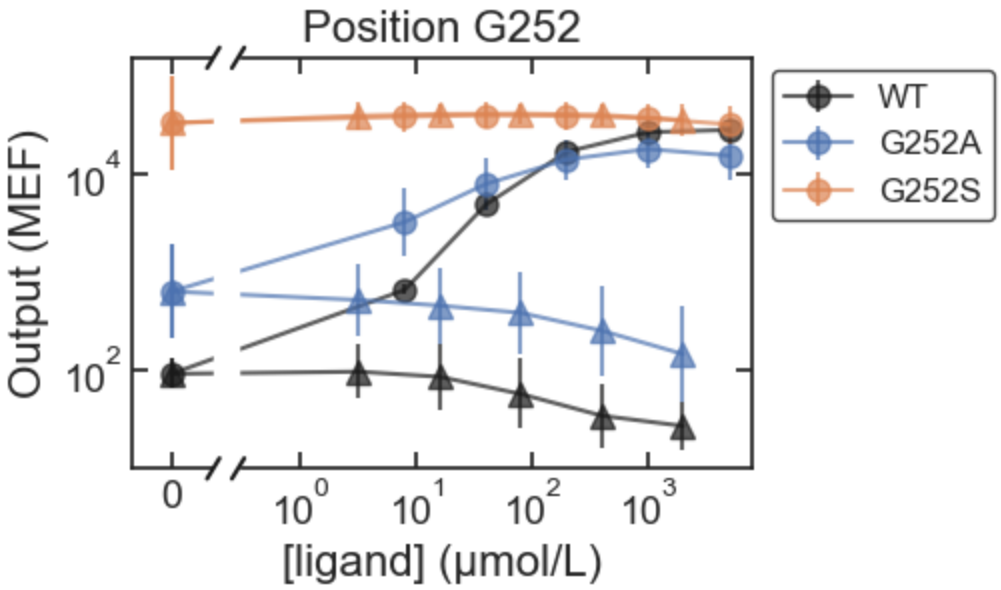

Position L243
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L243P
L243Q

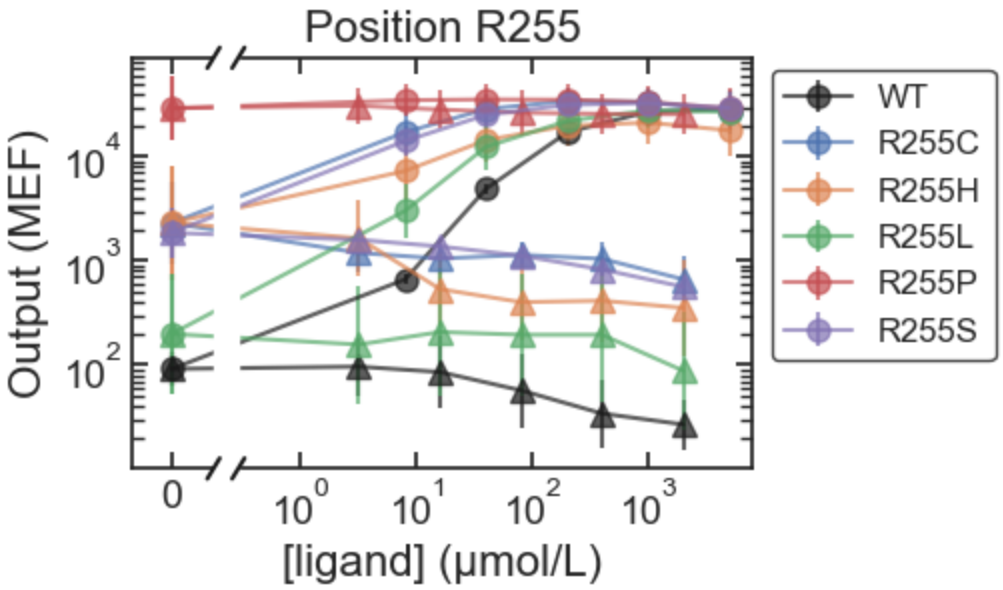

Position N246
Output (MEF)
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

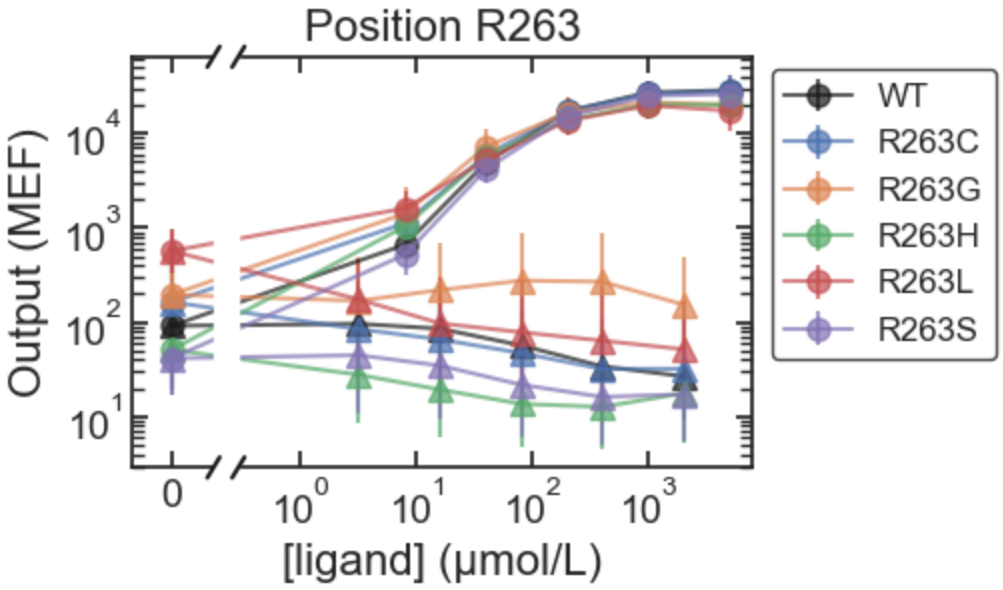

Position D247
Output (MEF)
[ligand] (μmol/L)
WT
D247A
D247E
D247V

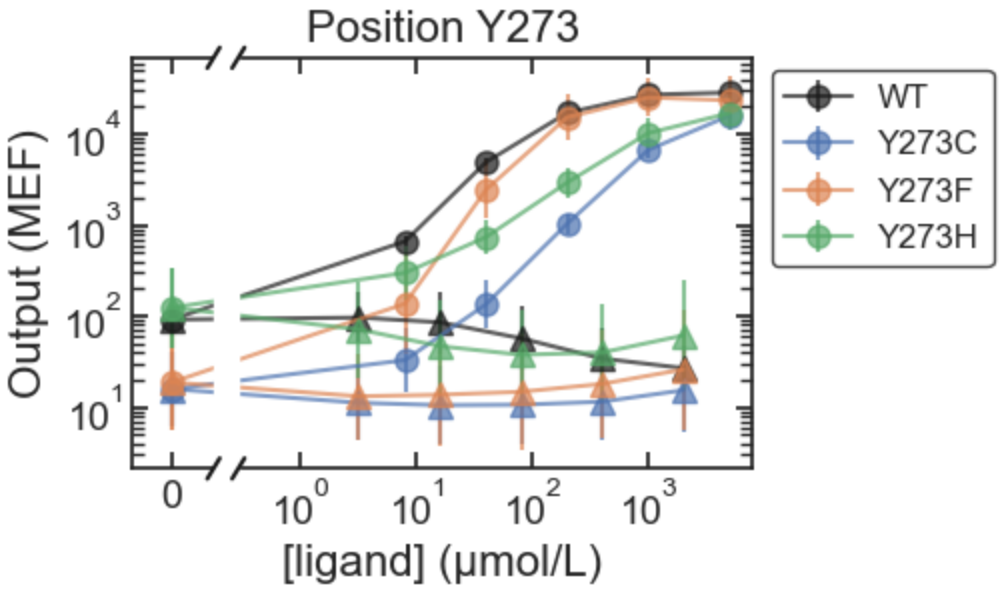

Position M249
Output (MEF)
[ligand] (μmol/L)
WT
M249K
M249R

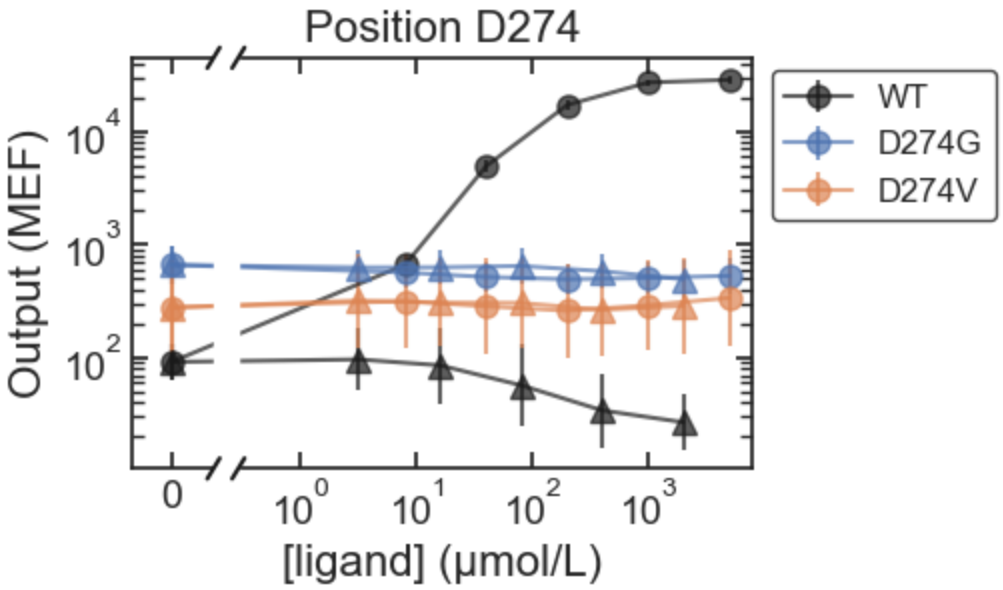

Position A250
Output (MEF)
[ligand] (μmol/L)
WT
A250T
A250V

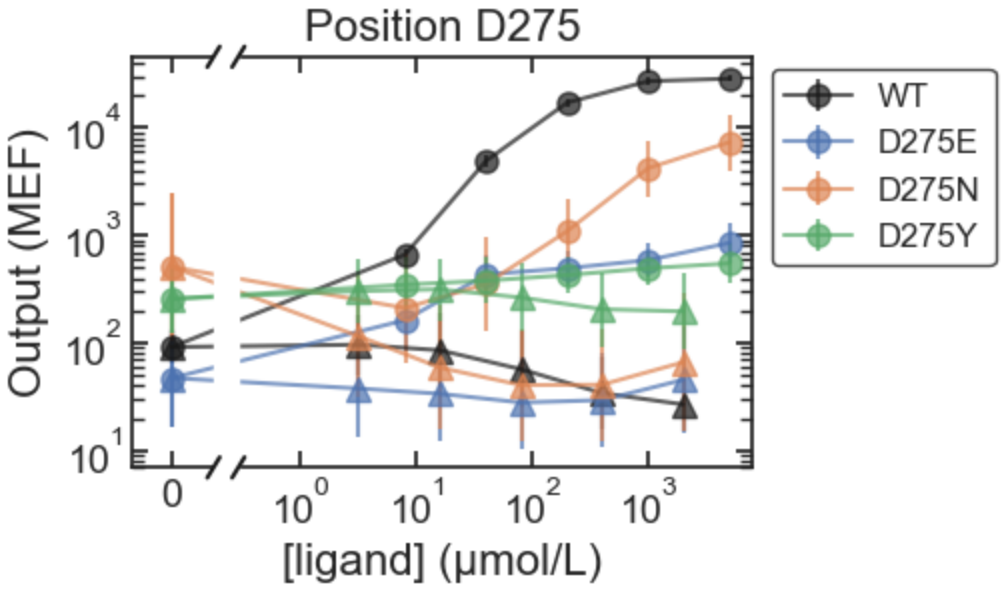

Position G252
Output (MEF)
[ligand] (μmol/L)
WT
G252A
G252S

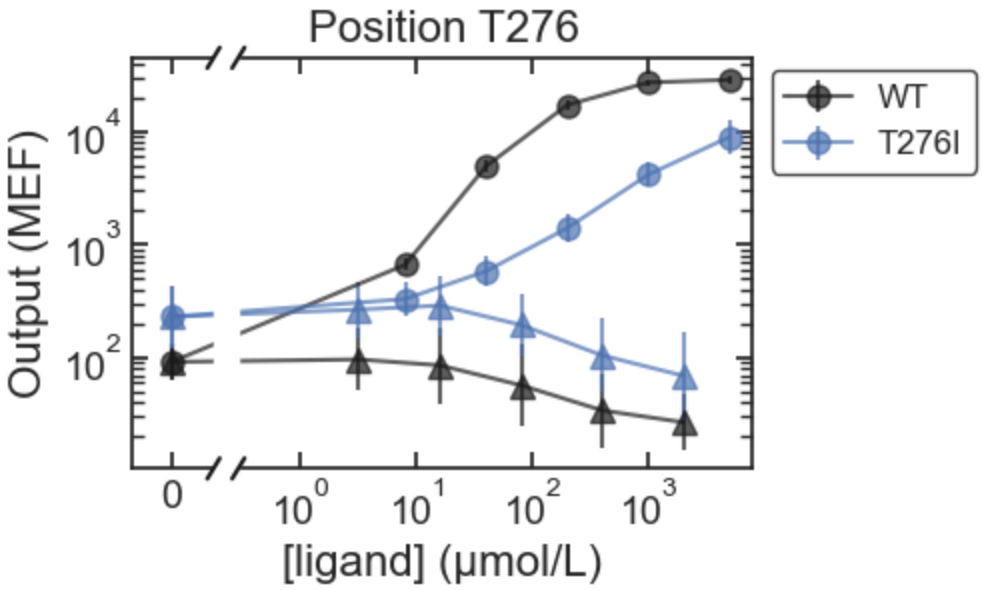

Position R255
Output (MEF)
[ligand] (μmol/L)
WT
R255C
R255H
R255L
R255P
R255S

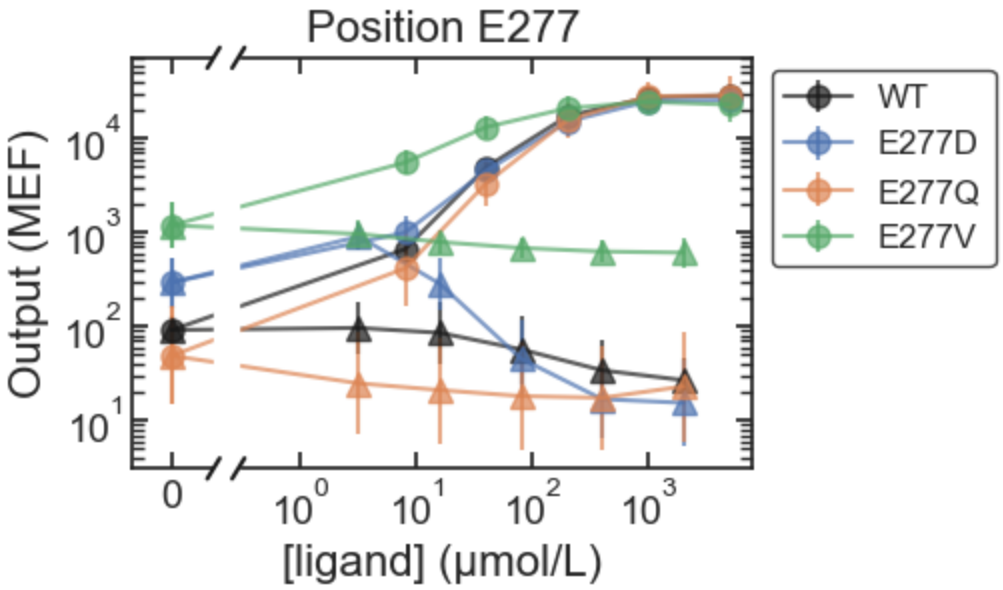

Position R263
Output (MEF)
[ligand] (μmol/L)
WT
R263C
R263G
R263H
R263L
R263S

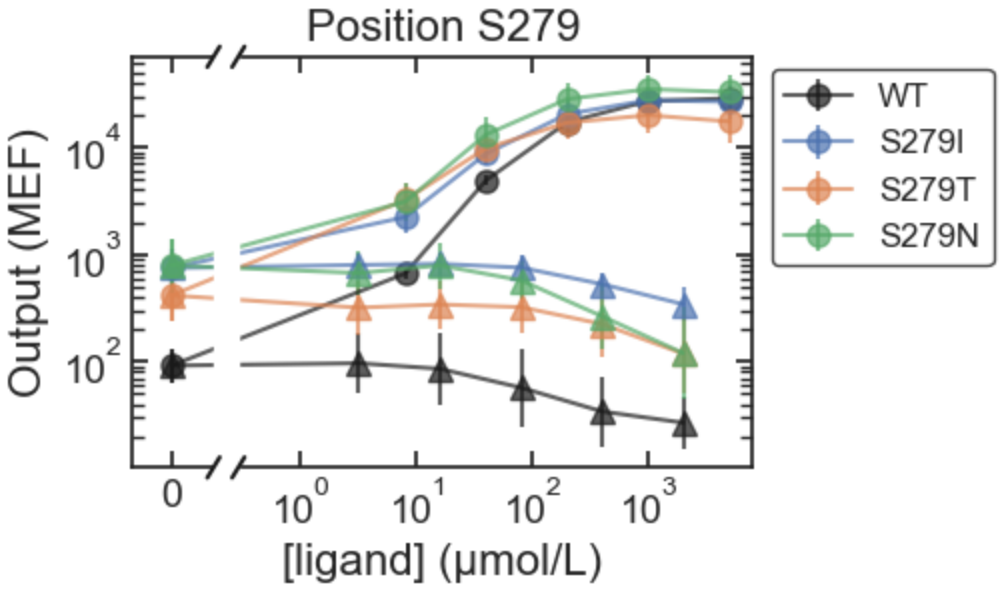

Position Y273
Output (MEF)
[ligand] (μmol/L)
WT
Y273C
Y273F
Y273H

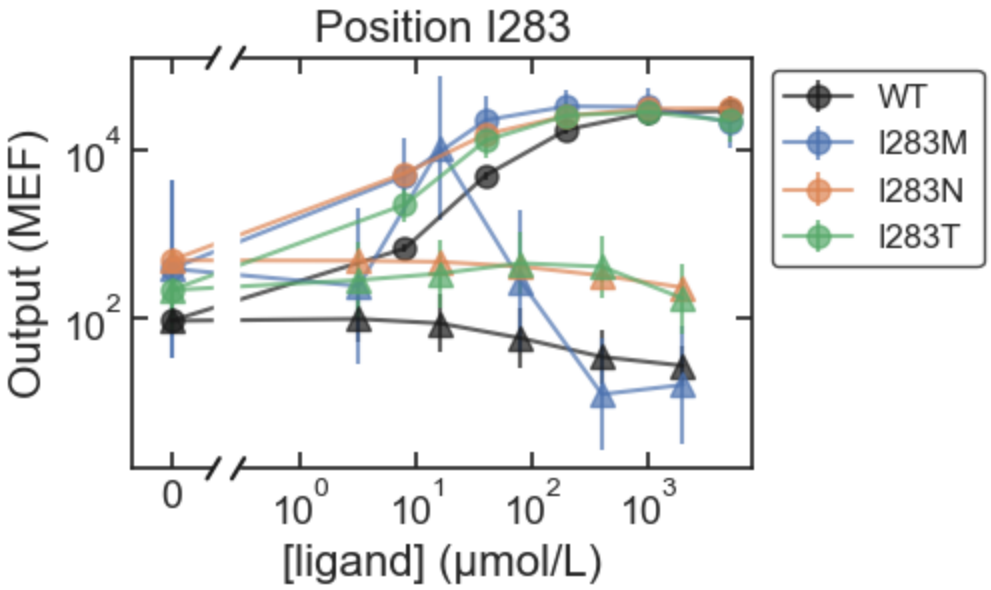

Position D274
Output (MEF)
[ligand] (μmol/L)
WT
D274G
D274V

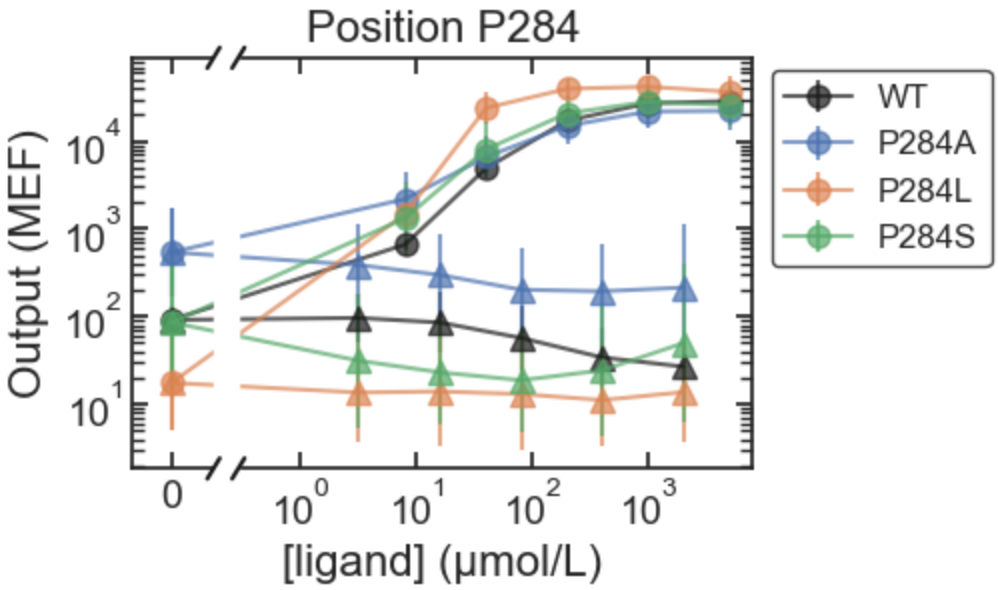

Position D275
Output (MEF)
[ligand] (μmol/L)
WT
D275E
D275N
D275Y

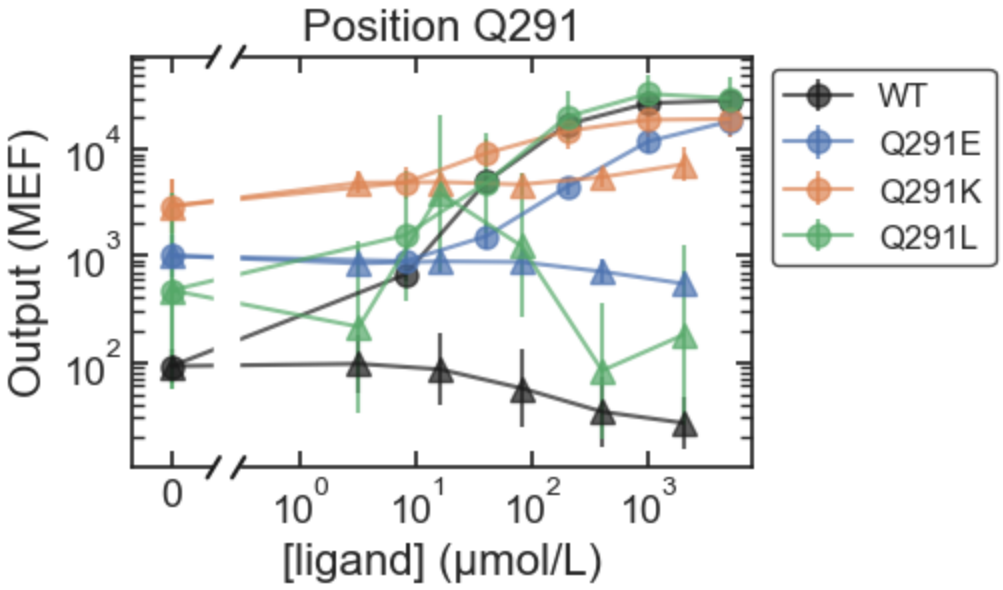

Position T276
Output (MEF)
[ligand] (μmol/L)
WT
T276I

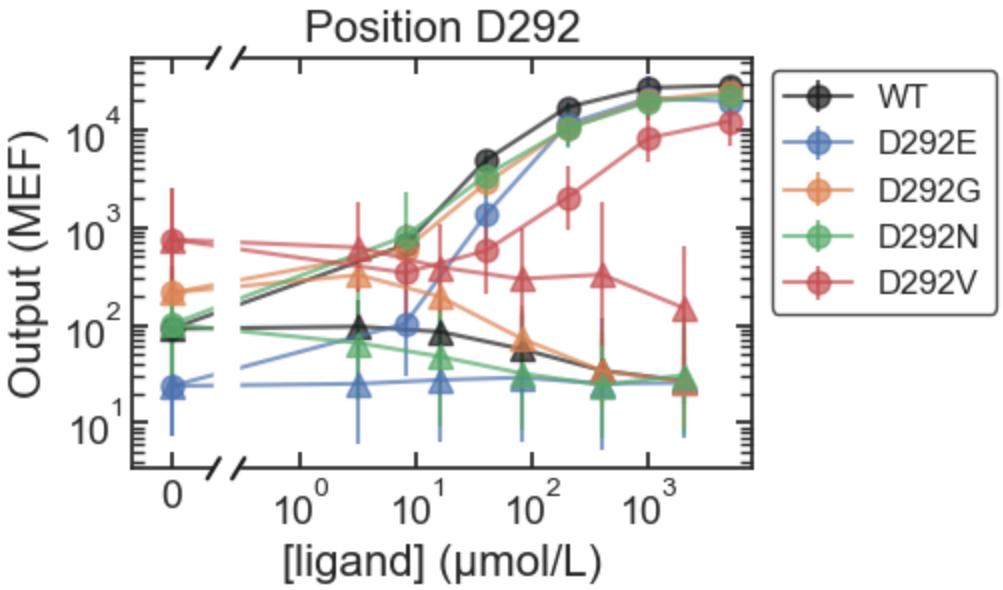

Position E277
Output (MEF)
[ligand] (μmol/L)
WT
E277D
E277Q
E277V

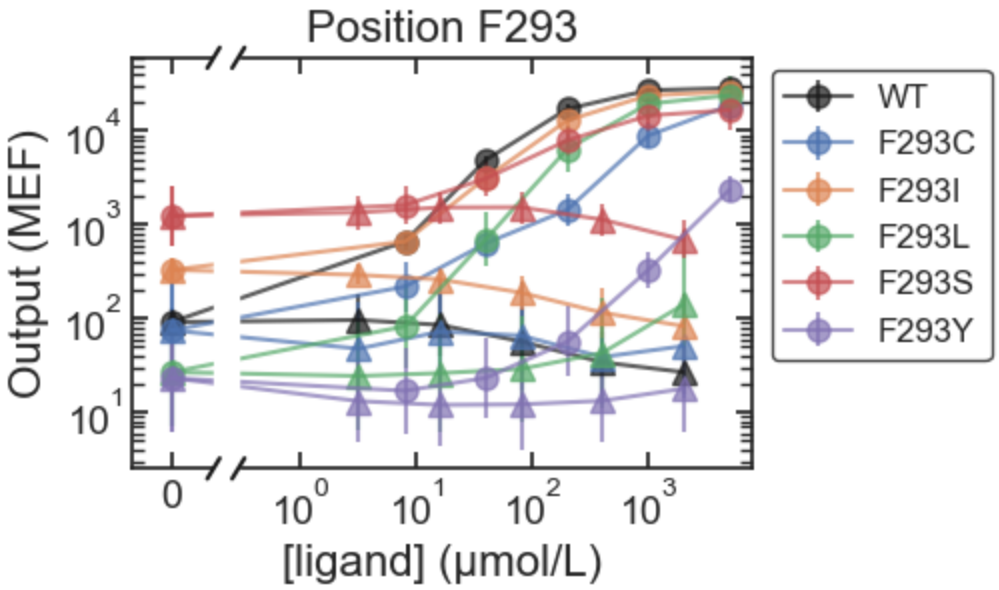

Position S279
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S279I
S279T
S279N

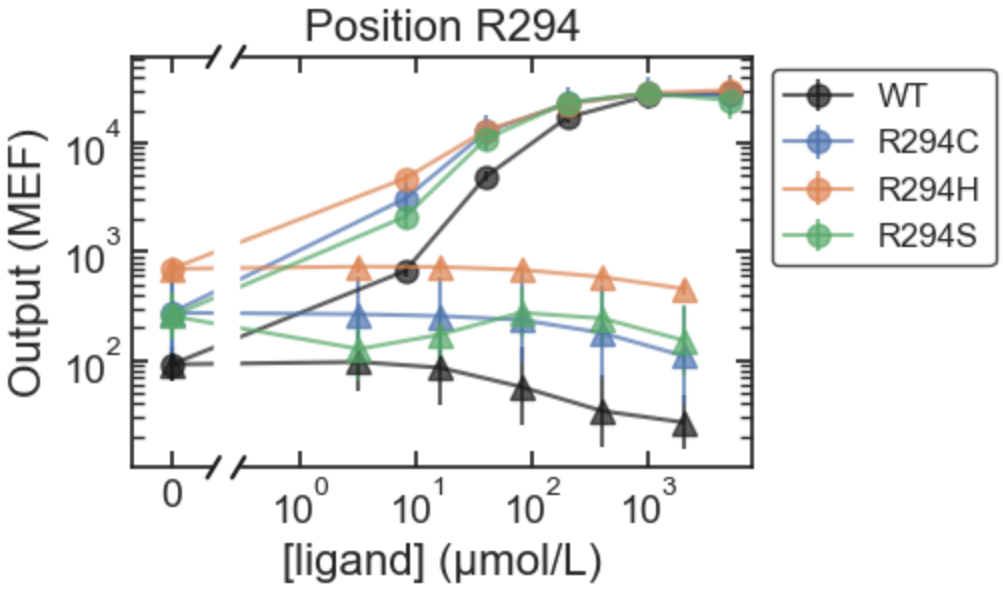

Position I283
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I283M
I283N
I283T

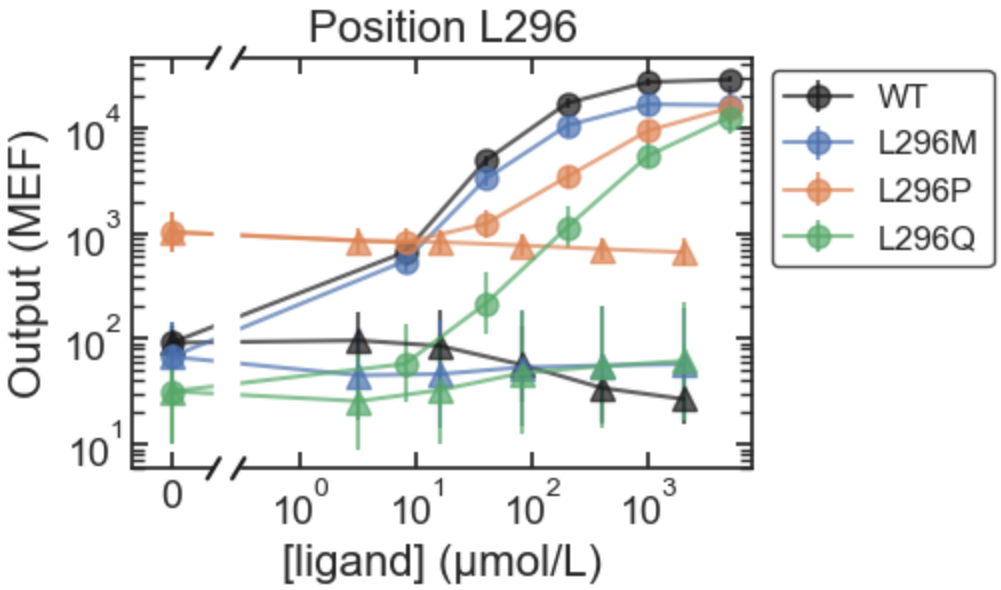

Position P284
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
P284A
P284L
P284S

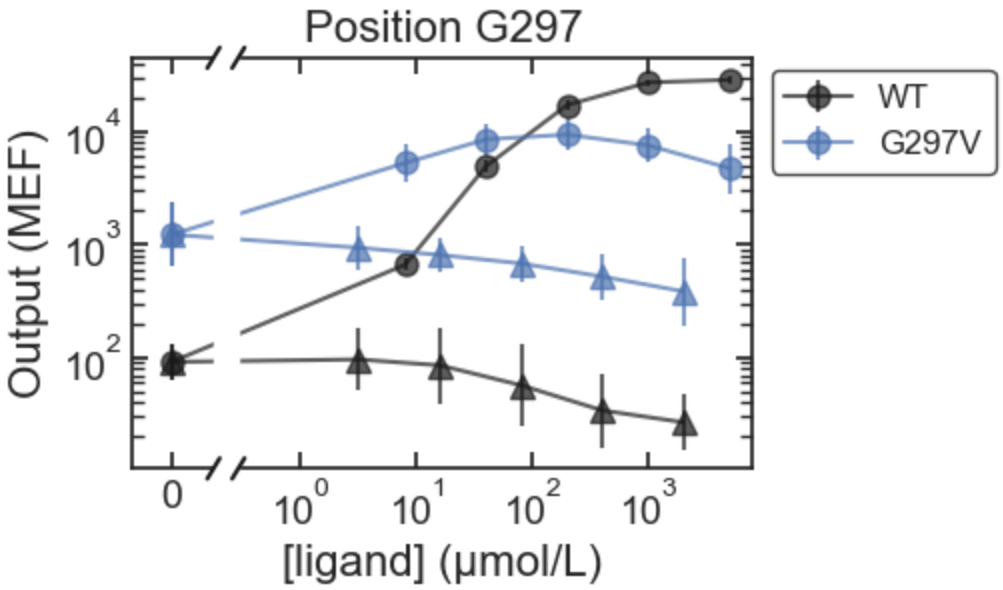

Position Q291
Output (MEF)
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L

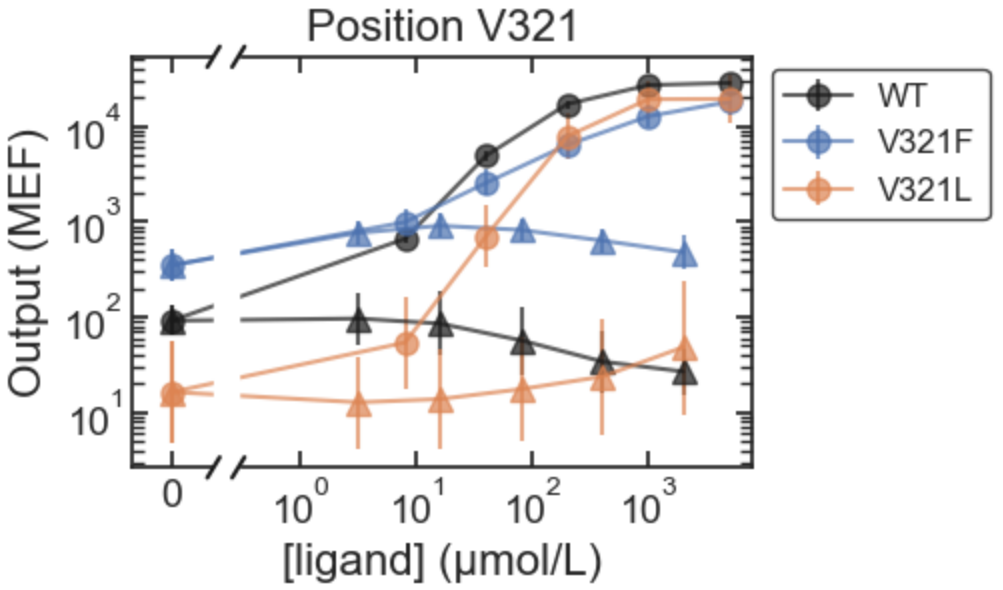

Position D292
Output (MEF)
[ligand] (μmol/L)
WT
D292E
D292G
D292N
D292V

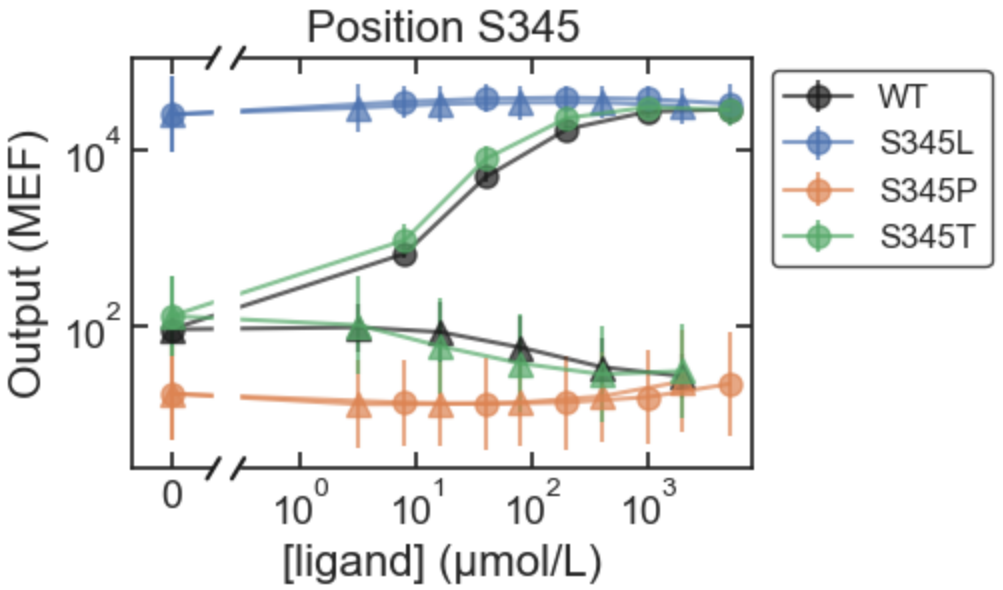

Position F293
Output (MEF)
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

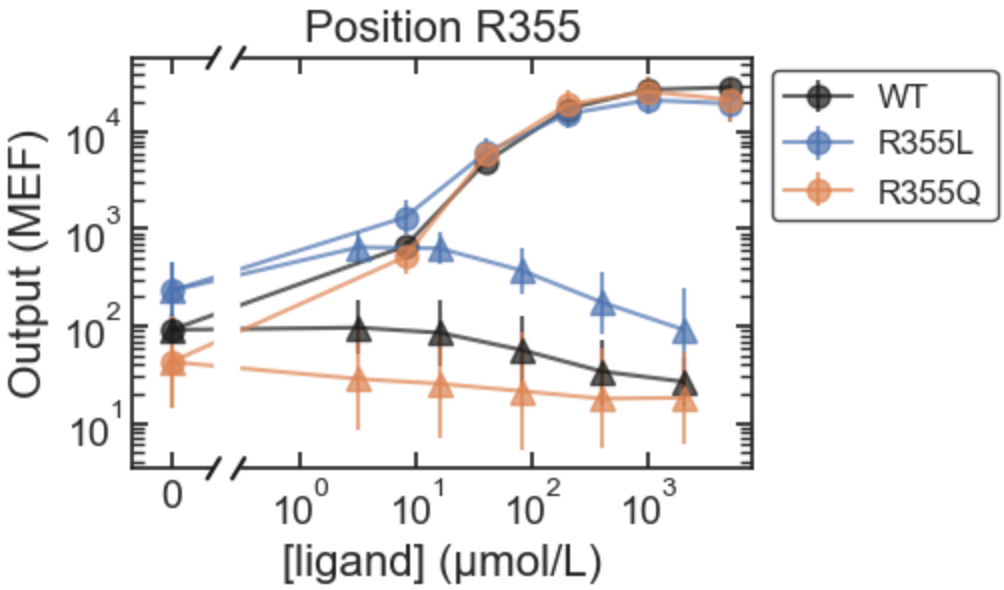

Position R294
Output (MEF)
[ligand] (μmol/L)
WT
R294C
R294H
R294S

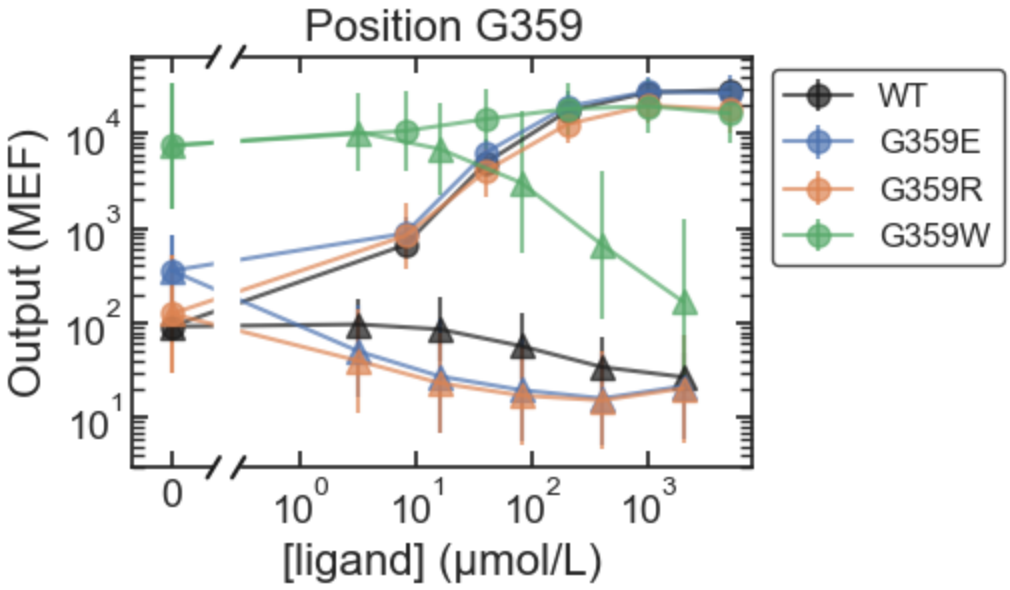

Position L296
Output (MEF)
[ligand] (μmol/L)
WT
L296M
L296P
L296Q


Position G297
Output (MEF)
[ligand] (μmol/L)
WT
G297V

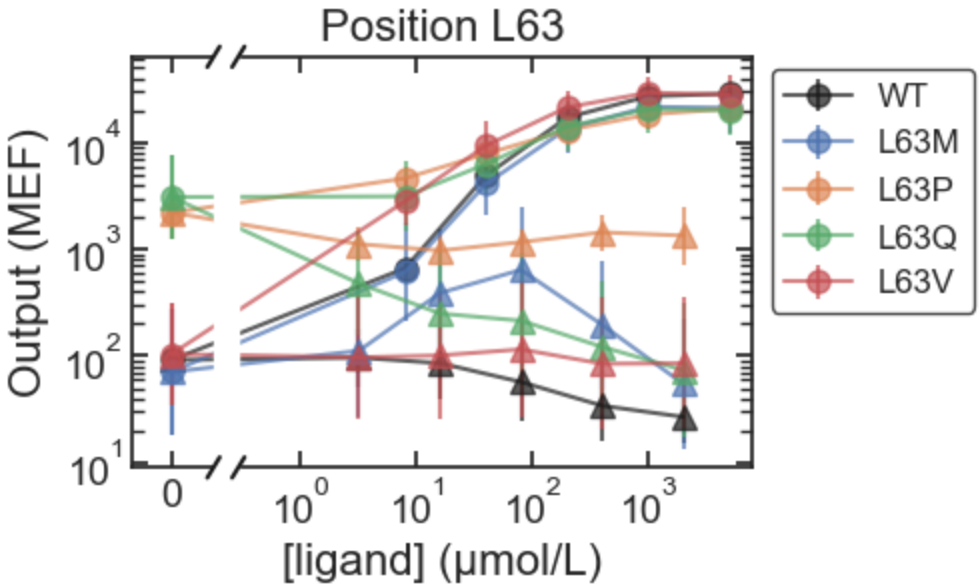

Position V321
Output (MEF)
[ligand] (μmol/L)
WT
V321F
V321L

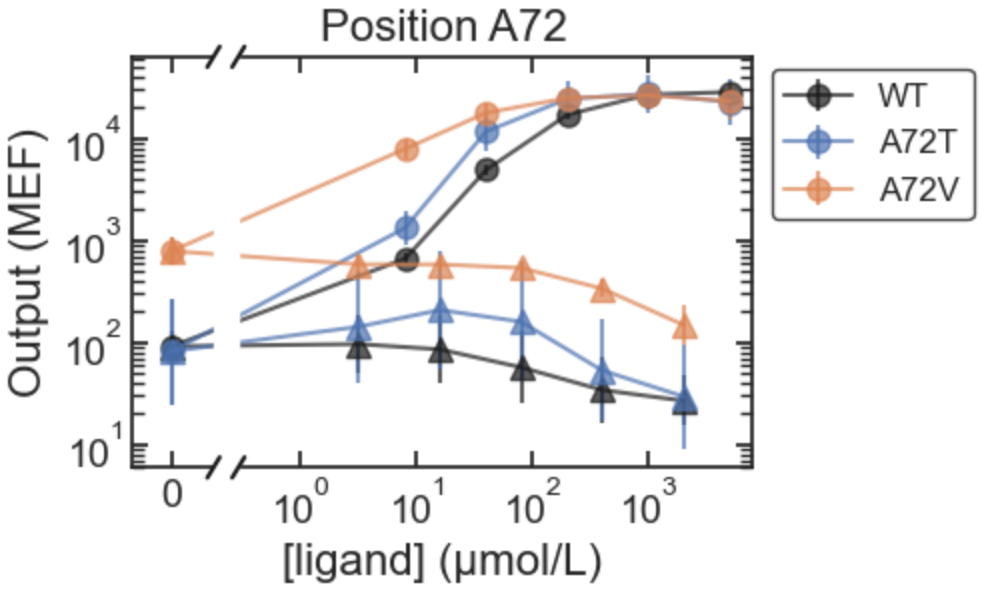

Position S345
Output (MEF)
[ligand] (μmol/L)
WT
S345L
S345P
S345T

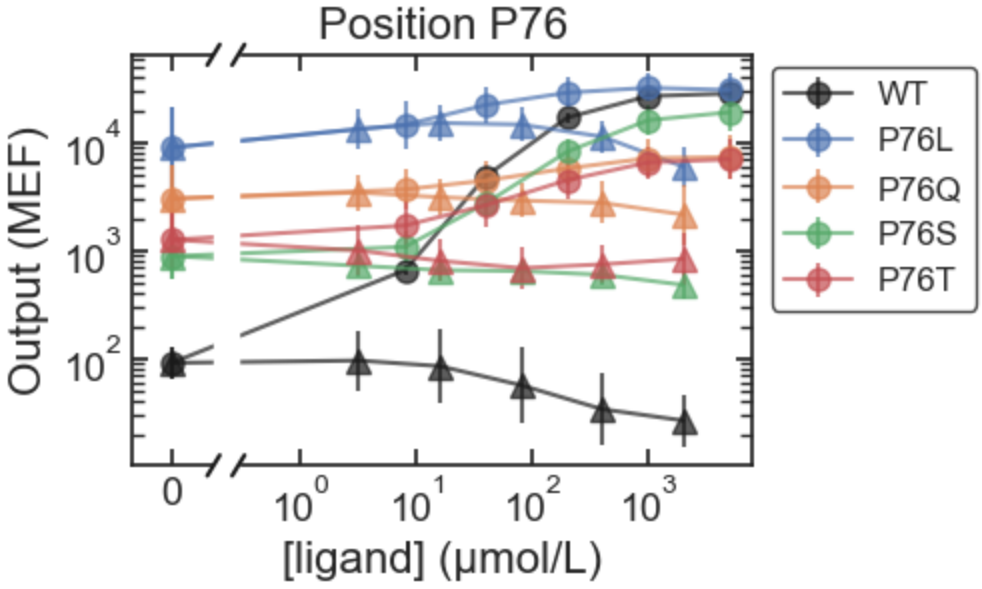

Position R355
Output (MEF)
[ligand] (μmol/L)
WT
R355L
R355Q

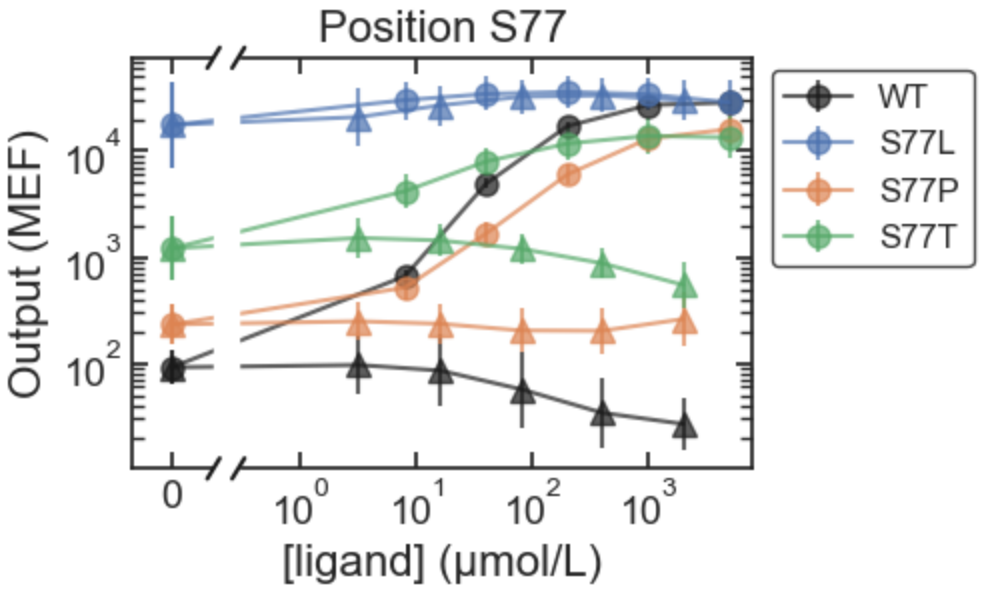

Position G359
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G359E
G359R
G359W

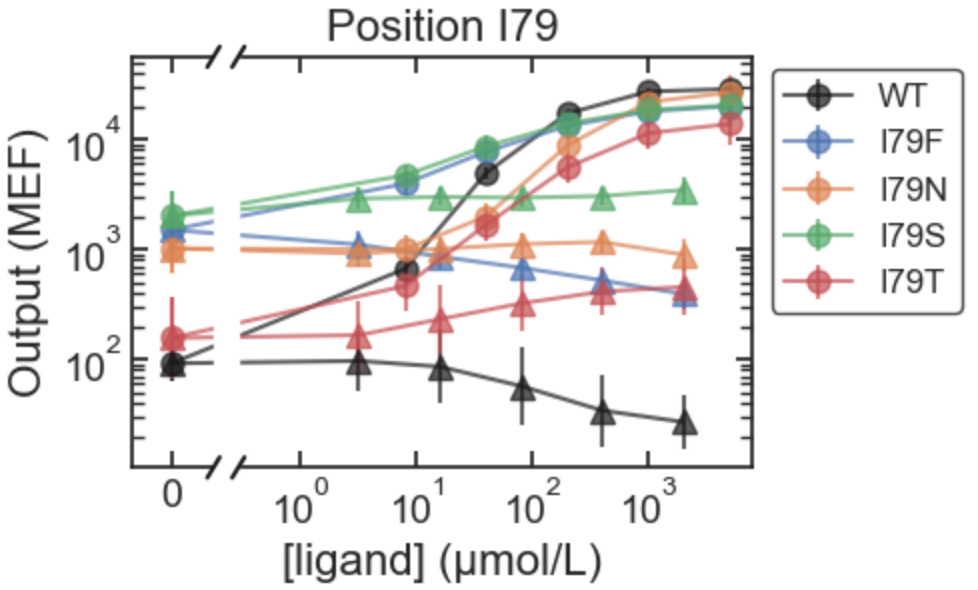
MWC_ONPF

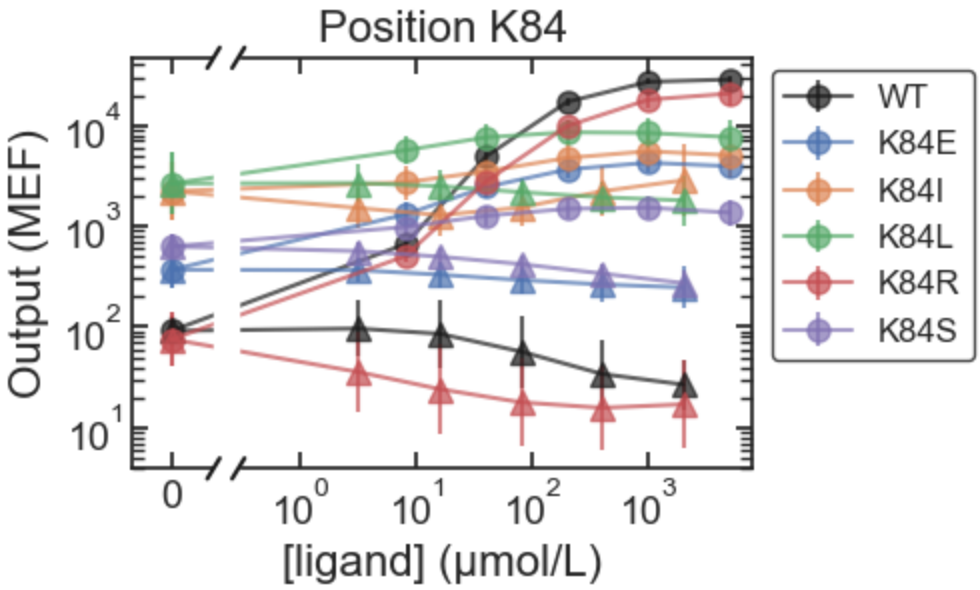

Position L63
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

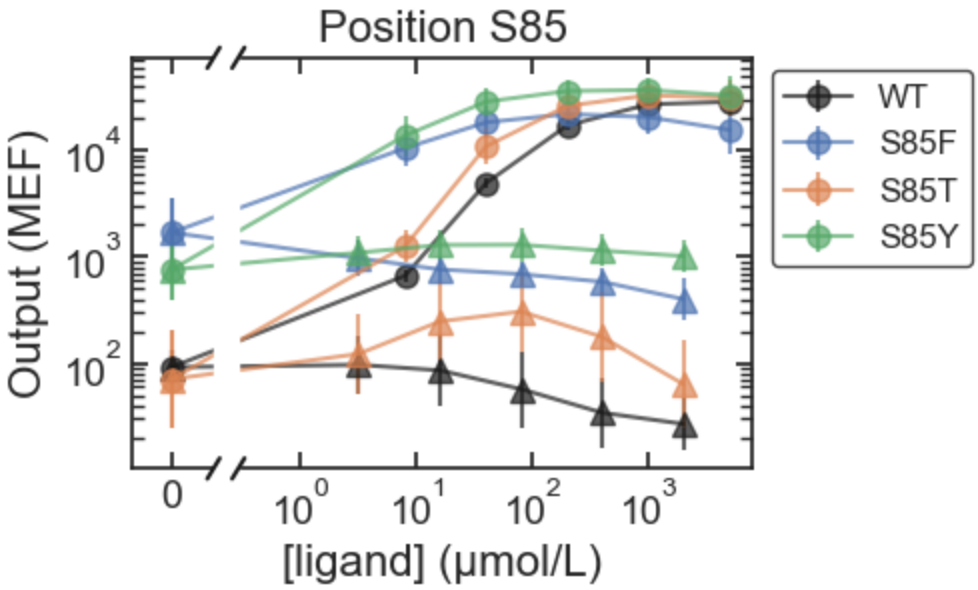

Position A72
Output (MEF)
[ligand] (μmol/L)
WT
A72T
A72V

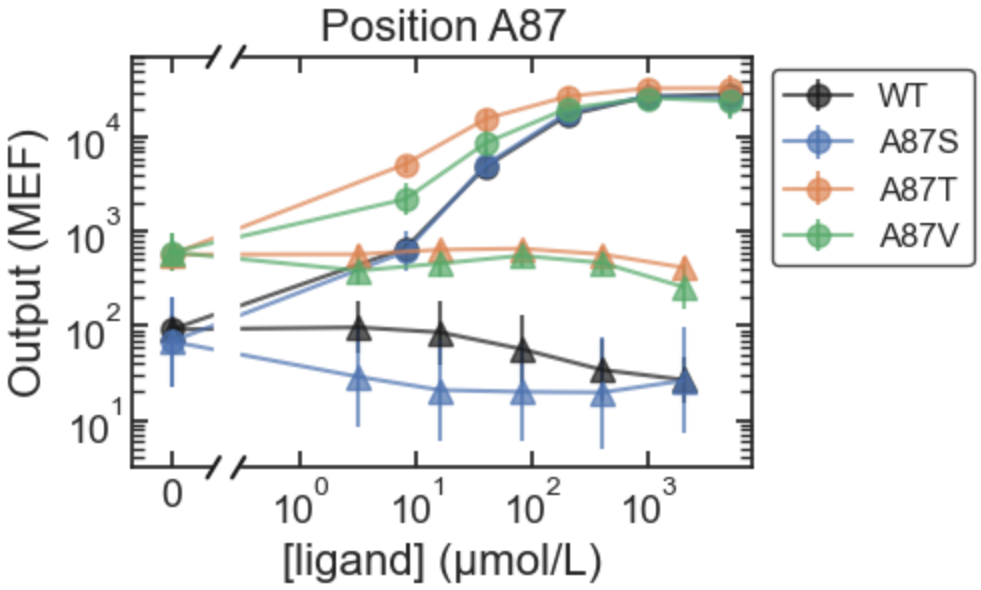

Position P76
Output (MEF)
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

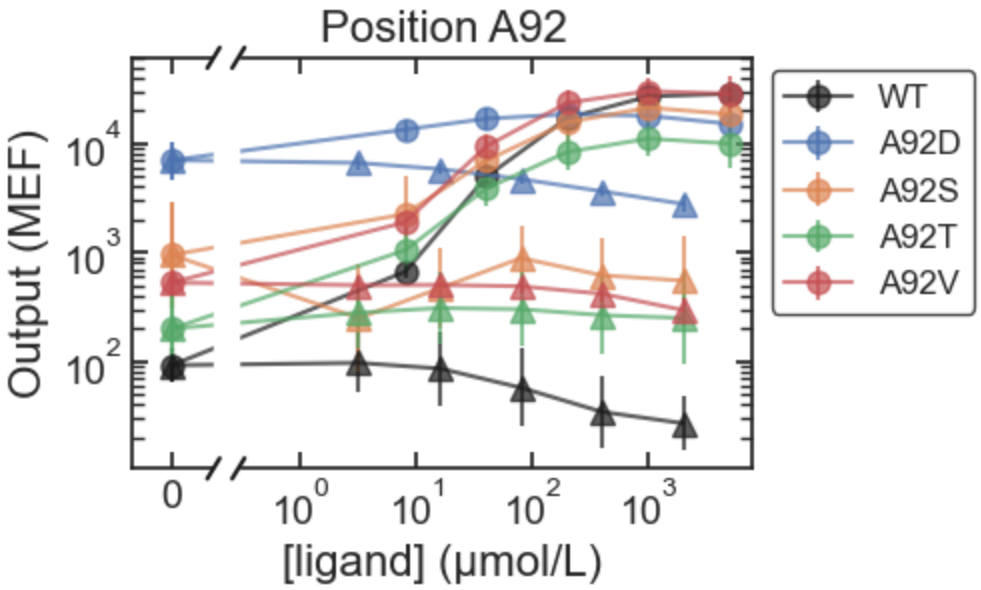

Position S77
Output (MEF)
[ligand] (μmol/L)
WT
S77L
S77P
S77T

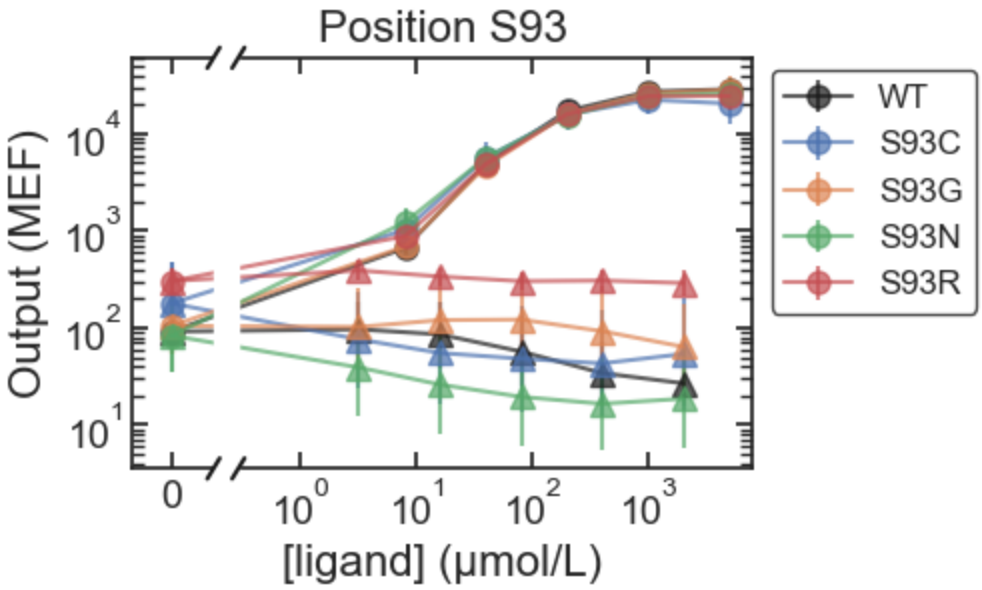

Position I79
Output (MEF)
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

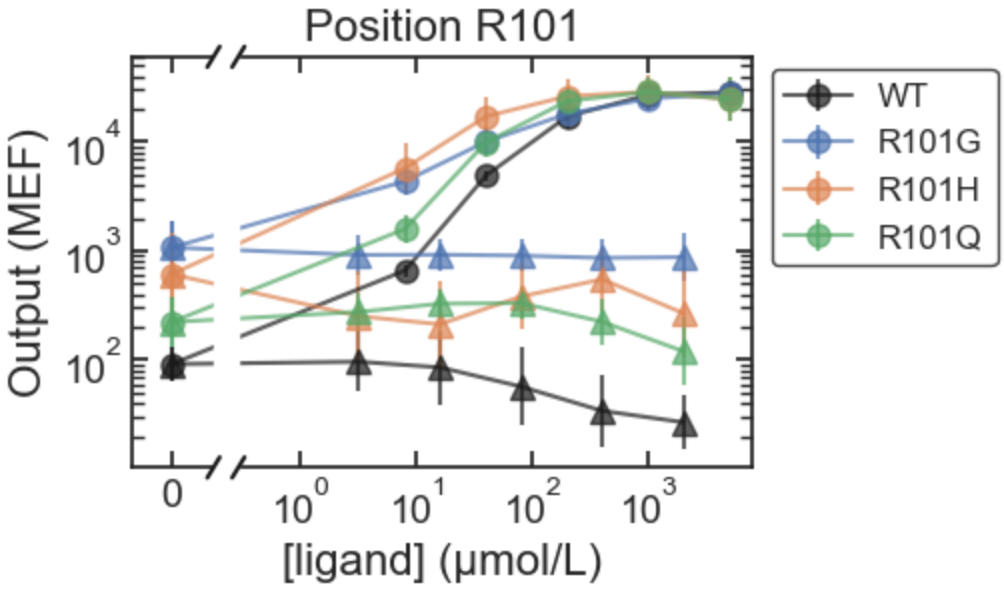

Position K84
Output (MEF)
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

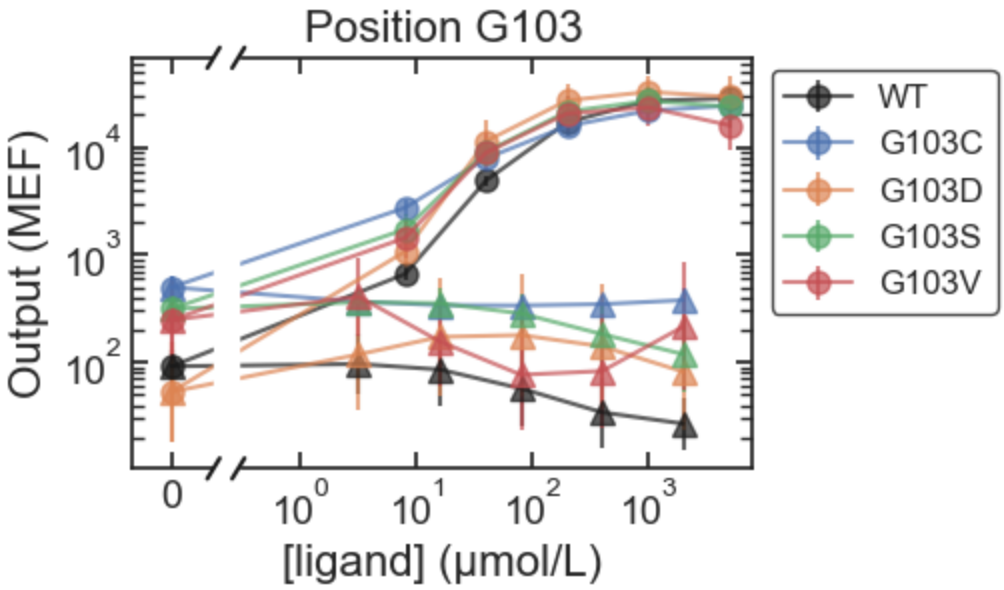

Position S85
Output (MEF)
[ligand] (μmol/L)
WT
S85F
S85T
S85Y

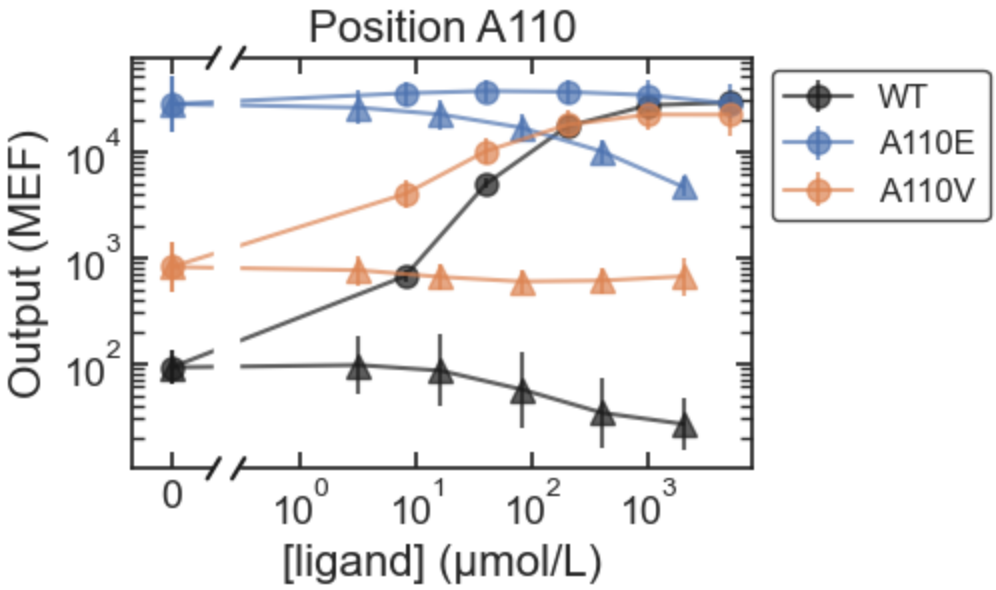

Position A87
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A87S
A87T
A87V

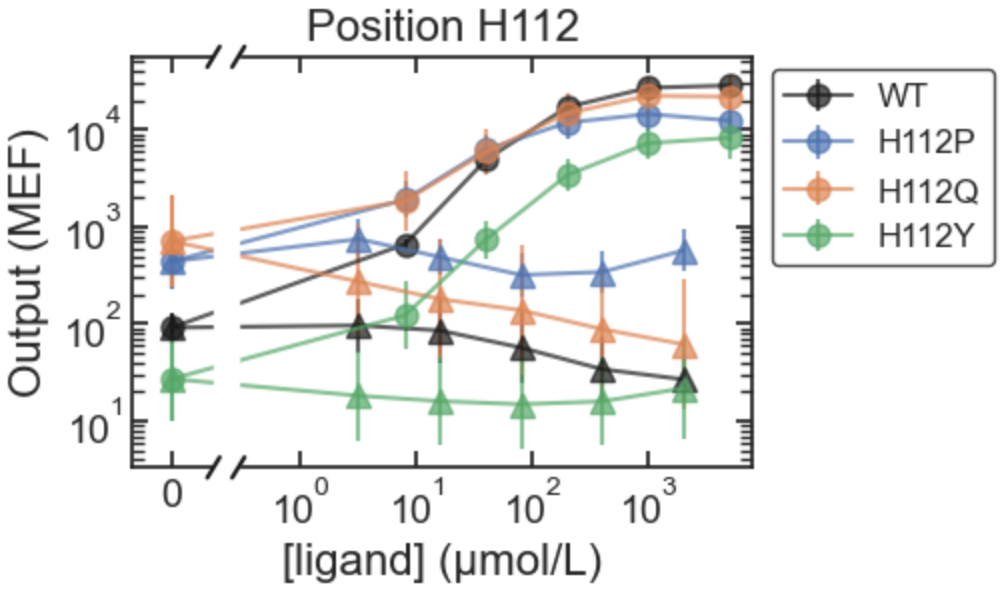

Position A92
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V

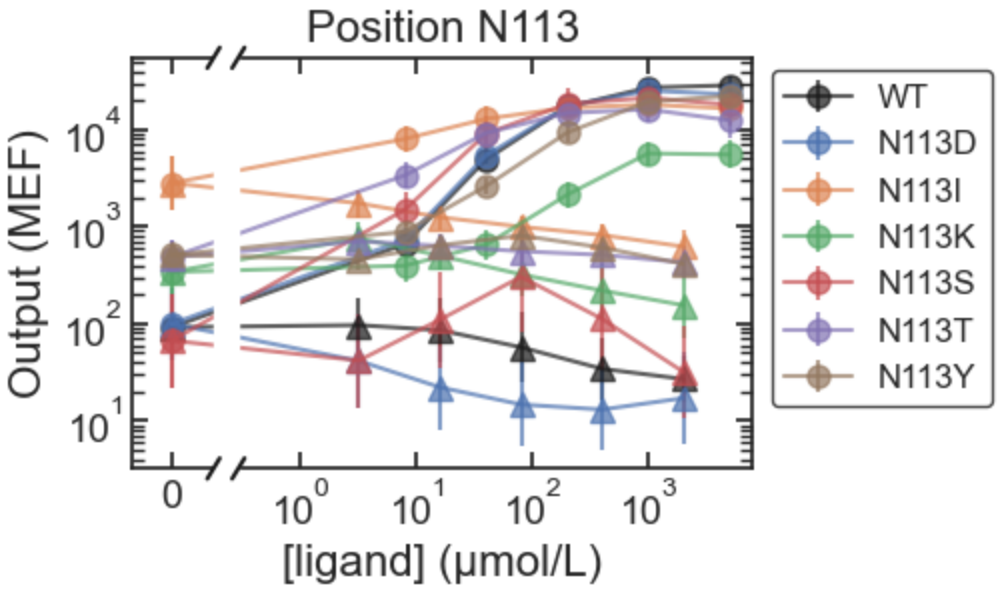

Position S93
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S93C
S93G
S93N
S93R

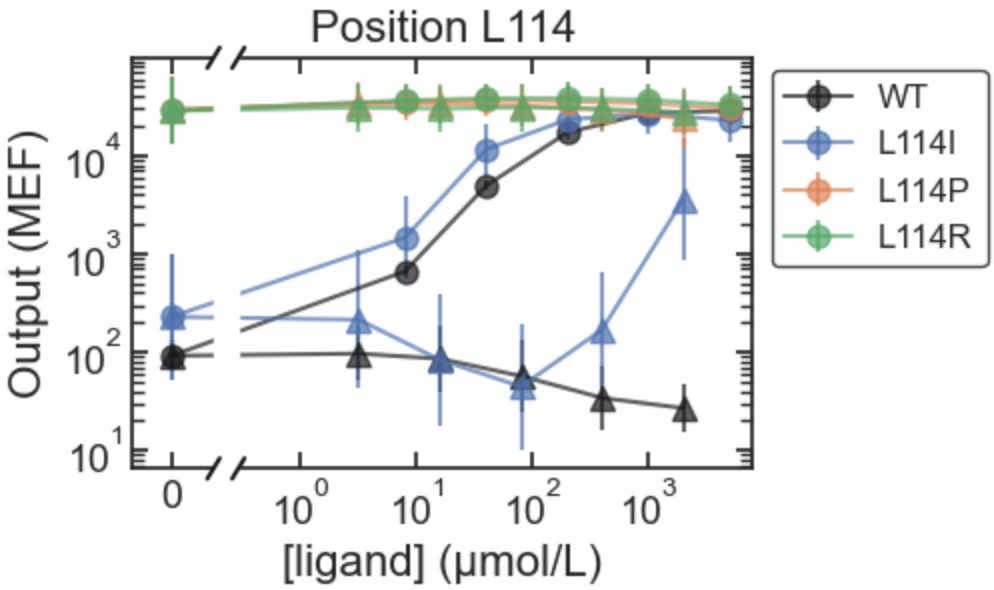

Position R101
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

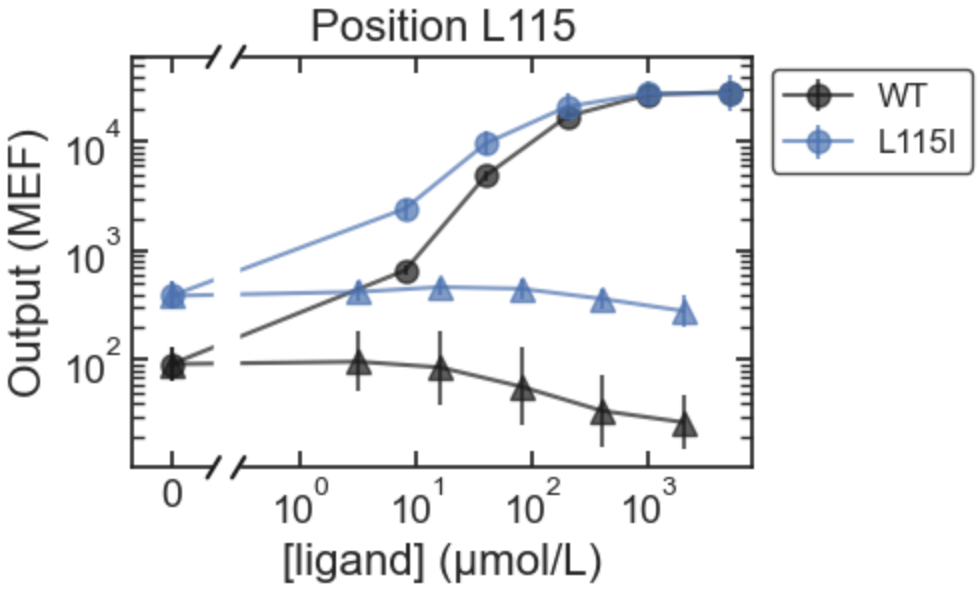

Position G103
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

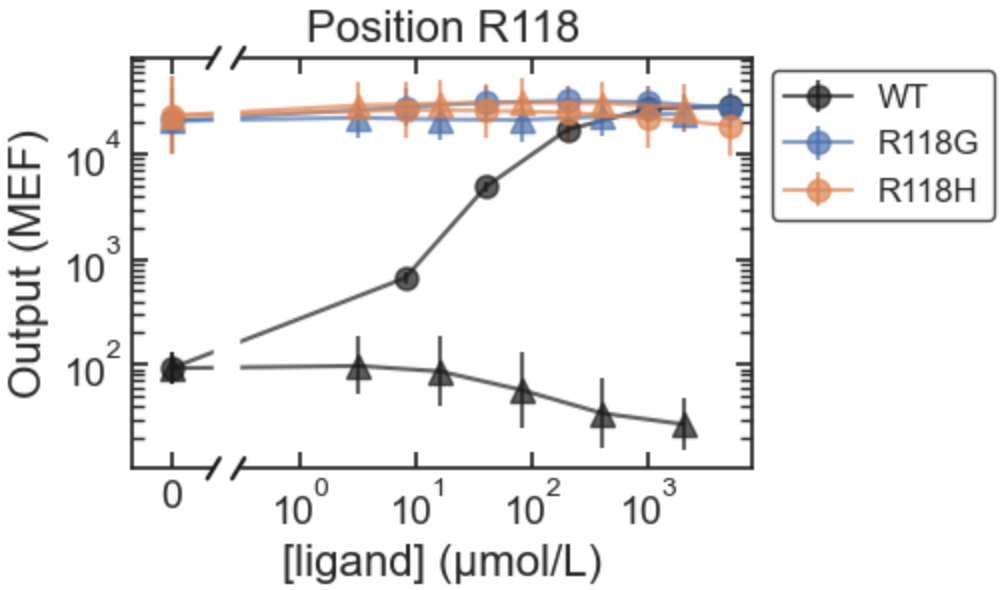

Position A110
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A110E
A110V

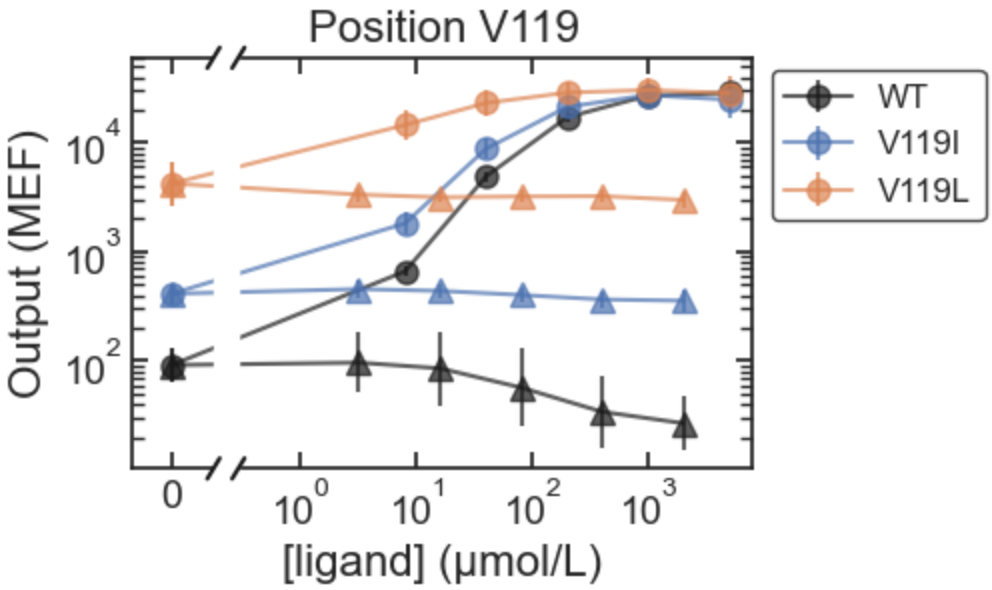

Position H112
Output (MEF)
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

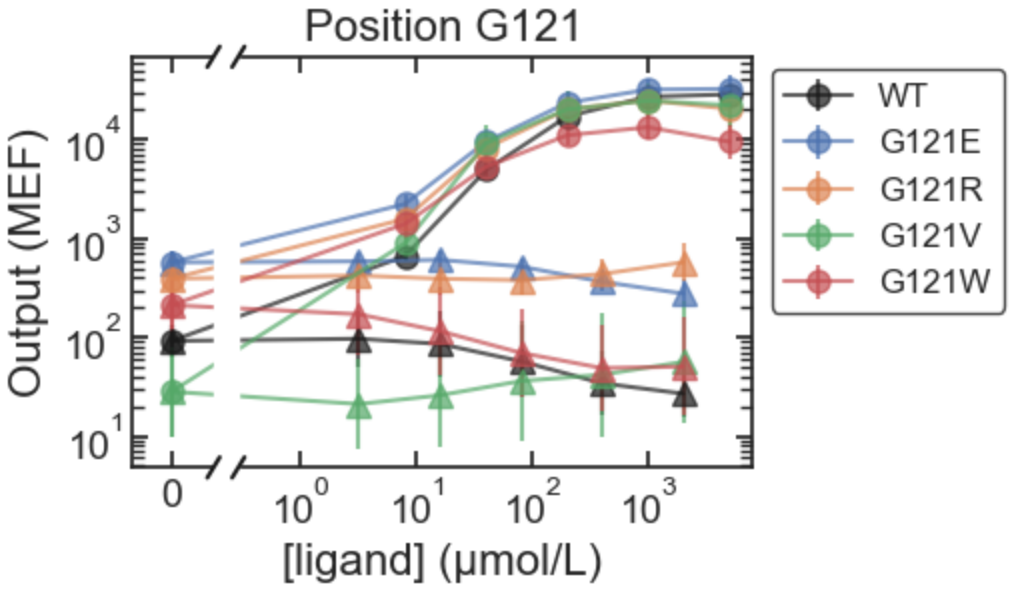

Position N113
Output (MEF)
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

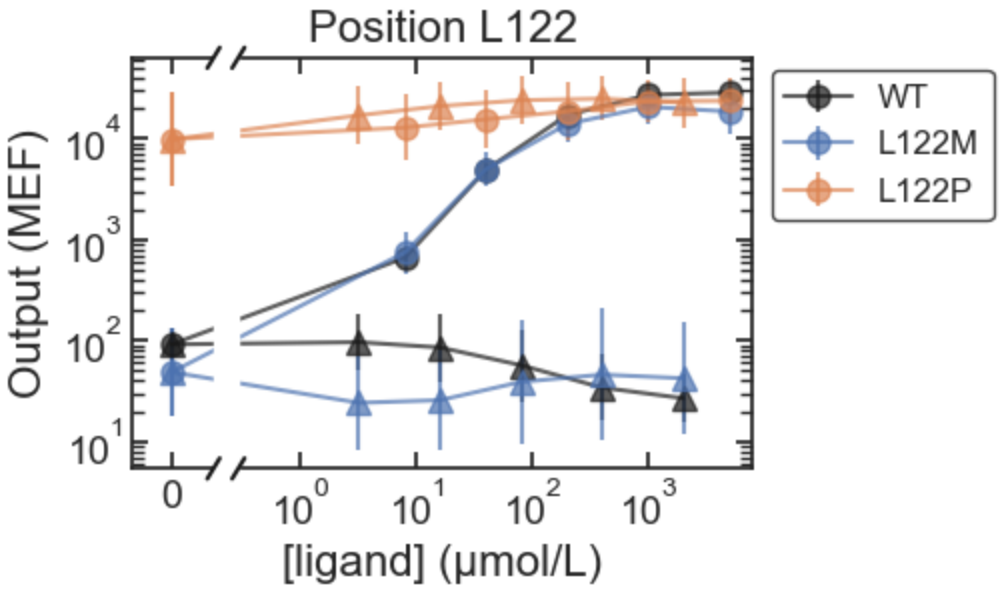

Position L114
Output (MEF)
[ligand] (μmol/L)
WT
L114I
L114P
L114R

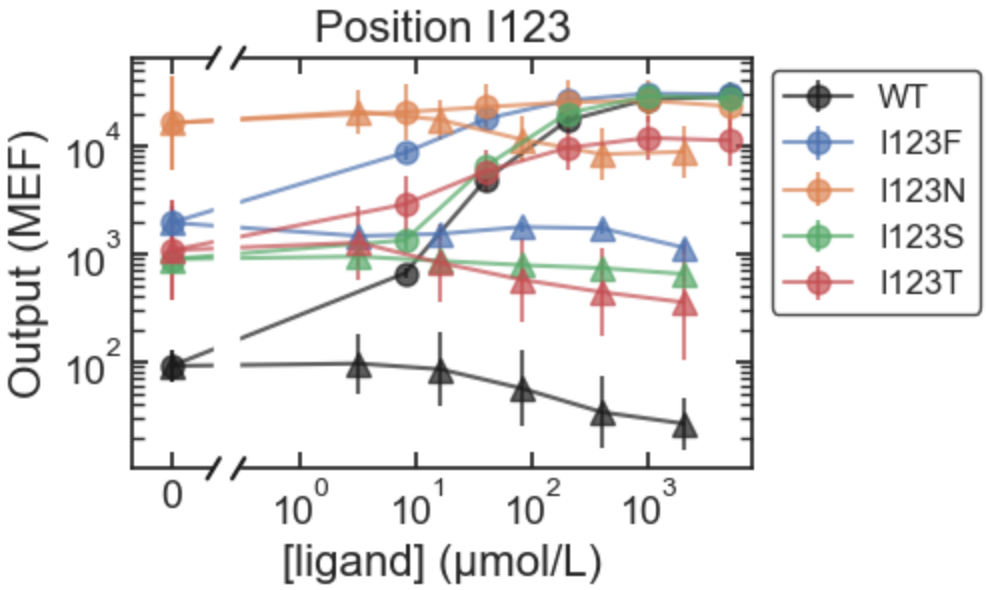

Position L115
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L115I

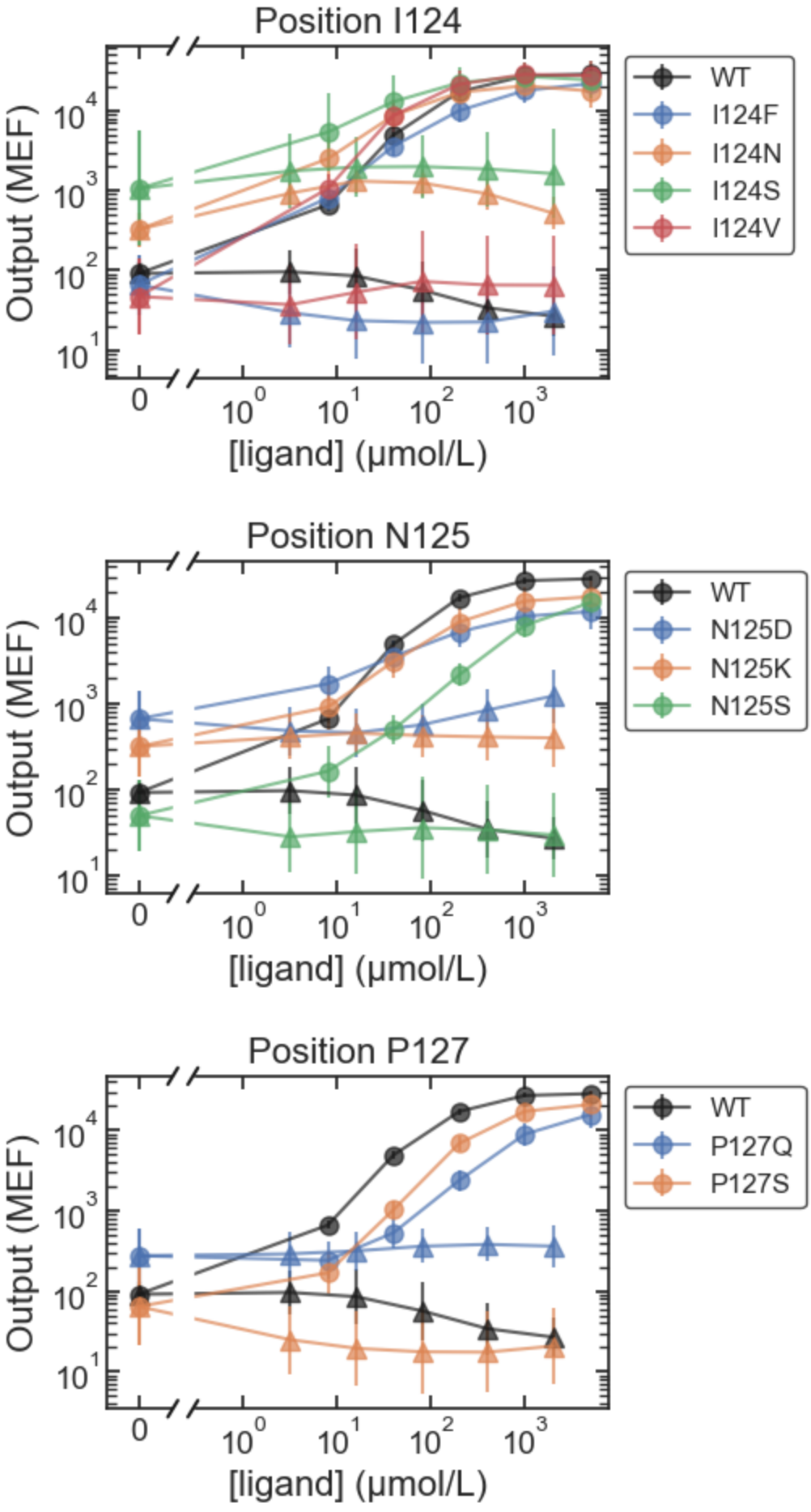

Position R118
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R118G
R118H

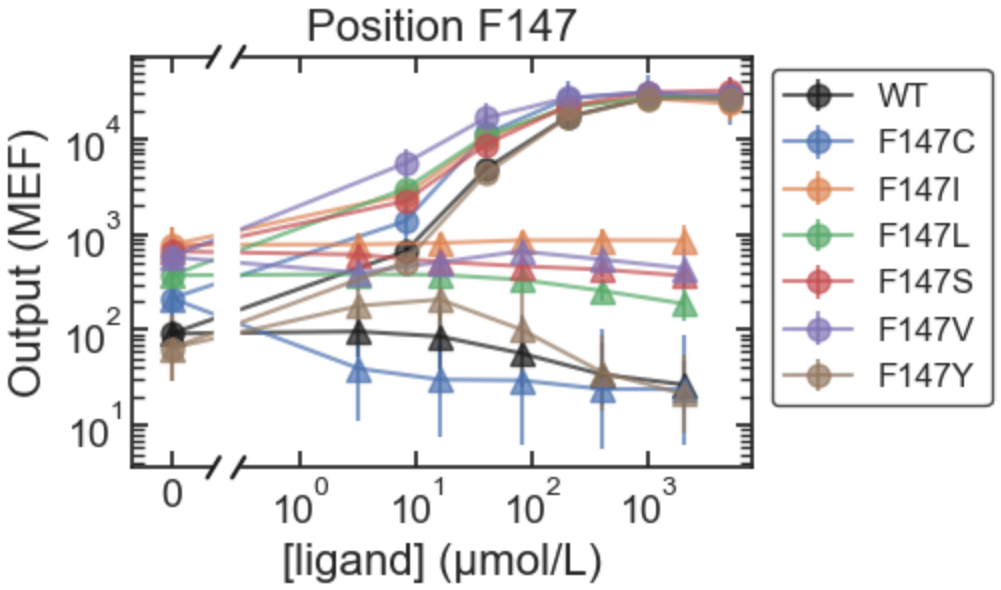

Position V119
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V119I
V119L

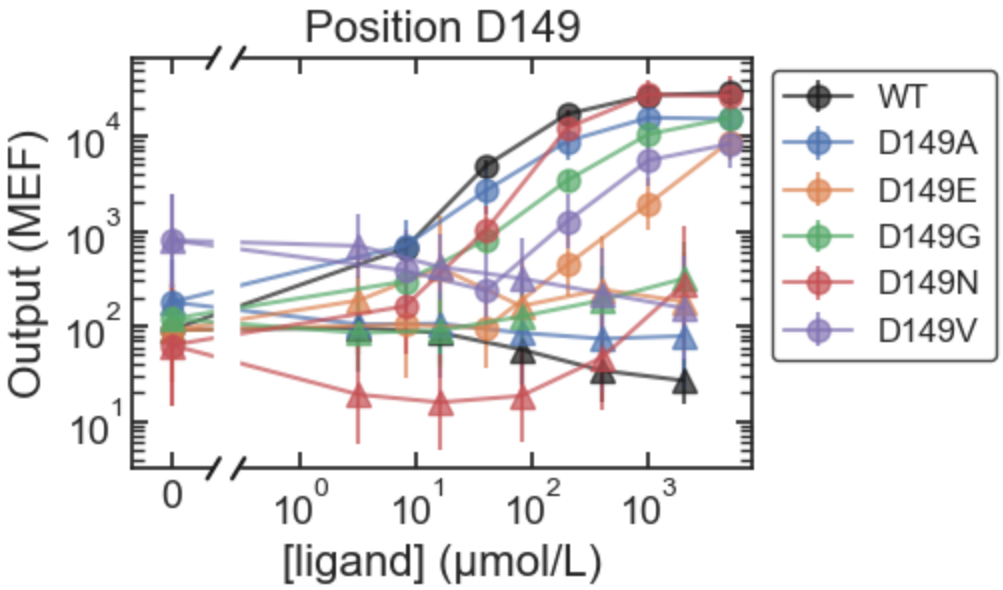

Position G121
Output (MEF)
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

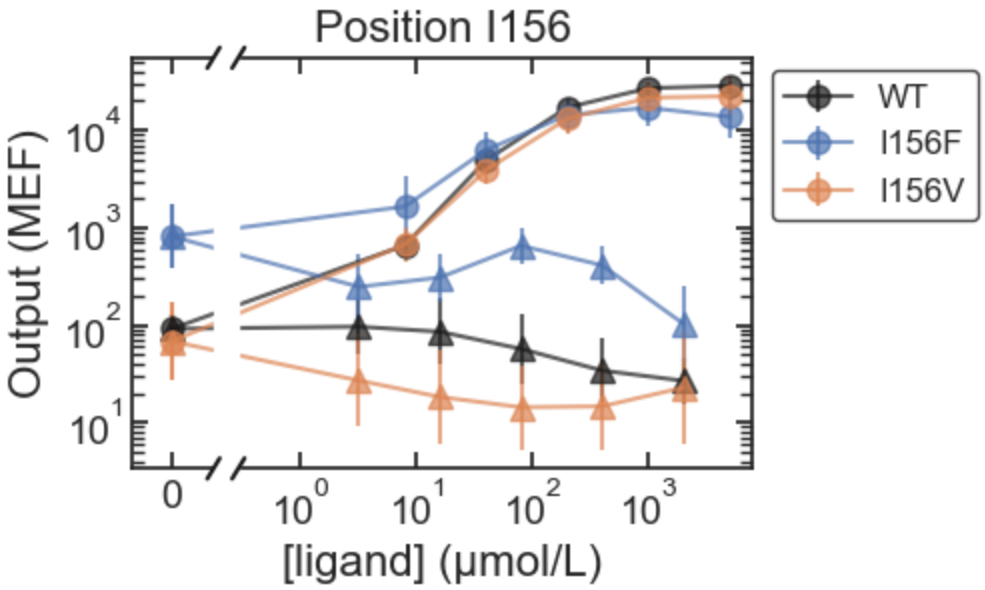

Position L122
Output (MEF)
[ligand] (μmol/L)
WT
L122M
L122P

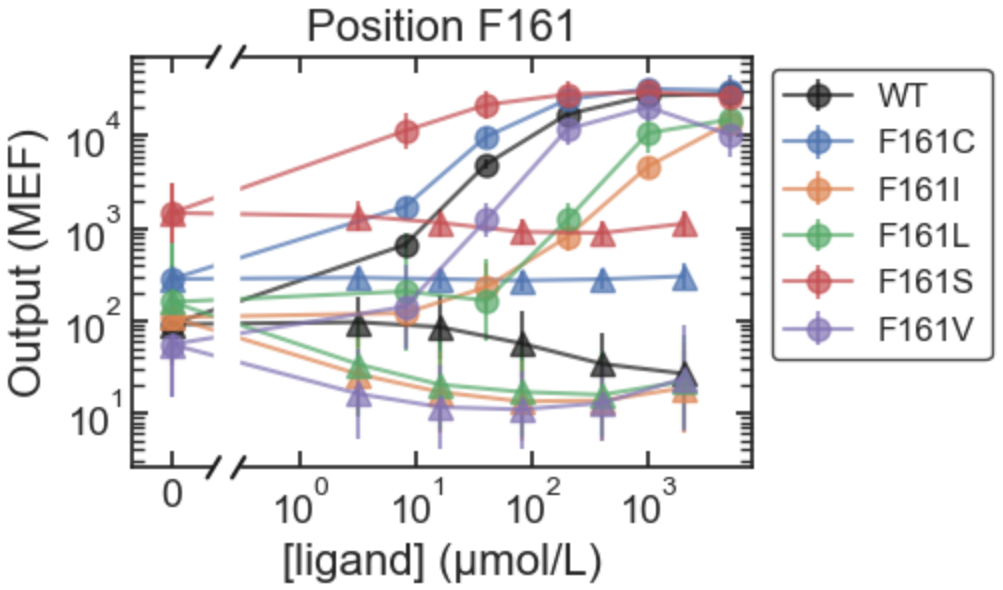

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

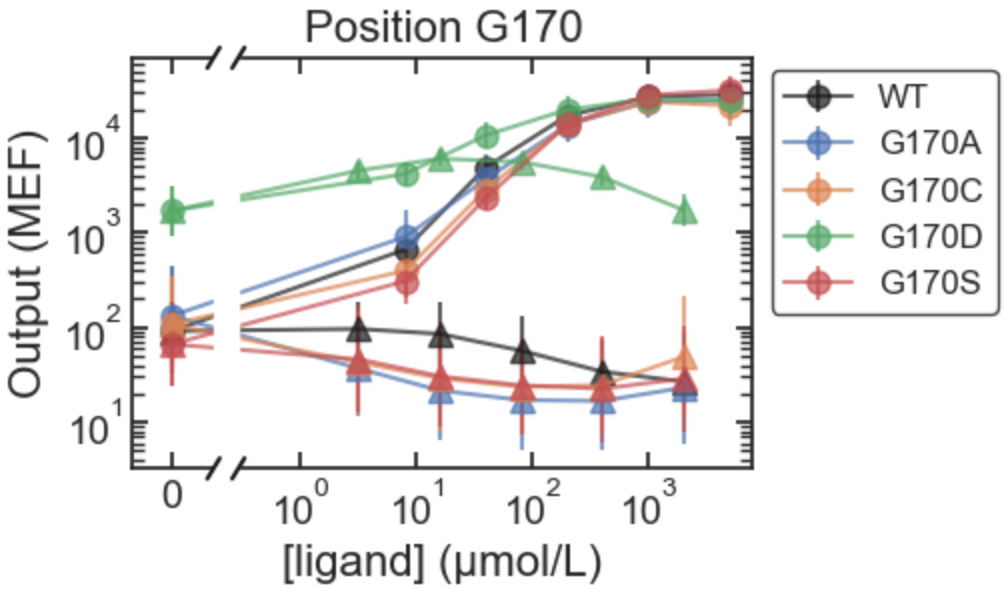

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V
Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S
Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

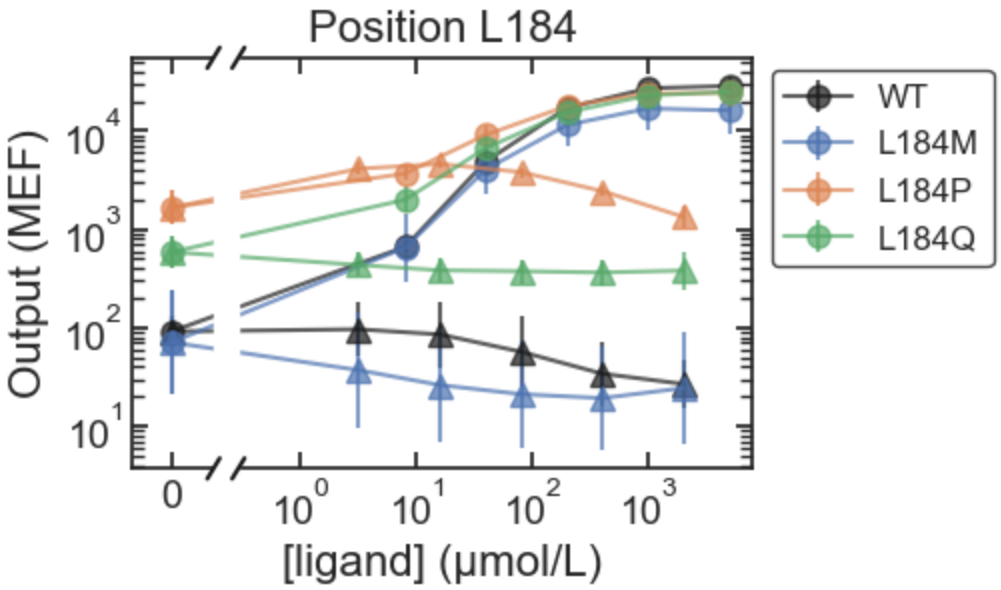

Position F147
Output (MEF)
[ligand] (μmol/L)
WT
F147C
F147I
F147L
F147S
F147V
F147Y

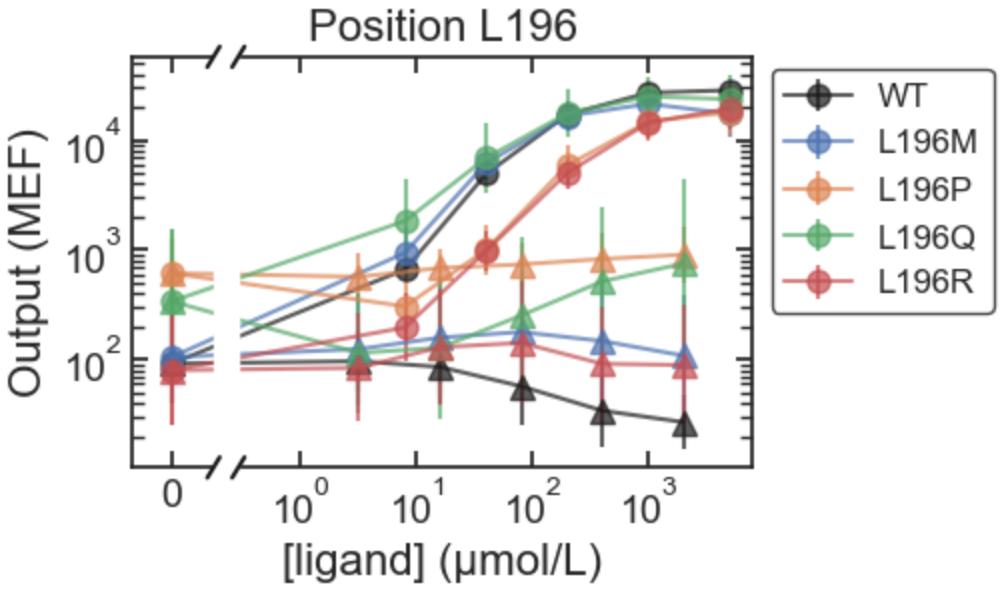

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

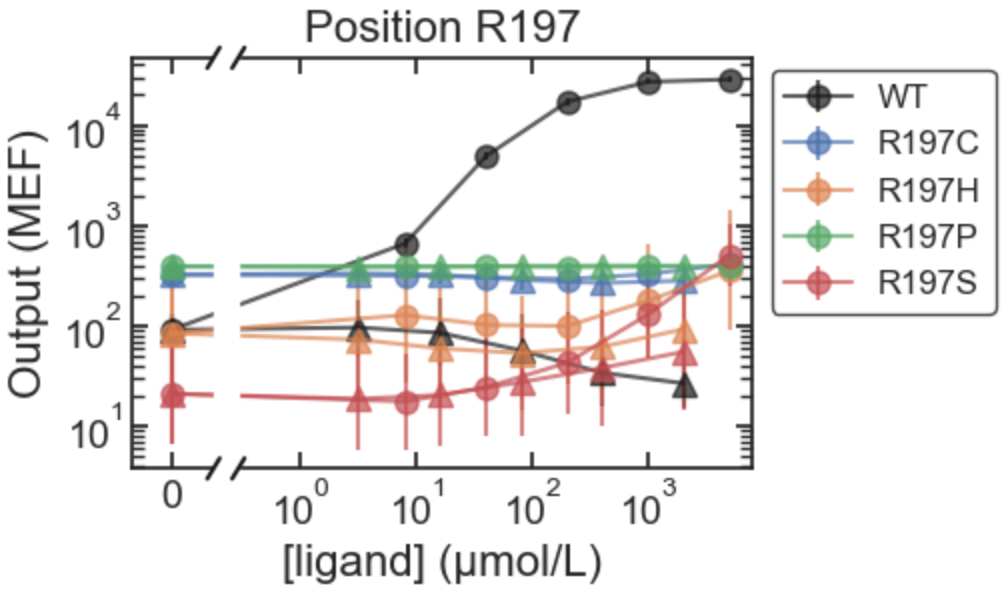

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

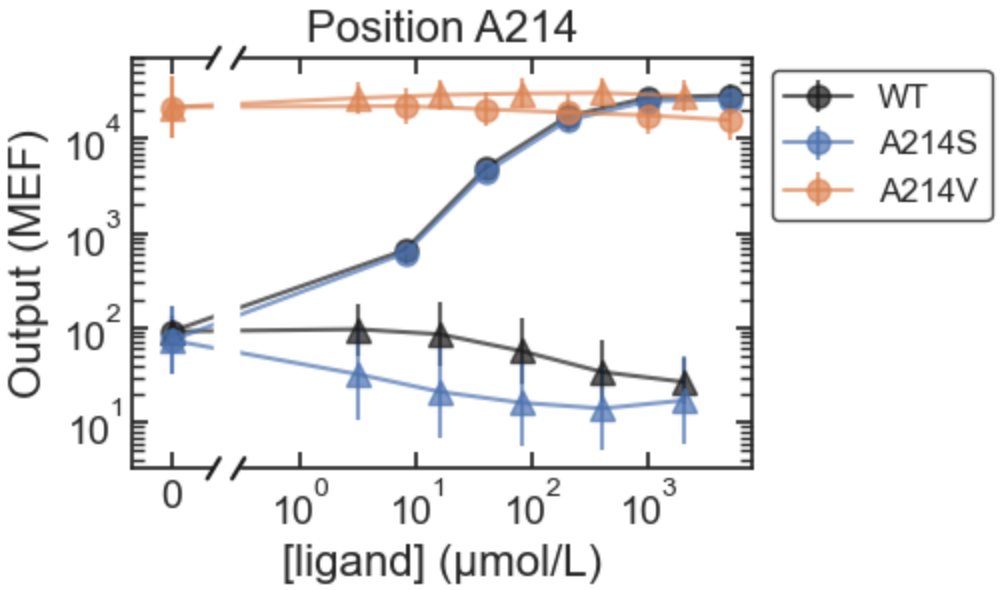

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

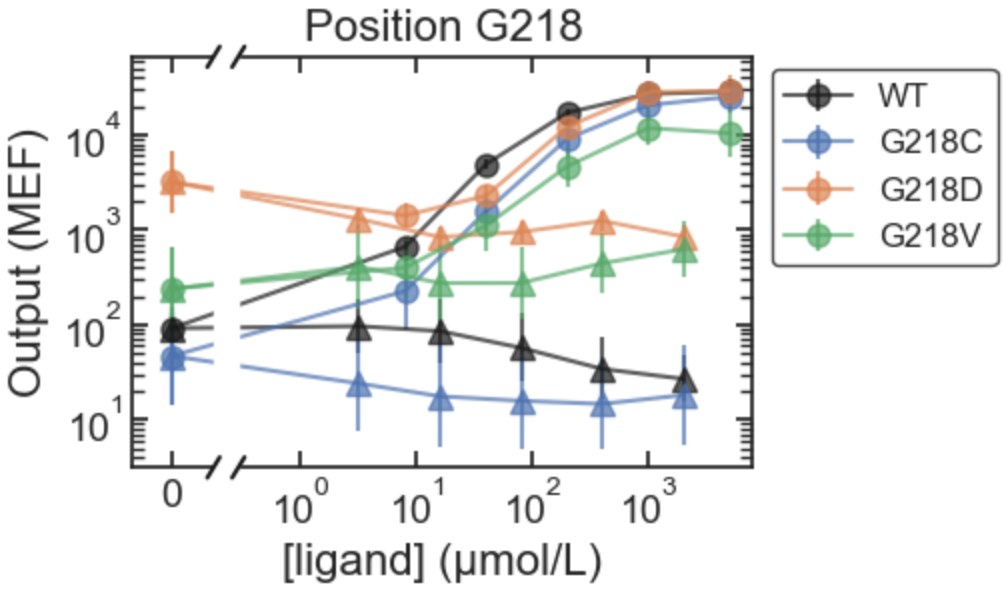

Position G170
Output (MEF)
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

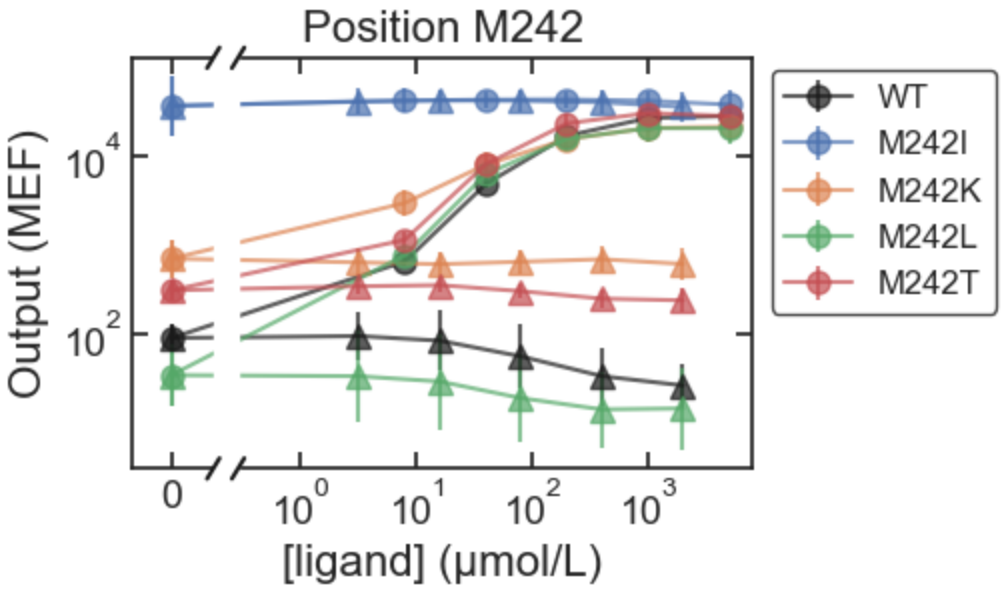

Position L184
Output (MEF)
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

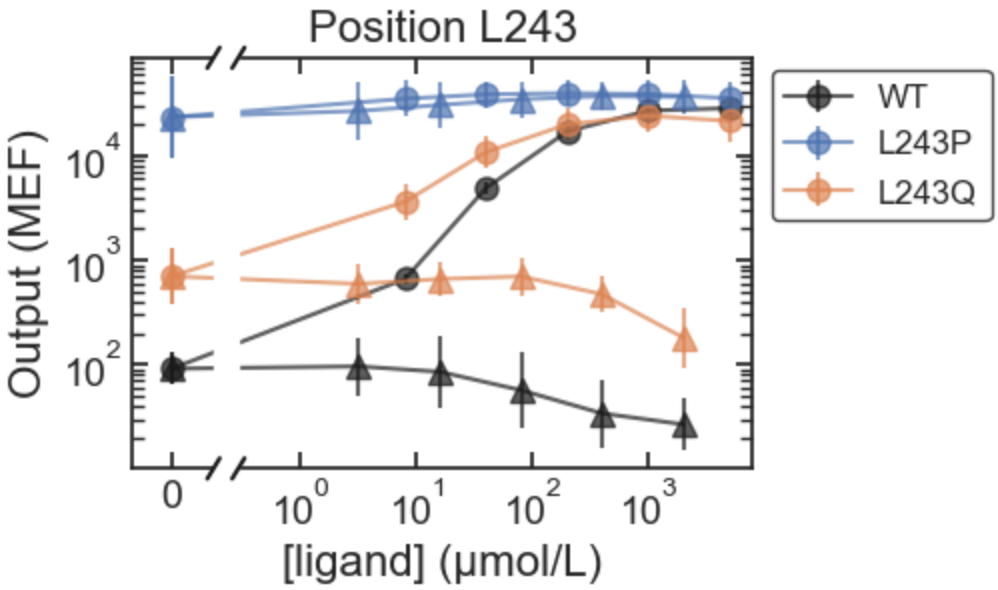

Position L196
Output (MEF)
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

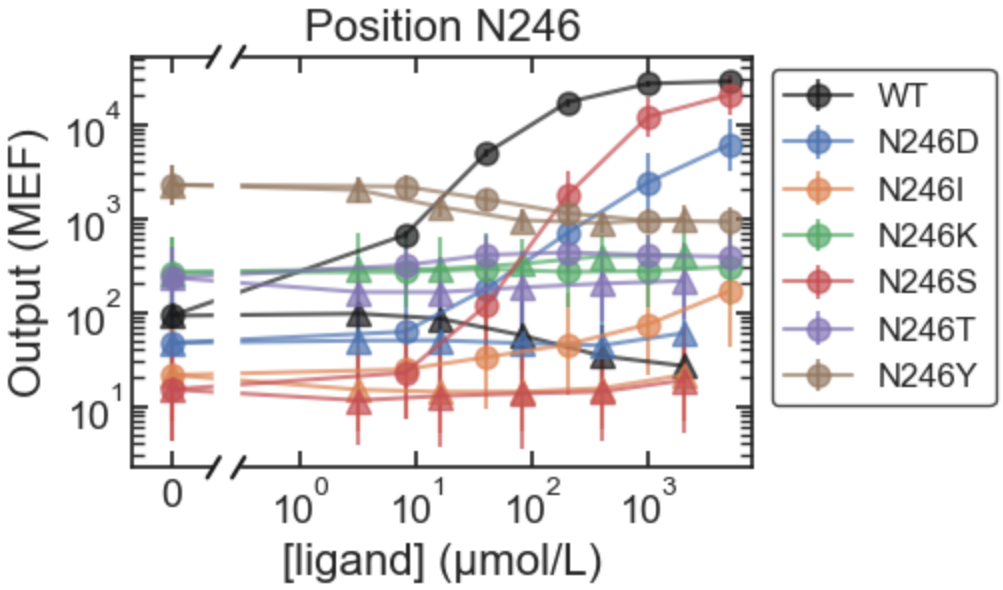

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

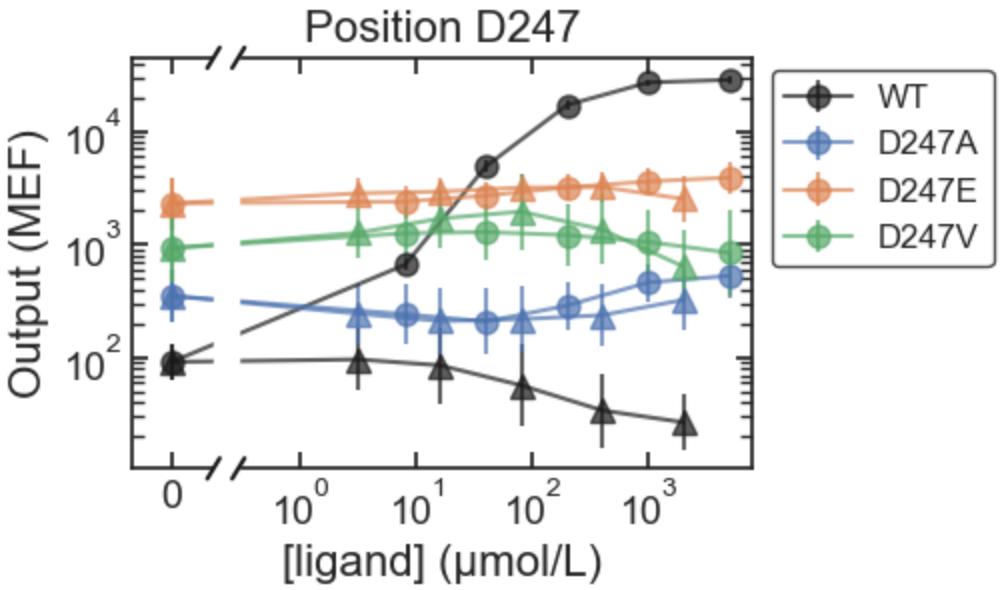

Position A214
Output (MEF)
[ligand] (μmol/L)
WT
A214S
A214V

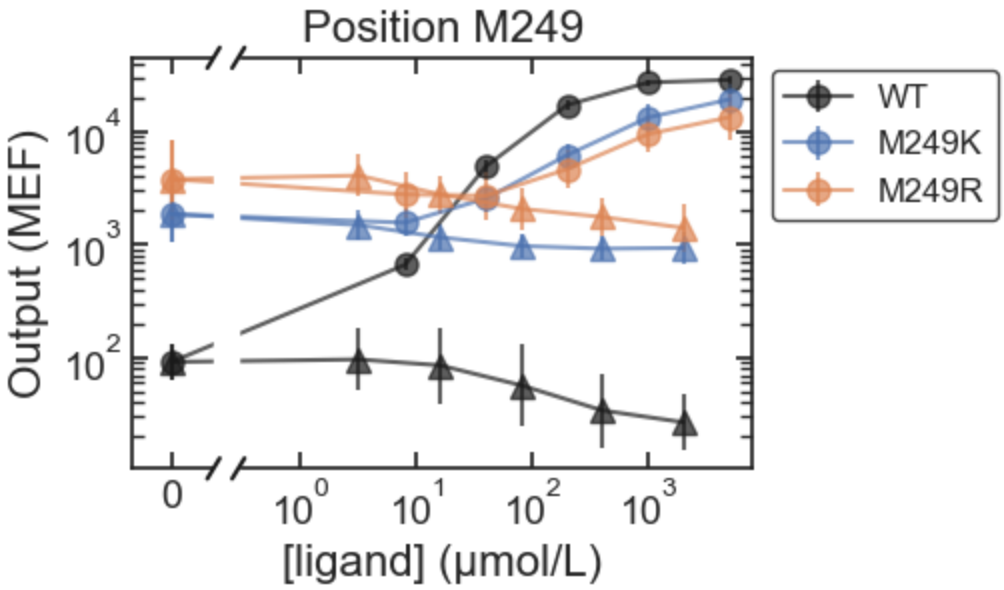

Position G218
Output (MEF)
[ligand] (μmol/L)
WT
G218C
G218D
G218V

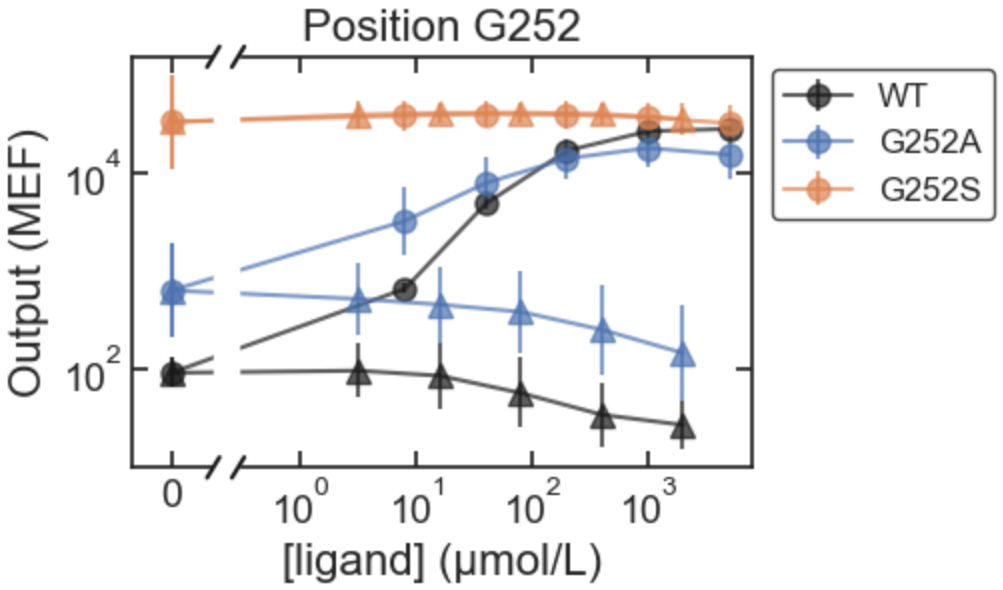

Position M242
Output (MEF)
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

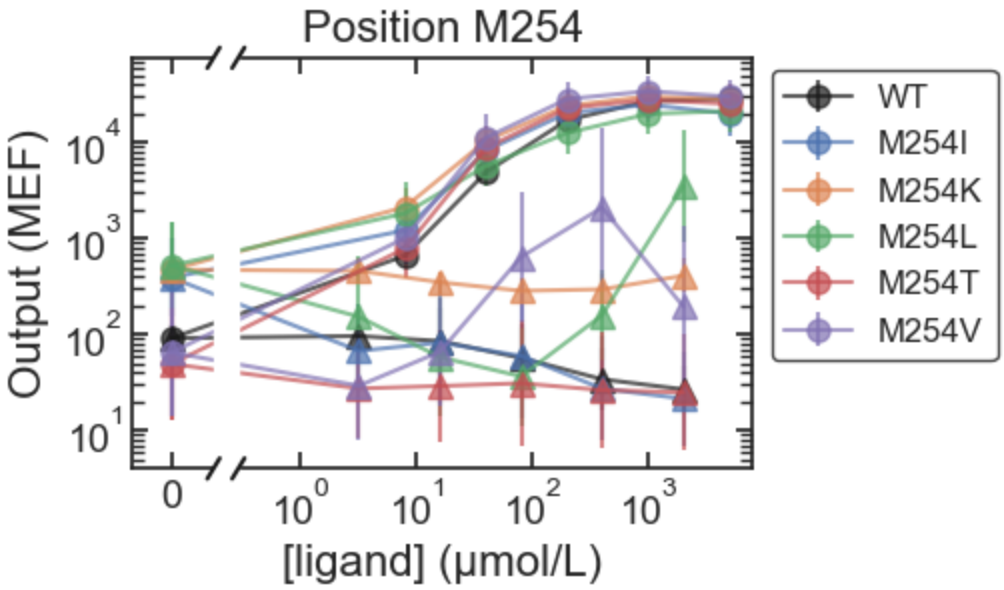

Position L243
Output (MEF)
[ligand] (μmol/L)
WT
L243P
L243Q

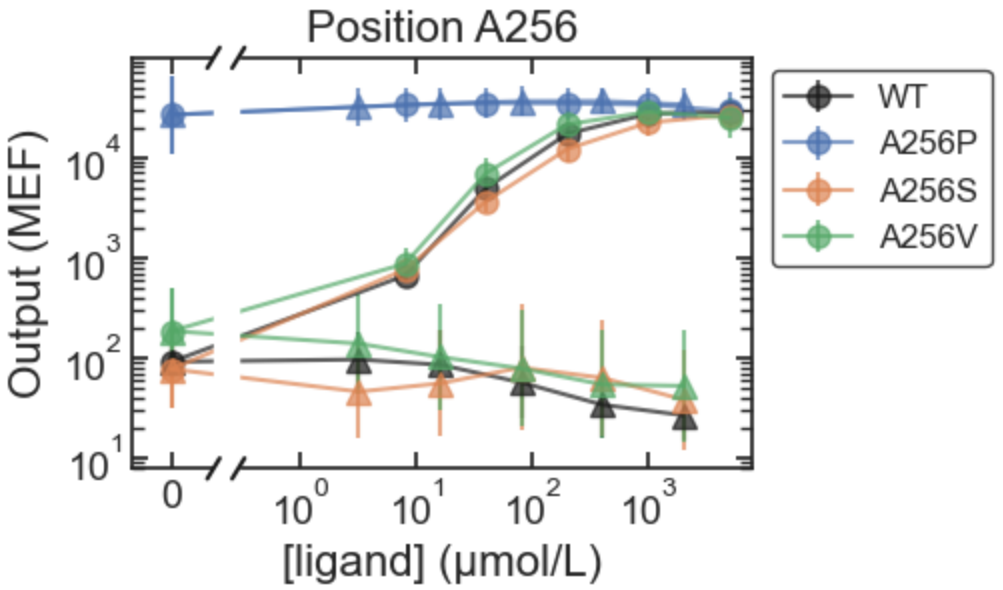

Position N246
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

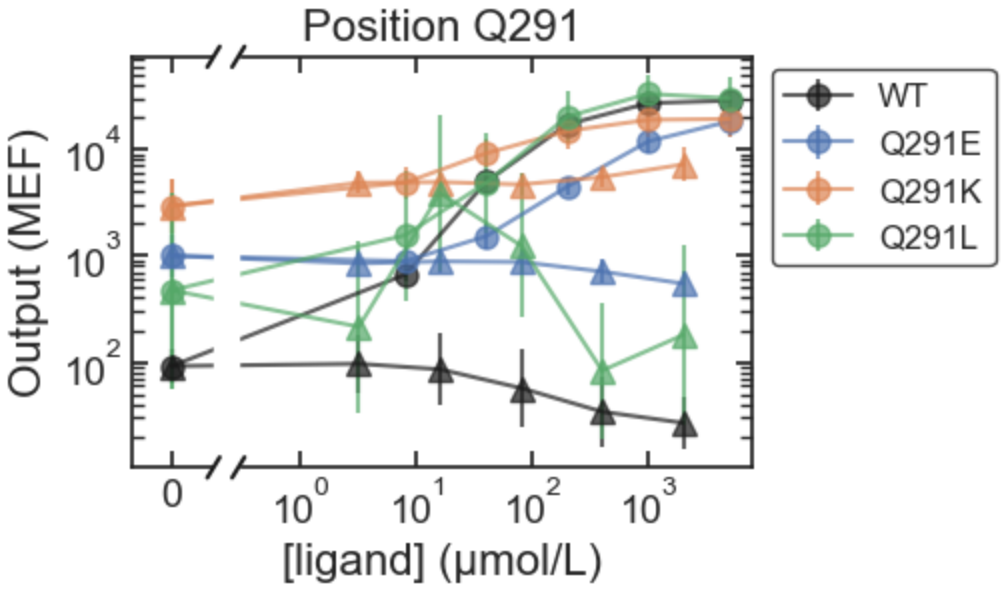

Position D247
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D247A
D247E
D247V

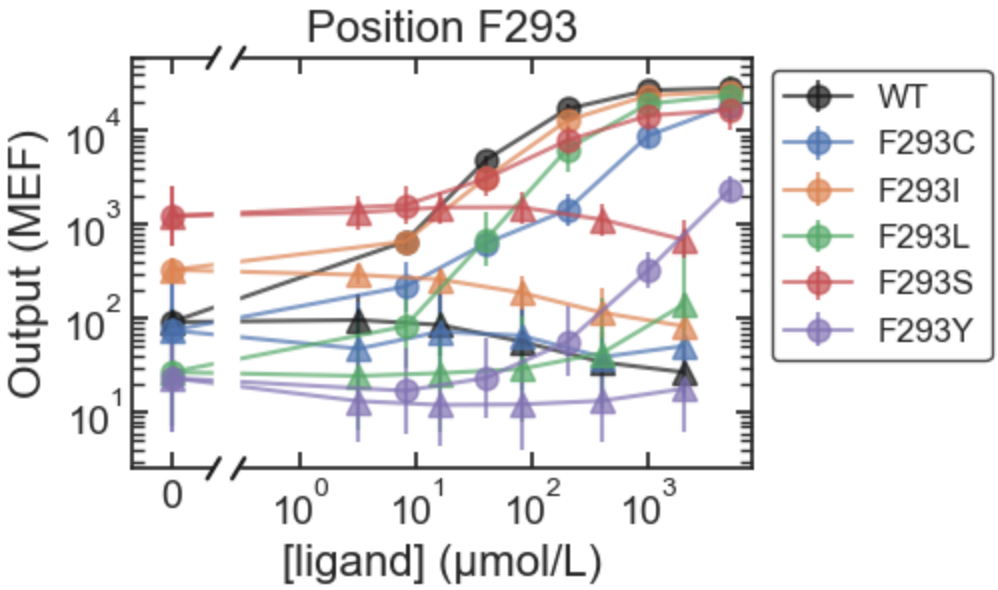

Position M249
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M249K
M249R

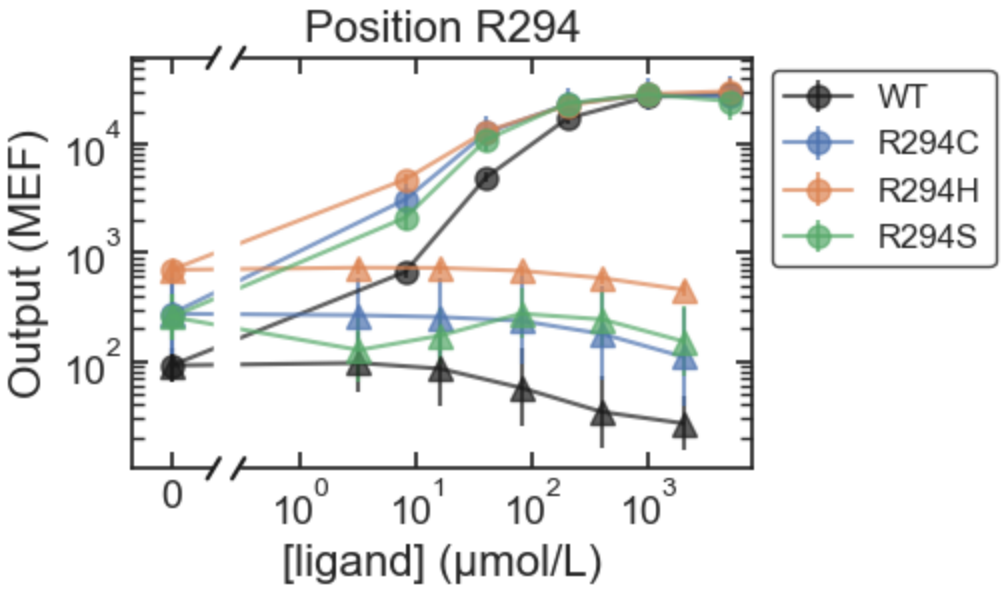

Position G252
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G252A
G252S

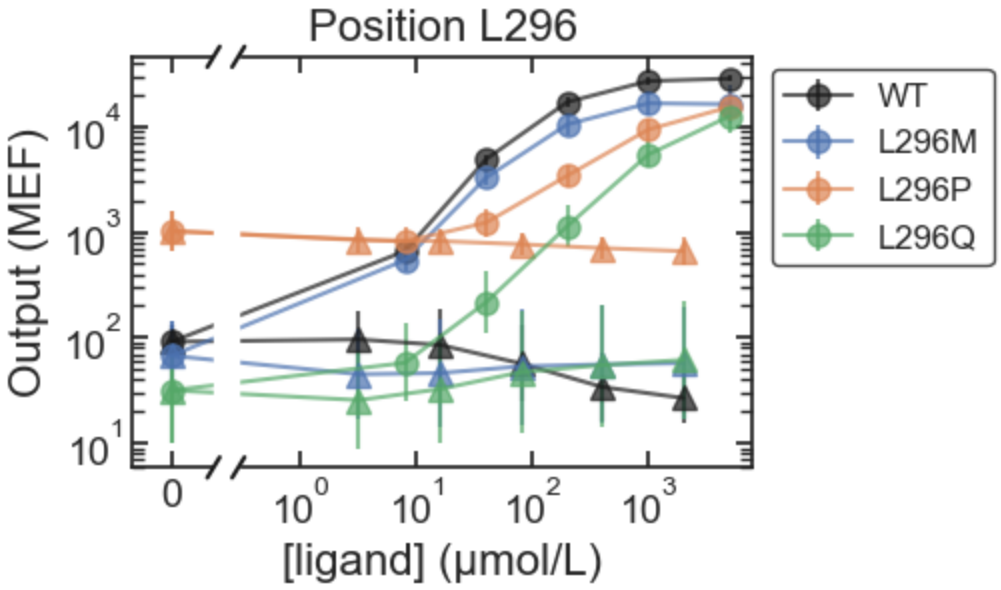

Position M254
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M254I
M254K
M254L
M254T
M254V

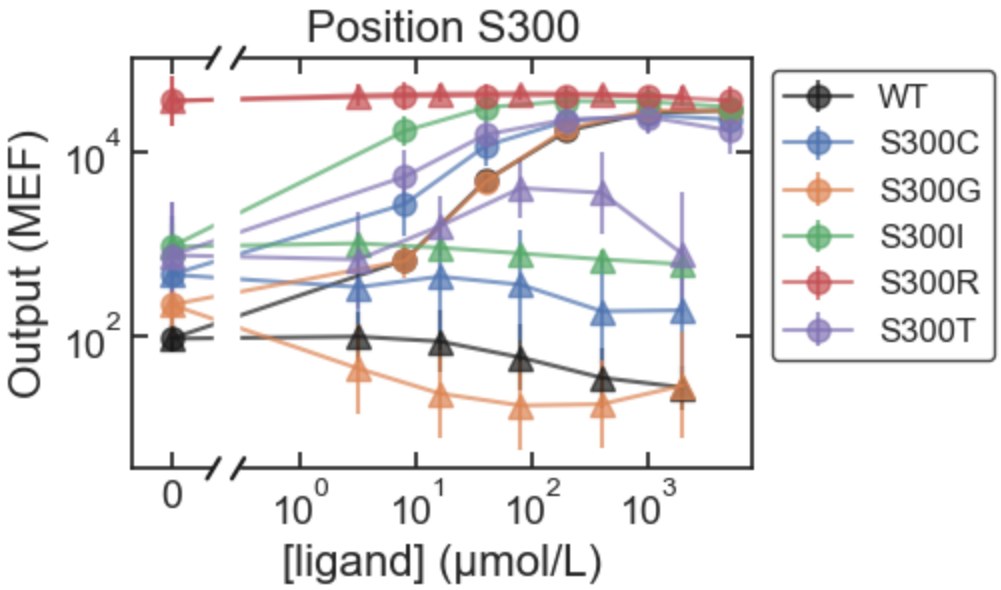

Position A256
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A256P
A256S
A256V

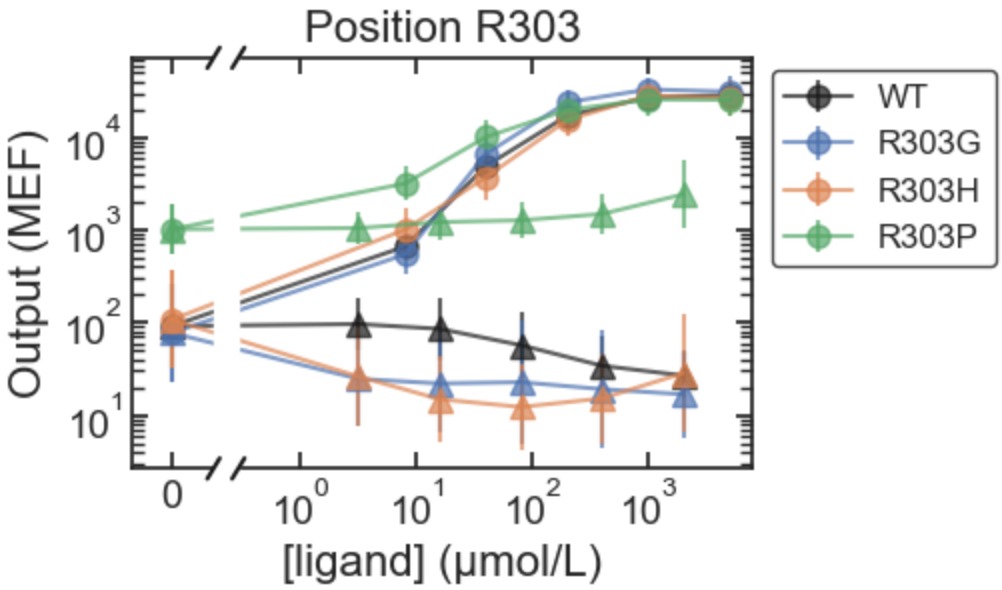

Position Q291
Output (MEF)
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L

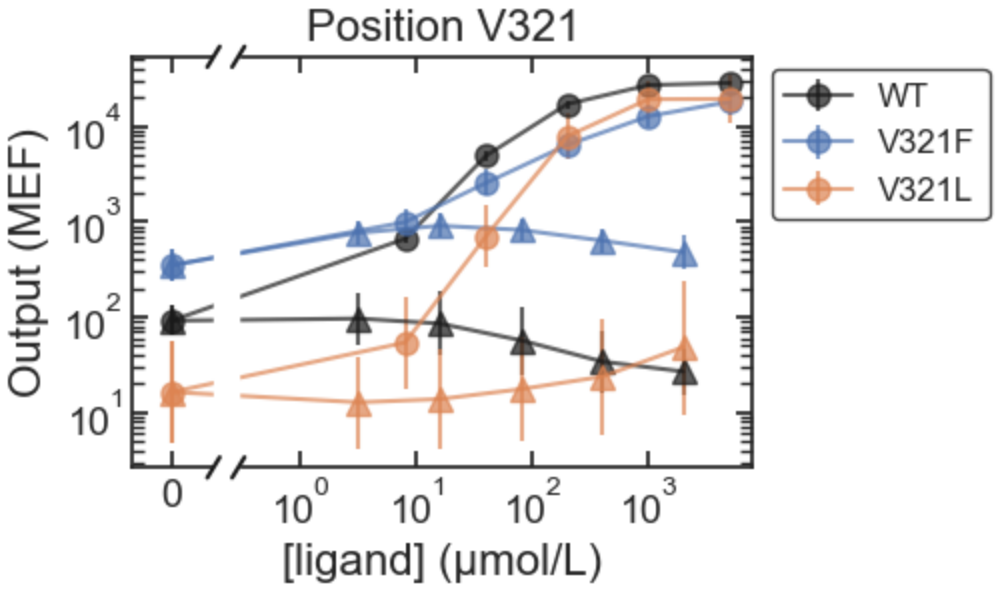

Position F293
Output (MEF)
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

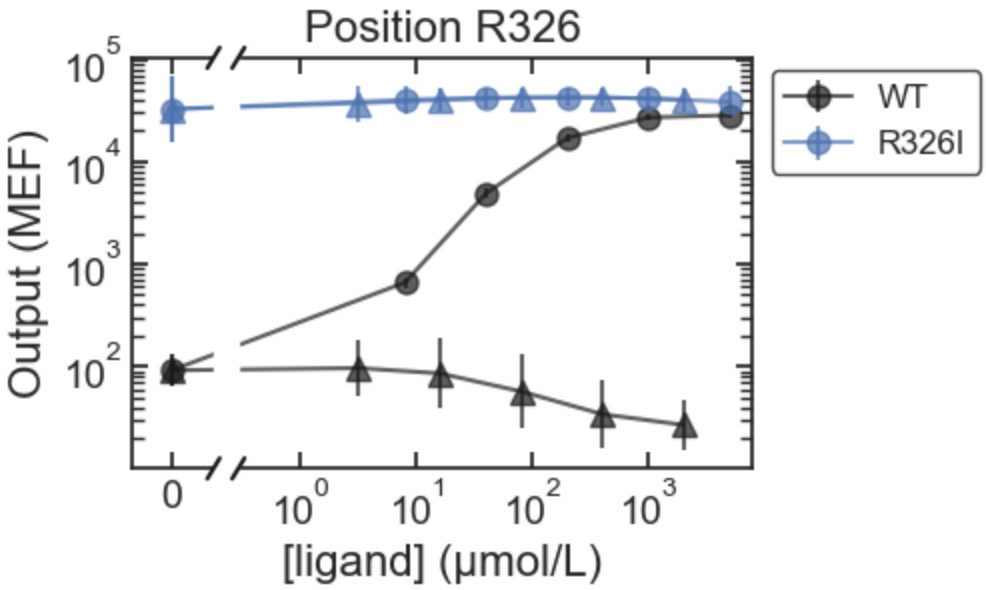

Position R294
Output (MEF)
[ligand] (μmol/L)
WT
R294C
R294H
R294S

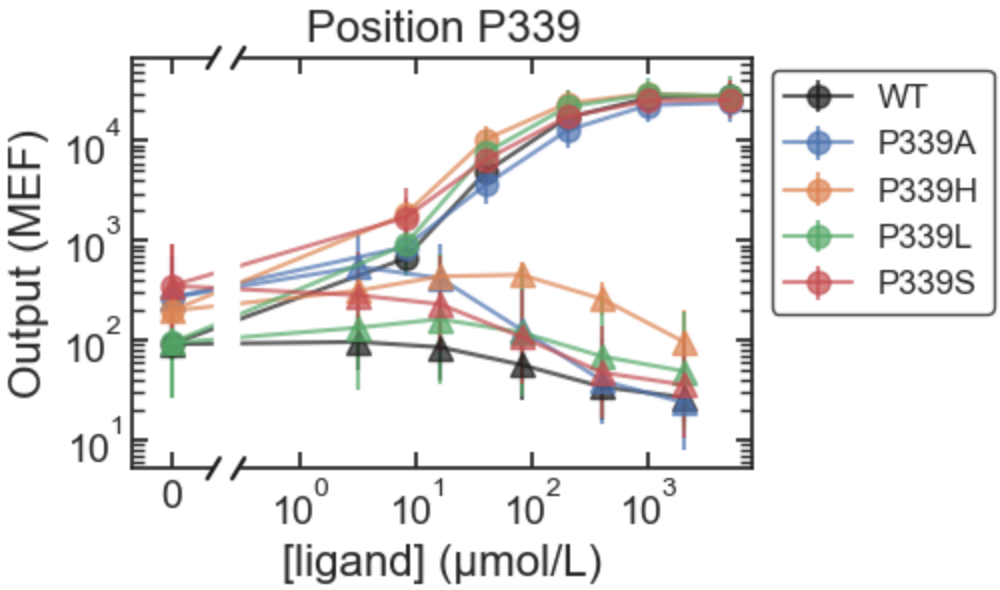

Position L296
Output (MEF)
[ligand] (μmol/L)
WT
L296M
L296P
L296Q

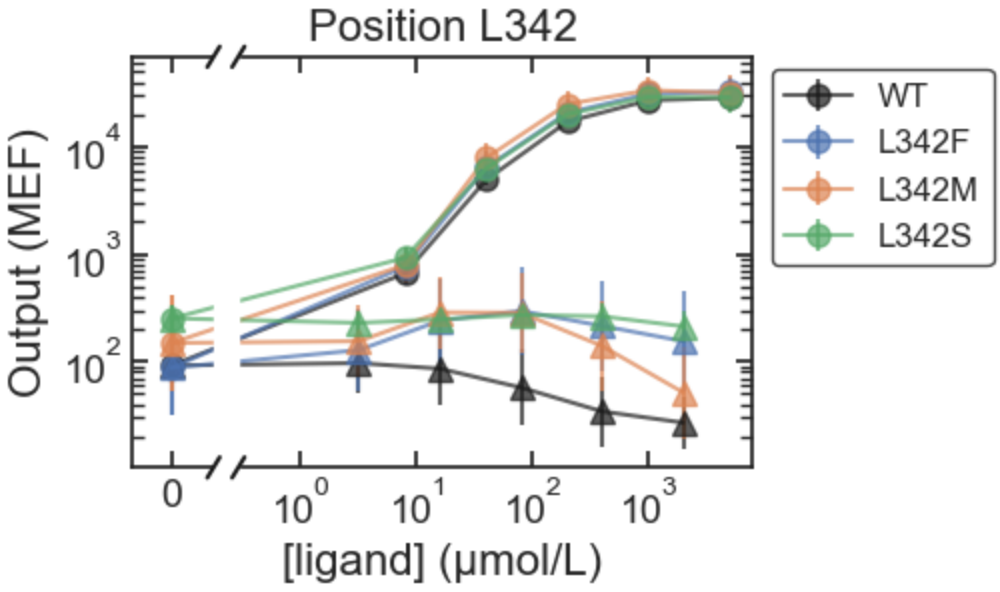

Position S300
Output (MEF)
[ligand] (μmol/L)
WT
S300C
S300G
S300I
S300R
S300T

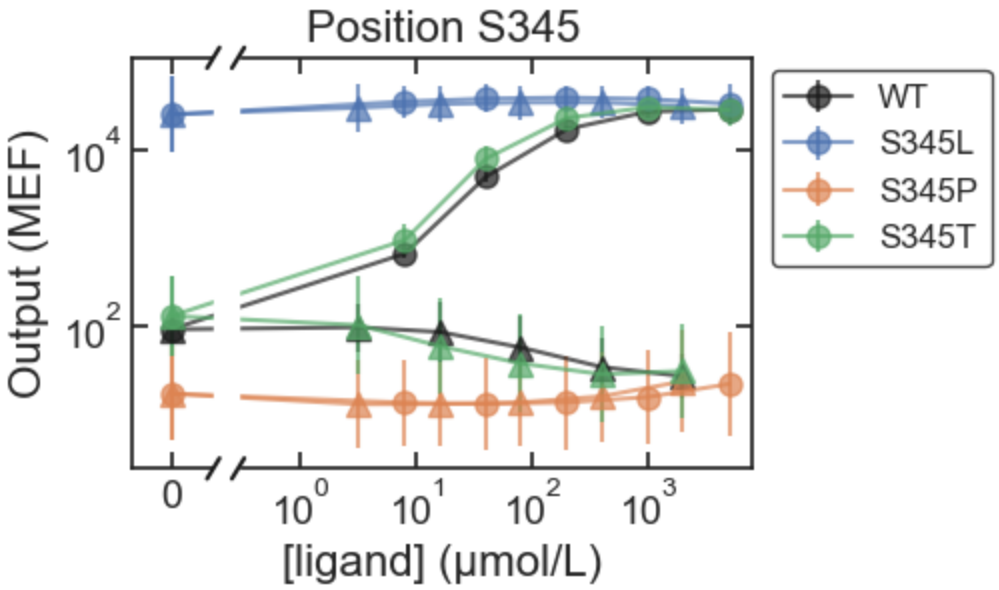

Position R303
Output (MEF)
[ligand] (μmol/L)
WT
R303G
R303H
R303P


Position V321
Output (MEF)
[ligand] (μmol/L)
WT
V321F
V321L

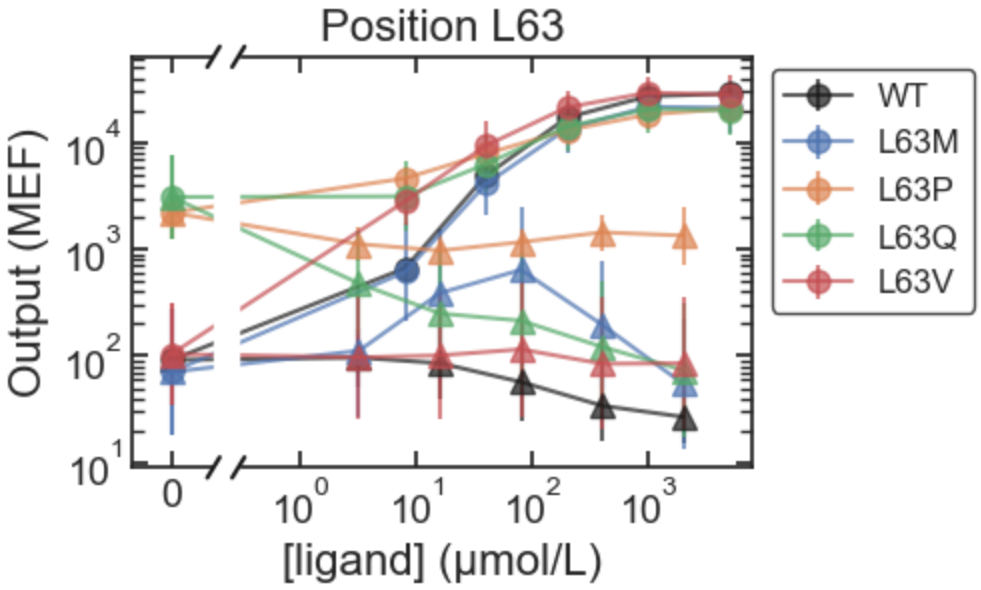

Position R326
Output (MEF)
[ligand] (μmol/L)
WT
R326I

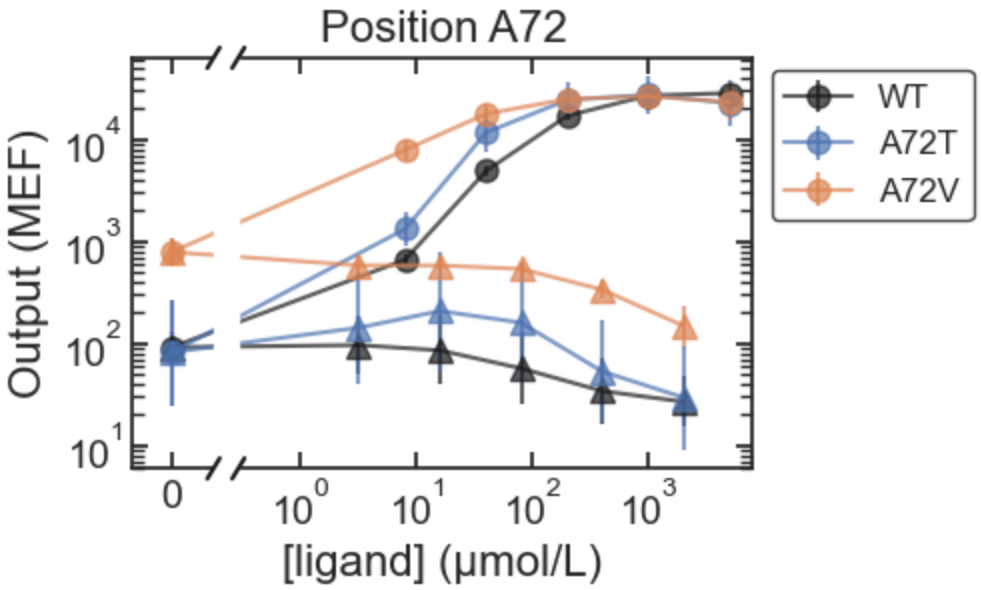

Position P339
Output (MEF)
[ligand] (μmol/L)
WT
P339A
P339H
P339L
P339S

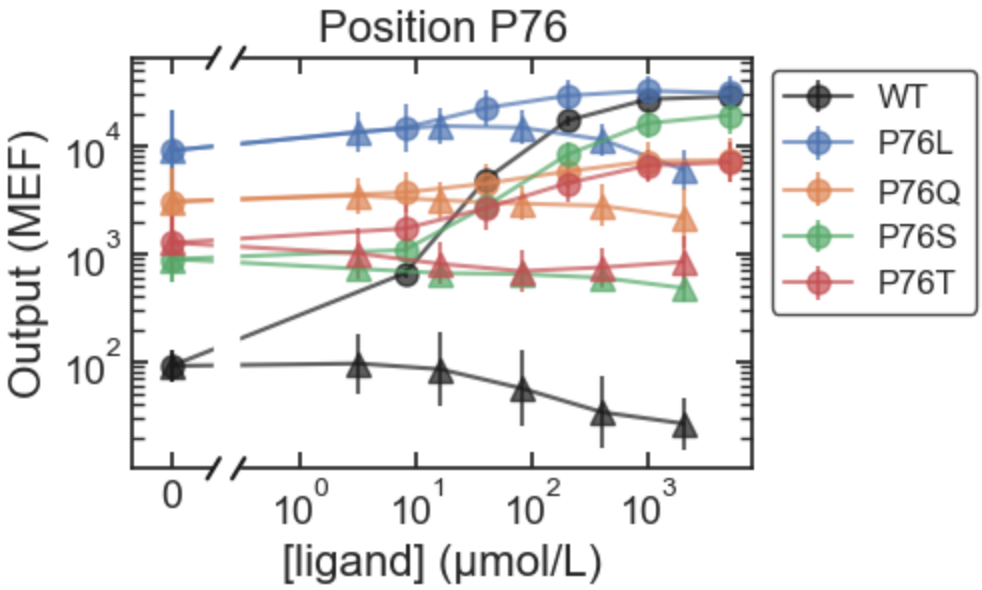


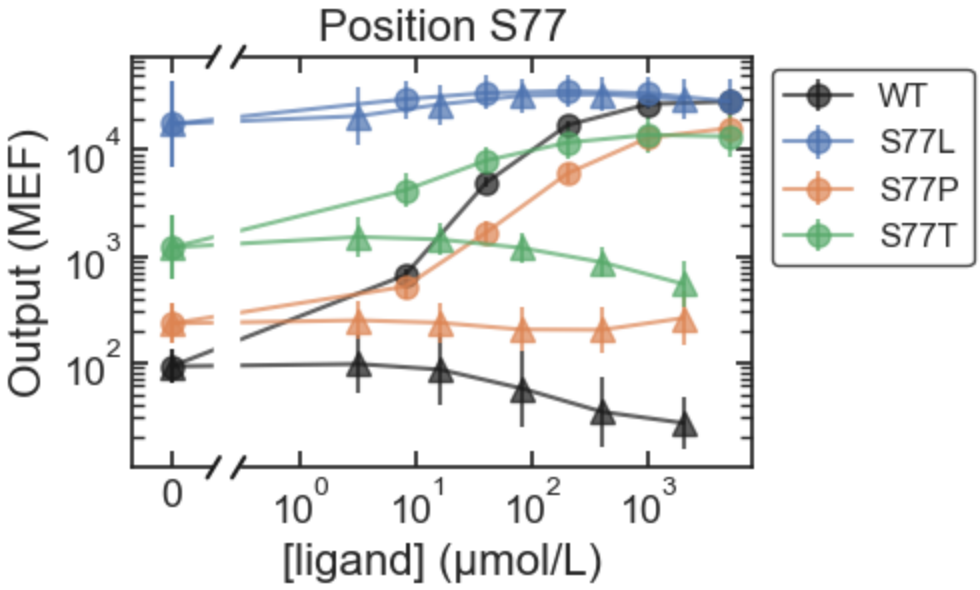


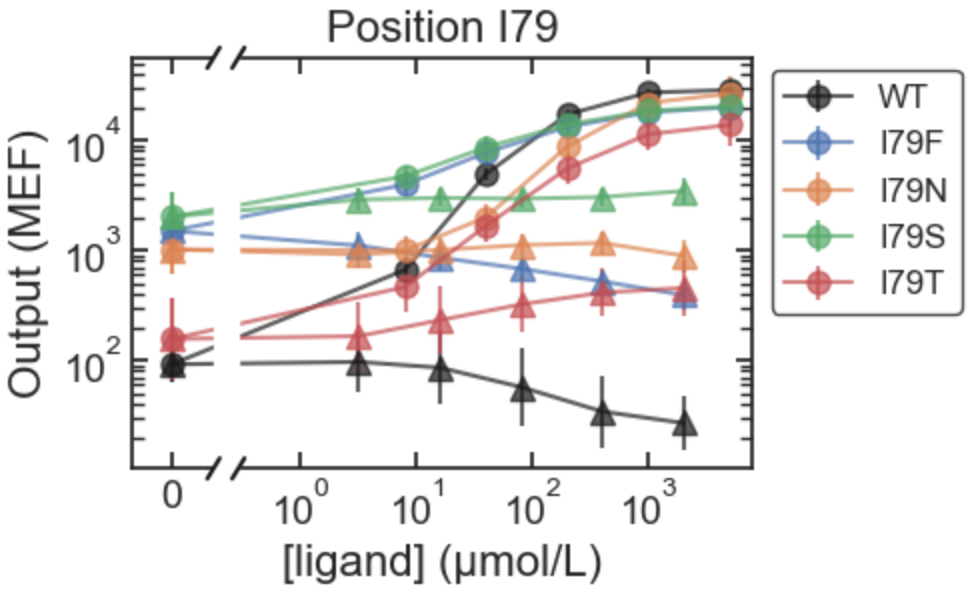
MWC_IPTG_ONPF

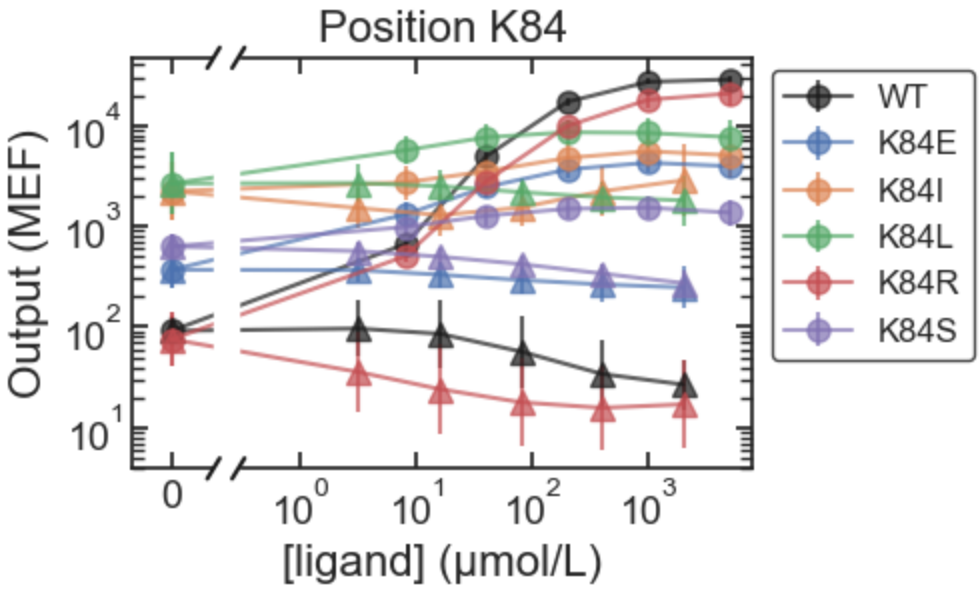

Position L63
Output (MEF)
[ligand] (μmol/L)
WT
L63M
L63P
L63Q
L63V

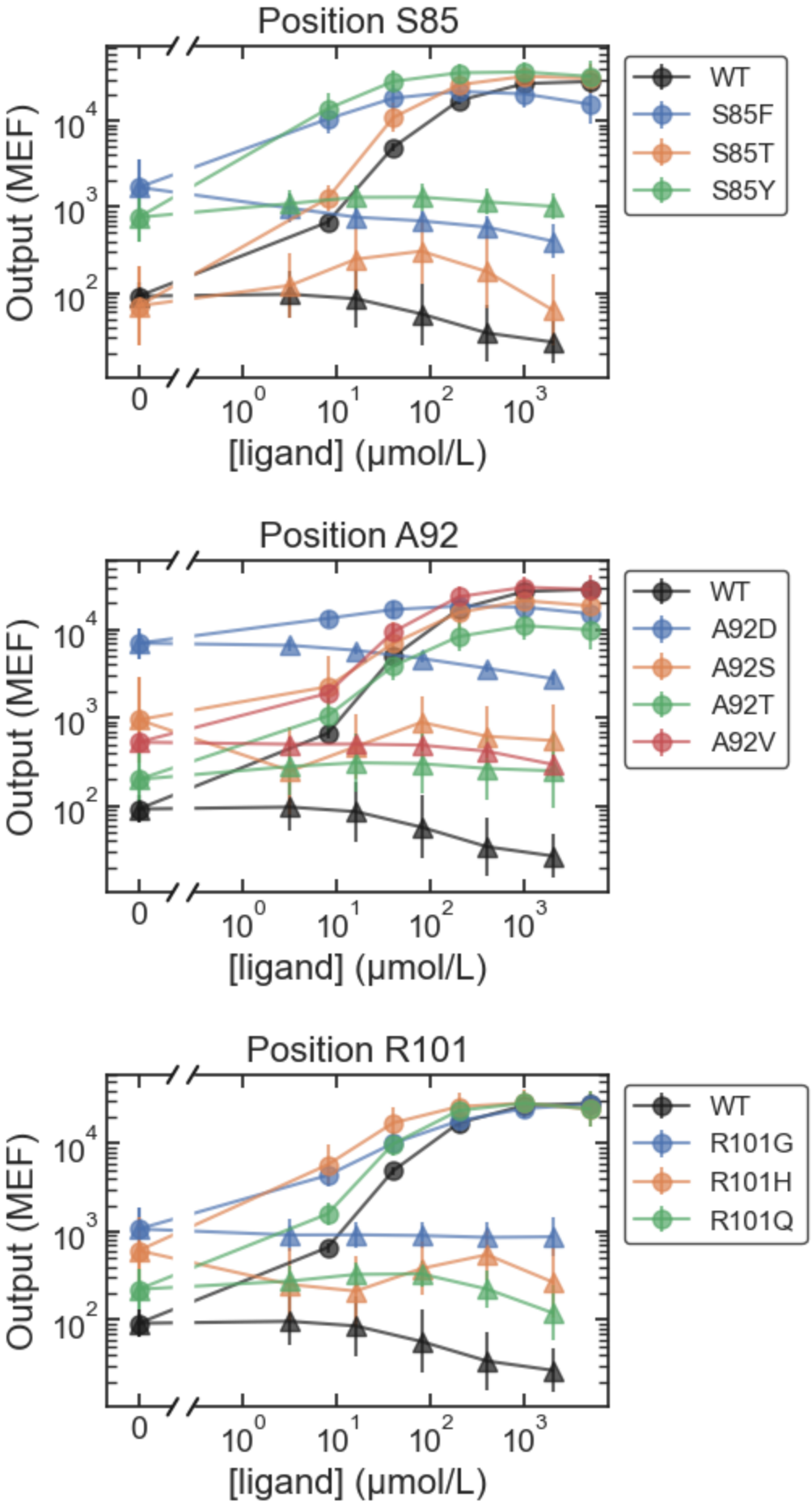

Position A72
Output (MEF)
[ligand] (μmol/L)
WT
A72T
A72V

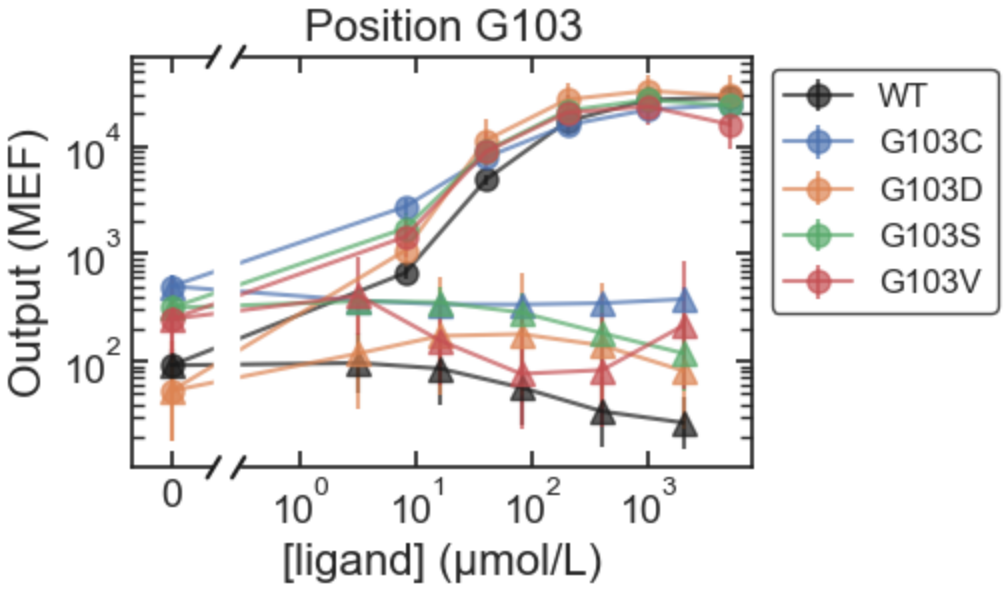

Position P76
Output (MEF)
[ligand] (μmol/L)
WT
P76L
P76Q
P76S
P76T

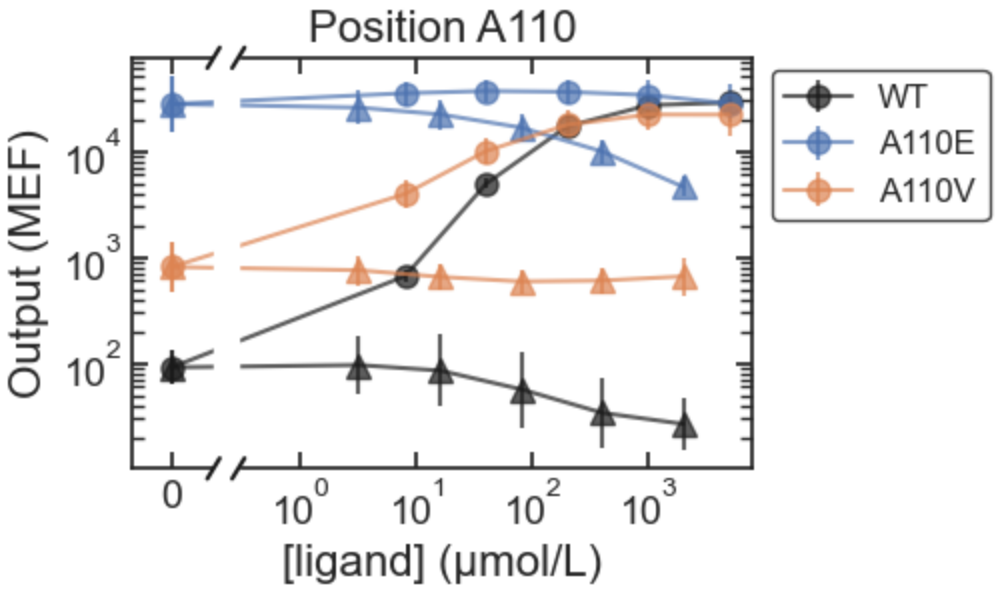

Position S77
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S77L
S77P
S77T

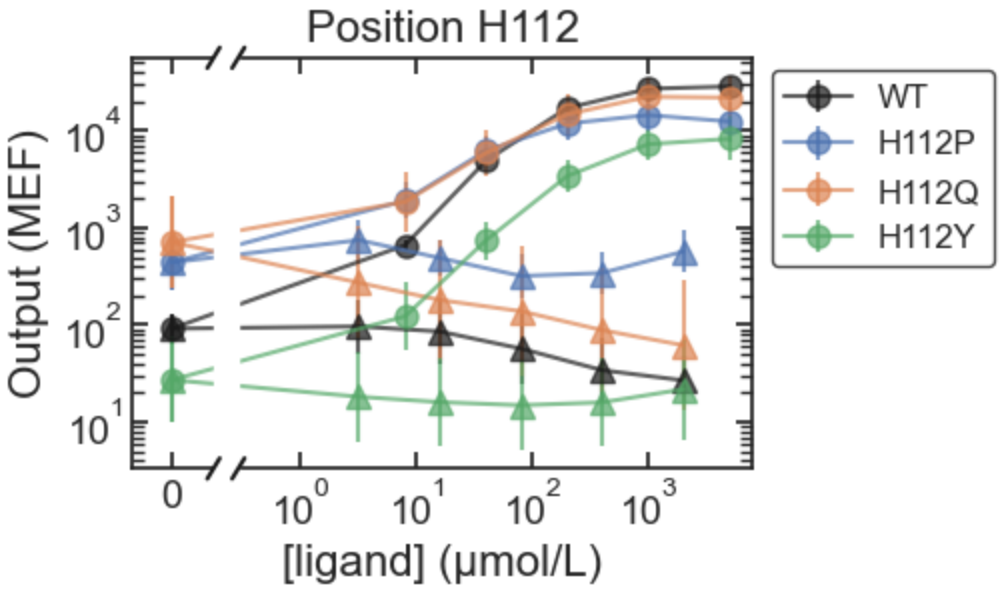

Position I79
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I79F
I79N
I79S
I79T

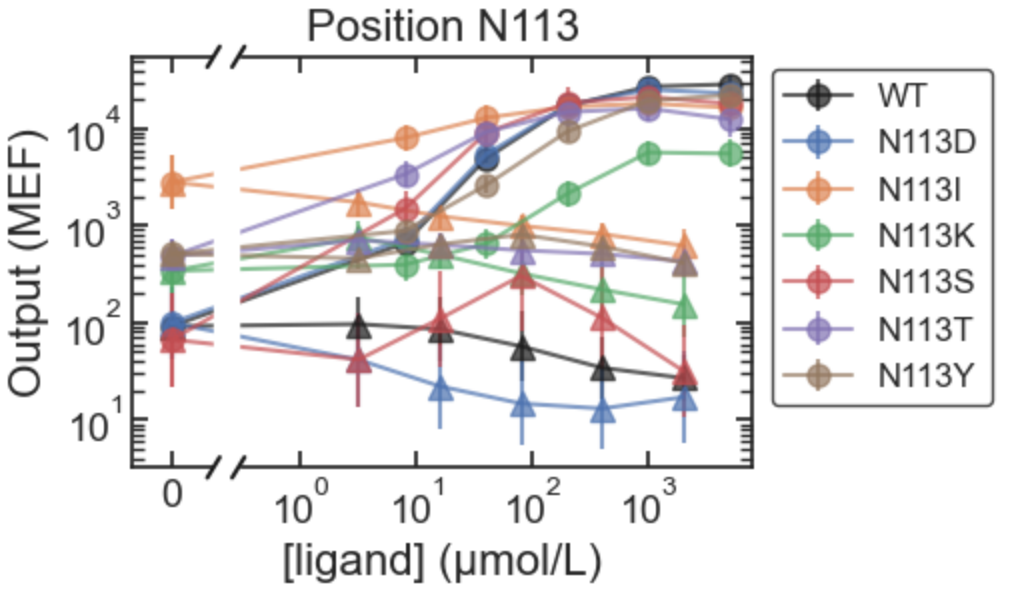

Position K84
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K84E
K84I
K84L
K84R
K84S

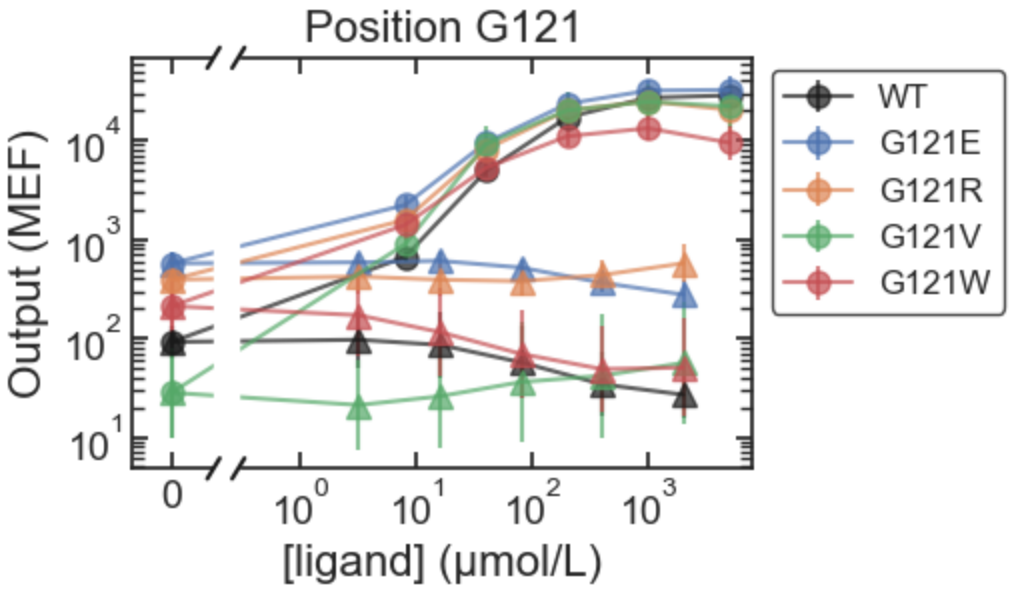

Position S85
Output (MEF)
[ligand] (μmol/L)
WT
S85F
S85T
S85Y
Position A92
Output (MEF)
[ligand] (μmol/L)
WT
A92D
A92S
A92T
A92V
Position R101
Output (MEF)
[ligand] (μmol/L)
WT
R101G
R101H
R101Q

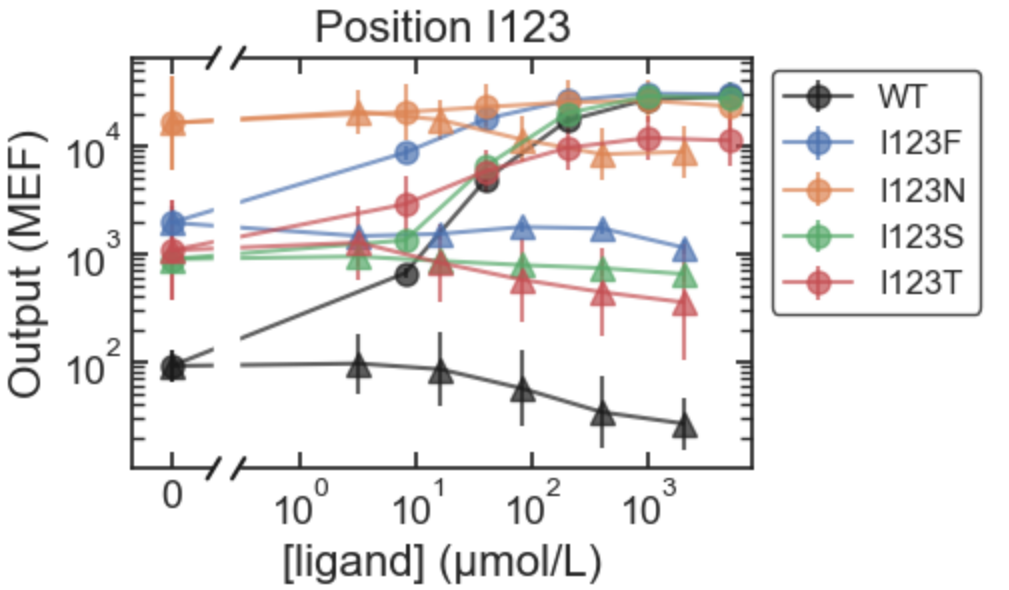

Position G103
Output (MEF)
[ligand] (μmol/L)
WT
G103C
G103D
G103S
G103V

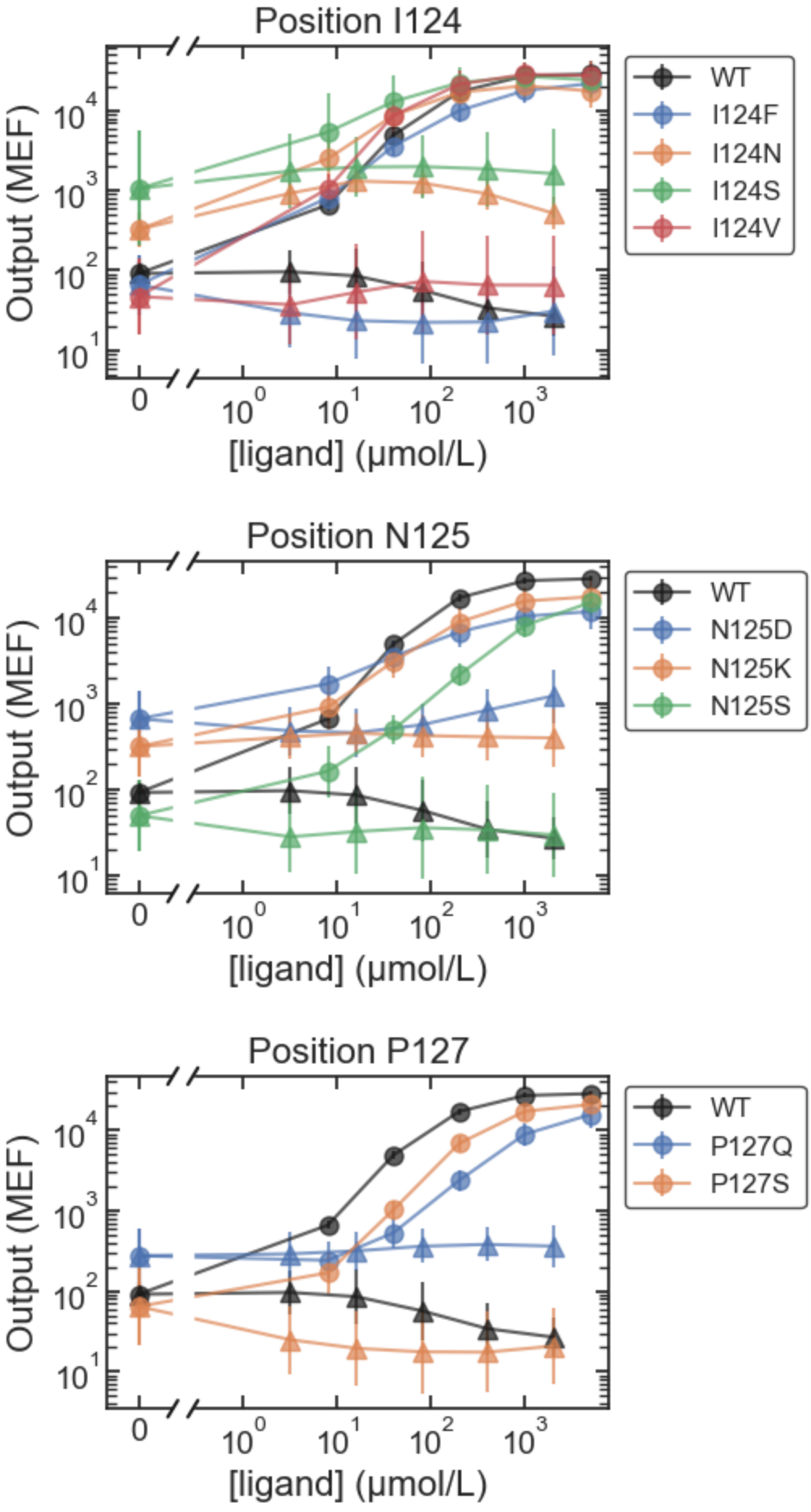

Position A110
Output (MEF)
[ligand] (μmol/L)
WT
A110E
A110V

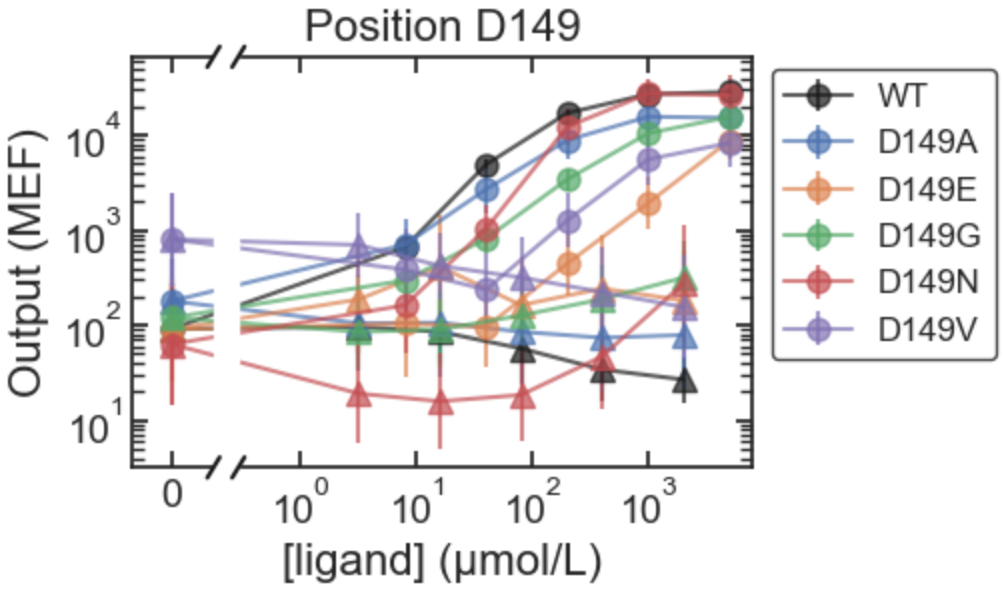

Position H112
Output (MEF)
[ligand] (μmol/L)
WT
H112P
H112Q
H112Y

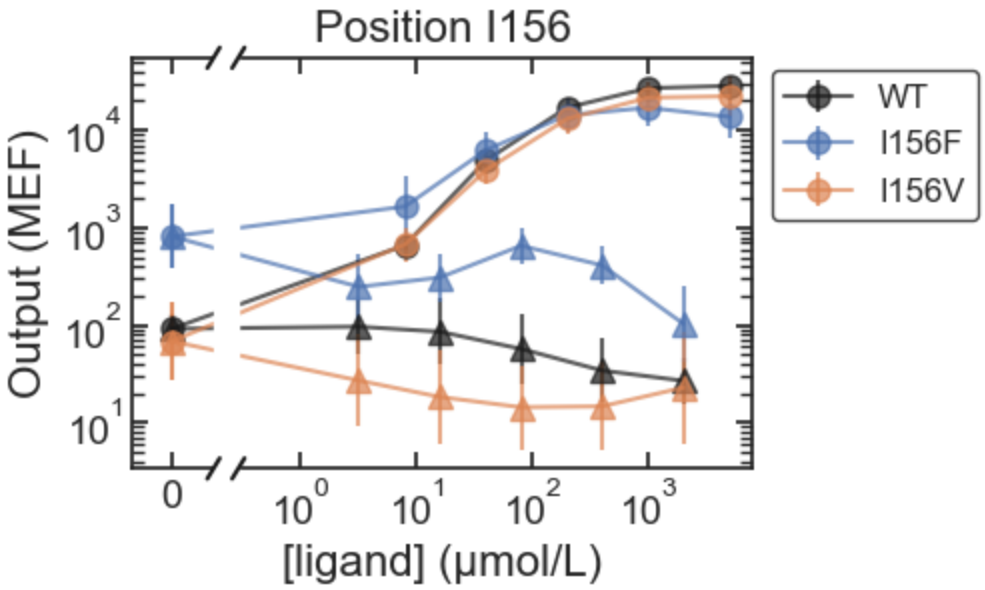

Position N113
Output (MEF)
[ligand] (μmol/L)
WT
N113D
N113I
N113K
N113S
N113T
N113Y

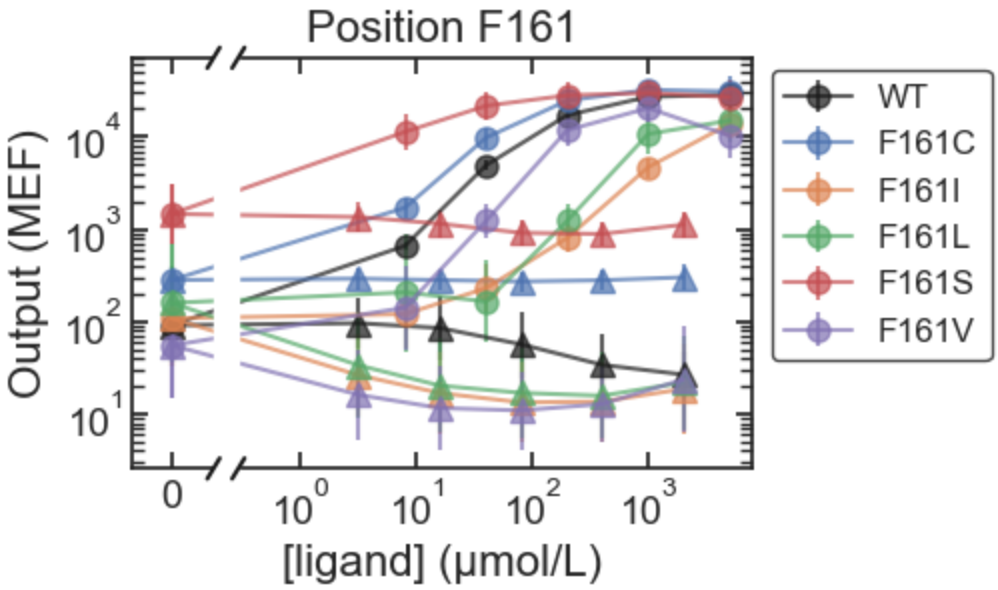

Position G121
Output (MEF)
[ligand] (μmol/L)
WT
G121E
G121R
G121V
G121W

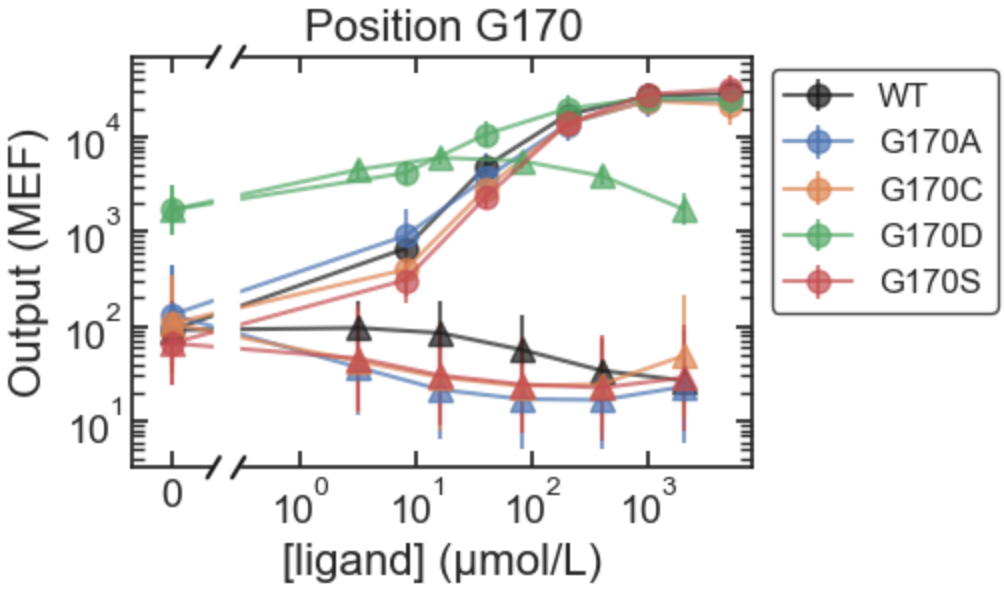

Position I123
Output (MEF)
[ligand] (μmol/L)
WT
I123F
I123N
I123S
I123T

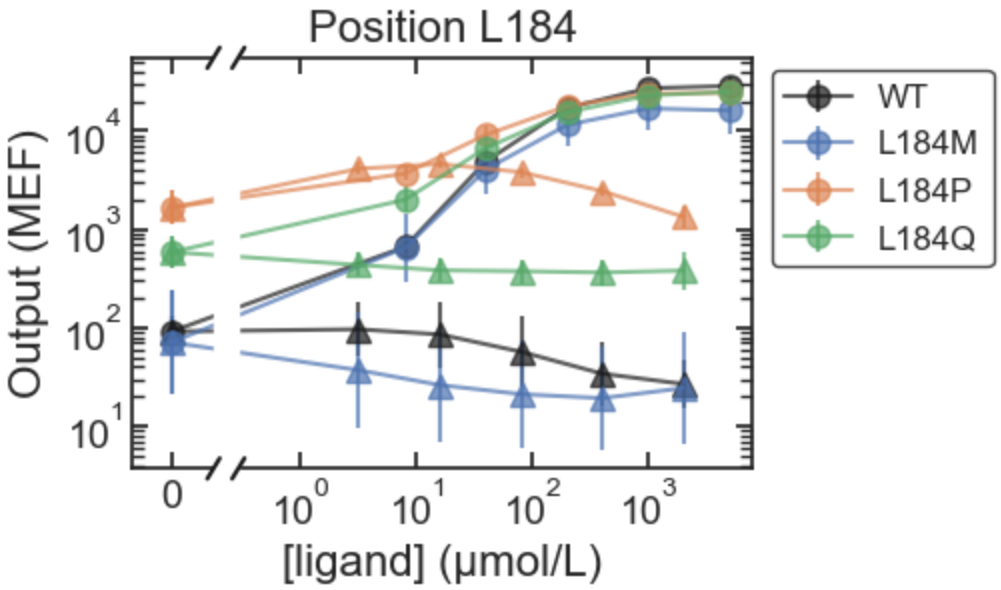

Position I124
Output (MEF)
[ligand] (μmol/L)
WT
I124F
I124N
I124S
I124V
Position N125
Output (MEF)
[ligand] (μmol/L)
WT
N125D
N125K
N125S
Position P127
Output (MEF)
[ligand] (μmol/L)
WT
P127Q
P127S

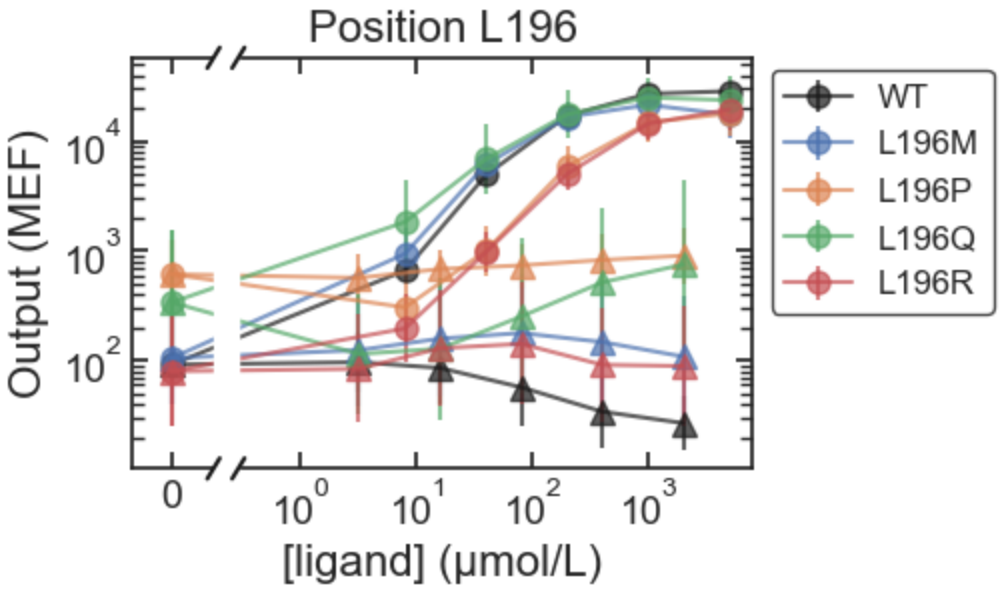

Position D149
Output (MEF)
[ligand] (μmol/L)
WT
D149A
D149E
D149G
D149N
D149V

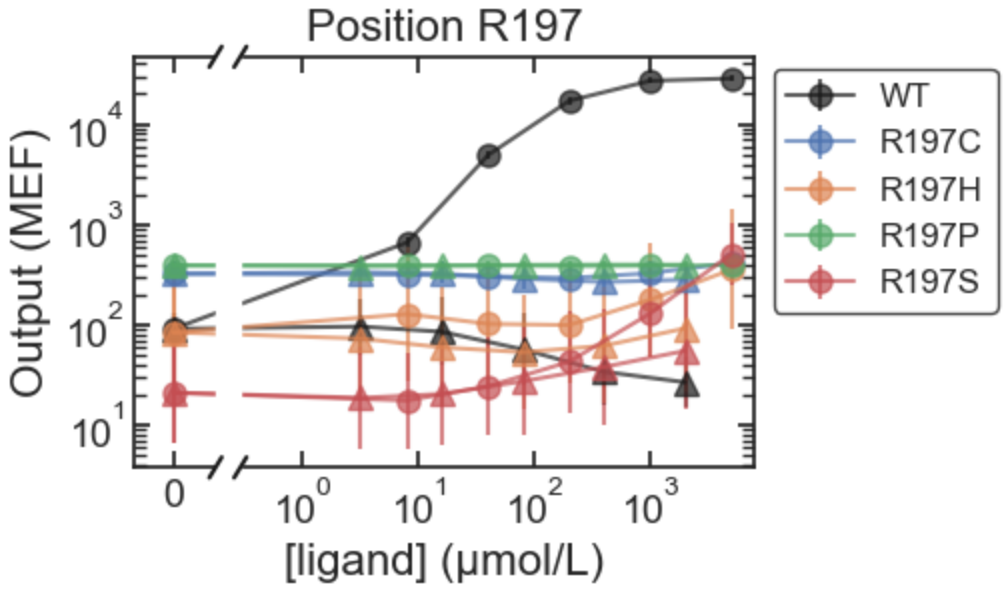

Position I156
Output (MEF)
[ligand] (μmol/L)
WT
I156F
I156V

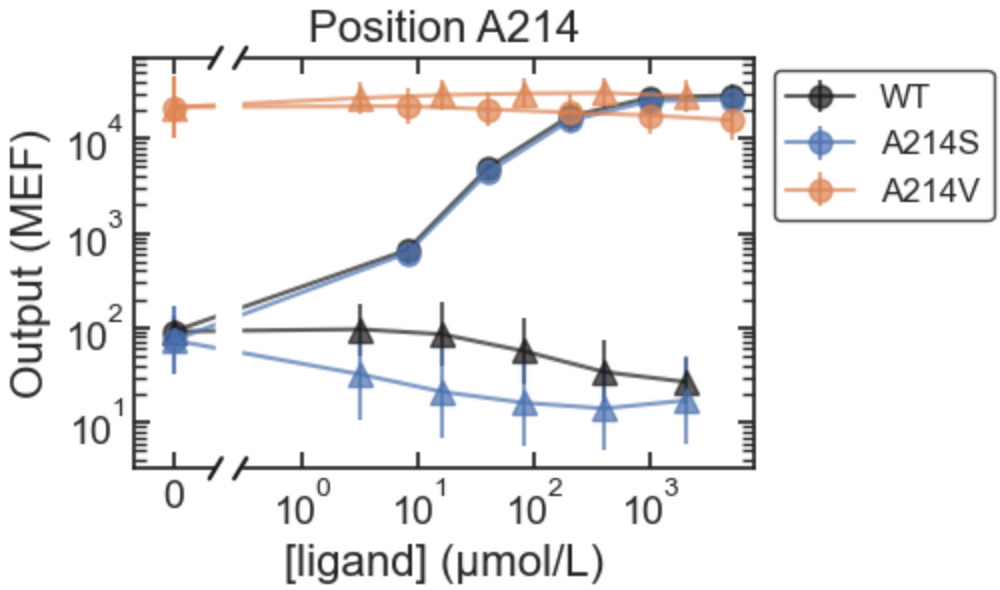

Position F161
Output (MEF)
[ligand] (μmol/L)
WT
F161C
F161I
F161L
F161S
F161V

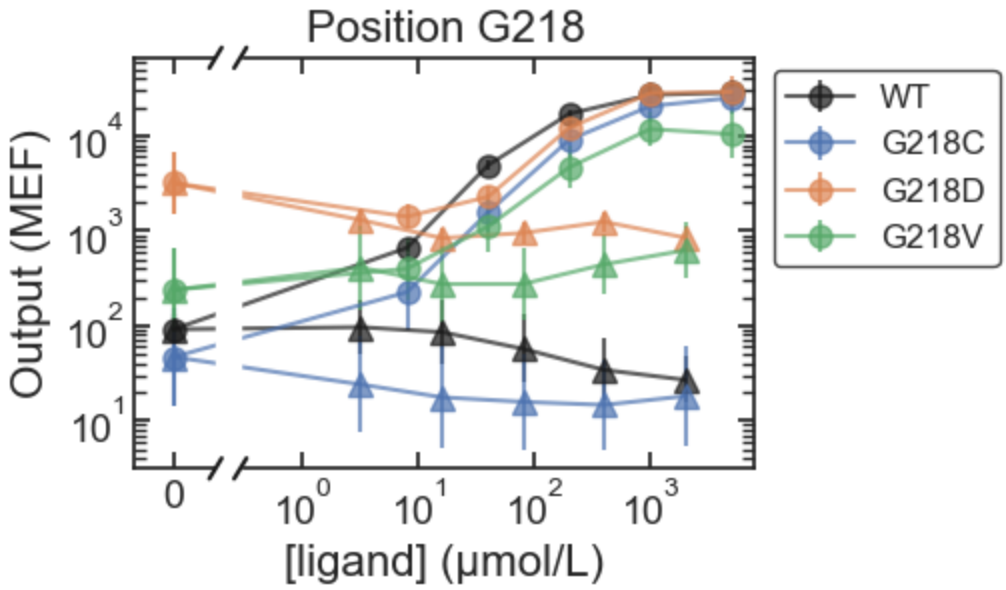

Position G170
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G170A
G170C
G170D
G170S

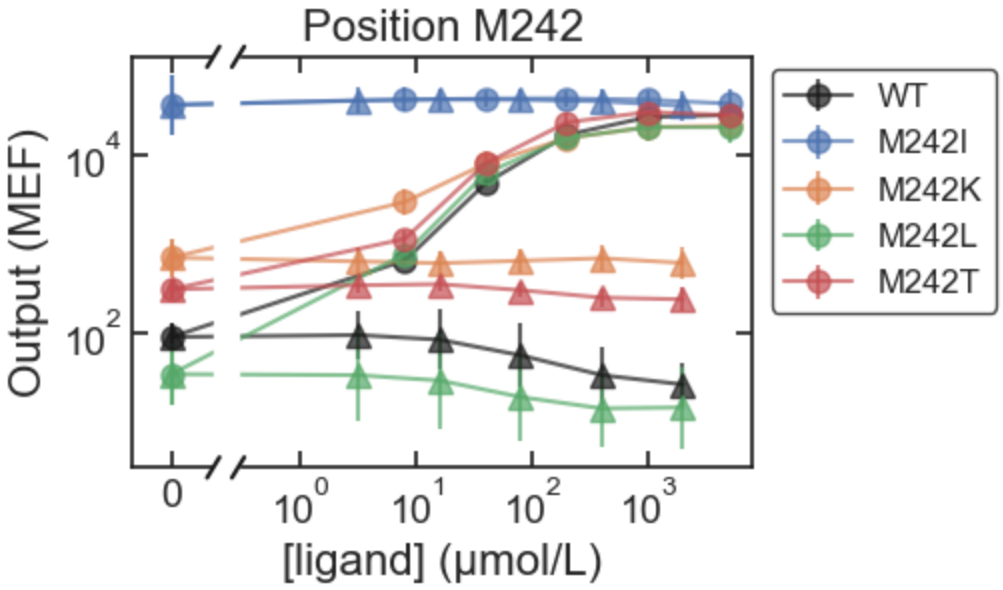

Position L184
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L184M
L184P
L184Q

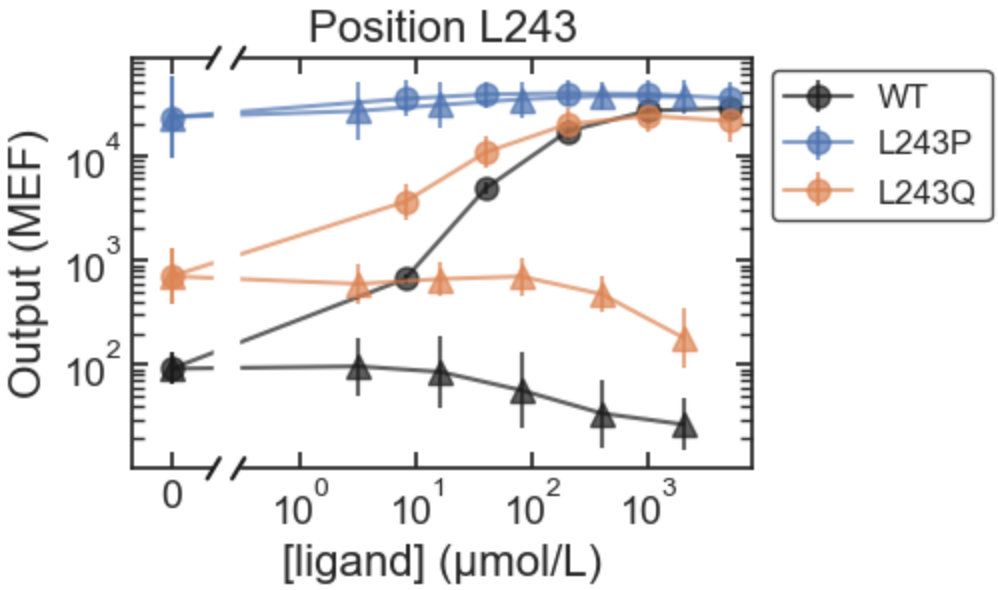

Position L196
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L196M
L196P
L196Q
L196R

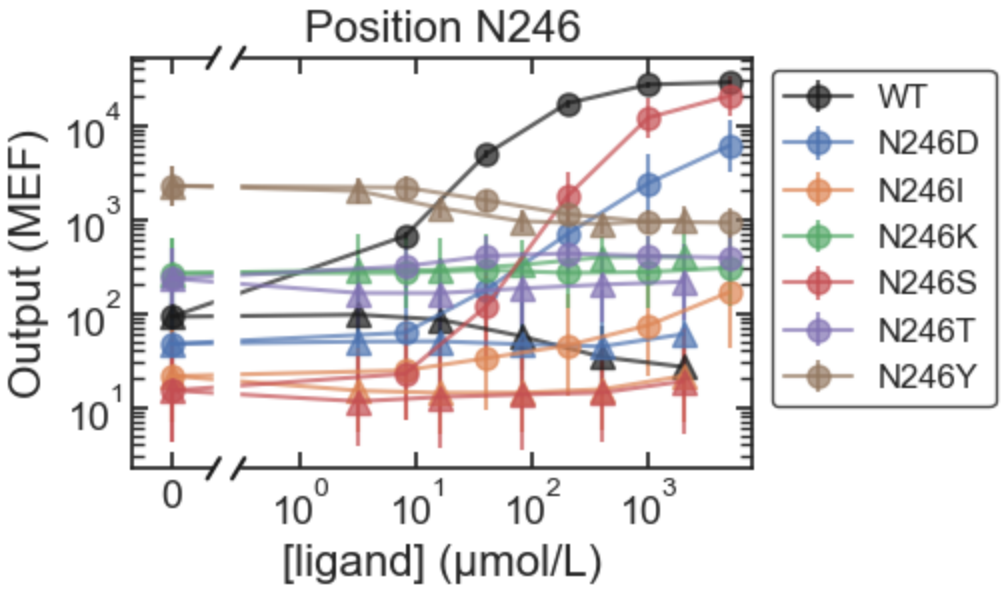

Position R197
Output (MEF)
[ligand] (μmol/L)
WT
R197C
R197H
R197P
R197S

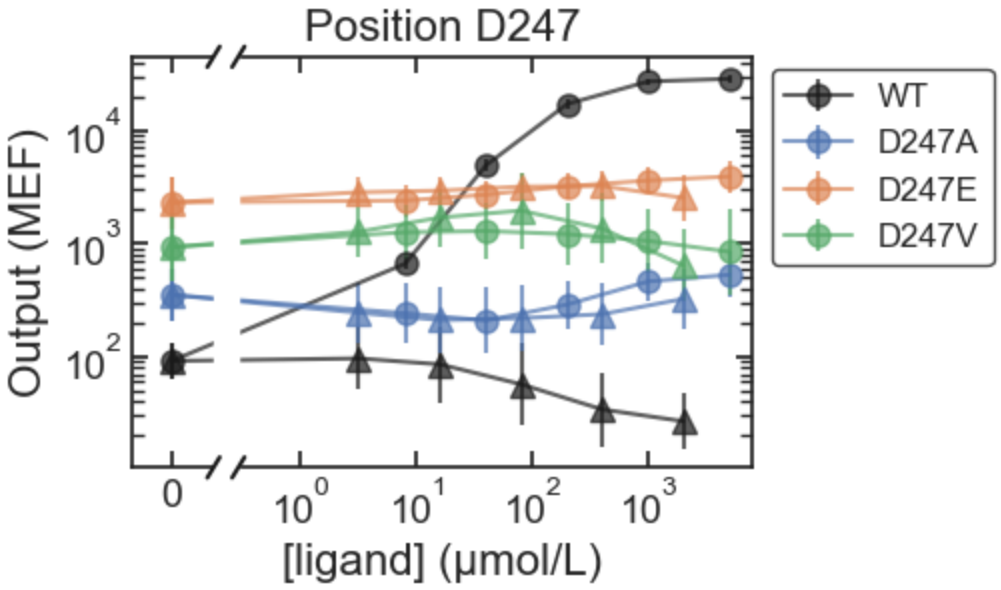

Position A214
Output (MEF)
[ligand] (μmol/L)
WT
A214S
A214V

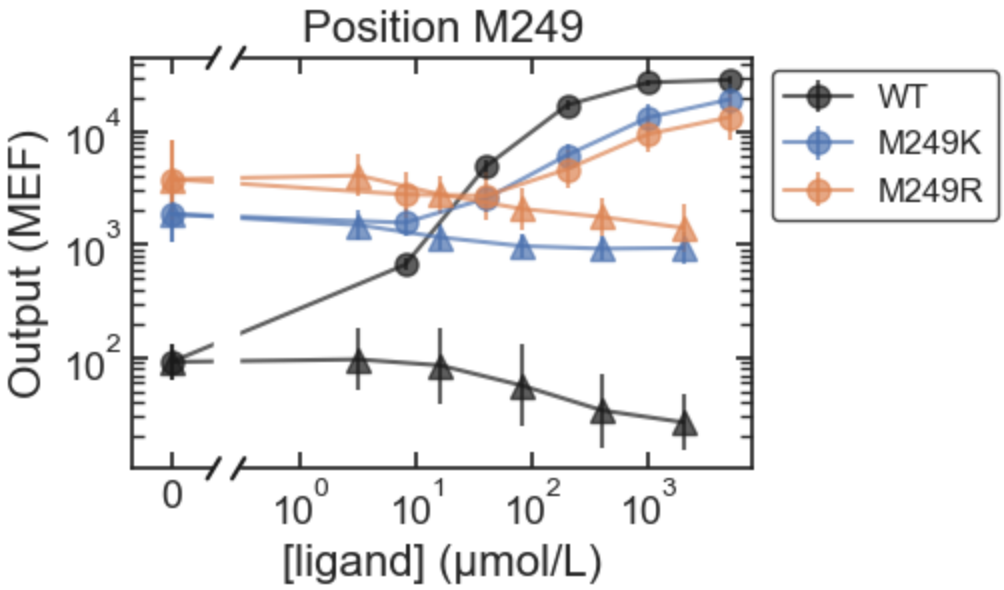

Position G218
Output (MEF)
[ligand] (μmol/L)
WT
G218C
G218D
G218V

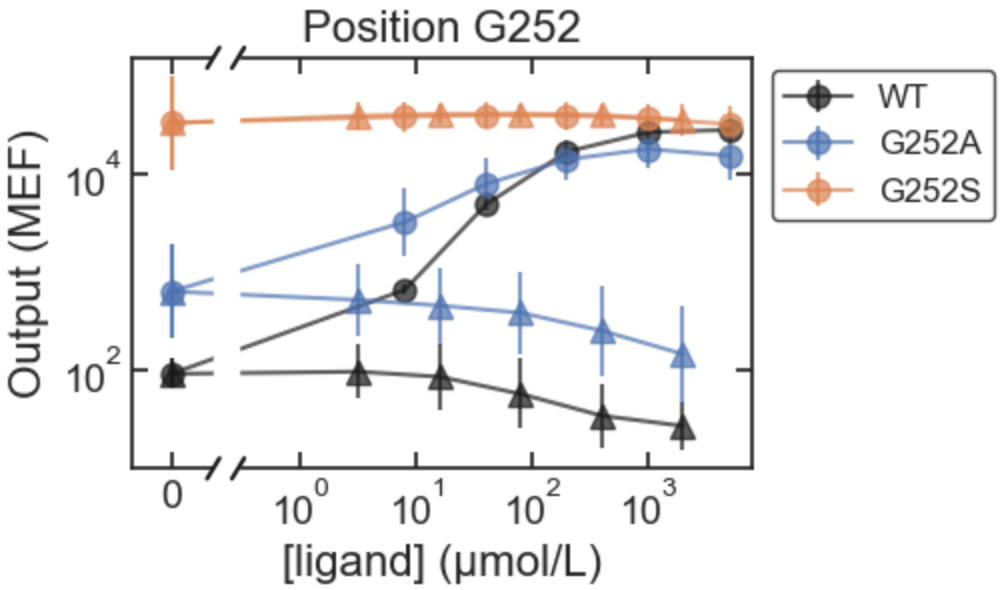

Position M242
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M242I
M242K
M242L
M242T

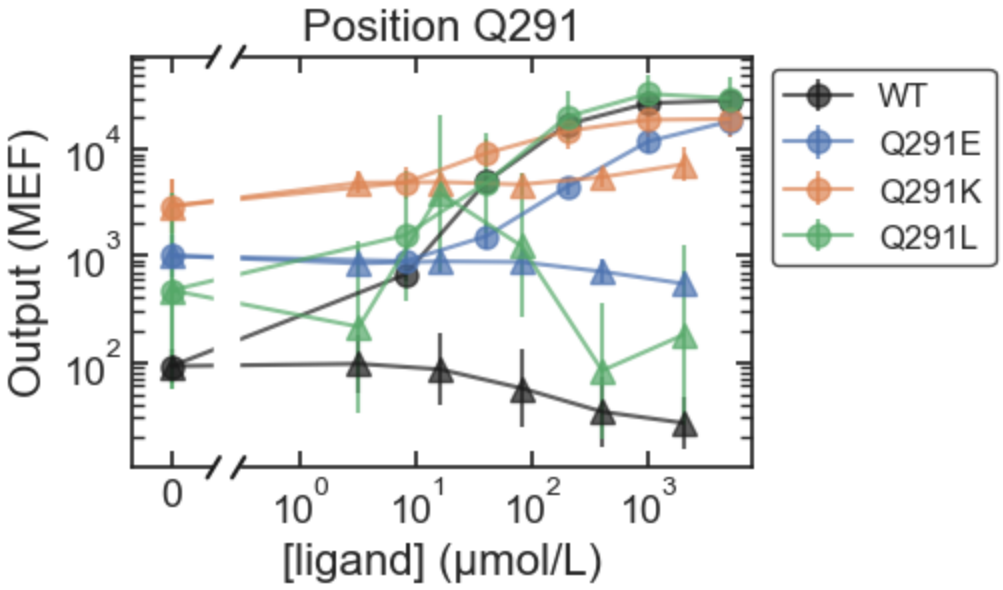

Position L243
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L243P
L243Q

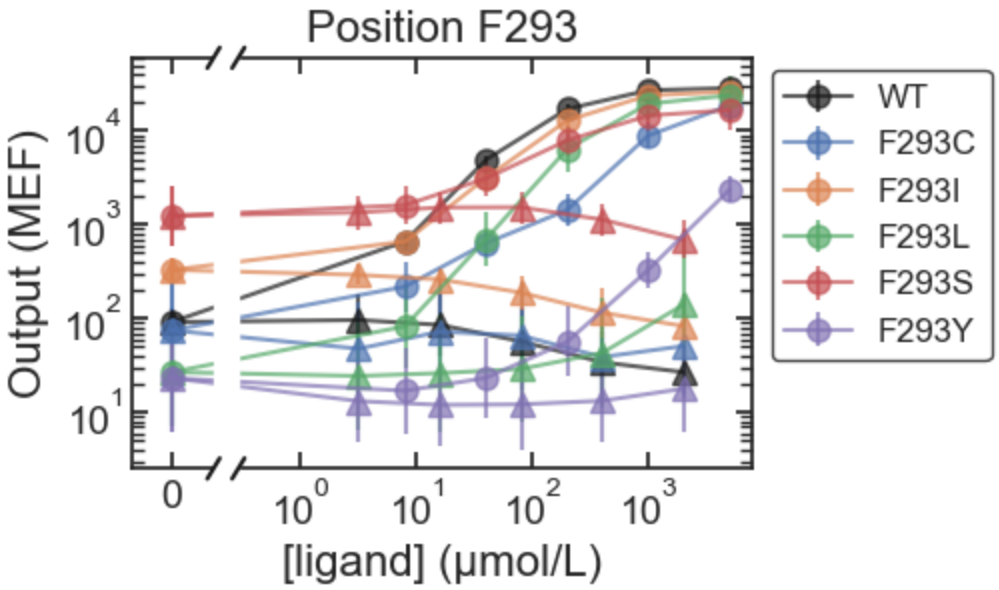

Position N246
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
N246D
N246I
N246K
N246S
N246T
N246Y

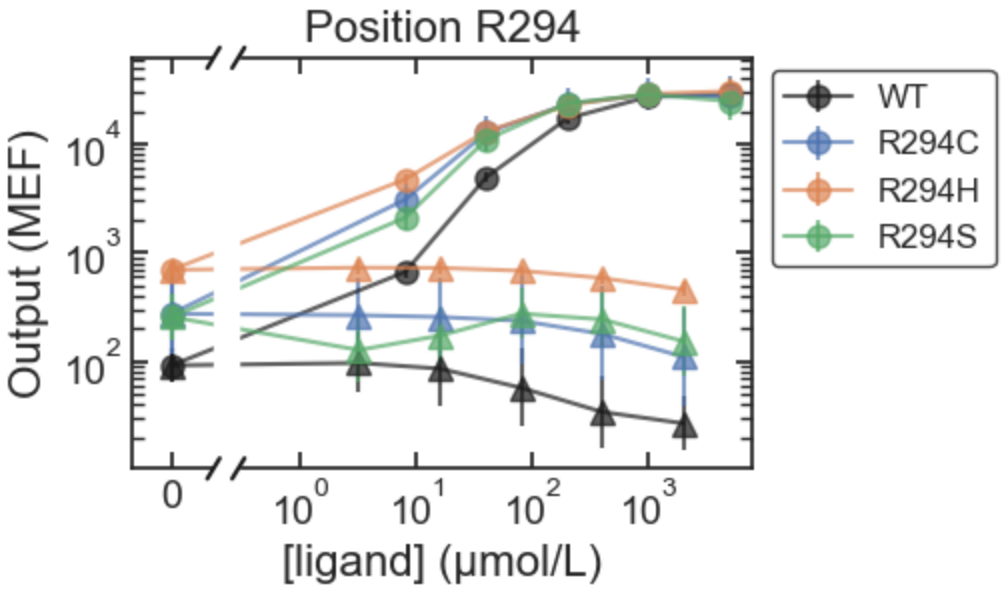

Position D247
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D247A
D247E
D247V

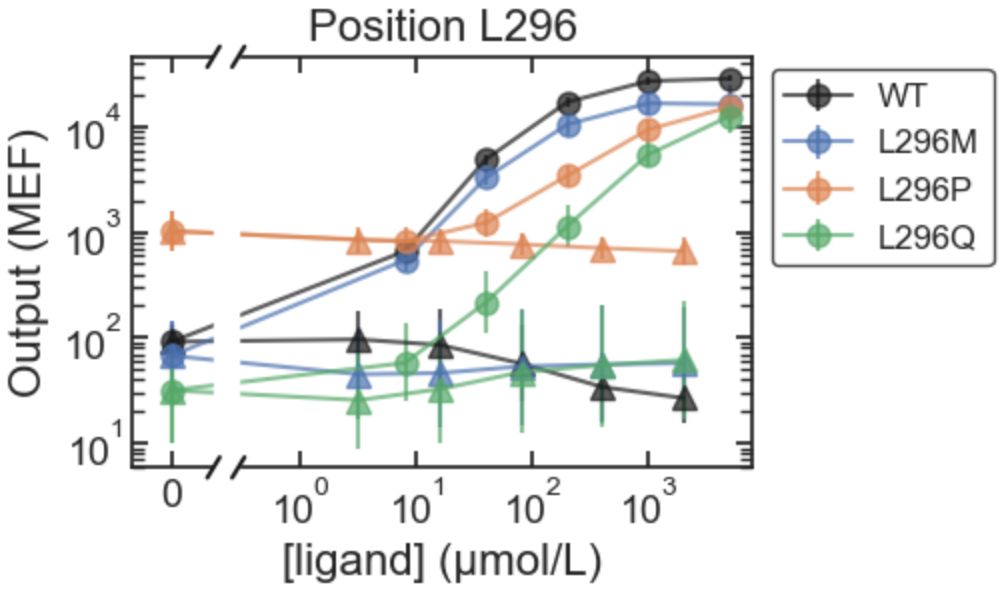

Position M249
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
M249K
M249R

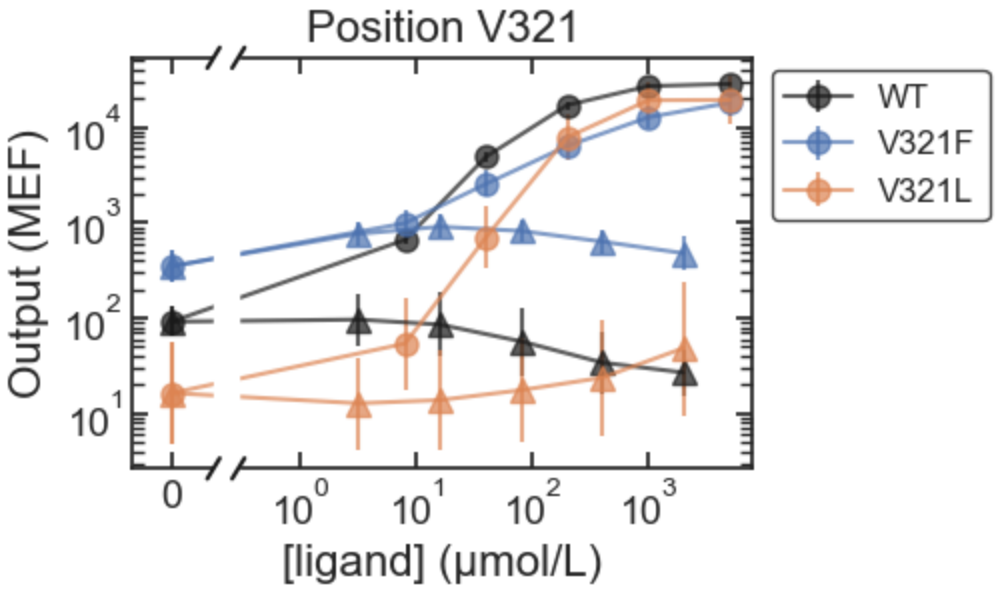

Position G252
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
G252A
G252S

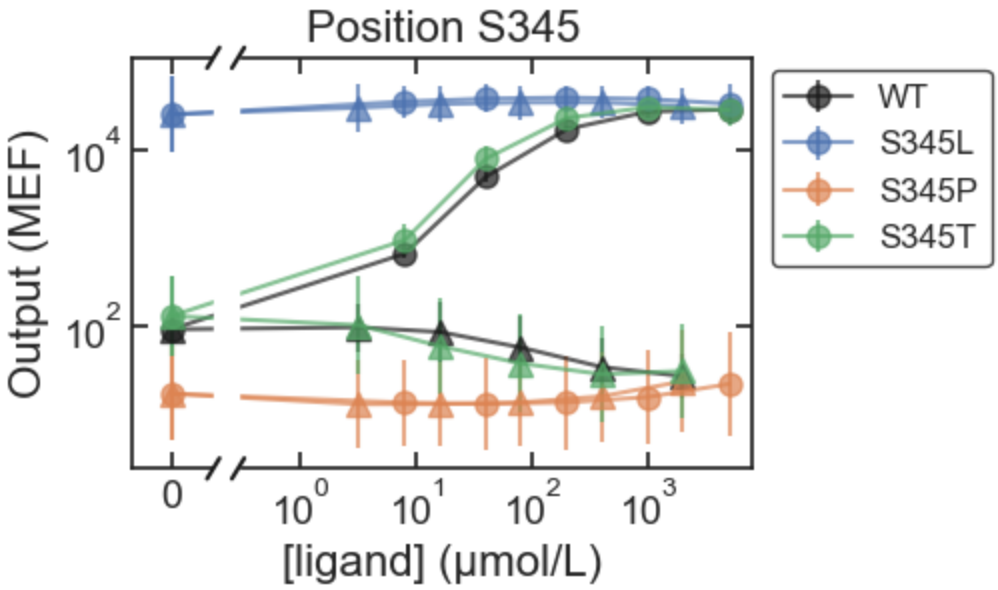

Position Q291
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q291E
Q291K
Q291L


Position F293
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
F293C
F293I
F293L
F293S
F293Y

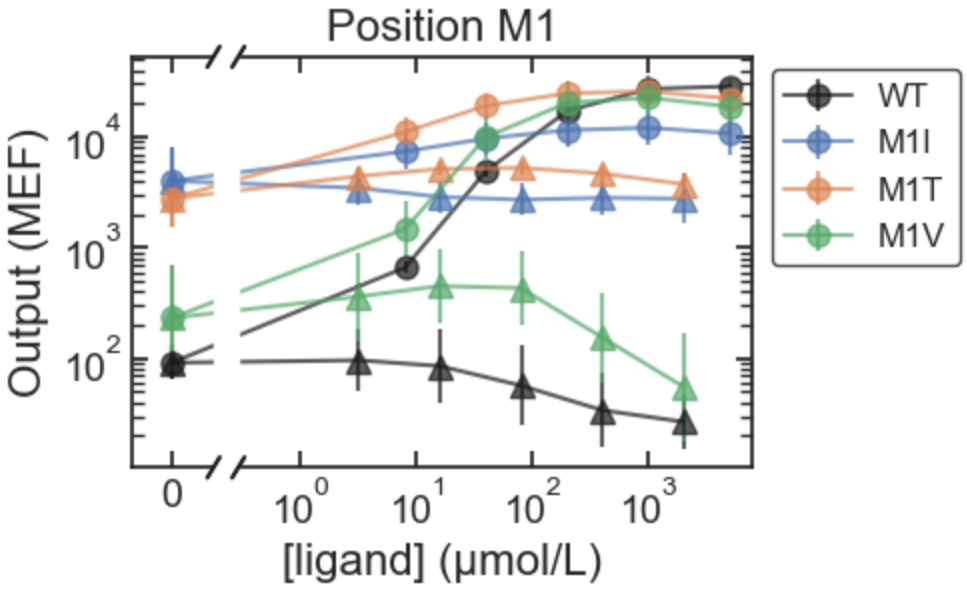

Position R294
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
R294C
R294H
R294S

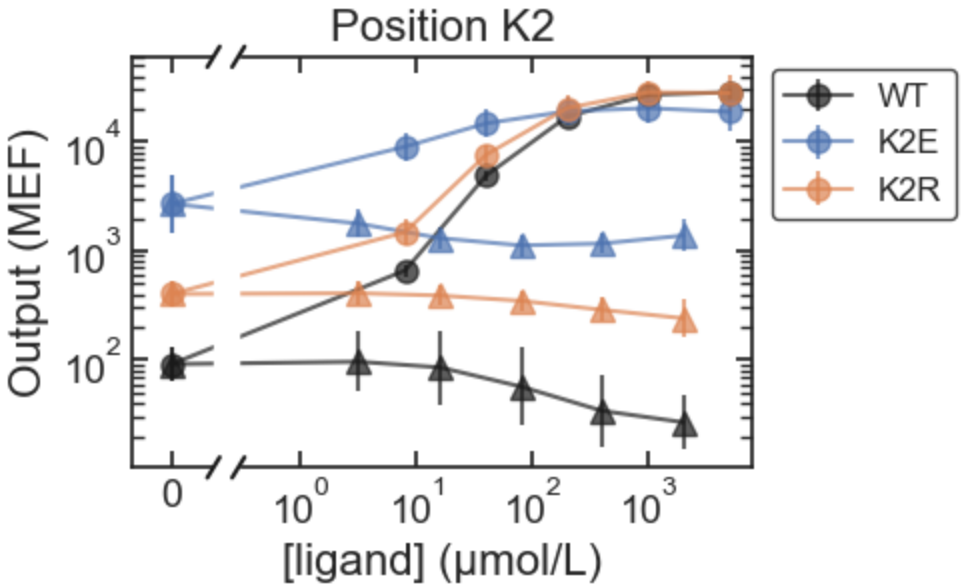

Position L296
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
L296M
L296P
L296Q

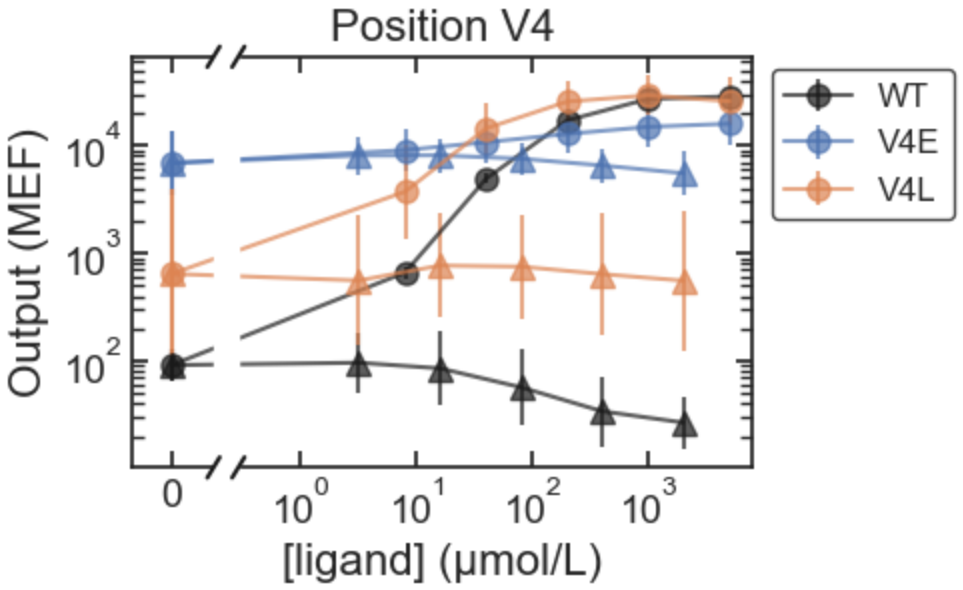

Position V321
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V321F
V321L

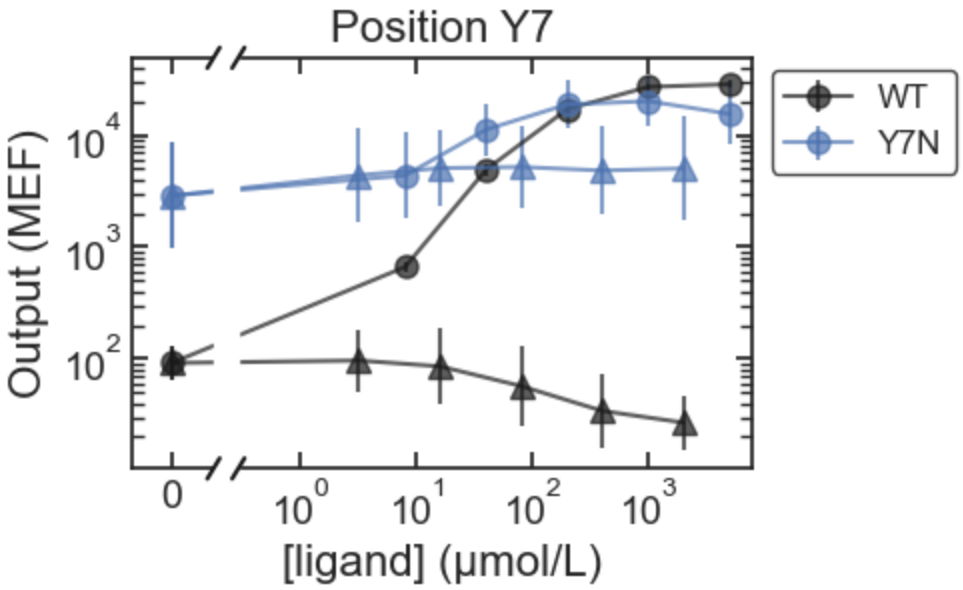

Position S345
Output (MEF)
10^4
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
S345L
S345P
S345T

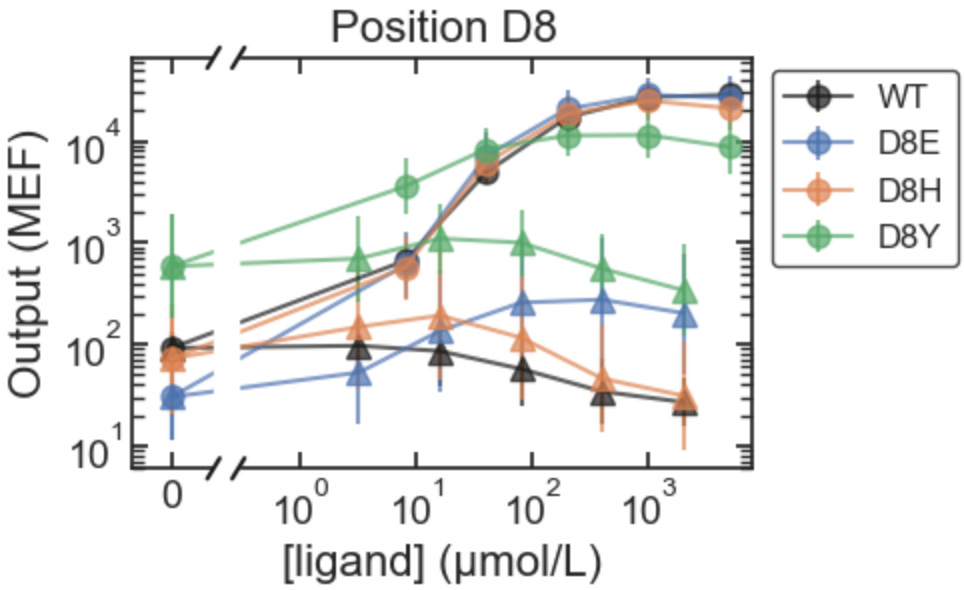
IPTG_only

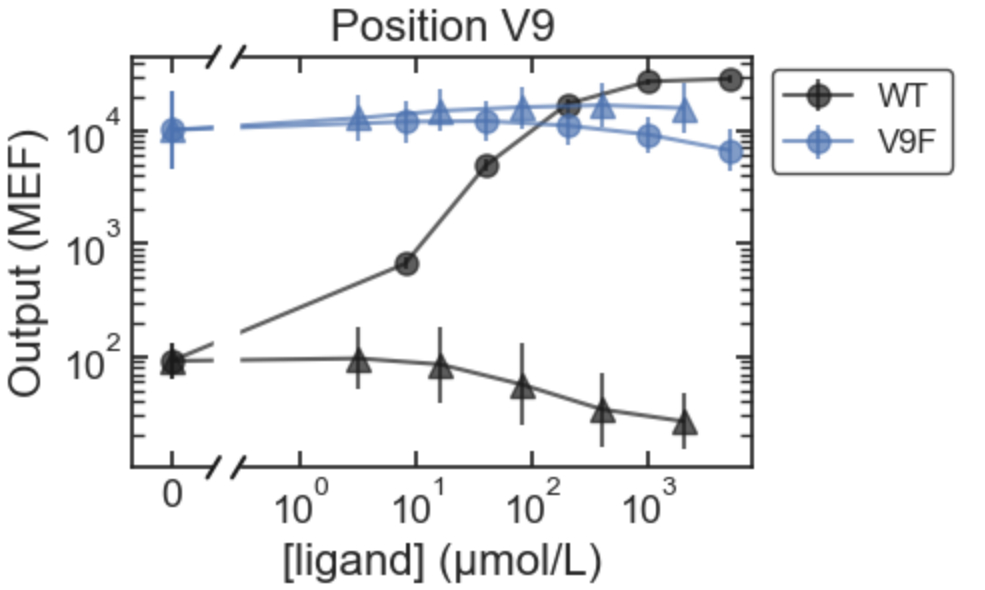

Position M1
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (µmol/L)
WT
M1I
M1T
M1V

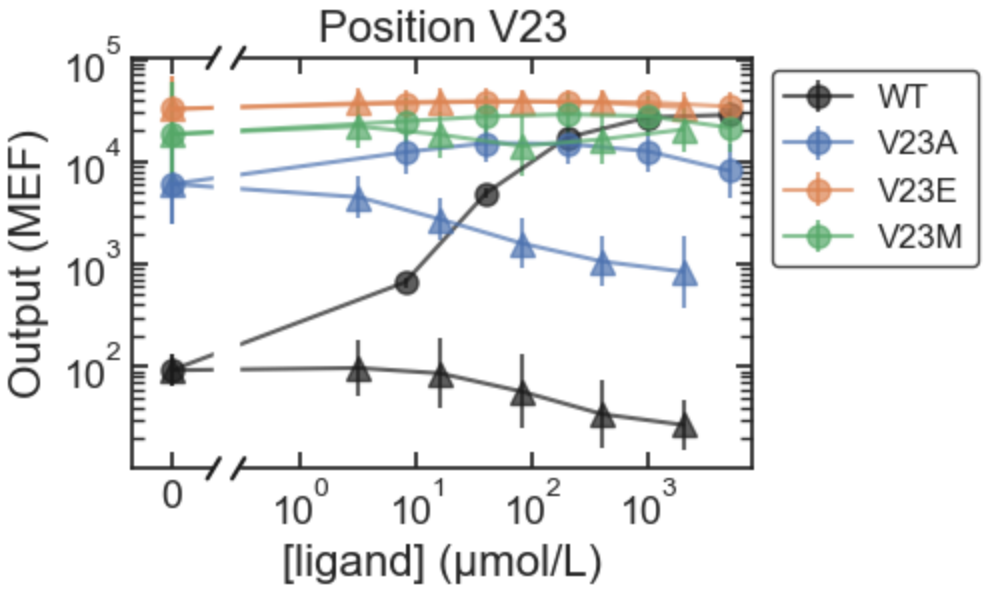

Position K2
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (µmol/L)
WT
K2E
K2R

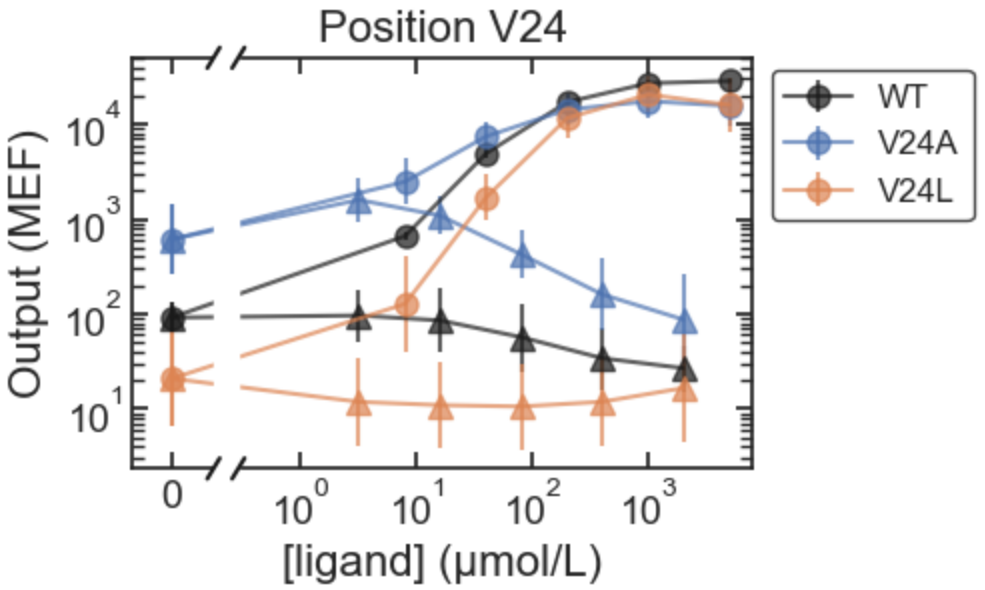

Position V4
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V4E
V4L

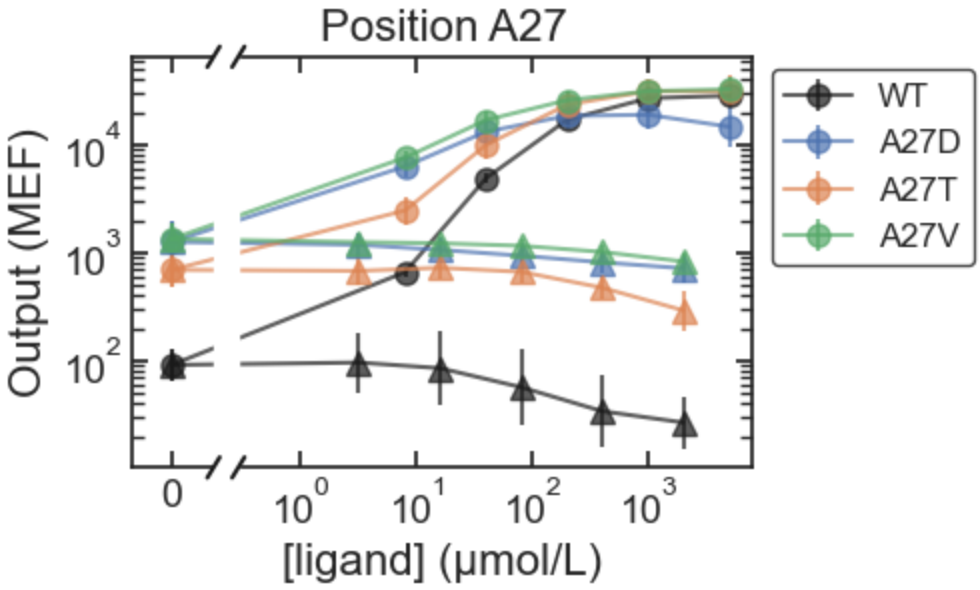

Position Y7
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Y7N

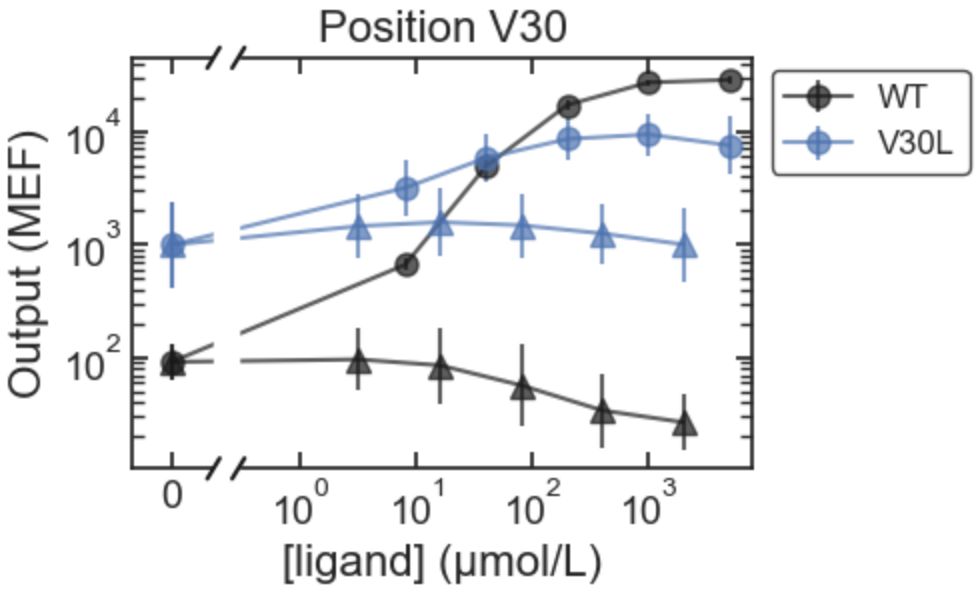

Position D8
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
D8E
D8H
D8Y

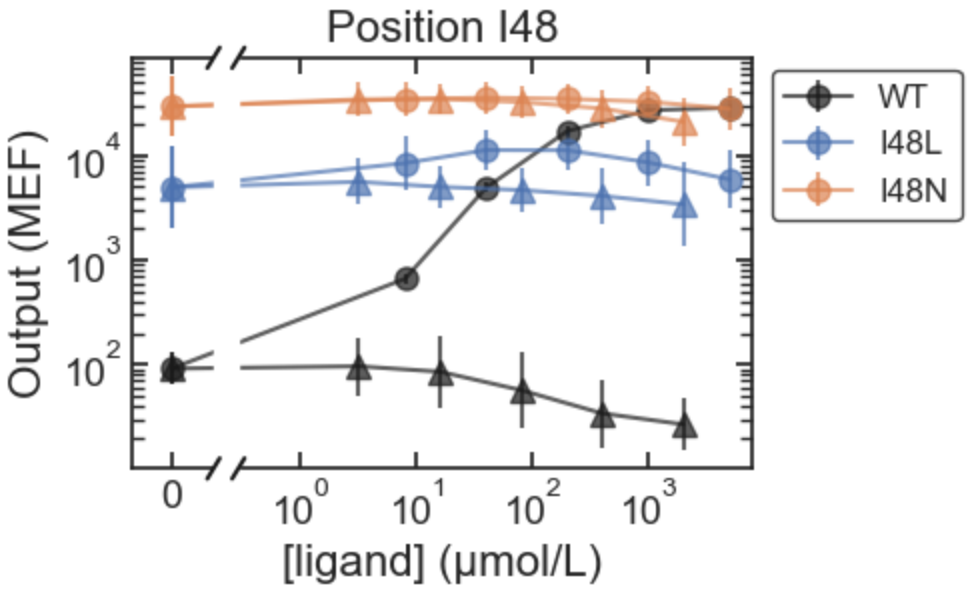

Position V9
Output (MEF)
[ligand] (μmol/L)
WT
V9F

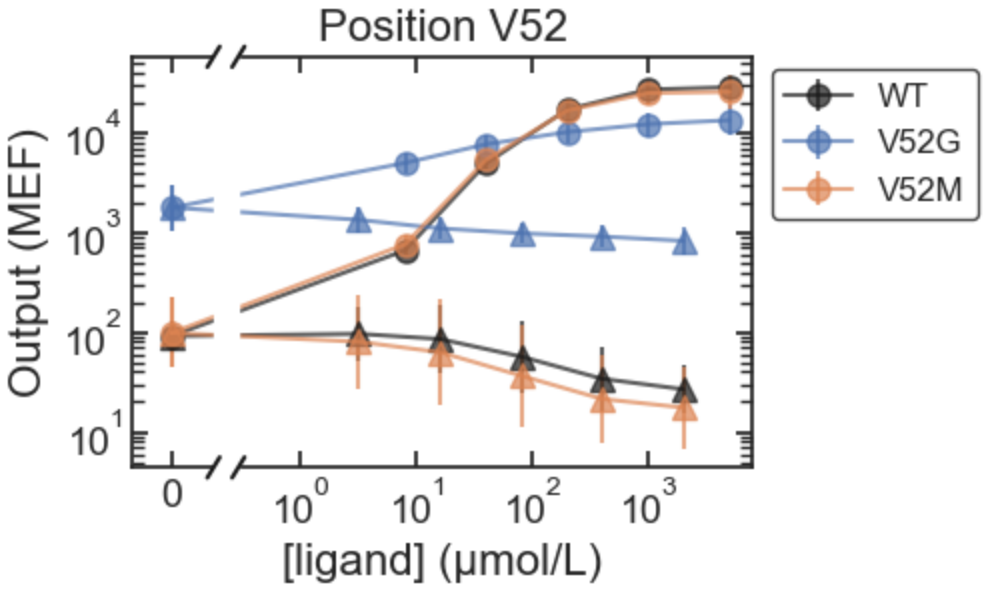

Position V23
Output (MEF)
[ligand] (μmol/L)
WT
V23A
V23E
V23M

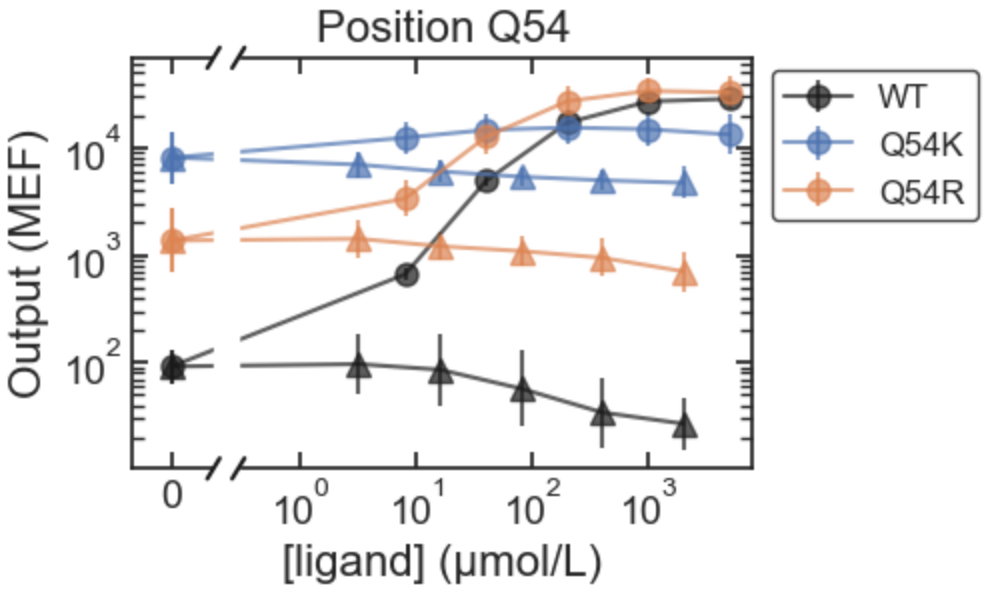

Position V24
Output (MEF)
[ligand] (μmol/L)
WT
V24A
V24L

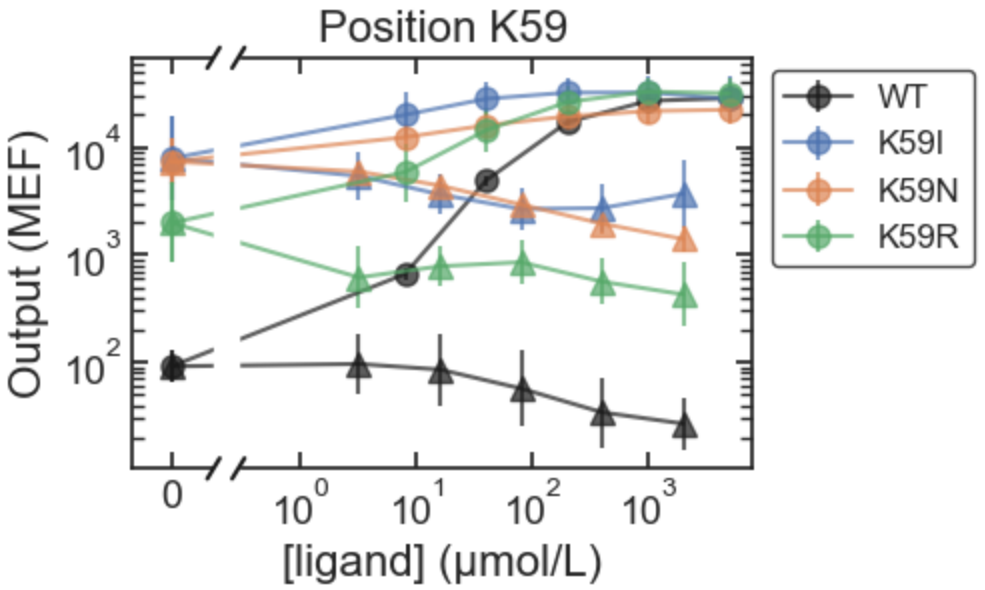

Position A27
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A27D
A27T
A27V

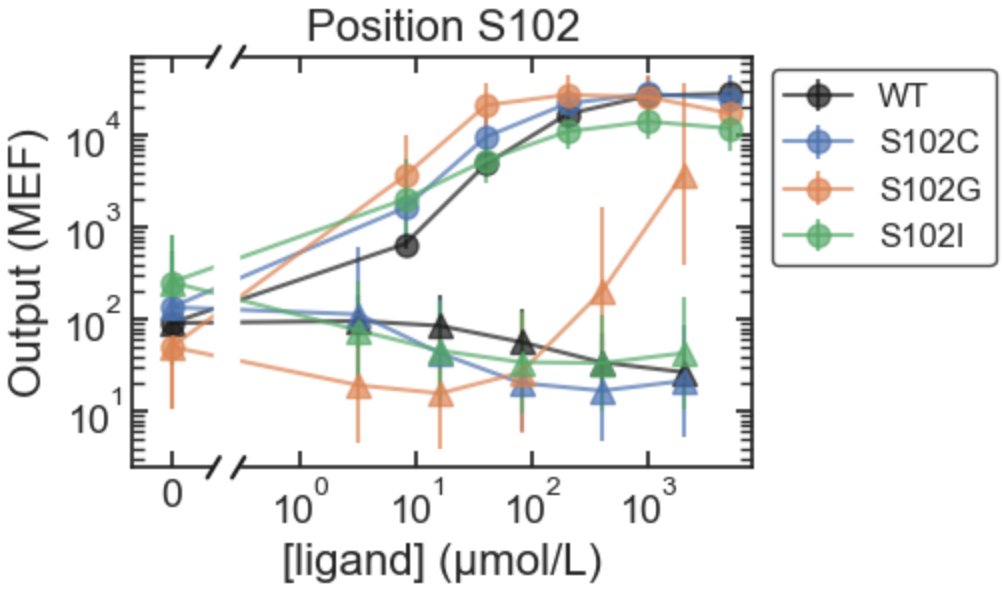

Position V30
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V30L

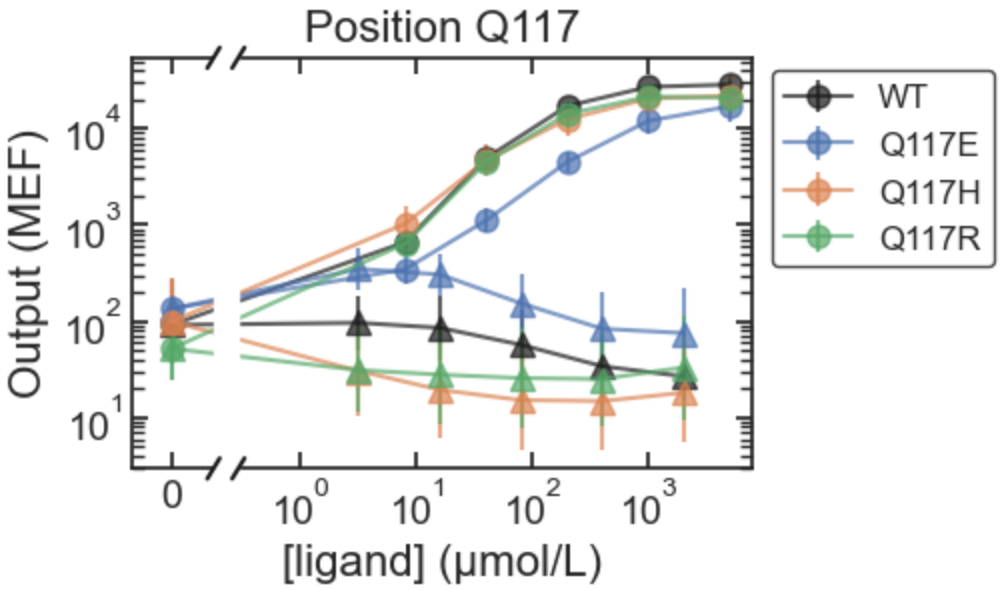

Position I48
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
I48L
I48N

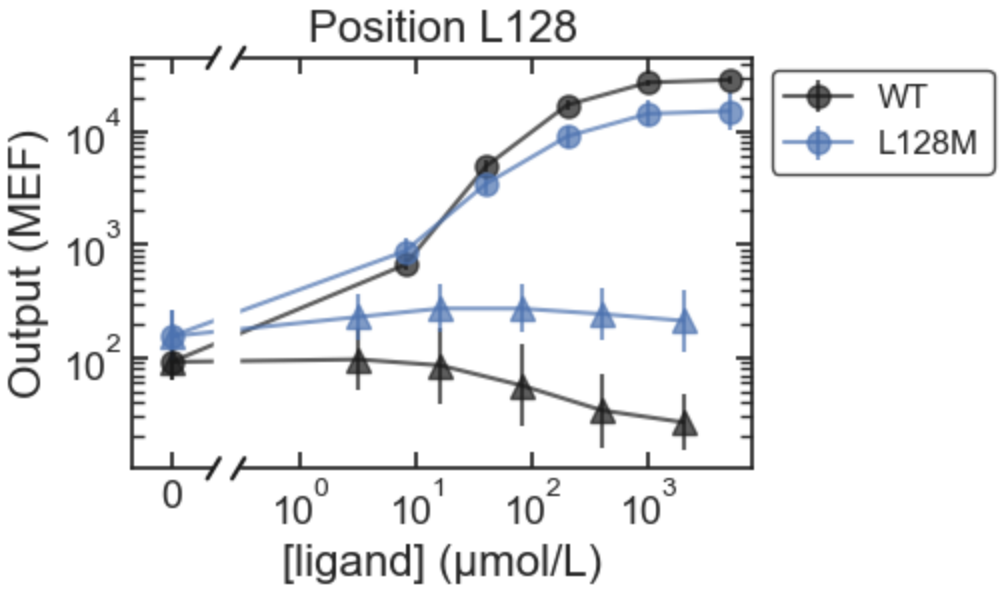

Position V52
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
V52G
V52M

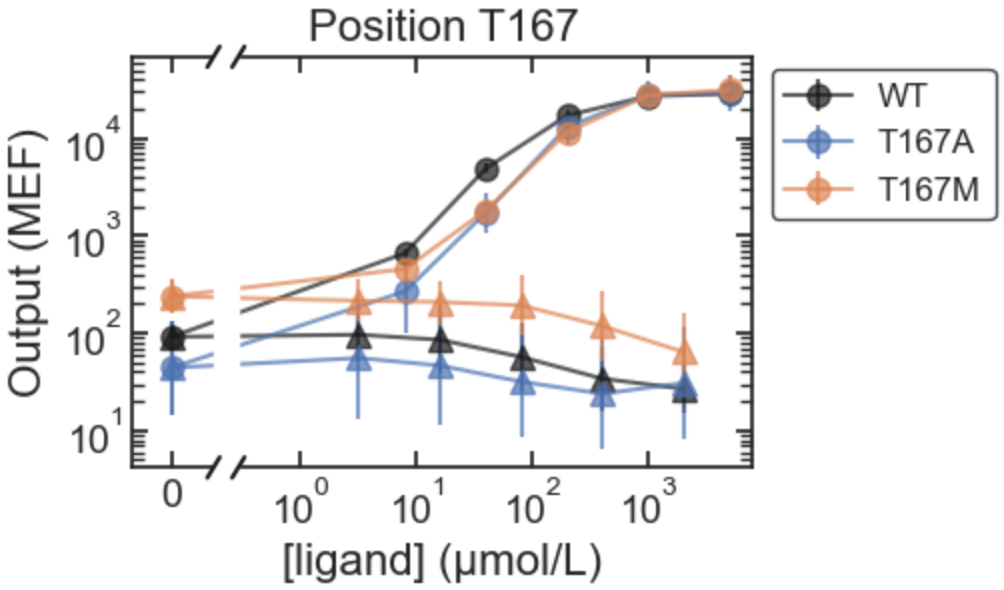

Position Q54
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
Q54K
Q54R

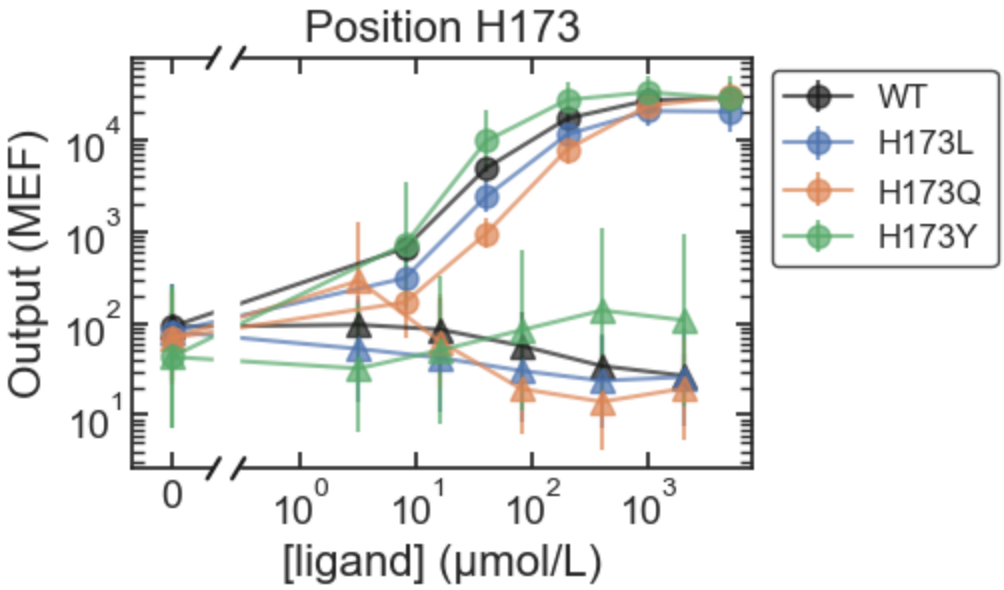

Position K59
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
K59I
K59N
K59R

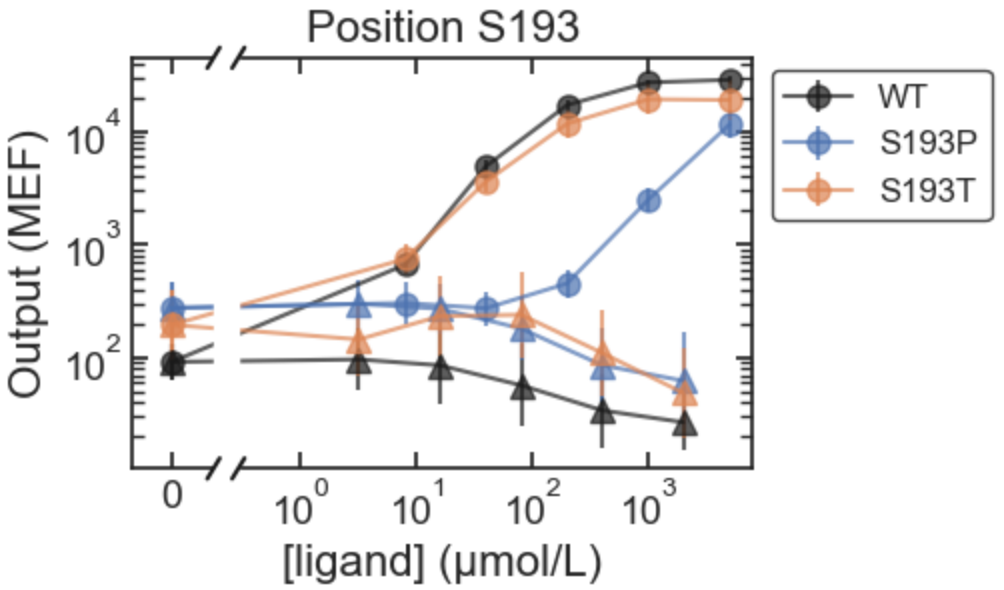

Position S102
Output (MEF)
[ligand] (μmol/L)
WT
S102C
S102G
S102I

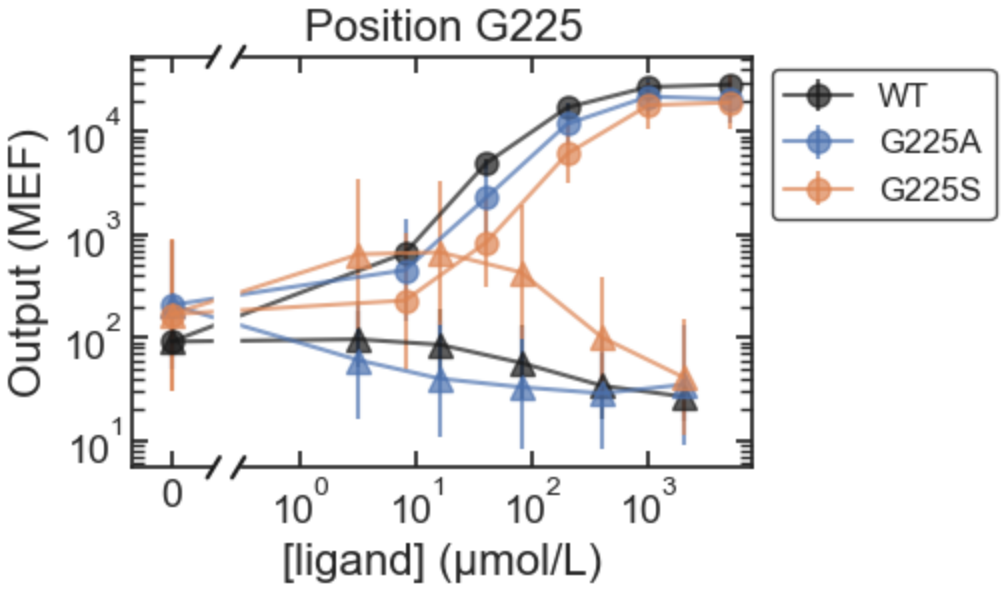

Position Q117
Output (MEF)
[ligand] (μmol/L)
WT
Q117E
Q117H
Q117R

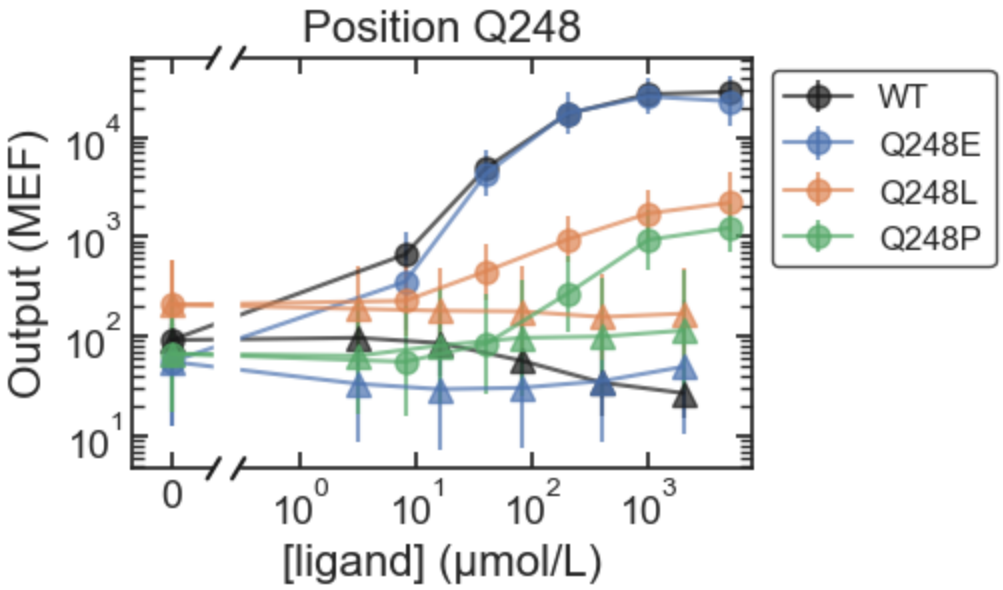

Position L128
Output (MEF)
[ligand] (μmol/L)
WT
L128M

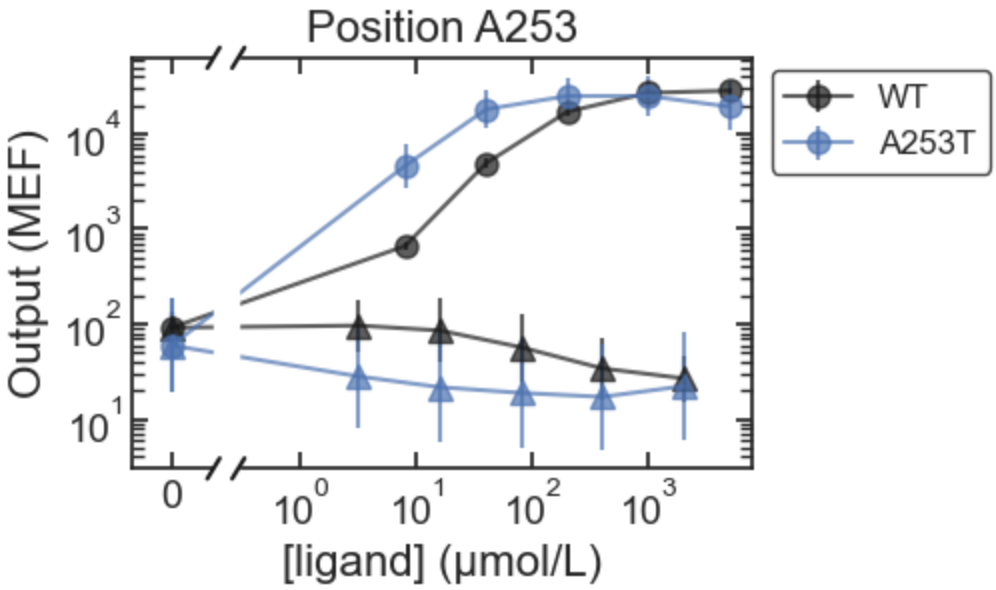

Position T167
Output (MEF)
[ligand] (μmol/L)
WT
T167A
T167M

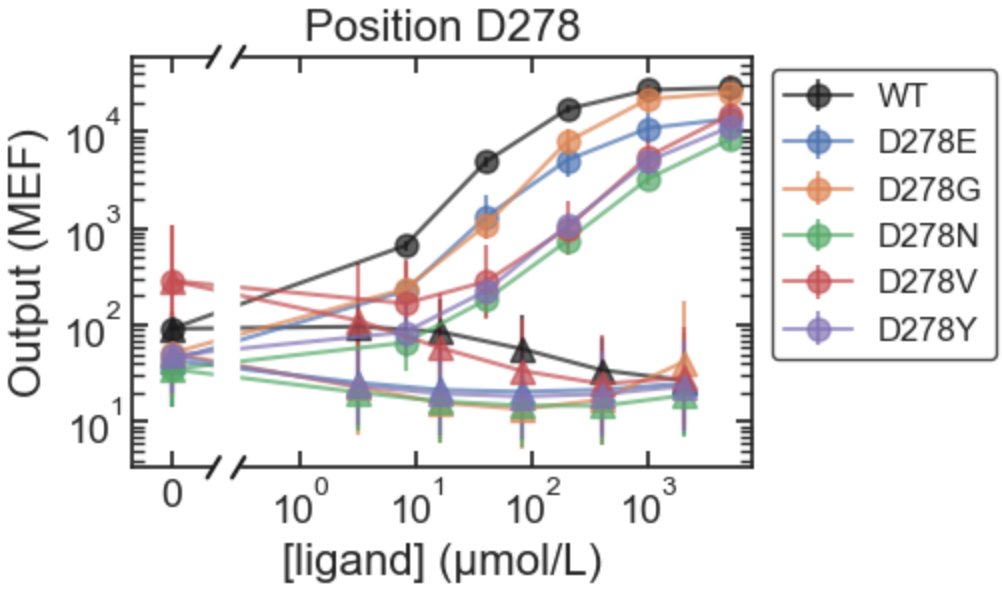

Position H173
Output (MEF)
[ligand] (μmol/L)
WT
H173L
H173Q
H173Y

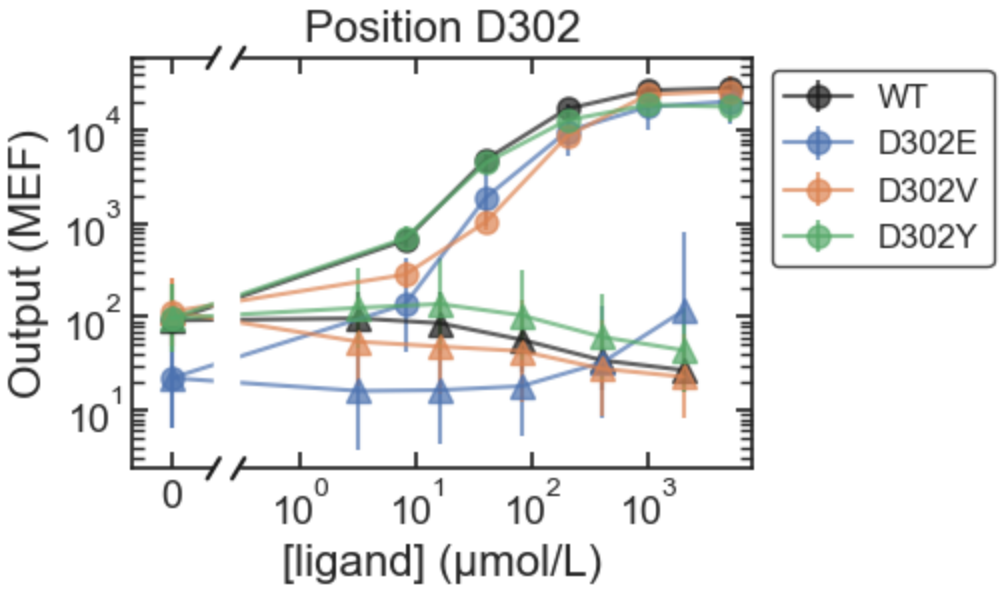

Position S193
Output (MEF)
[ligand] (μmol/L)
WT
S193P
S193T

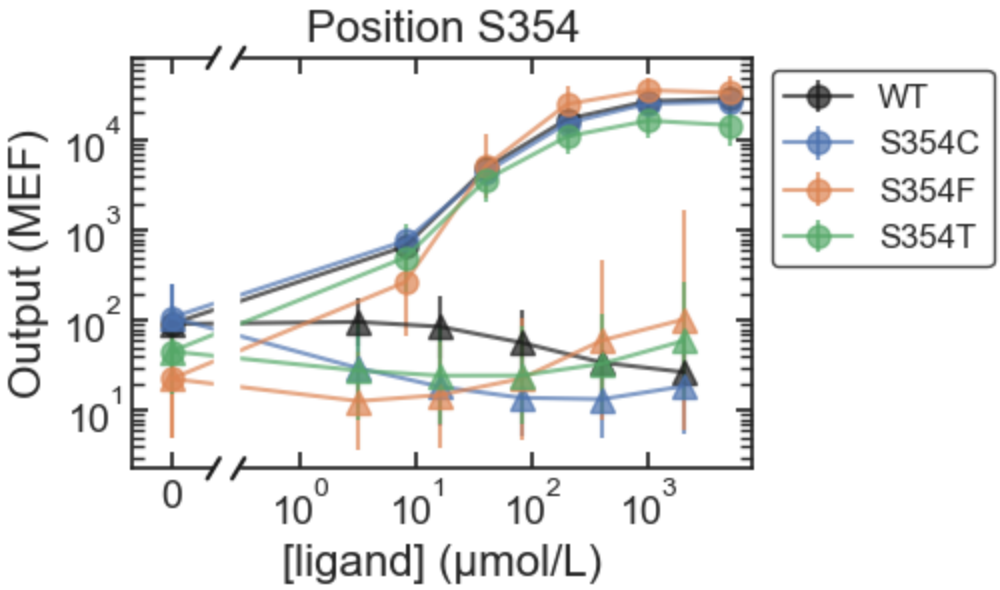

Position G225
Output (MEF)
[ligand] (µmol/L)
WT
G225A
G225S


Position Q248
Output (MEF)
[ligand] (µmol/L)
WT
Q248E
Q248L
Q248P

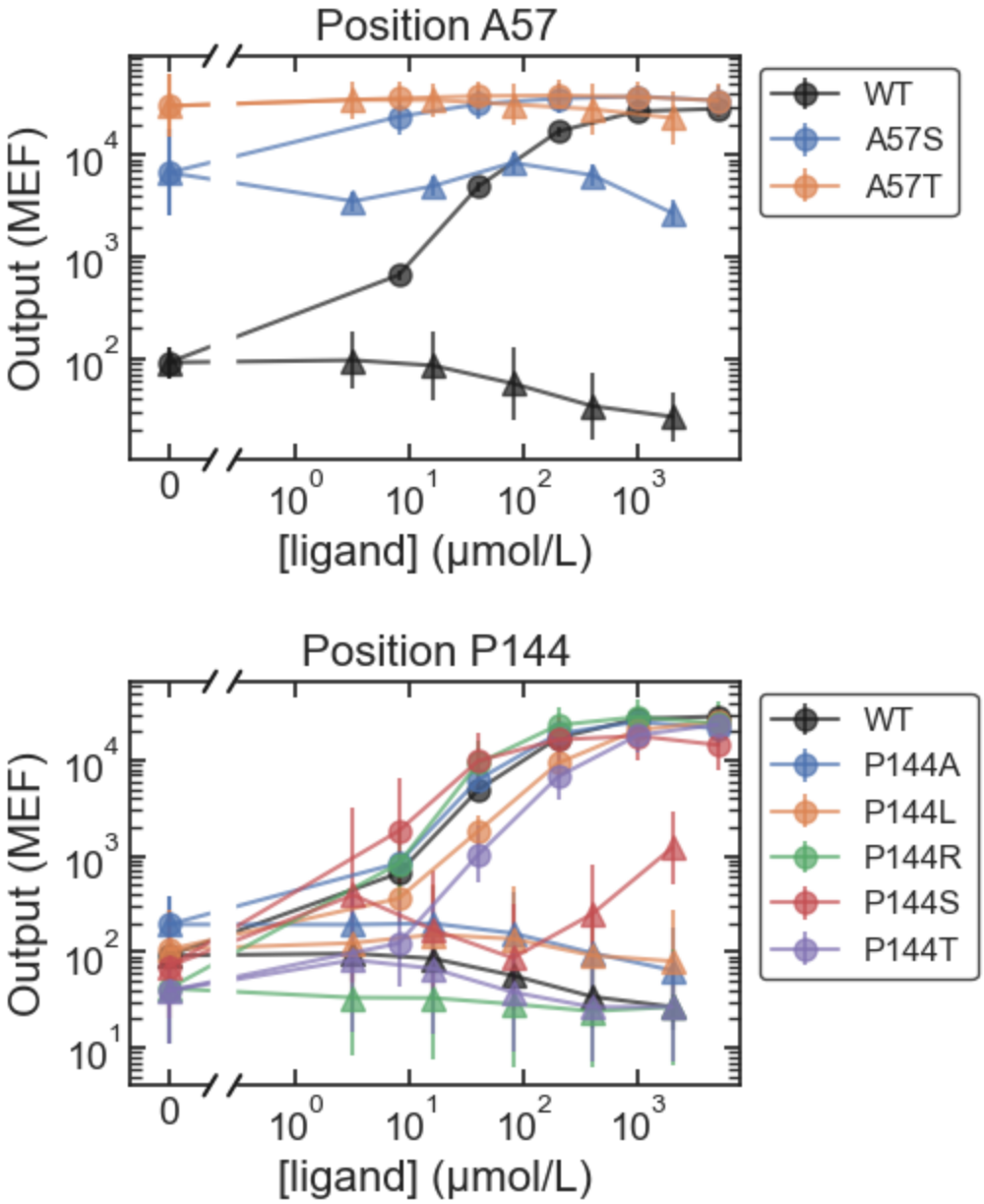

Position A253
Output (MEF)
[ligand] (µmol/L)
WT
A253T

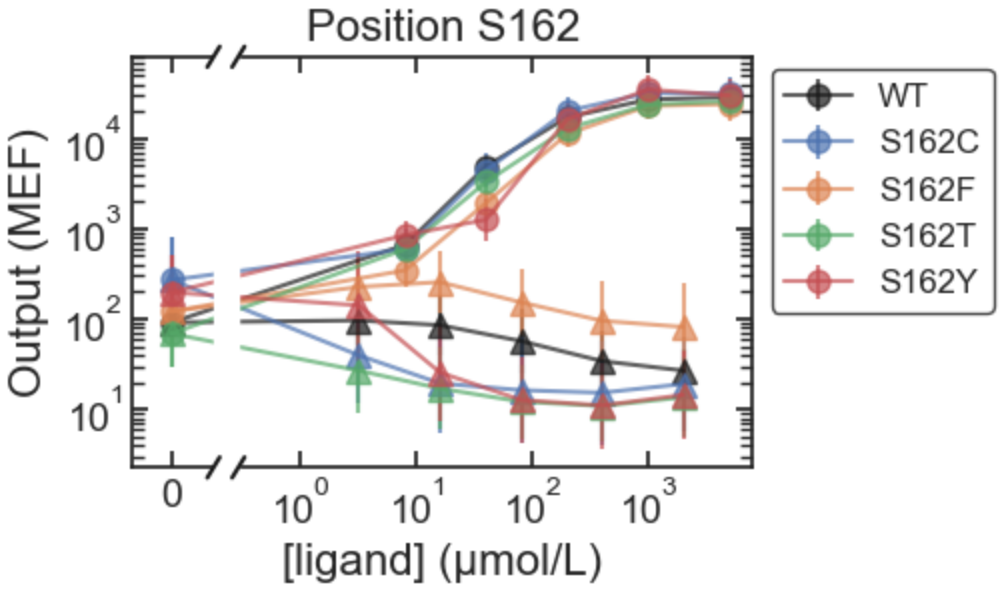

Position D278
Output (MEF)
[ligand] (μmol/L)
WT
D278E
D278G
D278N
D278V
D278Y


Position D302
Output (MEF)
[ligand] (μmol/L)
WT
D302E
D302V
D302Y

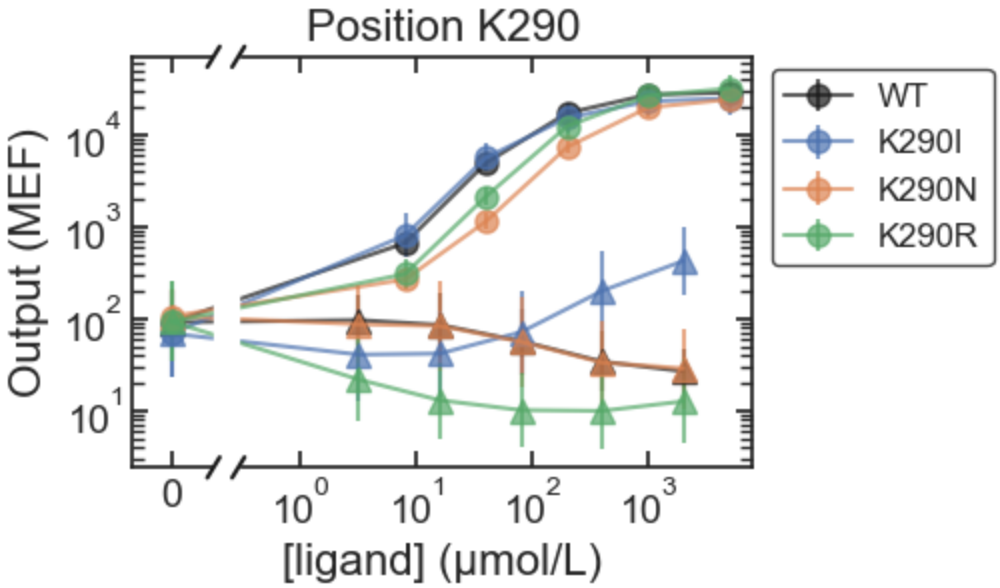

Position S354
Output (MEF)
[ligand] (μmol/L)
WT
S354C
S354F
S354T

ONPF_only


Position A57
Output (MEF)
10^4
10^3
10^2
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
A57S
A57T
Position P144
Output (MEF)
10^4
10^3
10^2
10^1
0
10^0
10^1
10^2
10^3
[ligand] (μmol/L)
WT
P144A
P144L
P144R
P144S
P144T



Both_noMWC



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